THE YANG–MILLS MASS GAP FROM ONE BOUND AT ONE SCALE

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Abstract. We prove the mass gap of four-dimensional lattice Yang–Mills theory at strong coupling for every compact simple gauge group, with explicit bounds. For the continuum theories built on tori we prove that a mass gap and a unique vacuum follow from one bound on partition functions at one physical scale. That bound is open at weak coupling, and the unrenormalised loop variables are shown to have a trivial limit.

1. Introduction

The Yang–Mills existence and mass gap problem asks for a quantum Yang–Mills theory on $\mathbb{R}^4$ with a compact simple gauge group that satisfies the axioms of constructive quantum field theory, and for a number $\Delta > 0$ such that the spectrum of the Hamiltonian lies in $\{0\} \cup [\Delta, \infty)$ [31]. In the Euclidean formulation the theory is a probability measure on gauge fields whose Schwinger functions satisfy the Osterwalder–Schrader axioms [46, 47]; the mass gap is then exponential decay of the truncated two-point functions of gauge-invariant observables at the rate Δ. The natural regularisation is Wilson's lattice gauge theory [61]: an element Q_e of $G \subset \mathrm{U}(N)$ sits on every link e of the lattice $a\mathbb{Z}^4$, and a configuration has the weight $\exp(N\beta \sum_p \operatorname{Re} \operatorname{Tr} Q_p)$, where Q_p is the ordered product around the plaquette p.[1] Asymptotic freedom forces the continuum limit $a \to 0$ to be taken with $\beta \to \infty$, that is, at weak coupling.

At small β the lattice theory has a unique state, exponential clustering and a transfer-matrix gap, by convergent expansions [48, 52]; Bałaban's renormalisation group [3, 4, 5] gives ultraviolet stability in finite volume at weak coupling (Dimock [16] presents the method for the scalar ϕ^4 model in three dimensions), and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [43]. No existing result gives a mass gap in physical units. This paper proves a lattice mass gap with explicit constants for every compact simple group, reduces the continuum gap to one estimate on one physical scale, proves that estimate at strong coupling, reduces it further to a decay of correlations across a slab of physical width $\ell/2$, and determines what the methods used here can and cannot reach. It also states precisely what is missing.

1.1. The lattice gap with explicit constants.

Let G be a compact connected Lie group with semisimple Lie algebra, identified with its image under a faithful unitary representation in $\mathrm{U}(N)$, and let $\kappa_G > 0$ be its least Ricci curvature for the bi-invariant metric $\langle X, Y \rangle = -\operatorname{Tr}(XY)$. For a simple group $\kappa_G = h^\vee/(2\ell_\rho)$, with $h^\vee$ the dual Coxeter number and ℓ_ρ the Dynkin index of the representation; for $\mathrm{SU}(N)$, $\kappa_G = N/2$.[2]

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[1] This is the normalisation of [56]. The conventional inverse coupling is $\beta_W = N^2\beta$, and for SU(2), identified with the unit quaternions, $\beta_W = 4\beta$.

[2] For a bi-invariant metric on a compact Lie group the Ricci curvature is $\mathrm{Ric}(X, X) = -\frac{1}{4}B(X, X)$ with B the Killing form, $B(X, Y) = \operatorname{Tr}(\mathrm{ad}_X \mathrm{ad}_Y)$. For $\mathfrak{su}(N)$, $B(X, Y) = 2N \operatorname{Tr}(XY)$, so $\mathrm{Ric}(X, X) = \frac{N}{2}\langle X, X \rangle$ in the metric $\langle X, Y \rangle = -\operatorname{Tr}(XY)$ of the defining representation, which is the value $\kappa_G = N/2$; in general

Theorem A. *Suppose $N|\beta| < \frac{2}{29}\kappa_G$. For every gauge-invariant local observable the expectations in boxes with free boundary converge, along every exhausting sequence of boxes, to a state ω that is invariant under the lattice symmetries; truncated correlations of Lipschitz observables decay at every rate below*

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0,$$

and for $\beta \geq 0$ the state is reflection positive and its transfer matrix has spectral gap $m \geq m_G(\beta)$. For $G = \mathrm{SU}(2)$ the same holds for $|\beta_W| < \frac{1}{3}$ with $m \geq -\log(3\beta_W)$.

Two ingredients carry the proof. The first is a gauge: a field of straight lines on $\mathbb{Z}^4$ that contains exactly one link of every plaquette (Sec. 3). Setting these links to the identity loses no gauge-invariant information, and every remaining link then interacts with exactly twelve others instead of eighteen. The second is a Hessian bound in this gauge, $|\operatorname{Hess} S| \leq \frac{29}{2}$ for the action $S = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$, proved by a divergence identity and a positivity statement for one plaquette (Sec. 5); with the Ricci curvature of G it gives a Bakry–Émery bound [7, 8] with an explicit decay rate. Without the gauge no Bakry–Émery constant is positive beyond $|\beta_W| = N^2/32$ (Theorem H.19), so the gauge enlarges the range of every group by the factor $32/29$ and supplies the rate. The SU(2) statement combines the gauge with the sharp Kantorovich modulus $\frac{1}{4}$ of the von Mises–Fisher family, proved in Appendix C, in Dobrushin's criterion [17, 18].

1.2. **A continuum mass gap from one cube.** The lattice gap of Theorem A is measured in lattice units and vanishes in physical units as $a \to 0$. The second result isolates a single estimate, posed on one scale, that controls the gap in physical units. On a periodic lattice let $\mathcal{P}_x$ resample the links of the cube of side n at x from their conditional law under the Wilson measure μ. The condition $\mathrm{BP}(n, \varpi)$ asks that

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4} \sum_x \mu\big((f - \mathcal{P}_x f)^2\big)$$

for every gauge-invariant f: the dynamics that resamples cubes, each link at rate one, has spectral gap at least ϖ on gauge-invariant functions. The condition involves no gauge fixing and no supremum over boundary conditions.

Theorem B. *Let $a_k \to 0$ and $\beta_k \geq 0$ be couplings, and suppose that there are $\varpi > 0$, $\ell > 0$ and integers $n_k \geq 2$ with $n_k a_k \leq \ell$ such that $\mathrm{BP}(n_k, \varpi)$ holds for the Wilson measure at β_k on all large tori. Then every continuum limit of the corresponding lattice theories has Hamiltonian H with $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$,*

$$\Delta = \frac{16\varpi}{625 e^2 \ell},$$

and 0 is a simple eigenvalue; if the limit is nontrivial, H has a mass gap of at least Δ.[3]

The proof (Sec. 6) turns the spectral gap into decay of covariances through a finite-speed bound for the cube dynamics, at a rate of order $1/n$ in lattice units, hence of order $1/\ell$ in physical units, and passes this decay through the Osterwalder–Schrader reconstruction.

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the Killing form is $2h^\vee$ times the trace form of the adjoint representation and $2\ell_\rho$ times that of ρ, which gives $h^\vee/(2\ell_\rho)$.

[3]The constant comes from the finite-speed bound: the cube chain decorrelates two observables at lattice distance s at the rate $\varpi/(e^2 q_n n)$ per unit of s, where $q_n = (2 + 1/n)^4$ is the number of cubes of side n that meet a given one, divided by n^4; $q_n \leq (5/2)^4 = 625/16$ for $n \geq 2$, and $na \leq \ell$ converts the lattice rate into the physical rate $\varpi/(e^2 q_n\, na) \geq 16\varpi/(625 e^2 \ell)$. The factor e^2 is the loss in the Poisson tail estimate of Lemma 6.2.

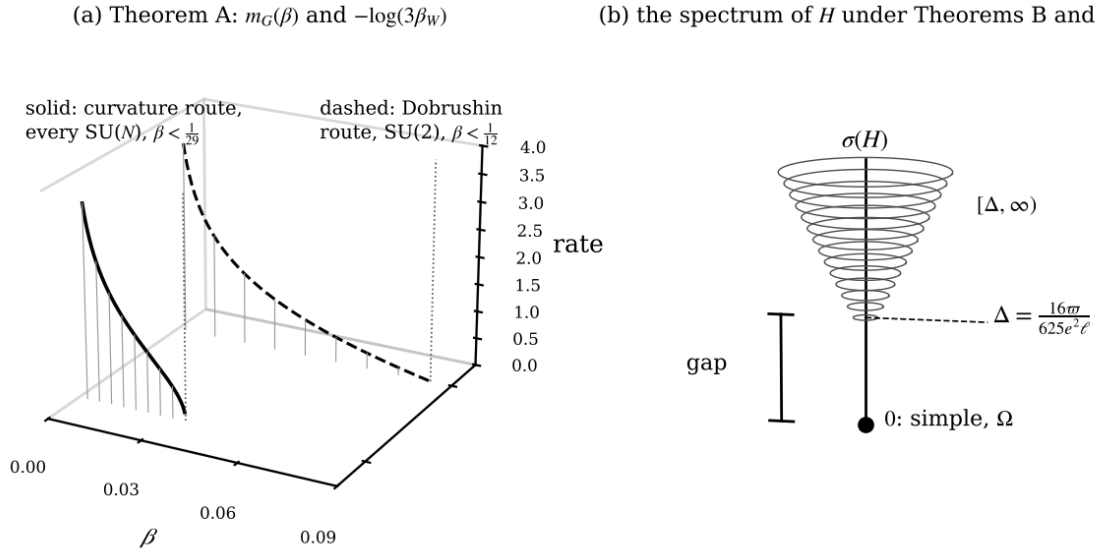


FIGURE 1. (a) The rates of Theorem A: the curvature route gives $m_G(\beta) = \operatorname{arccosh}(1 + \frac{1}{12}(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}))$ for $N|\beta| < \frac{2}{29}\kappa_G$, which for every SU($N$), where $\kappa_G = N/2$, is the one curve drawn for $\beta < \frac{1}{29}$; the Dobrushin route for SU(2) gives $-\log(3\beta_W) = -\log(12\beta)$ for $\beta < \frac{1}{12}$. (b) The spectrum of the Hamiltonian under Theorems B and E: 0 is a simple eigenvalue with eigenvector Ω, and the rest of the spectrum lies in $[\Delta, \infty)$ with $\Delta = 16\varpi/(625e^2\ell)$.

1.3. **The theories constructed on tori.** Theorem B takes the lattice theory to infinite volume before the lattice spacing is sent to zero. A continuum theory may also be built in the opposite order: at a fixed physical side L_p of a symmetric torus the mesh $a_N = 1/N$ is sent to zero, with couplings $\beta_{N,p}$, and the side is sent to infinity afterwards. Section 7 considers any such construction whose outcome is a mixture, over a compact space X of phases carrying a measure ν of full support, of theories S^χ each satisfying the Wightman axioms with a unique vacuum; this is hypothesis (W), stated in full in Definition 7.2; it is the shape of the conclusion claimed by the unrefereed Lean 4 construction [45, 62], on which nothing below depends. At finite side the lattice state is a thermal state at temperature $1/L_p$, and a decay estimate for its two-point functions is not what a gap controls, so a gap for such a theory needs more than (D). Call (U) the hypothesis that the same limit is obtained when the Euclidean time extent of the torus is sent to infinity first, at each mesh, and (D_ℓ) the cube condition $\mathrm{BP}(n_N, \varpi)$ with $n_N a_N \le \ell$ on all cylinders $(\mathbb{Z}/NL_p\mathbb{Z})^3 \times (\mathbb{Z}/M\mathbb{Z})$, $M \ge 2n_N + 4$, at the couplings $\beta_{N,p}$ of the construction.

Theorem E. *Under* (W), (U) *and* (D_ℓ) *the phases S^χ all coincide, and the Wightman theory of the construction has Hamiltonian H with $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$, $\Delta = 16\varpi/(625e^2\ell)$, and 0 a simple eigenvalue.*

This is Theorem E; it is a theorem about the three hypotheses, and it rests on nothing outside this paper. Two of the hypotheses can be weakened: (U) may be replaced by the single renormalisation-free statement (Z) that the partition function of the symmetric torus, divided by the NL_p-th power of the leading eigenvalue of the transfer matrix of the cylinder of the same side, tends to 1 (Corollary 7.11, through a sharp inequality bounding the thermal correction of any slab observable by that quantity, Lemma 7.8); and (D_ℓ) follows from the slab decorrelation of Theorem D at one physical separation, since that theorem holds on cylinders (Corollary 11.9, Proposition 7.13). Its proof builds the transfer matrix of a cylinder as a positive integral operator with a strictly positive kernel on the gauge-invariant

slice functions, deduces from the cube condition a gap of the cylinder's zero-temperature Hamiltonian, carries the decay of the renormalised two-point functions through the two limits with a constant that depends only on the test functions, and identifies the limit of the Euclidean vectors of the lattice theories with the vectors of the Wightman theory; the triviality of the phase decomposition is a consequence of the same decay. What Theorem E does not do is prove (D_ℓ): along the tuning of the construction the coupling tends to infinity, and the cube condition at weak coupling is exactly statement (D) of Sec. 1.8.

1.4. **One bound at one scale.** The hypotheses of Theorem E can be replaced by a single one that mentions neither cubes, nor slabs, nor a gap. For a spatial torus s with sides s_1, s_2, s_3 and volume $V(s)$ let $\mathcal{Z}_s(T)$ be the partition function of the torus $s \times (\mathbb{Z}/T\mathbb{Z})$ divided by the T-th power of the leading eigenvalue of the transfer matrix of s; it is the trace of the T-th power of the normalised transfer matrix, and $\log \mathcal{Z}_s(T)$ is the thermal part of the free energy at inverse temperature T. Hypothesis (B^*) asks that, at one time extent T_0 of a fixed physical size,

$$\log \mathcal{Z}_s(T_0) \le \gamma V(s), \qquad 2\gamma(96T_0)^3 \le \tfrac{1}{20},$$

for the spatial tori whose sides lie between $2T_0$ and $48T_0$.

Theorem F. *Under* (W) *and* (B^*) *at one physical time extent t_0, the phases S^χ coincide and the Wightman theory of the construction has $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$ with 0 a simple eigenvalue and $\Delta \ge 0.172/t_0$.*

The full statement is in Section 7.11. Its proof doubles the time extent: positivity of the transfer matrix and the exchange of the four directions of the torus turn the bound at T_0 into a smaller bound at $2T_0$ on a wider family of tori, and the iteration, which decreases doubly exponentially, produces the gap of every large torus together with (Z) (Figure 8). The hypothesis is equivalent, up to a factor two in γ, to a finite inequality between two partition functions, $Z(s, T_0)^2 \le e^{\gamma V(s)} Z(s, 2T_0)$, in which no eigenvalue appears (Proposition 7.57). It is not proved here at weak coupling, and it cannot be easy: it contains the lattice mass gap at the scale T_0 (Proposition 7.58), and for SU(2) the gap of a torus of fixed lattice size closes as the coupling grows (Theorem 7.59), so that T_0 must grow along the tuning (Corollary 7.60). Appendix I shows that the shortcuts which avoid the renormalised theory, among them the limit of the parallel transporters themselves, lead to trivial theories (Theorem I.3, Proposition I.4).

1.5. **Where the cube condition holds.**

Theorem C. (i) *At $\beta = 0$, $\mathrm{BP}(n, 1)$ holds for every n. For $21N|\beta| < \kappa_G$ and n large, $\mathrm{BP}(n, 1 - \alpha_n)$ holds with $\alpha_n = N|\beta|(18n - 3)/((\kappa_G - 21N|\beta|)n^2) \to 0$; for* SU(2) *the range is $|\beta_W| < \frac{5}{18}$, and for* SU(3) *it contains $0 \le \beta_W \le 0.45$.*

(ii) *For the free scalar field of lattice mass μ in d dimensions the best constant is $\varpi = \frac{\mu^2}{n^d}\langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1}\mathbf{1}\rangle$, with $-\Delta_D$ the Dirichlet Laplacian of the cube, and with $s = n\mu$ (the mass times the physical size of the cube)*

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2 + 10d} \le \varpi \le \frac{s^2}{s^2 + 16d/9}.$$

(iii) *For $G =$ SU(2) and $|\beta_W| \le \frac{7}{20}$, $\mathrm{BP}(n, \varpi)$ holds for every $n \ge 2$ with $\varpi = \frac{106}{1933} \cdot 517^{-4}$, independent of n and of the torus, and $\mathrm{BP}(n, 1 - 259/n)$ holds for $n \ge 260$. For $0 \le \beta_W \le \frac{9}{25}$ every torus limit is reflection positive and its transfer matrix has spectral gap at least $7/(625 \cdot 2957\, e^2)$.*

Part (ii) is the model case of the continuum tuning: the constant is uniform in the lattice spacing at fixed $m\ell$, and it vanishes as $m \to 0$. The cube condition is therefore a quantitative statement of mass generation at the scale ℓ. Part (i) is proved in Sec. 8 by the gauge of Theorem A and a weighted Dobrushin criterion for blocks of links. No argument

built on single-link updates in this gauge passes $\beta_W = \frac{1}{3}$ for SU(2) (Proposition 8.12). Part (iii) passes that value by resampling *stars*, the eight links at a vertex of one parity (Sec. 9): the perturbation of one plaquette at a vertex enters the law of a star only through the relative variable of two of its links, and no plaquette is internal to a star.

1.6. **Mass generation at one scale.** Statement (D) below asks for the cube condition along the continuum tuning. Section 11 reduces it to a hypothesis of a different kind, closer to physics. Split a box of the lattice, all of whose widths lie between $2d-1$ and $md-1$, into two outer parts of width d and a middle of width $d-1$ in one coordinate, fix an arbitrary configuration outside the box, and let ρ be the maximal correlation between the gauge-invariant observables of the two outer parts under the conditional law. The hypothesis $(\mathrm{M}_{d,m}(c))$ asks that $\rho \le c$ for all such boxes, splittings and boundary conditions. Along the tuning $d \approx \ell/(2a)$, so this is a decay of correlations across a slab of physical width about $\ell/2$, uniform in the boundary condition.

Theorem D proves that $(\mathrm{M}_{d,m}(c))$ with $c < \frac{1}{2}$ and m large in terms of c gives $\mathrm{BP}(2d-1, (1-2c)^4/32)$, hence, along the tuning, statement (D) and the continuum gap $(1-2c)^4/(1250\, e^2 \ell)$ in every nontrivial limit. The proof passes through a gap for the dynamics that resamples overlapping patches of a box, obtained from the decorrelation of the slabs by a Markov factorisation, and a finite-size criterion for gauge-invariant functions in the spirit of those of Knabe [34] and of Gosset and Mozgunov [25] for quantum spin systems. The hypothesis also gives exponential decay of gauge-invariant correlations under every boundary condition, at rate $\log(1/c)/d$ per lattice spacing (Proposition 11.11). It holds at strong coupling at fixed d and m, and in two dimensions along the weak-coupling tuning (Theorem H.8). In four dimensions it is not proved.

1.7. **The limits of the methods.** Appendix H proves that several natural routes to (D) cannot work, and locates the threshold of Theorem D.

(a) *The threshold $\frac{1}{2}$ is sharp.* For a massless Gaussian chain the maximal correlation of the slab mechanism equals $\frac{1}{2}$ exactly, and the gap of the corresponding block dynamics on M blocks is at most $16/(3M)$; so no decorrelation constant $c \ge \frac{1}{2}$ can give a gap uniform in the size (Proposition H.4).

(b) *Two dimensions.* For SU(2) in two dimensions the middle slab of a box integrates out, after an axial gauge, to a convolution power of the heat kernel, and $(\mathrm{M}_{d,m}(c))$ holds with $c = 20\, e^{-3g^2(da)^2/4}$, which tends to zero as the physical size of the box grows (Theorem H.8). The same condition for strips one link wide fails along the tuning: a pinned strip is a Brownian bridge on SU(2), and the antipodal boundary condition breaks a global symmetry, with correlation close to one (Proposition H.17).

(c) *Curvature.* The Bakry–Émery constant of the Wilson measure without gauge fixing is at most $N/2 - 16N|\beta|$ (Theorem H.19); in the gauge of Theorem A the constant $\frac{29}{2}$ cannot be replaced by anything below $\frac{38}{3}$, and no Bakry–Émery argument for the gauge-fixed measure reaches $|\beta_W| = \frac{3}{19}$ (Theorem H.20). Every smooth plaquette action, such as an effective action at a coarse scale, obeys a bound of the same kind (Proposition H.21).

(d) *Entropy, covariances and fixed blocks.* Block entropy factorisation implies the cube condition and so cannot be easier (Proposition H.22); the link covariances of every gauge-invariant measure are $\frac{1}{4}I$ and carry no information on the coupling (Proposition H.23); and a single-link spectral gap, or any gap for blocks of fixed lattice size, with constant C_k satisfying $C_k a_k \to 0$, forces every continuum limit to be trivial (Proposition H.24).

(e) *A gap in the limit is not enough.* If the lattice gaps m_k tend to zero, a choice of lattice spacings proportional to m_k produces a nontrivial continuum limit with a mass gap; this limit is kinematic and is not identified with Yang–Mills theory (Theorem H.27). The content of (D) and (E) is that the gap and the nontriviality hold along the tuning of asymptotic freedom.

1.8. **What remains.** Theorems A and C(i), (iii) are strong-coupling theorems. The Clay problem[4] needs the opposite end of the coupling range: the continuum limit lives at $\beta \to \infty$, which the ranges of these theorems do not reach. Two statements would complete the proof of a continuum mass gap, and neither is proved here:

(D) $\mathrm{BP}(n_k, \varpi)$ with $n_k a_k \le \ell$ along the asymptotic-freedom tuning $\beta_k \to \infty$, uniformly in k;

(E) convergence of the renormalised Schwinger functions to a nontrivial limit satisfying the Osterwalder–Schrader axioms.

Given (D) and (E), Theorem B gives the mass gap $\Delta \ge 16\varpi/(625e^2\ell)$, and by Theorem D statement (D) follows from $(\mathrm{M}_{d_k,m}(c))$ along the tuning with $c < \frac{1}{2}$. Section 12 explains why (D) is the heart of the matter: by Theorem C(ii) it cannot hold without a dynamically generated mass, it fails at fixed cube size at weak coupling, and it fails in the Coulomb phase of the abelian theory, so its proof has to use the non-abelian structure of G at every scale up to ℓ. For the theories built on tori, the statement of the Clay problem reduces, by Theorem E with the construction of [45] taken as the existence input, to (Z) or (U) together with (D_ℓ), or with the slab decorrelation $(\mathrm{M}_{d_N,m}(c))$, $c < \frac{1}{2}$, at the couplings of the construction; these remain open, and so does the mass gap. What each of them asks is located precisely: the slab decorrelation is the maximal correlation of two layers of a box, a quantity which in the vacuum of a cylinder equals $e^{-m_{\mathrm{lat}}s}$ exactly, so that only its uniformity in the boundary condition is missing (Section 11.6), a uniformity that holds at strong coupling in every box (Proposition 11.27), can fail at no fixed box (Proposition 11.14), only along the tuning, and is equivalent, up to constants and a change of scale, to a block Poincaré inequality uniform in the boundary condition (Section 11.7); and (Z) is invisible to the correlation functions, the gap and the positivity that (W) provides, and follows neither from (U) nor from the slab decorrelation nor from the cube condition, even taken together (Proposition 7.14, Proposition 7.17 and Proposition 7.19), while it does follow from a bound on the number of states of the cylinder per unit of physical energy (Proposition 7.20), holds at strong coupling (Proposition 7.23) and holds in two dimensions along the tuning, where that number is computed exactly (Proposition H.11); no bound of the strong-coupling shape can reach a tuning with a particle (Proposition 7.26), and (Z) is equivalent in strength to two conditions, the entropy bound at one physical scale and the vanishing, faster than the inverse side, of the Casimir defect of the vacuum energy under halving the spatial torus (Proposition 7.30, Lemma 7.31), so that a proof of it in four dimensions at weak coupling must count the states below the cutoff per unit of physical volume at one scale and the low-lying states of long tubes per unit of length; positivity in the four directions writes both conditions, and (Z) itself, as doubling defects of the free energies of finite tori, differences at an absolute precision $O(1)$ in which the extensive part cancels (Proposition 7.34), the gap and the free energy per site do not control them (Proposition 7.35), and the Casimir condition follows from a finite-size bound on the vacuum energies with an exponential decay in the physical sides, the form a massive expansion at that scale would take (Proposition 7.36). Once the spectral gap is available on the tori of the halving chain, the smallness of the Casimir coefficients is not needed at all: (Z) follows from the data at one scale and a physical Casimir energy below the gap (Theorem 7.39), the Casimir coefficient is itself bounded by the thermal corrections at aspect ratio one half (Lemma 7.38), and what remains of (C) is the exclusion, at every scale, of a branch in which those corrections grow linearly with the scale (Proposition 7.42, Theorem 7.43). Positivity alone makes the thermal corrections at a fixed time extensive

[4]The problem, as formulated by Jaffe and Witten [31], asks for a proof that for every compact simple gauge group G a nontrivial quantum Yang–Mills theory exists on $\mathbb{R}^4$, satisfying axioms at least as strong as the Wightman axioms, and that it has a mass gap: the Hamiltonian H of the theory satisfies $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$ for some $\Delta > 0$, with 0 the simple eigenvalue of the vacuum. Theorem B and Theorem E give the second half of this statement, with an explicit Δ, from the hypotheses named there; the first half, existence, is hypothesis (E), or (W).

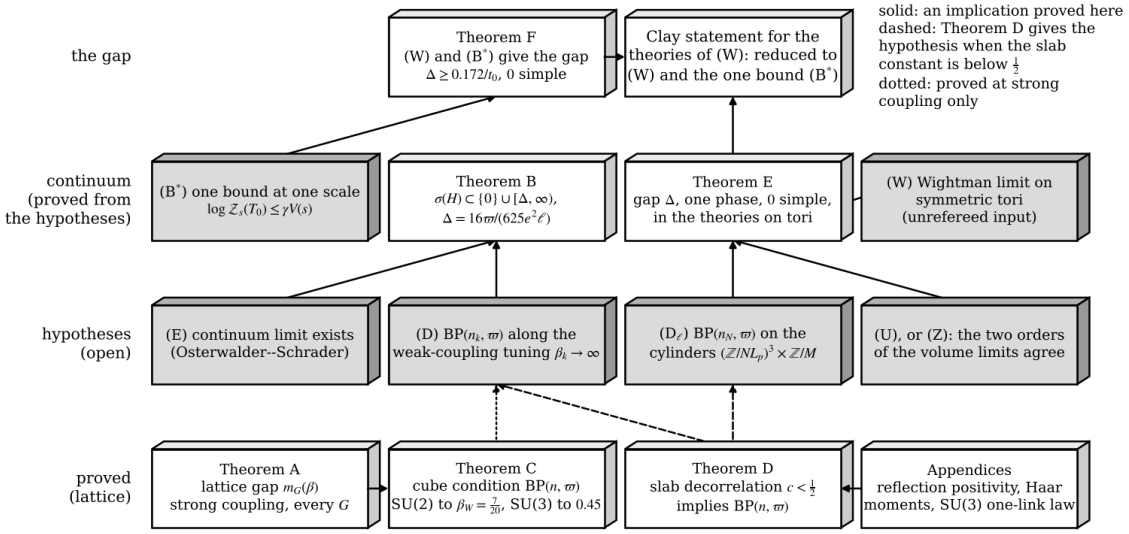


FIGURE 2. The logical structure of the paper. White boxes are theorems proved here, grey boxes are hypotheses; solid arrows are implications proved here, the dashed arrows are Theorem D, which yields the cube condition when the slab decorrelation constant is below $\frac{1}{2}$ (on cylinders by Corollary 11.9), and the dotted arrow records that Theorem C proves the cube condition at strong coupling only. With the construction of [45] as the existence input (W), the statement of the Clay problem for its theories reduces to (Z) or (U), and to (D_ℓ) or the slab decorrelation (Theorem E), and all of these follow from the single bound (B*) at one scale (Theorem F).

once the torus is cut into pieces much longer than the time, and the gap carries their density from one time to the double; so when the pressure bound at one scale is taken on boxes elongated up to twenty-four times with respect to the time, the gap excludes that branch and gives (Z), (K) and (C) with a rate, with no hypothesis at a second scale (Theorem 7.54, Corollary 7.55), and whether the bound on near-cubic boxes suffices is a question at one scale that remains open. The free field calibrates the condition exactly: the physical side times the Casimir defect converges to an explicit function $c(\mu s)$ of the mass times the scale, with $c(0) = 12.56$, so that a massless field, and the free gluons of the weak-coupling limit with the multiplicity $2 \dim G$, violate (C) at every scale, while a field of mass μ satisfies it exactly beyond the scale $17.7/\mu$ (Theorem 10.6, Corollary 10.7, Proposition 10.8); the Casimir condition is a statement about tori of about eighteen correlation lengths. The slab hypothesis, for its part, is uniform in the boundary condition by nature: an exact spin model with a gapped positive transfer matrix and a small vacuum slab correlation violates it under one boundary condition (Proposition 11.17), so the spectrum of the transfer matrix alone cannot feed Theorem D. The slab correlation is continuous in the boundary condition, so the hypothesis is an almost sure bound on a conditional correlation under the torus measure (Lemma 11.15); in the model it fails at every scale when the field is strong, and the cube condition itself excludes the bad boundary state there by a tunnelling function (Proposition 11.18); and the patch inequality behind the finite-size criterion is equivalent, boundary condition by boundary condition, to the local Knabe inequality on the torus (Proposition G.5).

1.9. **Related work.** Strong-coupling clustering goes back to Osterwalder and Seiler [48] and Seiler [52]; probabilistic treatments include the survey of Chatterjee [12], the Langevin approach of Shen, Zhu and Zhu [56], whose Bakry–Émery argument we sharpen, and

correlation decay for finite gauge groups at weak coupling by Adhikari and Cao [1]. In two dimensions the continuum measure is known [24, 19, 37]. The transfer matrix of lattice gauge theory is Lüscher's [38], with the temporal gauge and the positivity of the transfer matrix in Creutz's form [14], and the Osterwalder–Schrader reconstruction on the lattice is due to Osterwalder and Seiler [48]. One of us (I. J. S.) has formalised in Lean 4 with Mathlib, in a separate development, without unproved statements and with only the standard axioms, several ingredients of the strong-coupling analysis on periodic lattices: the Haar probability on $\mathrm{SU}(N)$, an infinite-volume Gibbs state obtained as a limit of torus states, site reflection positivity and the transfer matrix, the Holley–Stroock comparison, Dobrushin's covariance bounds, and the passage from decay of correlations to a transfer-matrix gap (one direction of Proposition 2.1). For the periodic state it proves a transfer-matrix gap for every N at $\beta_W < 5.9 \cdot 10^{-5}$, and explicit single-link heat-bath gaps on boxes of side fifteen for SU(2) at $\beta_W \le 0.166$ and for SU(3) at $\beta_W \le 0.38$ (the coupling there is $\beta_W/2$). It also proves, by a Poincaré inequality and a light-cone argument different from Lemma 6.2, that an ungauged one-cube influence condition gives a lattice gap of order $1/n$; this is a variant of the lattice step, Theorem 6.3 and Corollary 6.4, at a smaller rate. The continuum step, the gauge fixing and the curvature bounds are not part of that development, which shares with this paper the open hypothesis (D).

2. Lattice gauge theory and the transfer matrix

Write $e_0, \dots, e_3$ for the unit vectors of $\mathbb{Z}^4$, (x, μ) for the link from x to $x + e_\mu$, and $Q_{(x,\mu)}^{-1}$ for the variable of the reversed link. A box Λ is a product of four finite intervals, with its set $E(\Lambda)$ of links and its set $P(\Lambda)$ of plaquettes. For a plaquette at x in the plane $\mu < \nu$ with links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, put $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$. With dQ the Haar probability measure of G, the Wilson measure of a box with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p \Big) \prod_{e \in E(\Lambda)} dQ_e,$$

and the Wilson measure μ_L of the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ is defined in the same way with all links and plaquettes of Λ_L. Both are invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$. A function of finitely many links is a local observable, and it is gauge invariant if it is invariant under these transformations. For $\beta \ge 0$ the measures are reflection positive for reflections in lattice hyperplanes through sites and in hyperplanes bisecting links, on boxes symmetric under the reflection and on tori of even side [48].

Let ω be a state on the gauge-invariant local observables of $\mathbb{Z}^4$ that is invariant under translations and reflection positive in this sense. The Osterwalder–Schrader construction [46, 47, 48] gives a Hilbert space $\mathcal{H}$ with a unit vector Ω, a map $F \mapsto \Psi_F$ from the observables at nonnegative times to $\mathcal{H}$ with dense range, and a self-adjoint transfer matrix $0 \le \mathbb{T} \le 1$ with $\mathbb{T}\Omega = \Omega$ and

$$\langle \Psi_F, \mathbb{T}^t \Psi_H \rangle = \omega\big(\vartheta(\bar{F})\, \tau_t H\big), \qquad t \in \mathbb{Z}_{\ge 0},$$

where ϑ is the time reflection and τ_t the time translation; Ψ_F is the class of F for the positive semidefinite form $\omega(\vartheta(\bar{F})H)$, and $\|\Psi_F\|^2 \le \omega(|F|^2)$. The *spectral gap* of $\mathbb{T}$ is m if its spectrum on $\Omega^\perp$ lies in $[0, e^{-m}]$.

Proposition 2.1. *Let $m > 0$ and let $\mathcal{F}$ be a class of gauge-invariant local observables, supported at nonnegative times, that is dense in $L^2(\omega)$ on each bounded time support. If for every $F \in \mathcal{F}$ there is $C_F < \infty$ with*

$$\omega(\vartheta(\bar{F})\tau_t F) - |\omega(F)|^2 \le C_F e^{-mt} \qquad (t \ge 0),$$

then $\mathbb{T}$ has spectral gap at least m.

Proof. Put $\Psi'_F = \Psi_F - \langle \Omega, \Psi_F \rangle \Omega$, so that the left side equals $\langle \Psi'_F, \mathbb{T}^t \Psi'_F \rangle = \int \lambda^t \, d\sigma_F(\lambda)$, with σ_F the spectral measure of $\mathbb{T}$ at Ψ'_F. For $\delta > 0$ this is at least $\sigma_F((e^{-m+\delta}, 1])e^{(-m+\delta)t}$,

which exceeds $C_F e^{-mt}$ for large t unless $\sigma_F((e^{-m+\delta}, 1]) = 0$. So the spectral projection of $(e^{-m}, 1]$ annihilates every Ψ'_F. These vectors are dense in $\Omega^\perp$, because $\|\Psi_F\|^2 \le \omega(|F|^2)$ and the class is dense, so the projection vanishes on $\Omega^\perp$. □

The decay of a single two-point function does not suffice: it places only one spectral measure in $[0, e^{-m}]$, and the density of the class is what gives the gap.

3. The line field

A *direction field* is a map $\sigma : \mathbb{Z}^4 \to \{0, 1, 2, 3\}$; its links are $F = \{(x, \sigma(x)) : x \in \mathbb{Z}^4\}$. At each site put $\pi_x(\mu) = \sigma(x + e_\mu)$. The plaquette at x in the plane $\{\mu, \nu\}$ contains

$$[\sigma(x) \in \{\mu, \nu\}] + [\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu] \tag{3.1}$$

links of F.

Lemma 3.1. *Every plaquette contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s = \sigma(x)$ and let R be the other three directions. For the plane $\{s, \nu\}$ the count (3.1) is $1 + [\pi_x(s) = \nu] + [\pi_x(\nu) = s]$, which equals one for all $\nu \in R$ exactly when π_x fixes s and maps R into R. For a plane $\{\mu, \nu\} \subset R$ the count is $[\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu]$. Draw an arrow $\mu \to \pi_x(\mu)$ whenever $\pi_x(\mu) \in R \setminus \{\mu\}$; each vertex has at most one outgoing arrow, and the condition asks for exactly one arrow on each of the three pairs. Three arrows on three vertices with out-degree at most one give out-degree one everywhere, and such a tournament is a directed three-cycle. □

From now on σ is the field of period two of Table 2 in Appendix A; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, so every plaquette contains exactly one link of F.[5] Call the links outside F *active* and write A for their set.

Lemma 3.2.

(i) *F is a disjoint union of bi-infinite straight lines, one through every site; hence $F \cap E(\Lambda)$ is a forest for every box Λ.*

(ii) *An active link $e = (x, \mu)$ lies in six plaquettes. In four of them the link of F shares an end point with e; in the other two it is opposite to e.*

(iii) *Every active link shares a plaquette with exactly twelve other active links, two in each of its plaquettes, and two links share at most one plaquette.*

Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x + e_{\sigma(x)}) = \sigma(x)$, and since σ has period two, also $\sigma(x - e_{\sigma(x)}) = \sigma(x)$. If $(x - e_\nu, \nu) \in F$, straightness at $x - e_\nu$ gives $\sigma(x) = \nu$, so the incoming link of F at x is unique and continues the outgoing one. A line meets a box in a segment.

(ii) A link of F sharing an end point with e is the outgoing or incoming link of F at x, of direction $s = \sigma(x) \neq \mu$, or at $x + e_\mu$, of direction $t = \pi_x(\mu) \in R \setminus \{\mu\}$. They lie in the two plaquettes of e in the plane $\{\mu, s\}$ and the two in the plane $\{\mu, t\}$, four distinct plaquettes as $s \neq t$. In the two plaquettes of the remaining plane $\{\mu, \rho\}$ the unique link of F touches neither end of e, so it is the opposite link.

(iii) Each plaquette of e contains e, one link of F and two other links. □

4. Gauge fixing and the free-boundary state

For a function O of the links let O^F be O with every link of F set to the identity.

Lemma 4.1 (gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ with density proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$, $W_\Lambda(O) = \mu_\Lambda^F(O^F)$.*

[5] A field with one link of F at every site and one in every plaquette can exist only in four dimensions: a site carries $d(d-1)/2$ plaquettes at which it is the base point and a link lies in $2(d-1)$ plaquettes, and the two counts agree only for $d = 4$.

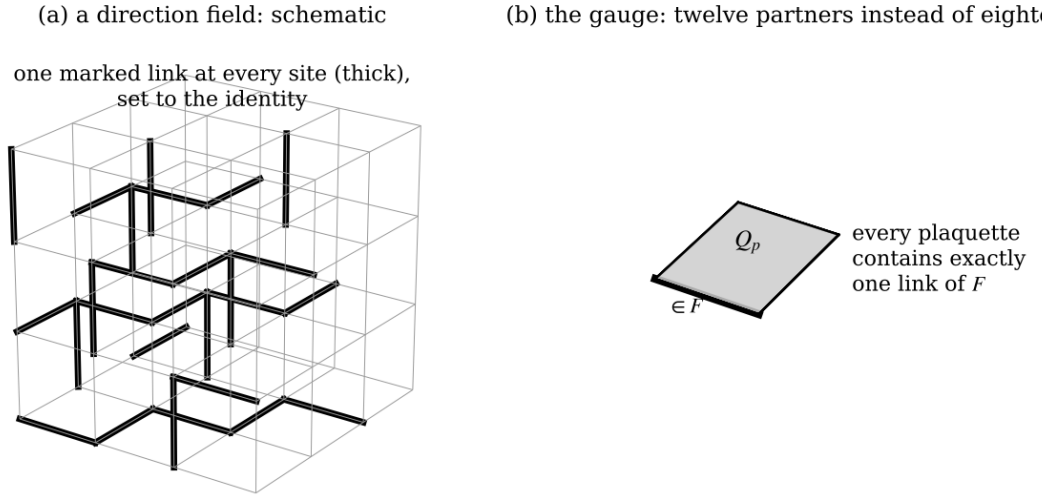


FIGURE 3. (a) A direction field on a three-dimensional slice, drawn schematically: one marked link at every site (thick), set to the identity by a gauge transformation; the marked links form lines. (b) In this gauge every plaquette contains exactly one link of F, so an active link shares a plaquette with twelve other active links instead of eighteen (Lemma 3.2); the actual field is four-dimensional and is listed in Appendix A.

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, \dots, x_k$. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ along every segment. The transformed field equals the identity on $F \cap E(\Lambda)$, and for fixed values on F the map from the active variables to the transformed ones is a product of translations, which preserve Haar measure. Gauge invariance of the action and of O gives the claim. □

No holonomy remains because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A in which γ_e^F resamples an active Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr} Q_p^F)$ given the other active links. Fix a bi-invariant metric δ on G of diameter D and define the Kantorovich coefficients

$$C_{ee'} = \sup \frac{W_1(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q'))}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over configurations that differ only at e'; W_1 is the Kantorovich distance [58], and $C_{ee'} = 0$ unless e and e' share a plaquette.

Lemma 4.2 (comparison)**.** *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$, and let $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$. Let $S \subset A$ and let μ_1, μ_2 be probability measures on G^A invariant under γ_e^F for $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$ with the distance of link midpoints, and suppose $\sum_{z \in S} w_z < \infty$. There is a coupling of μ_1 and μ_2 with $\varrho(b) = \mathbb{E}\, \delta(\sigma_b, \sigma'_b)$ satisfying*

$$\sum_{b \notin S} w_b \varrho(b) \le \frac{\theta D}{1 - \theta} \sum_{z \in S} w_z,$$

and $\mu_1 = \mu_2$ if $S = \emptyset$.

Proof. The couplings form a weakly compact set on which $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is continuous; let π^* minimise it. For $e \notin S$ replace (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [58, Cor. 5.22]. Both μ_i are invariant, so this is again a coupling, and changing the links that enter γ_e^F one at a time gives

$\varrho(e) \le \sum_{e'} C_{ee'}\varrho(e')$ by minimality. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^{\lambda} w_{e'}$, so $\sum_{e\notin S} w_e\varrho(e) \le \theta[\sum_{e'\notin S} w_{e'}\varrho(e') + D\sum_{z\in S} w_z]$. If $S = \emptyset$ this forces $\varrho \equiv 0$. □

Conditioning on the links of the support of H [33, Ch. 3] and applying the lemma to the two conditional measures gives, for the Gibbs measure μ of γ^F, a local F' that is L-Lipschitz in each link and a bounded H with finite support disjoint from that of F',

$$|\operatorname{Cov}_\mu(F', H)| \le \|H\|_\infty L \frac{\theta D}{1-\theta} |\operatorname{supp} H| \, e^{-\lambda \operatorname{dist}(\operatorname{supp} F', \operatorname{supp} H)}. \tag{4.1}$$

Proposition 4.3. *Suppose every row and column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then γ^F has a unique Gibbs measure μ_F; for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$; ω is invariant under translations and the hypercubic group; truncated correlations of Lipschitz gauge-invariant observables decay at every rate below $-\log\alpha$; and for $\beta \ge 0$, ω is reflection positive and its transfer matrix has spectral gap at least $-\log\alpha$.*

Proof. Gibbs measures exist by compactness and coincide by Lemma 4.2 with $S = \emptyset$. Extend μ^F_Λ by Haar measure outside Λ and let S be the set of active links with a plaquette outside $P(\Lambda)$; off S the extended measure is invariant under γ^F_e. Lemma 4.2 with T the support of O^F bounds $|\mu^F_\Lambda(O^F) - \mu_F(O^F)|$ by $C_T R^3 e^{-\lambda R}$, R the distance from T to the complement of Λ, for Lipschitz O, and uniform approximation handles continuous O; Lemma 4.1 identifies $\mu^F_\Lambda(O^F)$ with $W_\Lambda(O)$. Since $W_\Lambda(\tau_v O) = W_{\Lambda - v}(O)$ and the limit does not depend on the exhausting sequence, ω is invariant; the line field is a device for computing W_Λ and is never moved. Reflection positivity of W_Λ on symmetric boxes passes to the limit. The decay is (4.1), since $\operatorname{Cov}_\omega(O, O') = \operatorname{Cov}_{\mu_F}(O^F, O'^F)$. Lipschitz gauge-invariant functions are dense in the gauge-invariant part of $L^2(\omega)$ on every support, being Lipschitz functions on the compact quotient by the gauge group, so Proposition 2.1 gives the gap. □

Proof of Theorem A for SU(2). Identify SU(2) with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p \ni e$ let s_p be the product of the other three variables, ordered so that $\operatorname{Re}(Q_e s_p) = \frac{1}{2}\operatorname{Tr} Q_p$, with links of F equal to one. Since $N\beta \operatorname{Re}\operatorname{Tr} Q_p = 4\beta\langle Q_e, \bar{s}_p\rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx) \propto e^{\langle x,b\rangle}dx$ on S^3 with $b = 4\beta\sum_{p\ni e}\bar{s}_p$. Moving a partner e' by δ moves one staple by an isometric image of that displacement, hence b by $4|\beta|\delta$, and the sharp quarter (Theorem C.4 in Appendix C), $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$, gives $C_{ee'} \le |\beta|$. By Lemma 3.2(iii) every row and column has twelve nonzero entries, so $\alpha \le 12|\beta| = 3|\beta_W|$, and Proposition 4.3 applies for $|\beta_W| < \frac{1}{3}$. □

The value $\frac{1}{3}$ is the end of this argument: at $b = 0$ the derivative of ν_b has Kantorovich norm exactly $\frac{1}{4}$, the six staples of an active link can be arranged with sum zero while one of them moves freely, and so every row sum of the matrix is exactly $3|\beta_W|$. Without the gauge every link has eighteen partners, each entry of the matrix is again at most $|\beta|$, and the same argument stops at $18|\beta| = 1$, that is at $\beta_W = \frac{2}{9}$.

5. Curvature in the gauge

In this section $G \subset \mathrm{U}(N)$ is closed and connected with Lie algebra $\mathfrak{g}$ and the inner product $\langle X, Y\rangle = -\operatorname{Tr}(XY)$. A tangent vector at a field Q is $v = (X_e Q_e)$ with $X_e \in \mathfrak{g}$, and $|v|^2 = \sum_e |X_e|^2$. For the plaquette p at x in the plane $\mu < \nu$, with links $e_1, \dots, e_4$ as in Sec. 2, transport the vectors to x,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4},$$

and write $W = Q_p$. Along the geodesics $Q_e(t) = e^{tX_e}Q_e$ the plaquette variable is $Q_p(t) = e^{tY_1}e^{tY_2}We^{-tY_3}e^{-tY_4}$, and differentiating twice gives the second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$,

(5.1)
$$f_p = \sum_{i=1}^{4} \operatorname{Re}\operatorname{Tr}(Y_i^2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_1Y_2W) + 2\operatorname{Re}\operatorname{Tr}(Y_3Y_4W) - 2\sum_{i\in\{1,2\},\, j\in\{3,4\}} \operatorname{Re}\operatorname{Tr}(Y_iWY_j),$$

so that $\operatorname{Hess} S_0(v,v) = \sum_p f_p$ for $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$. The *corner form* of p is

(5.2)
$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, WY_3W^*\rangle - 2\langle Y_3, Y_4\rangle;$$

it collects the inner products of the vectors of two links that meet at a right angle at a corner of p.

Lemma 5.1 (divergence identity)**.** *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\sum_y \Big[|D_yX|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] = 4|X|^2 + \sum_p c_p,$$

where $D_yX = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y let o_μ be the vector of the outgoing link of direction μ and i_μ the transported vector of the incoming one, so that $D_yX = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which occur once in $|D_yX|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_yX|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. Every remaining cross term of $|D_yX|^2$ pairs two perpendicular ends at y, and such a pair is a corner of exactly one plaquette, with the sign and transport of (5.2); in a box two perpendicular ends at y span a plaquette of the box. □

Lemma 5.2 (one plaquette)**.** *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k+2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}(|Y_{k-1}|^2 + |Y_{k+1}|^2) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign + suffices, and conjugation by a unitary matrix preserves both sides, so we may take $W = \operatorname{diag}(w_a)$. The difference $\mathcal{F}$ of the two sides then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \neq b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix B shows that every block is positive definite. □

Theorem 5.3 (Hessian bound)**.** *For every closed subgroup $G \subset \mathrm{U}(N)$, every box or $\mathbb{Z}^4$, every field Q and every tangent vector v that vanishes on the links of F,*

$$|\operatorname{Hess} S_0(v,v)| \le \tfrac{29}{2}|v|^2.$$

Proof. Every plaquette has exactly one link of F, on which X vanishes, and by Lemma 3.2(ii) each active link is adjacent to that link in four of its plaquettes and opposite to it in two. Summing Lemma 5.2 over the plaquettes,

$$\sum_p (\pm f_p + \tfrac{9}{10} c_p) \le (4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4})|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 5.1, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$ with $\Sigma \ge 0$ the left side of that lemma. Hence $\pm \operatorname{Hess} S_0(v,v) \le (\frac{109}{10} + \frac{36}{10})|X|^2 = \frac{29}{2}|X|^2$; links near the boundary of a box lie in fewer plaquettes and receive less. Both lemmas concern vectors in $\mathfrak{u}(N)$, so the bound holds on every closed subgroup, whose geodesics $e^{tX}Q_e$ with $X \in \mathfrak{g}$ are geodesics of $\mathrm{U}(N)$. □

TABLE 1. The curvature $\kappa_G = h^\vee/(2\ell_\rho)$ in a faithful representation of least dimension N, and the range of Theorem A for $\beta_W = N^2\beta$.

Group	ρ	$h^\vee$	ℓ_ρ	κ_G	$\beta_W <$
$\mathrm{SU}(N)$, $N \geq 2$	$\mathbb{C}^N$	N	1	$\frac{N}{2}$	$\frac{N^2}{29}$
$\mathrm{SO}(N)$, $N \geq 5$	$\mathbb{R}^N$	$N-2$	2	$\frac{N-2}{4}$	$\frac{N(N-2)}{58}$
$\mathrm{Sp}(n)$, $n \geq 2$	$\mathbb{C}^{2n}$	$n+1$	1	$\frac{n+1}{2}$	$\frac{2n(n+1)}{29}$
G_2	7	4	2	1	$\frac{14}{29}$
F_4	26	9	6	$\frac{3}{4}$	$\frac{39}{29}$
E_6	27	12	6	1	$\frac{54}{29}$
E_7	56	18	12	$\frac{3}{4}$	$\frac{84}{29}$
E_8	248	30	60	$\frac{1}{4}$	$\frac{124}{29}$

The factor $\frac{9}{10}$ lets the divergence identity pay for the corners at the price $\frac{36}{10}$ while the blocks of one plaquette stay positive. Without the gauge the Hessian attains the value 16 (Theorem H.19).

Lemma 5.4 (Ricci curvature)**.** *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra, and let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra. Write the eigenvalues of* $\operatorname{ad} H$ *on $\mathfrak{g} \otimes \mathbb{C}$ and of H on $\mathbb{C}^N$, $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, over the roots α and the weights μ counted with multiplicity. Then* $\mathrm{Ric}_G = \kappa_G\langle\cdot,\cdot\rangle$ *with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4\sum_\mu \mu(H)^2} = \frac{h^\vee}{2\ell_\rho} \qquad (0 \neq H \in \mathfrak{t}).$$

Proof. For a bi-invariant metric $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [41, Sect. 7], so for an orthonormal basis (E_i), $\mathrm{Ric}(X,X) = \frac{1}{4}\sum_i |[X,E_i]|^2 = -\frac{1}{4}\operatorname{Tr}(\operatorname{ad} X)^2$. The Killing form and $\langle\cdot,\cdot\rangle$ are invariant and $\mathfrak{g}$ is simple, so the Killing form is $-4\kappa\langle\cdot,\cdot\rangle$ for a constant κ, and on $\mathfrak{t}$ the two forms are $-\sum_\alpha \alpha(H)^2$ and $\sum_\mu \mu(H)^2$. For H the coroot of the highest root, $\sum_\alpha \alpha(H)^2 = 4h^\vee$ and $\sum_\mu \mu(H)^2 = 2\ell_\rho$ [32, Ch. 6]. □

For two links $e \neq e'$ of a plaquette p write $\operatorname{Hess}\operatorname{Re}\operatorname{Tr} Q_p(v,v) = \sum_{e,e'\in p} H^p_{ee'}(X_e, X_{e'})$ with $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. By (5.1) a mixed block is, after isometric transport, $\pm\operatorname{Re}\operatorname{Tr}(abW)$ or $\pm\operatorname{Re}\operatorname{Tr}(aWb)$, so

$$|H^p_{ee'}(a,b)| \leq |a|\,|b|. \tag{5.3}$$

5.1. **Decay with an explicit rate.** Let A' be a finite set of active links and P' a finite set of plaquettes whose active links lie in A'. Call $S = \sum_{p\in P'} c_p \operatorname{Re}\operatorname{Tr} Q^F_p$, $0 \leq c_p \leq 1$, *admissible* if $|\operatorname{Hess} S(v,v)| \leq \frac{29}{2}|v|^2$ for v tangent to $G^{A'}$. By Theorem 5.3 the action $S_\Lambda = \sum_{p\in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q^F_p$ of a box is admissible, and so is every convex combination of admissible actions; by (5.3) and Lemma 3.2(iii) the mixed blocks of $\operatorname{Hess} S$ have norm at most one and each link has at most twelve partners. Let μ have density proportional to $e^{N\beta S}$ on $G^{A'}$, let $L = \Delta + N\beta\langle\nabla S, \nabla\cdot\rangle$ be its Langevin generator, self-adjoint in $L^2(\mu)$, and $P_t = e^{tL}$. For weights $w > 0$ on A' put $\Gamma^w(f) = \sum_e w_e|\nabla_e f|^2$, with ∇_e the gradient in the factor of e; Γ is the case $w \equiv 1$.

Lemma 5.5. *Let G have semisimple Lie algebra, $\lambda \geq 0$, $w_e = e^{\lambda d_e}$ with $|d_e - d_{e'}| \leq 1$ for partners, and*

$$K_\lambda = \kappa_G - N|\beta|\Big(\tfrac{29}{2} + 12\big(\cosh\tfrac{\lambda}{2} - 1\big)\Big).$$

Then $\Gamma^w(P_t f) \leq e^{-2K_\lambda t} P_t \Gamma^w(f)$ for smooth f and $t \geq 0$.

Proof. Put $V = -N\beta S$. On the product $G^{A'}$ the projection to one factor is parallel and the Ricci tensor is block diagonal, so Bochner's formula applied factor by factor gives

$$\tfrac{1}{2}L\Gamma^w(g) - \Gamma^w(g, Lg) = \sum_e w_e\big(|\nabla\nabla_e g|^2 + \mathrm{Ric}_G(\nabla_e g, \nabla_e g)\big) + \sum_{e,e'} w_e \operatorname{Hess} V_{ee'}(\nabla_e g, \nabla_{e'} g).$$

With $u_e = \nabla_e g$, $v_e = \sqrt{w_e}u_e$ and the symmetry of $\operatorname{Hess} V$, the last sum is

$$\operatorname{Hess} V(v,v) + \sum_{e\neq e'} \Big(\cosh\tfrac{\lambda(d_e - d_{e'})}{2} - 1\Big) \operatorname{Hess} V_{ee'}(v_e, v_{e'}),$$

since $(w_e + w_{e'})/(2\sqrt{w_e w_{e'}}) = \cosh\frac{\lambda(d_e-d_{e'})}{2}$. The first term is at least $-\frac{29}{2}N|\beta||v|^2$; in the second only partners contribute, each with coefficient at most $\cosh\frac{\lambda}{2} - 1$ and a block of norm at most $N|\beta|$, and with twelve partners it is at least $-12(\cosh\frac{\lambda}{2} - 1)N|\beta||v|^2$. As $\mathrm{Ric}_G \geq \kappa_G$ and $|v|^2 = \Gamma^w(g)$, the left side is at least $K_\lambda \Gamma^w(g)$. For $\varphi(s) = P_s\Gamma^w(P_{t-s}f)$ this gives $\varphi' \geq 2K_\lambda\varphi$, hence $\varphi(0) \leq e^{-2K_\lambda t}\varphi(t)$. □

Lemma 5.6. *Suppose $K_\lambda > 0$. Let f, g be smooth functions of the links of A_f and B_g, and let D be the graph distance from A_f to B_g in the graph on A' whose edges join partners. Then*

$$|\operatorname{Cov}_\mu(f,g)| \leq \frac{e^{-\lambda D/2}}{K_\lambda}\,\|\nabla f\|_\infty \sum_{e\in B_g} \|\nabla_e g\|_\infty.$$

Proof. Lemma 5.5 with $\lambda = 0$ shows that $P_t f \to \mu(f)$ uniformly, so $\operatorname{Cov}_\mu(f,g) = \int_0^\infty \mu(\Gamma(P_t f, g))\,dt$. Let d_e be the smaller of D and the graph distance from e to A_f; then $w = 1$ on A_f, $w = e^{\lambda D}$ on B_g, and $\Gamma^w(f) \leq \|\nabla f\|_\infty^2$. For $e \in B_g$, Lemma 5.5 gives $|\nabla_e P_t f|^2 \leq e^{-\lambda D}e^{-2K_\lambda t}\|\nabla f\|_\infty^2$, and integration in t gives the bound. □

Proof of Theorem A. Let λ^* solve $K_{\lambda^*} = 0$, so that $\lambda^*/2 = m_G(\beta)$, and let $0 < \lambda < \lambda^*$. Lemma 5.6 with $S = S_\Lambda$ gives $|\operatorname{Cov}_{\mu_\Lambda^F}(f,g)| \leq C_{f,g}e^{-\lambda D/2}$ uniformly in the box. For boxes $\Lambda \subset \Lambda'$ interpolate $S_s = (1-s)S_\Lambda + sS_{\Lambda'}$, which is admissible; for smooth f of the active links of Λ,

$$\frac{d}{ds}\mu_s(f) = N\beta \sum_{p\in P(\Lambda')\setminus P(\Lambda)} \operatorname{Cov}_{\mu_s}(f, \operatorname{Re}\operatorname{Tr} Q_p^F),$$

with $|\nabla_e \operatorname{Re}\operatorname{Tr} Q_p| \leq \sqrt{N}$. A plaquette outside $P(\Lambda)$ is at graph distance at least $R-1$ from the support of f, R the distance to the complement of Λ, and at most $C(k+1)^4$ plaquettes lie at graph distance k, so $|\mu_{\Lambda'}^F(f) - \mu_\Lambda^F(f)| \leq C_f R^4 e^{-\lambda R/2}$. The limit exists along every increasing sequence of boxes, two sequences are compared through their intersections, and continuous gauge-invariant observables are uniform limits of smooth ones $e^{\tau\Delta}O$, the heat semigroup commuting with translations of G. With Lemma 4.1 the limit ω exists; invariance and reflection positivity follow as in Proposition 4.3. Since $\mathrm{Ric}_G \geq 0$, $e^{\tau\Delta}O$ has the Lipschitz constant of O [8, Sect. 3.3], so the decay passes to Lipschitz observables; two links of a plaquette have times differing by at most one, so observables at time separation t are at graph distance at least $t - c$, and truncated correlations decay at every rate below $m_G(\beta)$. Proposition 2.1 gives the gap. □

The rate behaves like $\log\frac{\kappa_G}{6N|\beta|}$ as $\beta \to 0$. The proof also gives a Poincaré inequality with constant $1/(\kappa_G - \frac{29}{2}N|\beta|)$ for every μ_Λ^F, uniformly in the box [8, Cor. 5.7.2].

6. One cube and the continuum gap

Setting. Let $G \subset \mathrm{U}(N)$ be compact and nontrivial and φ a positive continuous class function on G, such as $e^{N\beta\operatorname{Re}\operatorname{Tr}(\cdot)}$. On a torus $\Lambda = \prod_{i=1}^4(\mathbb{Z}/L_i\mathbb{Z})$ with all sides $L_i \geq 2n+4$,

with links E and plaquettes P, let

$$\mu(dU) = \frac{1}{Z} \prod_{p \in P} \varphi(U_p) \prod_{e \in E} dU_e.$$

A link (y, ν) has base point y; differences of base points are taken in $\prod_i (-L_i/2, L_i/2]$ and $|\cdot|_\infty$ is the corresponding distance, which satisfies the triangle inequality. For $x \in \Lambda$ let V_x be the set of links with base point in $x + [0, n)^4$; every link lies in exactly n^4 of the $|\Lambda|$ cubes V_x. The symmetric torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, L even, is the case $L_i = L$, and the cylinders of Section 7 have three equal sides and a fourth, the Euclidean time, that may differ from them. For $V \subset E$ let ∂V be the links outside V that share a plaquette with a link of V; the conditional law $\gamma^0_V(\cdot \mid U)$ of U_V given the other links has density proportional to $\prod_{p \cap V \neq \emptyset} \varphi(U_p)$ and depends only on $U_{\partial V}$. Let $\mathcal{P}_x f(U) = \int f \, d\gamma^0_{V_x}(\cdot \mid U)$, the conditional expectation given the links outside V_x, an orthogonal projection on $L^2(\mu)$, and let $\mathcal{A}$ be the closed subspace of gauge-invariant functions. By gauge covariance of $\gamma^0_{V_x}$, $\mathcal{P}_x$ maps $\mathcal{A}$ into itself.

Definition 6.1. For $n \geq 2$ and $\varpi > 0$, the condition $\mathrm{BP}(n, \varpi)$ for μ is

$$\varpi \operatorname{Var}_\mu(f) \leq \frac{1}{n^4} \sum_{x \in \Lambda} \mu((f - \mathcal{P}_x f)^2) \qquad \text{for every real } f \in \mathcal{A}.$$

Put $q_n = (2+1/n)^4$, so that $16 < q_n \leq \frac{625}{16}$ for $n \geq 2$, and let $\mathcal{K} = |\Lambda|^{-1} \sum_x \mathcal{P}_x$ be the chain that resamples one uniformly chosen cube; for $Y \subset \Lambda$ let $\mathcal{K}_Y = |\Lambda|^{-1}(\sum_{x \in Y} \mathcal{P}_x + (|\Lambda| - |Y|)I)$ be the same chain with the moves outside Y suppressed. Write $y \rightsquigarrow z$ if $\partial V_z \cap V_y \neq \emptyset$. A link of ∂V_z has its base point in $z + [-1, n]^4$, so $y \rightsquigarrow z$ implies $|z - y|_\infty \leq n$, and for each z at most $(2n+1)^4 = q_n n^4$ points y have $y \rightsquigarrow z$ (the box $z + [-n, n]^4$ has that many points, as every side exceeds $2n+1$). For finite $S \subset E$ let $Y_1(S) = \{y : V_y \cap S \neq \emptyset\}$ and $Y_{j+1}(S) = Y_j(S) \cup \{y : y \rightsquigarrow z \text{ for some } z \in Y_j(S)\}$, $Y_0(S) = \emptyset$.

Lemma 6.2 (finite speed)**.** *Let f be a bounded function of the links of a finite set S, let $m_0 \geq 0$ and $Y = Y_{m_0}(S)$. Then $\mathcal{K}^t_Y f$ depends only on the links of $S \cup \bigcup_{y \in Y} \partial V_y$, and for every integer $t \geq 0$*

$$\|\mathcal{K}^t f - \mathcal{K}^t_Y f\|_\infty \leq \frac{2|S|}{q_n} \|f\|_\infty \sum_{m > m_0} \frac{(q_n \tau)^m}{m!}, \qquad \tau = \frac{t n^4}{|\Lambda|}.$$

Proof. If h depends only on the links of W, then $\mathcal{P}_x h = h$ when $V_x \cap W = \emptyset$, and otherwise $\mathcal{P}_x h$ depends only on the links of $(W \setminus V_x) \cup \partial V_x$; induction gives the first claim. For the second run two chains from the same configuration with the same uniformly chosen cubes $x_1, x_2, \ldots$. If $x_s \notin Y$ the first chain resamples V_{x_s} and the second does nothing; if $x_s \in Y$ and the two configurations agree on ∂V_{x_s}, both receive one common sample, and otherwise they resample independently. The marginals are the chains $\mathcal{K}$ and $\mathcal{K}_Y$. A link where the configurations differ at time t lies in $V_{x_{s_m}}$ for some times $s_1 < \cdots < s_m \leq t$ with $x_{s_1} \notin Y$ and $x_{s_l} \rightsquigarrow x_{s_{l+1}}$, by induction on t. If the set of such links meets S, then $x_{s_m} \in Y_1(S)$ and $x_{s_{m-l}} \in Y_{l+1}(S)$, so $m > m_0$. For fixed m there are at most $t^m/m!$ choices of times, $n^4|S|$ of x_{s_m} and $q_n n^4$ of each earlier cube, each sequence having probability $|\Lambda|^{-m}$; the difference of f is at most $2\|f\|_\infty$. □

For finite $S, S' \subset E$ let $d(S, S')$ be the least $|b - b'|_\infty$ over base points b of links of S and b' of links of S'.

Theorem 6.3 (decay from one cube)**.** *Suppose $\mathrm{BP}(n, \varpi)$ holds for μ. Then $\varpi \leq 4$, and for bounded real $f, g \in \mathcal{A}$ depending on the links of finite sets S_f, S_g with $d = d(S_f, S_g) \geq 3n - 2$,*

$$|\operatorname{Cov}_\mu(f, g)| \leq \Big(e + \frac{k|S_f||S_g|}{64e^2}\Big) \|f\|_\infty \|g\|_\infty \, e^{-\varpi k/(e^2 q_n)}, \qquad k = \Big\lfloor \frac{d+2}{n} \Big\rfloor - 2.$$

(a) one cube V_x and the links that enter its law (b) the sets $Y_m(S)$: each step moves at most n links

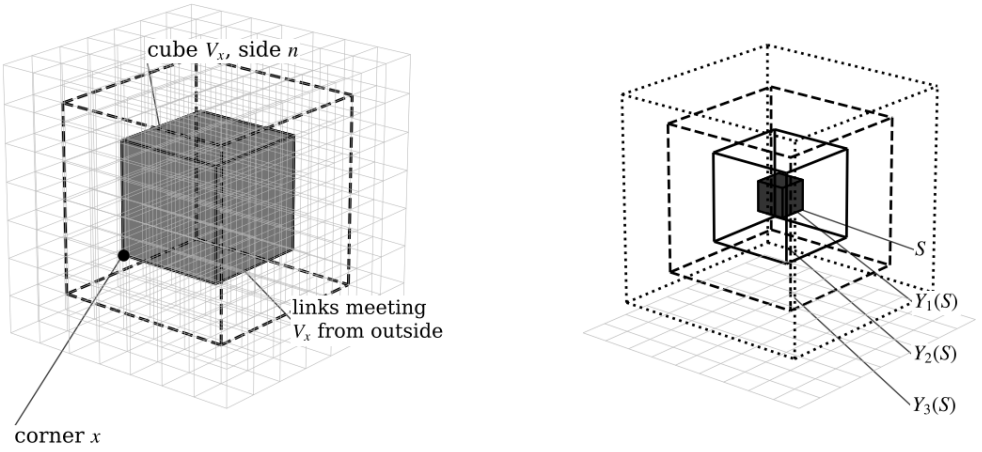


FIGURE 4. (a) The cube V_x of side n at x: its links are resampled by $\mathcal{P}_x$ from their conditional law given the boundary links ∂V_x, which lie in the plaquettes meeting V_x. (b) Finite speed: starting from a set S of links, the sets $Y_1(S) \subset Y_2(S) \subset Y_3(S)$ of cubes reached by one, two and three steps of the relation $y \rightsquigarrow z$; each step moves the dependence by at most n links in every coordinate, and at most $q_n n^4$ cubes feed one cube (Lemma 6.2).

In particular $|\operatorname{Cov}_\mu(f,g)| \le C(1+d)e^{-\varpi d/(e^2 q_n n)}$ *for every* d*, with* C *depending only on* $|S_f|, |S_g|, \|f\|_\infty, \|g\|_\infty$.

Proof. The bound $\varpi \le 4$. Take $f = \operatorname{Re}\operatorname{Tr} U_p$ for one plaquette; since $G \neq \{1\}$, $\operatorname{Re}\operatorname{Tr} g < N$ for $g \neq 1$, and μ has a positive continuous density with respect to Haar measure, so $\operatorname{Var}_\mu(f) > 0$. Each $I - \mathcal{P}_x$ is an orthogonal projection annihilating constants, so $\mu((f-\mathcal{P}_x f)^2) \le \operatorname{Var}_\mu(f)$, and it vanishes unless V_x contains a link of p, which happens for at most $4n^4$ points x.

Spectral part. On the functions of $\mathcal{A}$ with mean zero, $\mathcal{K}$ is self-adjoint with $0 \le \mathcal{K} \le I$, and $\mathrm{BP}(n,\varpi)$ reads $\langle h, (I-\mathcal{K})h\rangle \ge \epsilon\|h\|^2$ with $\epsilon = \varpi n^4/|\Lambda|$. So $\|\mathcal{K}^T h\|_2 \le e^{-\epsilon T}\|h\|_2$, and from $f - \mu(f) = \mathcal{K}^T(f-\mu(f)) + \sum_{t<T}\mathcal{K}^t(I-\mathcal{K})f$,

$$\operatorname{Cov}_\mu(f,g) = \langle \mathcal{K}^T(f-\mu f), g-\mu g\rangle + \sum_{t=0}^{T-1}\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle,$$

whose first term is at most $e^{-\epsilon T}\|f\|_\infty\|g\|_\infty$.

Local part. Let $Z_g = Y_1(S_g)$, so $|Z_g| \le n^4|S_g|$ and $(I-\mathcal{K})g = |\Lambda|^{-1}\sum_{x\in Z_g}(g-\mathcal{P}_x g)$; if h does not depend on the links of V_x then $\langle h, g-\mathcal{P}_x g\rangle = 0$. Put $m_0 = k-1$ and $Y = Y_{m_0}(S_f)$. A link of V_x, $x \in Z_g$, lying in S_f or in ∂V_y with $y \in Y$ would join base points of S_f and S_g at distance at most $n(m_0+3)-3 \le n\lfloor (d+2)/n\rfloor - 3 < d$, which is impossible: if $y \in Y_{m_0}(S_f)$ there is a chain $y = y_j \rightsquigarrow \cdots \rightsquigarrow y_1$, $j \le m_0$, with $y_1 \in Y_1(S_f)$, so $|y-y_1|_\infty \le n(m_0-1)$; a link of ∂V_y has base point in $y + [-1,n]^4$, a link of $V_{y_1} \cap S_f$ in $y_1 + [0, n-1]^4$, two links of V_x have base points differing by at most $n-1$, and the triangle inequality gives at most $n(m_0+1)-1 \le n(m_0+3)-3$. So $\mathcal{K}^t_Y f$ does not depend on the links of V_x, $x \in Z_g$, and by Lemma 6.2

$$|\langle \mathcal{K}^t f, (I-\mathcal{K})g\rangle| = |\langle \mathcal{K}^t f - \mathcal{K}^t_Y f, (I-\mathcal{K})g\rangle| \le \frac{4|S_f||S_g|n^4}{q_n|\Lambda|}\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau_t)^m}{m!}.$$

Summing over $t < T$ with $\tau = Tn^4/|\Lambda|$, the second term is at most

$$\frac{4|S_f||S_g|\tau}{q_n}\,\|f\|_\infty\|g\|_\infty \sum_{m\ge k}\frac{(q_n\tau)^m}{m!}.$$

Choice of T. Take $T = \lfloor k|\Lambda|/(e^2 q_n n^4)\rfloor$, so that $z = q_n\tau \le k/e^2$ and $\epsilon T \ge \varpi k/(e^2 q_n) - 1$, using $\varpi \le 4$ and $|\Lambda| \ge 4n^4$. Since $k! \ge (k/e)^k$, $\sum_{m\ge k} z^m/m! \le (ez/k)^k e^z \le e^{-k(1-e^{-2})}$, and $\tau/q_n \le k/(256e^2)$. As $\varpi/(e^2 q_n) < 1 - e^{-2}$, both terms are bounded as stated. For the second form use $k > d/n - 3$, and for $d < 3n - 2$ the trivial bound $\|f\|_\infty\|g\|_\infty$. □

A *torus limit* at β is a state ω on the gauge-invariant local functions of $\mathbb{Z}^4$ with $\omega(O) = \lim_j \mu_{L_j}(O)$ along some $L_j \to \infty$; by compactness every sequence has a subsequence along which one exists.

Corollary 6.4 (lattice gap)**.** *Let $\beta \ge 0$ and suppose $\mathrm{BP}(n, \varpi)$ holds for μ_L for all L in an infinite set of even integers $L \ge 2n + 4$. Then every torus limit along this set is invariant under translations, reflection positive, and its transfer matrix has spectral gap at least*

$$m_n = \frac{\varpi}{e^2 q_n n} \ge \frac{16\varpi}{625e^2 n}.$$

Proof. Translation invariance and reflection positivity are identities and inequalities among finitely many local functions of μ_L [48], and pass to ω. For a continuous gauge-invariant F at positive times the supports of $\vartheta(\bar F)$ and $\tau_t F$ are at distance d with $|d - t| \le c_F$, and Theorem 6.3 on Λ_L gives $|\operatorname{Cov}_{\mu_L}(\vartheta(\bar F), \tau_t F)| \le C_F(1 + t)e^{-m_n t}$ uniformly in L. In the limit $|\omega(\vartheta(\bar F)\tau_t F) - |\omega(F)|^2| \le C_F(1 + t)e^{-m_n t}$, and Proposition 2.1, applied with every $m < m_n$, gives the gap. □

6.1. **The continuum limit.** Let $a_k \to 0$ be lattice spacings with $a_k = 2^{-k}a_0$, and β_k couplings. Let ω_k be lattice states with transfer matrices $\mathbb{T}_k$ on spaces $\mathcal{H}_k$ with vacua Ω_k. Let $\mathcal{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t > 0\}$, and for each k let $f \mapsto O_k(f)$ be a linear map to gauge-invariant local observables at positive times; the example is $O_k(f) = Z_k \sum_x a_k^4 f(a_k x)\Phi(x)$ with a local field Φ and a renormalisation constant Z_k.

Definition 6.5. A *continuum limit* of (ω_k, O_k) is a Hilbert space $\mathcal{H}$ with a unit vector Ω, a self-adjoint $H \ge 0$ with $H\Omega = 0$, and a linear map $\psi : \mathcal{T} \to \mathcal{H}$ with dense range such that for all $f, g \in \mathcal{T}$ and dyadic $\tau \ge 0$

$$\langle \psi(f), e^{-\tau H}\psi(g)\rangle = \lim_k \omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k} O_k(g)\big), \qquad \langle \Omega, \psi(f)\rangle = \lim_k \omega_k(O_k(f)),$$

over the k with $\tau/a_k \in \mathbb{Z}$. It is *nontrivial* if $\mathcal{H} \ne \mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction produces such a limit from convergent renormalised Schwinger functions satisfying the axioms [46, 47]; the definition records only what is used below.

Proof of Theorem B. Let ω_k be torus limits at β_k along the tori on which $\mathrm{BP}(n_k, \varpi)$ holds. By Corollary 6.4, $\mathbb{T}_k$ has gap at least $16\varpi/(625e^2 n_k)$ on $\Omega_k^\perp$. For $f \in \mathcal{T}$ put $O = O_k(f)$ and $\varphi_k = \Psi_O - \omega_k(O)\Omega_k$. For dyadic τ and k large, $t = \tau/a_k$ is an integer and $16\varpi t/(625e^2 n_k) \ge \Delta\tau$ since $n_k a_k \le \ell$, so

$$\omega_k(\vartheta(\bar O)\tau_t O) - |\omega_k(O)|^2 = \langle \varphi_k, \mathbb{T}_k^t \varphi_k\rangle \le e^{-\Delta\tau}\|\varphi_k\|^2.$$

Both sides converge by Definition 6.5: with $\varphi = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$, $\langle \varphi, e^{-\tau H}\varphi\rangle \le e^{-\Delta\tau}\|\varphi\|^2$. These vectors form a subspace dense in $\Omega^\perp$, and the inequality extends to $\Omega^\perp$ and to all $\tau \ge 0$ by continuity. For $\varphi \in \Omega^\perp$ with spectral measure μ_φ of H this says $\int e^{-\tau E} d\mu_\varphi(E) \le e^{-\Delta\tau}\mu_\varphi([0, \infty))$ for all τ, which forces $\mu_\varphi([0, \Delta - \eta]) = 0$ for every $\eta > 0$. So 0 is a simple eigenvalue and the spectrum on $\Omega^\perp$ lies in $[\Delta, \infty)$, and is not empty if $\Omega^\perp \ne \{0\}$. □

The proof uses the cube condition only through the decay of two-point functions of gauge-invariant observables, and the constant Δ depends on ϖ and ℓ alone.

7. The bridge to a constructed continuum theory

Theorem B takes the lattice states first to infinite time extent and only then to the continuum. A construction of a continuum theory may proceed in a different order: on a symmetric torus of physical side L_p the lattice spacing is sent to zero, and the side is sent to infinity afterwards. The state at finite side is a thermal state at temperature $1/L_p$, and its two-point functions carry contributions of pairs of excited states that no single estimate on one scale controls (Remark 7.7). This section proves that the cube condition on cylinders, together with the hypothesis that the order of the two volume limits does not matter, gives every theory constructed in this way, whose outcome satisfies the Wightman axioms phase by phase, the mass gap of Theorem B, and that it forces the phase decomposition to be trivial. No property of any particular construction is used: the three hypotheses below are the whole input. Throughout, $G \subset \mathrm{U}(N)$ is a compact simple group and the Wilson weight is $\exp(N\beta \sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p)$.

7.1. **Continuum families.** For integers $L, M \geq 2$ let $\Lambda_{L,M} = (\mathbb{Z}/L\mathbb{Z})^3 \times (\mathbb{Z}/M\mathbb{Z})$ be the cylinder with spatial side L and time period M, with Wilson measure $\mu_{L,M}$ at coupling β; $\Lambda_{L,L} = \Lambda_L$ is the symmetric torus. A site is $(\vec{x}, t)$, a link is spatial or temporal, the time slice E_t is the set of spatial links with base time t, and the slab $[t, t']$ is the set of links with both ends at times in $[t, t']$.

Definition 7.1 (continuum family)**.** A continuum family consists of lattice spacings $a_N = 1/N$, physical sides $L_p \in \mathbb{N}$ with $L_p \to \infty$, couplings $\beta_{N,p} \geq 0$, a countable set V of *tokens*, and for every N, p and $v \in V$ a complex function $\Phi_v^{N,p}(x, Q)$ of a site x of $\Lambda_{NL_p,M}$ and a configuration Q, defined for every M by the same formula, such that

(F1) $\Phi_v^{N,p}(x, \cdot)$ is continuous, gauge invariant, and depends only on the links in $x + R_v$ for a finite set $R_v \subset \mathbb{Z}^4$ independent of N, p;

(F2) $\Phi_v^{N,p}(x + y, Q) = \Phi_v^{N,p}(x, \tau_{-y}Q)$ for every site y;

(F3) there is an involution $v \mapsto v^\vartheta$ of V with $\Phi_{v^\vartheta}^{N,p}(x, Q) = \overline{\Phi_v^{N,p}(\vartheta x, \vartheta Q)}$, where ϑ is the reflection $t \mapsto -t$ of the time coordinate, acting on configurations by $(\vartheta Q)_{(x,i)} = Q_{(\vartheta x,i)}$ for spatial links and $(\vartheta Q)_{(x,0)} = Q^{-1}_{(\vartheta x - e_0, 0)}$.[6]

The insertions may carry renormalisation constants depending on N; no bound on $\sup|\Phi_v^{N,p}|$ uniform in N is assumed. For $m \geq 1$ let $D_m \subset (\mathbb{R}^4)^m$ be the open set of configurations of m pairwise distinct points, and let $\mathcal{T}_m$ be the complex-valued C^∞ functions with compact support in D_m; a *cluster* is a pair $(\mathbf{v}, t)$ with $\mathbf{v} \in V^m$ and $t \in \mathcal{T}_m$. For a cluster supported in the box $(-L_p/2, L_p/2)^3 \times (-M/(2N), M/(2N))$ the lattice smear on $\Lambda_{NL_p,M}$ is

$$S_{N,p,M}(\mathbf{v}, t) = a_N^{4m} \sum_{x_1,\dots,x_m} t(a_N x_1, \dots, a_N x_m)\, \mu_{NL_p,M}\Big(\prod_{i=1}^m \Phi_{v_i}^{N,p}(x_i, \cdot)\Big), \tag{7.1}$$

the sum over sites, with $a_N x$ read in the box. Two clusters $(\mathbf{v}, t)$, $(\mathbf{v}', t')$ with $t \in \mathcal{T}_m$, $t' \in \mathcal{T}_{m'}$ have the product cluster $(\mathbf{v}\mathbf{v}', t \otimes t')$ when $t \otimes t'$ is supported in $D_{m+m'}$. The reflected cluster is $(\mathbf{v}^\vartheta, \Theta\bar{t})$ with $(\Theta\bar{t})(x_1, \dots, x_m) = \overline{t(\vartheta x_1, \dots, \vartheta x_m)}$, and $\tau_\sigma t(x) = t(x - \sigma e_0)$ is the translate by the Euclidean time σ. By (F3) the smear of the reflected cluster is the lattice expectation of $\vartheta\bar{F}$ when that of $(\mathbf{v}, t)$ is the expectation of F.

[6] The temporal link $(x, 0)$ runs from x to $x + e_0$; its image under ϑ runs from ϑx to $\vartheta x - e_0$, which is the link $(\vartheta x - e_0, 0)$ traversed backwards, whence the inverse. With this convention ϑ maps plaquette variables to plaquette variables of the reflected plaquette, up to conjugation and inversion, so the Wilson action and the Haar measure are invariant and ϑ is a measure preserving involution of every $\Lambda_{L,M}$ with M even, and of the infinite cylinder.

Definition 7.2 (hypotheses on the family)**.** Let $\ell > 0$, $\varpi > 0$ and $n_N \geq 2$ with $n_N a_N \leq \ell$.

(W) *Wightman limit on symmetric tori.* There are a compact metric space X of *phases* with a Borel probability measure ν of full support, and for every $\chi \in X$ and every cluster a number $S^\chi(\mathbf{v}, t) = \int t\, S^\chi_{\mathbf{v}}$ with $S^\chi_{\mathbf{v}}$ a smooth function on D_m, continuous in χ at every point of D_m, such that for every cluster

$$\lim_{p\to\infty} \lim_{N\to\infty} S_{N,p,NL_p}(\mathbf{v}, t) = \int_X S^\chi(\mathbf{v}, t)\, d\nu(\chi),$$

with $\chi \mapsto S^\chi(\mathbf{v}, t)$ ν-integrable, the inner limit along N in an arithmetic progression and the outer along a sequence of p, and for ν-almost every χ the functions $S^\chi_{\mathbf{v}}$ are the Schwinger functions of a Wightman theory with a unique vacuum, in the sense of Lemma 7.6 below, in which the field of v^ϑ is the adjoint of the field of v.

(U) *Order of the volume limits.* For every N, p and every cluster, $S^0_{N,p}(\mathbf{v}, t) = \lim_{M\to\infty} S_{N,p,M}(\mathbf{v}, t)$ satisfies

$$\lim_{p\to\infty} \lim_{N\to\infty} S^0_{N,p}(\mathbf{v}, t) = \int_X S^\chi(\mathbf{v}, t)\, d\nu(\chi)$$

along the same progression and sequence.

(D_ℓ) *The cube condition on cylinders.* For all large N and p and every $M \geq 2n_N + 4$, $\mathrm{BP}(n_N, \varpi)$ holds for $\mu_{NL_p,M}$ at coupling $\beta_{N,p}$.

The limit in (U) exists by Lemma 7.3(v) below; (U) asserts that the volume limit does not depend on whether the time extent is sent to infinity before or after the lattice spacing. Hypothesis (W) lists what a construction by scaling limits on symmetric tori delivers when it succeeds: a fixed finite stencil for every insertion, gauge invariance, convergence of the separated smears first in the mesh and then in the volume, a phase measure of full support, continuity of the separated correlations in the phase, and a Wightman theory with a unique vacuum for every phase, whose Schwinger functions at Euclidean points are the phase correlations. The reflection property (F3) and the identification of v^ϑ with the adjoint field hold when the insertions are real and reflection symmetric, as for Hermitian fields built from plaquettes. Hypothesis (U) is one instance of the uniqueness of the volume limit, which such a construction need not provide. A construction of this kind, for every compact simple group, is claimed in the Lean 4 development [45], assessed in [62]; neither is a refereed publication, the theorem below does not depend on them, and they are cited only as the source of the shape of (W).

Theorem E (the bridge)**.** *Let a continuum family satisfy* (W)*,* (U) *and* (D_ℓ)*. Then S^χ does not depend on χ, and the Wightman theory of* (W) *has Hamiltonian H with*

$$\sigma(H) \subset \{0\} \cup [\Delta, \infty), \qquad \Delta = \frac{16\varpi}{625 e^2 \ell},$$

with 0 *a simple eigenvalue.*

The proof uses (W) only through the phase structure, the continuity in χ, the integrability of the phase correlations and the Wightman property of the phases; the limit on symmetric tori enters through (U), or through the weaker hypothesis (Z) of Section 7.7, which implies the form of (U) that the proof needs. None of (W), (U), (Z) or (D_ℓ) is established in this paper for any four-dimensional family: Theorem C gives the cube condition at fixed strong coupling, whereas (D_ℓ) asks for it with ϖ fixed and $n_N \leq \ell N$ along the couplings $\beta_{N,p}$ of a construction.

The proof occupies the rest of the section: the transfer matrix of a cylinder (Sec. 7.2), its gap from the cube condition (Sec. 7.3), decay of the renormalised two-point functions at zero temperature with a constant that survives the limit (Sec. 7.4), the Euclidean vectors of a Wightman theory (Sec. 7.5), and the assembly (Sec. 7.6).

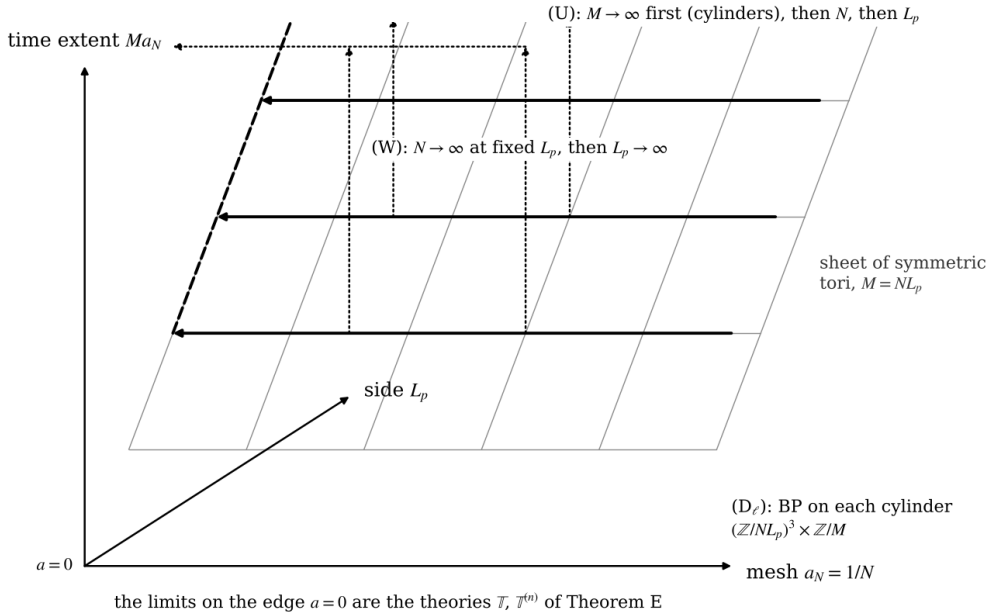


FIGURE 5. The two orders of the volume limits, in the space of mesh a_N, physical side L_p and physical time extent Ma_N. The symmetric tori of (W) lie on the sheet where the time extent equals the side: the mesh is sent to zero at fixed L_p along the sheet, then $L_p \to \infty$ along $a = 0$ (solid and dashed arrows). The zero-temperature order sends the time extent to infinity first, on the cylinders where (D_ℓ) is assumed, then the mesh, then the side (dotted arrows). Hypothesis (U), or its consequence of (Z) in Corollary 7.11, is the statement that the two routes have the same end point.

7.2. **The transfer matrix of a cylinder.** Fix L and β. Let $\mathcal{K} = L^2(G^{E_0}, \prod dQ_e)$ be the square integrable functions of the spatial links of one time slice. For configurations Q, Q' of a slice and a family $U = (U_{\vec{x}})$ of temporal link variables put

$$S_s(Q) = \sum_{p \subset E_0} \operatorname{Re} \operatorname{Tr} Q_p, \qquad S_0(Q, U, Q') = \sum_{\vec{x},\, i} \operatorname{Re} \operatorname{Tr} \big(Q_{(\vec{x},i)} U_{\vec{x}+e_i} Q'^{*}_{(\vec{x},i)} U^{*}_{\vec{x}} \big),$$

the sums over the spatial plaquettes of the slice and over the timelike plaquettes between two consecutive slices, and define the kernel

$$\mathbb{T}(Q, Q') = \int_{G^{(\mathbb{Z}/L\mathbb{Z})^3}} \exp\Big(N\beta \big[\tfrac{1}{2} S_s(Q) + \tfrac{1}{2} S_s(Q') + S_0(Q, U, Q') \big] \Big) \prod_{\vec{x}} dU_{\vec{x}}. \tag{7.2}$$

For a function F of the links of the slab $[0, w]$, $w \geq 1$, let $\hat{F}$ be the operator on $\mathcal{K}$ with kernel

$$\begin{aligned} \hat{F}(Q_0, Q_w) = \int & F(Q_0, U^{(0)}, Q_1, \dots, U^{(w-1)}, Q_w) \\ & \times \prod_{t=0}^{w-1} e^{N\beta[\frac{1}{2} S_s(Q_t) + \frac{1}{2} S_s(Q_{t+1}) + S_0(Q_t, U^{(t)}, Q_{t+1})]} \prod_{t=1}^{w-1} dQ_t \prod_{t=0}^{w-1} dU^{(t)}, \end{aligned} \tag{7.3}$$

and for $w = 0$ let $\hat{F}$ be multiplication by F. Gauge transformations $g = (g_{\vec{x}})$ of the slice act unitarily on $\mathcal{K}$ by $(R_g f)(Q) = f(g \cdot Q)$.

Lemma 7.3 (transfer matrix)**.** (i) $\mathbb{T}$ *is a self-adjoint Hilbert–Schmidt operator with a continuous strictly positive kernel; its largest eigenvalue* $\lambda_0 > 0$ *is simple, with a*

continuous eigenfunction $\Omega_0 > 0$, and every eigenfunction with nonzero eigenvalue is gauge invariant. Every other eigenvalue has modulus less than λ_0.

(ii) *For $M \geq 2$ the partition function of $\Lambda_{L,M}$ is $\operatorname{Tr} \mathbb{T}^M$.*[7]

(iii) *Let F be a bounded measurable function of the slab $[0, w]$ and H one of the slab $[0, w']$, and let $s \geq w$ and $M \geq s + w'$. Then*

$$\mu_{L,M}(F\, \tau_s H) = \frac{\operatorname{Tr}(\hat{F}\, \mathbb{T}^{s-w} \hat{H}\, \mathbb{T}^{M-s-w'})}{\operatorname{Tr} \mathbb{T}^M}.$$

(iv) *The function $\vartheta \bar{F}$ is a function of the slab $[-w, 0]$, and for $s \geq 0$ and $M \geq s + w + w'$*

$$\mu_{L,M}(\vartheta \bar{F}\, \tau_s H) = \frac{\operatorname{Tr}(\hat{F}^*\, \mathbb{T}^{s} \hat{H}\, \mathbb{T}^{M-s-w-w'})}{\operatorname{Tr} \mathbb{T}^M}.$$

(v) *For every bounded measurable function F of a slab the limit $\mu_L^0(F) = \lim_{M\to\infty} \mu_{L,M}(F)$ exists, and with $\Psi_F = \lambda_0^{-w} \hat{F} \Omega_0$ and $\mathbb{R} = \mathbb{T}/\lambda_0$,*

$$\mu_L^0(\vartheta \bar{F}\, \tau_s H) = \langle \Psi_F, \mathbb{R}^s \Psi_H \rangle, \qquad \mu_L^0(H) = \langle \Omega_0, \Psi_H \rangle \qquad (s \geq 0).$$

(vi) *If F is gauge invariant then $R_g \hat{F} = \hat{F} R_g$ for every g.*

(vii) *If $\beta \geq 0$ then $\mathbb{T} \geq 0$ as an operator on $\mathcal{K}$.*

Proof. (i) The integrand of (7.2) is continuous and positive on a compact space, so the kernel is continuous and strictly positive, and $\mathbb{T}$ is Hilbert–Schmidt. Replacing $U_{\vec{x}}$ by $U_{\vec{x}}^{-1}$, which preserves Haar measure, and using $\operatorname{Re}\operatorname{Tr} A = \operatorname{Re}\operatorname{Tr} A^*$ with the cyclicity of the trace turns $S_0(Q, U, Q')$ into $S_0(Q', U, Q)$, so the kernel is symmetric and $\mathbb{T}$ is self-adjoint. A simultaneous gauge transformation $Q \mapsto g \cdot Q$, $Q' \mapsto g' \cdot Q'$ is absorbed by $U_{\vec{x}} \mapsto g_{\vec{x}}^{-1} U_{\vec{x}} g'_{\vec{x}}$, so $\mathbb{T}(g \cdot Q, g' \cdot Q') = \mathbb{T}(Q, Q')$; hence $\mathbb{T} f$ is gauge invariant for every f, and an eigenfunction with nonzero eigenvalue, being in the range of $\mathbb{T}$, is gauge invariant and continuous. By Jentzsch's[8] theorem [30], in the form of [50, Thm. V.6.6], a compact operator with a strictly positive kernel has its spectral radius as a simple eigenvalue, with an almost everywhere positive eigenfunction, and every other eigenvalue has smaller modulus; $\Omega_0 = \lambda_0^{-1} \mathbb{T} \Omega_0$ is continuous and strictly positive.

(ii) Write the Wilson weight of $\Lambda_{L,M}$ as the product over $t \in \mathbb{Z}/M\mathbb{Z}$ of $\exp(N\beta[\frac{1}{2} S_s(Q_t) + \frac{1}{2} S_s(Q_{t+1}) + S_0(Q_t, U^{(t)}, Q_{t+1})])$: every spatial plaquette of slice t appears with weight $\frac{1}{2}$ in the factors $t-1$ and t, and every timelike plaquette once. Integrating the temporal links gives the kernel of $\mathbb{T}$ between consecutive slices, and the integral over the slices is the trace of the M-fold product.

(iii) With the same factorisation, the slab $[0, w]$ contributes $\hat{F}(Q_0, Q_w)$ after integrating its interior slices and temporal links, the slab $[s, s + w']$ contributes $\hat{H}(Q_s, Q_{s+w'})$, and the remaining $s - w$ and $M - s - w'$ factors contribute the powers of $\mathbb{T}$; when $s = w$ the shared slice $Q_w = Q_s$ is the integration variable of the composition $\hat{F}\hat{H}$.

(iv) Under ϑ a temporal link from time t to $t + 1$ becomes the link from $-t$ to $-t - 1$, traversed backwards, which is the inversion in (F3); the timelike plaquette weight is invariant, as in (i). So $\vartheta \bar{F}$ is the function of the slab $[-w, 0]$ whose kernel, in the variables (Q_{-w}, Q_0), is $\overline{\hat{F}(Q_0, Q_{-w})} = \hat{F}^*(Q_{-w}, Q_0)$. Now apply (iii) to the slabs $[-w, 0]$ and $[s, s + w']$.

(v) Let P_0 be the projection on Ω_0 and $\mathbb{R} = \mathbb{T}/\lambda_0 = P_0 + \mathbb{S}$ with $\|\mathbb{S}\| < 1$ by (i). Since $\mathbb{T}^2$ is trace class, $\|\mathbb{S}^k\|_1 \leq \|\mathbb{S}^2\|_1 \|\mathbb{S}\|^{k-2} \to 0$, so $\operatorname{Tr}(A\mathbb{R}^{M-c}) / \operatorname{Tr} \mathbb{R}^M \to \langle \Omega_0, A\Omega_0 \rangle$ for every

[7] The spatial plaquettes of the slice t carry weight $\frac{1}{2}$ in each of the two kernels $\mathbb{T}(Q_{t-1}, Q_t)$ and $\mathbb{T}(Q_t, Q_{t+1})$, so they carry the full Wilson weight exactly once around the cycle of M slices, and the temporal plaquettes between the slices t and $t + 1$ appear in $S_0(Q_t, U^{(t)}, Q_{t+1})$ only; the trace integrates over the M slices and the M families of temporal links, which are all the links of $\Lambda_{L,M}$.

[8] Jentzsch's theorem [30], in the form of [50, Ch. V, §6]: an integral operator on $L^2(X, m)$, with m a finite measure on a compact space X and a continuous kernel $k(x, y) > 0$ everywhere, is compact, its spectral radius $\lambda_0 > 0$ is an eigenvalue, simple, with an eigenfunction that is strictly positive, and every other eigenvalue has modulus strictly less than λ_0. Here $X = G^{E_0}$ with the product Haar measure, and the kernel (7.2) is positive because the integrand is.

bounded A and fixed c. Apply this to (iii) with $H = 1$ for the existence of the limit and to (iv) for the formula: $\operatorname{Tr}(\hat F^{*}\mathbb{T}^{s}\hat H\mathbb{T}^{M-s-w-w'})/\operatorname{Tr}\mathbb{T}^{M} \to \lambda_0^{-s-w-w'}\langle\Omega_0, \hat F^{*}\mathbb{T}^{s}\hat H\Omega_0\rangle$.

(vi) The kernel (7.3) is invariant under simultaneous gauge transformations of all slices of the slab, by the invariance of F and the substitution of (i) for the temporal links, so $\hat F(g\cdot Q_0, g\cdot Q_w) = \hat F(Q_0, Q_w)$.

(vii) Put $f(g) = e^{N\beta \operatorname{Re}\operatorname{Tr} g}$ on G and $c = N\beta \ge 0$. Since $\operatorname{Re}\operatorname{Tr} g = \frac12(\chi(g) + \overline{\chi(g)})$ with $\chi = \operatorname{Tr}$ the character of the defining representation,

$$f = \sum_{a,b\ge 0} \frac{(c/2)^{a+b}}{a!\,b!}\,\chi^{a}\bar\chi^{b},$$

uniformly on G, and $\chi^a\bar\chi^b$ is the character of $(\mathbb{C}^N)^{\otimes a}\otimes(\overline{\mathbb{C}^N})^{\otimes b}$, a sum of irreducible characters with nonnegative integer coefficients. Hence $f = \sum_\lambda f_\lambda\chi_\lambda$ with $f_\lambda \ge 0$, and as $\chi_\lambda(x^{-1}y) = \sum_{a,b}\overline{D^\lambda_{ab}(x)}D^\lambda_{ab}(y)$ in a unitary basis, $f(x^{-1}y)$ is a positive definite kernel on G; so is $f(xy^{-1}) = f(y^{-1}x) = f(x^{-1}y)$, by conjugation invariance and $\operatorname{Re}\operatorname{Tr} g = \operatorname{Re}\operatorname{Tr} g^{-1}$. By Schur's product theorem[9] the kernel $K(Q,Q') = \prod_{e\in E_0} f(Q_eQ_e'^{*})$ is positive definite on G^{E_0}, and being continuous it defines a nonnegative Hilbert–Schmidt operator K on $\mathcal{K}$. The substitution $Q' \mapsto U\cdot Q'$, $(U\cdot Q')_{(\vec x,i)} = U_{\vec x}Q'_{(\vec x,i)}U^{*}_{\vec x+e_i}$, turns $S_0(Q,U,Q')$ into $\sum\operatorname{Re}\operatorname{Tr}(Q_{(\vec x,i)}(U\cdot Q')^{*}_{(\vec x,i)})$, so

$$\mathbb{T} = D\Big(\int K\circ R_U^{-1}\prod_{\vec x} dU_{\vec x}\Big)D = D\,K\,\Pi\,D = D\,\Pi K\Pi\,D,$$

where D is multiplication by $e^{N\beta S_s/2}$, $\Pi = \int R_U\,dU$ is the orthogonal projection on the gauge-invariant functions, and K commutes with every R_g because $K(g\cdot Q, g\cdot Q') = K(Q,Q')$. Thus $\mathbb{T} = (K^{1/2}\Pi D)^{*}(K^{1/2}\Pi D) \ge 0$. □

The state μ_L^0 is the zero-temperature state of the cylinder of spatial side L: by (v) it is the expectation in the vector Ω_0 of the time-ordered products of slab operators, and $\mu_{L,M}$ is the corresponding thermal state at temperature $1/M$.

7.3. **The gap of the transfer matrix from the cube condition.**

Proposition 7.4 (cylinder gap)**.** *Let $n \ge 2$, $\varpi > 0$, $L \ge 2n+4$, and suppose* $\mathrm{BP}(n,\varpi)$ *holds for $\mu_{L,M}$ for every $M \ge 2n+4$. Then every eigenvalue $\lambda \ne \lambda_0$ of $\mathbb{T}$ satisfies*

$$\frac{|\lambda|}{\lambda_0} \le e^{-m_n}, \qquad m_n = \frac{\varpi}{e^2 q_n n}.$$

Proof. Let F be a real continuous gauge-invariant function of the slice E_0 with $|F| \le 1$, depending on a set S of links, and let $s \ge 3n-2$ be even. On $\Lambda_{L,M}$ with $M \ge 2s$ the functions F and τ_sF depend on links whose base points differ by s in time, so $d(S,\tau_sS) = s$; as $s \ge 3n-2$ and $n \ge 2$, $M \ge 2s \ge 6n-4 \ge 2n+4$, so all sides of $\Lambda_{L,M}$ are at least $2n+4$, $\mathrm{BP}(n,\varpi)$ holds for $\mu_{L,M}$, and Theorem 6.3 gives

$$\big|\mu_{L,M}(F\,\tau_sF) - \mu_{L,M}(F)^2\big| \le \Big(e + \frac{k|S|^2}{64e^2}\Big)e^{-\varpi k/(e^2q_n)}, \qquad k = \Big\lfloor\frac{s+2}{n}\Big\rfloor - 2,$$

uniformly in M. Let $M\to\infty$: by Lemma 7.3(v) with $w = w' = 0$ and $\vartheta\bar F = F$, the left side becomes $\langle F\Omega_0, \mathbb{R}^sF\Omega_0\rangle - \langle\Omega_0, F\Omega_0\rangle^2 = \sum_{j\ge1} r_j^s|\langle\phi_j, F\Omega_0\rangle|^2$, where $(\phi_j)_{j\ge0}$ is an

[9]The entrywise product of two positive semidefinite matrices (a_{ij}), (b_{ij}) is positive semidefinite, because it is the principal submatrix of the tensor product $(a_{ij}b_{kl})$ on the diagonal pairs (i,i), and the tensor product of positive semidefinite matrices is positive semidefinite. Applied to finite sections of continuous kernels this shows that the product of positive definite kernels is positive definite.

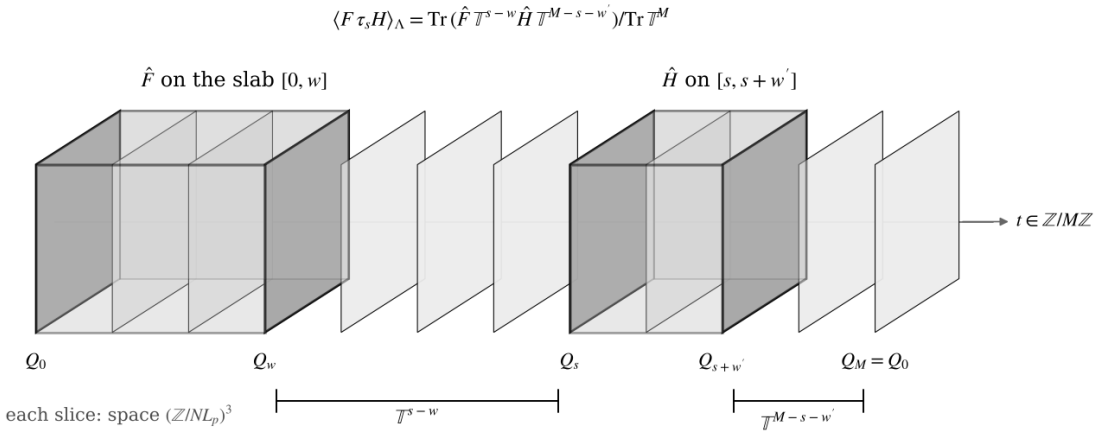


FIGURE 6. The cylinder $\Lambda_{L,M}$ sliced in time. A function F of the slab $[0,w]$ becomes the integral operator $\hat F$ on the slice space $\mathcal{K}$, a function H of the slab $[s,s+w']$ becomes $\hat H$, and the time steps between the slabs are powers of the transfer matrix $\mathbb{T}$; the trace formula is Lemma 7.3(iii), and the reflected form, with $\hat F^*$, is Lemma 7.3(iv).

orthonormal eigenbasis of $\mathbb{T}$ with real eigenvalues $\lambda_j = r_j\lambda_0$, $\phi_0 = \Omega_0$. As s is even every term is nonnegative, and $s/n - 3 < k \le s/n$, so for each $j \ge 1$

$$|r_j|^s\,|\langle\phi_j, F\Omega_0\rangle|^2 \le \Big(e + \frac{(s/n)|S|^2}{64e^2}\Big)e^{3\varpi/(e^2q_n)}e^{-m_n s}.$$

Letting $s\to\infty$ through even values gives $|r_j| \le e^{-m_n}$ whenever $\langle\phi_j, F\Omega_0\rangle \ne 0$ for some such F. It remains to see that the vectors $F\Omega_0$, F real, span a dense subspace of the gauge-invariant part of $\mathcal{K}$, which contains every ϕ_j with $\lambda_j \ne 0$ by Lemma 7.3(i). Let h be gauge invariant and square integrable. Since Ω_0 is continuous, strictly positive and gauge invariant, h/Ω_0 is gauge invariant and square integrable for the finite measure $\Omega_0^2\prod dQ_e$, and the continuous gauge-invariant functions are dense in that space, because averaging a continuous approximant over the compact gauge group preserves the approximation. If $F_k \to h/\Omega_0$ there, then $\|F_k\Omega_0 - h\|_{\mathcal{K}} \to 0$; the real and imaginary parts of F_k are real continuous gauge-invariant functions. □

The proposition needs the cube condition on cylinders of every time period and fixed spatial side: on a single symmetric torus the two-point function of a bounded observable at separation $s \le M/2$ contains the thermal terms $r_i^{M-s}r_j^s$, and a bound of the form Ce^{-ms} for $s \le M/2$ at one value of M only gives $r_1 \le e^{-m}(Cz)^{2/M}$ with $z = \mathrm{Tr}\,\mathbb{R}^M$, which tends to e^{-m} as $M\to\infty$ at fixed coupling but not along a tuning in which the coupling changes with M.

7.4. **Decay of renormalised two-point functions at zero temperature.**

Lemma 7.5 (gap to decay, with a surviving constant)**.** *Suppose every eigenvalue $\lambda \ne \lambda_0$ of $\mathbb{T}$ satisfies $|\lambda| \le e^{-m}\lambda_0$. Let F and H be bounded measurable functions of the slabs $[0,w]$ and $[0,w']$, let $\Psi'_F = \Psi_F - \langle\Omega_0,\Psi_F\rangle\Omega_0$, and put $C_F(s) = \mu_L^0(\vartheta\bar F\,\tau_s F) - |\mu_L^0(F)|^2 = \langle\Psi'_F, \mathbb{R}^s\Psi'_F\rangle \ge 0$. Then for all integers $s \ge s_1 \ge 0$,*

$$|\mu_L^0(\vartheta\bar F\,\tau_s H) - \overline{\mu_L^0(F)}\,\mu_L^0(H)| \le e^{-m(s-s_1)}\sqrt{C_F(s_1)\,C_H(s_1)}.$$

Proof. By Lemma 7.3(v) the left side is $|\langle \Psi'_F, \mathbb{R}^s \Psi'_H\rangle|$, since $\mathbb{R}\Omega_0 = \Omega_0$ and $\Psi'_F \perp \Omega_0$. By Lemma 7.3(vii), $\mathbb{R} \geq 0$, so $\mathbb{R}^{s_1/2}$ is defined by the functional calculus, and $\mathbb{R}^s = \mathbb{R}^{s_1/2}\,\mathbb{R}^{s-s_1}\,\mathbb{R}^{s_1/2}$. The subspace $\Omega_0^\perp$ is invariant under $\mathbb{R}$, on which $\|\mathbb{R}\| \leq e^{-m}$, so the Cauchy–Schwarz inequality gives $|\langle \Psi'_F, \mathbb{R}^s \Psi'_H\rangle| \leq e^{-m(s-s_1)} \|\mathbb{R}^{s_1/2}\Psi'_F\|\,\|\mathbb{R}^{s_1/2}\Psi'_H\|$, and $\|\mathbb{R}^{s_1/2}\Psi'_F\|^2 = \langle \Psi'_F, \mathbb{R}^{s_1}\Psi'_F\rangle = C_F(s_1)$. □

The point of the lemma is the constant. For a renormalised cluster the norms $\|\Psi_F\|$, the suprema $\|F\|_\infty$ and the variances of F under the lattice measure diverge as $N \to \infty$, so the covariance bound of Theorem 6.3 is useless for such F; the quantity $C_F(s_1)$ is the connected two-point function of the cluster with its reflection at the positive separation s_1, which under (U) converges to a Schwinger function at separated points.

7.5. **Euclidean vectors of a Wightman theory.** Let $(\mathcal{H}, \Omega, U)$ carry a Wightman theory in the sense of [54, Ch. 3]: a unitary representation $U(a,\Lambda)$ of the Poincaré group with energy–momentum P satisfying the spectrum condition $\sigma(P) \subset \overline{V_+}$, a unique invariant unit vector Ω, and operator-valued tempered distributions Φ_v, $v \in V$, on a common dense invariant domain containing Ω, covariant under U and local, with Ω cyclic for the polynomials in the fields, and with $\Phi_{v^\vartheta}(f) = \Phi_v(\bar f)^*$ on the domain. Let $H = P^0 \geq 0$. We use the metric $p \cdot z = p^0 z^0 - \vec p \cdot \vec z$ and $U(a) = e^{iP\cdot a}$, and the forward tube

$$\mathfrak{T}_m = \{ z \in (\mathbb{C}^4)^m : \ \operatorname{Im} z_1 \in V_+, \ \operatorname{Im}(z_{j+1} - z_j) \in V_+ \ (1 \leq j < m)\}.$$

A Euclidean point $x = (x_1, \dots, x_m) \in (\mathbb{R}^4)^m$ with times $0 < \tau_1 < \cdots < \tau_m$ corresponds to $\hat x = ((i\tau_1, \vec x_1), \dots, (i\tau_m, \vec x_m)) \in \mathfrak{T}_m$.

Lemma 7.6 (Euclidean vectors)**.** (i) *For $\mathbf{v} \in V^m$ there is a holomorphic function $\Psi_{\mathbf{v}} : \mathfrak{T}_m \to \mathcal{H}$ whose boundary value as $\operatorname{Im} z \to 0$ inside the tube is the vector-valued distribution $\Phi_{v_1}(x_1)\cdots\Phi_{v_m}(x_m)\Omega$.*

(ii) *For a cluster $(\mathbf{v}, t)$ with t supported in the configurations with $0 < \tau_1 < \cdots < \tau_m$, the vector $\psi(\mathbf{v},t) = \int t(x)\Psi_{\mathbf{v}}(\hat x)\,dx$ satisfies $\langle \Omega, \psi(\mathbf{v},t)\rangle = \int t\, S_{\mathbf{v}}$ and, for $\sigma \geq 0$ and a second such cluster,*

$$\langle \psi(\mathbf{v},t), e^{-\sigma H}\psi(\mathbf{v}',t')\rangle = \int (\Theta \bar t \otimes \tau_\sigma t')\, S_{\mathbf{v}^\vartheta \mathbf{v}'}, \qquad \mathbf{v}^\vartheta = (v_1^\vartheta, \dots, v_m^\vartheta),$$

where $S_{\mathbf{u}}$ are the Schwinger functions, defined as follows. The vacuum expectation $W_{\mathbf{u}}(x) = \langle \Omega, \Phi_{u_1}(x_1)\cdots\Phi_{u_k}(x_k)\Omega\rangle$ is the boundary value of a function $W_{\mathbf{u}}(\zeta)$ holomorphic in the difference tube

$$\mathfrak{T}'_k = \{\zeta \in (\mathbb{C}^4)^k : \ \operatorname{Im}(\zeta_{j+1} - \zeta_j) \in V_+ \ (1 \leq j < k)\},$$

and $S_{\mathbf{u}}(x) = W_{\mathbf{u}}(\hat x)$ when $\tau_1 < \cdots < \tau_k$, extended to the configurations with distinct times by permutation symmetry.

(iii) *The vectors $\psi(\mathbf{v},t)$ of (ii), over all $m \geq 1$, $\mathbf{v}$ and t, together with Ω, span a dense subspace of $\mathcal{H}$.*

Proof. (i) The map $(f_1, \dots, f_m) \mapsto \Phi_{v_1}(f_1)\cdots\Phi_{v_m}(f_m)\Omega$ is multilinear and separately continuous from $\mathcal{S}(\mathbb{R}^4)^m$ to $\mathcal{H}$, hence jointly continuous, and extends to a continuous linear map $f \mapsto \Psi_{\mathbf{v}}(f)$ on $\mathcal{S}(\mathbb{R}^{4m})$ by the nuclear theorem. Write $x_j = x_1 + \xi_1 + \cdots + \xi_{j-1}$. Since $\Phi_{v_j}(x_j)\cdots\Phi_{v_m}(x_m)\Omega = U(x_j)\Phi_{v_j}(0)\Phi_{v_{j+1}}(x_{j+1} - x_j)\cdots\Omega$ in the sense of distributions, the Fourier transform of $\Psi_{\mathbf{v}}$ in the variable x_1 is supported in $\overline{V_+}$, and so is its Fourier transform in each ξ_j, by the spectrum condition applied to the vector to the right of $U(\xi_j)$. For every $\chi \in \mathcal{H}$ the scalar distribution $\langle \chi, \Psi_{\mathbf{v}}(\cdot)\rangle$ therefore has Fourier transform supported in the closed convex cone $\overline{V_+}^m$ in these variables, so it is the boundary value of a function holomorphic in $\mathfrak{T}_m$ [54, Thm. 2-8]; the family of these distributions over $\|\chi\| \leq 1$ is bounded in $\mathcal{S}'$, so the holomorphic functions are uniformly bounded on compact subsets of the tube, and they define a weakly, hence strongly, holomorphic $\Psi_{\mathbf{v}} : \mathfrak{T}_m \to \mathcal{H}$ with $\langle \chi, \Psi_{\mathbf{v}}(z)\rangle$ equal to these continuations.

(ii) The vacuum expectation $W_{\mathbf{u}}$ is translation invariant, and the argument of (i) applied to the differences shows that it is the boundary value of a function holomorphic in $\mathfrak{T}'_k$; for x with increasing times, $\hat{x} \in \mathfrak{T}'_k$ and $W_{\mathbf{u}}(\hat{x}) = \langle \Omega, \Psi_{\mathbf{u}}(\hat{x}) \rangle$ when the times are positive, by (i) and translation invariance, so the first identity holds. For the second, let $z \in \mathfrak{T}_m$ and $z' \in \mathfrak{T}_{m'}$. The function $(z, z') \mapsto \langle \Psi_{\mathbf{v}}(z), \Psi_{\mathbf{v}'}(z') \rangle$ is antiholomorphic in z and holomorphic in z', and by $\Phi_{v^\vartheta}(f) = \Phi_v(\bar{f})^*$ its boundary value is the distribution $W_{\check{\mathbf{v}}^\vartheta \mathbf{v}'}(x_m, \dots, x_1, y_1, \dots, y_{m'})$, $\check{\mathbf{v}}^\vartheta = (v_m^\vartheta, \dots, v_1^\vartheta)$. The map $(z, z') \mapsto (\bar{z}_m, \dots, \bar{z}_1, z'_1, \dots, z'_{m'})$ sends $\mathfrak{T}_m \times \mathfrak{T}_{m'}$ into $\mathfrak{T}'_{m+m'}$: the imaginary parts of the consecutive differences are $\mathrm{Im}(z_{j+1} - z_j)$, $\mathrm{Im}\, z_1 + \mathrm{Im}\, z'_1$ and $\mathrm{Im}(z'_{j+1} - z'_j)$, all in V_+. Two functions holomorphic in $(\bar{z}, z')$ on this product of tubes with the same distributional boundary value coincide [54, Ch. 2], so

$$\langle \Psi_{\mathbf{v}}(z), \Psi_{\mathbf{v}'}(z') \rangle = W_{\check{\mathbf{v}}^\vartheta \mathbf{v}'}(\bar{z}_m, \dots, \bar{z}_1, z'_1, \dots, z'_{m'}).$$

Next, $U(a)\Psi_{\mathbf{v}'}(z') = \Psi_{\mathbf{v}'}(z' + a)$ for real a by covariance; both sides are holomorphic in a on the tube $\mathrm{Im}\, a \in V_+$ with these boundary values, the left by the spectrum condition and the right because $z' + a \in \mathfrak{T}_{m'}$, so they agree there, and $e^{-\sigma H}\Psi_{\mathbf{v}'}(\hat{y}) = \Psi_{\mathbf{v}'}(\widehat{\tau_\sigma y})$. At the Euclidean points $z = \hat{x}$, $z' = \widehat{\tau_\sigma y}$ one has $\bar{z}_j = \widehat{\vartheta x_j}$, and the times $-\tau_m < \dots < -\tau_1 < \sigma + \tau'_1 < \dots$ increase, so

$$\langle \Psi_{\mathbf{v}}(\hat{x}), e^{-\sigma H}\Psi_{\mathbf{v}'}(\hat{y}) \rangle = S_{\check{\mathbf{v}}^\vartheta \mathbf{v}'}(\vartheta x_m, \dots, \vartheta x_1, \tau_\sigma y_1, \dots, \tau_\sigma y_{m'}).$$

Integrate against $\bar{t}(x)t'(y)$ and substitute $u_i = \vartheta x_{m+1-i}$; the permutation symmetry

$$S_{\check{\mathbf{v}}^\vartheta \mathbf{v}'}(u_m, \dots, u_1, y) = S_{\mathbf{v}^\vartheta \mathbf{v}'}(u_1, \dots, u_m, y)$$

turns the result into $\int \overline{t(\vartheta u_1, \dots, \vartheta u_m)}\, (\tau_\sigma t')(y)\, S_{\mathbf{v}^\vartheta \mathbf{v}'}(u, y)\, du\, dy$, which is the claim.

(iii) Let $\chi \perp \Omega$ and $\chi \perp \psi(\mathbf{v}, t)$ for all clusters of (ii). Then the holomorphic function $\langle \chi, \Psi_{\mathbf{v}}(z) \rangle$ vanishes on the open subset $\{\hat{x}\}$ of the totally real submanifold $\{(i\tau_j, \vec{x}_j)\} \cap \mathfrak{T}_m$, which has real dimension $4m$; after the complex-linear change of coordinates $z_j^0 \mapsto -iz_j^0$ this is an open subset of $\mathbb{R}^{4m}$, so the function vanishes on the connected tube. Its boundary value $\langle \chi, \Phi_{v_1}(x_1) \cdots \Phi_{v_m}(x_m)\Omega \rangle$ then vanishes as a distribution, for all m and $\mathbf{v}$, so χ is orthogonal to the polynomials in the fields applied to Ω, which are dense, and to Ω itself. □

7.6. **Proof of Theorem E.** *Step 1: the lattice inequality.* Fix N and p large, put $L = NL_p$, $n = n_N$ and $a = a_N$, and let $\mathbb{T}$ be the transfer matrix of the cylinder of side L at coupling $\beta_{N,p}$. Since $n_N \le \ell N$, one has $L \ge 2n_N + 4$ as soon as $L_p \ge 2\ell + 1$ and $N \ge 4$, which holds for p and N large; so by (D_ℓ) and Proposition 7.4, $\mathbb{T}$ has gap $m_n = \varpi/(e^2 q_n n)$. Let $(\mathbf{v}, t)$ and $(\mathbf{v}', t')$ be clusters with t, t' supported in configurations whose times lie in (δ, ω), $0 < \delta < \omega$, and let $\rho \ge 0$ be such that $R_v \subset \mathbb{Z}^3 \times [-\rho, \rho]$ for the finitely many tokens v of $\mathbf{v}, \mathbf{v}'$, a link being in $x + R_v$ when its base point is. For $N > \rho/\delta$ a site x with $t(ax) \ne 0$ has time $x^0 \in (\delta N, \omega N)$, so the links of $x + R_v$ have end points at times in $[x^0 - \rho, x^0 + \rho + 1] \subset [0, \lceil \omega N \rceil + \rho]$; hence the lattice functions

$$F = a^{4m} \sum_x t(ax) \prod_i \Phi^{N,p}_{v_i}(x_i, \cdot), \qquad H = a^{4m'} \sum_y t'(ay) \prod_j \Phi^{N,p}_{v'_j}(y_j, \cdot)$$

are bounded, gauge invariant, and depend on the links of the slab $[0, w]$ with $w = \lceil \omega N \rceil + \rho$. Let $\sigma \ge \sigma_1 \ge 1$ be integers. Lemma 7.5 with $s = \sigma N$, $s_1 = \sigma_1 N$ gives

$$|\mu_L^0(\vartheta \bar{F}\, \tau_{\sigma N} H) - \overline{\mu_L^0(F)}\mu_L^0(H)| \le e^{-m_n N(\sigma - \sigma_1)} \sqrt{C_F(\sigma_1 N) C_H(\sigma_1 N)}, \tag{7.4}$$

and $m_n N = \varpi/(e^2 q_n n a) \ge \Delta$ because $na \le \ell$ and $q_n \le 625/16$. Each quantity in (7.4) is a zero-temperature smear $S^0_{N,p}$ of a cluster: by (F3), $\vartheta \bar{F} = a^{4m} \sum_y (\Theta \bar{t})(ay) \prod_i \Phi^{N,p}_{v_i^\vartheta}(y_i, \cdot)$, and by (F2), $\tau_{\sigma N} H = a^{4m'} \sum_y (\tau_\sigma t')(ay) \prod_j \Phi^{N,p}_{v'_j}(y_j, \cdot)$ since $\sigma N \in \mathbb{Z}$; so $\mu_L^0(\vartheta \bar{F}\, \tau_{\sigma N} H)$ is the smear of $c_1 = (\mathbf{v}^\vartheta \mathbf{v}', \Theta \bar{t} \otimes \tau_\sigma t')$, $\mu_L^0(F)$ and $\mu_L^0(H)$ those of $c_2 = (\mathbf{v}, t)$ and $c_3 = (\mathbf{v}', t')$, and $C_F(\sigma_1 N)$ is the smear of $c_4 = (\mathbf{v}^\vartheta \mathbf{v}, \Theta \bar{t} \otimes \tau_{\sigma_1} t)$ minus $|S^0_{N,p}(\mathbf{v}, t)|^2$. The product clusters are supported in $D_{m+m'}$ because reflected and translated times are separated by at least 2δ.

Step 2: the limit. Let $N \to \infty$ along the progression and then $p \to \infty$ along the sequence of (W). By (U) every smear in (7.4) converges to the ν-average of the corresponding phase correlation. Writing $\bar S(\cdot) = \int_X S^\chi(\cdot)\, d\nu$ and $\bar S^c(t,t';\sigma) = \bar S(\mathbf{v}^\vartheta \mathbf{v}', \Theta\bar t \otimes \tau_\sigma t') - \overline{\bar S(\mathbf{v},t)}\, \bar S(\mathbf{v}',t')$,

$$|\bar S^c(t,t';\sigma)| \le e^{-\Delta(\sigma-\sigma_1)} \sqrt{\bar S^c(t,t;\sigma_1)\, \bar S^c(t',t';\sigma_1)} \tag{7.5}$$

for all integers $1 \le \sigma_1 \le \sigma$, the right side being nonnegative as a limit of nonnegative numbers.

Step 3: the phases coincide. For ν-almost every χ the phase is a Wightman theory, and by Lemma 7.6(ii) its own connected function $S^{\chi,c}(t,t;\sigma) = \|e^{-\sigma H_\chi/2}\psi'_\chi\|^2 \ge 0$, with $\psi'_\chi = \psi_\chi(\mathbf{v},t) - \langle \Omega_\chi, \psi_\chi(\mathbf{v},t)\rangle \Omega_\chi$. Hence

$$\bar S^c(t,t;\sigma) = \int_X S^{\chi,c}(t,t;\sigma)\, d\nu(\chi) + \mathrm{Var}_\nu\big(S^\chi(\mathbf{v},t)\big) \ge \mathrm{Var}_\nu\big(S^\chi(\mathbf{v},t)\big),$$

the variance being finite because $|S^\chi(\mathbf{v},t)|^2 \le \|\psi_\chi(\mathbf{v},t)\|^2 = S^\chi(\mathbf{v}^\vartheta\mathbf{v}, \Theta\bar t \otimes t)$ is ν-integrable by (W); while by (7.5) with $t' = t$ the left side tends to 0 as $\sigma \to \infty$. So $S^\chi(\mathbf{v},t) = \bar S(\mathbf{v},t)$ for ν-almost every χ, for every cluster whose times are ordered in some (δ,ω). Fix m, $\mathbf{v} \in V^m$ and the open set $\mathcal{O} = \{0 < \tau_1 < \cdots < \tau_m\} \subset D_m$, and a countable set $\mathcal{D} \subset C_c^\infty(\mathcal{O})$ such that every $t \in C_c^\infty(\mathcal{O})$ is a uniform limit of elements of $\mathcal{D}$ supported in a fixed compact subset of $\mathcal{O}$. As V is countable, there is a ν-null set X_1 such that for $\chi \notin X_1$ the phase is a Wightman theory and $S^\chi(\mathbf{v},t) = \bar S(\mathbf{v},t)$ for all m, $\mathbf{v}$ and $t \in \mathcal{D}$. For $\chi, \chi' \notin X_1$ the continuous function $S^\chi_{\mathbf{v}} - S^{\chi'}_{\mathbf{v}}$ integrates to 0 against every $t \in \mathcal{D}$, hence against every $t \in C_c^\infty(\mathcal{O})$, hence vanishes on $\mathcal{O}$; by translation invariance in time and the permutation symmetry of the Schwinger functions it vanishes at every configuration with distinct times, and by continuity on D_m. Fix $\chi_0 \notin X_1$. For $x \in D_m$ the set $\{\chi : S^\chi_{\mathbf{v}}(x) = S^{\chi_0}_{\mathbf{v}}(x)\}$ is closed by (W) and contains $X \setminus X_1$, which is dense because ν has full support and $\nu(X_1) = 0$; so it is X. Thus $S^\chi = S^{\chi_0}$ for every χ, $\bar S = S^{\chi_0}$, and (7.5) holds for the Schwinger functions of the Wightman theory of (W), call it $(\mathcal{H}, \Omega, H)$.

Step 4: the spectrum. By Lemma 7.6(ii), (7.5) with $t' = t$ reads, for the vector $\psi' = \psi(\mathbf{v},t) - \langle \Omega, \psi(\mathbf{v},t)\rangle \Omega \perp \Omega$,

$$\langle \psi', e^{-\sigma H}\psi'\rangle \le e^{-\Delta(\sigma-\sigma_1)} \langle \psi', e^{-\sigma_1 H}\psi'\rangle \qquad (\sigma \ge \sigma_1 \ge 1 \text{ integers}).$$

If κ is the spectral measure of H at ψ' and $0 < \eta < \Delta$, then $e^{-\sigma(\Delta-\eta)}\kappa([0,\Delta-\eta]) \le e^{-\Delta(\sigma-\sigma_1)}\kappa([0,\infty))$, so $\kappa([0,\Delta-\eta]) \le e^{\Delta\sigma_1}\|\psi'\|^2 e^{-\eta\sigma} \to 0$. Hence the spectral projection $E_H([0,\Delta-\eta])$ annihilates ψ', that is $E_H([0,\Delta-\eta])\psi(\mathbf{v},t) = \langle \Omega, \psi(\mathbf{v},t)\rangle\Omega$, for every cluster with ordered times in some interval (δ,ω); every cluster of Lemma 7.6(ii) is of this form. These vectors and Ω, for which the identity is trivial, span a dense subspace by Lemma 7.6(iii), so $E_H([0,\Delta-\eta])$ is the projection on $\mathbb{C}\Omega$ for every $\eta > 0$: the spectrum of H lies in $\{0\} \cup [\Delta,\infty)$ and 0 is simple. □

Remark 7.7 (why the symmetric order needs (U)). On the symmetric torus the thermal two-point function of a slab function is

$$\mu_{L,L}(\vartheta\bar F\, \tau_s F) = \frac{\lambda_0^{-2w}}{\mathrm{Tr}\, \mathbb{R}^L} \sum_{i,j} r_i^{L-s-2w} r_j^s |\langle \phi_j, \hat F \phi_i\rangle|^2,$$

a sum of exponentials $e^{-s(E_j-E_i)}$ in which both signs of $E_j - E_i$ occur. It is convex, symmetric about $s = L/2 - w$ when $\hat F$ is self-adjoint, and it is not the Laplace transform of a positive measure on $[0,\infty)$ unless the terms with $i \ge 1$ vanish. Those terms are bounded, by the log-convexity of $\theta \mapsto \mathrm{Tr}(\mathbb{T}^\theta A \mathbb{T}^{1-\theta} A^*)$, by the two-point functions at adjacent separation, but no bound on their sum that is uniform in N follows from the gap of $\mathbb{T}$ alone: it would require a bound on $\sum_{i\ge1} r_i^L$, the thermal partition function of the torus relative to its vacuum, uniformly in the lattice spacing, which is a statement about the density of states up to the cutoff. Hypothesis (U) replaces that estimate by the statement that the two orders of the volume limits agree, and nothing in Theorem E uses more. The next subsection shows

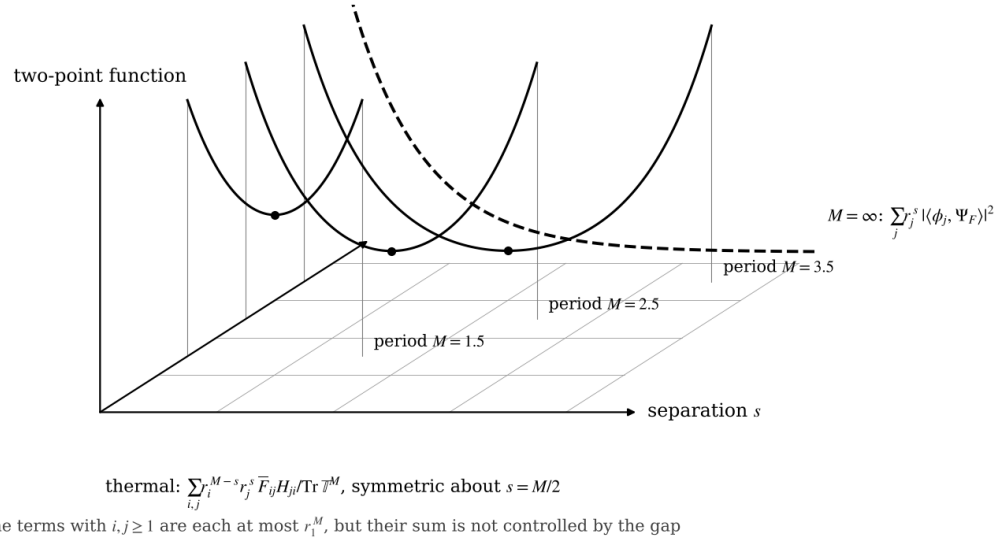


FIGURE 7. Two-point functions of one slab observable on cylinders of physical period M and at zero temperature ($M=\infty$), for one excited level of $\mathbb{R}$: the thermal function contains the terms $r_i^{M-s}r_j^s$ with both signs of E_j-E_i and is symmetric about $s=M/2$ (Remark 7.7), while the zero-temperature function is the Laplace transform of a positive measure, which is what Lemma 7.5 uses. The terms with $i,j\geq 1$ are each at most r_1^M, but their number is not controlled by the gap; Lemma 7.8 bounds their total by $\mathcal{Z}_L(M)-1$.

that the estimate is also sufficient: a bound on $\sum_{i\geq 1} r_i^L$ alone, hypothesis (Z) below, implies the form of (U) that the proof uses.

7.7. **Thermal states against the vacuum.** Fix L and β, let $\mathbb{T}$, λ_0, Ω_0 and $\mathbb{R}=\mathbb{T}/\lambda_0$ be as in Lemma 7.3, and for $M\geq 2$ put

$$\mathcal{Z}_L(M)=\mathrm{Tr}\,\mathbb{R}^M=\frac{Z(\Lambda_{L,M})}{\lambda_0^M}\ \geq 1,$$

the partition function of $\Lambda_{L,M}$ relative to the vacuum of the cylinder. By Lemma 7.3(i),(ii), $\lambda_0=\lim_{M\to\infty}Z(\Lambda_{L,M})^{1/M}$, so $\mathcal{Z}_L(M)$ is determined by the Wilson partition functions alone and involves no insertion.[10]

Lemma 7.8 (thermal discrepancy)**.** *Let G be a bounded measurable complex function of the slab $[0,w]$, $w\geq 0$, and let $M\geq 2w+2$. Then $C_M(G)=\mu_{L,M}(\vartheta\bar{G}\,G)-|\mu_{L,M}(G)|^2\geq 0$, and*

$$\big|\mu_{L,M}(G)-\mu_L^0(G)\big|^2\leq\big(\mathcal{Z}_L(M)-1\big)\,C_M(G),\qquad \mu_L^0(\vartheta\bar{G}\,G)\leq\mathcal{Z}_L(M)\,\mu_{L,M}(\vartheta\bar{G}\,G).$$

Proof. By Lemma 7.3(i),(vii) there is an orthonormal basis $(\phi_i)_{i\geq 0}$ of $\mathcal{K}$ of eigenvectors of $\mathbb{R}$, with $\phi_0=\Omega_0$, $\mathbb{R}\phi_i=r_i\phi_i$, $r_0=1$ and $0\leq r_i<1$ for $i\geq 1$ (the eigenvectors for the nonzero eigenvalues completed by a basis of $\ker\mathbb{T}$). Since $\mathbb{T}$ is Hilbert–Schmidt, $\mathbb{R}^k$ is trace class for $k\geq 2$, and $\mathcal{Z}_L(M)=\sum_i r_i^M=1+\varepsilon$ with $\varepsilon=\sum_{i\geq 1}r_i^M\geq 0$.

Put $X=\lambda_0^{-w}\hat{G}$. For $w=0$ this is multiplication by G; for $w\geq 1$ the weights in (7.3) are positive and with $G=1$ they give the kernel of $\mathbb{T}^w$, so $|\hat{G}(Q,Q')|\leq\|G\|_\infty\mathbb{T}^w(Q,Q')$ and

[10] In the language of statistical mechanics, $-\log\lambda_0$ is the ground-state energy of the cylinder Hamiltonian $H_L=-\log\mathbb{T}$ restricted to the range of $\mathbb{T}$, $\log Z(\Lambda_{L,M})=\log\mathrm{Tr}\,e^{-MH_L}$ is minus M times the free energy at temperature $1/M$, and $\log\mathcal{Z}_L(M)=\log\mathrm{Tr}\,e^{-M(H_L-E_0)}$ is the thermal correction to the free energy, which vanishes at zero temperature. Hypothesis (Z) is the statement that this correction tends to zero when the temperature $1/L_p$ does, uniformly in the mesh; for a theory with a gap Δ and finitely many states per unit of energy and volume one expects $\mathcal{Z}-1$ of order $L_p^4e^{-\Delta L_p}$.

$\hat G$ is Hilbert–Schmidt. In both cases X is bounded, and $X\mathbb{R}^{M-w}$, $X^*X\mathbb{R}^{M-2w}$ are trace class because $M-2w\ge 2$. Lemma 7.3(iii) with $H=1$, $w'=0$, $s=w$, Lemma 7.3(iv) with $F=H=G$, $s=0$, and Lemma 7.3(v) give, with $X_{ii}=\langle\phi_i, X\phi_i\rangle$,

$$\begin{aligned}\mathcal{Z}_L(M)\,\mu_{L,M}(G) &= \sum_i r_i^{M-w} X_{ii}, \qquad \mu_L^0(G) = X_{00},\\ \mathcal{Z}_L(M)\,\mu_{L,M}(\vartheta\bar G\, G) &= \sum_i r_i^{M-2w}\|X\phi_i\|^2.\end{aligned} \tag{7.6}$$

The term $i=0$ of the second identity is $\|X\Omega_0\|^2=\mu_L^0(\vartheta\bar G\,G)$ by Lemma 7.3(v), which gives the second inequality of the lemma.

Let $c=X_{00}$ and $G'=G-c$. The constant c on the slab $[0,w]$ has operator $c\,\mathbb{T}^w$, so $X'=\lambda_0^{-w}\hat G'=X-c\,\mathbb{R}^w$ and $X'_{00}=0$, while $D=\mu_{L,M}(G)-\mu_L^0(G)=\mu_{L,M}(G')-\mu_L^0(G')=\mu_{L,M}(G')$. By (7.6) for G', the Cauchy–Schwarz inequality over $i\ge 1$ with the weights $r_i^{M-w}=r_i^{M/2}r_i^{M/2-w}$, and $|X'_{ii}|\le\|X'\phi_i\|$,

$$\begin{aligned}\mathcal{Z}_L(M)\,|D| = \Big|\sum_{i\ge1} r_i^{M/2}\,r_i^{M/2-w}\,X'_{ii}\Big| &\le \varepsilon^{1/2}\Big(\sum_{i\ge1} r_i^{M-2w}\|X'\phi_i\|^2\Big)^{1/2}\\ &\le \varepsilon^{1/2}\big(\mathcal{Z}_L(M)\,\mu_{L,M}(\vartheta\bar G'G')\big)^{1/2}.\end{aligned}$$

The torus measure is invariant under ϑ, so $\mu_{L,M}(\vartheta\bar G)=\overline{\mu_{L,M}(G)}$, and expanding the product,

$$\begin{aligned}\mu_{L,M}(\vartheta\bar G'G') &= \mu_{L,M}(\vartheta\bar G\,G) - 2\Re\big(\bar c\,\mu_{L,M}(G)\big) + |c|^2\\ &= C_M(G) + |\mu_{L,M}(G)-c|^2 = C_M(G)+|D|^2.\end{aligned}$$

Hence $|D|^2\le\frac{\varepsilon}{1+\varepsilon}(C_M(G)+|D|^2)$, that is $|D|^2\le\varepsilon\,C_M(G)$. Finally $C_M(G)=\mu_{L,M}\big(\vartheta\overline{G''}\,G''\big)$ with $G''=G-\mu_{L,M}(G)$, by the same expansion, and the second identity of (7.6) applied to G'' shows that this is nonnegative. □

The inequality is sharp: for $\mathcal{K}=\mathbb{C}^2$, $\mathbb{R}=\mathrm{diag}(1,r)$, $w=0$ and $G=\mathrm{diag}(x_0,x_1)$ one has $D=r^M(x_1-x_0)/(1+r^M)$, $\mathcal{Z}-1=r^M$ and $C_M=r^M(x_1-x_0)^2/(1+r^M)^2$, so $|D|^2=(\mathcal{Z}-1)C_M$. The lemma uses no gap, and no bound on G beyond boundedness at fixed N: the renormalisation constants of the insertions cancel between the two sides. Conversely, no bound on the thermal correction follows from the gap and operator norms alone: for $m>0$, $M\ge1$ and $c\in(0,1)$, let $d=\lceil ce^{mM}/(1-c)\rceil$, $r=(c/((1-c)d))^{1/M}\le e^{-m}$, $\mathbb{R}=\mathrm{diag}(1,r,\dots,r)$ on $\mathbb{C}^{1+d}$ and $X=I-P_{\Omega_0}$; then $\mathrm{Tr}(X\mathbb{R}^M)/\,\mathrm{Tr}\,\mathbb{R}^M=dr^M/(1+dr^M)=c$ while $\langle\Omega_0,X\Omega_0\rangle=0$. The quantity $\mathcal{Z}_L(M)-1$ is therefore exactly the missing information of Remark 7.7.

Definition 7.9. For a continuum family put $\mathcal{Z}_{N,p}=\mathcal{Z}_{NL_p}(NL_p)$ at coupling $\beta_{N,p}$, the partition function of the symmetric torus Λ_{NL_p} relative to the vacuum of the cylinder of the same spatial side.

(Z) *Vacuum dominance on the symmetric torus.* $\lim_{p\to\infty}\limsup_{N\to\infty}(\mathcal{Z}_{N,p}-1)=0$, the limits along the progression and the sequence of (W).

(U_w) *Weak order of the volume limits.* For every cluster,

$$\lim_{p\to\infty}\limsup_{N\to\infty}\Big|S^0_{N,p}(\mathbf{v},t)-\int_X S^\chi(\mathbf{v},t)\,d\nu(\chi)\Big|=0.$$

Hypothesis (U) implies (U_w); (U_w) does not require the inner limit of $S^0_{N,p}$ to exist. Hypothesis (Z) is one number per (N,p), with no insertion and no renormalisation: it says that the free energy of the symmetric torus is its time extent times the free energy per unit time of the cylinder, up to $o(1)$, uniformly in the mesh. For a gapped theory with finite thermodynamics one expects $\mathcal{Z}_{N,p}-1$ of order $L_p^4e^{-\Delta L_p}$; (Z) is not proved here for any four-dimensional family.

Proposition 7.10 (vacuum dominance gives the order of the limits)**.** *Let a continuum family satisfy* (W)*. Let* $(\mathbf{v},t)$ *be a cluster with* t *supported in configurations whose times lie in* $(-T_0,T_0)$*, put* $\sigma_0=\lceil T_0\rceil+1\in\mathbb{N}$*,* $t_0=\tau_{\sigma_0}t$*, and let* $c^\vartheta=(\mathbf{v}^\vartheta\mathbf{v},\Theta\bar{t}_0\otimes t_0)$*. Then for all* p *with* $L_p>4T_0+6$ *and all large* N*,*

$$\left|S_{N,p,NL_p}(\mathbf{v},t)-S^0_{N,p}(\mathbf{v},t)\right|^2\le(\mathcal{Z}_{N,p}-1)\Big(S_{N,p,NL_p}(c^\vartheta)-\left|S_{N,p,NL_p}(\mathbf{v},t)\right|^2\Big).\tag{7.7}$$

Consequently (W) *and* (Z) *imply* (U_w)*.*

Proof. Translation. Since $\sigma_0N\in\mathbb{Z}$, the translation by σ_0N time steps is a bijection of the sites of every $\Lambda_{NL_p,M}$. By (F2) and the invariance of $\mu_{NL_p,M}$ and of $\mu^0_{NL_p}$ under lattice translations, $S_{N,p,M}(\mathbf{v},t_0)=S_{N,p,M}(\mathbf{v},t)$ and $S^0_{N,p}(\mathbf{v},t_0)=S^0_{N,p}(\mathbf{v},t)$, the box reading being consistent because t, t_0 and $\Theta\bar{t}_0$ are supported in the box when $L_p>4T_0+6$. The times of t_0 lie in $(1,\omega)$ with $\omega=2T_0+2$.

Slab function. Let ρ bound the time extent of the stencils R_v, and let $N>\rho$ be so large that $NL_p\ge2(\lceil\omega N\rceil+\rho)+2$, which is possible since $L_p>2\omega$. The lattice function $G=a^{4m}\sum_x t_0(ax)\prod_i\Phi^{N,p}_{v_i}(x_i,\cdot)$ is bounded, by (F1) and compactness, and depends on the slab $[0,w]$ with $w=\lceil\omega N\rceil+\rho$; moreover $S_{N,p,NL_p}(\mathbf{v},t)=\mu_{NL_p,NL_p}(G)$, $S^0_{N,p}(\mathbf{v},t)=\mu^0_{NL_p}(G)$, and by (F3), as in Step 1 of Section 7.6, $\mu_{NL_p,NL_p}(\vartheta\bar{G}\,G)=S_{N,p,NL_p}(c^\vartheta)$; the cluster c^ϑ is supported in D_{2m} since reflected and original times are separated by at least 2, and in the box. Lemma 7.8 with $L=M=NL_p$ gives (7.7).

Limit. By (W) applied to c^ϑ and to $(\mathbf{v},t)$, the second factor on the right of (7.7) converges as $N\to\infty$, for every large p, to a number k_p, and $k_p\to\int_X S^\chi(c^\vartheta)\,d\nu-|\int_X S^\chi(\mathbf{v},t)\,d\nu|^2<\infty$. Hence $\limsup_N|S^0_{N,p}-S_{N,p,NL_p}|\le(\limsup_N(\mathcal{Z}_{N,p}-1))^{1/2}k_p^{1/2}$, which tends to 0 as $p\to\infty$ by (Z); since $\lim_p\lim_N S_{N,p,NL_p}(\mathbf{v},t)=\int_X S^\chi(\mathbf{v},t)\,d\nu$ by (W), (U_w) follows. □

Corollary 7.11. *The proof of Theorem E remains valid with* (U) *replaced by* (U_w)*. Hence a continuum family satisfying* (W)*,* (Z) *and* (D_ℓ) *has the conclusion of Theorem E:* S^χ *does not depend on* χ*, and* $\sigma(H)\subset\{0\}\cup[\Delta,\infty)$ *with* $\Delta=16\varpi/(625e^2\ell)$ *and* 0 *simple.*

Proof. Step 1 is a statement at fixed N,p and uses only (D_ℓ). In Step 2 write $x_{N,p}(c)=S^0_{N,p}(c)$ and $x(c)=\int_X S^\chi(c)\,d\nu$ for a cluster c; by (U_w), $\eta_p(c)=\limsup_N|x_{N,p}(c)-x(c)|\to0$. The quantities in (7.4) are $A_{N,p}=x_{N,p}(c_1)-\overline{x_{N,p}(c_2)}x_{N,p}(c_3)$, $B_{N,p}=x_{N,p}(c_4)-|x_{N,p}(c_2)|^2\ge0$ and the analogous $C_{N,p}\ge0$ for t', with $c_1,\dots,c_4$ the clusters of Step 1; let A, B, C be the same expressions in x. From $|yy'-zz'|\le|y-z||y'|+|z||y'-z'|$ and the boundedness of the $x_{N,p}$ along the progression (they converge up to η_p), $\limsup_N|A_{N,p}-A|\le\eta_p$, and similarly for B and C, with $\eta_p\to0$. Hence $B,C\ge-\eta_p$ for every large p, so $B,C\ge0$, and

$$|A|\le\limsup_N|A-A_{N,p}|+e^{-\Delta(\sigma-\sigma_1)}\limsup_N\sqrt{B_{N,p}C_{N,p}}\le\eta_p+e^{-\Delta(\sigma-\sigma_1)}\sqrt{(B+\eta_p)(C+\eta_p)},$$

using $e^{-m_nN(\sigma-\sigma_1)}\le e^{-\Delta(\sigma-\sigma_1)}$. Letting $p\to\infty$ gives (7.5); Steps 3 and 4 use only (7.5) and (W). The second statement follows from Proposition 7.10. □

Lemma 7.12. $M\mapsto\mathcal{Z}_L(M)-1=\sum_{i\ge1}r_i^M$ *is nonincreasing and log-convex on* $\{M\ge2\}$*. If every eigenvalue* $\lambda\ne\lambda_0$ *of* $\mathbb{T}$ *satisfies* $|\lambda|\le e^{-m}\lambda_0$*, then* $\mathcal{Z}_L(M)-1\le e^{-m(M-M')}(\mathcal{Z}_L(M')-1)$ *for* $M\ge M'\ge2$*. In particular, under* (D_ℓ)*, for integers* $2\le b\le L_p$ *and large* N*,*

$$\mathcal{Z}_{N,p}-1\le e^{-\Delta(L_p-b)}(\mathcal{Z}_{NL_p}(bN)-1).$$

Proof. Each $r_i\in[0,1)$ for $i\ge1$, and a sum of exponentials $\sum_i e^{M\log r_i}$ is nonincreasing and log-convex in M. Under the gap, $r_i^M\le e^{-m(M-M')}r_i^{M'}$. By Proposition 7.4 and (D_ℓ), $m=m_{n_N}$ with $m_{n_N}N\ge\Delta$, as in Step 1 of Section 7.6. □

Thus (Z) follows from a bound on the relative partition function at the higher temperature $1/b$ that grows slower than $e^{\Delta(L_p-b)}$. Hypothesis (D_ℓ) can likewise be replaced by the slab decorrelation of Theorem D, through its form on cylinders.

Proposition 7.13 (the bridge from the slab decorrelation)**.** *Let a continuum family satisfy* (W) *and* (U)*, or* (W) *and* (Z)*, with the inner limits along the progression* $\{N_0 + kP\}$*. Let* K_0 *be a prime not dividing* P*, let* $\mathcal{N} = \{N_0 + kP\} \cap K_0\mathbb{Z}$*, and for* $N \in \mathcal{N}$ *put* $d_N = N/K_0$ *and* $n_N = 2d_N - 1$*, so that* $n_N a_N < \ell = 2/K_0$*. Suppose there are* $0 \le c < \frac{1}{2}$ *and* $m \ge 4$ *with* $m - 1 \ge 148/(1-2c)^4$ *such that* $(\mathrm{M}_{d_N,m}(c))$ *holds for the Wilson weight at* $\beta_{N,p}$ *for all large* $N \in \mathcal{N}$ *and all large* p*. Then the conclusion of Theorem E holds with*

$$\Delta = \frac{(1-2c)^4 K_0}{2500\, e^2}.$$

Proof. Since $K_0 \nmid P$, the congruence $N_0 + kP \equiv 0 \pmod{K_0}$ has a solution, and $\mathcal{N}$ is an arithmetic progression of difference K_0P, a subsequence of the original one; so (W) and (U), or (Z), hold along it with the same limits. For $N \ge 2K_0$ one has $d_N \ge 2$ and $n_N a_N = (2N/K_0 - 1)/N < \ell$; also $d_N \mid NL_p$, and $NL_p \ge 2md_N$ once $K_0 L_p \ge 2m$. By Corollary 11.9, $\mathrm{BP}(n_N, \varpi)$ with $\varpi = (1-2c)^4/32$ holds for $\mu_{NL_p,M}$ for every $M \in d_N\mathbb{Z}$ with $M \ge 2md_N$, and each such M is at least $2n_N + 4$. In the proof of Theorem E, (D_ℓ) enters only in Step 1, through Proposition 7.4, whose proof uses $\mathrm{BP}(n, \varpi)$ on $\Lambda_{L,M}$ for $M \ge 2s$ and then lets $M \to \infty$, a limit that exists along every sequence by Lemma 7.3(v); so the cylinders with $M \in d_N\mathbb{Z}$ suffice, and $\mathbb{T}$ has gap $m_n = \varpi/(e^2 q_n n)$. Steps 2 to 4 are unchanged, or replaced by Corollary 7.11 under (Z). The gap is $16\varpi/(625e^2\ell) = (1-2c)^4 K_0/(2500e^2)$. □

The proposition replaces the cube condition on cylinders by the slab decorrelation at the physical separation $(d_N - 1)a_N \approx 1/(2K_0)$, uniform in the boundary condition, which is the form in which Section 12 discusses the remaining hypothesis; none of (W), (U), (Z) or $(\mathrm{M}_{d_N,m}(c))$ along a tuning is proved here. What the cube condition does not give is a bound of this kind: (Z) is a statement per unit of physical volume, while Theorem 6.3 controls each local term of $\log \mathcal{Z}$ by a constant per lattice link or block, with sup-norm factors that grow with N. Whether (U) implies (Z) is not known to us; whether (Z) holds for Yang–Mills is open, and nothing in this subsection bears on (W) or on (D_ℓ). Section 11.6 shows what $(\mathrm{M}_{d_N,m}(c))$ asks beyond the gap it produces: computed in the vacuum of the cylinder instead of under an arbitrary boundary condition, the same maximal correlation equals $e^{-m_{\mathrm{lat}} s}$ exactly (Proposition 11.13).

Two facts delimit what a proof of (Z) can use. The first is that (Z) is not determined by the data that (W) and the gap provide: the correlation functions of the observables at every period and at zero temperature, the gap, the positivity of the transfer operator and the uniqueness of its leading eigenvector. The second is that the agreement of the two orders of limits for all local correlations, hypothesis (U), does not imply (Z) in the class of reflection-positive lattice models with a gapped transfer matrix.

Proposition 7.14 (decoupled sectors)**.** *Let* $\mathbb{R} \ge 0$ *be a self-adjoint operator on a separable Hilbert space* $\mathcal{K}$ *with* $\mathbb{R}^2$ *of trace class, such that* 1 *is a simple eigenvalue of* $\mathbb{R}$ *with unit eigenvector* Ω *and* $\|\mathbb{R}|_{\Omega^\perp}\| \le e^{-m}$*. Let* $d \ge 1$*,* $0 < \rho \le e^{-m}$*,* $\tilde{\mathcal{K}} = \mathcal{K} \otimes \mathbb{C}^{d+1}$*,* $\tilde{\mathbb{R}} = \mathbb{R} \otimes \mathrm{diag}(1, \rho, \dots, \rho)$*,* $\tilde{\Omega} = \Omega \otimes e_0$*, and* $\tilde{A} = A \otimes I$ *for a bounded operator* A *on* $\mathcal{K}$*. Then* $\tilde{\mathbb{R}}$ *has the same properties, with the same* m*, and for all* $M \ge 2$*, all bounded* $A_1, \dots, A_k$ *and all integers* $s_i \ge 0$ *with* $\sum_i s_i = M$*,*

$$\frac{\mathrm{Tr}(\tilde{A}_1 \tilde{\mathbb{R}}^{s_1} \cdots \tilde{A}_k \tilde{\mathbb{R}}^{s_k})}{\mathrm{Tr}\, \tilde{\mathbb{R}}^M} = \frac{\mathrm{Tr}(A_1 \mathbb{R}^{s_1} \cdots A_k \mathbb{R}^{s_k})}{\mathrm{Tr}\, \mathbb{R}^M},$$
$$\langle \tilde{\Omega}, \tilde{A}_1 \tilde{\mathbb{R}}^{s_1} \cdots \tilde{A}_k \tilde{\Omega} \rangle = \langle \Omega, A_1 \mathbb{R}^{s_1} \cdots A_k \Omega \rangle,$$

while $\mathrm{Tr}\, \tilde{\mathbb{R}}^M = (1 + d\rho^M)\, \mathrm{Tr}\, \mathbb{R}^M$*.*

Proof. Write $D = \mathrm{diag}(1, \rho, \dots, \rho)$ and $B = A_1 \mathbb{R}^{s_1} \cdots A_k \mathbb{R}^{s_k}$. The operator $\tilde{A}_1 \tilde{\mathbb{R}}^{s_1} \cdots \tilde{A}_k \tilde{\mathbb{R}}^{s_k}$ equals $B \otimes D^M$, and $\mathrm{Tr}(B \otimes D^M) = \mathrm{Tr}\, B\ \mathrm{Tr}\, D^M$ with $\mathrm{Tr}\, D^M = 1 + d\rho^M$; the case $k = 1$, $A_1 = I$ gives the last identity, and the quotient of the two cancels the factor $1 + d\rho^M$. Since $De_0 = e_0$, the vector $\tilde{\Omega}$ satisfies $\tilde{A}_1 \tilde{\mathbb{R}}^{s_1} \cdots \tilde{A}_k \tilde{\Omega} = (A_1 \mathbb{R}^{s_1} \cdots A_k \Omega) \otimes e_0$, which gives the

second identity. The spectrum of $\tilde{\mathbb{R}}$ is the set of products of eigenvalues of $\mathbb{R}$ and of D; as $\rho < 1$ the eigenvalue 1 arises only as $1 \cdot 1$, so it is simple with eigenvector $\tilde{\Omega}$, and every other eigenvalue is at most $\max(e^{-m}, \rho) = e^{-m}$. Positivity and the trace class property of $\tilde{\mathbb{R}}^2$ are inherited. □

Every quantity that enters (W), (U), the decay statements of Section 7.3, the reflection positivity and the uniqueness of the vacuum is of one of the two forms in the proposition, and is the same for $\mathbb{R}$ and for $\tilde{\mathbb{R}}$, whereas $\operatorname{Tr}\tilde{\mathbb{R}}^M - 1 = (1 + d\rho^M)\operatorname{Tr}\mathbb{R}^M - 1$ can be made as large as one wishes by the choice of d. For the Wilson transfer matrix the Hilbert space is fixed, $\mathcal{K} = L^2(G^{E_0})$, so the proposition does not refute the implication from (W) and (D_ℓ) to (Z); it says that a proof of (Z) must use something that sees all of $\mathcal{K}$, such as the completeness of the gauge-invariant functions of a slice, and not only the correlations of the tokens, the gap and the positivity.

The second fact is an example. Let $s \in \{\pm 1\}$ be a spin on each site of $\Lambda_{L,M}$ with weight $\exp(J\sum_{\vec{x}}\sum_t s_{\vec{x},t}s_{\vec{x},t+1})$, $J > 0$: L^3 independent Ising rings in the time direction, a translation-invariant and reflection-positive lattice model. Its transfer matrix is $\mathbb{T}_1^{\otimes L^3}$ with $\mathbb{T}_1 = \left(\begin{smallmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{smallmatrix}\right)$, of eigenvalues $2\cosh J$ and $2\sinh J$; so $\lambda_0 = (2\cosh J)^{L^3}$, the gap of $\mathbb{R}$ is $m = -\log\tanh J$, and

$$\mathcal{Z}_L(M) = \left(1 + \tanh^M J\right)^{L^3} = \left(1 + e^{-mM}\right)^{L^3}.$$

Take $L = M = NL_p$ and tune J so that $m = \Delta/N$, a physical gap Δ. For two spins on one ring at time distance t, $\mu_{L,M}(s_0 s_t) = (\tanh^t J + \tanh^{M-t} J)/(1 + \tanh^M J)$ and $\mu_L^0(s_0 s_t) = \tanh^t J$, so the two differ by at most $2e^{-m(M-t)} = 2e^{-\Delta(L_p - t/N)}$, and the correlations of local observables factor over the rings: the two orders of the limits agree for every local observable, uniformly in N, with an error $O(e^{-\Delta L_p})$ at fixed physical times. Yet $\mathcal{Z}_{N,p} - 1 \geq L^3 \tanh^M J = N^3 L_p^3 e^{-\Delta L_p} \to \infty$ as $N \to \infty$ for every p. So a gap together with the agreement of the two orders of limits for all local correlations does not give (Z); (Z) is a bound on the number of states below the cutoff per unit of physical volume, and in this example that number is $N^3 L_p^3$. The example says nothing about Yang–Mills; it shows what a proof of (Z) must contain. It also satisfies the slab decorrelation hypothesis of Section 11, in the exact form of Definition 11.3 below, uniformly in the mesh, so that hypothesis is blind to (Z) as well. Two lemmas on maximal correlations give this.[11]

Lemma 7.15 (maximal correlation of independent pairs)**.** *Let (X_k, Y_k), $1 \leq k \leq K$, be pairs of random variables with finitely many values, the pairs independent of one another, and $X = (X_1, \dots, X_K)$, $Y = (Y_1, \dots, Y_K)$. Then $\rho(X;Y) = \max_k \rho(X_k;Y_k)$.*

Proof. With $T_k : f \mapsto E[f \mid Y_k]$ on $L^2(\sigma(X_k))$ and $T : f \mapsto E[f \mid Y]$ on $L^2(\sigma(X)) = \bigotimes_k L^2(\sigma(X_k))$, the independence of the pairs gives $E[\prod_k f_k(X_k) \mid Y] = \prod_k E[f_k(X_k) \mid Y_k]$, and the product functions span, so $T = \bigotimes_k T_k$. A tensor product of singular value decompositions is a singular value decomposition, so the singular values of T with multiplicity are the products $\prod_k \sigma_{j_k}(T_k)$ of singular values of the factors, each $\sigma_1(T_k) = 1$ with the constants as singular vectors. The product with every $j_k = 1$ is 1; every other product has some $j_k \geq 2$ and is at most $\sigma_{j_k}(T_k) \leq \max_k \sigma_2(T_k)$, and this value is attained. So $\sigma_2(T) = \max_k \sigma_2(T_k)$, which is the claim by the footnote. □

Lemma 7.16 (two blocks of a chain)**.** *Let $J > 0$, $\kappa = \tanh J$, $n \geq 2$, $a, b \in \{\pm 1\}$, and let $(s_0, \dots, s_{n-1}) \in \{\pm 1\}^n$ have the law with weight $\exp\big(J(as_0 + \sum_{j<n-1} s_j s_{j+1} + s_{n-1}b)\big)$, a*

[11] The maximal correlation of two random variables X, Y with finitely many values is $\rho(X;Y) = \sup E[fg]$ over f a function of X and g a function of Y with $Ef = Eg = 0$ and $Ef^2 = Eg^2 = 1$, the supremum of the empty set being 0; this is Definition 11.2 below with the sites in place of the links and no gauge group. If $T : L^2(\sigma(X)) \to L^2(\sigma(Y))$ is the conditional expectation $f \mapsto E[f \mid Y]$, then $T\mathbf{1} = \mathbf{1}$, $T^* g = E[g \mid X]$, and T maps the orthogonal complement of the constants into itself, so $\rho(X;Y)$ is the norm of T on that complement, which is the second largest singular value of T when T has rank at least two and 0 otherwise.

chain whose two boundary spins are fixed. Split $\{0,\dots,n-1\}$ *into consecutive blocks of lengths* $n_- \ge 1$, $w \ge 0$, $n_+ \ge 1$, *let* A *be the spins of the first block and* B *those of the last, and put* $p = \kappa^{n_-}$, $y = \kappa^{w+1}$, $q = \kappa^{n_+}$. *Then, whatever* a *and* b,

$$\rho(A;B) = y\sqrt{\frac{(1-p^2)(1-q^2)}{(1-y^2q^2)(1-p^2y^2)}} \;\le\; y = \kappa^{w+1}.$$

For $n_- = n_+ = d$ *and* $w = d-1$ *the value is* $x/(1+x^2)$ *with* $x = \kappa^d$.

Proof. Let $u = n_- - 1$ and $v = n_- + w$, the last spin of A and the first of B. The weight is a product of nearest-neighbour factors, so the law is Markov: for a function f of A, $E[f \mid s_u,\dots,s_{n-1}] = E[f \mid s_u] =: \tilde f$, and for a function g of B, $E[g \mid s_0,\dots,s_v] = E[g \mid s_v] =: \tilde g$. For f, g centred of unit norm, $E[fg] = E[\tilde f g] = E[\tilde f \tilde g]$, and $\tilde f = \alpha(s_u - Es_u)$, $\tilde g = \beta(s_v - Es_v)$ with $\alpha^2 \operatorname{Var} s_u \le 1$ and $\beta^2 \operatorname{Var} s_v \le 1$, so $|E[fg]| \le |\operatorname{Cov}(s_u,s_v)|/\sqrt{\operatorname{Var} s_u \operatorname{Var} s_v}$, with equality for the normalised spins themselves. Hence $\rho(A;B)$ is the correlation coefficient of s_u and s_v. Write $T_1 = \left(\begin{smallmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{smallmatrix}\right) = \lambda_+ P_+ + \lambda_- P_-$ with $\lambda_\pm = 2\cosh J,\ 2\sinh J$, $P_\pm$ the projections onto $(1,\pm1)/\sqrt{2}$, and $S = \operatorname{diag}(1,-1)$, which exchanges the two eigenvectors. The vector of the spin value a is $(|+\rangle + a|-\rangle)/\sqrt{2}$. The partition function is $\langle a|T_1^{n+1}|b\rangle = \frac12(\lambda_+^{n+1} + ab\lambda_-^{n+1})$, and inserting S at the positions u, v, at distances n_- from the left end, $w+1$ from each other and n_+ from the right end, gives, with $D = 1 + ab\,pyq$,

$$Es_u = \frac{ap + byq}{D}, \qquad Es_v = \frac{apy + bq}{D}, \qquad E[s_u s_v] = \frac{y + ab\,pq}{D}.$$

Then $D^2 \operatorname{Cov}(s_u,s_v) = (y+abpq)(1+abpyq) - (ap+byq)(apy+bq) = y(1-p^2)(1-q^2)$, $D^2 \operatorname{Var} s_u = (1-p^2)(1-y^2q^2)$ and $D^2 \operatorname{Var} s_v = (1-q^2)(1-p^2y^2)$, both positive, which gives the formula; the inequality holds because $1-p^2 \le 1-p^2y^2$ and $1-q^2 \le 1-y^2q^2$. For $p = q = y = x$ the root is $(1-x^2)/(1-x^4) = 1/(1+x^2)$. □

Proposition 7.17 (the rings satisfy the slab hypothesis)**.** *For the Ising rings with* $\kappa = \tanh J$ *and* $m = -\log\kappa$, *for every box* Q, *every exterior configuration* ω *and every splitting* $Q = Q_- \cup Q_0 \cup Q_+$ *in a coordinate* i *with widths* n_-, w, n_+, *the maximal correlation of Definition* 11.2, *read with sites in place of links and no gauge group, is* $\rho^\omega(Q_-,Q_+) = 0$ *when* i *is a spatial coordinate, and when* i *is the time coordinate it is the value of Lemma* 7.16, *at most* $e^{-m(w+1)}$, *for every* ω. *In particular* $(\mathrm{M}_{d,m}(c))$ *of Definition* 11.3 *holds for all* $d \ge 2$ *and* $m \ge 4$ *with* $c = x/(1+x^2) < \frac12$, $x = e^{-md}$, *and* $(\mathrm{M}^*_w(e^{-m(w+1)}))$ *holds for every* $w \ge 1$. *Along the tuning* $m = \Delta/N$ *with* $d = d_N$ *as in Proposition* 7.13, $x = e^{-\Delta d_N/N}$ *and* c *are independent of* N.

Proof. Under the law of the spins of Q given ω the rings are independent, and the segment of a ring inside Q has the law of Lemma 7.16 with the two boundary spins read from ω, since the conditional law of a segment of a nearest-neighbour chain given the other spins depends only on the two adjacent ones. If i is spatial, Q_- and Q_+ meet disjoint sets of rings, so a function of one is independent of a function of the other and $\rho^\omega = 0$. If i is the time coordinate, Q_- and Q_+ consist, ring by ring, of the first and last blocks of the segment with w spins between them; Lemma 7.15 and Lemma 7.16 give the value, the same for every ring. For $(\mathrm{M}_{d,m}(c))$ take $n_- = n_+ = d$, $w = d-1$. □

The rings also satisfy the cube condition of Definition 6.1, read with sites in place of links, $\mathcal{P}_x$ being the conditional expectation given the spins off $x + [0,n)^4$. The proof runs a chain of windows around each ring, and the chain lemma of Section 11 is needed in a cyclic form.

Lemma 7.18 (cyclic chain)**.** *Let* $K \ge 3$ *and let* Q_u, $u \in \mathbb{Z}/K\mathbb{Z}$, *be orthogonal projections on a real Hilbert space with* $Q_uQ_{u'} = Q_{u'}Q_u$ *whenever* $u' \notin \{u-1,u,u+1\}$ *and* $Q_uQ_{u+1} + Q_{u+1}Q_u \ge -c_u(Q_u + Q_{u+1})$, *where* $c_u \ge 0$ *and* $c_{u-1} + c_u \le 2c < 1$ *for every* u. *Then* $H = \sum_u Q_u$ *satisfies* $H^2 \ge (1-2c)H$, *and* $\langle g, Hg\rangle \ge (1-2c)\|g\|^2$ *for every* $g \perp \ker H$.

Proof. $H^2 = \sum_u Q_u + \sum_{u' \neq u} Q_u Q_{u'}$, the second sum over ordered pairs. As $K \geq 3$, the $2K$ ordered pairs $(u, u \pm 1)$ are distinct and contribute $\sum_u (Q_u Q_{u+1} + Q_{u+1} Q_u) \geq -\sum_u (c_{u-1} + c_u) Q_u \geq -2cH$; every other ordered pair is a product of commuting projections, hence nonnegative. So $H^2 \geq (1 - 2c)H$, and the spectrum of $H \geq 0$ lies in $\{0\} \cup [1 - 2c, \infty)$. □

Proposition 7.19 (the rings satisfy the cube condition)**.** *Let $d \geq 2$, $n = 2d-1$, $\kappa = \tanh J$, and for $1 \leq s \leq 2d-1$ put*

$$\rho_s = \kappa^{2d-s} \frac{1 - \kappa^{2s}}{1 - \kappa^{4d}},$$

the value of Lemma 7.16 for two blocks of length s separated by $2d-1-s$ spins; ρ_s increases with s, and $\rho_d = x/(1+x^2) = c$ with $x = \kappa^d$. For $M \geq 4d$ let $K = \lfloor M/d \rfloor$, $r_M = \lceil (M - Kd)/K \rceil$ and $c_M = \rho_{d+r_M}$; thus $c_M = c$ when $d \mid M$, $r_M \leq 1$ when $M \geq d(d-1)$, and always $c_M \leq \kappa^{(3d-3)/4}$. If $L \geq 2d-1$, $M \geq 4d$ and $c_M < \frac{1}{2}$, then for every real $f \in L^2(\mu_{L,M})$

$$\frac{d\,(1 - 2c_M)}{2d-1} \operatorname{Var}(f) \leq \frac{1}{n^4} \sum_{x \in \Lambda_{L,M}} \mu_{L,M}((f - \mathcal{P}_x f)^2),$$

that is, $\mathrm{BP}(2d-1, \varpi_M)$ *holds with $\varpi_M = d(1-2c_M)/(2d-1) \geq \frac{1}{2}(1-2c_M)$. In particular* $\mathrm{BP}(2d-1, (1-2c)d/(2d-1))$ *holds whenever $d \mid M$, for every $J > 0$; and if $2\kappa^{(3d-3)/4} < 1$, then* $\mathrm{BP}(2d-1, \frac{1}{2}(1 - 2\kappa^{(3d-3)/4}))$ *holds on every cylinder with $M \geq 4d$.*

Proof. One ring. Fix a ring, a periodic chain of length M, and for $w \in \mathbb{Z}/M\mathbb{Z}$ let P_w be the conditional expectation given the spins off $[w, w+n)$. Choose $s_0, \dots, s_{K-1} \in \{d + \lfloor j/K \rfloor, d + \lceil j/K \rceil\}$, $j = M - Kd$, with $\sum_u s_u = M$, so that $d \leq s_u \leq d + r_M \leq 2d-1$, as $r_M \leq \lceil (d-1)/4 \rceil \leq d-1$; put $w_0 = 0$, $w_{u+1} = w_u + s_u$, $R_u = [w_u, w_u + n)$ and $Q_u = I - P_{w_u}$, $u \in \mathbb{Z}/K\mathbb{Z}$, $K \geq 4$. Two windows $R_u, R_{u'}$ with $u' \notin \{u, u \pm 1\}$ are separated by at least $2d - n = 1$ site on each side, so no bond joins them; given the other spins they are conditionally independent, and Q_u, $Q_{u'}$ commute, as in Lemma G.1(i). For an adjacent pair let $Q = R_u \cup R_{u+1}$, of length $s_u + n \leq 4d - 2 \leq M - 2$, and let $\mathcal{N}$ be the functions of the spins η off Q. For $x \in \operatorname{ran} P_{w_u} \ominus \mathcal{N}$ and $y \in \operatorname{ran} P_{w_{u+1}} \ominus \mathcal{N}$ one has $E[x \mid \eta] = E[y \mid \eta] = 0$, and given η the spins of Q have the law of Lemma 7.16 with the two neighbours of Q as boundary spins, x being a function of the last block $R_{u+1} \setminus R_u$ and y of the first block $R_u \setminus R_{u+1}$, both of length s_u, with the overlap of length $n - s_u$ between them. Hence $|E[xy \mid \eta]| \leq \rho_{s_u} E[x^2 \mid \eta]^{1/2} E[y^2 \mid \eta]^{1/2}$, and by the Cauchy–Schwarz inequality $|\langle x, y \rangle| \leq \rho_{s_u} \|x\| \|y\|$. Lemma 11.4 gives $Q_u Q_{u+1} + Q_{u+1} Q_u \geq -\rho_{s_u}(Q_u + Q_{u+1})$ with $\rho_{s_u} \leq c_M$, and Lemma 7.18 gives $\sum_u \langle g, Q_u g \rangle \geq (1 - 2c_M)\|g\|^2$ for $g \perp \ker H$. If $Hg = 0$ then $g = P_{w_u} g$ for every u; the windows cover the ring, and the Fubini argument of Lemma 11.6 shows that g is constant. The translated chains $\{R_u + a\}_u$, $a \in \mathbb{Z}/M\mathbb{Z}$, satisfy the same bound, and each window occurs in exactly K of them; summing over a,

$$\sum_{w \in \mathbb{Z}/M\mathbb{Z}} \langle g, (I - P_w) g \rangle \geq \frac{M}{K}(1 - 2c_M)\|g\|^2 \geq d\,(1 - 2c_M)\|g\|^2 \qquad (g \perp \mathbf{1}).$$

The product. $\mu = \bigotimes_k \mu_k$ over the L^3 rings and $L^2(\mu) = \bigotimes_k L^2(\mu_k)$. For a set A of rings let $\mathcal{H}_A = \bigotimes_{k \in A} L^2_0(\mu_k) \otimes \bigotimes_{k \notin A} \mathbb{R}\mathbf{1}$; these subspaces are orthogonal, $f = \sum_A f_A$ and $\operatorname{Var} f = \sum_{A \neq \emptyset} \|f_A\|^2$. The cube at x meets the rings of a set F_x of n^3 spatial sites in the window $w_x = x_4$, and under μ the conditional law of its spins given the others is the product of the one-ring conditional laws, so $\mathcal{P}_x = \bigotimes_{k \in F_x} P^{(k)}_{w_x} \otimes I$. Each $P^{(k)}_w$ fixes $\mathbf{1}$ and is self-adjoint, so it preserves $L^2_0(\mu_k)$; hence $\mathcal{P}_x$ preserves each $\mathcal{H}_A$ and acts there as $\bigotimes_{k \in A \cap F_x} Q^{(k)}_{w_x} \otimes I$ with $Q^{(k)}_w = P^{(k)}_w|_{L^2_0}$, $0 \leq Q^{(k)}_w \leq I$. For positive contractions $Q_1 \otimes Q_2 \leq Q_1 \otimes I$, since $Q_1 \otimes (I - Q_2) \geq 0$; so for $g \in \mathcal{H}_A$, a ring $k_0 \in A$ and every x with $k_0 \in F_x$, $\langle g, (I - \mathcal{P}_x) g \rangle \geq \langle g, (I - Q^{(k_0)}_{w_x} \otimes I) g \rangle$, while the other x contribute ≥ 0. As $L \geq n$, each window w is the window of exactly n^3 cubes with $k_0 \in F_x$, and the one-ring bound in the factor k_0 gives $\sum_x \langle g, (I - \mathcal{P}_x) g \rangle \geq n^3 d (1 - 2c_M)\|g\|^2$. Since every $I - \mathcal{P}_x$ is an orthogonal

projection preserving the decomposition, $\sum_x \mu((f-\mathcal{P}_x f)^2) = \sum_A \sum_x \langle f_A, (I-\mathcal{P}_x) f_A\rangle \geq n^3 d(1-2c_M)\operatorname{Var} f$, which is the claim, as $n^4\varpi_M = n^3 d(1-2c_M)$. The last assertions follow from $\sup_{M\geq 4d} c_M \leq \kappa^{(3d-3)/4}$, which holds because $\rho_s \leq \kappa^{2d-s}$ and $r_M \leq \lceil (d-1)/4\rceil$. □

Along the tuning $\kappa = e^{-\Delta/N}$ with $d = d_N = N/K_0$ and $L = NL_p$ as in Proposition 7.13, the proposition gives $\mathrm{BP}(n_N, \frac{1}{2}(1-2c))$ with the N-independent $c = e^{-\Delta/K_0}/(1+e^{-2\Delta/K_0}) < \frac{1}{2}$ on every cylinder with $d_N \mid M$, $M \geq 4d_N$, the cylinders used in the proof of that proposition; and since $\kappa^{(3d_N-3)/4} = e^{-3\Delta/(4K_0)} e^{3\Delta/(4N)}$, if $K_0 < 3\Delta/(4\log 2)$ it gives, for all large N and every $M \geq 4d_N$, $\mathrm{BP}(n_N, \varpi)$ with a fixed $\varpi > 0$: this is hypothesis (D_ℓ) of Definition 7.2 with $\ell = 2/K_0$. So the rings satisfy, uniformly in the mesh, the agreement of the two orders of the limits for every local observable, the slab hypothesis $(\mathrm{M}_{d,m}(c))$ with $c < \frac{1}{2}$ and $(\mathrm{M}^*_w(c))$, the cube condition (D_ℓ), the physical gap Δ with a simple leading eigenvalue, translation invariance and reflection positivity; and (Z) fails. The only hypotheses of Theorem E they do not satisfy are (W), which has no meaning for them, and (Z) itself.

So in the class of reflection-positive lattice models with a gapped transfer matrix and a unique vacuum, neither the agreement of the two orders of the limits for all local observables, nor the slab hypothesis, nor the cube condition, nor all of them together, imply (Z): the maximal correlation of a product of independent systems does not see the number of factors (Lemma 7.15), while $\mathcal{Z}_L(M) - 1$ is a trace, which does. The same lemma shows that the slab hypothesis taken in the time direction, under exterior configurations or under the periodic measure itself and uniformly in the period, holds for the rings and does not give (Z) either. In this model the boundary condition lowers the layer correlation: by Lemma 7.16 it is at most the vacuum value $e^{-m(w+1)}$ for every ω, with equality only as the outer blocks grow.

Two positive statements delimit the hypothesis from the other side: a bound on the density of states gives (Z), and (Z) holds at strong coupling, with the mechanism visible. For a continuum family let $r_0 = 1 > r_1 \geq r_2 \geq \dots$ be the eigenvalues of $\mathbb{R}$ on the cylinder of side NL_p at coupling $\beta_{N,p}$, counted with multiplicity, let $\varepsilon_i = -N\log r_i \in (0,\infty]$ for $i \geq 1$ be the excitation energies in physical units, and let $\mathcal{N}_{N,p}(\varepsilon)$ be the number of $i \geq 1$ with $\varepsilon_i \leq \varepsilon$, which is finite since $\mathcal{N}_{N,p}(\varepsilon) \leq e^{2\varepsilon/N}(\mathcal{Z}_{NL_p}(2) - 1)$.

Proposition 7.20 (an entropy bound gives vacuum dominance)**.** *Suppose that $\varepsilon_i \geq \Delta > 0$ for all $i \geq 1$ and all large N, p, as under (D_ℓ) by Proposition 7.4, and that there are $\delta \in (0,1]$ and p_0 such that*

$$\mathcal{N}_{N,p}(\varepsilon) \leq e^{(1-\delta)L_p\varepsilon} \qquad \text{for all } \varepsilon > 0,\ p \geq p_0 \text{ and all large } N. \tag{7.8}$$

Then $\mathcal{Z}_{N,p} - 1 \leq \delta^{-1} e^{-\delta\Delta L_p}$ for $p \geq p_0$ and large N, and (Z) holds. A bound $\mathcal{N}_{N,p}(\varepsilon) \leq \exp\bigl(C(L_p^3\varepsilon^3)^{1/4}\bigr)$ implies (7.8) as soon as $L_p\Delta \geq (C/(1-\delta))^4$.

Proof. $\mathcal{Z}_{N,p} - 1 = \sum_{i\geq 1} r_i^{NL_p} = \sum_{i\geq 1} e^{-L_p\varepsilon_i} = \int_{[\Delta,\infty)} e^{-L_p\varepsilon}\, d\mathcal{N}_{N,p}(\varepsilon)$. Integrating by parts on $[\Delta, E]$, the boundary term at E is at most $e^{-\delta L_p E}$ by (7.8) and tends to 0, so

$$\mathcal{Z}_{N,p} - 1 = L_p \int_\Delta^\infty e^{-L_p\varepsilon}\mathcal{N}_{N,p}(\varepsilon)\, d\varepsilon \leq L_p \int_\Delta^\infty e^{-\delta L_p\varepsilon}\, d\varepsilon = \delta^{-1} e^{-\delta\Delta L_p}.$$

For the last claim, $C(L_p^3\varepsilon^3)^{1/4} \leq (1-\delta)L_p\varepsilon$ is $(L_p\varepsilon)^{1/4} \geq C/(1-\delta)$, which holds for $\varepsilon \geq \Delta$ under the stated condition. □

The display in the proof shows that (Z) is exactly the statement

$$\lim_{p\to\infty} \limsup_{N\to\infty} L_p \int_\Delta^\infty e^{-L_p\varepsilon}\, \mathcal{N}_{N,p}(\varepsilon)\, d\varepsilon = 0,$$

that is, the number of states of the cylinder with physical energy at most ε must grow slower than $e^{L_p\varepsilon}$, uniformly in the mesh. A relativistic theory with finitely many species has entropy of order $(L_p^3\varepsilon^3)^{1/4}$ and satisfies (7.8) with room; the rings violate it, since there

$\mathcal{N}_{N,p}(\Delta) = N^3 L_p^3$. States of energy of the order of the cutoff, $\varepsilon \sim N$, are harmless, their number being at most $e^{cN^3L_p^3}$ by the bound before the proposition and their weight e^{-cNL_p}; what (7.8) excludes is a multiplicity growing with the lattice volume at physical energies, a flat band up to the cutoff. The gap fixes $\inf_i \varepsilon_i$ and nothing else, and by Proposition 7.14 the multiplicities are invisible to every correlation function, so (7.8) is a hypothesis on the whole of $\mathcal{K}$.

At strong coupling the bound holds, by a polymer expansion of the partition function. Let $G \subset \mathrm{U}(N)$, $c_0 = \int_G e^{N\beta \operatorname{Re}\operatorname{Tr} g}\, dg$, and for a plaquette p let $f_p = e^{N\beta \operatorname{Re}\operatorname{Tr} Q_p}/c_0 - 1$, so that $\|f_p\|_\infty \le \epsilon(\beta) = e^{2N^2\beta} - 1$ since $|\operatorname{Re}\operatorname{Tr} g| \le N$. On $\Lambda_{L,M}$ with $L, M \ge 3$, and on the infinite cylinder $C_L = (\mathbb{Z}/L\mathbb{Z})^3 \times \mathbb{Z}$, two distinct plaquettes share at most one link, a link lies in six plaquettes, so a plaquette has at most 20 neighbours sharing a link with it, and the base points of two such plaquettes differ by at most one in each coordinate. A *polymer* is a finite nonempty set γ of plaquettes that is connected for this neighbour relation; two polymers are incompatible if they share a link; the activity of γ is $z(\gamma) = \int \prod_{p\in\gamma} f_p \prod_{e\in E(\gamma)} dQ_e$, the integral over the links of γ.

Lemma 7.21 (polymers)**.** (i) $Z(\Lambda_{L,M}) = c_0^{6L^3M}\, \Xi$, *where* $\Xi = \sum \prod_i z(\gamma_i)$ *is the sum over families of pairwise compatible polymers, the empty family contributing* 1.

(ii) $z(\gamma) = 0$ *unless every link of* γ *lies in at least two of its plaquettes; such a* γ *has at least five plaquettes; and* $|z(\gamma)| \le \epsilon(\beta)^{|\gamma|}$.

(iii) *The number of polymers of* k *plaquettes containing a given plaquette is at most* 400^{k-1}.

(iv) *The base times of a polymer of* k *plaquettes form an interval of at most* k *consecutive values.*

Proof. (i) $\prod_p (1 + f_p) = \sum_S \prod_{p\in S} f_p$ over subsets S of the plaquettes; the integral of $\prod_{p\in S} f_p$ factorises over the connected components of S, which share no link, and the families of components are exactly the families of pairwise compatible polymers. (ii) If a link e lies in exactly one plaquette p of γ, integrate Q_e first: $Q_e \mapsto Q_p$ is a composition of translations of G and possibly an inversion, which preserve Haar measure, so $\int f_p\, dQ_e = \int_G e^{N\beta \operatorname{Re}\operatorname{Tr} g} dg/c_0 - 1 = 0$. If every link of γ lies in two of its plaquettes, each plaquette shares each of its four links with another plaquette of γ and at most one link with any given one, so it has at least four neighbours in γ. The bound on z is the bound on $\|f_p\|_\infty$. (iii) A polymer $\gamma \ni p$ with k plaquettes has a spanning tree in its neighbour graph, and a depth-first traversal of the tree from p is a closed walk of length $2(k-1)$ in a graph of maximal degree 20 that visits every plaquette of γ; distinct polymers give distinct walks, and there are at most $20^{2(k-1)}$ walks. (iv) Neighbouring plaquettes have base times differing by at most one. □

The cluster expansion of $\log \Xi$ is the theorem of Kotecký and Preiss [36].[12] Take $a(\gamma) = |\gamma|$ and $d(\gamma) = d\,|\gamma|$ with $e^{1+d} = 1/(800\,\epsilon(\beta))$, which needs $\epsilon(\beta) \le 1/(800e)$. A polymer incompatible with γ contains one of the at most six plaquettes through one of the at most $4|\gamma|$ links of γ, so by Lemma 7.21(ii),(iii), with $u = 400\,\epsilon(\beta)e^{1+d} = \frac{1}{2}$,

$$\sum_{\gamma' \not\sim \gamma} |z(\gamma')| e^{(1+d)|\gamma'|} \le 24|\gamma| \sum_{k\ge 5} 400^{k-1} \big(\epsilon(\beta) e^{1+d}\big)^k = 24|\gamma|\, \frac{u^5}{400(1-u)} = \frac{24}{6400}\, |\gamma| \le |\gamma|.$$

[12] Let $\mathcal{P}$ be a finite set of polymers with a symmetric reflexive incompatibility relation $\not\sim$, activities $z : \mathcal{P} \to \mathbb{C}$, and $\Xi = \sum \prod_i z(\gamma_i)$ over families of pairwise compatible polymers. If $a, d : \mathcal{P} \to [0, \infty)$ satisfy $\sum_{\gamma' \not\sim \gamma} |z(\gamma')| e^{a(\gamma') + d(\gamma')} \le a(\gamma)$ for every γ, then $\log \Xi = \sum_X w(X)$, the absolutely convergent sum over clusters, that is, multisets $X = \{\gamma_1, \dots, \gamma_n\}$ whose incompatibility graph is connected, of $w(X) = \frac{1}{n!} \phi^T(\gamma_1, \dots, \gamma_n) \prod_i z(\gamma_i)$ with ϕ^T the Ursell coefficient, and $\sum_{X \not\sim \gamma} |w(X)| e^{d(X)} \le a(\gamma)$ for every γ, where $d(X) = \sum_i d(\gamma_i)$ and $X \not\sim \gamma$ means that some $\gamma_i \not\sim \gamma$ [36, Theorem 1]. The union of the polymers of a cluster is connected, and $w(X)$ depends only on which subsets of that union are polymers, which pairs share a link, and the activities.

So the expansion converges on every $\Lambda_{L,M}$ and on every finite time interval of C_L, where the incidence counts are the same or smaller, and the weights $W(S) = \sum_{X:\bigcup\gamma_i=S} w(X)$ of a connected plaquette set S do not depend on the lattice containing S. Summing the bound over the finite intervals, for every plaquette p of $\Lambda_{L,M}$ or of C_L,

$$\sum_{S\ni p} |W(S)|\, e^{d|S|} \le 1, \qquad \log Z(\Lambda_{L,M}) = 6L^3 M \log c_0 + \sum_{S\subset P(\Lambda_{L,M})} W(S), \tag{7.9}$$

the second by Lemma 7.21(i) and the regrouping of the absolutely convergent cluster sum by supports.

Lemma 7.22 (winding)**.** *Let $\pi : C_L \to \Lambda_{L,M}$ be the covering by the time translations by multiples of $M \ge 3$, and let $S \subset P(\Lambda_{L,M})$ be connected with $|S| \le M-1$. Then every connected component $\tilde S$ of $\pi^{-1}(S)$ is finite, π maps $\tilde S$ bijectively onto S preserving which subsets are polymers and which pairs share a link, so that $W(\tilde S) = W(S)$; the components are the translates $\tilde S + nM$, $n \in \mathbb{Z}$, and exactly one of them has its least base time in $[0, M)$. Conversely every finite connected $\tilde S \subset P(C_L)$ with $|\tilde S| \le M-1$ and least base time in $[0, M)$ arises in this way from $S = \pi(\tilde S)$.*

Proof. π is injective on the six plaquettes through a link and on any set of plaquettes whose base times lie in M consecutive values. If $\tilde p \in \pi^{-1}(S)$ and $q \in S$ shares the link e with $\pi(\tilde p)$, the lift $\tilde e$ of e in $\tilde p$ lies in a unique plaquette $\tilde q$ with $\pi(\tilde q) = q$, and $\tilde q$ is a neighbour of $\tilde p$; so every component of $\pi^{-1}(S)$ maps onto S. If a component were infinite, its base times would form an infinite interval by Lemma 7.21(iv) and connectedness, hence contain M consecutive values, whose plaquettes have distinct images, giving $|S| \ge M$. So $\tilde S$ is finite; a finite set is not invariant under a nonzero translation, so $\tilde S \cap (\tilde S + nM) = \emptyset$ for $n \ne 0$, and two plaquettes of $\tilde S$ with the same image, which differ by such a translation, coincide: π is injective on $\tilde S$. If $\pi(\tilde p)$ and $\pi(\tilde q)$ share a link, the lift of $\pi(\tilde q)$ adjacent to $\tilde p$ lies in $\tilde S$ and has the same image as $\tilde q$, hence equals $\tilde q$; so sharing a link, and with it connectedness of subsets, is preserved, and two links of plaquettes of $\tilde S$ with the same image coincide. The activities are preserved because they are Haar integrals over the links of the subset of functions of its plaquette variables. Any other component contains a lift $\tilde p + nM$ of $\pi(\tilde p)$ and so equals $\tilde S + nM$; the least base times of the translates run through one residue class modulo M. For the converse, $S = \pi(\tilde S)$ is connected with $|S| \le M-1$, and the component of $\pi^{-1}(S)$ containing $\tilde S$ maps bijectively onto S while $\tilde S$ already maps onto S, so the two are equal. □

Proposition 7.23 (vacuum dominance at strong coupling)**.** *Let $G \subset \mathrm{U}(N)$, $\beta \ge 0$ with $800e\,\epsilon(\beta) < 1$, where $\epsilon(\beta) = e^{2N^2\beta} - 1$, and $L, M \ge 3$. Then*

$$0 \le \log \mathcal{Z}_L(M) \le 6L^3(M+1)\big(800e\,\epsilon(\beta)\big)^M.$$

In particular a continuum family whose couplings satisfy $\sup_{N,p} \beta_{N,p} = \beta_0$ with $800e\,\epsilon(\beta_0) < 1$ satisfies (Z)*, with $\mathcal{Z}_{N,p} - 1 \to 0$ as $N \to \infty$ for every p.*

Proof. Split the sum in (7.9) according to $|S| \le M-1$ or $|S| \ge M$. By Lemma 7.22 and the invariance of W under time translation on C_L, the first part equals $M\,A_{L,M}$, where $A_{L,M} = \sum W(\tilde S)$ over the connected $\tilde S \subset P(C_L)$ with least base time 0 and $|\tilde S| \le M-1$. For the second part, each S with $|S| \ge M$ is counted once for each of its plaquettes when one sums over p, so

$$\sum_{|S|\ge M} |W(S)| \le \sum_{p\in P(\Lambda_{L,M})} \sum_{S\ni p,\,|S|\ge M} \frac{|W(S)|}{|S|} \le \frac{6L^3M}{M}\, e^{-dM} \sup_p \sum_{S\ni p} |W(S)| e^{d|S|} \le 6L^3 e^{-dM}$$

by (7.9). Let $a_L = \sum W(\tilde S)$ over all connected $\tilde S \subset P(C_L)$ with least base time 0; the sum converges absolutely, since every such $\tilde S$ contains one of the $6L^3$ plaquettes of base time 0, and

the same count gives $|A_{L,M} - a_L| \le 6L^3 e^{-dM}$. Hence $M^{-1}\log Z(\Lambda_{L,M}) \to 6L^3 \log c_0 + a_L$, and as $\lambda_0 = \lim_M Z(\Lambda_{L,M})^{1/M}$ by Lemma 7.3(i),(ii), $\log \lambda_0 = 6L^3 \log c_0 + a_L$. Therefore

$$\log \mathcal{Z}_L(M) = \log Z(\Lambda_{L,M}) - M \log \lambda_0 = M(A_{L,M} - a_L) + \sum_{|S| \ge M} W(S),$$

of modulus at most $6L^3(M+1)e^{-dM}$ with $e^{-d} = 800e\,\epsilon(\beta)$; and $\log \mathcal{Z}_L(M) \ge 0$ because $\mathcal{Z}_L(M) = \operatorname{Tr} \mathbb{R}^M \ge 1$. For a family with $L = M = NL_p$ the exponent is $6N^3 L_p^3 (NL_p + 1)(800e\,\epsilon(\beta_0))^{NL_p} \to 0$. □

The proposition shows where $\log \mathcal{Z}_L(M)$ comes from: the connected plaquette sets with at least M plaquettes, the surfaces that wind around the time circle, each of weight $(C\beta)^M$ and with L^3 positions per unit of time. This is the exact counterpart of the rings, $\mathcal{Z} - 1 \approx L^3 e^{-mM}$ with $e^{-m} = \tanh J$. At strong coupling the lattice gap $m = -\log(C\beta)$ is of order one in lattice units and $L^3 e^{-mM} = N^3 L_p^3 (C\beta)^{NL_p} \to 0$; along a tuning with a physical gap, $m = \Delta/N$, the same expression is $N^3 L_p^3 e^{-\Delta L_p}$, and (Z) asks that the entropy of the winding surfaces be counted per unit of physical, not lattice, volume. The constant $1/(800e)$ is far below the range of Theorem A and is not optimised; no fixed coupling reaches the continuum, so the proposition gives (Z) for no continuum family of Yang–Mills theory, and (Z) at weak coupling remains open.

The constant can be improved by a factor of about twenty-four, and the rate in M by a factor four, by using what the winding surfaces are; this is as far as the expansion goes, and the proposition after it shows that no bound of this shape, at any fixed coupling, can reach a continuum family. Call a set S of plaquettes *closed* if every link of S lies in at least two plaquettes of S; for a set S of plaquettes of $\Lambda_{L,M}$ or of C_L let S_t be the set of its temporal plaquettes with base time t.

Lemma 7.24 (closed polymers)**.** *Let $L \ge 4$ and $M \ge 4$.*

(i) *If $\int_G \operatorname{Tr} g\, dg = 0$, which holds for $G = \mathrm{SU}(N)$, then $\|f_p\|_\infty \le \epsilon'(\beta) = e^{N^2\beta} - 1$.*
(ii) *A closed polymer has at least six plaquettes.*
(iii) *The number of polymers of k plaquettes containing a given plaquette is at most $(20e)^{k-1}$.*
(iv) *If S is closed and $S_t \ne \emptyset$, then $|S_t| \ge 4$.*

Proof. (i) Put $x = N^2\beta$ and $X = \operatorname{Re}\operatorname{Tr} g/N \in [-1,1]$, so that $\int X\, dg = 0$ and $c_0 = \int e^{xX} dg$. By Jensen's inequality $c_0 \ge 1$, and by convexity $e^{xX} \le \frac{1+X}{2} e^x + \frac{1-X}{2} e^{-x}$, so $c_0 \le \cosh x$. Hence $f_p \le e^x/c_0 - 1 \le e^x - 1$ and $-f_p \le 1 - e^{-x}/\cosh x = \tanh x \le x \le e^x - 1$.

(ii) Let γ be closed and $p \in \gamma$, with links $\ell_1, \dots, \ell_4$, ℓ_1 and ℓ_3 opposite. Each ℓ_i lies in a plaquette $q_i \in \gamma$ other than p, and $q_i \ne q_j$ for $i \ne j$ because two distinct plaquettes share at most one link; so $|\gamma| \ge 5$. If $|\gamma| = 5$ then $\gamma = \{p, q_1, \dots, q_4\}$, and each of the three links of q_i other than ℓ_i lies in some q_j with $j \ne i$, no two of them in the same q_j; so q_1 and q_3 share a link. Let p be spanned at the site x by the directions i and j, with ℓ_1 the link in the direction i at x and ℓ_3 the one at $x + e_j$. The plaquette q_1 is spanned by ℓ_1 and a direction other than $+e_j$, so the j-th coordinates of its sites lie in $\{x_j - 1, x_j\}$, and those of q_3 lie in $\{x_j + 1, x_j + 2\}$; as the side in the direction j is at least 4, the two sets are disjoint modulo it and q_1, q_3 share no site. So $|\gamma| \ge 6$, and the boundary of a cube attains it.

(iii) The neighbour graph of the plaquettes has degree at most 20. Fix for every polymer a spanning tree of its neighbour graph, and lift the tree of a polymer $\gamma \ni p$ to the universal cover of the neighbour graph rooted at p, a tree in which the root has 20 children and every other vertex at most 19; the lift is a subtree with k vertices containing the root, and it determines γ by projection. Let t_k be the number of subtrees with k vertices containing the root of the tree in which the root has Δ children and every other vertex $\Delta - 1$. Its generating function $R(x) = \sum_k t_k x^k$ satisfies $R = x(1+A)^\Delta$ with $A = x(1+A)^{\Delta-1}$, and Lagrange inversion gives $t_k = \frac{\Delta}{k-1}\binom{(\Delta-1)k}{k-2}$ for $k \ge 2$. With $\binom{n}{m} \le (en/m)^m$ and $(1 + \frac{2}{k-2})^{k-2} \le e^2$,

$t_k \le \frac{\Delta}{k-1}\, e^2\big(e(\Delta-1)\big)^{k-2} \le \frac{e}{k-1}(e\Delta)^{k-1} \le (e\Delta)^{k-1}$ for $k \ge 4$, and $t_1 = 1$, $t_2 = \Delta$, $t_3 = \binom{\Delta}{2} + \Delta(\Delta-1) \le (e\Delta)^2$ directly. Subtrees of the cover with more children at the root bound the subtrees of the actual neighbour graph, so the count is at most $t_k \le (20e)^{k-1}$.

(iv) A vertical link between the times t and $t+1$ lies in temporal plaquettes of base time t only, so each vertical link of a plaquette of S_t lies in at least two plaquettes of S_t, and each plaquette of S_t contains exactly two vertical links, at the sites $\vec{x}$ and $\vec{x}+e_i$ of the slice. The graph whose vertices are the vertical links of the plaquettes of S_t and whose edges are the plaquettes of S_t is then a subgraph of the spatial lattice $(\mathbb{Z}/L\mathbb{Z})^3$ of minimal degree at least 2, so it contains a cycle, of length at least 4 since that lattice has girth 4 for $L \ge 4$. Hence $|S_t| \ge 4$. □

Proposition 7.25 (the strong-coupling range)**.** *Let $G \subset \mathrm{U}(N)$, let $\epsilon'(\beta) = e^{N^2\beta} - 1$ if $\int_G \operatorname{Tr} g\, dg = 0$ and $\epsilon'(\beta) = \epsilon(\beta)$ otherwise, and let $\beta \ge 0$ with $24e^2\epsilon'(\beta) \le 1$. Then for $L, M \ge 4$*

$$0 \le \log \mathcal{Z}_L(M) \le 6L^3(M+1)\big(24e^2\epsilon'(\beta)\big)^{4M}.$$

For $G = \mathrm{SU}(2)$ the condition is $\beta \le \frac{1}{4}\log(1 + 1/(24e^2)) = 1.405 \times 10^{-3}$, that is $\beta_W = 4\beta \le 5.62 \times 10^{-3}$, against $\beta \le 5.74 \times 10^{-5}$ in Proposition 7.23.

Proof. By Lemma 7.21(ii) only closed polymers have nonzero activity, and a closed polymer has at least six plaquettes by Lemma 7.24(ii); its activity is at most $\epsilon'^{\,|\gamma|}$ by Lemma 7.24(i). Take $a(\gamma) = |\gamma|$ and $d(\gamma) = d|\gamma|$ with $e^d = 1/(24e^2\epsilon') \ge 1$, so that $v = 20e\,\epsilon' e^{1+d} = \frac{5}{6}$. A polymer incompatible with γ contains one of the at most $24|\gamma|$ plaquettes through a link of γ, so by Lemma 7.24(iii)

$$\sum_{\gamma' \not\sim \gamma} |z(\gamma')| e^{(1+d)|\gamma'|} \le 24|\gamma| \sum_{k \ge 6} (20e)^{k-1}\big(\epsilon' e^{1+d}\big)^k = \frac{6}{5e}\,\frac{v^6}{1-v}\,|\gamma| \le 0.89\,|\gamma|,$$

and (7.9) holds with this d, on $\Lambda_{L,M}$ and on the finite intervals of C_L. The support S of a cluster is a union of closed polymers, hence closed and connected. Call $S \subset P(\Lambda_{L,M})$ *wrapping* if $S_t \ne \emptyset$ for every $t \in \mathbb{Z}/M\mathbb{Z}$; by Lemma 7.24(iv) a wrapping S has $|S| \ge 4M$. If S is not wrapping, pick t^* with $S_{t^*} = \emptyset$ and lift every plaquette of S to its preimage in C_L with base time in $[t^*+1, t^*+M]$. A vertical link between the times t and $t+1$ lies in temporal plaquettes of base time $t \ne t^*$ only, whose lifts have base time t; a spatial link at time t lies in spatial plaquettes at time t, lifted to the representative of t in $[t^*+1, t^*+M]$, in temporal plaquettes of base time t, absent if $t \equiv t^*$ and otherwise lifted to that representative, and in temporal plaquettes of base time $t-1$, lifted to the representative of $t-1$ in $[t^*, t^*+M-1]$, which is in $[t^*+1, t^*+M-1]$ since $S_{t^*} = \emptyset$, with their top at the representative of t. So the plaquettes of S through a link lift to plaquettes through one lifted link, the lift $\tilde{S}$ is connected and closed, $\pi : \tilde{S} \to S$ preserves which pairs share a link in both directions and which subsets are polymers, and $W(\tilde{S}) = W(S)$ as in Lemma 7.22; a connected set does not cross an empty layer, so a different choice of t^* gives a time translate of $\tilde{S}$, and the lifts of S are the translates $\tilde{S} + nM$, exactly one of which has least base time in $[0, M)$. The base times of the temporal plaquettes of a connected set form an interval, because a spatial plaquette at time t shares links with temporal plaquettes of base times $t-1$ and t only; so $\tilde{S}$ has its temporal base times in at most $M-1$ consecutive values. Conversely a finite connected closed $\tilde{S} \subset P(C_L)$ whose temporal base times lie in at most $M-1$ consecutive values, or which has no temporal plaquette, has all its base times in M consecutive values, projects bijectively onto a non-wrapping S with the same structure, and $\pi(\tilde{S})_{t_0-1} = \emptyset$ for t_0 its least temporal base time. Hence, with the time translation invariance of W on C_L, the non-wrapping part of the sum in (7.9) is $MA'_{L,M}$, where $A'_{L,M} = \sum W(\tilde{S})$ over the connected closed $\tilde{S} \subset P(C_L)$ with least base time 0 and temporal base times in at most $M-1$ consecutive values. Let $a_L = \sum W(\tilde{S})$ over all connected closed $\tilde{S} \subset P(C_L)$ with least base time 0, as in the proof of Proposition 7.23; a set counted in a_L and not in $A'_{L,M}$ has at least M nonempty temporal

layers, each with at least four plaquettes by Lemma 7.24(iv), which holds on C_L with the same proof, so $|\tilde{S}| \geq 4M$ and $|A'_{L,M} - a_L| \leq 6L^3 e^{-4dM}$ by (7.9). The wrapping part is at most $\sum_p \sum_{S \ni p,\, |S| \geq 4M} |W(S)|/|S| \leq \frac{6L^3 M}{4M} e^{-4dM}$. As $\log \lambda_0 = 6L^3 \log c_0 + a_L$,

$$\log \mathcal{Z}_L(M) = M(A'_{L,M} - a_L) + \sum_{S \text{ wrapping}} W(S)$$

has modulus at most $(6M + \frac{3}{2})L^3 e^{-4dM} \leq 6L^3(M+1)e^{-4dM}$, and $\log \mathcal{Z}_L(M) \geq 0$ as before. For SU(2), $N^2\beta = 4\beta$ and $\int \operatorname{Tr} g \, dg = 0$. □

Proposition 7.26 (no fixed rate along a tuning)**.** *Suppose that along a continuum family there are $C, m > 0$ with $\log \mathcal{Z}_{NL_p}(M) \leq CN^3 L_p^3 M e^{-mM}$ for all $M \geq 4$ and all large N and p, as Proposition 7.25 gives whenever $24e^2\epsilon'(\sup \beta_{N,p}) < 1$. Then the smallest excitation energy $\varepsilon_1 = -N \log r_1$ of the cylinder, in physical units, satisfies $\varepsilon_1 \geq Nm$ for all large N and p. So no bound of this shape holds along a tuning on whose cylinders some excitation has bounded physical energy: along such a tuning (Z) must be reached with a rate of order $1/N$ in lattice units, and the entropy of the winding surfaces must be counted per unit of physical volume.*

Proof. $r_1^M \leq \mathcal{Z}_{NL_p}(M) - 1 \leq e^{\log \mathcal{Z}} - 1 \leq 2 \log \mathcal{Z}$ once $\log \mathcal{Z}_{NL_p}(M) \leq 1$, which holds for large M. So $-M\varepsilon_1/N \leq \log(2CN^3 L_p^3 M) - mM$; divide by M and let $M \to \infty$. □

7.8. **The thermal hypothesis at one scale.** Hypothesis (Z) is a statement about tori whose four sides all grow with p. Positivity of the transfer matrix in each of the four directions reduces it to two statements: one at a single physical scale, where the state count of Proposition 7.20 is needed, and one about ground-state energies only, which carries the growth of the volume. The construction of Lemma 7.3 uses neither that the slice is a cube nor that the direction is the time axis: for a torus $\Lambda(n_1, n_2, n_3, n_4) = \prod_\mu \mathbb{Z}/n_\mu\mathbb{Z}$ with all $n_\mu \geq 3$ and each direction μ, the same kernel on the links of a transverse slice $\prod_{\nu \neq \mu} \mathbb{Z}/n_\nu\mathbb{Z}$ defines a self-adjoint Hilbert–Schmidt operator $\mathbb{T}^{(\mu)}$ with a simple largest eigenvalue, nonnegative when $\beta \geq 0$, with $Z(n_1, n_2, n_3, n_4) = \operatorname{Tr}(\mathbb{T}^{(\mu)})^{n_\mu}$ for the partition function Z, which is symmetric in its arguments because the Wilson weight and the Haar measure are invariant under a relabelling of the axes. For a spatial torus $\vec{\ell} = (\ell_1, \ell_2, \ell_3)$ with all $\ell_i \geq 3$ let $\mathbb{T}_{\vec{\ell}}$ be the time transfer matrix, $E_0(\vec{\ell}) = -\log \lambda_0(\vec{\ell})$ its vacuum energy, $\mathbb{R} = \mathbb{T}_{\vec{\ell}}/\lambda_0(\vec{\ell})$, and for $M \geq 2$

$$\mathcal{Z}_{\vec{\ell}}(M) = \operatorname{Tr} \mathbb{R}^M = Z(\vec{\ell}, M)\, e^{ME_0(\vec{\ell})}, \qquad D_{\vec{\ell}}(M) = \mathcal{Z}_{\vec{\ell}}(M) - 1 = \sum_{i \geq 1} r_i^M \geq 0,$$

so that $\mathcal{Z}_L(M) = \mathcal{Z}_{(L,L,L)}(M)$; throughout this subsection $\beta \geq 0$, so that all the transfer matrices are nonnegative. Since $\int_G \operatorname{Re} \operatorname{Tr} g \, dg \geq 0$ and each plaquette variable is Haar distributed when one of its links is integrated, Jensen's inequality gives $Z \geq 1$ and $E_0 \leq 0$.

Lemma 7.27 (cutting a side)**.** *Let $\beta \geq 0$. For $n_1', n_1'' \geq 3$ and $n_2, n_3, n_4 \geq 3$,*

$$Z(n_1' + n_1'', n_2, n_3, n_4) \leq Z(n_1', n_2, n_3, n_4)\, Z(n_1'', n_2, n_3, n_4),$$

and likewise in every direction; and $D_{\vec{\ell}}(M' + M'') \leq D_{\vec{\ell}}(M')\, D_{\vec{\ell}}(M'')$ for $M', M'' \geq 2$.

Proof. With the eigenvalues $\lambda_i \geq 0$ of $\mathbb{T}^{(1)}$, $\sum_i \lambda_i^{a+b} \leq (\sum_i \lambda_i^a)(\sum_i \lambda_i^b)$, the sums being finite for $a, b \geq 2$ because $\mathbb{T}^{(1)}$ is Hilbert–Schmidt; the second claim is the same inequality for $\sum_{i \geq 1} r_i^{M'+M''}$. □

Proposition 7.28 (halving)**.** *Let $\ell_i = \ell_i' + \ell_i''$ with $\ell_i', \ell_i'' \geq 3$ for $i = 1, 2, 3$, let $\mathcal{S}$ be the eight tori $(\ell_1^a, \ell_2^b, \ell_3^c)$ with $a, b, c \in \{\prime, \prime\prime\}$, and let $M = M' + M''$ with $M', M'' \geq 2$. Then*

$$\kappa(\vec{\ell}) := E_0(\vec{\ell}) - \sum_{s \in \mathcal{S}} E_0(s) \geq 0 \quad \text{and} \quad \log(1 + D_{\vec{\ell}}(M)) \leq M\kappa(\vec{\ell}) + \sum_{s \in \mathcal{S}} D_s(M')\, D_s(M'').$$

Proof. Lemma 7.27 in the direction 1, then in the direction 2 on both factors, then in the direction 3, gives $Z(\vec{\ell},M)\le\prod_{s\in\mathcal{S}}Z(s,M)$ for every $M\ge 2$. Taking M-th roots and $M\to\infty$, $\lambda_0(\vec{\ell})\le\prod_s\lambda_0(s)$ by Lemma 7.3(i),(ii), which is $\kappa(\vec{\ell})\ge 0$. Multiplying the inequality by $e^{ME_0(\vec{\ell})}=e^{M\kappa(\vec{\ell})}\prod_s e^{ME_0(s)}$ gives $1+D_{\vec{\ell}}(M)\le e^{M\kappa(\vec{\ell})}\prod_s(1+D_s(M))$; take logarithms, use $\log(1+x)\le x$ and Lemma 7.27 for $D_s(M)$. □

The defect $\kappa(\vec{\ell})$ is the vacuum energy of the spatial torus minus the vacuum energies of its eight halves, the extensive part cancelling exactly; in physical units it is a Casimir energy. Fix (N,p), the coupling $\beta_{N,p}$ and $n_0=NL_p$, and put $m_j=\lfloor n_0 2^{-j}\rfloor$, so that $m_j\in\{2m_{j+1},2m_{j+1}+1\}$ and both $\lfloor n/2\rfloor$ and $\lceil n/2\rceil$ lie in $\{m_{j+1},m_{j+1}+1\}$ when $n\in\{m_j,m_j+1\}$. Let

$$d_j=\max\{D_{\vec{\ell}}(M):\ell_1,\ell_2,\ell_3,M\in\{m_j,m_j+1\}\},$$
$$\eta_j=(m_j+1)\max\{\kappa(\vec{\ell}):\ell_i\in\{m_j,m_j+1\}\},$$

the halving in κ being $\ell_i=\lfloor\ell_i/2\rfloor+\lceil\ell_i/2\rceil$. Proposition 7.28 with $M'=\lfloor M/2\rfloor$, $M''=\lceil M/2\rceil$ gives, whenever $m_{j+1}\ge 3$,

$$d_j\le\exp\bigl(\eta_j+8d_{j+1}^2\bigr)-1. \tag{7.10}$$

Lemma 7.29 (contraction)**.** *If $d_{j+1}\le\frac{1}{10}$ and $\eta_j\le\frac{1}{70}$, then $d_j\le\frac{1}{10}$ and $d_j\le 0.84\,d_{j+1}+1.05\,\eta_j$.*

Proof. $y=\eta_j+8d_{j+1}^2\le\frac{1}{70}+\frac{2}{25}<0.0943$, and by convexity $e^y-1\le y\,(e^{0.0943}-1)/0.0943\le 1.049\,y$, with $e^{0.0943}-1=0.0989\le\frac{1}{10}$; and $8d_{j+1}^2\le 0.8\,d_{j+1}$. □

For a physical scale $s\ge 1$ put, at the coupling $\beta_{N,p}$,

$$D_0(N,p;s)=\sup\{D_{\vec{\ell}}(M):\ell_1,\ell_2,\ell_3,M\in[\lfloor Ns\rfloor,2Ns+1]\},$$
$$\eta(N,p;s)=\sup\{(m+1)\kappa(\vec{\ell}):\ell_i\in\{m,m+1\},\ m\ge Ns\};$$

$\eta(N,p;s)$ is nonincreasing in s, and $(m+1)\kappa(\vec{\ell})$ is the physical side of the torus times the physical Casimir defect $N\kappa$.

Proposition 7.30 ((Z) from one scale)**.** *Suppose that there are $s_0\ge 1$, p_0 and $N_0(p)$ such that for $p\ge p_0$ and $N\ge N_0(p)$*

(Z$_0$) *$D_0(N,p;s_0)\le\frac{1}{10}$, and*
(C) *$\eta(N,p;s_0)\le\frac{1}{70}$, and moreover $\lim_{s\to\infty}\limsup_{p\to\infty}\limsup_{N\to\infty}\eta(N,p;s)=0$.*

Then the family satisfies (Z).

Proof. Fix $p\ge p_0$ with $L_p\ge 2s_0$ and $N\ge\max\{N_0(p),3\}$, and let J be the least j with $m_j<2Ns_0$; then $J\ge 1$, $m_J=\lfloor m_{J-1}/2\rfloor\ge\lfloor Ns_0\rfloor\ge 3$, and $2^J>L_p/(2s_0+1)$. The sides and periods in d_J lie in $\{m_J,m_J+1\}\subset[\lfloor Ns_0\rfloor,2Ns_0+1]$, since $m_J<2Ns_0$ is an integer, so $d_J\le D_0(N,p;s_0)\le\frac{1}{10}$; for $j<J$, $m_j\ge 2Ns_0$, so $\eta_j\le\eta(N,p;s_0)\le\frac{1}{70}$ and (7.10) applies. By Lemma 7.29, downwards from $j=J-1$, $d_j\le\frac{1}{10}$ and $d_j\le 0.84d_{j+1}+1.05\eta_j$ for all $j<J$, so

$$\mathcal{Z}_{N,p}-1=D_{(n_0,n_0,n_0)}(n_0)\le d_0\le\frac{0.84^J}{10}+1.05\sum_{j<J}0.84^j\eta_j.$$

For $j<J$, $m_j\ge n_0 2^{-j}-1\ge n_0 2^{-j-1}$ because $n_0 2^{-j-1}\ge m_j/2\ge 1$, so $\eta_j\le\eta(N,p;L_p2^{-j-1})$. Given $\varepsilon>0$ choose, by (C), a scale s with $\limsup_p\limsup_N\eta(N,p;s)<\varepsilon$, then j_0 with $0.84^{j_0}<\varepsilon$. For p so large that $L_p2^{-j_0}\ge s$, splitting the sum at j_0 and using that η is nonincreasing in the scale,

$$\mathcal{Z}_{N,p}-1\le\frac{0.84^J}{10}+\frac{1.05}{0.16}\,\eta(N,p;s)+\frac{1.05}{70\cdot 0.16}\,0.84^{j_0}\le\frac{0.84^J}{10}+6.6\,\eta(N,p;s)+\frac{\varepsilon}{10}.$$

Take $\limsup_N$, then $\limsup_p$, using $J \to \infty$: the result is at most 6.7ε, and ε was arbitrary. □

The entropy bound (7.8) is thus needed at one physical scale only, through (Z_0); the passage from the scale s_0 to the scale $L_p \to \infty$, where the rings fail and where the count per unit of physical volume is decisive, is carried entirely by (C), a statement about ground-state energies and not about the excited spectrum. The constants $\frac{1}{10}$ and $\frac{1}{70}$ are not optimised. Condition (C) has an exact expression as a deficit per unit length of long tubes.

Lemma 7.31 (deficits of tubes)**.** *Let $b, c \geq 3$ and $e(b,c) = \lim_{M\to\infty} E_0(b,c,M)/M$, a limit in $(-\infty, 0]$ that exists because $M \mapsto E_0(b,c,M)$ is superadditive by Lemma 7.27 and nonpositive, and for $\ell \geq 3$ let $\zeta(\ell; b, c) = \ell\, e(b,c) - E_0(\ell, b, c)$, which is nonnegative. Write $\mathcal{Z}^{(1)}_{(b,c,M)}(\ell) = \mathrm{Tr}(\mathbb{R}^{(1)})^\ell$ for the relative partition function of the transfer matrix in the direction 1 on the transverse torus (b, c, M).*

(i) $\zeta(\ell; b, c) = \lim_{M\to\infty} M^{-1} \log \mathcal{Z}^{(1)}_{(b,c,M)}(\ell)$.

(ii) *For $\ell = \ell' + \ell''$ with $\ell', \ell'' \geq 3$,*

$$0 \leq E_0(\ell, b, c) - E_0(\ell', b, c) - E_0(\ell'', b, c) = \zeta(\ell'; b, c) + \zeta(\ell''; b, c) - \zeta(\ell; b, c)$$
$$\leq \lim_{M\to\infty} \frac{\log \mathcal{Z}^{(1)}_{(b,c,M)}(\ell') + \log \mathcal{Z}^{(1)}_{(b,c,M)}(\ell'')}{M}.$$

(iii) *$\kappa(\vec{\ell})$ is a sum of seven differences of the form* (ii)*, one in the direction 1, two in the direction 2 and four in the direction 3, so $\kappa(\vec{\ell}) \leq 14 \max \zeta(\ell^\pm; b, c)$, the maximum over the tubes whose cross-section (b, c) has sides in $\{\ell'_i, \ell''_i, \ell_i\}$ and whose length $\ell^\pm$ is one of the halves.*

Proof. (i) By the symmetry of Z, $Z(\ell, b, c, M) = \mathrm{Tr}(\mathbb{T}^{(1)})^\ell = e^{-\ell E_0(b,c,M)} \mathcal{Z}^{(1)}_{(b,c,M)}(\ell)$, while $M^{-1} \log Z(\ell, b, c, M) \to -E_0(\ell, b, c)$ by Lemma 7.3(i),(ii) for the time transfer matrix on the slice (ℓ, b, c); divide by M and let $M \to \infty$. (ii) The identity is (i) three times. The ratio $\mathcal{Z}^{(1)}(\ell')\mathcal{Z}^{(1)}(\ell'')/\mathcal{Z}^{(1)}(\ell) = (\sum_i r_i^{\ell'})(\sum_i r_i^{\ell''})/\sum_i r_i^{\ell}$, with $r_0 = 1$, lies between 1 and $\mathcal{Z}^{(1)}(\ell')\mathcal{Z}^{(1)}(\ell'')$. (iii) Telescope $E_0(\ell_1, \ell_2, \ell_3) - \sum_{\mathcal{S}} E_0$ by halving first ℓ_1, then ℓ_2 in both halves, then ℓ_3 in the four quarters, and bound each of the seven terms by (ii), dropping $-\zeta(\ell; b, c) \leq 0$. □

So (C) follows from a bound on the deficit ζ of a tube of physical cross-section $s \times s$ and length s, per unit of physical length, that is $o(1/s)$: by (i), ζ is the growth rate of the relative partition function of a long tube at inverse temperature ℓ, an entropy of the tube per unit of its length. For a theory with a mass gap Δ one expects the Casimir defect to be of order $e^{-\Delta s}$, far below what (C) asks; what the rings show is that (Z_0) is where they fail, since their κ vanishes identically while $D_{\vec{\ell}}(M) = (1 + e^{-mM})^{\ell_1\ell_2\ell_3} - 1$ exceeds $\frac{1}{10}$ at every physical scale. The one-scale condition (Z_0) in turn follows from a gap and a pressure bound at that scale.

Proposition 7.32 ((Z_0) from a gap and a pressure bound)**.** *Suppose that for some $s_0 \geq 1$, $\Delta > 0$, $c \geq 0$ and all large p and N, on every spatial torus with sides in $[\lfloor Ns_0 \rfloor, 2Ns_0 + 1]$ at the coupling $\beta_{N,p}$:* (a) *every eigenvalue $\lambda \neq \lambda_0$ of the time transfer matrix satisfies $|\lambda| \leq e^{-\Delta/N}\lambda_0$, and* (b) $\log \mathcal{Z}_{\vec{\ell}}(\lfloor Ns_0/2 \rfloor) \leq c$. *If $s_0 > \frac{2}{\Delta}(c + \log 10)$ then (Z_0) holds at the scale s_0 for all large N.*

Proof. Let $M \in [\lfloor Ns_0 \rfloor, 2Ns_0 + 1]$ and $M_1 = \lfloor Ns_0/2 \rfloor$, so $M - M_1 \geq \lfloor Ns_0 \rfloor - \lfloor Ns_0/2 \rfloor \geq (Ns_0 - 1)/2$. By Lemma 7.12 and (a), $D_{\vec{\ell}}(M) \leq e^{-\Delta(M-M_1)/N} D_{\vec{\ell}}(M_1)$, and $D_{\vec{\ell}}(M_1) \leq e^c - 1 < e^c$ by (b); so $D_{\vec{\ell}}(M) \leq e^{c - \Delta s_0/2 + \Delta/(2N)}$, which is at most $\frac{1}{10}$ once $N \geq \Delta/(\Delta s_0 - 2c - 2\log 10)$. □

Condition (b) is a bound on the physical pressure of the box of side s_0 at the temperature $2/s_0$, uniform in the mesh, and (a) is the gap on the tori of that side; hypothesis (D_ℓ) gives (a) on the cylinder of side NL_p only, so on the smaller tori it is a separate assumption unless the cube condition is assumed there too. Neither (Z_0) nor (C) is proved here for Yang–Mills theory at weak coupling: by Lemma 7.31 the second is a bound on the low-lying state count of long tubes per unit length, and the first is the entropy bound at one scale.

Positivity in the four directions turns (Z), (Z_0) and condition (b) into statements about partition functions of finite tori alone, with no transfer matrix, vacuum energy or infinite cylinder left in them.

Lemma 7.33 (doubling)**.** *Let $\mathbb{T} \geq 0$ be a self-adjoint Hilbert–Schmidt operator with a simple largest eigenvalue $\lambda_0 > 0$, let $\mathbb{R} = \mathbb{T}/\lambda_0$ have the eigenvalues $1 = r_0 > r_1 \geq r_2 \geq \cdots \geq 0$, and for $M \geq 2$ let $Z(M) = \operatorname{Tr} \mathbb{T}^M$, $\mathcal{Z}(M) = \operatorname{Tr} \mathbb{R}^M$ and $\psi(M) = 2\log Z(M) - \log Z(2M)$. Then*

$$\log \mathcal{Z}(M) \leq \psi(M) \leq 2\log \mathcal{Z}(M), \qquad \log \mathcal{Z}(M) = \sum_{k\geq 0} 2^{-k-1} \psi(2^k M),$$

and neither constant in the first display can be improved.

Proof. From $Z(M) = \lambda_0^M \mathcal{Z}(M)$, $Z(M)^2/Z(2M) = \mathcal{Z}(M)^2/\mathcal{Z}(2M)$. The sums $\sum_i r_i^M$ are finite for $M \geq 2$, and since $0 \leq r_i < 1$ for $i \geq 1$, $1 \leq \mathcal{Z}(2M) = 1 + \sum_{i\geq 1} r_i^{2M} \leq 1 + \sum_{i\geq 1} r_i^M = \mathcal{Z}(M)$; dividing gives the two inequalities. For the series put $F_k = \log Z(2^k M)$. Then $\sum_{k<K} 2^{-k-1}(2F_k - F_{k+1}) = F_0 - 2^{-K} F_K$, and $2^{-K} F_K = M \log \lambda_0 + 2^{-K} \log \mathcal{Z}(2^K M)$ with $0 \leq \log \mathcal{Z}(2^K M) \leq \log \mathcal{Z}(M)$, so the partial sums tend to $F_0 - M \log \lambda_0 = \log \mathcal{Z}(M)$. For the constants take $\mathbb{R} = \operatorname{diag}(1, r, \ldots, r)$ with q entries equal to r, so that $\psi/\log \mathcal{Z} = 2 - \log(1 + qr^{2M})/\log(1 + qr^M)$: at fixed q and $r \uparrow 1$ the ratio tends to 1, and at fixed M, $q = \lceil r^{-3M/2} \rceil$ and $r \downarrow 0$ one has $qr^M \to \infty$, $qr^{2M} \to 0$ and the ratio tends to 2. □

For a spatial torus $\vec{\ell}$ and $M \geq 2$ put $\psi_{\vec{\ell}}(M) = 2\log Z(\vec{\ell}, M) - \log Z(\vec{\ell}, 2M)$; for a torus $n = (n_1, n_2, n_3, n_4)$ with all $n_\mu \geq 3$ let $|n| = n_1 n_2 n_3 n_4$, $f(n) = |n|^{-1} \log Z(n)$ its free energy per site, and

$$\delta(n) = 16 \log Z(n) - \log Z(2n) = 16|n|\big(f(n) - f(2n)\big).$$

Both are unchanged when $Z(n)$ is multiplied by $e^{a|n|}$ for a constant a, a vacuum-energy counterterm, since $|(\vec{\ell}, 2M)| = 2|(\vec{\ell}, M)|$ and $|2n| = 16|n|$. By the symmetry of Z, the relative partition function of the transfer matrix in the direction μ on the transverse torus t is $\mathcal{Z}_t(m) = Z(n) e^{mE_0(t)}$, n being the torus with side m in the direction μ and t transverse to it; it is the quantity written $\mathcal{Z}^{(1)}$ in Lemma 7.31.

Proposition 7.34 ((Z) and (Z_0) as doubling defects)**.** *Let $\beta \geq 0$.*

(i) $\log \mathcal{Z}_{\vec{\ell}}(M) \leq \psi_{\vec{\ell}}(M) \leq 2\log \mathcal{Z}_{\vec{\ell}}(M)$ *for $M \geq 2$. Hence* (Z) *holds if and only if*

$$\lim_{p\to\infty} \limsup_{N\to\infty} \big(2\log Z(\Lambda_{NL_p}) - \log Z(NL_p, NL_p, NL_p, 2NL_p)\big) = 0;$$

(Z_0) *holds at the scale s_0 whenever $\psi_{\vec{\ell}}(M) \leq \log \frac{11}{10}$ for all $\ell_1, \ell_2, \ell_3, M \in [\lfloor Ns_0 \rfloor, 2Ns_0 + 1]$; and condition* (b) *of Proposition 7.32 holds with the constant c whenever $\psi_{\vec{\ell}}(\lfloor Ns_0/2 \rfloor) \leq c$ on those tori and $Ns_0 \geq 4$.*

(ii) *For every torus n with all $n_\mu \geq 3$,*

$$8\log \mathcal{Z}_{(n_1,n_2,n_3)}(n_4) \leq \delta(n) \leq 16\log \mathcal{Z}_{(n_1,n_2,n_3)}(n_4) + 8\log \mathcal{Z}_{(n_1,n_2,2n_4)}(n_3) + 4\log \mathcal{Z}_{(n_1,2n_3,2n_4)}(n_2) + 2\log \mathcal{Z}_{(2n_2,2n_3,2n_4)}(n_1).$$

Hence (Z_0) *holds at s_0 if $|n|\,(f(n) - f(2n)) \leq \frac{1}{2} \log \frac{11}{10}$ for all n with $n_\mu \in [\lfloor Ns_0 \rfloor, 2Ns_0 + 1]$, and* (Z) *holds if*

$$\lim_{p\to\infty} \limsup_{N\to\infty} (NL_p)^4 \big(f(\Lambda_{NL_p}) - f(\Lambda_{2NL_p})\big) = 0.$$

Proof. (i) is Lemma 7.33 for $\mathbb{T}_{\vec\ell}$: $1 + D_{\vec\ell}(M) = \mathcal{Z}_{\vec\ell}(M) \le e^{\psi_{\vec\ell}(M)} \le \frac{11}{10}$ on the range of (Z$_0$), $\log \mathcal{Z}_{\vec\ell}(\lfloor Ns_0/2\rfloor) \le \psi_{\vec\ell}(\lfloor Ns_0/2\rfloor) \le c$ for (b), and for (Z) the two inequalities at $\vec\ell = (n_0, n_0, n_0)$, $M = n_0 = NL_p$ show that $\log \mathcal{Z}_{N,p}$ and ψ tend to 0 together in the iterated limits, $\mathcal{Z}_{N,p} - 1 \to 0$ being the same as $\log \mathcal{Z}_{N,p} \to 0$. (ii) Let $n^{(0)} = n$, $n^{(1)} = (n_1, n_2, n_3, 2n_4)$, $n^{(2)} = (n_1, n_2, 2n_3, 2n_4)$, $n^{(3)} = (n_1, 2n_2, 2n_3, 2n_4)$ and $n^{(4)} = 2n$, and $F = \log Z$. Expanding,

$$16F(n) - F(2n) = \sum_{j=0}^{3} 2^{3-j}\big(2F(n^{(j)}) - F(n^{(j+1)})\big),$$

and the j-th bracket doubles the side in the direction $4 - j$ with the transverse torus t_j equal to (n_1, n_2, n_3), $(n_1, n_2, 2n_4)$, $(n_1, 2n_3, 2n_4)$, $(2n_2, 2n_3, 2n_4)$ in turn; by Lemma 7.33 for the transfer matrix in that direction it lies in $[\log \mathcal{Z}_{t_j}(n_{4-j}), 2\log \mathcal{Z}_{t_j}(n_{4-j})]$. All brackets being nonnegative, the lower bound keeps $j = 0$ alone and the upper bound sums the four. For $n = (\vec\ell, M)$ in the range of (Z$_0$), $\log \mathcal{Z}_{\vec\ell}(M) \le \delta(n)/8 = 2|n|(f(n) - f(2n)) \le \log \frac{11}{10}$; at $n = \Lambda_{n_0}$, $\log \mathcal{Z}_{N,p} \le 2n_0^4(f(n_0) - f(2n_0))$. □

In (i) the upper bound $Z(\vec\ell, 2M) \le Z(\vec\ell, M)^2$ is positivity (Lemma 7.27) and the lower bound is $\mathcal{Z}_{\vec\ell}(2M) \ge 1$: positivity gives no lower bound on $Z(\vec\ell, 2M)/Z(\vec\ell, M)^2$ that does not contain $\mathcal{Z}$ itself, which is why halving descends in scale only and Proposition 7.28 needs κ, that is (C), to ascend. In (ii) the thermal correction at one scale is bounded by free energies per site of two finite tori, n and $2n$, but by their *difference* at the precision $|n|^{-1} = (Ns_0)^{-4}$ per site, an absolute precision $O(1)$ in the total: the extensive part and every counterterm cancel exactly, and what is left is the finite-size part of the free energy. The gap and the limit of the free energy per site do not control it; the example is the model of Proposition 7.17 made symmetric in the four directions.

Proposition 7.35 (symmetrised rings)**.** *Let $\Delta > 0$, $r = e^{-\Delta/N}$ and $K = \operatorname{artanh} r$. On the torus $\Lambda(n)$ place at every site x and for every direction μ a spin $\sigma_x^\mu \in \{\pm 1\}$, with the uniform a priori measure and the weight $\prod_{x,\mu} e^{K\sigma_x^\mu \sigma_{x+e_\mu}^\mu}$, an Ising ring on every lattice line. Then:*

(i) *$Z(n) = (\cosh K)^{4|n|} \prod_\mu (1+r^{n_\mu})^{|n|/n_\mu}$, so that $f(n) = 4\log\cosh K + \sum_\mu n_\mu^{-1}\log(1+r^{n_\mu})$ converges to $4\log\cosh K$ as $\min_\mu n_\mu \to \infty$, for every N;*

(ii) *in every direction the transfer matrix between consecutive slices is self-adjoint, nonnegative and of finite rank, with a simple largest eigenvalue λ_0 and every other eigenvalue at most $e^{-\Delta/N}\lambda_0$: condition* (a) *of Proposition 7.32 holds with the gap Δ on every torus and for every N;*

(iii) *$D_{\vec\ell}(M) = (1 + r^M)^{\ell_1\ell_2\ell_3} - 1$, so that for every $s \ge 1$*

$$D_0(N, p; s) \ge \lfloor Ns\rfloor^3 e^{-\Delta(2s+1/N)} \quad \text{and} \quad |n|(f(n) - f(2n)) \ge \tfrac{1}{2}\lfloor Ns\rfloor^3 \log\big(1 + e^{-\Delta(2s+1/N)}\big)$$

for n with sides in $[\lfloor Ns\rfloor, 2Ns + 1]$; both tend to infinity with N.

Proof. (i) A periodic ring of length m with the uniform a priori measure has partition function $2^{-m}\operatorname{Tr} A^m$ with $A = \left(\begin{smallmatrix} e^K & e^{-K} \\ e^{-K} & e^K \end{smallmatrix}\right)$, of eigenvalues $2\cosh K$ and $2\sinh K$, so it equals $\cosh^m K\,(1 + \tanh^m K) = \cosh^m K\,(1 + r^m)$; there are $|n|/n_\mu$ rings of length n_μ in the direction μ, and the species are independent. (ii) Take the direction 4, the others by symmetry, and a transverse torus t. The variables of a slice are the σ_x^ν, $x \in t$, all ν; the rings in the direction 4 couple consecutive slices through $\prod_{x\in t} \frac{1}{2}A(\sigma_x^4, \sigma_x'^4)$, and the spins σ_x^i, $i \le 3$, interact inside their slice only, with a weight $w(\sigma^{\mathrm{sp}}) > 0$, which is split as $w^{1/2}(\sigma^{\mathrm{sp}})w^{1/2}(\sigma'^{\mathrm{sp}})$ between the two kernels. The transfer matrix is therefore $(\frac{1}{2}A)^{\otimes|t|} \otimes |w^{1/2}\rangle\langle w^{1/2}|$, a nonnegative operator of finite rank whose nonzero eigenvalues are $\|w^{1/2}\|^2(\cosh K)^{|t|}r^k$ with multiplicity $\binom{|t|}{k}$, $0 \le k \le |t|$: the largest, $k = 0$, is simple and the others are at most $r\lambda_0$. (iii) By (ii), $\mathcal{Z}_{\vec\ell}(M) = \sum_k \binom{|\vec\ell|}{k} r^{kM} = (1 + r^M)^{|\vec\ell|}$. With $\ell_i = \lfloor Ns\rfloor$

and $M = \lfloor 2Ns+1 \rfloor$, $r^M \geq e^{-\Delta(2s+1/N)}$ and $(1+x)^k - 1 \geq kx$. For the second bound, by (i), $|n|(f(n) - f(2n)) = \sum_\mu (|n|/n_\mu)\big(\log(1+r^{n_\mu}) - \tfrac{1}{2}\log(1+r^{2n_\mu})\big)$, each term is at least $\tfrac{1}{2}(|n|/n_\mu)\log(1+r^{n_\mu})$ because $1 + r^{2n_\mu} \leq 1 + r^{n_\mu}$, and the term $\mu = 4$ with $n_4 \leq 2Ns+1$ and $|n|/n_4 \geq \lfloor Ns \rfloor^3$ gives the claim. □

So a family may have nonnegative transfer matrices in all four directions, a uniform physical gap on every torus, and a free energy per site that converges, and still violate (Z_0) at every physical scale by a factor of order N^3: no bound on $D_0(N, p; s_0)$ in terms of the gap and the infinite-volume free energy per site exists, and the hypothesis of Proposition 7.34(ii), the finite-size difference of the free energies of n and $2n$, is the quantity that separates the gauge theory, if (Z_0) holds for it, from this model. Condition (C), in turn, follows from a finite-size bound on the vacuum energies of the spatial tori with the extensive part subtracted. Write $V(\vec{\ell}) = \ell_1\ell_2\ell_3$ and $\sigma_*(\vec{\ell}) = \sum_{i=1}^3 e^{-m_*\ell_i/N}$.

Proposition 7.36 ((C) from a finite-size bound)**.** *Let $m_* > 0$, $K \geq 0$, $s_* \geq 1$, and suppose that at the coupling $\beta_{N,p}$ there is a number $f_{N,p}$ such that*

$$\big|E_0(\vec{\ell}) + f_{N,p}V(\vec{\ell})\big| \leq KN^{-4}V(\vec{\ell})\,\sigma_*(\vec{\ell}) \tag{7.11}$$

for every spatial torus with all $\ell_i \geq Ns_*$. Then for $N \geq \max\{m_*, 3/s_*\}$ and $s \geq \max\{2s_* + 1, 8/m_*\}$,

$$0 \leq \eta(N, p; s) \leq 6eK\, s^4 e^{-m_* s/2},$$

so that, if (7.11) *holds with K and m_* independent of $p \geq p_0$ and $N \geq N_0(p)$,* (C) *holds at every $s_0 \geq \max\{2s_* + 1, 8/m_*\}$ with $6eKs_0^4 e^{-m_* s_0/2} \leq \frac{1}{70}$. If the four-dimensional form*

$$\big|\log Z(\vec{\ell}, M) - f_{N,p}V(\vec{\ell})M\big| \leq KN^{-4}V(\vec{\ell})M\big(\sigma_*(\vec{\ell}) + e^{-m_*M/N}\big) \tag{7.12}$$

holds for all $\vec{\ell}$ with $\ell_i \geq Ns_*$ and all $M \geq Ns_*$, then $0 \leq \log \mathcal{Z}_{\vec{\ell}}(M) \leq KN^{-4}V(\vec{\ell})M\big(2\sigma_*(\vec{\ell}) + e^{-m_*M/N}\big)$ there; in particular $\log \mathcal{Z}_{N,p} \leq 7KL_p^4 e^{-m_* L_p}$ once $L_p \geq s_*$, and (7.12) *with constants independent of N and p gives* (Z) *directly.*

Proof. Put $w(\vec{\ell}) = E_0(\vec{\ell}) + f_{N,p}V(\vec{\ell})$. Since $V(\vec{\ell}) = \prod_i(\ell_i' + \ell_i'') = \sum_{s\in\mathcal{S}} V(s)$ exactly, the terms in $f_{N,p}$ cancel in κ: $\kappa(\vec{\ell}) = w(\vec{\ell}) - \sum_{s\in\mathcal{S}} w(s)$, so $|\kappa(\vec{\ell})| \leq KN^{-4}\big(V(\vec{\ell})\sigma_*(\vec{\ell}) + \sum_s V(s)\sigma_*(s)\big)$. Let $\ell_i \in \{m, m+1\}$ with $m \geq Ns$. The halves have sides at least $\lfloor m/2 \rfloor \geq (m-1)/2 \geq (Ns-1)/2 \geq Ns_*$ because $s \geq 2s_* + 1$, and $Ns_* \geq 3$; so (7.11) applies to $\vec{\ell}$ and to every $s \in \mathcal{S}$, with $\sigma_* \leq 3e^{-m_*(m-1)/(2N)}$, and $\sum_s V(s) = V(\vec{\ell}) \leq (m+1)^3$. Hence

$$(m+1)\kappa(\vec{\ell}) \leq 6K\Big(\frac{m+1}{N}\Big)^4 e^{-m_*(m-1)/(2N)} = 6Ke^{m_*/N}x^4 e^{-m_* x/2}, \qquad x = \frac{m+1}{N} \geq s.$$

For $N \geq m_*$, $e^{m_*/N} \leq e$; and $x \mapsto x^4 e^{-m_* x/2}$ decreases on $x \geq 8/m_*$, since for $x \geq y \geq 8/m_*$ one has $e^{m_*(x-y)/2} \geq e^{4(x-y)/y} \geq (1 + (x-y)/y)^4 = (x/y)^4$. So the supremum over $m \geq Ns$ is at most $6eKs^4 e^{-m_* s/2}$; nonnegativity is Proposition 7.28. For the last part, $\log \mathcal{Z}_{\vec{\ell}}(M) = \big(\log Z(\vec{\ell}, M) - f_{N,p}V(\vec{\ell})M\big) + Mw(\vec{\ell})$. Dividing (7.12) by M and letting $M \to \infty$, with $M^{-1}\log Z(\vec{\ell}, M) \to -E_0(\vec{\ell})$ by Lemma 7.3(i),(ii), gives (7.11), so $|Mw(\vec{\ell})| \leq KN^{-4}V(\vec{\ell})M\sigma_*(\vec{\ell})$, and (7.12) bounds the first bracket. On the symmetric torus $V(\vec{\ell})M = n_0^4$, $n_0/N = L_p$ and $2\sigma_* + e^{-m_*M/N} = 7e^{-m_* L_p}$. □

Bound (7.11) says that the vacuum energy of a spatial torus is extensive up to a term exponentially small in its physical sides, with a rate m_* and a prefactor K in physical units; a bound of the same shape without the decay, $|E_0(\vec{\ell}) + f_{N,p}V(\vec{\ell})| \leq KN^{-4}V(\vec{\ell})$, gives only $(m+1)\kappa(\vec{\ell}) \leq 2Kx^4$, which satisfies (C) at no scale. The decay in the sides is the mass; so (C), like (Z_0) and like the slab hypothesis, is a statement at a fixed physical scale about a theory which is not free there, and the finite-size bound (7.11) is the form in which a

massive expansion of the theory at that scale would deliver it. None of the three is proved here for Yang–Mills theory in four dimensions.

At strong coupling the bound (7.12) holds on every torus, with the rate $4d$ per lattice spacing of Proposition 7.25, so that every hypothesis of Proposition 7.30 is satisfied there and the one-scale reduction closes on itself in that range. Write $\Lambda(n) = \Lambda(n_1, n_2, n_3, n_4)$, call a plaquette spanned by the directions μ and ν a *μ-plaquette* with base μ-coordinate the μ-coordinate of its base point, and for a plaquette set S let $S_t^{(\mu)}$ be its μ-plaquettes of base μ-coordinate t; *S wraps in μ* if $S_t^{(\mu)} \neq \emptyset$ for every $t \in \mathbb{Z}/n_\mu\mathbb{Z}$.

Proposition 7.37 (the finite-size bound at strong coupling)**.** *Let G, ϵ', β and d be as in Proposition 7.25, with $24e^2\epsilon'(\beta) \le 1$ and $e^{-d} = 24e^2\epsilon'(\beta)$. There is a number $f(\beta)$ such that for every torus with all $n_\mu \ge 4$*

$$\big|\log Z(\Lambda(n)) - f(\beta)\,|n|\big| \le 3|n| \sum_{\mu=1}^{4} \frac{e^{-4dn_\mu}}{n_\mu} \le \frac{3}{4}\,|n| \sum_{\mu=1}^{4} e^{-4dn_\mu}.$$

Consequently $|E_0(\vec{\ell}) + f(\beta)V(\vec{\ell})| \le \frac{3}{4}V(\vec{\ell})\sum_i e^{-4d\ell_i}$ for all $\ell_i \ge 4$; for $\ell_i \in \{m, m+1\}$ with $m \ge 8$, $0 \le (m+1)\kappa(\vec{\ell}) \le \frac{9}{2}(m+1)^4 e^{-2d(m-1)}$; and $0 \le \log \mathcal{Z}_{\vec{\ell}}(M) \le \frac{21}{4}V(\vec{\ell})M\, e^{-4d\min\{\ell_1,\ell_2,\ell_3,M\}}$ for all $\ell_i, M \ge 4$. Hence along every family with $\sup_{N,p}\beta_{N,p} \le \beta_0$ and $24e^2\epsilon'(\beta_0) < 1$, both clauses of (C) *and* (Z_0) *at every scale hold, with $\eta(N,p;s)$ and $D_0(N,p;s_0)$ tending to zero as $N \to \infty$ uniformly in p.*

Proof. The proofs of Lemma 7.21, of Lemma 7.24 and of the Kotecký–Preiss condition in Proposition 7.25 use only the local incidences of the lattice and sides at least 4, so they hold verbatim on $\Lambda(n)$: with W the weights of the closed connected supports,

$$\log Z(\Lambda(n)) = 6|n|\log c_0 + \sum_{S \subset P(\Lambda(n))} W(S), \qquad \sum_{S \ni p} |W(S)| e^{d|S|} \le 1$$

for every plaquette p; the same argument in every finite box of $\mathbb{Z}^4$ gives weights $W(\tilde{S})$ independent of the box, and by monotone convergence the same bound for every plaquette $\tilde{p}$ of $\mathbb{Z}^4$. In the direction μ the proof of Lemma 7.24(iv) gives $|S_t^{(\mu)}| \ge 4$ whenever S is closed and $S_t^{(\mu)} \neq \emptyset$, on $\Lambda(n)$ and on $\mathbb{Z}^4$: the μ-links from the coordinate t to $t+1$ lie in μ-plaquettes of base μ-coordinate t only, each such plaquette contains two of them, so by closedness the graph on the transverse sites whose edges are the plaquettes of $S_t^{(\mu)}$ has minimal degree at least 2 and contains a cycle, of length at least 4, the girth of $\mathbb{Z}^3$ and of a three-torus with sides at least 4. So a support wrapping in μ has $|S| \ge 4n_\mu$, and

$$\sum_{S \text{ wrapping}} |W(S)| \le \sum_{\mu} \sum_{p} \sum_{S \ni p,\ |S| \ge 4n_\mu} \frac{|W(S)|}{|S|} \le 6|n| \sum_{\mu} \frac{e^{-4dn_\mu}}{4n_\mu}.$$

Let S be a connected support wrapping in no direction, choose t_μ^* with $S_{t_\mu^*}^{(\mu)} = \emptyset$ for each μ, let $B = \prod_\mu \{t_\mu^* + 1, \dots, t_\mu^* + n_\mu\} \subset \mathbb{Z}^4$, on whose sites the covering $\pi : \mathbb{Z}^4 \to \Lambda(n)$ is a bijection, and lift each plaquette of S to the plaquette with the lifted base point. A plaquette of S spanned by μ and ν with base x has base μ-coordinate $x_\mu \not\equiv t_\mu^*$ and base ν-coordinate $x_\nu \not\equiv t_\nu^*$, so all its corners lift into B. Let e be a link of S in the direction ν with tail x; every plaquette of S through e is a ν-plaquette of base ν-coordinate x_ν, so $x_\nu \not\equiv t_\nu^*$ and the lift $\tilde{e}$ of e from the lift $\tilde{x}$ of x has both ends in B, and such a plaquette has base x or $x - e_\mu$ for some $\mu \neq \nu$, in the second case with base μ-coordinate $x_\mu - 1 \not\equiv t_\mu^*$, so that the lift of its base is $\tilde{x} - e_\mu$; hence the plaquettes of S through e lift to plaquettes through $\tilde{e}$. On plaquettes and links with corners in B the covering is injective. So the lift $\tilde{S}$ is connected and closed, $\pi : \tilde{S} \to S$ preserves in both directions which pairs share a link and which subsets are polymers, and the activities, Haar integrals over the links of the subset, correspond; thus $W(\tilde{S}) = W(S)$ as in Lemma 7.22, and the sites of $\tilde{S}$ have μ-extent at most

$n_\mu - 1$. A connected set does not cross an empty layer, so another choice of the t^*_μ gives a translate of $\tilde S$ by a period, and for a fixed lift $\tilde p$ of $p \in S$ exactly one lift of S contains $\tilde p$. Conversely a finite connected closed $\tilde S \subset P(\mathbb{Z}^4)$ whose sites have μ-extent at most $n_\mu - 1$ for every μ lies in a translate of such a box B, so it projects bijectively onto a connected closed S with the same structure, and S wraps in no direction because the μ-plaquettes of $\tilde S$ have base μ-coordinates in an interval of length at most $n_\mu - 1$. For $\tilde S$ connected the base μ-coordinates of its μ-plaquettes form an interval $[u, v]$, since a plaquette at the coordinate t not spanned by μ shares links with μ-plaquettes of base μ-coordinates $t - 1$ and t only, and all sites of $\tilde S$ have μ-coordinates in $[u, v + 1]$, or all the same μ-coordinate if $\tilde S$ has no μ-plaquette; so a μ-extent at least n_μ forces at least n_μ nonempty layers $\tilde S^{(\mu)}_t$ and $|\tilde S| \geq 4n_\mu$. Writing the non-wrapping part of the sum as $\sum_p \sum_{S \ni p} W(S)/|S|$ and lifting through a fixed $\tilde p$ over p, it equals $\sum_p (a(\tilde p) - T(\tilde p))$, where $a(\tilde p) = \sum_{\tilde S \ni \tilde p} W(\tilde S)/|\tilde S|$ over all finite connected closed $\tilde S \subset P(\mathbb{Z}^4)$, absolutely convergent, and $T(\tilde p)$ is the part of that sum over the $\tilde S$ with some μ-extent at least n_μ, so that $|T(\tilde p)| \leq \sum_\mu e^{-4dn_\mu}/(4n_\mu)$. By the translation invariance of W on $\mathbb{Z}^4$, $a(\tilde p) = a_o$ depends on the orientation o of p only, and each of the six orientations has $|n|$ plaquettes; with $f(\beta) = 6 \log c_0 + \sum_o a_o$ the two error terms add up to $12|n| \sum_\mu e^{-4dn_\mu}/(4n_\mu)$, and $e^{-4dn_\mu}/n_\mu \leq e^{-4dn_\mu}/4$.

For the consequences, divide the bound on $Z(\vec\ell, M)$ by M and let $M \to \infty$, as in the proof of Proposition 7.36: $|E_0(\vec\ell) + f(\beta)V(\vec\ell)| \leq \frac34 V(\vec\ell) \sum_i e^{-4d\ell_i}$. For $\ell_i \in \{m, m+1\}$, $m \geq 8$, the sides of $\vec\ell$ and of its halves are at least $h = \lfloor m/2 \rfloor \geq 4$, so with $V(\vec\ell) = \sum_s V(s)$, $\kappa(\vec\ell) \leq \frac34 \cdot 2V(\vec\ell) \cdot 3e^{-4dh} \leq \frac92 (m+1)^3 e^{-2d(m-1)}$. For $\log \mathcal{Z}_{\vec\ell}(M) = (\log Z(\vec\ell, M) - f(\beta)V(\vec\ell)M) + M(E_0(\vec\ell) + f(\beta)V(\vec\ell))$ the two bounds give $\frac34 V(\vec\ell) M(2 \cdot 3 + 1) e^{-4d \min\{\ell_i, M\}}$. Along the family, ϵ' increases with β, so $d \geq d_0 = -\log(24e^2\epsilon'(\beta_0)) > 0$; then $\eta(N, p; s) \leq \sup_{m \geq Ns} \frac92 (m + 1)^4 e^{-2d_0(m-1)}$, and on the range of (Z$_0$), $\log(1 + D_0(N, p; s_0)) \leq \frac{21}{4}(2Ns_0 + 1)^4 e^{-4d_0 \lfloor Ns_0 \rfloor}$; both tend to zero as $N \to \infty$, uniformly in p. □

So in the range of Proposition 7.25 the hypotheses of Proposition 7.30 hold and (Z) follows from it a second time, with the rate $4d$ per lattice spacing, a physical mass $4dN \to \infty$: this is the situation that Proposition 7.26 excludes along a tuning with a particle, and what (7.11) asks there is the same bound with a rate per unit of physical length. In two dimensions the bound is empty: for every compact G the transfer matrix of a circle of L sites has $\lambda_0 = b_0^L$ by Proposition H.11(i), so $E_0(L) = -L \log b_0$ exactly, the Casimir defect κ vanishes identically, and (Z) is a one-scale statement there, which is why its proof along the tuning in Appendix H needs no halving.

The gap against the Casimir defect. Proposition 7.30 carries the passage from the scale s_0 to the scale L_p by the Casimir coefficients alone, and asks them to be small, $\eta \leq \frac{1}{70}$, and to tend to zero. Both demands disappear once the spectral gap is available on the tori of the halving chain: the gap gains a factor $e^{-\Delta/N}$ for each unit of time, the cut costs the factor $e^{M_1\kappa}$ over half the period, and the recursion contracts as soon as the physical Casimir energy $N\kappa$ lies below the gap. The Casimir coefficient is moreover bounded by thermal data at the same scale, which closes the circle with Proposition 7.28 and shows exactly what positivity alone cannot do. Throughout, n_0, m_j, d_j, η_j and J are as before Proposition 7.30, $L_j = \{m_j, m_j + 1\}$, so that $L_j \cap L_{j+1} = \emptyset$ when $m_{j+1} \geq 2$, and $m_j + 1 \leq 2m_{j+1} + 2$. A *mixed torus of level j* is a spatial torus $t = (t_1, t_2, t_3)$ with every $t_i \in L_j \cup L_{j+1}$; $\mathcal{S}_j$ denotes their set, and $L_j^3 \subset \mathcal{S}_j$. The *level-$j$ halving* of t cuts every side $t_i \in L_j$ into $\lfloor t_i/2 \rfloor + \lceil t_i/2 \rceil$ and leaves the sides in L_{j+1} uncut; its pieces, at most eight, form the set $\mathcal{P}(t)$ and have all their sides in L_{j+1}. The partial defect is $\kappa_p(t) = E_0(t) - \sum_{s \in \mathcal{P}(t)} E_0(s)$, which is $\kappa(t)$ for $t \in L_j^3$, and we put

$$X_j = \max\{\log \mathcal{Z}_t(\tau) : t \in \mathcal{S}_j,\ \tau \in L_{j+1}\}, \qquad K_j = \max\{(m_j + 1)\kappa_p(t) : t \in \mathcal{S}_j\} \ \geq \eta_j.$$

When $m_{j+1} \geq 3$, Lemma 7.27 applied to the sides in L_j gives $Z(t,\tau) \leq \prod_{s\in\mathcal{P}(t)} Z(s,\tau)$ for every $\tau \geq 2$, hence, exactly as in the proof of Proposition 7.28, $\kappa_p(t) \geq 0$ and

$$\log \mathcal{Z}_t(\tau) \leq \tau\kappa_p(t) + \sum_{s\in\mathcal{P}(t)} \log \mathcal{Z}_s(\tau) \qquad (t \in \mathcal{S}_j,\ \tau \geq 2). \tag{7.13}$$

Since Z is symmetric in its four arguments, for a 4-torus n and a direction μ with transverse 3-torus u,

$$\log Z(n) = -n_\mu E_0(u) + \log \mathcal{Z}_u(n_\mu), \tag{7.14}$$

by the definition of $\mathcal{Z}_u$: the relative partition function in the direction μ is the one of u regarded as a spatial torus.

Lemma 7.38 (finite-time duality)**.** *Let $t \in \mathcal{S}_j$ with $m_{j+1} \geq 3$, and let $M \geq 3$.*

(i) *With $\Phi(t,M) = \sum_{s\in\mathcal{P}(t)} \log Z(s,M) - \log Z(t,M) \geq 0$,*

$$M\kappa_p(t) = \Phi(t,M) + \log \mathcal{Z}_t(M) - \sum_{s\in\mathcal{P}(t)} \log \mathcal{Z}_s(M).$$

(ii) *Cutting the sides of t that lie in L_j one direction after another, $\Phi(t,M)$ is a sum over the $2^k - 1 \leq 7$ single cuts, k the number of such sides, and each single cut, which halves a side $a+b \in L_j$ of the current torus into $a,b \in L_{j+1}$ and has the transverse torus u formed by its two other current sides and M, contributes at most $\log \mathcal{Z}_u(a) + \log \mathcal{Z}_u(b)$.*

(iii) *$m_{j+1}\kappa_p(t) \leq 15X_j$, and therefore*

$$K_j \leq 30\Big(1 + \frac{1}{m_{j+1}}\Big)X_j.$$

In particular $(m+1)\kappa(\vec{\ell}) \leq 30(1+1/m')X$ for every $\vec{\ell} \in \{m, m+1\}^3$ with $m' = \lfloor m/2 \rfloor \geq 3$, where X is the largest $\log \mathcal{Z}_t(\tau)$ over spatial tori t with sides in $\{m, m+1, m', m'+1\}$ and $\tau \in \{m', m'+1\}$.

Proof. (i) $\log Z(t,M) = -ME_0(t) + \log \mathcal{Z}_t(M)$, and the same for every $s \in \mathcal{P}(t)$; subtract and use the definition of κ_p. That $\Phi \geq 0$ is Lemma 7.27. (ii) The cutting of t proceeds in one direction, then in a second direction on both pieces, then in a third on all four, so $\Phi(t,M)$ is the sum of $1+2+4$ single-cut defects when $k=3$, and of fewer when $k<3$. By (7.14) a single cut of a torus n with $n_\mu = a+b$ has the defect $\log \mathcal{Z}_u(a) + \log \mathcal{Z}_u(b) - \log \mathcal{Z}_u(a+b) \leq \log \mathcal{Z}_u(a) + \log \mathcal{Z}_u(b)$, because $\mathcal{Z} \geq 1$. (iii) Take $M = m_{j+1} \in L_{j+1}$. Every transverse torus u in (ii) has its sides in $L_j \cup L_{j+1}$, so $u \in \mathcal{S}_j$, and $a,b \in L_{j+1}$: each of the at most fourteen terms is at most X_j, and $\log \mathcal{Z}_t(m_{j+1}) \leq X_j$ as well; the sum $\sum_s \log \mathcal{Z}_s(M) \geq 0$ in (i) is dropped. Finally $(m_j+1)/m_{j+1} \leq (2m_{j+1}+2)/m_{j+1}$. For the last claim apply this with $m_j = m$, so that $L_{j+1} = \{m', m'+1\}$. □

Lemma 7.38 is the converse of (7.13): the ground-state quantity κ is bounded by the thermal corrections at aspect ratio $\frac{1}{2}$ and at the same scale, with no limit $M \to \infty$. In the notation of Proposition 7.30, $\eta(N,p;s) \leq 30(1+\frac{2}{Ns-2})\sup_{m\geq Ns} X(m)$ whenever $\lfloor Ns/2 \rfloor \geq 3$, where $X(m)$ is the quantity of (iii), so that (C) follows from vacuum dominance at aspect ratio $\frac{1}{2}$ at all scales. The two bounds together read $K_j \leq 30.6X_j$ and $X_j \leq \frac{1}{2}K_j + 8X_{j+1}$, up to terms of order $1/m_{j+1}$: the product of the coefficients is $15 > 1$, so positivity alone does not make the recursion contract, and indeed for a massless field nothing decays. The spectral gap supplies the missing factor.

Theorem 7.39 ((Z) from one scale, a gap, and a Casimir energy below the gap)**.** *Fix (N,p) with $L_p \geq 2s_0$ and $\lfloor Ns_0 \rfloor \geq 3$, and let $\Delta > 0$ and $\theta \in (0,1]$ with $\theta\Delta s_0 \geq \log(10 \cdot 1.1^8) = 3.0651\ldots$. Suppose that*

(G) *for $0 \leq j < J$ and every $\vec{\ell} \in L_j^3$, every eigenvalue $\lambda \neq \lambda_0$ of $\mathbb{T}_{\vec{\ell}}$ satisfies $|\lambda| \leq e^{-\Delta/N}\lambda_0$;*

(Z_0) *$d_J \leq \frac{1}{10}$, which holds when $D_0(N,p;s_0) \leq \frac{1}{10}$;*

(K) *for* $0 \le j < J$ *and every* $\vec{\ell} \in L_j^3$, $(m_j+1)\kappa(\vec{\ell}) \le (1-\theta)\Delta m_j/N$.
Then $d_j \le 1.1^8 e^{-\theta\Delta m_j/(2N)} \le \frac{1}{10}$ *for* $0 \le j < J$, *and in particular*

$$\mathcal{Z}_{N,p} - 1 \le 2.144\, e^{-\theta\Delta L_p/2}.$$

If (G), (Z_0) *and* (K) *hold with fixed* s_0, Δ, θ *for all* $p \ge p_0$ *and* $N \ge N_0(p)$, *the family satisfies* (Z).

Proof. Downward induction on $j < J$, the hypothesis being $d_{j+1} \le \frac{1}{10}$, which for $j+1 = J$ is (Z_0). Let $\vec{\ell} \in L_j^3$ and $M \in L_j$, and put $M_1 = \lfloor M/2 \rfloor \in L_{j+1}$, so that $M_1 \ge m_{j+1} \ge m_J \ge 3$, $M - M_1 = \lceil M/2 \rceil \ge m_j/2$ and $M_1 \le (m_j+1)/2$. By (G) and Lemma 7.12, $D_{\vec{\ell}}(M) \le e^{-\Delta(M-M_1)/N} D_{\vec{\ell}}(M_1)$. By (7.13) with $t = \vec{\ell}$, whose eight halves have their sides in L_{j+1}, and the time $M_1 \in L_{j+1}$,

$$1 + D_{\vec{\ell}}(M_1) \le e^{M_1\kappa(\vec{\ell})} \prod_{s\in\mathcal{P}(\vec{\ell})} (1 + D_s(M_1)) \le e^{M_1\kappa(\vec{\ell})}(1+d_{j+1})^8 \le 1.1^8 e^{M_1\kappa(\vec{\ell})}.$$

Hence, by (K),

$$\begin{aligned} D_{\vec{\ell}}(M) &\le 1.1^8 \exp\Big(-\frac{\Delta m_j}{2N} + \frac{(m_j+1)\kappa(\vec{\ell})}{2}\Big) \\ &\le 1.1^8 \exp\Big(-\frac{\Delta m_j}{2N} + \frac{(1-\theta)\Delta m_j}{2N}\Big) = 1.1^8 e^{-\theta\Delta m_j/(2N)}. \end{aligned}$$

As $m_j \ge 2Ns_0$ for $j < J$, this is at most $1.1^8 e^{-\theta\Delta s_0} \le \frac{1}{10}$. At $j = 0$, $\mathcal{Z}_{N,p} - 1 = D_{(n_0,n_0,n_0)}(n_0) \le d_0$ and $m_0 = NL_p$; and $1.1^8 = 2.1436\ldots$. The last statement follows by letting $N \to \infty$, then $p \to \infty$. □

Neither the quadratic term of (7.10) nor Lemma 7.29 is used: the factor $(1+d_{j+1})^8$ need only stay bounded. Condition (K) is the exact threshold of the argument, the gain $e^{-\Delta m_j/(2N)}$ of the gap over half the period against the cost $e^{M_1\kappa}$ of the cut, and it allows $(m+1)\kappa$ to grow like $(1-\theta)\Delta s$ with the physical side s. Only (Z_0) on the tori of L_J is used, a subset of the window of Proposition 7.30.

Corollary 7.40. (i) *Suppose* (G) *and* (Z_0) *hold and* $\eta(N,p;2s_0) \le H$ *for some* $H \ge 0$, *with* $\Delta s_0 \ge \frac{H}{2} + 3.0651$. *Then* (K) *holds with* $\theta = 1 - H/(2\Delta s_0)$, *and* $\mathcal{Z}_{N,p} - 1 \le 2.144\, e^{-(\Delta s_0 - H/2)L_p/(2s_0)}$. *With the first clause of* (C), $H = \frac{1}{70}$, *the condition is* $\Delta s_0 \ge 3.0723$, *and the limit clause of* (C) *is not needed.*

(ii) *Let a continuum family satisfy* (W) *and* (D_ℓ), *the latter for all* $N \ge N_*$, $p \ge p_*$, *and suppose that* $L_p = 2^p$ *and that* $\beta_{N,p} = \beta_N$ *does not depend on* p *for* $p \ge p_*$. *Let* $\Delta_E = 16\varpi/(625e^2\ell)$, *and suppose there are* $r_0 \ge p_*$ *with* $2^{r_0} \ge \ell + 1$, $\theta \in (0,1]$ *with* $\theta\Delta_E 2^{r_0} \ge 3.0651$, *and* N_1 *such that for all* $N \ge N_1$, *with* $c = N2^{r_0}$,

(Z_0^{dy}) $D_{(c,c,c)}(c) \le \frac{1}{10}$, *and*

(K^{dy}) $N\kappa(N2^r, N2^r, N2^r) \le (1-\theta)\Delta_E$ *for all* $r > r_0$, *the halving being into the eight cubes of side* $N2^{r-1}$.

Then $\limsup_{N\to\infty}(\mathcal{Z}_{N,p} - 1) \le 2.144\, e^{-\theta\Delta_E 2^{p-1}}$ *for* $p > r_0$, *so* (Z) *holds and the family has the conclusion of Theorem E by Corollary 7.11. Condition* (Z_0^{dy}) *holds for all* $N \ge N_*$ *as soon as* $\log \mathcal{Z}_{(c,c,c)}(c/2) \le c_b$ *with* $\Delta_E 2^{r_0} \ge 2(c_b + \log 10)$.

Proof. (i) For $j < J$, $m_j \ge 2Ns_0$, so $(m_j+1)\kappa(\vec{\ell}) \le \eta_j \le \eta(N,p;2s_0) \le H$, while $(1-\theta)\Delta m_j/N \ge (1-\theta)2\Delta s_0 = H$; and $\theta\Delta s_0 = \Delta s_0 - H/2 \ge 3.0651$. Apply Theorem 7.39; with $H = \frac{1}{70}$, $\frac{1}{70} + 3.0651 < 3.0723$.

(ii) Fix $p > r_0$ and $N \ge \max\{N_*, N_1, 2\}$, put $c_j = N2^{p-j}$ and $J = p - r_0$, so that $c_j = 2c_{j+1}$: the chain is exact and consists of cubes. For $j < J$, $p' = p - j$ lies in $(r_0, p]$, so $p' \ge p_*$ and $c_j = NL_{p'}$; by (D_ℓ) at p', BP(n_N, ϖ) holds for $\mu_{c_j,M}$ at $\beta_{N,p'} = \beta_N$ for all $M \ge 2n_N + 4$, and $c_j \ge N2^{r_0+1} \ge 2(\ell+1)N \ge 2n_N + 4$ because $n_N \le \ell N$ and $N \ge 2$.

By Proposition 7.4 the cube c_j has the gap $e^{-m_{n_N}}$ with $m_{n_N}N \geq \Delta_E$, as in Step 1 of Section 7.6. Write $d_j = D_{(c_j,c_j,c_j)}(c_j)$; then $d_J \leq \frac{1}{10}$ by $(\mathrm{Z}_0^{\mathrm{dy}})$, and the computation in the proof of Theorem 7.39, with $M = c_j$, $M_1 = c_{j+1} = c_j/2$ and the eight halves all equal to the cube c_{j+1}, gives

$$\begin{aligned} d_j &\leq e^{-\Delta_E c_j/(2N)}\big(1.1^8 e^{c_{j+1}\kappa(c_j)} - 1\big) \leq 1.1^8 \exp\Big(-\frac{c_j}{2N}\big(\Delta_E - N\kappa(c_j)\big)\Big) \\ &\leq 1.1^8 e^{-\theta\Delta_E 2^{p-j-1}} \leq \tfrac{1}{10} \end{aligned}$$

by $(\mathrm{K}^{\mathrm{dy}})$ at $r = p - j > r_0$ and $2^{p-j-1} \geq 2^{r_0}$. At $j = 0$ this is the bound on $\mathcal{Z}_{N,p} - 1$, and (Z) follows since $\theta\Delta_E 2^{p-1} \to \infty$. For the last claim, c is even, the cube $c = NL_{r_0}$ has the gap by (D_ℓ) at $p' = r_0$, and Lemma 7.12 from the time $c/2$ to the time c gives $D_{(c,c,c)}(c) \leq e^{-\Delta_E 2^{r_0}/2}(e^{c_b} - 1) < e^{c_b - \Delta_E 2^{r_0}/2} \leq \frac{1}{10}$. □

For such a family the conclusion of Theorem E thus follows from (W), (D_ℓ), a pressure bound at one physical scale 2^{r_0}, namely a thermal correction at the temperature 2^{1-r_0} bounded by a constant uniformly in N, and the Casimir energy of the dyadic cubes above that scale below the gap that (D_ℓ) provides; no hypothesis on non-cubic tori enters. The scale is large, $2^{r_0} \geq 3.0651 \cdot 625 e^2 \ell/(16\theta\varpi)$, about $885\ell/(\theta\varpi)$, because $\Delta_E\ell$ is about 0.0035ϖ. What $(\mathrm{K}^{\mathrm{dy}})$ asks at every scale above 2^{r_0} is made precise by the following dichotomy: given the gap on the mixed tori and small thermal corrections at the previous scale, the thermal correction at a scale is either exponentially small or at least linear in the scale, and nothing lies between.

Lemma 7.41 (convex alternative)**.** *Let $A, b > 0$, $B \geq 1$ and $x \geq 0$ with $x \leq A(Be^{bx} - 1)$. If $x \leq x_* = b^{-1}\log\big(1/(2AbB)\big)$, then $x \leq 2A(B-1)$.*

Proof. From $1 - e^{-bx} \leq bx$ one has $e^{bx} - 1 \leq bx\, e^{bx}$, and $ABb\, e^{bx} \leq ABb\, e^{bx_*} = \frac{1}{2}$; so $x \leq A(B-1) + AB(e^{bx} - 1) \leq A(B-1) + \frac{x}{2}$. □

Proposition 7.42 (dichotomy of the thermal corrections)**.** *Let $j \geq 0$ with $m_{j+2} \geq 3$, suppose that every $t \in \mathcal{S}_j$ has the gap $e^{-\Delta/N}$ in the sense of* (G)*, and that $X_{j+1} \leq x$. Put $a_j = \Delta(m_j - 1)/(4N)$ and*

$$b_j = \frac{30\,(m_j+2)(m_{j+1}+1)}{4\,(m_j+1)\,m_{j+1}}, \qquad b_j \leq 7.8 \text{ once } m_{j+1} \geq 50.$$

Then $X_j \leq e^{-a_j}\big(e^{b_j X_j + 8x} - 1\big)$, and consequently either

$$X_j \leq 2e^{-a_j}\big(e^{8x} - 1\big) \qquad \text{or} \qquad X_j > \frac{a_j - \log(2b_j) - 8x}{b_j}.$$

Proof. Let $t \in \mathcal{S}_j$, $\tau \in L_{j+1}$ and $\tau_2 = \lfloor \tau/2 \rfloor \in L_{j+2}$, so that $\tau_2 \geq m_{j+2} \geq 3$, $\tau - \tau_2 = \lceil \tau/2 \rceil \geq m_{j+1}/2 \geq (m_j - 1)/4$ and $\tau_2 \leq (m_{j+1} + 1)/2 \leq (m_j + 2)/4$. By the gap and Lemma 7.12, $\log \mathcal{Z}_t(\tau) \leq D_t(\tau) \leq e^{-a_j} D_t(\tau_2)$. By (7.13) at the time τ_2, $\log \mathcal{Z}_t(\tau_2) \leq \tau_2 \kappa_p(t) + \sum_{s \in \mathcal{P}(t)} \log \mathcal{Z}_s(\tau_2)$, where every s lies in $L_{j+1}^3 \subset \mathcal{S}_{j+1}$ and $\tau_2 \in L_{j+2}$, so that each of the at most eight terms is at most $X_{j+1} \leq x$; and by Lemma 7.38(iii), $\tau_2\kappa_p(t) \leq \frac{m_j+2}{4(m_j+1)} K_j \leq b_j X_j$. Hence $D_t(\tau_2) \leq e^{b_j X_j + 8x} - 1$, and the maximum over t and τ gives the inequality. The alternative is Lemma 7.41 with $A = e^{-a_j}$, $B = e^{8x}$ and $b = b_j$. For the bound on b_j, $m_{j+1} \geq 50$ gives $m_j \geq 100$, and $\frac{30}{4} \cdot \frac{102}{101} \cdot \frac{51}{50} < 7.8$. □

In physical terms, with $s = m_j/N$, the thermal correction of a mixed torus of side about s at the temperature about $2/s$ is either of order $e^{-\Delta s/4}$ or at least about $\Delta s/31$. Excluding the second branch at every scale is exactly what (K) asks beyond the gap and the data at one scale, and it yields (C) itself, with a rate.

Theorem 7.43 ((C) with a rate)**.** *Fix N, p and a chain with n_0 arbitrary, and let $\lfloor N s_0 \rfloor \geq 6$, $\theta \in (0, 1]$ and $\Delta > 0$ with $\theta\Delta(s_0/2 - 1/(4N)) \geq \log(10e^{0.8}) = 3.1025\ldots$. Suppose that*

(G′) *every* $t \in \mathcal{S}_j$, $0 \le j < J$, *has the gap* $e^{-\Delta/N}$;
(X$_0$) $X_J \le \frac{1}{10}$;
(K′) $(m_j+1)\kappa_p(t) \le (1-\theta)\Delta(m_j-4)/N$ *for* $0 \le j < J$ *and* $t \in \mathcal{S}_j$.
Then for $0 \le j < J$

$$X_j \le e^{0.8} e^{-\theta\Delta(m_j-1)/(4N)} \le \tfrac{1}{10}, \qquad K_j \le 30 e^{0.8}\Big(1+\frac{1}{m_{j+1}}\Big) e^{-\theta\Delta(m_j-1)/(4N)}.$$

If (G′) *holds for all spatial tori with sides at least* $\lfloor Ns_0 \rfloor$, (X$_0$) *holds in the form* $\log \mathcal{Z}_t(\tau) \le \frac{1}{10}$ *for all* t *with sides in* $[\lfloor Ns_0/2 \rfloor, 2Ns_0+1]$ *and* $\tau \in [\lfloor Ns_0/2 \rfloor, Ns_0+1]$, *and* (K′) *holds for every* $m \ge 2Ns_0$ *in place of* m_j, *then for* $s \ge 2s_0$ *with* $Ns \ge 200$

$$\eta(N,p;s) \le 67.5\, e^{-\theta\Delta(s-1/N)/4},$$

so that (C) *holds at every scale* $s_1 \ge \max\{2s_0,\, 4\log(70\cdot 67.5)/(\theta\Delta) + 1/N\}$ *when these hypotheses hold for all large* p *and* N *with fixed constants.*

Proof. Downward induction with the hypothesis $X_{j+1} \le \frac{1}{10}$, which for $j+1 = J$ is (X$_0$); note $m_{j+2} \ge m_{J+1} \ge \lfloor Ns_0/2 \rfloor \ge 3$. As in the proof of Proposition 7.42, for $t \in \mathcal{S}_j$ and $\tau \in L_{j+1}$, $\log \mathcal{Z}_t(\tau) \le e^{-\Delta(m_j-1)/(4N)} \exp(\tau_2 \kappa_p(t) + 0.8)$, and by (K′)

$$\tau_2\kappa_p(t) \le \frac{(m_j+2)(m_j-4)}{4(m_j+1)} \cdot \frac{(1-\theta)\Delta}{N} \le \frac{(1-\theta)\Delta(m_j-2)}{4N},$$

because $(m+2)(m-4) \le (m+1)(m-2)$. So $X_j \le e^{0.8} e^{-\theta\Delta(m_j-1)/(4N)}$, which is at most $\frac{1}{10}$ because $(m_j-1)/(4N) \ge s_0/2 - 1/(4N)$ for $m_j \ge 2Ns_0$; Lemma 7.38(iii) gives the bound on K_j. For the uniform statement apply this to the chain with $n_0 = m$ for each $m \ge \max\{Ns, 2Ns_0\}$: then $j = 0$, $\mathcal{S}_0 \supset \{m, m+1\}^3$ with $\kappa_p = \kappa$, every side occurring in $\mathcal{S}_j$ for $j < J$ is at least $m_J \ge \lfloor Ns_0 \rfloor$, the tori of $\mathcal{S}_J$ with the times of L_{J+1} lie in the stated window, and (K′) holds along the chain. So $(m+1)\kappa(\vec{\ell}) \le 30e^{0.8}(1+1/\lfloor m/2 \rfloor) e^{-\theta\Delta(m-1)/(4N)}$, with $30e^{0.8}(1+1/99) < 67.5$ for $m \ge 200$ and $(m-1)/(4N) \ge (s-1/N)/4$. □

Given (G′) and (X$_0$), condition (K′) can be replaced by the exclusion of the large branch, $X_j \le (a_j - \log(2b_j) - 0.8)/b_j$ for $j < J$: by Proposition 7.42 then $X_j \le 2(e^{0.8}-1)e^{-a_j} < 2.452\, e^{-a_j} \le \frac{1}{10}$ whenever $a_j \ge 3.1997$, and the induction runs as before. That exclusion allows the thermal correction at aspect ratio $\frac{1}{2}$ to grow linearly with the scale, about $\Delta s/31$, where the expected size is $e^{-\Delta s/4}$; whether positivity, the gap on all spatial tori with sides at least $\lfloor Ns_0 \rfloor$ and $X_J \le \frac{1}{10}$ at one scale already exclude it is taken up in Section 7.10: with the one-scale datum taken on boxes elongated up to twenty-four times with respect to the time, the gap excludes it (Corollary 7.55), and whether $X_J \le \frac{1}{10}$ itself suffices is a question at one scale that remains open there. For product systems, $Z(n) = \exp \sum_\mu (|n|/n_\mu)\varphi(n_\mu)$ with $\varphi(n) = \log \operatorname{Tr} \mathsf{t}^n$ for a nonnegative matrix t of largest eigenvalue t_0, of which Proposition 7.35 is the case $\mathsf{t} = \left(\begin{smallmatrix} e^K & e^{-K} \\ e^{-K} & e^K \end{smallmatrix}\right)$, the transverse factors cancel in $\mathcal{Z}$, $\log \mathcal{Z}_{\vec{\ell}}(M) = V(\vec{\ell})\psi(M)$ with $\psi(n) = \log \operatorname{Tr}(\mathsf{t}/\mathsf{t}_0)^n$, and the gap propagates the smallness of ψ from s_0 to every larger scale, so no example arises there; an example would need excitations whose degeneracy appears only above a scale. In this form the content of (C), beyond the gap and the data at one scale, is a single a priori bound at aspect ratio $\frac{1}{2}$, and the recursion of Theorem 7.39 shows that it is the physical Casimir energy against the gap, not the smallness of the Casimir coefficient, that decides (Z).

7.9. **The pressure bound at one scale.** Corollary 7.40(ii) reduces the base of the dyadic chain, condition ($\mathrm{Z}_0^{\mathrm{dy}}$), to a single bound: with $s = 2^{r_0}$, $c = Ns$ and $h = c/2$,

$$X_N(s) := \log \mathcal{Z}_{(c,c,c)}(h) \le c_b \qquad \text{uniformly in } N, \tag{7.15}$$

the thermal correction of the cube of physical side s at the temperature $2/s$. At fixed N it is finite, since $h \ge 2$ and the transfer matrix is Hilbert–Schmidt; its whole content is the uniformity in N. This subsection places it among the other hypotheses. Positivity and

symmetry express it, up to a thermal term that the gap controls, through the vacuum energy of a tube of cross-section $s \times s$ (Proposition 7.45); it is implied by none of (D_ℓ), $(\mathrm{K}^{\mathrm{dy}})$, the gap, positivity and symmetry, and it is not necessary for (Z) (Proposition 7.47); it is equivalent to the extensivity of two tori up to a constant (Proposition 7.48); it holds in two dimensions along the tuning (Proposition 7.50); and for the free field it is computed exactly in Theorem 10.10, with the threshold $\mu s > 5.21$ for the clause of Corollary 7.40(ii). Taken on elongated boxes, it is the one datum from which, with the gap, (Z) follows with no condition at a second scale (Section 7.10). Throughout, $\beta \geq 0$, all sides are at least 3, and $\mathcal{Z}_t(m)$ denotes the relative partition function of the transfer matrix on the transverse torus t at the time m, as after Lemma 7.33; by the symmetry of Z, $E_0(\ell_1, \ell_2, \ell_3) = -\lim_M M^{-1} \log Z(\ell_1, \ell_2, \ell_3, M)$ is symmetric in the three sides.

Lemma 7.44 (convexity in a side)**.** *Fix all sides but one. Then:*

(i) $n_\mu \mapsto \log Z(n)$ *is convex and subadditive on the integers* ≥ 3*;*
(ii) $\ell \mapsto E_0(\ell, b, c)$ *is concave and superadditive, and* $\ell \mapsto \zeta(\ell; b, c)$ *of Lemma 7.31 is convex and nonincreasing, on the integers* ≥ 3*;*
(iii) $M \mapsto \log \mathcal{Z}_{\vec{\ell}}(M)$ *is convex and nonincreasing on the integers* ≥ 2*.*

Proof. (i) By the symmetry of Z, $\log Z(n) = \log \sum_i \lambda_i^{n_\mu}$ with the eigenvalues $\lambda_i \geq 0$ of the transfer matrix in the direction μ, the sum being finite for $n_\mu \geq 2$; a logarithm of a sum of exponentials of linear functions of n_μ is convex, and subadditivity is Lemma 7.27. (ii) $E_0(\ell, b, c) = -\lim_M M^{-1} \log Z(\ell, b, c, M)$ by Lemma 7.3(i),(ii) for the time transfer matrix on the slice (ℓ, b, c); by (i) in the direction of ℓ it is a pointwise limit of concave functions of ℓ, and second differences pass to the limit. Superadditivity is Lemma 7.31(ii), and $\zeta(\ell; b, c) = \ell e(b, c) - E_0(\ell, b, c)$ is then convex. It is nonincreasing because by Lemma 7.31(i) it is a limit of $M^{-1} \log \mathcal{Z}_{(b,c,M)}(\ell)$, and $\mathcal{Z}_{(b,c,M)}(\ell) = \sum_i r_i^\ell$ with $0 \leq r_i \leq 1$ is nonincreasing in ℓ. (iii) is the same for $\log \sum_i r_i^M$. □

Proposition 7.45 (exchange of directions)**.** *Let* $b, c, h \geq 3$ *and let* $t = (b, c)$ *be the cross-section.*

(i) *On the torus with sides* $(b, c, 2h, h)$*,*

$$\log \mathcal{Z}_{(b,c,2h)}(h) = \log \mathcal{Z}_{(b,c,h)}(2h) + h\,\delta, \qquad \delta = E_0(b, c, 2h) - 2E_0(b, c, h) = 2\zeta(h; t) - \zeta(2h; t),$$

and $\zeta(h; t) \leq \delta \leq 2\zeta(h; t)$*.*

(ii) *For integers* $\ell > \tau \geq 3$*,* $\zeta(\tau; b, c) \leq \log \mathcal{Z}_{(b,c,\ell)}(\tau)/(\ell - \tau)$*.*

(iii) *For the cube,* $b = c = 2h$*, with* $X = \log \mathcal{Z}_{(c,c,c)}(h)$ *and* $Y = \log \mathcal{Z}_{(c,c,h)}(c) \geq 0$*,*

$$Y + h\,\zeta(h; c, c) \leq X \leq Y + 2h\,\zeta(h; c, c), \qquad \zeta(h; c, c) \leq X/h.$$

In physical units, with $c = Ns$*,* $h = Ns/2$ *and* $\mathsf{f}(s) = N\zeta(Ns/2; Ns, Ns)$*, the vacuum energy of the spatial torus* $(Ns, Ns, Ns/2)$ *relative to the infinite tube of cross-section* $s \times s$*, per unit of physical length, this reads* $\frac{s}{2}\mathsf{f}(s) \leq X - Y \leq s\,\mathsf{f}(s)$*.*

(iv) *If every eigenvalue* $\lambda \neq \lambda_0$ *of the time transfer matrix of the spatial torus* (c, c, h) *satisfies* $|\lambda| \leq e^{-\Delta/N}\lambda_0$*, then* $Y \leq e^{-\Delta s/2}\big(\mathcal{Z}_{(c,c,h)}(h) - 1\big)$*.*

Proof. (i) By (7.14) in the direction of length h, whose transverse torus is $(b, c, 2h)$, and in the direction of length $2h$, whose transverse torus is (b, c, h),

$$\log Z(b, c, 2h, h) = -hE_0(b, c, 2h) + \log \mathcal{Z}_{(b,c,2h)}(h) = -2hE_0(b, c, h) + \log \mathcal{Z}_{(b,c,h)}(2h),$$

which is the identity. Lemma 7.31(ii) with $\ell' = \ell'' = h$ gives $\delta = 2\zeta(h; t) - \zeta(2h; t)$, and $0 \leq \zeta(2h; t) \leq \zeta(h; t)$ by Lemma 7.44(ii).

(ii) Let $k \geq 1$. Lemma 7.27 in the third direction gives $Z(b, c, k\ell, \tau) \leq Z(b, c, \ell, \tau)^k$, and (7.14) in the direction of length τ gives $\log Z(b, c, m, \tau) = -\tau E_0(b, c, m) + \log \mathcal{Z}_{(b,c,m)}(\tau)$

for every m. By Lemma 7.31(i) along $M = k\ell$ and the definition of $e(b,c)$,

$$\zeta(\tau;b,c) = \lim_{k\to\infty} \frac{\log Z(b,c,k\ell,\tau) + \tau E_0(b,c,k\ell)}{k\ell} \le \frac{\log Z(b,c,\ell,\tau)}{\ell} + \tau e(b,c)$$
$$= \frac{\log \mathcal{Z}_{(b,c,\ell)}(\tau)}{\ell} + \frac{\tau}{\ell}\,\zeta(\ell;b,c).$$

Since $\zeta(\ell;b,c) \le \zeta(\tau;b,c)$, this gives $(1-\tau/\ell)\zeta(\tau;b,c) \le \ell^{-1}\log \mathcal{Z}_{(b,c,\ell)}(\tau)$.

(iii) is (i) with $b = c = 2h$ and (ii) with $\ell = c$, $\tau = h$, so that $\ell - \tau = h$; multiply by N for the physical form. (iv) Lemma 7.12 from the time h to the time c gives $D_{(c,c,h)}(c) \le e^{-\Delta(c-h)/N} D_{(c,c,h)}(h)$, and $\log(1+D) \le D$. □

So $X \ge Y$ always, and the excess $X - Y$ is, up to a factor 2, the physical tube energy $s\,\mathsf{f}(s)$, a ground-state quantity of the same kind as the Casimir defect κ: by Lemma 7.31(iii), $\kappa(c) \le 14 \max \zeta$ over tubes of the same cross-sections. The pressure bound (7.15) implies $s\,\mathsf{f}(s) \le 2c_b$; conversely $s\,\mathsf{f}(s) \le t$ and $Y \le y$ give $X \le t + y$, and (iv) controls Y by the thermal correction of the smaller torus (c,c,h) at the time h, with the factor $e^{-\Delta s/2}$. What these inequalities do not give is smallness: Proposition 7.28 with $M = h$ gives $X(c) \le h\kappa(c) + 8\log(1 + D_{(h,h,h)}(h))$, which under the gap on the cube h reduces the bound at the scale s to the same bound at the scale $s/2$ with the factor 8, the volume count of the rings; and iterating (i)–(iv) gives an exponential tower with no gain below the gap scale. That no combination of positivity, symmetry, the gap, (D_ℓ) and (K^{dy}) can succeed is shown by the rings. The cube condition for the symmetrised rings is proved first.

Proposition 7.46 (the symmetrised rings satisfy the cube condition)**.** *Let $d \ge 2$, $n = 2d-1$, $K > 0$, $\kappa = \tanh K$, and let μ be the measure of Proposition 7.35 on a torus whose four sides are at least $4d$; let c_m be the quantity of Proposition 7.19 for a ring of length m, and c_* its maximum over the four sides. Let $\mathcal{P}_x$ be the conditional expectation given the spins at the sites off $x + [0,n)^4$. If $c_* < \frac12$, then for every real $f \in L^2(\mu)$*

$$\frac{d\,(1-2c_*)}{2d-1}\,\mathrm{Var}\, f \le \frac{1}{n^4}\sum_x \mu((f - \mathcal{P}_x f)^2),$$

that is, $\mathrm{BP}(2d-1,\varpi)$ *holds with $\varpi = d(1-2c_*)/(2d-1) \ge \frac12(1-2c_*)$, read with sites in place of links as in Proposition 7.19; in particular* $\mathrm{BP}(2d-1, \frac12(1-2\kappa^{(3d-3)/4}))$ *holds whenever $2\kappa^{(3d-3)/4} < 1$.*

Proof. The weight factorises over the lattice lines: the ring of species ν along a line in the direction ν is a periodic Ising chain of length equal to the side m_ν, and the rings are independent, so $L^2(\mu) = \bigotimes_k L^2(\mu_k)$ over all rings k. The one-ring part of the proof of Proposition 7.19 uses only the chain and its length $m \ge 4d$: for $g \perp \mathbf{1}$ in $L^2(\mu_k)$, $\sum_{w\in\mathbb{Z}/m\mathbb{Z}} \langle g, (I - P_w^{(k)})g\rangle \ge d(1-2c_m)\|g\|^2 \ge d(1-2c_*)\|g\|^2$, $P_w^{(k)}$ being the conditional expectation given the spins of the ring off the window $[w, w+n)$. The cube $x + [0,n)^4$ meets, for each ν, exactly the n^3 rings of species ν whose lines pass through its projection transverse to ν, each in the window $[x_\nu, x_\nu + n)$, and these rings are distinct because every side exceeds n; by independence the conditional law of the spins of the cube given the others is the product of the one-ring window laws, so $\mathcal{P}_x = \bigotimes_{k\in F_x} P^{(k)}_{w_x(k)} \otimes I$ with $|F_x| = 4n^3$. The product part of the proof of Proposition 7.19 now applies word for word: decompose $f = \sum_A f_A$ into the orthogonal spaces $\mathcal{H}_A$, fix $g \in \mathcal{H}_A$ and a ring $k_0 \in A$; a given window w of k_0 is the window $w_x(k_0)$ of exactly n^3 cubes, namely $x_\nu = w$ and n choices of each of the three transverse coordinates, so $\sum_x \langle g, (I - \mathcal{P}_x)g\rangle \ge n^3 d(1-2c_*)\|g\|^2$, and summing over A gives the claim, as $n^4\varpi = n^3 d(1-2c_*)$. The last statement is $c_m \le \kappa^{(3d-3)/4}$ for $m \ge 4d$. □

Proposition 7.47 (the pressure bound against the other hypotheses)**.** *Fix $s = 2^{r_0}$ and put $c = Ns$, $h = c/2$, $X_N = \log \mathcal{Z}_{(c,c,c)}(h)$.*

(i) *The time rings of Proposition 7.19, with* $\tanh J = e^{-\Delta/N}$ *and* $L_p = 2^p$*, satisfy* (D_ℓ) *with* $\ell = 2/K_0$ *for every* $K_0 < 3\Delta/(4\log 2)$*, with a coupling independent of* p*; their Casimir defect vanishes identically, so* $(\mathrm{K}^{\mathrm{dy}})$ *holds for every* $\theta \in (0,1]$*; all their transfer matrices are nonnegative, the spatial ones of rank one, and the time one has the gap* $e^{-\Delta/N}$ *on every torus. Yet*

$$X_N = c^3\log\bigl(1+e^{-\Delta s/2}\bigr) \ge \tfrac12 N^3 s^3 e^{-\Delta s/2} \to \infty, \qquad D_{(c,c,c)}(c) = \bigl(1+e^{-\Delta s}\bigr)^{c^3} - 1 \to \infty.$$

(ii) *The symmetrised rings of Proposition 7.35, with* $r = e^{-\Delta/N}$*, are symmetric in the four directions, positive, and have the gap* $e^{-\Delta/N}$ *on every torus in every direction; by Proposition 7.46 they satisfy* (D_ℓ) *as in* (i) *for* $L_p \ge 4/K_0$*. Yet* $X_N = c^3\log(1+e^{-\Delta s/2}) \to \infty$*, and they violate* $(\mathrm{K}^{\mathrm{dy}})$*: with* $\varphi(m) = \log(1+r^m)$*,*

$$\kappa(2h,2h,2h) = 12h^2\bigl(2\varphi(h) - \varphi(2h)\bigr) \ge 12h^2\varphi(h),$$

so $N\kappa(N2^r, N2^r, N2^r) \ge 6N^3 4^{r-1} e^{-\Delta 2^{r-1}} \to \infty$ *for every fixed* r*.*

(iii) *The symmetrised rings with* $r = e^{-E_N/N}$*,* $E_N = (4/s)\log N$*, are symmetric and positive, have the gap* $e^{-E_N/N} \le e^{-\Delta/N}$ *for every fixed* Δ *once* $N \ge e^{\Delta s/4}$*, satisfy* (D_ℓ) *with* $\varpi = \frac14$*,* $\ell = 2/K_0$ *and any* $K_0 \ge 1$ *for large* N*, and*

$$\sup_{r>r_0} N\kappa(N2^r, N2^r, N2^r) \le \frac{24\cdot 4^{r_0}}{N}, \quad D_{(c,c,c)}(c) \le e^{s^3/N} - 1, \quad \mathcal{Z}_{N,p} - 1 \le e^{L_p^3 N^{3-4L_p/s}} - 1$$

for $N \ge 2$*, so* $(\mathrm{K}^{\mathrm{dy}})$ *holds for every* $\theta < 1$*,* $(\mathrm{Z}_0^{\mathrm{dy}})$ *holds, and* $\mathcal{Z}_{N,p} - 1 \to 0$ *as* $N \to \infty$ *for every* p *with* $L_p > 3s/4$*, which is* (Z)*; while*

$$\tfrac12 s^3 N \le X_N = N^3 s^3 \log\bigl(1+N^{-2}\bigr) \le s^3 N \to \infty.$$

Proof. (i) The cube condition is the paragraph after Proposition 7.19. For the time rings $Z(\vec\ell, M) = \bigl(\cosh^M J\,(1+r^M)\bigr)^V$ with $V = \ell_1\ell_2\ell_3$ and $r = \tanh J$, by the computation of Proposition 7.35(i) for one species; so $E_0(\vec\ell) = -V\log\cosh J$ is exactly additive under halving, $\kappa \equiv 0$, and $\mathcal{Z}_{\vec\ell}(M) = (1+r^M)^V$, which is the formula for X_N and $D_{(c,c,c)}(c)$ with $r^h = e^{-\Delta s/2}$, $r^c = e^{-\Delta s}$; the lower bound is $\log(1+y) \ge y/2$ for $0 \le y \le 1$. The time transfer matrix is the tensor product over the V sites of the chain matrix, with the eigenvalues $\cosh^V J\, r^k$, $0 \le k \le V$, so the gap is $r = e^{-\Delta/N}$; a spatial transfer matrix couples no two slices and equals $|w^{1/2}\rangle\langle w^{1/2}|$ as in the proof of Proposition 7.35(ii), nonnegative of rank one.

(ii) Positivity, symmetry and the gap are Proposition 7.35(ii), and $\mathcal{Z}_{\vec\ell}(M) = (1+r^M)^V$ is its part (iii). Along the tuning $d_N = \lfloor N/K_0\rfloor$, $n_N = 2d_N - 1 \le \ell N$, Proposition 7.46 needs all four sides at least $4d_N$, which holds for $NL_p \ge 4N/K_0$ and $M \ge 2n_N + 4 = 4d_N + 2$, and $c_* \le \kappa^{(3d_N-3)/4} = e^{-\Delta(3d_N-3)/(4N)} \to e^{-3\Delta/(4K_0)} < \frac12$ when $K_0 < 3\Delta/(4\log 2)$, so ϖ is bounded below for large N. From Proposition 7.35(i), $E_0(\vec\ell) = -\lim_M M^{-1}\log Z(\vec\ell, M) = -4V\log\cosh K - \sum_{i\le 3}(V/\ell_i)\varphi(\ell_i)$, since $M^{-1}\log(1+r^M)^V \to 0$; with $V = 8h^3$ for the cube $2h$ and $V = h^3$ for its eight halves, $\kappa(2h,2h,2h) = -12h^2\varphi(2h) + 24h^2\varphi(h)$, and $\varphi(2h) \le \varphi(h)$. With $h = N2^{r-1}$, $12Nh^2\varphi(h) \ge 6Nh^2 r^h = 6N^3 4^{r-1} e^{-\Delta 2^{r-1}}$.

(iii) Now $r^h = N^{-2}$ and $r^c = N^{-4}$, so $X_N = c^3\log(1+N^{-2})$ and $D_{(c,c,c)}(c) = (1+N^{-4})^{N^3s^3} - 1 \le e^{s^3/N} - 1$, using $\frac{y}{2} \le \log(1+y) \le y$. For $h = N2^{r-1}$ with $r = r_0 + 1 + k$, $k \ge 0$, one has $r^h = N^{-4\cdot 2^k}$ and, by (ii),

$$N\kappa(2h,2h,2h) \le 24Nh^2 r^h = 24\cdot 4^{r_0}\, 4^k N^{3-4\cdot 2^k} \le \frac{24\cdot 4^{r_0}}{N}\, 4^k\, 2^{-4(2^k-1)} \le \frac{24\cdot 4^{r_0}}{N},$$

because $N^{4-4\cdot 2^k} \le 2^{-4(2^k-1)}$ for $N \ge 2$ and $2k + 4 \le 4\cdot 2^k$. On the symmetric torus, $\mathcal{Z}_{N,p} = (1+r^{NL_p})^{N^3L_p^3}$ with $r^{NL_p} = N^{-4L_p/s}$, and $(1+y)^V \le e^{Vy}$. The gap $e^{-E_N/N}$ is at most $e^{-\Delta/N}$ once $E_N \ge \Delta$. For (D_ℓ), $c_* \le e^{-E_N(3d_N-3)/(4N)} \to 0$ because $E_N \to \infty$ and

$(3d_N-3)/(4N)\to 3/(4K_0)$; so eventually $c_*\le\frac14$ and $\varpi\ge\frac14$ by Proposition 7.46, whose side conditions hold as in (ii). □

Part (i) shows that in Corollary 7.40(ii) the pressure bound is not implied by (D_ℓ), $(\mathrm{K}^{\mathrm{dy}})$, positivity and the gap, (W) having no meaning for a spin model; there (Z) fails too, as it must. Part (ii) adds the symmetry in the four directions. Part (iii) shows that the pressure bound is not necessary: $(\mathrm{Z}_0^{\mathrm{dy}})$, $(\mathrm{K}^{\mathrm{dy}})$, (D_ℓ) and (Z) all hold while $X_N\to\infty$. The mechanism is a band of N^3s^3 states at the physical energy $E_N=(4/s)\log N$, invisible at aspect ratio one, where $c^3e^{-E_Ns}=s^3/N$, and divergent at aspect ratio $\frac12$, where $c^3e^{-E_Ns/2}=s^3N$. In a symmetric product system with a fixed gap Δ both X_N and $N\kappa$ grow like N^3, as in (ii); whether a symmetric system with a fixed gap, $(\mathrm{K}^{\mathrm{dy}})$ and $X_N\to\infty$ exists is not decided here. By Proposition 7.14 a proof of the pressure bound for the gauge theory must use the completeness of the gauge-invariant slice functions; by Proposition 7.45 it is equivalent, up to the thermal term Y, to a bound $s\,\mathsf{f}(s)\le 2c_b$ on the physical tube energy at one scale.

Proposition 7.48 (one-scale equivalent form)**.** *Let $c\ge 4$ be even and, for $f\in\mathbb{R}$, $g_f(M)=\log Z(c,c,c,M)-fc^3M$.*

(a) *If $|g_f(M)|\le A$ for $M\in\{c/2,c\}$ and some f, then $X=\log\mathcal{Z}_{(c,c,c)}(c/2)\le 3A$.*

(b) *With $f=-E_0(c,c,c)/c^3$, $g_f(M)=\log\mathcal{Z}_{(c,c,c)}(M)\in[0,X]$ for every $M\ge c/2$.*

So the pressure bound (7.15) *holds uniformly in N if and only if the two tori $(c,c,c,c/2)$ and (c,c,c,c) are extensive up to a constant, $|\log Z(n)-f_N|n||\le A$, with one common density f_N, which may depend on N, and a constant A independent of N.*

Proof. (a) By Lemma 7.33 for the time transfer matrix of the cube, $X\le\psi_{(c,c,c)}(c/2)=2\log Z(c,c,c,c/2)-\log Z(c,c,c,c)=2g_f(c/2)-g_f(c)\le 3A$, the terms fc^3M cancelling because the second torus has twice the volume of the first. (b) is (7.14) and Lemma 7.44(iii). □

Corollary 7.49 (stability bounds at one scale)**.** *Suppose that for all large N there are $e_N\in\mathbb{R}$ and constants $E_-\le E_+$ independent of N with*

$$E_-\,v(n)\le\log Z(n)-e_N|n|\le E_+\,v(n),\qquad v(n)=|n|/N^4,$$

for the two tori $n=(c,c,c,c/2)$ and $n=(c,c,c,c)$ at the coupling β_N. Then $X_N\le(E_+-E_-)s^4$ for all large N, and under the other hypotheses of Corollary 7.40(ii) its conclusion follows as soon as $\Delta_E2^{r_0}\ge 2\big((E_+-E_-)16^{r_0}+\log 10\big)$. For $E=E_+-E_->0$ a real $s>0$ with $\Delta_Es\ge 2(Es^4+\log 10)$ exists only if $\Delta_E^4\ge\frac{4096}{27}(\log 10)^3E>1852\,E$.

Proof. With $g(M)=\log Z(c,c,c,M)-e_Nc^3M$, $g(c/2)\le E_+s^4/2$ and $g(c)\ge E_-s^4$, so $X_N\le 2g(c/2)-g(c)\le(E_+-E_-)s^4$ by the proof of Proposition 7.48(a); then apply the last clause of Corollary 7.40(ii). For the last statement put $s_*=(\Delta_E/8E)^{1/3}$ and $s=ts_*$: $\Delta_Es-2Es^4=\Delta_Es_*(t-t^4/4)\le\frac34\Delta_Es_*$, since $\frac34-t+\frac{t^4}{4}=\frac14(t-1)^2(t^2+2t+3)\ge 0$; so $2\log 10\le\frac34\Delta_Es_*$ is necessary, and raising it to the third power gives $8(\log 10)^3\le\frac{27}{64}\Delta_E^3\cdot\Delta_E/(8E)$. □

The hypothesis of the corollary is a statement at one torus shape and one physical scale, with no decay rate, much weaker than the finite-size condition (FS) of Proposition 7.36; no such bound is proved here for the Wilson action at weak coupling. Its constants must depend on the shape, since for the six free gluon modes $\log Z-e|n|$ tends to the Stefan–Boltzmann value $\frac{\pi^2}{15}\alpha^{-3}$ as the aspect ratio α of the time extent to the side tends to 0, and the scale it asks for, $2/s\le\Delta_E/(c_b+\log 10)$, is a temperature below the gap, where the theory is not weakly coupled. In two dimensions the bound holds at every scale.

Proposition 7.50 (the pressure bound in two dimensions)**.** *Let $G=\mathrm{SU}(2)$, $\beta=\kappa N^2$ with a fixed $\kappa>0$, and $\gamma=\kappa_0\min\{3/(8\kappa),\frac12\}$ as in Proposition H.11(iii). For all $N\ge 1$ and all*

$L, M \geq 2$,

$$\log \mathcal{Z}_L(M) \leq -\log\bigl(1 - e^{-\gamma LM/(2N^2)}\bigr);$$

in particular $\log \mathcal{Z}_{Ns}(Ns/2) \leq -\log(1 - e^{-\gamma s^2/4})$ *for every* N.

Proof. By Proposition H.11(i), $\mathcal{Z}_L(M) - 1 = \sum_{j \geq 1/2} (b_j/b_0)^{LM}$, and by part (iii)(a) of that proposition, which is a statement about b_j/b_0 at $\beta = \kappa N^2$ alone, $-\log(b_j/b_0) \geq \gamma j/N^2$ for all $j \in \frac{1}{2}\mathbb{N}$ and $N \geq 1$. Hence $\mathcal{Z}_L(M) - 1 \leq \sum_{m \geq 1} e^{-\gamma mLM/(2N^2)} = (e^{\gamma LM/(2N^2)} - 1)^{-1}$, and $\log(1 + (e^y - 1)^{-1}) = -\log(1 - e^{-y})$. □

In two dimensions every energy is proportional to the side and $\kappa \equiv 0$, so this is a one-scale statement at every scale; it says nothing about four dimensions. The free field, where every quantity of this subsection is explicit, is treated in Theorem 10.10: there the pressure bound holds uniformly in N at every mass $\mu > 0$ and every scale, with the limit $F(\mu s)$ of X_N, and the clause of Corollary 7.40(ii) holds exactly when $\mu s > 5.2063$, against $\mu s > 2.4851$ for $(\mathrm{Z}_0^{\mathrm{dy}})$ itself and $\mu s > 17.71$ for the Casimir condition (C).

7.10. **Extensivity of the thermal corrections and the ascent in time.** The paragraph after Theorem 7.43 left open whether positivity, the gap and data at one scale exclude the large branch of Proposition 7.42. The obstruction was quantitative: in Lemma 7.38 the pieces of a halving have the size of the scale and the time is half of it, so a cut costs as much as the thermal correction it is meant to control, and the coefficient 15 does not contract. The cost of a cut is a tube deficit per unit length times the time, and the deficit per unit length is a density of thermal correction divided by the length of the tube on which it is measured (Proposition 7.45(ii)). Cutting into pieces much longer than the time therefore costs a small fraction of the thermal correction, and positivity alone then makes the thermal correction extensive, with a density controlled by boxes of bounded size. The gap carries this density from one time to the double time, at a fixed box size where the total correction is of order one, and the two steps alternate. The result is that (Z), (K′) and (C) with a rate follow from the gap and from a single bound at one scale, the pressure bound of Section 7.9 taken on boxes elongated up to twenty-four times with respect to the time, and that nothing at more than one scale is used. Throughout, $\beta \geq 0$ and all sides are at least 3, as in Section 7.9; $V(t) = t_1 t_2 t_3$.

For a spatial torus t and integers $T \geq 3$ and $P \geq 2T$, the *cut of mesh* P of t is defined as follows. A side $n \geq 2P$, written $n = qP + r$ with $q \geq 2$ and $0 \leq r < P$, is cut into $q - 1$ pieces of length P and one piece of length $P + r < 2P$; a side $n < 2P$ is not cut. The cuts are made in the direction 1, then in the direction 2 on every slab, then in the direction 3 on every column. The pieces form the set $\mathcal{P}^{[P]}(t)$, and $\kappa^{[P]}(t) = E_0(t) - \sum_{s \in \mathcal{P}^{[P]}(t)} E_0(s)$ is the defect of the cut. If all sides of t are at least $2T$, then all pieces have their sides in $[2T, 2P)$ and $\sum_s V(s) = V(t)$. For a bound $S \geq 2P$ put

$$f_S(T) = \max\Bigl\{ \frac{\log \mathcal{Z}_t(T)}{V(t)} : \ t \text{ has all its sides in } [2T, S] \Bigr\},$$

a maximum over a finite set of nonnegative numbers.

Lemma 7.51 (the cost of a cut of mesh P)**.** *Let $T \geq 3$, $P \geq 2T$, $S \geq 2P$, and let t have all its sides in $[2T, S]$. Then $\kappa^{[P]}(t) \geq 0$,*

$$\log \mathcal{Z}_t(T) \leq T\kappa^{[P]}(t) + \sum_{s \in \mathcal{P}^{[P]}(t)} \log \mathcal{Z}_s(T) \qquad \text{and} \qquad \kappa^{[P]}(t) \leq \frac{12\, f_S(T)\, V(t)}{P}.$$

Proof. All pieces have sides at least $2T \geq 6$, so repeated use of Lemma 7.27 gives $Z(t, \tau) \leq \prod_s Z(s, \tau)$ for every $\tau \geq 2$; letting $\tau \to \infty$ gives $\kappa^{[P]}(t) \geq 0$, and multiplying by $e^{TE_0(t)} = e^{T\kappa^{[P]}(t)} \prod_s e^{TE_0(s)}$ and taking logarithms gives the first inequality, exactly as for (7.13).

The defect $\kappa^{[P]}(t)$ is the sum of the defects of the single cuts, by telescoping. A single cut acts on a current torus with sides A, B, C, after relabelling the axes, and splits A into a and $A-a$ with $a \in [P, 2P)$ and $A-a \geq P$, since what remains contains at least one piece. By Lemma 7.31(ii) its defect is at most $\zeta(a; B, C) + \zeta(A-a; B, C)$, and since $a, A-a \geq P \geq T$ and ζ is nonincreasing in its first argument (Lemma 7.44), this is at most $2\zeta(T; B, C)$. Proposition 7.45(ii) with $\ell = 2T$ and $\tau = T$ gives

$$\zeta(T; B, C) \leq \frac{\log \mathcal{Z}_{(B,C,2T)}(T)}{T} \leq \frac{f_S(T) \cdot 2T\, BC}{T} = 2 f_S(T)\, BC,$$

because the torus $(B, C, 2T)$ has all its sides in $[2T, S]$: B and C are sides of t or lengths of pieces already cut, hence in $[2T, 2P) \cup [2T, S]$. So a single cut of cross-section $B \times C$ costs at most $4 f_S(T) BC$. In the direction 1 there are at most t_1/P cuts, each of cross-section $t_2 \times t_3$, together at most $4 f_S(T) V(t)/P$. In the direction 2 the slabs have widths a_i with $\sum_i a_i = t_1$ and receive at most t_2/P cuts each, of cross-section $a_i \times t_3$, together at most $4 f_S(T) \sum_i a_i t_3\, t_2/P = 4 f_S(T) V(t)/P$; the direction 3 is the same. Adding the three directions gives the bound. □

Proposition 7.52 (extensivity at a fixed time)**.** *Let $T \geq 3$ and $P > 12T$ be integers, and put*

$$g(T, P) = \max\Big\{ \frac{\log \mathcal{Z}_s(T)}{V(s)} : \ s \text{ has all its sides in } [2T, 2P) \Big\}.$$

Then every spatial torus t with all its sides at least $2T$ satisfies

$$\log \mathcal{Z}_t(T) \leq \frac{g(T, P)}{1 - 12T/P}\, V(t),$$

in particular $\log \mathcal{Z}_t(T) \leq 2 g(T, P) V(t)$ when $P = 24T$.

Proof. Fix $S \geq 2P$ at least as large as every side of t. For a torus t' with all sides in $[2T, S]$, Lemma 7.51 and the bound $\log \mathcal{Z}_s(T) \leq g(T, P) V(s)$ on its pieces give

$$\log \mathcal{Z}_{t'}(T) \leq \Big(\frac{12T}{P} f_S(T) + g(T, P) \Big) V(t');$$

if all sides of t' are below $2P$, then t' is its own only piece and the bound holds as well. The maximum over t' gives $f_S(T) \leq \frac{12T}{P} f_S(T) + g(T, P)$, and since $f_S(T)$ is finite and $12T/P < 1$, $f_S(T) \leq g(T, P)/(1 - 12T/P)$ for every S. □

No gap and no smallness of a Casimir coefficient enter: positivity and the symmetry of Z make the thermal correction at a fixed time extensive, with a density bounded by the density on boxes whose sides lie between $2T$ and $48T$. What is given up is the aspect ratio: the boxes that control the density are up to twenty-four times longer than the time, where Lemma 7.38 works with pieces of the size of the time. With the finite-box bound taken at $\ell = cT$ instead of $2T$, the coefficient $12T/P$ becomes $6cT/((c-1)P)$, so boxes elongated by a little more than twelve times already suffice; the constants are not optimised here.

Theorem 7.53 (the ascent in time)**.** *Let $T_0 \geq 3$ be an integer, $T_i = 2^i T_0$, and let $\delta > 0$ satisfy $\delta T_0 \geq \log(16(e-1)) = 3.3139\ldots$. Suppose that*

- (G_{24}) *for every $i \geq 1$, every spatial torus with all its sides in $[2T_i, 48T_i)$ has the gap $e^{-\delta}$ in the sense of* (G) *of Theorem 7.39;*
- (B_{24}) *$\log \mathcal{Z}_s(T_0) \leq \gamma V(s)$ for every spatial torus s with all its sides in $[2T_0, 48T_0)$, where $0 \leq \gamma \leq \frac{1}{2}(96T_0)^{-3}$.*

Put $F_0 = 2\gamma$ and $F_{i+1} = 2(e-1)e^{-\delta T_i} F_i$, so that $F_i = 2\gamma\,(2(e-1))^i\, e^{-\delta(T_i - T_0)}$ with $2(e-1) = 3.4365\ldots$. Then for every $i \geq 0$ one has $(96T_i)^3 F_i \leq 1$, and every spatial torus t with all its sides at least $2T_i$ satisfies $\log \mathcal{Z}_t(T_i) \leq F_i V(t)$. If moreover t has the gap $e^{-\delta}$, then for every $\tau \geq T_i$

$$\log \mathcal{Z}_t(\tau) \leq D_t(\tau) \leq e^{-\delta(\tau - T_i)} \big(e^{F_i V(t)} - 1 \big). \tag{7.16}$$

Proof. Induction on i. For $i = 0$, Proposition 7.52 with $T = T_0$, $P = 24T_0$ and (B_{24}) gives $\log \mathcal{Z}_t(T_0) \le 2\gamma V(t)$, and $(96T_0)^3 F_0 \le 1$ by the choice of γ. Suppose the statement holds for i, and let s have all its sides in $[2T_{i+1}, 48T_{i+1}) = [4T_i, 96T_i)$. By (G_{24}) and Lemma 7.12 from the time $T_i \ge 3$ to the time T_{i+1}, and by the induction hypothesis, since the sides of s are at least $2T_i$,

$$\log \mathcal{Z}_s(T_{i+1}) \le D_s(T_{i+1}) \le e^{-\delta T_i} D_s(T_i) \le e^{-\delta T_i}\big(e^{F_i V(s)} - 1\big).$$

The function $v \mapsto (e^{F_i v} - 1)/v$ is increasing, $V(s) < (96T_i)^3$ and $y_i = (96T_i)^3 F_i \le 1$, so

$$\frac{\log \mathcal{Z}_s(T_{i+1})}{V(s)} \le e^{-\delta T_i} F_i \, \frac{e^{y_i} - 1}{y_i} \le (e-1) e^{-\delta T_i} F_i.$$

This bounds $g(T_{i+1}, 24T_{i+1})$, and Proposition 7.52 at the time T_{i+1} gives $\log \mathcal{Z}_t(T_{i+1}) \le F_{i+1} V(t)$ for every t with sides at least $2T_{i+1}$. Finally $y_{i+1} = 8(96T_i)^3 F_{i+1} = 16(e-1) e^{-\delta T_i} y_i \le y_i \le 1$, because $\delta T_i \ge \delta T_0 \ge \log(16(e-1))$. The last claim is Lemma 7.12 from T_i to τ together with $\log(1+D) \le D$. □

The gap is used only on tori whose sides lie in $[2T_i, 48T_i)$, of aspect ratio below 24. Positivity carries each step in space, and the gap each step in time at a box size below $96T_i$, where the total thermal correction is at most $y_i \le 1$; the density gains $e^{-\delta T_i}$ per doubling of the time while the box volume grows by the factor 8, which is the origin of the threshold $\log(16(e-1))$. In dimension d the same argument runs with 12 replaced by $4(d-1)$, 96^3 by 96^{d-1} and the threshold by $\log(2^d(e-1))$, the window $[2T, 48T)$ becoming $[2T, 16(d-1)T)$.

Theorem 7.54 ((Z) from the gap and one scale)**.** *Let a continuum family have the following property: there are $t_0 > 0$ and $\Delta > 0$ with $\Delta t_0 \ge \log(16(e-1))$ such that, for all $p \ge p_0$ and $N \ge N_0(p)$, with $\delta = \Delta/N$ and $T_0 = \lceil N t_0 \rceil \ge 3$, the hypotheses (G_{24}) and (B_{24}) of Theorem 7.53 hold at the coupling $\beta_{N,p}$. Then, whenever $NL_p \ge 4T_0$,*

$$\mathcal{Z}_{N,p} - 1 \le 7.24 \cdot 10^{-5} \, e^{-\Delta L_p/2},$$

so the family satisfies (Z)*; with* (W) *and* (D_ℓ)*, Corollary 7.11 gives the conclusion of Theorem E. No Casimir condition, no (K) and no halving chain enters.*

Proof. $\delta T_0 = \Delta \lceil N t_0 \rceil / N \ge \Delta t_0$, so Theorem 7.53 applies. Put $n_0 = NL_p$ and let i be the largest index with $2T_i \le n_0$; then $i \ge 1$ and $n_0 < 4T_i$. The cube (n_0, n_0, n_0) has its sides in $[2T_i, 48T_i)$, so it has the gap by (G_{24}), and by (7.16) with $\tau = n_0$, $n_0 - T_i \ge n_0/2$ and

$$F_i n_0^3 < 64\, T_i^3 F_i = \frac{64}{96^3} (96T_i)^3 F_i \le \frac{64}{96^3} < 7.234 \cdot 10^{-5},$$

one gets $\mathcal{Z}_{N,p} - 1 = D_{(n_0,n_0,n_0)}(n_0) \le e^{-\delta n_0/2}(e^{7.234 \cdot 10^{-5}} - 1) \le 7.24 \cdot 10^{-5} e^{-\Delta L_p/2}$, as $\delta n_0/2 = \Delta L_p/2$. Since $L_p \to \infty$, (Z) follows. □

Corollary 7.55 (the large branch is excluded)**.** *Assume the hypotheses of Theorem 7.53, with (G_{24}) also for $i = 0$, at fixed N and p, in the chain notation m_j, L_j, $\mathcal{S}_j$, X_j, K_j, a_j, b_j of Section 7.8.*

(i) *For every j with $m_{j+1} \ge 2T_0$,*

$$X_j \le 5.8 \cdot 10^{-4} \, e^{-\delta m_{j+1}/2}, \qquad K_j \le 0.021 \, e^{-\delta m_{j+1}/2}.$$

(ii) *Hence X_j lies on the small branch of Proposition 7.42, $X_j \le (a_j - \log(2b_j) - 0.8)/b_j$, whenever $a_j \ge \log(2b_j) + 0.8 + 5.8 \cdot 10^{-4} b_j$, which for $b_j \le 7.8$ holds once $a_j \ge 3.56$; and* (K′) *of Theorem 7.43 holds for $t \in \mathcal{S}_j$ with every $\theta \le 1 - 0.021N/(\Delta(m_j - 4))$.*

(iii) *If the hypotheses hold for all $p \ge p_0$ and $N \ge N_0(p)$ as in Theorem 7.54, then*

$$\eta(N, p; s) \le 0.021 \, e^{-\Delta(s - 1/N)/4} \qquad \text{whenever } \lfloor Ns/2 \rfloor \ge 2T_0,$$

so (C) *holds, in both its clauses, at every scale $s_0 \ge 1/N + 1.55/\Delta$ with $\lfloor N s_0/2 \rfloor \ge 2T_0$.*

Proof. (i) Let $t \in \mathcal{S}_j$ and $\tau \in L_{j+1}$, and let i be the largest index with $2T_i \le m_{j+1}$, so that $m_{j+1} \le 4T_i - 1$. The sides of t lie in $[m_{j+1}, m_j + 1]$ and $m_j + 1 \le 2m_{j+1} + 2 \le 8T_i$, so they lie in $[2T_i, 48T_i)$; thus t has the gap by (G_{24}) and $V(t) \le (8T_i)^3$. By (7.16), with $\tau - T_i \ge \tau/2 \ge m_{j+1}/2$ and $F_i V(t) \le (8/96)^3 (96T_i)^3 F_i \le 512/96^3 < 5.79 \cdot 10^{-4}$, one gets $\log \mathcal{Z}_t(\tau) \le e^{-\delta m_{j+1}/2}(e^{5.79\cdot 10^{-4}} - 1) \le 5.8 \cdot 10^{-4} e^{-\delta m_{j+1}/2}$. The bound on K_j is Lemma 7.38(iii), $K_j \le 30(1 + 1/m_{j+1})X_j$ with $m_{j+1} \ge 6$, and $35 \cdot 5.8 \cdot 10^{-4} < 0.021$. (ii) The first claim compares the two numbers; $\log 15.6 + 0.8 + 0.0046 < 3.56$. The second is $(m_j + 1)\kappa_p(t) \le K_j \le 0.021$. (iii) Use the last claim of Lemma 7.38(iii): $(m+1)\kappa(\vec{\ell}) \le 30(1 + 1/m')X$ for $\vec{\ell} \in \{m, m+1\}^3$ with $m' = \lfloor m/2 \rfloor$, where X is the largest $\log \mathcal{Z}_t(\tau)$ over tori with sides in $\{m, m+1, m', m'+1\}$ and $\tau \in \{m', m'+1\}$. The proof of (i) with $m' \ge 2T_0$ in place of m_{j+1} and $m + 1 \le 2m' + 2$ gives $X \le 5.8 \cdot 10^{-4} e^{-\delta m'/2}$, and $m' \ge (m-1)/2 \ge (Ns - 1)/2$. The last clause is $0.021 e^{-\Delta(s_0 - 1/N)/4} \le \frac{1}{70}$, as $4 \log(1.47) < 1.55$, together with the decay in s. □

What this changes, and what it does not. Theorem 7.39, Proposition 7.42 and Theorem 7.43 carry the passage between scales by (K), or equivalently by the exclusion of the large branch, and the question after Theorem 7.43 was whether the gap and the one-scale datum $X_J \le \frac{1}{10}$ exclude it by themselves. With the datum (B_{24}) in place of $X_J \le \frac{1}{10}$ the answer is yes, and (Z) follows with no hypothesis at any scale but t_0: the multi-scale content of (C) and (K) is carried by positivity in space and by the gap in time. The two data differ at one scale in two respects. (B_{24}) controls boxes with sides in $[2T_0, 48T_0)$ at the time T_0, elongated by up to the factor 24 (about 12 after optimising the constants), where X_J controls boxes of aspect ratio at most about 4; and (B_{24}) asks a total thermal correction at most $\frac{1}{2}$ on a box of side $96T_0$, a density at most $\frac{1}{2}(96T_0)^{-3}$, where $X_J \le \frac{1}{10}$ is a bound on one box of the size of the scale. Positivity does not convert one into the other: the cutting coefficient of Lemma 7.51 is $12T/P$, below 1 only when the pieces are more than twelve times longer than the time, which is the same obstruction as the coefficient 15 after Lemma 7.38. Whether the gap and $X_J \le \frac{1}{10}$ imply (B_{24}) at some T_0 comparable to Ns_0 is therefore a question at one scale, and it is not decided here; for a theory with a gap and finite thermodynamics, $\log \mathcal{Z}_s(T_0)$ is the physical volume times t_0 times the pressure at the temperature $1/t_0$, so (B_{24}) is of the same nature as the pressure bounds of Proposition 7.32(b) and of (7.15), with the shape dependence that Proposition 7.48 and Theorem 10.10 describe. Second, (G_{24}) asks the gap on every spatial torus of aspect ratio below 24 with sides between $2T_0$ and NL_p, where (D_ℓ) gives it on the cubes $(NL_p)^3$ alone; so Corollary 7.40(ii) is not superseded, and whether the large branch is excluded with the gap on dyadic cubes only is not decided either. No system satisfying (G_{24}) and (B_{24}) lies on the large branch at any scale at least $2T_0$; the symmetrised rings of Proposition 7.35 violate both, by the factor N^3. For Yang–Mills at weak coupling neither (G_{24}) nor (B_{24}) is proved, any more than (D_ℓ) or (W).

7.11. **The ascent without a gap: one hypothesis at one scale.** Theorem 7.53 carries the step in time by the gap, with the invariant $y_i \le 1$, at which the thermal correction of the largest window box is of order one. With a smaller invariant the gap is not needed: the second inequality of Lemma 7.27, $D_s(2T) \le D_s(T)^2$, gains the factor $D_s(T)$ itself per doubling of the time, and once this factor is below $\frac{1}{16}$ it beats the growth of the window volume. The pressure bound at one scale, with a constant twenty times smaller than in (B_{24}), then gives the gap on every larger torus of bounded aspect ratio, hypothesis (Z), and the cylinder gap that Section 7.6 takes from (D_ℓ). Throughout, $\beta \ge 0$, all sides are at least 3, and the notation is that of Sections 7.8 and 7.10.

Theorem 7.56 (the ascent without a gap hypothesis)**.** *Let $T_0 \ge 3$ be an integer, $T_i = 2^i T_0$, and suppose that*

(B*) $\log \mathcal{Z}_s(T_0) \le \gamma V(s)$ *for every spatial torus* s *with all its sides in* $[2T_0, 48T_0)$*, where* $y_0 = 2\gamma(96T_0)^3 \le \frac{1}{20}$.

Put $y_{i+1} = 16(e^{y_i}-1)^2$ *and* $u_0 = 16.83\, y_0 \le 0.8415$*. Then:*

(a) $y_i \le \frac{1}{20}$ *and* $16.83\, y_i \le u_0^{2^i}$ *for every* $i \ge 0$*, and every spatial torus* t *with all its sides at least* $2T_i$ *satisfies* $\log \mathcal{Z}_t(T_i) \le F_i V(t)$ *with* $F_i = y_i(96T_i)^{-3}$.

(b) *If all the sides of* t *lie in* $[2T_i, 96T_i)$ *for some* $i \ge 0$*, then* $r_1(t) \le u_0^{1/T_0}$*: the torus* t *has the gap* $e^{-\delta_*}$ *with* $\delta_* = T_0^{-1}\log(1/u_0) \ge 0.172/T_0$*. This holds for every cube of side at least* $2T_0$*, and for every torus whose smallest side* a *is at least* $2T_0$ *and whose largest side is less than* $24a$.

(c) *For a cube of side* $n \ge 2T_0$*,*

$$\mathcal{Z}_n(n) - 1 \le 1.9 \cdot 10^{-11}\, u_0^{n/(2T_0)}.$$

Proof. (a) Induction on i. For $i = 0$, Proposition 7.52 with $T = T_0$ and $P = 24T_0$, and (B*), give $\log \mathcal{Z}_t(T_0) \le 2\gamma V(t) = F_0 V(t)$ for every t with sides at least $2T_0$. Suppose the statement holds for i, and let s have all its sides in $[2T_{i+1}, 48T_{i+1}) = [4T_i, 96T_i)$. Its sides are at least $2T_i$, so $D_s(T_i) = \mathcal{Z}_s(T_i) - 1 \le e^{F_i V(s)} - 1$ by the induction hypothesis. Lemma 7.27 with $M' = M'' = T_i$ and $\log(1+x) \le x$ give

$$\log \mathcal{Z}_s(T_{i+1}) \le D_s(2T_i) \le D_s(T_i)^2 \le \big(e^{F_i V(s)} - 1\big)^2.$$

The function $v \mapsto (e^{F_i v} - 1)^2/v$ is the product of the two positive increasing functions $e^{F_i v} - 1$ and $(e^{F_i v} - 1)/v$ of $v > 0$, and $V(s) < (96T_i)^3$ with $F_i(96T_i)^3 = y_i$; hence

$$\frac{\log \mathcal{Z}_s(T_{i+1})}{V(s)} \le \frac{(e^{y_i} - 1)^2}{(96T_i)^3}.$$

This bounds $g(T_{i+1}, 24T_{i+1})$, and Proposition 7.52 at the time T_{i+1} gives $\log \mathcal{Z}_t(T_{i+1}) \le 2(e^{y_i} - 1)^2 (96T_i)^{-3} V(t)$ for every t with sides at least $2T_{i+1}$. The coefficient is F_{i+1}, because $(96T_{i+1})^3 = 8(96T_i)^3$ and $8 \cdot 2(e^{y_i} - 1)^2 = y_{i+1}$. For the numbers, convexity of the exponential gives $e^y - 1 \le 20(e^{1/20} - 1)\, y \le 1.0255\, y$ on $[0, \frac{1}{20}]$, so $16(e^y - 1)^2 \le 16.83\, y^2 \le 0.8415\, y$ there. Thus $y_{i+1} \le y_i \le \frac{1}{20}$ and $16.83\, y_{i+1} \le (16.83\, y_i)^2$, whence $16.83\, y_i \le u_0^{2^i}$.

(b) Let all sides of t lie in $[2T_i, 96T_i)$. By (a), $V(t) < (96T_i)^3$ and

$$r_1^{T_i} \le D_t(T_i) \le e^{F_i V(t)} - 1 \le e^{y_i} - 1 \le 1.0255\, y_i \le 0.061\, u_0^{2^i} \le u_0^{2^i},$$

so $r_1 \le u_0^{2^i/T_i} = u_0^{1/T_0}$, and $\log(1/0.8415) > 0.172$. A cube of side $n \ge 2T_0$ has $2T_i \le n < 4T_i$ for the largest i with $2T_i \le n$. If the smallest side is $a \ge 2T_0$ and the largest is $b < 24a$, take the largest i with $2T_i \le a$; then $a < 4T_i$ and $b < 96T_i$.

(c) Let $2T_i \le n < 4T_i$ and $t = (n, n, n)$. Then $F_i V(t) < 64 T_i^3 F_i = y_i/13824 \le 3.7 \cdot 10^{-6}$, so $D_t(T_i) \le e^{y_i/13824} - 1 \le y_i/13823$. Since D_t is nonincreasing in the time (Lemma 7.12) and $n \ge 2T_i$, Lemma 7.27 gives

$$D_t(n) \le D_t(2T_i) \le D_t(T_i)^2 \le \Big(\frac{u_0^{2^i}}{16.83 \cdot 13823}\Big)^2 \le 1.9 \cdot 10^{-11}\, u_0^{2^{i+1}} \le 1.9 \cdot 10^{-11}\, u_0^{n/(2T_0)},$$

because $n/(2T_0) < 2^{i+1}$ and $u_0 < 1$. □

The recursion $y \mapsto 16(e^y - 1)^2$ contracts below its fixed point $y_c = 0.0589\ldots$, and every $y_0 < y_c$ gives the three conclusions with other constants; $\frac{1}{20}$ is chosen for the numbers. Hypothesis (B*) contains a gap on the base window, since on the cube of side $2T_0$ it says $D \le e^{8\gamma T_0^3} - 1 \le 2.3 \cdot 10^{-7}$; the theorem is a finite-size criterion in the sense of Knabe [34], in which positivity in the four directions propagates a datum at one scale to a gap at all larger ones.

Theorem F (the mass gap from one scale)**.** *Let a continuum family satisfy* (W)*, and suppose there are* $t_0 > 0$*,* $y_0 \le \frac{1}{20}$ *and* p_0*,* $N_0(p)$ *such that for all* $p \ge p_0$ *and* $N \ge N_0(p)$

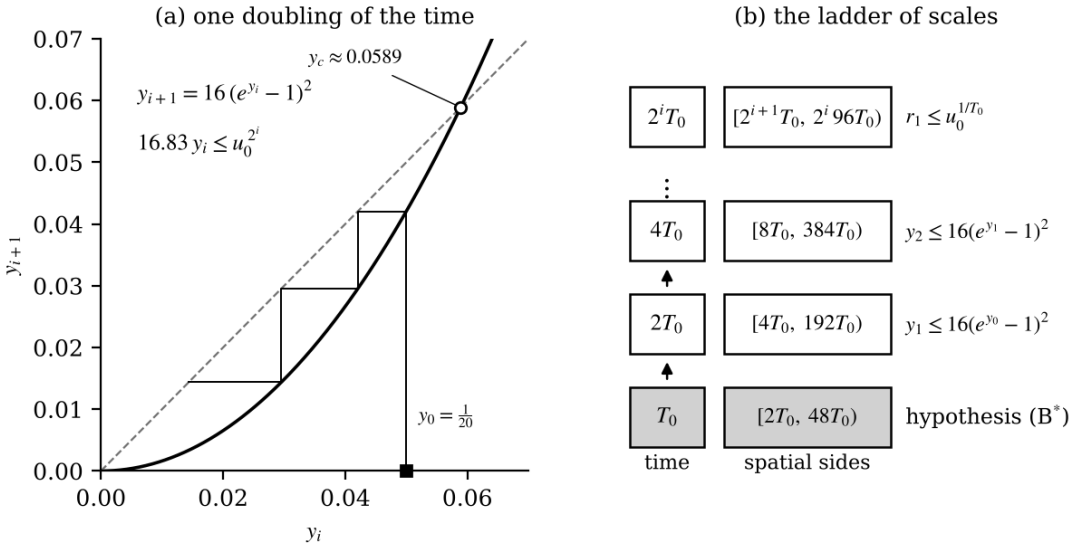


FIGURE 8. The ascent of Theorem 7.56. (a) One doubling of the time extent maps the normalised pressure y_i to at most $16(e^{y_i} - 1)^2$; below the fixed point y_c the iteration decreases doubly exponentially, and the starting value $y_0 \le \frac{1}{20}$ lies below it. (b) The hypothesis is made at the single time extent T_0 (grey); every larger scale, with its gap, is a consequence.

there is an integer T_0 with $3 \le T_0 \le Nt_0$ for which (B*) *holds at the coupling $\beta_{N,p}$ with $\gamma = y_0/(2(96T_0)^3)$. Put $u_0 = 16.83\,y_0$ and*

$$\Delta = \frac{\log(1/u_0)}{t_0} \ge \frac{0.172}{t_0}.$$

Then, for $L_p \ge 2t_0$:

(i) *$\mathcal{Z}_{N,p} - 1 \le 1.9 \cdot 10^{-11} e^{-\Delta L_p/2}$, so the family satisfies* (Z);
(ii) *every eigenvalue $\lambda \ne \lambda_0$ of the transfer matrix of the cylinder of spatial side NL_p satisfies $0 \le \lambda \le e^{-\Delta/N}\lambda_0$;*
(iii) *the conclusion of Theorem E holds with this Δ: S^χ does not depend on χ, and the Wightman theory of* (W) *has $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$ with 0 a simple eigenvalue.*

Neither (U)*, nor* (D_ℓ)*, nor the slab hypothesis* (M)*, nor* (G_{24}) *is assumed.*

Proof. Put $n = NL_p$; then $n \ge 2Nt_0 \ge 2T_0$. (i) is Theorem 7.56(c), with $u_0^{n/(2T_0)} \le u_0^{L_p/(2t_0)} = e^{-\Delta L_p/2}$ because $T_0 \le Nt_0$ and $u_0 < 1$; and $L_p \to \infty$. (ii) is Theorem 7.56(b) for the cube of side n: $r_1 \le u_0^{1/T_0} \le u_0^{1/(Nt_0)} = e^{-\Delta/N}$, the eigenvalues being nonnegative by Lemma 7.3(vii). (iii) In Step 1 of Section 7.6, hypothesis (D_ℓ) enters only through Proposition 7.4, which gives the transfer matrix of the cylinder of side NL_p a gap m with $mN \ge \Delta$, and Lemma 7.5 then gives (7.4) with the factor $e^{-mN(\sigma-\sigma_1)} \le e^{-\Delta(\sigma-\sigma_1)}$. Here (ii) is that gap, with $m = \Delta/N$, so (7.4) holds with the factor $e^{-\Delta(\sigma-\sigma_1)}$. By (i) and Proposition 7.10, (W) gives (U_w), and the proof of Corollary 7.11 derives (7.5) from (7.4) and (U_w), using the gap only through that factor. Steps 3 and 4 use (7.5) and (W) alone. □

Theorem F replaces the hypotheses (D_ℓ) or (M), (G_{24}), (Z) and (U) by one bound at one scale. That bound does not involve the side L_p, an insertion or a renormalisation constant, and it has a finite form.

Proposition 7.57 (finite form)**.** *With $\psi_s(T) = 2\log Z(s,T) - \log Z(s,2T)$ as in Section 7.8, hypothesis* (B*) *with the constant γ follows from*

(B**) $\quad Z(s,T_0)^2 \le e^{\gamma V(s)}\, Z(s,2T_0) \quad$ *for every s with all its sides in* $[2T_0, 48T_0)$,

and (B*) *with* γ *implies* (B**) *with* 2γ. (B**) *compares the partition functions of* $2\cdot(46T_0)^3$ *finite four-dimensional tori and is unchanged when* $Z(n)$ *is multiplied by* $e^{a|n|}$.

Proof. Lemma 7.33 for the time transfer matrix of s gives $\log \mathcal{Z}_s(T_0) \le \psi_s(T_0) \le 2\log \mathcal{Z}_s(T_0)$, and (B**) is $\psi_s(T_0) \le \gamma V(s)$. □

With $\rho = \mathbb{R}^{T_0}/\operatorname{Tr}\mathbb{R}^{T_0}$, the thermal state of the torus s at the time extent T_0, the ratio $Z(s,2T_0)/Z(s,T_0)^2$ is $\operatorname{Tr}\rho^2$, so (B**) says that this state is nearly pure: $\operatorname{Tr}\rho^2 \ge e^{-\gamma V(s)}$. In physical terms (B**) asks that the thermal part of the free energy at the temperature $1/t_0$, in boxes of sides between $2t_0$ and $48t_0$, be at most $y_0/(2\cdot 96^3 t_0^3) \le 2.9\cdot 10^{-8}\, t_0^{-3}$ per unit volume. Two results delimit what a proof of it must contain.

Proposition 7.58 ((B*) contains the lattice mass gap)**.** *Suppose* (B*) *holds at a coupling* β *for some* T_0*, and let* μ_n^0 *be the zero-temperature state of the cylinder of spatial side* n*. Then for every* $n \ge 2T_0$*, every bounded measurable function* F *of a slab* $[0,w]$ *and every integer* $s \ge 0$*,*

$$0 \le \mu_n^0(\vartheta\bar F\,\tau_s F) - |\mu_n^0(F)|^2 \le \|F\|_\infty^2\, e^{-\delta_* s}, \qquad \delta_* = \frac{\log(1/u_0)}{T_0},$$

and the same bound holds, for local F*, for every weak-∗ limit point of* $(\mu_n^0)_n$ *on* $\mathbb{Z}^3\times\mathbb{Z}$.

Proof. By Lemma 7.3(v) the middle term is $\langle \Psi'_F, \mathbb{R}^s\Psi'_F\rangle$ with $\Psi'_F \perp \Omega_0$. By Theorem 7.56(b), $0 \le \mathbb{R} \le e^{-\delta_*}$ on $\Omega_0^\perp$, and $\|\Psi'_F\|^2 \le \|\Psi_F\|^2 = \mu_n^0(\vartheta\bar F\, F) \le \|F\|_\infty^2$. The bound is uniform in n, so it passes to limit points. □

A proof of (B*) along couplings $\beta\to\infty$ therefore proves exponential clustering of four-dimensional lattice Yang–Mills theory at arbitrarily weak coupling, with a rate $\log(1/u_0)/T_0(\beta)$. The base scale $T_0(\beta)$ cannot stay bounded, by the next theorem: at a fixed lattice size the gap closes as $\beta\to\infty$. We state it for SU(2), where the weight of Section 2 is $\exp(c\sum_p \operatorname{Re}\operatorname{Tr} Q_p)$ with $c = 2\beta$.

Theorem 7.59 (the gap closes at a fixed lattice size)**.** *Let* $G = \mathrm{SU}(2)$*,* $c = 2\beta \ge \frac{1}{32}$ *and*

$$\varepsilon(c) = \frac{12(\log c + 5.42)}{c}.$$

For every spatial torus $t = (\ell_1,\ell_2,\ell_3)$ *with sides at least* 3 *and* $(\ell_1\ell_2)^2\varepsilon(c) < 3$*,*

$$1 - r_1(t) \le \frac{2(\ell_1+\ell_2)^2\,\varepsilon(c)}{1-(\ell_1\ell_2)^2\varepsilon(c)/3}.$$

Proof. Write $A_p = 2 - \operatorname{Tr} Q_p = \frac12\|Q_p - 1\|_F^2 \in [0,4]$, with the Frobenius norm, since $\|U-1\|_F^2 = 4 - 2\operatorname{Tr} U$ for $U\in\mathrm{SU}(2)$.

Step 1: an energy bound uniform in the torus. On a torus with $|n|$ sites, $4|n|$ links and $6|n|$ plaquettes let $A = \sum_p A_p$ and $\tilde Z(c) = \int e^{-cA}\prod_e dQ_e \le 1$, so that the Wilson measure is $\tilde Z(c)^{-1}e^{-cA}\prod_e dQ_e$ and $\mathbb{E}_c[A] = -(\log\tilde Z)'(c)$. Since $(\log\tilde Z)'' = \operatorname{Var}_c(A) \ge 0$, the mean $\mathbb{E}_c[A]$ is nonincreasing in c, and

$$\frac{c}{2}\,\mathbb{E}_c[A] \le \int_{c/2}^{c} \mathbb{E}_{c'}[A]\,dc' = \log\tilde Z(c/2) - \log\tilde Z(c) \le -\log\tilde Z(c).$$

If every link has $\|Q_e - 1\|_F \le r$, then $\|Q_p - 1\|_F \le 4r$ and $A_p \le 8r^2$. In the class angle $\theta\in[0,\pi]$, with $\operatorname{Tr} U = 2\cos\theta$, Haar measure is $\frac{2}{\pi}\sin^2\theta\,d\theta$ and $\|U-1\|_F = \sqrt{8}\sin(\theta/2)$. The ball of radius $r\le 2$ is $\{\theta\le\theta_r\}$ with $\theta_r = 2\arcsin(r/\sqrt8)\in[r/\sqrt2,\pi/2]$, and by $\sin\theta\ge 2\theta/\pi$ on $[0,\pi/2]$ its measure is at least $\frac{2}{\pi}\int_0^{\theta_r}(2\theta/\pi)^2 d\theta = 8\theta_r^3/(3\pi^3) \ge 2\sqrt2\, r^3/(3\pi^3) \ge r^3/33$. With $r^2 = 1/(8c)$, which has $r\le 2$ because $c\ge\frac{1}{32}$, restricting the integral to these balls gives $\tilde Z(c) \ge e^{-6|n|}(r^3/33)^{4|n|}$, so

$$\mathbb{E}_c[A] \le \frac{2}{c}\,|n|(6 + 4\log 33 + 6\log(8c)) \le \frac{12}{c}\,|n|\,(\log c + 5.42).$$

The $|n|$ translates of a plaquette have equal expectation and $A_p \geq 0$, so $\mathbb{E}_c[A_p] \leq \varepsilon(c)$ for every plaquette on every torus, and by Lemma 7.3(v) the same bound holds for $\mu_t^0 = \lim_M \mu_{t,M}$.

Step 2: three slice functions of mean zero. On the slice at time 0 let P_1 and P_2 be the ordered products of the links from the origin once around the directions 1 and 2, and $F_1 = \operatorname{Tr} P_1$, $F_2 = \operatorname{Tr} P_2$, $F_3 = \operatorname{Tr}(P_1 P_2)$; they are real, continuous and gauge invariant, being traces of closed loops. The centre flip z_1, which replaces $Q_{(x,1)}$ by $-Q_{(x,1)}$ on all links in the direction 1 with $x_1 = 0$, at all times, preserves Haar measure and every plaquette variable, because the two links in the direction 1 of a plaquette have the same coordinate x_1, so a plaquette contains none or two of the flipped links. It therefore preserves $\mu_{t,M}$ and μ_t^0, and it maps F_1 to $-F_1$ and F_3 to $-F_3$. The flip z_2 maps F_2 to $-F_2$. So $\mu_t^0(F_k) = 0$ for $k = 1, 2, 3$, that is, $F_k \Omega_0 \perp \Omega_0$.

Step 3: the three functions are not all small. For $A, B \in \mathrm{SU}(2)$ with $x = \operatorname{Tr} A$, $y = \operatorname{Tr} B$ and $z = \operatorname{Tr} AB$, Fricke's identity is

$$\operatorname{Tr}(ABA^{-1}B^{-1}) = x^2 + y^2 + z^2 - xyz - 2;$$

it follows from $\operatorname{Tr}(AB) + \operatorname{Tr}(AB^{-1}) = xy$, applied three times.[13] Put $\sigma = x^2 + y^2 + z^2$ and $\eta = 2 - \operatorname{Tr}(ABA^{-1}B^{-1}) \geq 0$. Since $|xyz| \leq (\sigma/3)^{3/2}$, the function $\phi(\sigma) = \sigma + (\sigma/3)^{3/2}$ satisfies $\phi(\sigma) \geq 4 - \eta$. As $\phi(3) = 4$ and $\phi' \geq 1$, $\sigma < 3$ gives $3 - \sigma \leq \phi(3) - \phi(\sigma) \leq \eta$; so $\sigma \geq 3 - \eta$ in every case. Apply this to $A = P_1$ and $B = P_2$. The commutator $K = P_1 P_2 P_1^{-1} P_2^{-1}$ is the holonomy of the boundary of the sheet of the $\ell_1 \ell_2$ plaquettes of the $(1,2)$-plane through the origin, cut open along the two cycles. Adding the plaquettes of the sheet one at a time, row by row, so that each step keeps a disc, multiplies the boundary holonomy by a conjugate of $Q_p^{\pm 1}$; so K is a product of $\ell_1 \ell_2$ conjugates of the $Q_p^{\pm 1}$, and $\|UV - 1\|_F \leq \|U - 1\|_F + \|V - 1\|_F$ for unitary matrices gives $\|K - 1\|_F \leq \sum_p \|Q_p - 1\|_F$. Hence, by the Cauchy–Schwarz inequality,

$$\eta = \tfrac{1}{2}\|K - 1\|_F^2 \leq \tfrac{1}{2}\Big(\sum_p \sqrt{2A_p}\Big)^2 \leq \ell_1 \ell_2 \sum_p A_p, \qquad \mu_t^0(\eta) \leq (\ell_1 \ell_2)^2 \varepsilon,$$

the sums over the plaquettes of the sheet. So $\sum_k \mu_t^0(F_k^2) \geq 3 - (\ell_1 \ell_2)^2 \varepsilon$, and some k has $\|F_k \Omega_0\|^2 = \mu_t^0(F_k^2) \geq 1 - (\ell_1 \ell_2)^2 \varepsilon / 3$.

Step 4: the Dirichlet form. For a real function F of the slice, Lemma 7.3(v) with $w = w' = 0$ and $s = 1$, and the invariance of μ_t^0 under time translations, give

$$\langle F\Omega_0, (1 - \mathbb{R}) F \Omega_0 \rangle = \mu_t^0(F^2) - \mu_t^0(F\, \tau_1 F) = \tfrac{1}{2}\, \mu_t^0\big((F - \tau_1 F)^2\big).$$

The loop of F_k at time 0 and its translate at time 1, joined by the temporal link at the origin, bound a strip of m_k temporal plaquettes, with $m_1 = \ell_1$, $m_2 = \ell_2$ and $m_3 = \ell_1 + \ell_2$. As in Step 3 the two holonomies X and Y, the second conjugated by the temporal link, satisfy $\|X - Y\|_F \leq \sum_p \|Q_p - 1\|_F$ over the strip, and $|\operatorname{Tr} X - \operatorname{Tr} Y| \leq \sqrt{2}\, \|X - Y\|_F$. So $(F_k - \tau_1 F_k)^2 \leq 4 m_k \sum_p A_p$ and

$$\tfrac{1}{2}\, \mu_t^0\big((F_k - \tau_1 F_k)^2\big) \leq 2 m_k^2\, \varepsilon \leq 2(\ell_1 + \ell_2)^2 \varepsilon.$$

Step 5. On $\Omega_0^{\perp}$ one has $\mathbb{R} \leq r_1$, so for the k of Step 3, $1 - r_1 \leq \langle F_k \Omega_0, (1 - \mathbb{R}) F_k \Omega_0 \rangle / \|F_k \Omega_0\|^2$, which is the claim. □

Corollary 7.60 (the base scale diverges)**.** *If* (B*) *holds for* SU(2) *with the base scale* T_0 *at a coupling with* $\varepsilon = \varepsilon(2\beta) \leq 10^{-3}$*, then*

$$T_0 > \Big(\frac{3}{32\, \varepsilon}\Big)^{1/4} = \Big(\frac{\beta}{64(\log 2\beta + 5.42)}\Big)^{1/4}.$$

[13] The Cayley–Hamilton identity $B + B^{-1} = y \cdot 1$ gives $\operatorname{Tr}(AB) + \operatorname{Tr}(AB^{-1}) = xy$ for $A, B \in \mathrm{SU}(2)$. Hence $\operatorname{Tr}(ABA^{-1}B^{-1}) = \operatorname{Tr}(AB)\operatorname{Tr}(A^{-1}B^{-1}) - \operatorname{Tr}(ABBA)$, $\operatorname{Tr}(A^{-1}B^{-1}) = \operatorname{Tr}(BA) = z$, and $\operatorname{Tr}(ABBA) = \operatorname{Tr}(A^2B^2) = \operatorname{Tr}(A^2B)y - \operatorname{Tr}(A^2)$, with $\operatorname{Tr}(A^2B) = xz - y$ and $\operatorname{Tr}(A^2) = x^2 - 2$; so the commutator has trace $z^2 - (xz - y)y + x^2 - 2$.

Proof. On the cube t of side $2T_0$, (B*) gives $r_1^{T_0} \le D_t(T_0) \le e^{8\gamma T_0^3} - 1 = e^{4y_0/96^3} - 1 \le 2.3 \cdot 10^{-7}$, so $r_1 \le e^{-x}$ with $x = 15.28/T_0$, and by concavity $1 - r_1 \ge (1 - e^{-1}) \min\{1, x\} \ge \min\{0.632,\, 9.6/T_0\}$. Suppose $T_0^4 \varepsilon \le \frac{3}{32}$. Then $(2T_0)^4 \varepsilon \le \frac{3}{2}$, and Theorem 7.59 with $\ell_1 = \ell_2 = 2T_0$ gives $1 - r_1 \le 64 T_0^2 \varepsilon$. If $T_0 \le 15$ this is at most $64\sqrt{3/32}\,\sqrt{\varepsilon} \le 0.62 < 0.632$, and if $T_0 \ge 16$ it gives $T_0^3 \varepsilon \ge 0.15$, while $T_0^3 \varepsilon \le \frac{3}{32 \cdot 16} < 0.006$. Both are impossible. □

Thus no tuning with $\beta_N \to \infty$ has its base scale at a bounded number of lattice units, and a bound of the shape of Proposition 7.25, with a rate fixed in lattice units, cannot serve along it (Proposition 7.26); the mechanism is the centre symmetry, which gives the Polyakov loops mean zero while the action no longer suppresses their fluctuation at a fixed lattice size. The proof of Theorem 7.56 uses of the Wilson action only that Z is symmetric in the four sides and is, in each direction, the trace of a power of a nonnegative self-adjoint Hilbert–Schmidt operator with a simple largest eigenvalue. What remains open for four-dimensional Yang–Mills theory at weak coupling is therefore (W) and the one bound (B**), for integers $T_0(\beta) \to \infty$ along a sequence $\beta \to \infty$:

$$Z(s, T_0)^2 \le \exp\Big(\frac{V(s)}{40\,(96T_0)^3}\Big) Z(s, 2T_0) \qquad \text{for every } s \text{ with all its sides in } [2T_0, 48T_0).$$

Neither is proved here.

8. The cube condition at strong coupling

Keep the setting of Sec. 6, with a bi-invariant metric δ of diameter D on G, invariant under inversion, and $d_E(U, U') = \sum_e \delta(U_e, U'_e)$. Two distinct links are *partners* if they share a plaquette; two links share at most one plaquette, and every link has eighteen partners, three in each of its six plaquettes. The direction field σ of Sec. 3 has period two, so it is defined on Λ_L. By Lemma 3.2(i) the links of F with an end point v are exactly $(v, \sigma(v))$ and $(v - e_{\sigma(v)}, \sigma(v))$, and $\sigma(v - e_{\sigma(v)}) = \sigma(v)$.

Proposition 8.1. *At $\beta = 0$,* BP$(n, 1)$ *holds for every n.*

Proof. For $\varphi \equiv 1$, μ is a product of Haar measures. Write $f = \sum_S f_S$ with f_S a function of the links of S and $\int f_S \, dU_e = 0$ for $e \in S$; the f_S are orthogonal and $\mathcal{P}_x f = \sum_{S \cap V_x = \emptyset} f_S$, so $\sum_x \mu((f - \mathcal{P}_x f)^2) = \sum_{S \ne \emptyset} \|f_S\|^2 \#\{x : S \cap V_x \ne \emptyset\} \ge n^4 \operatorname{Var}_\mu(f)$. □

Gauge fixing inside a cube. For a cube $Q_x = x + [0, n)^4$ write $\hat{v} = v - x$ and call v *interior* if $\hat{v} \in [1, n-1]^4$; let I_x be the set of interior vertices. For $z \in I_x$ put $\tau(z) = (z - e_{\sigma(z)}, \sigma(z))$, a link of F with head z, and $T_x = \{\tau(z) : z \in I_x\}$. Every link with an end point in I_x has its base point in Q_x, so $T_x \subset V_x$, the map τ is injective, and no link outside V_x touches I_x. Call the links of $V_x \setminus T_x$ *free*. For $\psi \in G^{V_x \setminus T_x}$ let $[\psi, U]$ be the configuration equal to ψ on the free links, to 1 on T_x and to U outside V_x, and let $\gamma^T_{x,U}$ be the probability measure on $G^{V_x \setminus T_x}$ with density proportional to $\prod_{p \cap V_x \ne \emptyset} \varphi([\psi, U]_p)$.

Lemma 8.2. *For every bounded measurable $f \in \mathcal{A}$, $\mathcal{P}_x f(U) = \int f([\psi, U])\, \gamma^T_{x,U}(d\psi)$.*

Proof. Let $\sigma' \in G^{V_x}$ and write (σ', U) for the configuration equal to σ' on V_x and to U elsewhere. Order I_x by the sum of the coordinates of $\hat{z}$; then $z - e_{\sigma(z)}$ precedes z or is not interior. Put $g_v = 1$ for $v \notin I_x$ and recursively $g_z = g_{z - e_{\sigma(z)}} \sigma'_{\tau(z)}$. The gauge transformation g sets every link of T_x to 1 and fixes every link outside V_x, so $g \cdot (\sigma', U) = [\psi, U]$ with $\psi = (g \cdot \sigma')|_{V_x \setminus T_x}$. Since g depends only on σ'_{T_x}, for fixed σ'_{T_x} the map $\sigma'_{V_x \setminus T_x} \mapsto \psi$ multiplies each coordinate on both sides by fixed elements, so $\sigma' \mapsto (\psi, \sigma'_{T_x})$ preserves Haar measure. Plaquette variables change by conjugation and φ is a class function, so the density of $\gamma^0_{V_x}(\cdot \mid U)$ at σ' is proportional to $\prod_{p \cap V_x \ne \emptyset} \varphi([\psi, U]_p)$, while $f(\sigma', U) = f([\psi, U])$. Integrating out σ'_{T_x} gives the claim. □

A link (b,μ) of V_x is a *face link* if $\hat b_i = 0$ for some $i \neq \mu$, that is, if both its end points lie in one lower face $\{\hat v_i = 0\}$ of the cube.

Lemma 8.3. (i) *If all four links of a plaquette are free, its link of F is a face link.*
(ii) *Every free link has at most fifteen free partners.*

Proof. (i) Let $\phi = (y,s)$ be the link of F of the plaquette p and $h = y + e_s$. If h were interior, then $\sigma(h) = s$ and $\phi = \tau(h) \in T_x$. Since $y \in Q_x$, $\hat h_i = \hat y_i \in [0, n-1]$ for $i \neq s$ and $\hat h_s \in [1,n]$, so $\hat h_s = n$ or $\hat y_i = 0$ for some $i \neq s$. If $\hat h_s = n$, the link of p parallel to its second direction ν through h has a base point with s-coordinate n and is not in V_x. Hence ϕ is a face link.

(ii) Let $k = (b,\mu)$ be free and $m(p)$ the number of free links of a plaquette $p \ni k$; k has $\sum_{p\ni k}(m(p)-1)$ free partners. If k is a face link, $\hat b_i = 0$ with $i \neq \mu$, the plaquette of k with base point $b - e_i$ in the plane $\{\mu, i\}$ has its other three links based at i-coordinate -1, outside V_x; the other five plaquettes give at most 3 each, 15 in all. Otherwise k has at most $12 + M$ free partners, where M counts the plaquettes $p \ni k$ with $m(p) = 4$, whose link ϕ of F is a face link by (i), so $\phi \neq k$. If $k \in F$ then $M = 0$; let k be active and ϕ lie in the face $\{\hat v_i = 0\}$. If ϕ shares an end point v with k, then $v = b$ (for $v = b + e_\mu$ either k is a face link or $\hat b_i = -1$), $\hat b_i = 0$ and $i = \mu$; there are two links of F at b, each in at most one plaquette of k. If ϕ is opposite to k, it is $(b \pm e_\nu, \mu)$ in a plaquette of the plane $\{\mu,\nu\}$; as k is not a face link, $\nu = i$, the sign is $-$ and $\hat b_i = 1$, and by Lemma 3.2(ii) the plane $\{\mu,\nu\}$ is the one plane in which the link of F is opposite to k, so this happens for at most one plaquette. Hence $M \le 3$. □

For a free link e the conditional law $\gamma^T_e(\cdot \mid \psi, U)$ of ψ_e under $\gamma^T_{x,U}$ has density proportional to $\prod_{p\ni e}\varphi([\psi,U]_p)$. Let C^T_{ek} be the supremum of $W_1(\gamma^T_e(\cdot \mid \psi,U), \gamma^T_e(\cdot \mid \psi',U'))/\delta([\psi,U]_k, [\psi',U']_k)$ over pairs for which $[\psi,U]$ and $[\psi',U']$ differ only at a link k, free or outside V_x; it vanishes unless e and k are partners. For $j \notin V_x$ let $b(V_x,j)$ be the number of links of V_x that are partners of j, and

$$R_x(j) = \sup \frac{W_1^{d_x}(\gamma^T_{x,U}, \gamma^T_{x,U'})}{\delta(U_j, U'_j)}, \qquad d_x(\psi,\psi') = \sum_{e \in V_x \setminus T_x} \delta(\psi_e, \psi'_e),$$

the supremum over configurations that differ only at j.

Lemma 8.4. *If $C^T_{ek} \le c$ for all free e and all k, with $15c < 1$, then $R_x(j) \le c\, b(V_x,j)/(1-15c)$.*

Proof. Let U, U' differ only at j, let π minimise $\int d_x\, d\pi$ over couplings of $\gamma^T_{x,U}$ and $\gamma^T_{x,U'}$, and let $\varrho(e) = \int \delta(\psi_e,\psi'_e)\, d\pi$. Replacing (ψ_e,ψ'_e) by a measurably chosen optimal coupling of the two conditional laws of ψ_e [58, Cor. 5.22] gives another coupling, so by minimality and the triangle inequality $\varrho(e) \le c\sum_k \varrho(k) + c[e \text{ a partner of } j]\delta(U_j, U'_j)$, the sum over the free partners k of e. Summing over e and using Lemma 8.3(ii), $(1-15c)\sum_e \varrho(e) \le c\, b(V_x,j)\delta(U_j,U'_j)$. □

Lemma 8.5. *For $n \ge 2$, $L \ge 2n+4$ and a link j, $\sum_{x:\, j\notin V_x} b(V_x,j) = 18n^3 - 3n^2$.*

Proof. Let b be the base point of j and ν its direction. For each $\mu \neq \nu$ the partners in the two plaquettes of the plane $\{\mu,\nu\}$ have base points $b + \Delta$, $\Delta \in \{e_\nu, e_\mu, 0, -e_\mu, -e_\mu, e_\nu - e_\mu\}$. With $z = b - x$ and $Q = [0,n)^4$, a link based at $b+\Delta$ lies in V_x exactly when $z + \Delta \in Q$, and $j \in V_x$ when $z \in Q$. The sum is $\sum_\Delta (n^4 - |Q \cap (Q - \Delta)|)$, which is 0 for $\Delta = 0$, n^3 for $\Delta = \pm e_\mu, e_\nu$ and $2n^3 - n^2$ for $\Delta = e_\nu - e_\mu$, that is $6n^3 - n^2$ for each of the three μ. □

Lemma 8.6. (i) *For $G = \mathrm{SU}(2)$, the chordal metric and $\varphi = e^{2\beta \operatorname{Re}\operatorname{Tr}(\cdot)}$, $C^T_{ek} \le |\beta| = |\beta_W|/4$.*
(ii) *Let G have semisimple Lie algebra, δ the geodesic distance of $\langle X,Y\rangle = -\operatorname{Tr}(XY)$, and $6N|\beta| < \kappa_G$. Then $C^T_{ek} \le N|\beta|/(\kappa_G - 6N|\beta|)$.*

Proof. (i) The conditional law of ψ_e is the von Mises–Fisher law ν_b with $b = 4\beta \sum_{p \ni e} \bar{s}_p$, where s_p is the product of the other three variables of p, as in Sec. 4. A partner k enters one s_p, and moving it by δ_k moves $\bar{s}_p$ by δ_k, since left and right multiplication and inversion are isometries; the sharp quarter, Theorem C.4, gives the bound. Links of T_x do not move.

(ii) The conditional law is $\nu \propto e^{-V} d\psi_e$ with $V = -N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr}[\psi, U]_p$; by (5.1) each term has second derivative $\operatorname{Re} \operatorname{Tr}(Y^2 W)$ with $|Y| = |X|$, so $\operatorname{Ric}_G + \operatorname{Hess} V \geq K_1 = \kappa_G - 6N|\beta|$. Lemma 5.5 with one factor and the argument of Lemma 5.6 give $|\operatorname{Cov}_\nu(g,h)| \leq \|\nabla g\|_\infty \|\nabla h\|_\infty / K_1$. Move the partner k along a unit-speed geodesic; the derivative of $\nu_s(g)$ is $-\operatorname{Cov}_{\nu_s}(g, \partial_s V_s)$, and $\partial_s V_s$ comes from the one plaquette containing e and k, with gradient in ψ_e of norm at most $N|\beta|$ by (5.3). For smooth 1-Lipschitz g this gives $|\nu_0(g) - \nu_\delta(g)| \leq N|\beta|\delta/K_1$; convolution on G preserves Lipschitz constants for a bi-invariant metric, and Kantorovich–Rubinstein duality concludes. □

Proposition 8.7. *If $\sum_{x:\, i \notin V_x} R_x(i) \leq \alpha n^4$ for every link i, with $\alpha < 1$, then $\mathrm{BP}(n, 1-\alpha)$ holds.*

Proof. For a function h let $\operatorname{Lip}(h)$ be the least constant with $|h(U) - h(U')| \leq \operatorname{Lip}(h)\delta(U_i, U_i')$ whenever U, U' differ only at one link i; then $|h(U) - h(U')| \leq \operatorname{Lip}(h) d_E(U, U')$. Let $h \in \mathcal{A}$ be Lipschitz and U, U' differ only at i. If $i \in V_x$, $\mathcal{P}_x h(U) = \mathcal{P}_x h(U')$. If $i \notin V_x$, Lemma 8.2 and an optimal coupling π of $\gamma^T_{x,U}$ and $\gamma^T_{x,U'}$ give

$$|\mathcal{P}_x h(U) - \mathcal{P}_x h(U')| \leq \int |h([\psi, U]) - h([\psi', U'])| \, d\pi \leq \operatorname{Lip}(h)(R_x(i) + 1)\delta(U_i, U_i').$$

Since i lies in n^4 of the cubes, $\operatorname{Lip}(\mathcal{K}h) \leq (1 - \kappa_0)\operatorname{Lip}(h)$ with $\kappa_0 = (1-\alpha)n^4/L^4$, and as $\mathcal{K}$ preserves $\mathcal{A}$, $\operatorname{Var}_\mu(\mathcal{K}^t h) \leq \frac{1}{2}(1-\kappa_0)^{2t} \operatorname{Lip}(h)^2 D^2 |E|^2$. The restriction of $\mathcal{K}$ to the mean-zero part of $\mathcal{A}$ is self-adjoint with $0 \leq \mathcal{K} \leq I$, and $\operatorname{Var}_\mu(\mathcal{K}^t h) = \int \lambda^{2t} d\nu_h$ for its spectral measure ν_h at $h - \mu(h)$, so ν_h lives on $[0, 1-\kappa_0]$. Lipschitz functions are dense in $L^2(\mu)$, and averaging over the compact gauge group, the orthogonal projection onto $\mathcal{A}$, does not increase Lip, because gauge transformations act link by link by isometries; so Lipschitz functions are dense in $\mathcal{A}$, and $\langle f, (I - \mathcal{K}) f \rangle \geq \kappa_0 \operatorname{Var}_\mu(f)$ for real $f \in \mathcal{A}$, which is $\mathrm{BP}(n, 1-\alpha)$. □

Theorem 8.8. *Let $n \geq 2$ and $L \geq 2n + 4$. If $C^T_{ek} \leq c$ for all cubes, free links e and links k, with $15c < 1$, then $\mathrm{BP}(n, 1 - \alpha_n)$ holds whenever*

$$\alpha_n = \frac{c\,(18n - 3)}{(1 - 15c)\, n^2} < 1.$$

In particular:

(i) *for $G = \mathrm{SU}(2)$ and $|\beta_W| < \frac{4}{15}$, with $\alpha_n = |\beta_W|(18n-3)/((4 - 15|\beta_W|)n^2)$;*

(ii) *for $G \subset \mathrm{U}(N)$ with semisimple Lie algebra and $21N|\beta| < \kappa_G$, with $\alpha_n = N|\beta|(18n - 3)/((\kappa_G - 21N|\beta|)n^2)$.*

For fixed β in these ranges α_n decreases to 0, so $\mathrm{BP}(n, \varpi)$ holds with $\varpi \to 1$ as $n \to \infty$, uniformly in the torus.

Proof. Lemmas 8.4 and 8.5 give $\sum_{x:\, i \notin V_x} R_x(i) \leq c(18n^3 - 3n^2)/(1 - 15c) = \alpha_n n^4$, and Proposition 8.7 applies. Lemma 8.6 gives $c = |\beta_W|/4$ for SU(2), and $c = N|\beta|/(\kappa_G - 6N|\beta|)$ in general, for which $15c < 1$ reads $21N|\beta| < \kappa_G$ and $c/(1-15c) = N|\beta|/(\kappa_G - 21N|\beta|)$. □

For SU(2) the condition $\alpha_n < 1$ reads $|\beta_W|(15n^2 + 18n - 3) < 4n^2$; at $|\beta_W| = \frac{1}{4}$ it holds for $n \geq 18$, and its threshold increases to $\frac{4}{15}$ as $n \to \infty$. At strong coupling the condition is therefore *easier* for large cubes, because the response of a cube to one boundary link is spread over n^4 links while it touches only $O(n^3)$ of them.

A weighted count. For a free link e let $N(e)$ be its set of free partners. The proof of Lemma 8.4 gives, for each free e,

$$\varrho(e) \leq \sum_{k \in N(e)} C^T_{ek} \varrho(k) + C^T_{ej} [e \text{ a partner of } j] \, \delta(U_j, U_j'). \tag{8.1}$$

Lemma 8.9. *Let w be a function on the free links with values in $[w_-, w_+]$, $w_- > 0$, with $\sum_{k\in N(e)} w(k) \le \lambda w(e)$ for every free e. If $C^T_{ek} \le c$ for all free e and all k, and $c\lambda < 1$, then $R_x(j) \le (w_+/w_-)\, c\, b(V_x, j)/(1 - c\lambda)$.*

Proof. Multiply (8.1) by $w(e)$ and sum over e. Since $k \in N(e)$ exactly when $e \in N(k)$, $\sum_e w(e)\varrho(e) \le c\lambda \sum_k w(k)\varrho(k) + c\, w_+ b(V_x, j)\delta(U_j, U'_j)$, and $\sum_e \varrho(e) \le w_-^{-1} \sum_e w(e)\varrho(e)$. □

Write a link of V_x as $(x + b, \mu)$ with $b \in [0, n-1]^4$, let $z(e) = \#\{i \ne \mu : b_i = 0\}$ be the number of lower faces containing it, and call it a *foot* if $b_\mu = 0$. On free links put $w = 45$ if $z = 1$ and e is active and not a foot, $w = 47$ if $z = 1$ and $e \in F$ or e is a foot, $w = 48$ if $z = 0$ and e is an active foot, and $w = 40$ otherwise.

Proposition 8.10. *For every $n \ge 2$, every cube and every free link e, $\sum_{k\in N(e)} w(k) \le \frac{72}{5} w(e)$.*

Proof. Work in the coordinates b, so that $(y, \nu) \in F$ when $\sigma(x + y) = \nu$. Let $e = (b, \mu)$. For $\nu \ne \mu$ the other links of the plaquette p^+_ν of base point b in the plane $\{\mu, \nu\}$ are $P_1 = (b + e_\mu, \nu)$, $P_2 = (b + e_\nu, \mu)$, $P_3 = (b, \nu)$, and those of the plaquette p^-_ν of base point $b - e_\nu$ are $P_4 = (b - e_\nu, \mu)$, $P_5 = (b - e_\nu + e_\mu, \nu)$, $P_6 = (b - e_\nu, \nu)$. If e is active, with $s = \sigma(x + b)$, $t = \sigma(x + b + e_\mu)$ and ρ the fourth direction, the links of F among them are P_3, P_6 for $\nu = s$, P_1, P_5 for $\nu = t$ and P_2, P_4 for $\nu = \rho$: the first four by Lemma 3.2(i) at b and at $b + e_\mu$, and in the plane $\{\mu, \rho\}$ the link of F is opposite to e by Lemma 3.2(ii). If $e \in F$, none of them is in F.

Reduction to the orthant. Call a link with base point in $O = \mathbb{Z}^4_{\ge 0}$ *O-free* if it is active or has $z \ge 1$, extend w to such links by the same rule, and count as partners of e the O-free links of the plaquettes of e with base point in O. If $b_\nu = 0$, the links of p^-_ν other than e lie outside V_x. A link $(y, \mu') \in V_x$ lies in T_x exactly when it is in F and its head is interior, that is, $z = 0$ and $y_{\mu'} \le n - 2$; so free links are O-free except the *exceptional* ones, links of F with $z = 0$ and $y_{\mu'} = n - 1$, of weight 40. An exceptional link ϕ in a plaquette p is its link of F, and p has a link ℓ based at μ'-coordinate n, which lies outside V_x, is active, has its base point in O (as $z(\phi) = 0$), and has $w(\ell) \ge 40$; exchanging ϕ for ℓ shows that the orthant sum dominates the sum over $N(e)$ for non-exceptional e. An exceptional e has at most two partners in V_x in each plaquette, each of weight at most 48, and $12 \cdot 48 = 576 = \frac{72}{5} \cdot 40$.

The orthant count. Fix $\nu \ne \mu$, put $f = [b_\mu = 0]$, $\beta = b_\nu$, and let z' be the number of zeros among b_i, $i \notin \{\mu, \nu\}$. From the base points, $z(P_1) = z(P_2) = z(P_5) = z'$, $z(P_3) = z(P_6) = z' + f$, $z(P_4) = z' + [\beta = 1]$; P_2, P_4 are feet when $f = 1$, P_1, P_3 when $\beta = 0$, P_5, P_6 when $\beta = 1$; and p^-_ν counts only when $\beta \ge 1$. An O-free link has weight 40 if $z \ge 2$, 47 or 45 if $z = 1$ according as it is in F or a foot or neither, and 48 or 40 if $z = 0$ (then it is active) according as it is a foot or not; a link of F with $z = 0$ contributes nothing. Let $C(\text{role}, f, \beta, z')$ be the resulting weight of the O-free links among $P_1, \dots, P_6$, the role of ν being s, t, ρ, or F when $e \in F$. For $z' = 2$, $C = 120$ if $\beta = 0$ and $C = 240$ otherwise; for $z' \le 1$ the values are

		$z' = 0$, $\beta =$			$z' = 1$, $\beta =$		
role	f	0	1	≥ 2	0	1	≥ 2
s	0	88	173	160	139	271	274
s	1	143	277	270	134	259	264
t	0	88	173	160	139	271	274
t	1	95	187	186	134	261	268
ρ	0	96	223	160	141	271	274
ρ	1	95	227	170	134	259	264
F	0	136	261	240	139	269	270
F	1	143	275	266	134	259	264

For instance, for the role s, $f = 1$, $\beta = 1$, $z' = 0$ the weights of $P_1, \dots, P_6$ are $40, 48, 47, 47, 48, 47$. The orthant sum is $\sum_{\nu\ne\mu} C(\text{role}(\nu), f, b_\nu, z'_\nu)$, where $z'_\nu = z(e) - 1$ if

$b_\nu = 0$ and $z'_\nu = z(e)$ otherwise, and $\frac{72}{5}w(e)$ is 576, 648, 676.8 or 691.2 for $w(e) = 40, 45, 47, 48$. For active e with $z = 0$ the sum is at most $173 + 173 + 223 = 569$ ($f = 0$, $w = 40$) or $277 + 187 + 227 = 691$ ($f = 1$, $w = 48$). For active e with $z = 1$ it is at most $96 + 274 + 274 = 644$ ($f = 0$, $w = 45$), and for $f = 1$, $w = 47$, at most $143 + 268 + 264$, $95 + 264 + 264$ or $95 + 264 + 268$ as the zero direction has role s, t or ρ, all at most 675. For active e with $z = 2$ it is at most $141 + 139 + 240 = 520$, for $z = 3$ it is 360, with $w = 40$. For $e \in F$ (then $z \geq 1$) it is at most $136 + 270 + 270 = 676$ or $143 + 264 + 264 = 671$ when $z = 1$, $w = 47$; at most 518 when $z = 2$; and 360 when $z = 3$, with $w = 40$. □

Theorem 8.11. *Let $n \geq 2$ and $L \geq 2n + 4$. If $C^T_{ek} \leq c$ for all cubes, free links e and links k, with $\frac{72}{5}c < 1$, then $\mathrm{BP}(n, 1 - \alpha_n)$ holds whenever $\alpha_n = 6c(18n - 3)/((5 - 72c)n^2) < 1$. In particular:*

(i) *for $G = \mathrm{SU}(2)$ and $|\beta_W| < \frac{5}{18}$, with $\alpha_n = 3|\beta_W|(18n - 3)/(2(5 - 18|\beta_W|)n^2)$;*
(ii) *for $G = \mathrm{SU}(3)$, the Hilbert–Schmidt distance and $0 \leq \beta_W \leq 0.45$.*

Proof. Lemma 8.9 with Proposition 8.10, $\lambda = \frac{72}{5}$ and $w_+/w_- = \frac{6}{5}$, and Lemma 8.5 give $\sum_{x:\, i \notin V_x} R_x(i) \leq \frac{6}{5}c(18n^3 - 3n^2)/(1 - \frac{72}{5}c) = \alpha_n n^4$, and Proposition 8.7 applies; its proof uses only that δ is bi-invariant and invariant under inversion. For (i), $c = |\beta_W|/4$ by Lemma 8.6. For (ii) let $r = 2\beta_W$, $Z_*(r) = \int_{\mathrm{SU}(3)} e^{r \operatorname{Re}\operatorname{Tr} U} dU$, $\epsilon(r) = 1 - 5e^{-r/5}/(9Z_*(r))$ and $\gamma(r) = \frac{283}{128} - \frac{r}{2}$. The conditional law of a free link is $\nu_b \propto e^{\operatorname{Re}\operatorname{Tr}(Ub)} dU$ with b a sum of $N\beta = r/6$ times six products of elements of SU(3) and identities, the links of T_x being 1, so Lemma F.5 of Appendix F gives $c = r\sqrt{\epsilon(r)}/(6\gamma(r))$, increasing in r. With the moments $m_k = \int (\operatorname{Re}\operatorname{Tr} U)^k dU$, $(m_0, \ldots, m_8) = (1, 0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{15}{16}, \frac{65}{32}, \frac{231}{64}, \frac{973}{128})$, and $|m_k| \leq 3^{k-8} m_8$ for $k \geq 8$,

$$Z_*(r) \leq \sum_{k=0}^{8} \frac{m_k r^k}{k!} + \frac{m_8}{3^8} \frac{(3r)^9}{9!} \Big(1 - \frac{3r}{10}\Big)^{-1}.$$

At $r = 0.9$ the sum is at most 1.2599146 and the tail at most $3.4 \cdot 10^{-5}$, so $Z_*(0.9) \leq 1.2599486$; with $e^{-0.18} \geq 1 - 0.18 + 0.0162 - 0.000972$ this gives $\epsilon(0.9) \leq 0.63172 < 0.7949^2$, and with $\gamma(0.9) = \frac{1127}{640}$, $\frac{72}{5}c \leq 2.16 \cdot 0.7949 \cdot 640/1127 \leq 0.9751$. □

For fixed n the threshold of (i) exceeds that of Theorem 8.8(i) exactly when $n^2 - 6n + 1 > 0$, that is for $n \geq 6$; both theorems hold. The same arithmetic at $r = 0.88$, where $Z_* \leq 1.246559$, $\epsilon \leq 0.62627$ and $\gamma = \frac{5667}{3200}$, gives $15c \leq 0.9832$, so Theorem 8.8 holds for SU(3) with $0 \leq \beta_W \leq 0.44$.

Without the gauge fixing every link has eighteen partners, and the same argument needs $18c < 1$, that is $|\beta_W| < \frac{2}{9}$ for SU(2). No choice of gauge-fixed links and weights goes much further:

Proposition 8.12. *Let $T \subset V_x$ have at most $(n - 1)^4$ links and A be the partner matrix of $V_x \setminus T$. Then $\lambda_{\max}(A) \geq 6r(4 - r)(3 - r)/(4 - r^2)$ with $r = (1 - 1/n)^2$, which tends to 12. If $c\lambda_{\max}(A) \geq 1$, the inequalities $0 \leq \varrho \leq cA\varrho + s$ used in Lemma 8.4 admit solutions with $\sum_e \varrho(e)$ arbitrarily large, so for SU(2) the argument cannot pass $|\beta_W| = \frac{1}{3}$.*

Proof. The $P = 6n^2(n-1)^2 = 6rn^4$ plaquettes with all links in V_x contain at most $6|T| = aP$ links of T in all, $a \leq r$, so their numbers $m(p)$ of links outside T have mean at least $4 - a \geq 3$; by convexity $\mathbf{1}^\top A \mathbf{1} \geq \sum_p m(p)(m(p) - 1) \geq P(4 - a)(3 - a)$, and the Rayleigh quotient of $\mathbf{1}$ is at least $6r(4 - a)(3 - a)/(4 - ar)$, which decreases in a because $r/(4 - ar) \leq 1/(4 - a)$. For a Perron vector $v \geq 0$ of A, $\varrho = tv$ satisfies $cA\varrho \geq \varrho$ for every $t > 0$. □

9. Stars: the cube condition for SU(2) past $\beta_W = \frac{1}{3}$

Proposition 8.12 shows that no argument built on single-link updates and gauge-fixed links passes $\beta_W = \frac{1}{3}$ for SU(2). This section replaces single links by *stars*, the eight links at a vertex of one parity, and passes that value: the cube condition holds uniformly in the

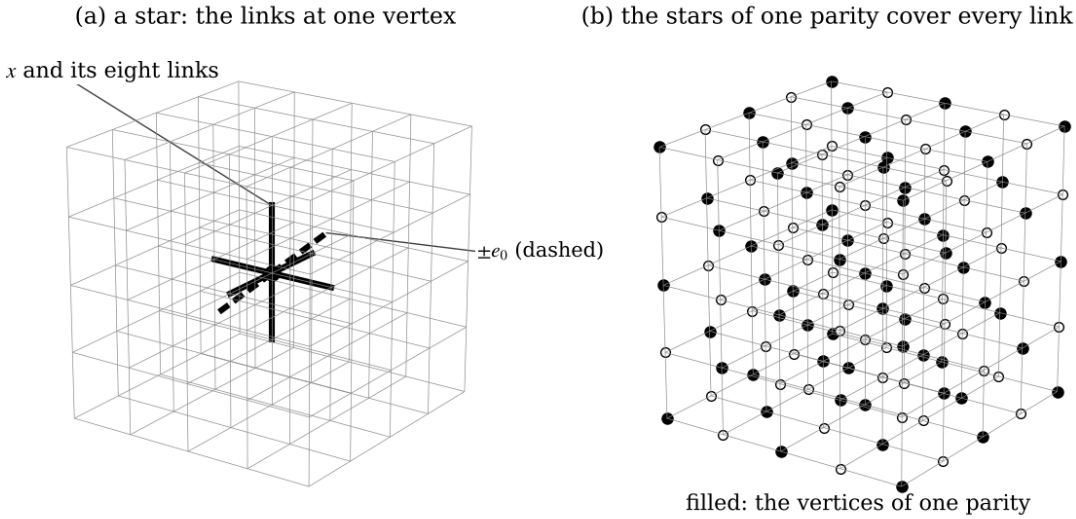


FIGURE 9. (a) A star: the eight links at a vertex x of the four-dimensional lattice, three spatial pairs drawn as axes and the pair $\pm e_0$ as a dashed stub. (b) The vertices of one parity (filled) on a three-dimensional slice of a torus of even side: every link has exactly one end point of each parity, so the stars of one parity partition the links and no plaquette is internal to a star.

cube size for $|\beta_W| \le \frac{7}{20}$ (Theorem 9.14), and a block condition giving a lattice gap holds for $|\beta_W| \le \frac{9}{25}$ (Theorem 9.5). The gain is visible at first order: the perturbation of one plaquette at a vertex enters the law of the star only through the relative variable of two of its links, so a star responds with the coefficient c where two single links respond with $2c$, and no plaquette is internal to a star.[14]

Setting. Throughout the section $G = \mathrm{SU}(2)$, identified with the unit quaternions $S^3 \subset \mathbb{R}^4$, with the Euclidean inner product $\langle \cdot, \cdot \rangle$ of $\mathbb{R}^4$ and the chordal metric $\delta(U, V) = |U - V|$, which is bi-invariant, invariant under inversion, and of diameter 2. For unit quaternions $\mathrm{Re}\,\mathrm{Tr}\,U = 2\langle U, 1\rangle$, and for unit p, q and $x, y \in \mathbb{R}^4$, $\langle px, y\rangle = \langle x, \bar{p}y\rangle$, $\langle xq, y\rangle = \langle x, y\bar{q}\rangle$ and $|pxq| = |x|$. The Wilson measure is $\mu(dU) = Z^{-1} \prod_p e^{2\beta\,\mathrm{Re}\,\mathrm{Tr}\,U_p} \prod_e dU_e$ on Λ_L, L even, with $\beta_W = 4\beta$, and $\mathcal{A}$, V_x, $\mathcal{P}_x$ and $\mathrm{BP}(n, \varpi)$ are as in Sec. 6. For a set V of links, $\mathcal{P}_V$ is the conditional expectation given the links outside V. For $b \in \mathbb{R}^4$, $\nu_b(dx) \propto e^{\langle x, b\rangle} dx$ is the von Mises–Fisher law on S^3, and the sharp quarter (Theorem C.4 in Appendix C) states

$$W_1(\nu_b, \nu_{b'}) \le \theta\,|b - b'| \quad \text{for all } b, b' \in \mathbb{R}^4, \qquad \theta = \tfrac{1}{4}, \tag{9.1}$$

with W_1 the Kantorovich distance for δ. This is the only property of ν_b used in the section. We put

$$c = 4\theta|\beta| = |\beta| = \tfrac{1}{4}|\beta_W|.$$

Every argument below uses only (9.1), and with a constant θ in place of $\frac{1}{4}$ the ranges become $4\theta|\beta| \le \frac{7}{80}$ and $4\theta|\beta| \le \frac{9}{100}$.

9.1. **Even stars and two-paths.** Call a vertex *even* if the sum of its coordinates is even. For an even vertex v the *star* B_v is the set of the eight links with an end point at v. Write the eight directions as $\pm 1, \dots, \pm 4$, and for the link of B_v in direction σ let X_σ be its parallel transporter from v outwards: $X_{+\mu} = U_{(v,\mu)}$ and $X_{-\mu} = U^{-1}_{(v-e_\mu,\mu)}$. Two directions are *adjacent* if they are distinct and not opposite; each direction is adjacent to six others, and there are 24 adjacent pairs.

[14] A Dobrushin certificate bounds the response of a block to a change of its boundary by a matrix of local influences, and contraction means that this matrix has spectral radius below one. Grouping links into blocks changes the matrix: interactions inside a block are treated exactly instead of being bounded.

Lemma 9.1. *Let $L \geq 4$ be even.*

(i) *Every link has exactly one even end point, so the stars partition E.*

(ii) *Every plaquette has exactly two even corners v, v', which are opposite corners, and two of its links lie in B_v, two in $B_{v'}$. The plaquettes with corner v correspond bijectively to the adjacent pairs $\{\sigma, \tau\}$: the plaquette $p_{\sigma\tau}$ has corners v, $v+\sigma$, $v+\sigma+\tau$, $v+\tau$.*

(iii) *For such a plaquette, $\operatorname{Re}\operatorname{Tr} U_p = 2\langle h_p, X_\sigma^{-1} X_\tau\rangle$, where the* two-path *$h_p$ is the transporter from $v+\sigma$ to $v+\tau$ through the opposite even corner $v' = v+\sigma+\tau$, the product of the two links of p in $B_{v'}$, suitably inverted. h_p is unchanged by a gauge transformation at v', and changing one of its two links by η in δ changes h_p by exactly η.*

(iv) *The conditional law of $(X_\sigma)_\sigma$ given the links outside B_v is*

$$\gamma_{v,h}(dX) = \frac{1}{Z_v} \prod_{\{\sigma,\tau\}} e^{4\beta\langle h_{\sigma\tau}, X_\sigma^{-1} X_\tau\rangle} \prod_\sigma dX_\sigma,$$

with one ordering fixed in each adjacent pair. It depends on the configuration only through the 24 two-paths $h = (h_{\sigma\tau})$, and it is invariant under $X_\sigma \mapsto gX_\sigma$, the gauge transformation at v.

(v) *A link i with even end point v_i lies in exactly six plaquettes; their even corners other than v_i are six distinct even vertices, and for each of them, v, the link i is a factor of exactly one two-path h_p with p at v.*

Proof. (i) The end points of a link differ by a unit vector, so their parities differ. (ii) The corners of a plaquette are y, $y+e_\mu$, $y+e_\mu+e_\nu$, $y+e_\nu$, of parities $\epsilon, \epsilon+1, \epsilon, \epsilon+1$, and each link joins consecutive corners. A plaquette with even corner v is spanned by two adjacent directions at v, and conversely; since $L \geq 4$ these 24 squares are distinct. (iii) Going around p from v, U_p is conjugate to $X_\sigma h_p X_\tau^{-1}$ or to its inverse, and $\operatorname{Re}\operatorname{Tr}$ is a class function, invariant under inversion on $\mathrm{SU}(2)$; so $\operatorname{Re}\operatorname{Tr} U_p = \operatorname{Re}\operatorname{Tr}(X_\sigma h_p X_\tau^{-1}) = 2\langle X_\sigma h_p, X_\tau\rangle = 2\langle h_p, X_\sigma^{-1} X_\tau\rangle$. A gauge transformation g at v' multiplies the first factor of h_p on the right by g^{-1} and the second on the left by g, and multiplication by unit quaternions is an isometry. (iv) The plaquettes containing a link of B_v are those with corner v, by (ii), and their weights are given by (iii); the gauge transformation at v changes no two-path. (v) If i joins v_i to $w = v_i + \sigma$, its plaquettes are spanned by σ and the six directions $\tau \perp \sigma$, with opposite even corners $v_i + \sigma + \tau$, which are distinct since $L \geq 4$; in the plaquette with corner v, the link i is one of the two links at v_i, hence a factor of the two-path. □

For the star at v define the quotient pseudometric

$$\bar{d}_v(X, X') = \min_{g \in G} \sum_\sigma \delta(gX_\sigma, X'_\sigma).$$

9.2. **The seven-spin system.** Fix an adjacent pair $\{a, b\}$, let S_a be the set of the seven directions other than a, A_7 its adjacency matrix and e_b the indicator of b. In S_a, the direction b is adjacent to $\bar{a} = -a$ and to the four directions C of the other two axes; $\bar{a}$ is adjacent to b, $\bar{b} = -b$ and C; $\bar{b}$ to $\bar{a}$ and C; and each element of C to b, $\bar{a}$, $\bar{b}$ and the two elements of C on the other axis. The largest degree is 6. For $0 \leq c < \frac{1}{6}$ put

$$s(c) = \mathbf{1}^{\mathsf{T}} (I - cA_7)^{-1} e_b = \sum_{k \geq 0} c^k \, \mathbf{1}^{\mathsf{T}} A_7^k e_b,$$

a nondecreasing function of c with $s(c) \geq 1$. The symmetries fixing a and b permute C transitively, so $x = (I - cA_7)^{-1} e_b$ is constant, equal to z, on C, and $(x_b, x_{\bar{a}}, x_{\bar{b}}, z)$ is the unique solution of

$$\begin{aligned} x_b &= 1 + c(x_{\bar{a}} + 4z), \qquad & x_{\bar{a}} &= c(x_b + x_{\bar{b}} + 4z), \\ x_{\bar{b}} &= c(x_{\bar{a}} + 4z), & z &= c(x_b + x_{\bar{a}} + x_{\bar{b}} + 2z), \end{aligned} \tag{9.2}$$

with $s(c) = x_b + x_{\bar{a}} + x_{\bar{b}} + 4z$.

Lemma 9.2. (i) $s(\frac{1}{12}) = \frac{26}{15}$, *with* $(x_b, x_{\bar a}, x_{\bar b}, z) = \frac{1}{210}(221, 28, 11, 26)$.
(ii) $s(\frac{7}{80}) = \frac{3480}{1933}$, *with* $(x_b, x_{\bar a}, x_{\bar b}, z) = \frac{1}{181702}(192531, 26320, 10829, 24360)$.
(iii) $s(\frac{9}{100}) = \frac{5450}{2957}$, *with* $(x_b, x_{\bar a}, x_{\bar b}, z) = \frac{1}{348926}(371363, 53100, 22437, 49050)$.
Hence $6c\,s(c)$ *equals* $\frac{13}{15}$ *at* $c = \frac{1}{12}$, $\frac{1827}{1933}$ *at* $c = \frac{7}{80}$ *and* $\frac{2943}{2957}$ *at* $c = \frac{9}{100}$, *and* $6c\,s(c) \le \frac{2943}{2957}$ *for* $0 \le c \le \frac{9}{100}$.

Proof. Since $c\lambda_{\max}(A_7) \le 6c < 1$, the matrix $I - cA_7$ is invertible and it suffices to check (9.2). (i) Multiplied by $12 \cdot 210$ the four equations read $12 \cdot 221 - (28 + 104) = 2520$, $12 \cdot 28 - (221 + 11 + 104) = 0$, $12 \cdot 11 - 132 = 0$ and $12 \cdot 26 - (221 + 28 + 11 + 52) = 0$; the sum is $364/210 = 26/15$. (ii) Multiplied by $80 \cdot 181702 = 14536160$:

$$\begin{aligned} 80 \cdot 192531 - 7(26320 + 4 \cdot 24360) &= 15402480 - 866320 = 14536160, \\ 80 \cdot 26320 - 7(192531 + 10829 + 4 \cdot 24360) &= 2105600 - 7 \cdot 300800 = 0, \\ 80 \cdot 10829 - 7(26320 + 4 \cdot 24360) &= 866320 - 7 \cdot 123760 = 0, \\ 80 \cdot 24360 - 7(192531 + 26320 + 10829 + 2 \cdot 24360) &= 1948800 - 7 \cdot 278400 = 0. \end{aligned}$$

The sum is $327120/181702 = 3480/1933$, as $327120 = 94 \cdot 3480$ and $181702 = 94 \cdot 1933$. (iii) Multiplied by $100 \cdot 348926$: $100 \cdot 371363 - 9(53100 + 196200) = 34892600$, $100 \cdot 53100 - 9(371363 + 22437 + 196200) = 5310000 - 9 \cdot 590000 = 0$, $100 \cdot 22437 - 9(53100 + 196200) = 0$ and $100 \cdot 49050 - 9(371363 + 53100 + 22437 + 98100) = 4905000 - 9 \cdot 545000 = 0$. The sum is $643100/348926 = 5450/2957$, as $348926 = 118 \cdot 2957$ and $643100 = 118 \cdot 5450$. The last claim follows from the monotonicity of s. □

Lemma 9.3 (one star)**.** *Assume* (9.1) *and* $c < \frac{1}{6}$. *Let* h, h' *be two families of two-paths at* v *that differ only at one adjacent pair* $\{a, b\}$. *Then*

$$W_1^{\bar d_v}(\gamma_{v,h}, \gamma_{v,h'}) \le c\,s(c)\,|h_{ab} - h'_{ab}|.$$

Proof. Under $\gamma_{v,h}$ put $Z_\sigma = X_a^{-1} X_\sigma$. The map $X \mapsto (X_a, Z)$ preserves Haar measure, and by Lemma 9.1(iv) the density depends only on Z, with $Z_a = 1$; so X_a is Haar distributed and independent of $Z_S = (Z_\sigma)_{\sigma \in S_a}$, whose law $\mu_h^{(a)}$ has density proportional to $\prod e^{4\beta \langle h_{\sigma\tau}, Z_\sigma^{-1} Z_\tau \rangle}$. Given a coupling of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$, take $X_a = X'_a$ Haar and independent, $X_\sigma = X_a Z_\sigma$ and $X'_\sigma = X_a Z'_\sigma$; this couples $\gamma_{v,h}$ and $\gamma_{v,h'}$, and with $g = 1$ in the definition, $\bar d_v(X, X') \le \sum_{\sigma \in S_a} \delta(Z_\sigma, Z'_\sigma)$. It suffices to bound the Kantorovich distance of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ for the cost $\sum_{\sigma \in S_a} \delta$.

For $\sigma \in S_a$ the conditional law of Z_σ given the others is ν_{B_σ} with $B_\sigma = 4\beta \sum_{\tau \text{ adj. } \sigma} w_{\sigma\tau}$, where $w_{\sigma\tau} = Z_\tau \bar h_{\sigma\tau}$ or $Z_\tau h_{\tau\sigma}$ according to the ordering of the pair, since $\langle h_{\sigma\tau}, Z_\sigma^{-1} Z_\tau \rangle = \langle Z_\sigma, Z_\tau \bar h_{\sigma\tau} \rangle$; changing Z_τ by η changes $w_{\sigma\tau}$ by η, and changing h_{ab} changes only B_b, by at most $4|\beta||h_{ab} - h'_{ab}|$. Let π minimise $\int \sum_\sigma \delta(Z_\sigma, Z'_\sigma)\, d\pi$ over couplings of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ (one exists by compactness) and put $\varrho_\sigma = \int \delta(Z_\sigma, Z'_\sigma)\, d\pi$. Replacing (Z_σ, Z'_σ) by a measurably chosen optimal coupling of the two conditional laws [58, Cor. 5.22] gives another coupling that changes only the σ-term of the cost, so by minimality and (9.1)

$$\varrho_\sigma \le \theta \int |B_\sigma - B'_\sigma|\, d\pi \le c \sum_{\tau \in S_a,\ \tau \text{ adj. } \sigma} \varrho_\tau + c\,|h_{ab} - h'_{ab}|\,[\sigma = b],$$

that is, $\varrho \le cA_7 \varrho + c|h_{ab} - h'_{ab}| e_b$ componentwise. Iterating, $\varrho \le \sum_{k<K} (cA_7)^k c|h_{ab} - h'_{ab}| e_b + (cA_7)^K \varrho$, and $(cA_7)^K \to 0$ since the spectral radius of cA_7 is at most $6c < 1$. Summing the components gives $\sum_\sigma \varrho_\sigma \le c\,s(c)|h_{ab} - h'_{ab}|$. □

9.3. **The star condition and a lattice gap up to** $\beta_W = \frac{9}{25}$**.** Let $\mathcal{P}_v = \mathcal{P}_{B_v}$ and let $N_s = L^4/2$ be the number of even vertices. For a function h on G^E let $\mathrm{Lip}(h)$ be the least constant with $|h(U) - h(U')| \le \mathrm{Lip}(h)\delta(U_i, U'_i)$ whenever U, U' differ only at one link i; changing links one at a time, $|h(U) - h(U')| \le \mathrm{Lip}(h) \sum_e \delta(U_e, U'_e)$.

Proposition 9.4 (star condition)**.** *Assume* (9.1) *and put* $\gamma = c\,s(c)$*, with* $c < \frac{1}{6}$ *and* $6\gamma < 1$*. Then for every real* $f \in \mathcal{A}$

$$\sum_{v \text{ even}} \mu((f - \mathcal{P}_v f)^2) \geq (1 - 6\gamma)\,\mathrm{Var}_\mu(f).$$

Proof. Let $h \in \mathcal{A}$ be Lipschitz, gauge invariant at every configuration, and let U, U' differ only at i. If $i \in B_v$, then $\mathcal{P}_v h(U) = \mathcal{P}_v h(U')$. Otherwise let π be a coupling of $\gamma_v(\cdot \mid U)$ and $\gamma_v(\cdot \mid U')$ minimising the integral of $\bar{d}_v$, a continuous pseudometric on a compact space. For $\sigma, \sigma' \in G^{B_v}$ and $g \in G$ the gauge transformation g at v changes only the links of B_v, so by gauge invariance $h(\sigma', U') = h(g\sigma', U')$ and

$$|h(\sigma, U) - h(\sigma', U')| \leq \mathrm{Lip}(h)\Big(\sum_{e \in B_v} \delta(\sigma_e, (g\sigma')_e) + \delta(U_i, U_i')\Big);$$

the infimum over g gives $\mathrm{Lip}(h)(\bar{d}_v(\sigma, \sigma') + \delta(U_i, U_i'))$. By Lemma 9.1(iv), (v), the laws $\gamma_v(\cdot \mid U)$ and $\gamma_v(\cdot \mid U')$ differ only through the two-path of one plaquette, which moves by $\delta(U_i, U_i')$, if v is one of the six vertices of Lemma 9.1(v), and coincide otherwise. Lemma 9.3 gives

$$|\mathcal{P}_v h(U) - \mathcal{P}_v h(U')| \leq \mathrm{Lip}(h)\,\big(1 + \gamma\,[v \text{ one of the six}]\big)\,\delta(U_i, U_i').$$

For $\mathcal{K}_s = N_s^{-1} \sum_v \mathcal{P}_v$, summing over v (exactly one star contains i) gives $\mathrm{Lip}(\mathcal{K}_s h) \leq (1 - \kappa_0)\,\mathrm{Lip}(h)$ with $\kappa_0 = (1 - 6\gamma)/N_s$. Each $\mathcal{P}_v$ maps $\mathcal{A}$ to itself, by gauge covariance of γ_v, and for a Lipschitz $g \in \mathcal{A}$, $\mathrm{Var}_\mu(g) \leq \frac{1}{2} \iint (g(U) - g(U'))^2\, \mu(dU)\mu(dU') \leq 2\,\mathrm{Lip}(g)^2 |E|^2$. So $\mathrm{Var}_\mu(\mathcal{K}_s^t h) \leq 2(1 - \kappa_0)^{2t}\,\mathrm{Lip}(h)^2 |E|^2$. On the mean-zero part of $\mathcal{A}$ the operator $\mathcal{K}_s$ is self-adjoint with $0 \leq \mathcal{K}_s \leq I$, and $\mathrm{Var}_\mu(\mathcal{K}_s^t h) = \int \lambda^{2t}\, d\nu_h(\lambda)$ for its spectral measure ν_h at $h - \mu(h)$; hence ν_h lives on $[0, 1 - \kappa_0]$. Lipschitz functions are dense in $L^2(\mu)$, and the orthogonal projection onto $\mathcal{A}$, averaging over the compact gauge group, which acts link by link by isometries, does not increase Lip; so such h are dense in $\mathcal{A}$, and the spectral projection of $\mathcal{K}_s$ on $(1 - \kappa_0, 1]$ vanishes on the mean-zero part of $\mathcal{A}$. Thus $\langle f, (I - \mathcal{K}_s) f \rangle \geq \kappa_0\,\mathrm{Var}_\mu(f)$, and $\langle f, (I - \mathcal{P}_v) f \rangle = \mu((f - \mathcal{P}_v f)^2)$. □

Theorem 9.5. *Let* $G = \mathrm{SU}(2)$ *and* $|\beta_W| \leq \frac{9}{25}$*. Then for every even* $L \geq 4$ *and every real* $f \in \mathcal{A}$*,*

$$\sum_{v \text{ even}} \mu((f - \mathcal{P}_v f)^2) \geq \frac{14}{2957}\,\mathrm{Var}_\mu(f),$$

and for every $n \geq 2$ *and even* $L \geq 2n + 4$*,* $\mathrm{BP}(n, \varpi_n)$ *holds with*

$$\varpi_n = \frac{14}{2957} \cdot \frac{(n-1)^4}{n^4 \lceil (n-1)^4/2 \rceil}, \qquad \varpi_2 = \frac{14}{16 \cdot 2957}.$$

For $0 \leq \beta_W \leq \frac{9}{25}$ *every torus limit is translation invariant and reflection positive, and its transfer matrix has spectral gap at least* $\varpi_2/(2e^2 q_2) = 7/(625 \cdot 2957\, e^2)$*.*

Proof. Here $c = |\beta| \leq \frac{9}{100}$, and by Lemma 9.2 $1 - 6c\,s(c) \geq \frac{14}{2957}$; the first claim is Proposition 9.4. For a cube V_x, an even vertex v with $v - x \in [1, n-1]^4$ has its eight links based in $x + [0, n)^4$, so $B_v \subset V_x$. The σ-algebra of the links outside V_x is contained in that of the links outside B_v, so $\mu((f - \mathcal{P}_x f)^2) = \mu(f^2) - \mu((\mathcal{P}_x f)^2) \geq \mu(f^2) - \mu((\mathcal{P}_v f)^2) = \mu((f - \mathcal{P}_v f)^2)$. Each cube contains at most $N = \lceil (n-1)^4/2 \rceil$ such vertices, and each even v is in this position in exactly $(n-1)^4$ cubes. Averaging over those v in each cube,

$$\sum_x \mu((f - \mathcal{P}_x f)^2) \geq \frac{(n-1)^4}{N} \sum_{v \text{ even}} \mu((f - \mathcal{P}_v f)^2) \geq \frac{(n-1)^4}{N} \cdot \frac{14}{2957}\,\mathrm{Var}_\mu(f),$$

which is $\mathrm{BP}(n, \varpi_n)$. The last claim is Corollary 6.4 with $n = 2$ and $q_2 = \frac{625}{16}$. □

The same proof works whenever $6c\,s(c) < 1$. The exact threshold is the root c_* of $6c\,s(c) = 1$, and Lemma 9.2(iii) shows $c_* > \frac{9}{100}$.[15] At $\beta_W = \frac{1}{3}$ the star condition holds with $6c\,s(c) = \frac{13}{15}$, while the single-link argument of Sec. 8 is exactly marginal there. The constant ϖ_n of Theorem 9.5 decays like $\frac{28}{2957}n^{-4}$, so that theorem is a cube condition for each fixed n, enough for the lattice gap but not for Theorem B. The lower faces of a cube are the reason, and the next proposition shows that they stop every certificate that treats their links one at a time.

Proposition 9.6 (the face barrier)**.** *For a lower face $\{\hat v_i = 0\}$ of the cube V_x, $\hat v = v - x$, let F_i be the set of links $(b,\mu) \in V_x$ with $\hat b_i = 0$ and $\mu \neq i$, and let A_F be the matrix on F_i with $(A_F)_{ek} = 1$ when $e \neq k$ lie in a common plaquette all of whose links are in F_i. Then $\lambda_{\max}(A_F) \geq 12(1-\frac{1}{n})^2$. Consequently, if a certificate for the response of a cube treats the links of F_i as single sites with influence matrix $C \geq cA_F$ on $F_i \times F_i$, the inequalities $0 \leq \varrho \leq C\varrho + \mathfrak{s}$ admit solutions with $\sum \varrho$ arbitrarily large as soon as $12c(1-\frac{1}{n})^2 \geq 1$; as $n \to \infty$ such a certificate cannot pass $c = \frac{1}{12}$, that is $|\beta_W| = \frac{1}{3}$.*

Proof. $|F_i| = 3n^3$. A plaquette in a plane $\{\mu,\nu\} \not\ni i$ with base point y, $\hat y_i = 0$, has all four links in F_i exactly when its three base points lie in $x + [0,n)^4$, that is $\hat y_\mu, \hat y_\nu \leq n-2$; there are $3n(n-1)^2$ of them. Two links share at most one plaquette, so $\mathbf{1}^{\mathsf T} A_F \mathbf{1} = 12 \cdot 3n(n-1)^2$, and the Rayleigh quotient of $\mathbf{1}$ is $12(1-\frac{1}{n})^2$. If $c\lambda_{\max}(A_F) \geq 1$ and $w \geq 0$ is a Perron vector of A_F, extended by 0, then $\varrho = tw$ satisfies $C\varrho \geq tcA_F w \geq \varrho$ on F_i for every $t > 0$. □

For single-link updates the hypothesis $C \geq cA_F$ holds with $c = |\beta|$: at $b = 0$ the derivative of $b \mapsto \nu_b$ has Kantorovich norm exactly $\frac{1}{4}$ (Lemma C.3(i)), and the staples of a face link can cancel while one of them moves. The forest of Sec. 8 does not help, since its links T_x have interior heads and the face links are free. What does help is to let the gauge transformation act at the vertices of the faces as well, and to pay for it.

9.4. **The cube as a block system.** Fix x and $n \geq 2$, and write $\hat v = v - x$, computed in a window $x + [-2, n+2]^4$ that embeds in $\mathbb{Z}^4$ because $L \geq 2n+4$. Let $V = V_x$ and $Q = \{v : \hat v \in [0,n-1]^4\}$. Call v *lower* if $v \in Q$, with zero set $Z(v) = \{a : \hat v_a = 0\}$ and $z(v) = |Z(v)|$, and *upper* if $\hat v \in [0,n]^4$ has exactly one coordinate equal to n. If k is a link with odd end point o and direction μ, its six plaquettes are indexed by (ν,ϵ), $\nu \neq \mu$, $\epsilon = \pm 1$; the even corner of the plaquette (ν,ϵ) other than the even end point of k is $o + \epsilon e_\nu$, and there k is a factor of the two-path of this plaquette only (Lemma 9.1(v)).

Lemma 9.7. *Let $n \geq 2$.*

(i) *If v is lower and even, $B_v \cap V$ consists of B_v minus the $z(v)$* outside links *$(v - e_a, a)$, $a \in Z(v)$; we call the directions $-a$, $a \in Z(v)$,* fixed *and the others* free*. If v is upper and even with $\hat v_a = n$, then $B_v \cap V = \{(v - e_a, a)\}$. Other even vertices carry no link of V. Hence V is partitioned into the* blocks *$B_v \cap V$, v even and lower or upper, and the* owner *$w(k)$ of $k \in V$ is its even end point.*

(ii) *Let $k \in V$ have odd end point o, direction μ and owner $o + \varsigma e_\mu$, $\varsigma = \pm 1$, and let p be its plaquette (ν,ϵ), $v = o + \epsilon e_\nu$. The links of p at v are in the directions $-\epsilon\nu$ and $\varsigma\mu$. If v is lower, both lie in V. If v is upper with $\hat v_a = n$, the links of p at v other than $(v - e_a, a)$ lie outside V.*

(iii) *Let v be lower and even and $a \in Z(v)$. The outside link $(v - e_a, a)$ is a factor of no two-path h_p at a lower or upper vertex. If a plaquette p at v has one of its links at v outside V, then both links of p at the opposite even corner lie outside V.*

Proof. (i) The outgoing links (v,b) of a lower v have base point in Q, and an incoming link $(v - e_b, b)$ lies in V if and only if $\hat v_b \geq 1$. If $\hat v_a = n$ for exactly one a, no outgoing link and no

[15] Solving (9.2) numerically gives $c_* \approx 0.09023$, that is $\beta_W \approx 0.3609$. No statement of the paper uses this value.

incoming link $(v-e_b,b)$, $b\neq a$, has its base point in Q, while $(v-e_a,a)$ has. If $\hat v\notin[0,n]^4$ or two coordinates equal n, no link at v has its base point in Q.

(ii) The corners of p are o, $o+\varsigma e_\mu$, $o+\varsigma e_\mu+\epsilon e_\nu$ and v, so the links at v go to o, in direction $-\epsilon\nu$, and to $v+\varsigma e_\mu$, in direction $\varsigma\mu$. Let v be lower. If $\epsilon=+1$ the first link is (o,ν), and $\hat o=\hat v-e_\nu$ lies in Q, its ν-coordinate being $\hat o_\nu\geq 0$; if $\epsilon=-1$ it is (v,ν), with base point $v\in Q$. The second link is (v,μ) if $\varsigma=+1$; if $\varsigma=-1$ it is $(v-e_\mu,\mu)$, which lies in V because $\hat v_\mu=\hat o_\mu\geq 1$, as $k=(o-e_\mu,\mu)\in V$. If v is upper, (i) gives the claim.

(iii) The plaquettes of $(v-e_a,a)$ have the even corners v and $v-e_a\pm e_\nu$, whose a-coordinate is -1, so these are neither lower nor upper; and a link is a factor of a two-path only at the even corner opposite to its own even end point. If p at v is spanned by $-a$ and τ, the opposite corner $v'=v-e_a+\tau$ and both its neighbours in p, $v-e_a$ and $v+\tau$, are joined to it by links based at a-coordinate -1. □

For a block vertex v the conditional law $\gamma_v(\cdot\mid\sigma,U)$ of the links of $B_v\cap V$, given the other links of V (values σ) and the links off V (values U), has density proportional to the product of the weights $e^{4\beta\langle h_p,X_\sigma^{-1}X_\tau\rangle}$ of the plaquettes p at v with a link at v in V. It depends on (σ,U) only through these two-paths and the values of the links at v off V.

9.5. **The soft quotient.** At a vertex of a lower face the gauge transformation moves the outside links at that vertex, and the conditional law of the cube does not see it. The following pseudometric allows it anyway, at the price of the displacement it causes on those links. The price is legitimate in the Lipschitz contraction argument of Proposition 8.7, because the test function depends on all links.

Definition 9.8. For a lower even v and configurations X,X' of $B_v\cap V$, written as outward transporters, put

$$\tilde d_v(X,X')=\min_{g\in G}\Big[\sum_{\sigma\text{ free}}\delta(gX_\sigma,X'_\sigma)+z(v)\,\delta(g,1)\Big],$$

and for an upper even v let $\tilde d_v$ be δ on its single link. For $\sigma,\sigma'\in G^V$ let $D(\sigma,\sigma')=\sum_v\tilde d_v(\sigma_{B_v},\sigma'_{B_v})$, the sum over the block vertices, and for $j\notin V$

$$\tilde R_x(j)=\sup\Big\{\frac{W^D(\gamma^0_{V,U},\gamma^0_{V,U'})}{\delta(U_j,U'_j)}:U,U'\text{ differ only at }j\Big\},$$

where $\gamma^0_{V,U}=\gamma^0_V(\cdot\mid U)$ and W^D is the Kantorovich distance for the cost D.

The minimum exists by compactness, and $\tilde d_v$ and D are continuous, symmetric, since $\delta(gX,X')=\delta(X,g^{-1}X')$ and $\delta(g,1)=\delta(g^{-1},1)$, and satisfy the triangle inequality, by composing the minimising elements. For $z(v)=0$, $\tilde d_v=\bar d_v$.

Lemma 9.9. *Let h be a function on G^E, gauge invariant at every configuration, with $|h(W)-h(W')|\leq\mathrm{Lip}(h)\sum_e\delta(W_e,W'_e)$. If U,U' differ only at $j\notin V$, then for all $\sigma,\sigma'\in G^V$*

$$|h(\sigma,U)-h(\sigma',U')|\leq\mathrm{Lip}(h)\big(D(\sigma,\sigma')+\delta(U_j,U'_j)\big).$$

Proof. For each lower even v choose g_v attaining $\tilde d_v(\sigma_{B_v},\sigma'_{B_v})$, and let g be the gauge transformation equal to g_v at these vertices and to 1 elsewhere; then $h(\sigma,U)=h(g\cdot(\sigma,U))$. A link of V has exactly one even end point v, and g acts on it by $X\mapsto g_vX$ on its outward transporter at v, or trivially if v is upper. A link $e\notin V$ has at most one end point where $g\neq 1$, namely an even lower v, and then e is one of the $z(v)$ outside links at v, by Lemma 9.7(i); there $\delta((g\cdot U)_e,U'_e)\leq\delta(g_v,1)+\delta(U_e,U'_e)$ by bi-invariance. Summing over all links, $|h(g\cdot(\sigma,U))-h(\sigma',U')|\leq\mathrm{Lip}(h)\big(\sum_v\tilde d_v+\delta(U_j,U'_j)\big)$. □

9.6. **Block responses.**

Lemma 9.10 (Dobrushin on S^3)**.** *Assume* (9.1). *Let S be finite and ν,ν' probability measures on G^S with continuous positive densities with respect to Haar measure, such that for every $s\in S$ the conditional law of x_s given $x_{S\setminus s}$ is $\nu_{B_s(x)}$ under ν and $\nu_{B'_s(x)}$ under ν', where $|B_s(x)-B_s(y)|\le 4|\beta|\sum_t a_{st}\delta(x_t,y_t)$ and $|B_s(x)-B'_s(x)|\le 4|\beta|\eta_s$ for all x,y, with $a_{st}\ge 0$, $a_{ss}=0$ and $\eta_s\ge 0$. Then there is a coupling π of ν,ν' whose $\varrho_s=\int\delta(x_s,x'_s)\,d\pi$ satisfy $\varrho\le cA\varrho+c\eta$ componentwise. Consequently:*

(a) *if $\sum_s a_{st}\le d$ for every t and $cd<1$, then $\sum_s\varrho_s\le c\sum_s\eta_s/(1-cd)$;*
(b) *if the spectral radius of cA is less than 1 and $y\in[0,\infty)^S$ satisfies $y\ge cAy+c\eta$, then $\varrho\le y$;*
(c) *if $y\ge 0$ satisfies $y\ge c_0Ay+\eta$ and $c\le c_0$, then $cy\ge cA(cy)+c\eta$, so $\varrho\le cy$ when the spectral radius of cA is less than 1.*

Proof. Let π minimise $\int\sum_s\delta(x_s,x'_s)\,d\pi$ over couplings. Fix s and replace under π the pair (x_s,x'_s) by a sample of an optimal coupling of $\nu_{B_s(x)}$ and $\nu_{B'_s(x')}$, chosen measurably in $(x_{S\setminus s},x'_{S\setminus s})$ [58, Cor. 5.22]. This is again a coupling of ν,ν', and only the s-term of the cost changes, so by minimality, (9.1) and the triangle inequality, $\varrho_s\le\theta\int|B_s(x)-B'_s(x')|\,d\pi\le c\sum_t a_{st}\varrho_t+c\eta_s$. (a) Summing over s, $\sum\varrho\le cd\sum\varrho+c\sum\eta$. (b) The vector $y-\varrho$ satisfies $y-\varrho\ge cA(y-\varrho)$, hence $y-\varrho\ge(cA)^K(y-\varrho)\to 0$. (c) $cy-c^2Ay\ge c(y-c_0Ay)\ge c\eta$; apply (b). □

From now on $c\le c_0=\frac{7}{80}$, that is $|\beta_W|\le\frac{7}{20}$, and $\kappa_0=s(c_0)=\frac{3480}{1933}<1.8004$, so that $6c_0\kappa_0=\frac{1827}{1933}$. For a direction a and a family h of two-paths let $\mu_h^{(a)}$ be the law on G^{S_a} with density proportional to $\prod_{\{\sigma,\tau\}}e^{4\beta\langle h_{\sigma\tau},Z_\sigma^{-1}Z_\tau\rangle}$, $Z_a=1$, as in the proof of Lemma 9.3. By that proof, Lemma 9.2(ii) and Lemma 9.10(c), if h,h' differ only at the pair $\{a,b\}$ there is a coupling of $\mu_h^{(a)}$ and $\mu_{h'}^{(a)}$ with

$$\int\sum_{\sigma\in S_a}\delta(Z_\sigma,Z'_\sigma)\le\kappa_0 c\,|h_{ab}-h'_{ab}|\qquad(c\le c_0).\tag{9.3}$$

Lemma 9.11. *Let $c\le c_0$ and v a block vertex, and write $\gamma^{h,Y}$ for its block law as a function of the two-paths h at v and of the values Y of the links at v off V. Put $\kappa_1=\kappa_0$, $\kappa_2=\frac{9}{4}$, $\kappa_3=\frac{40}{13}$ and $\kappa_4=\frac{160}{59}$.*

(a) *Let v be lower, and let (h,Y) and (h',Y) differ only at the two-path h_p of a plaquette p whose two links at v are free. Then $W^{\tilde d_v}(\gamma^{h,Y},\gamma^{h',Y})\le\kappa_{z(v)}c\,|h_p-h'_p|$, with κ_0 for $z(v)=0$.*
(b) *In particular, for $z(v)=1$ the constant is the full-star constant $\kappa_0 c$.*
(c) *Let v be lower and p a plaquette at v with one link at v free and one fixed. Then the bound of* (a) *holds with $\kappa_0 c$ if $z(v)=1$ and with $\frac{16}{9}c$ if $z(v)\ge 2$.*
(d) *Let v be lower and let (h,Y) and (h,Y') differ only at one outside link, by η in δ. Then $W^{\tilde d_v}(\gamma^{h,Y},\gamma^{h,Y'})\le\eta$.*
(e) *Let v be upper. If (h,Y) and (h',Y') differ only at one two-path h_p with $p\ni(v-e_a,a)$, or only at one outside link at v, the distance is at most $c|h_p-h'_p|$, respectively $c\eta$.*

Proof. Representation. Let v be lower and F its set of fixed directions, let $\Phi_h(X)=\prod e^{4\beta\langle h_{\sigma\tau},X_\sigma^{-1}X_\tau\rangle}$ over all 24 adjacent pairs, so that $\Phi_h(gX)=\Phi_h(X)$, and note that $\gamma^{h,Y}$ has density proportional to Φ_h evaluated at $X_F=Y_F$. Fix a free direction σ_0. If $F=\emptyset$, the map $X\mapsto(X_{\sigma_0},(X_{\sigma_0}^{-1}X_\rho)_{\rho\ne\sigma_0})$ preserves Haar measure and carries γ^h to $\mathrm{Haar}\otimes\mu_h^{(\sigma_0)}$, as in the proof of Lemma 9.3. If $F\ne\emptyset$, fix $f_1\in F$ and let $\Psi(X)$ consist of $W_\rho=X_{\sigma_0}^{-1}X_\rho$ for the free $\rho\ne\sigma_0$ and of $W_m=X_{\sigma_0}^{-1}Y_{f_1}$. Then Ψ is a bijection of the free coordinates, with inverse $X_{\sigma_0}=Y_{f_1}W_m^{-1}$, $X_\rho=Y_{f_1}W_m^{-1}W_\rho$, and it preserves Haar measure: $X_{\sigma_0}\mapsto X_{\sigma_0}^{-1}Y_{f_1}$ does, and for fixed X_{σ_0} so does each $X_\rho\mapsto X_{\sigma_0}^{-1}X_\rho$. Since $\Phi_h(X)=\Phi_h(X_{\sigma_0}^{-1}X)$ and $X_{\sigma_0}^{-1}X_f=W_mK_f$

with $K_f = Y_{f_1}^{-1} Y_f$, the image of $\gamma^{h,Y}$ has density proportional to Φ_h at $X_{\sigma_0} = 1$, $X_\rho = W_\rho$, $X_f = W_m K_f$. If (W, W') couples the images for (h, Y) and (h', Y) and $X = \Psi^{-1} W$, $X' = \Psi^{-1} W'$, then $g = Y_{f_1} W_m'^{-1} W_m Y_{f_1}^{-1}$ satisfies $g X_{\sigma_0} = X'_{\sigma_0}$, $\delta(g X_\rho, X'_\rho) = \delta(W_\rho, W'_\rho)$ and $\delta(g, 1) = \delta(W_m, W'_m)$, so

$$\tilde{d}_v(X, X') \le \sum_\rho \delta(W_\rho, W'_\rho) + z(v)\, \delta(W_m, W'_m). \tag{9.4}$$

(a), (b) for $z(v) \le 1$. Take σ_0 and τ to be the two directions of p at v. For $z(v) = 1$, $F = \{f_1\}$, the image is exactly $\mu_h^{(\sigma_0)}$, the coordinate W_m being Z_{f_1}, and the change of $h_p = h_{\sigma_0 \tau}$ is a change of the field on Z_τ; (9.3) and (9.4) give $\kappa_0 c |h_p - h'_p|$. For $z(v) = 0$ couple with the same X_{σ_0} and use $g = 1$; this is Lemma 9.3 with (9.3).

(a) for $z(v) = 2$. Let $F = \{-i, -j\}$; the free directions are $P = +i$, $Q = +j$, $K_\pm = \pm k$ and $L_\pm = \pm l$. Up to the symmetries $i \leftrightarrow j$, $k \leftrightarrow l$, $k \mapsto -k$ and $l \mapsto -l$ of the star, which fix F, the pair of p is one of $\{P, Q\}$, $\{P, K_+\}$ and $\{K_+, L_+\}$. In the image the terms of the density are: for adjacent free $\rho, \rho' \ne \sigma_0$, $\langle h, W_\rho^{-1} W_{\rho'} \rangle$; for $\{\sigma_0, \rho\}$, a field on W_ρ; for $\{f, \rho\}$, $\langle h, (W_m K_f)^{-1} W_\rho \rangle = \langle K_f h, W_m^{-1} W_\rho \rangle$, linear in W_m and in W_ρ; for $\{\sigma_0, f\}$, a field on W_m; and for $\{-i, -j\}$, a constant. So Lemma 9.10 applies with $a_{\rho\rho'} = 1$ for adjacent free $\rho, \rho' \ne \sigma_0$, $a_{\rho m} = a_{m\rho}$ equal to the number of $f \in F$ adjacent to ρ (one for P, Q, two for $K_\pm, L_\pm$), and $\eta = |h_p - h'_p| e_\tau$. Every row sum of A is at most 9, so $c_0 A$ has spectral radius at most $\frac{63}{80}$. By (9.4) the cost is at most $\sum_\rho \varrho_\rho + 2\varrho_m$. We give $\hat{y} \ge c_0 A \hat{y} + e_\tau$ and use Lemma 9.10(c); each item lists $\hat{y}$ and the inequalities multiplied by 80 and by the denominator of $\hat{y}$.

The pair $\{P, Q\}$, $\sigma_0 = P$, $\tau = Q$. The spins are Q, $K_\pm$, $L_\pm$, each of the last four adjacent to Q and to the two of the other axis, and m; $100\hat{y} = (Q, K_\pm, L_\pm, m) = (108, 16, 21)$. The inequalities are $8640 \ge 7(4 \cdot 16 + 21) + 8000 = 8595$, $1280 \ge 7(108 + 2 \cdot 16 + 2 \cdot 21) = 1274$ and $1680 \ge 7(108 + 8 \cdot 16) = 1652$, and the cost is at most $(108 + 4 \cdot 16 + 2 \cdot 21)/100 = 2.14$.

The pair $\{P, K_+\}$, $\sigma_0 = K_+$, $\tau = P$. The spins are P and Q, adjacent to each other and to K_-, $L_\pm$; K_-, adjacent to $P, Q, L_\pm$; $L_\pm$, adjacent to P, Q, K_-; and m; $200\hat{y} = (P, Q, K_-, L_\pm, m) = (216, 32, 35, 32, 40)$. The inequalities are $17280 \ge 7(32 + 35 + 64 + 40) + 16000 = 17197$, $2560 \ge 7(216 + 35 + 64 + 40) = 2485$, $2800 \ge 7(216 + 32 + 64 + 80) = 2744$, $2560 \ge 7(216 + 32 + 35 + 80) = 2541$ and $3200 \ge 7(216 + 32 + 70 + 128) = 3122$, and the cost is at most $(216 + 32 + 35 + 64 + 80)/200 = 2.135$.

The pair $\{K_+, L_+\}$, $\sigma_0 = K_+$, $\tau = L_+$, with the same spins and adjacencies; $200\hat{y} = (P, Q, K_-, L_+, L_-, m) = (32, 32, 36, 219, 19, 54)$. The inequalities are $2560 \ge 7(32 + 36 + 219 + 19 + 54) = 2520$ for P and for Q, $2880 \ge 7(64 + 219 + 19 + 108) = 2870$, $17520 \ge 7(64 + 36 + 108) + 16000 = 17456$, $1520 \ge 7(64 + 36 + 108) = 1456$ and $4320 \ge 7(64 + 72 + 2 \cdot 238) = 4284$, and the cost is at most $(64 + 36 + 219 + 19 + 108)/200 = 2.23$.

In all cases the response is at most $2.23\, c|h_p - h'_p| \le \kappa_2 c |h_p - h'_p|$.

(a) for $z(v) \ge 3$, and (c), (d) for $z(v) \ge 2$. Use the coordinates X_σ, σ free, directly, and $g = 1$ in $\tilde{d}_v$. Each conditional law is ν_B with $B = 4\beta$ times the sum of the unit vectors of the plaquettes at v through σ, those through free neighbours depending isometrically on them; so Lemma 9.10(a) applies with A the adjacency matrix of the free directions, whose largest degree is $d = 5, 4, 3$ for $z(v) = 2, 3, 4$ (for $z = 2$, $+i$ is adjacent to $+j$, $\pm k$, $\pm l$; for $z = 3$, $+i$ to $+j$, $+k$, $\pm l$; for $z = 4$ the free directions are the four positive ones). A change of h_p with both links free gives $\sum \eta = 2|h_p - h'_p|$, with one link free $\sum \eta = |h_p - h'_p|$, and a change of the outside link in direction f by η gives η on each free neighbour of f, at most d of them. With $c \le c_0$: $\frac{2c}{1-4c} \le \frac{40}{13} c$, $\frac{2c}{1-3c} \le \frac{160}{59} c$, $\frac{c}{1-cd} \le \frac{c}{1-5c_0} = \frac{16}{9} c$ and $\frac{dc}{1-dc} \le \frac{5c_0}{1-5c_0} = \frac{7}{9}$.

(c), (d) for $z(v) = 1$. Let $F = \{f\}$ and $X_f = Y$. The map $X \mapsto Z = (Y^{-1} X_\rho)_\rho$ carries $\gamma^{h,Y}$ to $\mu_h^{(f)}$, which does not depend on Y. A change of h_p, p spanned by f and τ, is a change of the field on Z_τ, so (9.3) gives a coupling with $\int \sum \delta(Y Z_\rho, Y Z'_\rho) \le \kappa_0 c |h_p - h'_p|$,

with $g = 1$. For a change $Y \to Y'$ couple $X = YZ$ and $X' = Y'Z$ and take $g = Y'Y^{-1}$; then $\tilde{d}_v(X, X') \le \delta(Y'Y^{-1}, 1) = \delta(Y, Y')$.

(e) The block law is ν_b with $b = 4\beta$ times the sum of the six vectors of the plaquettes through $(v - e_a, a)$; such a change moves b by at most $4|\beta||h_p - h'_p|$, respectively $4|\beta|\eta$, since an outside link lies in at most one plaquette with $(v - e_a, a)$. Apply (9.1). □

Part (b) is the point of the soft quotient: a star with one fixed link is exactly as good as a full star, and the price $z(v)\delta(g, 1)$ never feeds back into the system, because an outside link of a lower vertex is a factor of no two-path at a block (Lemma 9.7(iii)). The margin is lost at the edges of the cube, $z(v) = 2$, and the next lemma pays for them with weights.

9.7. **The column inequality.** For a block vertex v and a plaquette p at v let $\Gamma_v(p)$ be the constant of Lemma 9.11(a), (c) or (e), and $\Gamma_v(p) = 0$ if no link of p at v lies in V. Put $u_v = 1$ for upper v and $u_v = u_{z(v)}$ for lower v, with

$$u_0 = u_1 = 1, \quad u_2 = \tfrac{5}{6}, \quad u_3 = \tfrac{7}{10}, \quad u_4 = \tfrac{3}{5}; \qquad \lambda = \tfrac{97}{100}.$$

Lemma 9.12. *Let $n \ge 4$ and $c \le c_0$. For every $k \in V$, with plaquettes $p_{(\nu,\epsilon)}$ and even corners $v_{(\nu,\epsilon)} = o + \epsilon e_\nu$ as above,*

$$\sum_{(\nu,\epsilon)} u_{v_{(\nu,\epsilon)}} \Gamma_{v_{(\nu,\epsilon)}}(p_{(\nu,\epsilon)}) \le \lambda\, u_{w(k)}.$$

Proof. Put $f(z) = u_z \kappa_z c$: $f(0) = f(1) = \kappa_0 c < 1.8004\, c$, $f(2) = \frac{15}{8} c$, $f(3) = \frac{28}{13} c < 2.1539\, c$ and $f(4) = \frac{96}{59} c < 1.6272\, c$, all at least c. By Lemma 9.7(ii) a lower $v_{(\nu,\epsilon)}$ has both links of p free, so it contributes $f(z(v_{(\nu,\epsilon)}))$; an upper one contributes at most c, and other vertices contribute 0. Let o, μ and ς be as in Lemma 9.7(ii).

Case A: $\hat{o}$ has a coordinate equal to n. Since $k \in V$ is (o, μ), which needs $o \in Q$, or $(o - e_\mu, \mu)$, which needs $\hat{o} - e_\mu \in [0, n-1]^4$, this coordinate is $\hat{o}_\mu$, and $w(k) = o - e_\mu$ with $\hat{w}_\mu = n - 1 \ge 1$; so $z(w) \le 3$ and $u_w \ge \frac{7}{10}$. Every $v_{(\nu,\epsilon)}$ has μ-coordinate n, so the sum is at most $6c \le 6c_0 = 0.525 < \frac{7}{10} \cdot 0.97$.

Case B: $o \in Q$. Let $Z = z(o)$ be the number of zeros of $\hat{o}$, $a = \#\{\nu \ne \mu : \hat{o}_\nu = 0\}$ and $b = \#\{\nu \ne \mu : \hat{o}_\nu = 1\}$. As $n \ge 4$, the classes $\{0\}$, $\{1\}$, $[2, n-2]$ and $\{n-1\}$ are disjoint. For $\nu \ne \mu$: if $\hat{o}_\nu = 0$, $\epsilon = -1$ gives no block and $\epsilon = +1$ a lower vertex with $z = Z - 1$; if $\hat{o}_\nu = 1$, $\epsilon = +1$ gives $z = Z$ and $\epsilon = -1$ gives $z = Z + 1$; if $2 \le \hat{o}_\nu \le n - 2$ both give $z = Z$; and if $\hat{o}_\nu = n - 1$, $\epsilon = -1$ gives $z = Z$ and $\epsilon = +1$ an upper vertex, which contributes at most $c \le f(Z)$. Hence the sum is at most

$$a\, f(Z-1) + b\big(f(Z) + f(Z+1)\big) + 2(3 - a - b) f(Z).$$

If $\hat{o}_\mu = 0$, then $\varsigma = +1$, $Z = a + 1$ and $z(w) = a$, and the sum is at most $h(a, b) = a f(a) + b(f(a+1) + f(a+2)) + 2(3 - a - b) f(a+1)$, to be compared with λu_a. If $\hat{o}_\mu \ge 1$, then $Z = a$ and w is upper, with $u_w = 1$, or lower with $z(w) \in \{a, a+1\}$, so $u_w \ge u_{a+1}$; the sum is at most $g(a, b) = a f(a-1) + b(f(a) + f(a+1)) + 2(3 - a - b) f(a)$, to be compared with λu_{a+1}. Here $a + b \le 3$. Since $f(0) = f(1) \le f(2) \le f(3)$, the quantities $h(0, b)$, $g(1, b)$, $h(1, b)$ and $g(2, b)$ increase with b, and $g(0, b) = 6 f(0)$. Every term is c times a constant, so

it suffices to take $c = c_0$; with $\hat\lambda = \lambda/c_0 = \frac{388}{35} > 11.0857$ and f in units of c, the cases are

$$\begin{aligned}
&h(0,3) = 3f(0) + 3f(2) < 5.4012 + 5.625 = 11.0262 < \hat\lambda,\\
&g(0,b) = 6f(0) < 10.8024 < \hat\lambda,\\
&g(1,2) = 3f(0) + 2f(2) < 9.1512 < \tfrac{5}{6}\hat\lambda, \quad \tfrac{5}{6}\hat\lambda > 9.2380,\\
&h(1,2) = f(0) + 2f(2) + 2f(3) < 9.8582 < \hat\lambda,\\
&g(2,1) = 2f(0) + f(2) + f(3) < 7.6297 < \tfrac{7}{10}\hat\lambda, \quad \tfrac{7}{10}\hat\lambda > 7.7599,\\
&h(2,0) = 2f(2) + 2f(3) < 8.0578 < \tfrac{5}{6}\hat\lambda,\\
&h(2,1) = 2f(2) + f(3) + f(4) < 7.5311 < \tfrac{5}{6}\hat\lambda,\\
&g(3,0) = 3f(2) = 5.625 < \tfrac{3}{5}\hat\lambda, \quad \tfrac{3}{5}\hat\lambda > 6.6514,\\
&h(3,0) = 3f(3) < 6.4617 < \tfrac{7}{10}\hat\lambda.
\end{aligned}$$

All inequalities hold. □

9.8. **The cube condition uniformly in the cube size.**

Proposition 9.13. *Let $c \le c_0$, $n \ge 4$ and $L \ge 2n+4$ even. For every link j,*

$$\sum_{x:\, j \notin V_x} \tilde R_x(j) \le 259\, n^3.$$

Proof. Fix x, $j \notin V = V_x$ and U, U' differing only at j, and write $\eta = \delta(U_j, U'_j)$. Let π minimise $\int D\, d\pi$ over couplings of $\gamma^0_{V,U}$ and $\gamma^0_{V,U'}$, and put $\varrho_v = \int \tilde d_v\, d\pi$. As in Lemma 9.10, replacing the block of v by an optimal coupling for $\tilde d_v$ of $\gamma_v(\cdot \mid \sigma, U)$ and $\gamma_v(\cdot \mid \sigma', U')$ changes only the v-term, so $\varrho_v \le \int \Omega_v\, d\pi$ with $\Omega_v = W^{\tilde d_v}(\gamma_v(\cdot \mid \sigma, U), \gamma_v(\cdot \mid \sigma', U'))$.

Pointwise bound. Changing the two-paths at v one at a time, and then the outside link j if it is a link at v, the triangle inequality for $W^{\tilde d_v}$ and Lemma 9.11 give $\Omega_v \le \sum_p \Gamma_v(p) |h_p(\sigma, U) - h_p(\sigma', U')| + r_v \eta$, where $r_v \le 1$ if v is lower and j is an outside link at v, $r_v \le c$ if v is upper and j is a link at v, and $r_v = 0$ otherwise. For each block vertex w fix g^*_w attaining $\tilde d_w(\sigma_{B_w}, \sigma'_{B_w})$, with $g^*_w = 1$ for upper w, and put $\delta_k = \delta(g^*_{w(k)} X_k(\sigma), X_k(\sigma'))$ for $k \in V$, so that $\sum_{k \in B_w \cap V} \delta_k \le \tilde d_w$. If both links of p at the opposite corner v' lie in V, then v' is lower, an upper vertex carrying one link of V, and by invariance of h_p under $X \mapsto g^*_{v'} X$ at v', $|h_p(\sigma, U) - h_p(\sigma', U')|$ is at most the sum of their δ_k. If exactly one lies in V, the other is off V and is not an outside link of a lower vertex (Lemma 9.7(iii)), so v' is upper and the bound is $\delta_k + [j \in p]\eta$. If none lies in V, the bound is $[j \in p]\eta$. Each $k \in V$ is a factor of a two-path at v for at most one p at v, and these v are the vertices $v_{(\nu,\epsilon)}$ of k, so Lemma 9.12 gives, pointwise,

$$\sum_v u_v \Omega_v \le \lambda \sum_{k \in V} u_{w(k)} \delta_k + B\eta \le \lambda \sum_w u_w \tilde d_w + B\eta,$$

with $B = \sum_{p \ni j} u_{v(p)} \Gamma_{v(p)}(p) + \max_v u_v r_v$, where $v(p)$ is the even corner of p opposite to the even end point of j.

Constants. By Lemma 9.11, $u_v \Gamma_v(p) \le \frac{28}{13} c$ in all cases ($\max_z f(z) = f(3)$; with one fixed link $\kappa_0 c$ or $\frac{16}{9} c$; upper vertices c), and only the plaquettes $p \ni j$ containing a link of V contribute. Integrating, $\sum_v u_v \varrho_v \le \lambda \sum_v u_v \varrho_v + B_x(j)\eta$, with $B_x(j) = \frac{28}{13} c\, P_x(j) + R_x(j)$, where $P_x(j)$ is the number of plaquettes $p \ni j$ with a link in V_x, and $R_x(j) = 1$ if j is an outside link at a lower vertex of the cube, c if j is a link off V_x at an upper even vertex of the cube, and 0 otherwise. All terms are finite, so $\sum_v u_v \varrho_v \le B_x(j)\eta/(1-\lambda)$ and

$$W^D(\gamma^0_{V,U}, \gamma^0_{V,U'}) \le \sum_v \varrho_v \le \frac{B_x(j)\,\eta}{u_4(1-\lambda)} = \frac{500}{9} B_x(j)\,\eta.$$

Summation over x. A plaquette counted in $P_x(j)$ contains a partner k of j with $k \in V_x$, and distinct plaquettes give distinct partners. For $j = (b, \nu)$ the eighteen partners have base points $b + \Delta$ with $\sum \|\Delta\|_1 = 18$, $\Delta \in \{0, e_\mu, e_\nu, -e_\mu, -e_\mu, e_\nu - e_\mu\}$ for each $\mu \neq \nu$, and $\#\{x : k \in V_x,\ j \notin V_x\} = |Q_0 \setminus (Q_0 - \Delta)| \le n^3 \|\Delta\|_1$ with $Q_0 = [0, n)^4$, by moving one coordinate at a time; so $\sum_x P_x(j) \le 18n^3$. For R_x, let v be the even end point of j. If v is the head of $j = (v - e_a, a)$, it is a lower vertex with $a \in Z(v)$ for exactly n^3 cubes, and an upper vertex of the cube with j off V_x for at most $3n^3$ cubes, the coordinate equal to n not being a; if v is the base point of j, it is never lower with $j \notin V_x$ and is upper for at most $4n^3$ cubes. So $\sum_x R_x(j) \le (1 + 3c)n^3$. With $c \le c_0$,

$$\sum_x \tilde{R}_x(j) \le \frac{500}{9}\Big(\frac{28}{13} \cdot 18c_0 + 1 + 3c_0\Big)n^3 = \frac{500}{9} \cdot \frac{4841}{1040}\, n^3 < 258.61\, n^3. \qquad \square$$

Theorem 9.14. *Let $G = \mathrm{SU}(2)$ and $|\beta_W| \le \frac{7}{20}$.*

(i) *For every $n \ge 260$ and even $L \ge 2n+4$, $\mathrm{BP}(n, 1-259/n)$ holds; in particular $\mathrm{BP}(n, \frac{1}{2})$ holds for $n \ge 518$.*

(ii) *For every $n \ge 2$ and even $L \ge 2n + 4$, $\mathrm{BP}(n, \varpi)$ holds with $\varpi = \frac{106}{1933} \cdot 517^{-4}$, independent of n and L.*

Proof. (i) Let $\mathcal{K} = L^{-4} \sum_x \mathcal{P}_x$ and let h be gauge invariant at every configuration with $\mathrm{Lip}(h) < \infty$, and let U, U' differ only at i. If $i \in V_x$, then $\mathcal{P}_x h(U) = \mathcal{P}_x h(U')$. If $i \notin V_x$, Lemma 9.9 and any coupling π of $\gamma^0_{V_x,U}$ and $\gamma^0_{V_x,U'}$ give $|\mathcal{P}_x h(U) - \mathcal{P}_x h(U')| \le \mathrm{Lip}(h)(\int D\, d\pi + \delta(U_i, U_i'))$, and the infimum over π gives $\mathrm{Lip}(h)(\tilde{R}_x(i)+1)\delta(U_i, U_i')$. As i lies in n^4 cubes, Proposition 9.13 gives $\mathrm{Lip}(\mathcal{K}h) \le (1-\kappa)\,\mathrm{Lip}(h)$ with $\kappa = (1-259/n)n^4/L^4 > 0$, and $\mathcal{K}h$ is again gauge invariant at every configuration, by gauge covariance of $\gamma^0_{V_x,U}$. The spectral argument of Proposition 9.4 gives $\langle h, (I - \mathcal{K})h\rangle \ge \kappa\, \mathrm{Var}_\mu(h)$, first for such h and then, by density, for every real $f \in \mathcal{A}$; and $L^{-4} \sum_x \mu((f - \mathcal{P}_x f)^2) = \langle f, (I - \mathcal{K})f\rangle$, which is $\mathrm{BP}(n, 1 - 259/n)$.

(ii) For $n \ge 518$ use (i). For $2 \le n \le 517$ use the proof of Theorem 9.5, which gives $\mathrm{BP}(n, (1 - 6c\, s(c))(n-1)^4/(n^4\lceil (n-1)^4/2\rceil))$ whenever $6c\, s(c) < 1$; here $1 - 6c\, s(c) \ge 1 - \frac{1827}{1933} = \frac{106}{1933}$ by Lemma 9.2(ii), and $(n-1)^4 \ge \lceil (n-1)^4/2\rceil$, so $\mathrm{BP}(n, \frac{106}{1933}n^{-4})$ holds, and $\frac{106}{1933}n^{-4} \ge \varpi$. $\square$

Theorem 9.14 extends the uniform range $\frac{5}{18}$ of Theorem 8.11 past $\beta_W = \frac{1}{3}$, where Proposition 8.12 stops every gauge-fixed single-link scheme and Proposition 9.6 every scheme that treats face links as single sites. The faces cost nothing here; the margin is lost at the cube edges, through the column $h(0,3)$ of Lemma 9.12.[16] The value $\beta_W \le \frac{7}{20}$ is far below the scaling region of four-dimensional SU(2), where β_W is of order 2 and above, and the next proposition explains why no one-step contraction of this kind reaches it.

9.9. **Single links near flat configurations.** Let $A(\kappa) = \int x_1\, \nu_{\kappa e_1}(dx)$ be the mean resultant length of ν_b at $|b| = \kappa$. By symmetry $\int x\, \nu_b(dx) = A(|b|)b/|b|$; A is nondecreasing, its derivative being the variance of x_1, and $A(\kappa) \to 1$ as $\kappa \to \infty$.

Proposition 9.15. *Let $\beta > 0$, let e be a link whose six plaquettes contain no link fixed at a value other than 1, and let k be a partner of e. The single-link influence satisfies*

$$C_{ek} = \sup \frac{W_1(\nu_b, \nu_{b'})}{\delta(U_k, U_k')} \ge \frac{A(24\beta)}{6}.$$

Hence in any single-link scheme in which every link has at least twelve free partners, as for the forest of Theorem A, every row sum of the influence matrix is at least $2A(24\beta)$, which tends to 2 as $\beta \to \infty$.

[16] A linear programme over weights depending on the block, with the exact constants for $z(v) = 2$, suggests on cubes of side up to 8 that the method reaches the star threshold $\beta_W \approx 0.3609$ of Theorem 9.5; with weights depending only on $z(v)$ it stops near $\beta_W \approx 0.345$. These are numerical observations and no statement rests on them.

Proof. Take all links equal to 1, so every staple is 1 and $b = 24\beta \cdot 1$. Change U_k so that the staple of the plaquette containing e and k becomes $\cos\vartheta + t\sin\vartheta$ with a unit imaginary quaternion t; then $\delta(U_k, U_k') = 2\sin\frac{\vartheta}{2}$ and $b' = 4\beta(5+\cos\vartheta+t\sin\vartheta)$. The function $x \mapsto \langle x, t\rangle$ is 1-Lipschitz, so $W_1(\nu_b, \nu_{b'}) \geq \langle \int x\, d\nu_{b'} - \int x\, d\nu_b, t\rangle = A(|b'|)\, 4\beta \sin\vartheta/|b'|$. Divide by $2\sin\frac{\vartheta}{2}$ and let $\vartheta \to 0$. For the limit of A: the law of x_1 under $\nu_{\kappa e_1}$ has density proportional to $e^{\kappa t}\sqrt{1-t^2}$ on $[-1,1]$, and for $0 < \epsilon < \frac{1}{2}$ the mass of $[-1, 1-2\epsilon]$ divided by that of $[1-\epsilon, 1]$ is at most $e^{-\kappa\epsilon}\pi / \int_{1-\epsilon}^{1} \sqrt{1-t^2}\, dt \to 0$. □

At weak coupling the Gibbs measure concentrates near flat configurations, so even an influence matrix computed only on typical configurations does not contract there: per link it approaches $\frac{1}{6}$ per partner, and $12 \cdot \frac{1}{6} = 2$.[17] Larger blocks with better internal estimates push the threshold to some finite β_W, but the mass at weak coupling is non-perturbative, $m \sim \Lambda e^{-c\beta}$, and a contraction can appear only at block sizes of order $e^{c\beta}$ lattice units. Section 12 and Appendix H make this precise.

10. Free massive fields

Here $d \geq 1$, $n \geq 2$, $L \geq n+1$, $\mathbb{T} = (\mathbb{Z}/L\mathbb{Z})^d$, $\mu > 0$ is a lattice mass, $-\Delta$ is the lattice Laplacian and $Q = -\Delta + \mu^2$. For a positive definite Q on $\mathbb{R}^N$ let γ_Q be the centred Gaussian law with covariance $C = Q^{-1}$, and for $V \subset \{1, \dots, N\}$ let $\mathcal{P}_V$ be the conditional expectation given the coordinates outside V. For the cubes $V_x = x + \{0, \dots, n-1\}^d$, $x \in \mathbb{T}$, put

$$\varpi(Q) = \inf_f \frac{n^{-d} \sum_x \|f - \mathcal{P}_{V_x} f\|^2}{\mathrm{Var}_{\gamma_Q}(f)}, \tag{10.1}$$

the infimum over nonconstant $f \in L^2(\gamma_Q)$, the best constant in $\mathrm{BP}(n, \cdot)$. With $s = n\mu$ and $\mu = ma$, $n = \ell/a$, the parameter $s = m\ell$ is the mass times the physical size of the cube.

Lemma 10.1 (linear functions suffice)**.** *For a positive definite Q and any family of coordinate sets (V_x), the infimum in* (10.1) *equals its infimum over the linear functions $\ell_h(\phi) = \langle h, \phi\rangle$, and*

$$\|\ell_h - \mathcal{P}_V \ell_h\|^2 = \langle h_V, Q_{VV}^{-1} h_V\rangle, \qquad \mathrm{Var}(\ell_h) = \langle h, Q^{-1} h\rangle,$$

with h_V the restriction of h to V and Q_{VV} the block of Q on V.

Proof. The conditional law of ϕ_V given ϕ_{V^c} is Gaussian with covariance Q_{VV}^{-1}, which gives the formula for ℓ_h. Write $\phi = Q^{-1/2}\xi$ with ξ standard Gaussian. Conditioning on ϕ_{V^c} is conditioning on the Gaussian subspace $K_V = \{\langle u, \xi\rangle : u \in Q^{-1/2}\mathbb{R}^{V^c}\}$, and on the m-th Wiener chaos, identified with the symmetric tensors $\mathrm{Sym}^m(\mathbb{R}^N)$, it acts as $\pi_V^{\otimes m}$, with π_V the orthogonal projection onto $Q^{-1/2}\mathbb{R}^{V^c}$ [29, Ch. 4]. The chaoses are orthogonal and invariant under every $\mathcal{P}_V$, so it suffices to treat one chaos $m \geq 1$. The projection $I - \pi_V^{\otimes m}$ dominates $(I - \pi_V) \otimes I^{\otimes(m-1)}$, as both are projections that commute and the range of the second lies in that of the first. For a symmetric tensor F, therefore,

$$\sum_x \|F - \pi_{V_x}^{\otimes m} F\|^2 \geq \Big\langle F, \Big(\sum_x (I - \pi_{V_x})\Big) \otimes I^{\otimes(m-1)} F\Big\rangle \geq \lambda \|F\|^2,$$

where λ is the least eigenvalue of $\sum_x (I - \pi_{V_x})$ on $\mathbb{R}^N$, which is the infimum of the quotient over linear functions ($m = 1$). □

Theorem 10.2 (the exact constant)**.** *Let $V = \{0, \dots, n-1\}^d$, $-\Delta_D$ its Dirichlet Laplacian and $\mathbf{1}$ its constant function. Then*

$$\varpi(-\Delta + \mu^2) = \frac{\mu^2}{n^d} \langle \mathbf{1}, (-\Delta_D + \mu^2)^{-1} \mathbf{1}\rangle,$$

[17] Numerically $A(\kappa) = \frac{1}{2}$ at $\kappa \approx 2.45$, that is $\beta_W \approx 0.41$.

attained only at multiples of $\ell_{\mathbf{1}}$ among linear functions, and

$$\Big(\frac{5}{6}\Big)^d \frac{s^2}{s^2+10d} \le \varpi(-\Delta+\mu^2) \le \frac{s^2}{s^2+4dn^2\sin^2\frac{\pi}{2n+2}} \le \frac{s^2}{s^2+16d/9}.$$

Proof. Since $L \ge n+1$, the block $Q_{V_xV_x}$ is the translate of $Q_{VV} = -\Delta_D+\mu^2$; put $G_V = Q_{VV}^{-1}$. By Lemma 10.1, $n^d\varpi$ is the least value of $\langle h, Sh\rangle/\langle h, Q^{-1}h\rangle$ with $S = \sum_x R_{V_x}^{\mathsf T} Q_{V_xV_x}^{-1} R_{V_x}$, that is, the least eigenvalue of $Q^{1/2}SQ^{1/2}$. Translations commute with Q and S, so the plane waves $e_k(y) = e^{ik\cdot y}$ diagonalise it with eigenvalues $\lambda_k = \hat q(k)\langle e_k^V, G_V e_k^V\rangle$, where $\hat q(k) = \mu^2 + \sum_i(2-2\cos k_i)$ and e_k^V is the restriction to V.

Let B map f to the function $u \mapsto \sum_{w\sim u,\, w\notin V} f(w)$ on V. Then $Q_{VV}R_Vf = R_VQf + Bf$, so $Q_{VV}e_k^V = \hat q(k)e_k^V + Be_k$, and pairing G_V of this with e_k^V gives

$$\lambda_k = n^d - \sum_{u,u'\in V}\ \sum_{w\sim u',\, w\notin V} G_V(u,u')\cos\big(k\cdot(w-u)\big).$$

Write $Q_{VV} = (2d+\mu^2)(I-K)$ with K entrywise nonnegative and $\|K\| \le 2d/(2d+\mu^2) < 1$; the Neumann series shows that G_V has positive entries. Hence $\lambda_k - \lambda_0 = \sum G_V(u,u')(1-\cos(k\cdot(w-u))) \ge 0$, with equality only at $k=0$ (take $u=u'$ on a face and $w = u+e_i$). Since $Q\mathbf{1} = \mu^2\mathbf{1}$, $\lambda_0 = \mu^2\langle\mathbf{1}, G_V\mathbf{1}\rangle$, which is the formula.

Upper bound. $G_V \le (\lambda_1+\mu^2)^{-1}$ with $\lambda_1 = 4d\sin^2\frac{\pi}{2n+2}$ the least eigenvalue of $-\Delta_D$, and $\|\mathbf{1}\|^2 = n^d$; then $\sin t \ge 2t/\pi$ gives $n^2\lambda_1 \ge 4dn^2/(n+1)^2 \ge 16d/9$.

Lower bound. By the variational formula $\langle\mathbf{1}, G_V\mathbf{1}\rangle \ge \langle\mathbf{1}, w\rangle^2/\langle w, Q_{VV}w\rangle$, take $w(u) = \prod_i \tau(u_i)$ with the parabolic profile $\tau(s) = 6(s+1)(n-s)/(n(n+1)(n+2))$ on $\{0,\dots,n-1\}$, zero outside. Then $\sum\tau = 1$, $S_2 = \sum\tau^2 = \alpha_n/n$ and $D_2 = \sum_{s\in\mathbb Z}(\tau(s+1)-\tau(s))^2 = \beta_n/n^3$ with

$$\alpha_n = \frac{6(n^2+2n+2)}{5(n+1)(n+2)} \le \frac{6}{5}, \qquad \beta_n = \frac{12n^2}{(n+1)(n+2)} = 10\rho_n\alpha_n, \quad \rho_n = \frac{n^2}{n^2+2n+2} < 1,$$

by the power sums $\sum_{u=1}^n u(n+1-u) = n(n+1)(n+2)/6$, $\sum_{u=1}^n u^2(n+1-u)^2 = n(n+1)(n+2)(n^2+2n+2)/30$ and $\sum_{u=0}^n(n-2u)^2 = n(n+1)(n+2)/3$. The Dirichlet form of the product is $\langle w, -\Delta_D w\rangle = dS_2^{d-1}D_2$ and $\langle\mathbf{1}, w\rangle = 1$, so

$$\varpi \ge \frac{\mu^2}{n^d(\mu^2S_2^d + dS_2^{d-1}D_2)} = \frac{1}{\alpha_n^d + d\alpha_n^{d-1}\beta_n s^{-2}} = \alpha_n^{-d}\frac{s^2}{s^2+10d\rho_n} \ge \Big(\frac{5}{6}\Big)^d\frac{s^2}{s^2+10d}. \quad \square$$

At $n=2$ the two bounds coincide with $s^2/(s^2+4d)$. For $d=4$ the lower bound is 0.0118 at $s=1$ and 0.0438 at $s=2$; the upper bound vanishes as $s\to 0$, so a massless free field satisfies no cube condition uniform in the lattice spacing. The same computation applies, with the Hodge decomposition of the torus, to the free massive vector field on gauge-invariant functions; it is not used here.

10.1. **The Casimir defect of the free field.** The Casimir condition (C) of Proposition 7.30 asks that the physical side of a spatial torus times its Casimir defect under halving be at most $\frac{1}{70}$ at one scale and tend to zero at large scales. For the free field both quantities can be computed exactly in the continuum limit, and the computation calibrates the condition: it fails for a massless field at every scale, and it holds for a field of mass μ exactly beyond the scale $17.7/\mu$. Throughout this subsection the spatial torus is $\Lambda_{\vec\ell} = \prod_{i=1}^3 \mathbb Z/\ell_i\mathbb Z$ with all $\ell_i \ge 2$, $V = \ell_1\ell_2\ell_3$, its dual momenta are $k \in \prod_i \frac{2\pi}{\ell_i}\mathbb Z/\ell_i\mathbb Z$, $\hat k^2 = \sum_i 4\sin^2(k_i/2)$, and $\mu_0 \ge 0$ is a lattice mass. The transfer matrix of the free field of mass $\mu_0 > 0$ is the operator $\mathbb T_{\vec\ell}$ on $L^2(\mathbb R^{\Lambda_{\vec\ell}})$ with the kernel

$$\mathbb T_{\vec\ell}(\varphi,\varphi') = \exp\Big(-\tfrac12|\varphi'-\varphi|^2 - \tfrac14\langle\varphi, Q\varphi\rangle - \tfrac14\langle\varphi', Q\varphi'\rangle\Big), \qquad Q = -\Delta+\mu_0^2, \tag{10.2}$$

the spatial part of the action being split symmetrically between the two slices as in Lemma 7.3; $\operatorname{Tr}\mathbb{T}_{\vec\ell}^{M}$ is the partition function $Z(\vec\ell, M)$ of the free field on $\Lambda_{\vec\ell}\times\mathbb{Z}/M\mathbb{Z}$ for every $M\geq 1$, which is symmetric in its four arguments. Put

$$\omega(k)=\operatorname{arccosh}\Big(1+\frac{\mu_0^2+\hat k^2}{2}\Big),\qquad E_0(\vec\ell)=-\log\lambda_0(\mathbb{T}_{\vec\ell}),$$

and define $\kappa(\vec\ell)=E_0(\vec\ell)-\sum_{s\in\mathcal{S}}E_0(s)$ and $\eta(N;s)$ as in Section 7.8, with the eight halves $s\in\mathcal{S}$ of sides $\lfloor\ell_i/2\rfloor$ and $\lceil\ell_i/2\rceil$; there is no coupling and hence no index p. Let $p_t(x)=\frac{1}{2\pi}\int_{-\pi}^{\pi}e^{iqx-4t\sin^2(q/2)}\,dq=e^{-2t}I_{|x|}(2t)$, $x\in\mathbb{Z}$, be the heat kernel of the Laplacian of $\mathbb{Z}$, and $h_L(t)=\sum_{j\in\mathbb{Z}/L\mathbb{Z}}e^{-4t\sin^2(\pi j/L)}=\operatorname{Tr}e^{t\Delta_{\mathbb{Z}/L\mathbb{Z}}}$.

Lemma 10.3 (the free vacuum energy)**.** *Let $\mu_0>0$. Then $\mathbb{T}_{\vec\ell}$ is a positive Hilbert–Schmidt operator with a continuous strictly positive kernel, its largest eigenvalue is simple, and*

$$E_0(\vec\ell)=\frac12\sum_k\omega(k)-\frac{V}{2}\log 2\pi,$$

$$\kappa(\vec\ell)=\frac12\int_0^\infty\frac{dt}{t}\,e^{-\mu_0^2t}p_t(0)\Big[\prod_{j=1}^{3}\big(h_{\lfloor\ell_j/2\rfloor}(t)+h_{\lceil\ell_j/2\rceil}(t)\big)-\prod_{j=1}^{3}h_{\ell_j}(t)\Big]\geq 0.$$

The integral converges absolutely also at $\mu_0=0$, and $\kappa_0(\vec\ell):=\lim_{\mu_0\downarrow0}\kappa(\vec\ell)$ exists, equals that integral at $\mu_0=0$, and equals $\frac12\sum_{k\neq0}\omega_0(k)-\frac12\sum_s\sum_{k\neq0}\omega_0(k)$ with $\omega_0(k)=\operatorname{arccosh}(1+\hat k^2/2)$.

Proof. The kernel (10.2) is $e^{-\langle\varphi,Q\varphi\rangle/4}G(\varphi-\varphi')e^{-\langle\varphi',Q\varphi'\rangle/4}$ with G a Gaussian, a positive definite function, so $\mathbb{T}_{\vec\ell}$ is a positive operator; its kernel is continuous, strictly positive and square integrable because $Q>0$, so it is Hilbert–Schmidt and, by Jentzsch's theorem, its largest eigenvalue is simple with a positive eigenfunction. In the orthonormal basis of real Fourier modes the quadratic forms $|\varphi'-\varphi|^2$ and $\langle\varphi,Q\varphi\rangle$ are diagonal, so $\mathbb{T}_{\vec\ell}$ is the tensor product over the modes of the one-dimensional operators with kernel $K_a(x,y)=\exp(-\frac12(x-y)^2-\frac{a}{4}(x^2+y^2))$, $a=\mu_0^2+\hat k^2>0$. For $c>0$ and $b=c-1-a/2$,

$$\int_{\mathbb{R}}K_a(x,y)e^{-by^2/2}dy=\sqrt{\frac{2\pi}{c}}\exp\Big(-\frac{x^2}{2}+\frac{x^2}{2c}-\frac{a}{4}x^2\Big),$$

which equals $\sqrt{2\pi/c}\,e^{-bx^2/2}$ exactly when $c^2-(2+a)c+1=0$, that is $c=e^{\omega}$ with $\cosh\omega=1+a/2$, and then $b=e^{\omega}-\cosh\omega=\sinh\omega>0$. The positive function $e^{-bx^2/2}$ is therefore an eigenfunction of the positive kernel K_a, hence its Perron eigenfunction, with eigenvalue $\sqrt{2\pi}\,e^{-\omega/2}$; the largest eigenvalue of the tensor product is the product, which is the formula for E_0.

For the integral representation, $\frac{1}{2\pi}\int_{-\pi}^{\pi}\log(A-2\cos q)\,dq=\log\frac{A+\sqrt{A^2-4}}{2}$ for $A\geq2$ (write $A-2\cos q=c|1-e^{iq}/c|^2$ with $c+1/c=A$, $c\geq1$, and use $\int_{-\pi}^{\pi}\log|1-e^{iq}/c|\,dq=0$), so with $A=2+a$, $\omega(k)=\frac{1}{2\pi}\int_{-\pi}^{\pi}\log(a+4\sin^2\frac{q}{2})dq$. By Frullani's formula $\log\lambda=\int_0^\infty(e^{-t}-e^{-\lambda t})\frac{dt}{t}$, applied with $\lambda=a+4\sin^2\frac{q}{2}\in[a,a+4]$ and the two integrals exchanged, which is legitimate because λ ranges over a compact subset of $(0,\infty)$,

$$\omega(k)=\int_0^\infty\frac{dt}{t}\Big[e^{-t}-e^{-\mu_0^2t}p_t(0)\prod_{j=1}^{3}e^{-4t\sin^2(k_j/2)}\Big].$$

Summing over k gives $\sum_k\omega(k)=\int\frac{dt}{t}[Ve^{-t}-e^{-\mu_0^2t}p_t(0)\prod_j h_{\ell_j}(t)]$, each bracket being $O(t)$ at 0 and $O(e^{-t}+e^{-\mu_0^2t})$ at infinity. The extensive terms cancel in κ because $\sum_s V(s)=V$, and $\sum_s\prod_j h_{s_j}=\prod_j(h_{\lfloor\ell_j/2\rfloor}+h_{\lceil\ell_j/2\rceil})$, which is the formula. Positivity of κ is the proof of Proposition 7.28: Lemma 7.27, which uses only the positivity and the Hilbert–Schmidt property of the transfer matrices in the three spatial directions and the symmetry of Z, gives $Z(\vec\ell,M)\leq\prod_s Z(s,M)$, and the M-th root gives $\lambda_0(\vec\ell)\leq\prod_s\lambda_0(s)$.

At $\mu_0 = 0$ the integrand is still absolutely integrable: as $t \to 0$, $h_L(t) = L - 2Lt + O(t^2)$ for $L \geq 2$ because $\sum_j \sin^2(\pi j/L) = L/2$, so the bracket is $O(t^2)$; as $t \to \infty$ the bracket is bounded by $9 \max_j \ell_j^3$ and $p_t(0) \leq (4\pi t)^{-1/2}$ (the bound of Lemma 10.5(ii) below). Dominated convergence as $\mu_0 \downarrow 0$ identifies the limit of the integral with its value at $\mu_0 = 0$. On the other side, $\omega(0) = \operatorname{arccosh}(1 + \mu_0^2/2) \to 0$ and the other $\omega(k)$ are continuous in μ_0, which gives the mode sum. □

For $v > 0$ put $\vartheta(v) = \sum_{j\in\mathbb{Z}} e^{-4\pi^2 j^2 v}$, $\beta(v) = 8\vartheta(v)^3 - \vartheta(v/4)^3$, and for $x \geq 0$

$$c(x) = \frac{1}{\sqrt{4\pi}} \int_0^\infty v^{-3/2} e^{-x^2 v/4} \beta(v)\, dv. \tag{10.3}$$

Proposition 10.4 (the continuum Casimir function)**.** (i) *β is positive, with the Poisson form*

$$\beta(v) = 8(4\pi v)^{-3/2} \sum_{w\in\mathbb{Z}^3\setminus 0} \big(e^{-|w|^2/4v} - e^{-|w|^2/v}\big),$$

and c is finite, strictly decreasing on $[0,\infty)$ and tends to 0 at infinity.

(ii) *For $x > 0$, with K_2 the modified Bessel function of the second kind,*

$$c(x) = \frac{x^2}{4\pi^2} \sum_{w\in\mathbb{Z}^3\setminus 0} \frac{4K_2(x|w|/2) - K_2(x|w|)}{|w|^2},$$

$$c(0) = \frac{15}{2\pi^2} \sum_{w\in\mathbb{Z}^3\setminus 0} |w|^{-4} > 6.8, \qquad c(0) = 12.5630\ldots$$

(iii) *For $x > 0$, with $\varepsilon(x) = \sum_{|w|>1} |w|^{-5/2} e^{-x(|w|-1)/2}$,*

$$\frac{9}{2\pi^{3/2}}\, x^{3/2} e^{-x/2}\Big(1 + \frac{2}{x}\Big) \leq c(x) \leq \frac{6}{\pi^{3/2}}\, x^{3/2} e^{-x/2}\Big(1 + \frac{6}{x} + \frac{12}{x^2}\Big)\Big(1 + \frac{\varepsilon(x)}{6}\Big).$$

(iv) *The equation $c(x) = \frac{1}{70}$ has exactly one root X_*, and $16.75 \leq X_* \leq 17.99$.*

Proof. (i) Poisson summation gives $\vartheta(u) = (4\pi u)^{-1/2} \sum_j e^{-j^2/4u}$, so that $\vartheta(v)^3 = (4\pi v)^{-3/2} \sum_{w\in\mathbb{Z}^3} e^{-|w|^2/4v}$ and $\vartheta(v/4)^3 = 8(4\pi v)^{-3/2} \sum_{w\in\mathbb{Z}^3} e^{-|w|^2/v}$; the terms $w = 0$ cancel in β and every other term is positive. The integrand of (10.3) is positive and strictly decreasing in x; it is integrable because $\beta(v) = O(v^{-3/2} e^{-1/4v})$ at 0 and $\beta(v) \to 7$ at infinity, and dominated convergence gives $c(x) \to 0$.

(ii) For $p, q > 0$, $\int_0^\infty v^{-3} e^{-pv - q/v} dv = 2(p/q) K_2(2\sqrt{pq})$ [60, §6.22], and at $p = 0$ the integral is q^{-2}. Inserting the Poisson form of β into (10.3), the prefactor is $(4\pi)^{-1/2} \cdot 8(4\pi)^{-3/2} = 1/(2\pi^2)$, and the terms $p = x^2/4$, $q = |w|^2/4$ and $q = |w|^2$ give the series; at $x = 0$ each w contributes $(16-1)|w|^{-4}$. The sum $Z_4 = \sum_{w\neq 0} |w|^{-4}$ exceeds the six terms $|w| = 1$ and the twelve terms $|w|^2 = 2$, so $Z_4 > 9$ and $c(0) > 135/(2\pi^2) > 6.8$; the value $Z_4 = 16.5323\ldots$ gives $c(0) = 12.5630\ldots$.[18]

(iii) From $K_\nu(z) = \int_0^\infty e^{-z\cosh u} \cosh \nu u\, du$, $K_\nu(z)$ is decreasing in $z > 0$ and increasing in $\nu \geq 0$, and $K_{3/2}(z) = \sqrt{\pi/2z}\, e^{-z}(1 + 1/z)$, $K_{5/2}(z) = \sqrt{\pi/2z}\, e^{-z}(1 + 3/z + 3/z^2)$. For the lower bound keep the six terms $|w| = 1$ of the series, where $4K_2(x/2) - K_2(x) \geq 3K_2(x/2) \geq 3K_{3/2}(x/2)$. For the upper bound drop the negative terms, replace K_2 by $K_{5/2}$ and use, for $r = |w| \geq 1$ and $z = x/2$, $K_{5/2}(rz) \leq r^{-1/2} e^{-(r-1)z} K_{5/2}(z)$, which is read off from the closed form; the terms $|w| = 1$ give the factor 6 and the others give $\varepsilon(x)$.

(iv) Uniqueness is (i). The lower bound of (iii) is $\frac{9}{2\pi^{3/2}}(x^{3/2} + 2x^{1/2}) e^{-x/2}$, decreasing for $x > 2$ since $(x^{3/2} + 2x^{1/2})' - \frac{1}{2}(x^{3/2} + 2x^{1/2}) = x^{-1/2}(1 + \frac{x}{2} - \frac{x^2}{2})$; its value at $x = 16.75$

[18] $Z_4 = \int_0^\infty t\big(\theta(t)^3 - 1\big) dt$ with $\theta(t) = \sum_j e^{-tj^2}$, by $\int_0^\infty t e^{-t|w|^2} dt = |w|^{-4}$; the integral is computed with the Poisson form of θ for $t < 1$. Values of c from the Bessel series: $c(8) = 0.787$, $c(12) = 0.158$, $c(16) = 0.0297$, $c(18) = 0.0126$, $c(20) = 0.0053$, and the root of $c = \frac{1}{70}$ is $X_* = 17.7106$. Nothing below rests on these numbers; the bracket in (iv) is proved from the bounds (iii).

exceeds $\frac{1}{70}$ (it is $\frac{1}{70}$ at 16.7521), so $c > \frac{1}{70}$ on $[0, 16.75]$. The upper bound of (iii) is a product of decreasing functions of x (the factor $(x^{3/2} + 6x^{1/2} + 12x^{-1/2})e^{-x/2}$ has logarithmic derivative $-\frac{1}{2}\,\frac{x^3+3x^2+6x+12}{x^3+6x^2+12x} < 0$, and ε is decreasing); it equals $\frac{1}{70}$ at $x = 17.9870$, where $\varepsilon = 0.1246$, so $c < \frac{1}{70}$ on $[17.99, \infty)$. □

Lemma 10.5 (two bounds on the heat kernel)**.** (i) *For $0 \le t \le 2$ and $x \in \mathbb{Z}$, $p_t(x) \le t^{|x|}/|x|!$, and $p_t(0) \le 1$ for all t.*
(ii) *For $t > 0$, $p_t(0) \le (\pi/16t)^{1/2}$ and $p_t(x) \le (\pi/16t)^{1/2} \exp(-x^2/(4t + 2|x|))$.*
(iii) *$\sqrt{4\pi t}\, p_t(0) \to 1$ as $t \to \infty$.*

Proof. (i) $p_t(x) = e^{-2t}\sum_{j\ge 0} t^{2j+|x|}/(j!(j+|x|)!) \le e^{-2t}e^{t^2}t^{|x|}/|x|!$, and $t^2 \le 2t$ for $t \le 2$; $p_t(0) \le 1$ because $I_0(z) \le e^z$. (ii) The integrand of $p_t(x)$ is entire and 2π-periodic in q, so the contour may be shifted to $q + i\eta$, $\eta \ge 0$, which gives for $x \ge 0$ $p_t(x) \le e^{-\eta x}\frac{1}{2\pi}\int e^{-2t(1-\cos q\cosh\eta)}dq = e^{-\eta x}e^{2t(\cosh\eta-1)}\,e^{-z}I_0(z)$ with $z = 2t\cosh\eta$. Since $\sin(\theta/2) \ge \theta/\pi$ on $[0, \pi]$,

$$e^{-z}I_0(z) = \frac{1}{\pi}\int_0^\pi e^{-2z\sin^2(\theta/2)}d\theta \le \frac{1}{\pi}\int_0^\infty e^{-2z\theta^2/\pi^2}d\theta = \sqrt{\frac{\pi}{8z}} \le \sqrt{\frac{\pi}{16t}}.$$

With $\sinh\eta = y := x/2t$ the exponent is $-2t\Psi(y)$, $\Psi(y) = y\,\mathrm{arcsinh}\,y - \sqrt{1+y^2} + 1$, and $\Psi(y) \ge y^2/(2(1+y))$ because both vanish at 0 and $\Psi'(y) = \mathrm{arcsinh}\,y \ge \log(1+y) \ge \frac{1}{2}(1-(1+y)^{-2}) = \frac{d}{dy}\frac{y^2}{2(1+y)}$; and $2t\cdot\frac{y^2}{2(1+y)} = \frac{x^2}{4t+2x}$. (iii) is $I_0(z) \sim e^z/\sqrt{2\pi z}$ [60, §7.23]. □

Theorem 10.6 (the exact Casimir limit of the free field)**.** (i) *Let $m \to \infty$, $\ell_i \in \{m, m+1\}$ and $\mu_0 m \to x \in [0, \infty]$. Then $(m+1)\kappa(\vec{\ell}) \to c(x)$, with $c(\infty) = 0$.*
(ii) *For the free field of physical mass $\mu \ge 0$, lattice mass $\mu_0 = \mu/N$, and every $s > 0$: $\lim_{N\to\infty}\eta(N; s) = c(\mu s)$.*

Proof. (i) Substitute $t = m^2\tau$ in Lemma 10.3:

$$m\kappa(\vec{\ell}) = \frac{1}{2}\int_0^\infty \frac{d\tau}{\tau}\, e^{-\mu_0^2 m^2\tau}\,(m\,p_{m^2\tau}(0))\,\mathcal{B}_m(\tau),$$
$$\mathcal{B}_m(\tau) = \prod_j (h_{s_j'} + h_{s_j''})(m^2\tau) - \prod_j h_{\ell_j}(m^2\tau),$$

with $s_j' = \lfloor \ell_j/2 \rfloor$, $s_j'' = \lceil \ell_j/2 \rceil$. For fixed $\tau > 0$, $m\,p_{m^2\tau}(0) \to (4\pi\tau)^{-1/2}$ by Lemma 10.5 (iii). If $L/m \to \lambda > 0$, then, taking the representatives $|j| \le L/2$, $h_L(m^2\tau) = \sum_{|j|\le L/2} e^{-4m^2\tau\sin^2(\pi j/L)} \to \sum_j e^{-4\pi^2 j^2\tau/\lambda^2} = \vartheta(\tau/\lambda^2)$, by dominated convergence in j with the dominating sequence $e^{-16j^2m^2\tau/L^2} \le e^{-4j^2\tau}$, from $\sin(\pi j/L) \ge 2|j|/L$ and $m/L \ge \frac{1}{2}$. Here $\lambda = 1$ for the sides and $\lambda = \frac{1}{2}$ for the halves, so $\mathcal{B}_m(\tau) \to 8\vartheta(4\tau)^3 - \vartheta(\tau)^3$, and $e^{-\mu_0^2 m^2\tau} \to e^{-x^2\tau}$ (which is 0 for $x = \infty$). The substitution $\tau = v/4$ turns the limiting integral into $c(x)$. It remains to dominate the integrand by a function of τ alone. By the Poisson identity $h_L(t) = L\sum_{w\in\mathbb{Z}} p_t(Lw)$, the discrete Fourier inversion of $\sum_w p_t(x + Lw) = \frac{1}{L}\sum_k e^{ikx - 4t\sin^2(k/2)}$ at $x = 0$, the terms $w = 0$ cancel in $\mathcal{B}_m$ and

$$|\mathcal{B}_m(\tau)| \le \sum_{s\in\mathcal{S}} V(s) \sum_{w\ne 0}\prod_j p_t(s_j w_j) + V\sum_{w\ne 0}\prod_j p_t(\ell_j w_j), \qquad t = m^2\tau,$$

where every side involved lies in $[m/3, m+1]$. Write $r = \lceil m/3 \rceil$ for the least of them. *Region $t \le 1$.* By Lemma 10.5(i), for a side $L \ge r$ and $k \ge 1$, $p_t(Lk) \le t^{Lk}/(Lk)! \le (t^L/L!)^k \le \delta^k$ with $\delta = t^r/r! \le \frac{1}{2}$, so $\sum_{k\ge1} p_t(Lk) \le 2\delta$, and $\sum_{w\ne0}\prod_j p_t(L_j w_j) \le (p_t(0) + 4\delta)^3 - p_t(0)^3 \le 108\,\delta$. The region contributes at most $m\cdot 9(m+1)^3\cdot 108\int_0^1 t^r\frac{dt}{t\,r!} = \frac{972\,m(m+1)^3}{r\cdot r!} \to 0$. *Region $1 \le t \le m^2$, that is $m^{-2} \le \tau \le 1$.* Put $A = (\pi/16\tau)^{1/2}$, so that $m\,p_t(0) \le A$ and $p_t(0) \le A/m$. For a side $L \ge m/3$ and $k \ge 1$, Lemma 10.5(ii) and $a/(b+c) \ge \frac{1}{2}\min(a/b, a/c)$

give

$$\frac{L^2k^2}{4t+2Lk} \geq \frac{1}{2}\min\Big(\frac{k^2}{36\tau}, \frac{mk}{6}\Big) \geq \frac{k}{72\sqrt{\tau}},$$

using $m \geq \tau^{-1/2}$ and $\tau \leq 1$, so $\sum_{k\geq 1} p_t(Lk) \leq \frac{A}{m}\sum_{k\geq 1} e^{-k/72\sqrt{\tau}} \leq \frac{A}{m}\,B(\tau)$ with $B(\tau) = \frac{e^{-1/72\sqrt{\tau}}}{1-e^{-1/72}}$, and $\sum_{w\neq 0}\prod_j p_t(L_j w_j) \leq \big(\frac{A}{m}\big)^3\big[(1+2B)^3-1\big] \leq \big(\frac{A}{m}\big)^3 6B(1+2B)^2$. The integrand is therefore at most $\frac{1}{2}\cdot A\cdot 9\cdot 8m^3\cdot(A/m)^3\cdot 6B(1+2B)^2 = 216\,A^4B(\tau)(1+2B(\tau))^2$, a function of τ alone which is integrable against $d\tau/\tau$ on $(0,1]$, since $A^4 = (\pi/16\tau)^2$ and $B(\tau) \leq 73\,e^{-1/72\sqrt{\tau}}$. *Region* $\tau \geq 1$. For a side $L \leq m+1 \leq 2m$, $h_L(m^2\tau) \leq 1+2\sum_{j\geq 1} e^{-4j^2\tau} \leq 1+\frac{2e^{-4}}{1-e^{-4}} \leq 1.04$, so $|\mathcal{B}_m| \leq 9\cdot 1.04^3 \leq 10.2$ and the integrand is at most $5.1\,A \leq 5.1(\pi/16)^{1/2}\tau^{-1/2}$, integrable against $d\tau/\tau$ on $[1,\infty)$. Dominated convergence proves (i), as $(m+1)/m \to 1$.

(ii) Lower bound: the supremum defining $\eta(N;s)$ contains the cube of side $m = 2\lceil Ns/2\rceil$, whose halves are cubes, and $\mu_0 m = \mu m/N \to \mu s$ as $N \to \infty$, so $\liminf_N \eta(N;s) \geq \lim(m+1)\kappa = c(\mu s)$ by (i). Upper bound: if $\limsup_N \eta(N;s) > c(\mu s)+\epsilon$, there are $N_k \to \infty$, $m_k \geq N_k s$ and admissible $\vec{\ell}_k$ with $(m_k+1)\kappa(\vec{\ell}_k) \geq c(\mu s)+\epsilon$; along a subsequence $\mu m_k/N_k \to x \in [\mu s,\infty]$, and (i) gives the limit $c(x) \leq c(\mu s)$ because c is decreasing, a contradiction. □

Corollary 10.7 (the Casimir condition for the free field)**.** *For the free field of physical mass* $\mu \geq 0$*:*

(i) *if* $\mu = 0$*, then* $\eta(N;s) \geq c(0) > 6.8$ *(numerically* 12.56*) for every* N *and every* s*, so both clauses of* (C) *fail at every scale* s_0*;*

(ii) *if* $\mu > 0$*, then* (C) *holds at the scale* s_0 *when* $\mu s_0 > X_*$ *and fails when* $\mu s_0 < X_*$*; in particular* (C) *requires* $\mu s_0 \geq 16.75$*, that is, a torus of at least* 16.75 *Compton wavelengths;*

(iii) *the rate* $e^{-m_* s/2}$ *of Proposition 7.36 is attained: for the free field* $\lim_N \eta(N;s) = c(\mu s) \geq \frac{9}{2\pi^{3/2}}(\mu s)^{3/2}e^{-\mu s/2}$*.*

Proof. (i) The supremum defining $\eta(N;s)$ contains the even cubes of every side $m \geq Ns$, along which $(m+1)\kappa_0 \to c(0)$ by Theorem 10.6 (i) at $x = 0$. (ii) By Theorem 10.6(ii) and the monotonicity of c, $\lim_N \eta(N;s_0) = c(\mu s_0)$ is below $\frac{1}{70}$ exactly when $\mu s_0 > X_*$, and $\lim_s \lim_N \eta(N;s) = \lim_s c(\mu s) = 0$. (iii) is Proposition 10.4(iii). □

The free gluons of the weak-coupling limit give the same function with a multiplicity. Consider $\dim G$ independent copies of the Gaussian lattice Maxwell field: real link variables $A_{x,i}$ on the spatial links, the transfer kernel $\exp(-\frac{1}{2}\sum_{x,i}(A'_{x,i}-A_{x,i})^2 - \frac{1}{4}S(A) - \frac{1}{4}S(A'))$ with $S(A) = \sum_{x,i<j} F_{ij}(x)^2$, $F_{ij} = \partial_i A_j - \partial_j A_i$ the lattice curl, acting on functions invariant under the gauge transformations $A_{x,i} \mapsto A_{x,i} + \chi_{x+e_i} - \chi_x$; the three constant modes $\sum_x A_{x,i}$ of each colour, which the curl does not see, are taken on the circle $\mathbb{R}/2\pi\mathbb{Z}$, as they are for the gauge field.

Proposition 10.8 (free gluons)**.** *In this model* $E_0(\vec{\ell}) = \dim G\big[\sum_{k\neq 0}\omega_0(k) - \frac{3V}{2}\log 2\pi\big]$*, hence* $\kappa^{\mathrm{Max}}(\vec{\ell}) = 2\dim G\cdot\kappa_0(\vec{\ell})$ *and*

$$\lim_{N\to\infty}\eta^{\mathrm{Max}}(N;s) = 2\dim G\cdot c(0) = \begin{cases} 75.38\ldots & G = \mathrm{SU}(2), \\ 201.00\ldots & G = \mathrm{SU}(3), \end{cases}$$

at every scale s*. The free gluons violate* (C) *at every scale.*

Proof. In momentum space $F_{ij}(k) = q_i A_j(k) - q_j A_i(k)$ with $q_i = e^{ik_i} - 1$, $|q_i|^2 = 4\sin^2(k_i/2)$, and Lagrange's identity gives $\sum_{i<j}|q_iA_j - q_jA_i|^2 = |q|^2|A|^2 - |\langle q,A\rangle|^2$. So for $k \neq 0$ the form S has the eigenvalue $\hat{k}^2$ on the two-dimensional orthogonal complement of q and 0 on $A \propto q$, which is the Fourier transform of a gradient, that is, a gauge direction; the Fourier change of variables is orthogonal. A gauge-invariant function depends only on the transverse coordinates and the three constant modes, and the kinetic form $\frac{1}{2}\sum(A'-A)^2$ is

the sum of the kinetic forms of these orthogonal coordinates, so the transfer operator is the tensor product of the one-dimensional operators $K_{\hat{k}^2}$ of Lemma 10.3 over the two transverse modes of each $k \neq 0$, with largest eigenvalue $\sqrt{2\pi}e^{-\omega_0(k)/2}$ each, and of the three wrapped Gaussian kernels on the circle, whose Perron eigenfunction is the constant with eigenvalue $\sqrt{2\pi}$, the mass of the wrapped normalised Gaussian. The constants are proportional to V and cancel in κ; the two transverse modes give $2\kappa_0$ per colour; and Theorem 10.6 at $\mu = 0$ gives the limit. □

The pressure bound (7.15) of Section 7.9 is also explicit for the free field. It needs the whole spectrum of the one-mode kernel, not only its Perron eigenvalue.

Lemma 10.9 (the spectrum of the one-mode kernel)**.** *Let $a > 0$, $\cosh\omega = 1 + a/2$ with $\omega > 0$, and $K_a(x,y) = \exp(-\frac{1}{2}(x-y)^2 - \frac{a}{4}(x^2+y^2))$ on $L^2(\mathbb{R})$. The eigenvalues of K_a are $\sqrt{2\pi}\,e^{-\omega/2}e^{-j\omega}$, $j = 0,1,2,\dots$, each simple, with eigenfunctions $q_j(x)e^{-bx^2/2}$, q_j a polynomial of degree j and $b = \sinh\omega$. Hence K_a is trace class and $\operatorname{Tr} K_a^M = (2\pi)^{M/2}e^{-M\omega/2}/(1-e^{-M\omega})$ for $M \geq 1$, and for the transfer matrix (10.2) of the free field of mass $\mu_0 > 0$ on $\Lambda_{\vec{\ell}}$,*

$$\log \mathcal{Z}_{\vec{\ell}}(M) = -\sum_k \log\bigl(1 - e^{-M\omega(k)}\bigr), \qquad M \geq 1,$$

and every eigenvalue $\lambda \neq \lambda_0$ of $\mathbb{T}_{\vec{\ell}}$ satisfies $\lambda \leq e^{-\omega(0)}\lambda_0$, with equality for the one-mode excitation of the constant mode.

Proof. Put $c = e^{\omega}$, so that $b = c - 1 - a/2$ as in Lemma 10.3. For a polynomial q, completing the square in y,

$$\int_{\mathbb{R}} K_a(x,y)\,q(y)e^{-by^2/2}dy = e^{-bx^2/2}\int_{\mathbb{R}} e^{-\frac{c}{2}(y-x/c)^2}q(y)\,dy = \sqrt{\frac{2\pi}{c}}\,e^{-bx^2/2}\,\mathbb{E}\bigl[q(x/c + \xi/\sqrt{c})\bigr],$$

ξ a standard Gaussian variable, because $-\frac{1}{2}(x-y)^2 - \frac{a}{4}(x^2+y^2) - \frac{b}{2}y^2 = -\frac{c}{2}(y-x/c)^2 - \frac{b}{2}x^2$ when $c = 1 + a/2 + b$ and $c^2 - (2+a)c + 1 = 0$, which is the case for $c = e^{\omega}$, $b = \sinh\omega$. For $q(y) = y^j$ the right side is $\sqrt{2\pi/c}\,e^{-bx^2/2}$ times a polynomial of degree j with leading coefficient c^{-j}. So the finite-dimensional space of polynomials of degree at most j times $e^{-bx^2/2}$ is invariant, and K_a acts on it by a triangular matrix with the diagonal entries $\sqrt{2\pi/c}\,c^{-i} = \sqrt{2\pi}e^{-\omega/2}e^{-i\omega}$, $0 \leq i \leq j$, which are distinct; hence there are eigenfunctions $q_j e^{-bx^2/2}$ with $\deg q_j = j$ and these eigenvalues. Since K_a is self-adjoint, eigenfunctions of distinct eigenvalues are orthogonal, and the $q_j e^{-bx^2/2}$ span the polynomials times $e^{-bx^2/2}$, which are dense in $L^2(\mathbb{R})$; so they form an orthogonal basis of eigenfunctions, every eigenvalue is simple, and there are no others. The trace is the geometric series. The transfer matrix is the tensor product over the V real Fourier modes of the $K_{a(k)}$, $a(k) = \mu_0^2 + \hat{k}^2$, so its eigenvalues are $\prod_k \sqrt{2\pi}e^{-\omega(k)/2}e^{-j_k\omega(k)}$ over all $(j_k) \in \mathbb{N}_0^V$, the largest being $j \equiv 0$ and the second $j = 1$ at the k with the least $\omega(k)$, which is $k = 0$; dividing by λ_0^M and taking the trace gives the mode sum. □

For $\vartheta > 0$ and $x > 0$ put

$$F_\vartheta(x) = \sum_{n\in\mathbb{Z}^3} -\log\Bigl(1 - \exp\bigl(-\vartheta\sqrt{x^2 + 4\pi^2|n|^2}\bigr)\Bigr), \qquad F = F_{1/2}, \tag{10.4}$$

a convergent series of positive terms, strictly decreasing in x and in ϑ.

Theorem 10.10 (the pressure bound of the free field)**.** *Let $\mu > 0$ and $s > 0$, put $x = \mu s$, and for N with $c = Ns$ an even integer let the free field of mass $\mu_0 = \mu/N$ live on the cube of side c; let $X_N = \log \mathcal{Z}_{(c,c,c)}(c/2)$ and $D_N = D_{(c,c,c)}(c)$.*

(i) *$X_N = \sum_{n\in(-c/2,c/2]^3} -\log\bigl(1 - e^{-\frac{c}{2}\omega(2\pi n/c)}\bigr)$, and $\log(1 + D_N)$ is the same sum with $\frac{c}{2}$ replaced by c.*

(ii) *If* $N \geq \mu/2$, *then* $X_N \leq F_{0.2297}(x)$ *and* $\log(1 + D_N) \leq F_{0.4595}(x)$.
(iii) $\lim_{N\to\infty} X_N = F(x)$ *and* $\lim_{N\to\infty} D_N = e^{F_1(x)} - 1$; *and* $F(x) = \log(2/x) + O(1)$ *as* $x \downarrow 0$.
(iv) *On every spatial torus the second eigenvalue of the time transfer matrix is* $e^{-\Delta_N/N}\lambda_0$ *with* $\Delta_N = 2N \operatorname{arsinh}(\mu/2N)$, *which increases to* μ.

Consequently the pressure bound (7.15) *holds at every scale and every mass, with* $c_b = \sup_N X_N < \infty$; *the clause* $\Delta_N s \geq 2(X_N + \log 10)$ *of Corollary 7.40*(ii), *with the actual gap in place of* Δ_E, *holds for all large* N *if* $x > x_*$ *and fails for all large* N *if* $x < x_*$, *where* $x_* = 5.2063\ldots$ *is the root of* $x/2 = F(x) + \log 10$; *and* $(\mathrm{Z}_0^{\mathrm{dy}})$, $D_N \leq \frac{1}{10}$, *holds for all large* N *if* $x > x_1$ *and fails for all large* N *if* $x < x_1$, *where* $x_1 = 2.4851\ldots$ *is the root of* $F_1(x) = \log 1.1$.

Proof. (i) is Lemma 10.9 with $M = c/2$ and $M = c$, the dual momenta of the cube being $2\pi n/c$ with $n \in (-c/2, c/2]^3$.

(ii) For $|k_i| \leq \pi$, $\sin^2(k_i/2) \geq k_i^2/\pi^2$, so $\mu_0^2 + \hat{k}^2 \geq (4/\pi^2)(\mu_0^2 + |k|^2)$. Put $u = \frac{1}{2}\sqrt{\mu_0^2 + \hat{k}^2}$; as $\mu_0 \leq 2$ when $N \geq \mu/2$, $u \leq \frac{1}{2}\sqrt{4 + 12} = 2$, and the concavity of arsinh on $[0, \infty)$ with $\operatorname{arsinh} 0 = 0$ gives $\operatorname{arsinh} u \geq \frac{u}{2} \operatorname{arsinh} 2$. Since $\omega = \operatorname{arccosh}(1 + 2u^2) = 2 \operatorname{arsinh} u$,

$$\omega(k) \geq \operatorname{arsinh}(2)\, u \geq \frac{\operatorname{arsinh} 2}{\pi}\sqrt{\mu_0^2 + |k|^2} \geq 0.4595\sqrt{\mu_0^2 + |k|^2},$$

and with $c\mu_0 = x$, $c|k| = 2\pi|n|$ this gives $\frac{c}{2}\omega \geq 0.2297\sqrt{x^2 + 4\pi^2|n|^2}$ and $c\,\omega \geq 0.4595\sqrt{x^2 + 4\pi^2|n|^2}$. As $y \mapsto -\log(1 - e^{-y})$ is decreasing and the sums in (i) run over part of $\mathbb{Z}^3$, (ii) follows.

(iii) For fixed n, $c^2(\mu_0^2 + \hat{k}^2) \to x^2 + 4\pi^2|n|^2$ with $k = 2\pi n/c$, and $\omega = 2 \operatorname{arsinh}(\frac{1}{2}\sqrt{a}) = \sqrt{a}\,(1 + O(a))$, so $\frac{c}{2}\omega \to \frac{1}{2}\sqrt{x^2 + 4\pi^2|n|^2}$ and $c\,\omega \to \sqrt{x^2 + 4\pi^2|n|^2}$. Dominated convergence, with the termwise bounds of (ii), gives the limits. The term $n = 0$ of F is $-\log(1 - e^{-x/2}) = \log(2/x) + O(x)$, and the others are bounded uniformly for $x \leq 1$.

(iv) By Lemma 10.9 the second eigenvalue is $e^{-\omega(0)}\lambda_0$ with $\omega(0) = 2 \operatorname{arsinh}(\mu_0/2)$, so $\Delta_N = N\omega(0)$; and $y^{-1} \operatorname{arsinh} y$ decreases on $(0, \infty)$ to the limit 1 at 0, so $\Delta_N = \mu\, y^{-1} \operatorname{arsinh} y$ with $y = \mu/2N$ increases to μ.

For the consequences: by (ii), $\sup_{N \geq \mu/2} X_N \leq F_{0.2297}(x)$, and the finitely many smaller N give finite values, so $c_b < \infty$. Next, $\Delta_N s - 2(X_N + \log 10) \to x - 2F(x) - 2\log 10$, which is strictly increasing in x, since F decreases, tends to $-\infty$ at 0 and to $+\infty$ at ∞; so it has one root x_* and the sign of the limit decides for all large N. In the same way $D_N \to e^{F_1(x)} - 1$ is strictly decreasing in x, from $+\infty$ to 0, and crosses $\frac{1}{10}$ once, at x_1. □

The same three thresholds, 17.71 for (C) by Corollary 10.7, 5.21 for the clause of Corollary 7.40(ii) and 2.49 for $(\mathrm{Z}_0^{\mathrm{dy}})$ itself, measure the slack of the three routes on the one model where everything is explicit: the Casimir route pays for its generality in the scale it needs, the dyadic route with a pressure bound pays a factor of two, and $(\mathrm{Z}_0^{\mathrm{dy}})$ is what the free field actually provides beyond two and a half correlation lengths.

For the gauge theory itself nothing of this is proved. At weak coupling the analysis of Lüscher [39] (see also [35]) writes the vacuum energy of a torus of side ℓ, for $\ell\Lambda \ll 1$, as the sum of the free gluon energies and a zero-mode energy $\varepsilon_0 g^{2/3}/\ell + O(g^{4/3}/\ell)$ with $\varepsilon_0 > 0$, the ground energy of the quartic Hamiltonian of the constant modes; the latter enters $(m + 1)\kappa$ with the coefficient $1 - 8 \cdot 2 = -15$, so that in this regime η is $2 \dim G\, c(0)$ decreased by a term of order $g^{2/3}$, far above $\frac{1}{70}$. We do not control the exchange of the weak-coupling and large-volume limits on the lattice, and the statement is recorded as a heuristic only. Its consequence is a delimitation. For a theory whose lightest states have masses m_i and multiplicities n_i, the free-field picture gives $\eta \approx \sum_i n_i\, c(m_i s)$, so that (C) at the scale s_0 needs $m_{\mathrm{gap}} s_0 \gtrsim 17.7$: the Casimir condition is a statement about tori of about eighteen correlation lengths, beyond perturbation theory and of the same strength as a finite-size

bound on the gap itself. This is consistent with Theorem 7.39, where the condition is derived from the gap on the tori of the halving chain, and with Corollary 10.7(iii), where its rate is that of the gap.

11. Mass generation at one scale implies the cube condition

Theorem B reduces the continuum gap to statement (D), a cube condition uniform along the tuning. This section reduces (D) in turn to a hypothesis of a different kind, closer to the physics: a uniform decay of correlations between the gauge-invariant observables of two slabs of a box separated by a slab of physical width about $\ell/2$, for every boundary condition. If the maximal correlation is at most $c < \frac{1}{2}$, then the cube condition holds with $\varpi = (1-2c)^4/32$ (Theorem D). The value $\frac{1}{2}$ is the exact limit of the method (Proposition H.4 in Appendix H); the hypothesis implies exponential decay of gauge-invariant correlations under every boundary condition (Proposition 11.11); and it holds in two dimensions along the weak-coupling tuning (Theorem H.8). In four dimensions it is a form of mass generation at the physical scale and it is not proved here.

11.1. **Boxes, boundary conditions and gauge-invariant observables.** In this section $G \subset \mathrm{U}(N)$ is compact and nontrivial, φ is a positive continuous class function on G, λ denotes normalised Haar measure on G or on a product of copies of G, and $\omega_\varphi = \log(\max\varphi/\min\varphi)$. Links of $\mathbb{Z}^4$ are written (y,ν) with base point y, from y to $y+e_\nu$. A *box* is a set $Q = \prod_{i=1}^4 [\alpha_i, \alpha_i + s_i) \cap \mathbb{Z}$ with integers $s_i \geq 1$, its *widths*, and $V(Q)$ is the set of links with base point in Q. For an exterior configuration $\omega \in G^{E(\mathbb{Z}^4)\setminus V(Q)}$ let ν_Q^ω be the probability measure on $G^{V(Q)}$ with density proportional to $\prod_{p\cap V(Q)\neq\emptyset} \varphi((\sigma,\omega)_p)$ with respect to λ, a finite product. It is the conditional law of $U_{V(Q)}$ given the other links under any Gibbs measure with weight φ, and on a torus Λ_L with all $s_i \leq L-3$ it is the conditional law of the torus measure.[19] For $A \subset V(Q)$ write E_A for the conditional expectation under ν_Q^ω given $\sigma_{V(Q)\setminus A}$; its regular version is integration against the measure on G^A with density proportional to $\prod_{p\cap A\neq\emptyset} \varphi(U_p)$, which is jointly continuous in all variables. In particular, if $Q' \subset Q$ is a box, the conditional law of $\sigma_{V(Q')}$ under ν_Q^ω given $\sigma_{V(Q)\setminus V(Q')} = \eta$ is $\nu_{Q'}^{(\eta,\omega)}$.

A vertex v is *interior* to Q if the eight links incident to v lie in $V(Q)$. Let $\mathcal{G}_Q$ be the group of $h \in G^{\mathbb{Z}^4}$ with $h_v = 1$ at every vertex not interior to Q, acting by $(h\sigma)_{(y,\nu)} = h_y \sigma_{(y,\nu)} h_{y+e_\nu}^{-1}$; it fixes every link outside $V(Q)$, whose end points are not interior. Let $\mathcal{A}_Q^\omega$ be the set of real $F \in L^2(\nu_Q^\omega)$ with $F \circ h = F$ almost surely for every $h \in \mathcal{G}_Q$: the gauge-invariant observables of the box, the boundary condition having fixed the gauge on the boundary.

Lemma 11.1. *Let Q be a box, ω an exterior configuration and $\nu = \nu_Q^\omega$.*

(i) *ν is invariant under $\mathcal{G}_Q$.*

(ii) *For every $S \subset V(Q)$, each $h \in \mathcal{G}_Q$ maps $\sigma(\sigma_S)$ onto itself, and $E_A(F\circ h) = (E_A F)\circ h$ for $A \subset V(Q)$ and $F \in L^2(\nu)$.*

(iii) *$\mathcal{A}_Q^\omega$ is a closed subspace that contains the constants and is preserved by every E_A. Every $F \in \mathcal{A}_Q^\omega$ has a version $\tilde F$ with $\tilde F(h\sigma) = \tilde F(\sigma)$ for all h and σ, and if F is $\sigma(\sigma_S)$-measurable, $\tilde F$ can be taken $\sigma(\sigma_S)$-measurable.*

Proof. (i) h multiplies each link on the left and on the right by fixed elements, so it preserves λ. Since h fixes the links outside $V(Q)$, $(h\sigma,\omega) = h\cdot(\sigma,\omega)$, and the holonomy of a plaquette with base point y is conjugated by h_y; φ is a class function, so the density is invariant.

(ii) $(h\sigma)_e$ is a function of σ_e alone and $h^{-1} \in \mathcal{G}_Q$, so h maps $\sigma(\sigma_S)$ onto itself. For a bounded $\sigma(\sigma_{V(Q)\setminus A})$-measurable K, by (i), $\nu((F\circ h)K) = \nu(F\,(K\circ h^{-1})) = \nu((E_A F)(K\circ h^{-1})) = \nu(((E_A F)\circ h)K)$, and $(E_A F)\circ h$ is $\sigma(\sigma_{V(Q)\setminus A})$-measurable.

[19] The plaquettes meeting $V(Q)$ have their links based in $\prod_i [\alpha_i - 1, \alpha_i + s_i]$, which embeds in Λ_L when $s_i \leq L-3$.

(iii) By (i) each map $F \mapsto F \circ h$ is unitary, so its fixed space is closed, and so is the intersection of these spaces; (ii) gives the invariance under E_A. The group $\mathcal{G}_Q \cong G^I$, I the finite set of interior vertices, is compact and acts continuously. For a measurable version F_0 put $\tilde F(\sigma) = \int F_0(h\sigma)\,dh$ where $\int |F_0(h\sigma)|\,dh < \infty$, and $\tilde F(\sigma) = 0$ elsewhere. By the invariance of Haar measure under right translations, the set where the integral converges and $\tilde F$ are invariant under every $k \in \mathcal{G}_Q$. For each h, $F_0(h\sigma) = F_0(\sigma)$ for almost every σ, so by Fubini's theorem, for almost every σ the equality holds for almost every h, and $\tilde F = F_0$ almost surely. If F_0 is a function of σ_S, so is $\sigma \mapsto F_0(h\sigma)$ for each h, and so is $\tilde F$. □

Definition 11.2. Let Q be a box, i a coordinate, and $Q = Q_- \cup Q_0 \cup Q_+$ a splitting into three nonempty boxes, consecutive in the coordinate i in this order: the *outer parts* $Q_\pm$ and the *middle* Q_0. For an exterior configuration ω put $\nu = \nu_Q^\omega$ and

$$\rho^\omega(Q_-, Q_+) = \sup |\nu(XY)|,$$

the supremum over $X \in \mathcal{A}_Q^\omega$ measurable with respect to $\sigma_{V(Q_-)}$ and $Y \in \mathcal{A}_Q^\omega$ measurable with respect to $\sigma_{V(Q_+)}$ with $\nu(X) = \nu(Y) = 0$ and $\nu(X^2) = \nu(Y^2) = 1$ (the supremum of the empty set being 0).

This is the maximal correlation[20] between the gauge-invariant observables of the two outer parts, the middle being integrated out, under the boundary condition ω. By the Cauchy–Schwarz inequality $\rho^\omega \le 1$.

Definition 11.3. Let $d \ge 2$, $m \ge 4$ and $w \ge 1$ be integers and $0 \le c \le 1$.

(i) $(\mathrm{M}_{d,m}(c))$: $\rho^\omega(Q_-, Q_+) \le c$ for every box Q all of whose widths lie in $[2d-1, md-1]$, every coordinate i in which Q has width $3d-1$, the splitting in which Q_-, Q_0, Q_+ have widths d, $d-1$, d in the coordinate i, and every exterior configuration ω.

(ii) $(\mathrm{M}^*_w(c))$: $\rho^\omega(Q_-, Q_+) \le c$ for every box Q, every coordinate i, every splitting whose middle has width w in the coordinate i, the outer parts being of any widths, and every ω.

The boxes of $(\mathrm{M}_{d,m}(c))$ are fat: all their widths are at least $2d-1$. The restriction matters. In two dimensions the same condition for boxes of width one in a transverse direction fails along the weak-coupling tuning, through a broken global symmetry of a pinned strip, while the fat form holds there (Theorem H.8 and Proposition H.17 in Appendix H). At $\varphi \equiv 1$ the links are independent under ν_Q^ω, so $\rho^\omega = 0$ and both conditions hold with $c = 0$. Along the tuning, with $d \approx \ell/(2a)$, $(\mathrm{M}_{d,m}(c))$ is a statement at the physical separation $(d-1)a \approx \ell/2$ in boxes of physical size at most about $m\ell/2$, uniform in the boundary condition, and stated for all square-integrable gauge-invariant observables of the two regions rather than for pairs of local ones.

11.2. **Two projections and a chain of projections.**

Lemma 11.4. *Let P_1, P_2 be orthogonal projections on a real Hilbert space $\mathcal{K}$, $\mathcal{N}$ a closed subspace of* $\operatorname{ran} P_1 \cap \operatorname{ran} P_2$*, and suppose that $|\langle x, y\rangle| \le c\|x\|\|y\|$ for all $x \in \operatorname{ran} P_1 \ominus \mathcal{N}$ and $y \in \operatorname{ran} P_2 \ominus \mathcal{N}$. Put $Q_k = I - P_k$ and $X = Q_1 + Q_2$. Then $X^2 \ge (1-c)X$, that is, $Q_1Q_2 + Q_2Q_1 \ge -c(Q_1 + Q_2)$.*

Proof. For $n \in \mathcal{N}$, $P_k n = n$, so $Xn = 0$; as X is self-adjoint it also preserves $\mathcal{N}^\perp$. Let $g \in \mathcal{N}^\perp$. Then $P_k g \perp \mathcal{N}$, since $\langle P_k g, n\rangle = \langle g, P_k n\rangle = \langle g, n\rangle = 0$, so with $p = \|P_1 g\|$ and $q = \|P_2 g\|$, $|\langle P_1 g, P_2 g\rangle| \le cpq$. Put $s = \langle g, (P_1 + P_2)g\rangle = p^2 + q^2$. Then $\|(P_1 + P_2)g\|^2 = p^2 + q^2 + 2\langle P_1 g, P_2 g\rangle \le (1+c)(p^2+q^2)$ and $s \le \|g\|\|(P_1+P_2)g\| \le \|g\|((1+c)s)^{1/2}$, so $s \le (1+c)\|g\|^2$ and $\langle g, Xg\rangle = 2\|g\|^2 - s \ge (1-c)\|g\|^2$. So $X = 0$ on $\mathcal{N}$, and on $\mathcal{N}^\perp$ the

[20] The maximal correlation of two random variables, the supremum of the correlations of square-integrable functions of each, was introduced by Hirschfeld and by Gebelein [23]; for jointly Gaussian variables it equals the absolute value of the ordinary correlation.

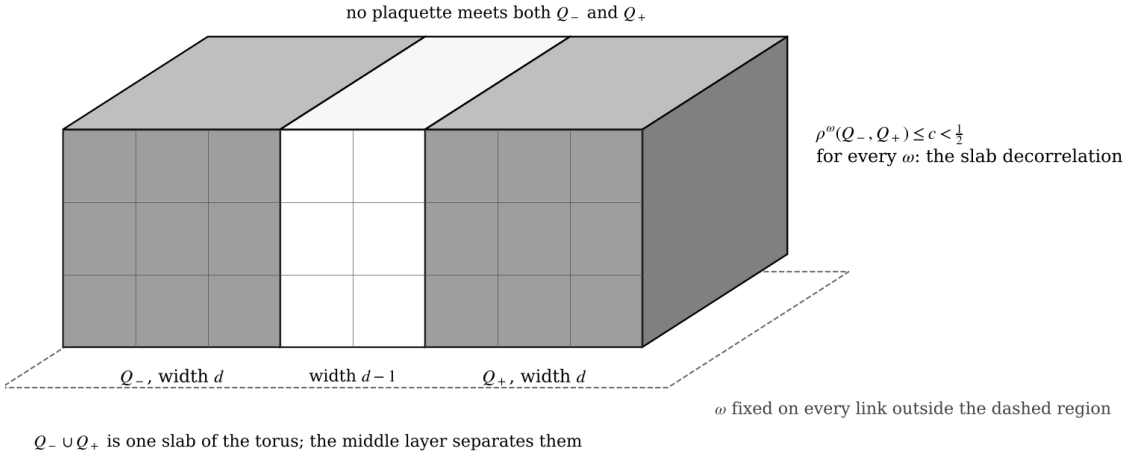


FIGURE 10. The splitting of Definition 11.3: in the coordinate i a box Q of width $3d-1$ is cut into Q_-, Q_0, Q_+ of widths d, $d-1$, d; the middle is integrated out, the exterior configuration ω is fixed, and $\rho^\omega(Q_-, Q_+)$ is the maximal correlation of gauge-invariant observables of the two outer parts. No plaquette meets both Q_- and Q_+, since the base points of the links of a plaquette differ by at most one in each coordinate and $d \ge 2$.

operator $X-(1-c)$ is nonnegative and commutes with $X \ge 0$; hence $X^2-(1-c)X = X^{1/2}(X-(1-c))X^{1/2} \ge 0$ there. Finally $X^2 - X = Q_1Q_2 + Q_2Q_1$. □

The angle between $\operatorname{ran} P_1 \ominus \mathcal{N}$ and $\operatorname{ran} P_2 \ominus \mathcal{N}$ is the Friedrichs angle of the two ranges, and the lemma is a form of the classical relation between that angle and the rate of the alternating projections [15].

Lemma 11.5. *Let $Q_0, \dots, Q_M$ be orthogonal projections on a real Hilbert space $\mathcal{K}$ with $Q_tQ_{t'} = Q_{t'}Q_t$ whenever $|t-t'| \ge 2$ and $Q_tQ_{t+1} + Q_{t+1}Q_t \ge -c(Q_t+Q_{t+1})$ for $0 \le t < M$, with $0 \le c < \frac{1}{2}$. Then $H = \sum_t Q_t$ satisfies $H^2 \ge (1-2c)H$, and $\langle g, Hg\rangle \ge (1-2c)\|g\|^2$ for every $g \perp \ker H$.*

Proof. $H^2 = \sum_t Q_t + \sum_{t<M}(Q_tQ_{t+1}+Q_{t+1}Q_t) + \sum Q_tQ_{t'}$, the last sum over the ordered pairs with $|t-t'| \ge 2$. Each such $Q_tQ_{t'}$ is a product of commuting projections, hence a projection and nonnegative. The middle sum is at least $-c\sum_{t<M}(Q_t+Q_{t+1}) \ge -2cH$. So $H^2-(1-2c)H \ge 0$, and by the spectral theorem the spectrum of the nonnegative operator H lies in $\{0\} \cup [1-2c, \infty)$, which gives the second claim. □

Lemma 11.6. *Let Q be a box, ω an exterior configuration, and $R_0, \dots, R_M$ subsets of $V(Q)$ with union $V(Q)$. If $F \in L^2(\nu_Q^\omega)$ is, for every t, almost surely equal to a $\sigma(\sigma_{V(Q)\setminus R_t})$-measurable function, then F is almost surely constant.*

Proof. The density of $\nu = \nu_Q^\omega$ with respect to λ is bounded above and below by positive constants, so $L^2(\nu) = L^2(\lambda)$ with the same null sets. Let Λ_t be the conditional expectation under λ given $\sigma_{V(Q)\setminus R_t}$, which integrates out σ_{R_t}. By hypothesis $F = F_t$ almost surely with F_t a function of $\sigma_{V(Q)\setminus R_t}$, so $\Lambda_t F = F_t = F$. By Fubini's theorem $\Lambda_0 \cdots \Lambda_M$ integrates out all of $\bigcup_t R_t = V(Q)$, so $F = \Lambda_0 \cdots \Lambda_M F$ is constant. □

11.3. **One direction.** Fix a box $R = \prod_i I_i$ and a coordinate i with $I_i = [\alpha, \alpha+(M+2)d-1)$ for an integer $M \ge 0$. For $0 \le t \le M$ let R_t be the box obtained from R by replacing I_i with $\alpha + dt + [0, 2d-1)$, and for $t < M$ let Q_t be the box obtained by replacing I_i with $\alpha + dt + [0, 3d-1)$, split into Q_t^-, Q_t^0, Q_t^+ by $\alpha+dt+[0,d)$, $\alpha+dt+[d,2d-1)$ and

$\alpha + dt + [2d-1, 3d-1)$. Then

(11.1)
$$V(Q_t) = V(R_t) \cup V(R_{t+1}), \quad V(Q_t^-) = V(R_t) \setminus V(R_{t+1}), \quad V(Q_t^+) = V(R_{t+1}) \setminus V(R_t),$$

and $\bigcup_t V(R_t) = V(R)$, since the intervals $\alpha + dt + [0, 2d-1)$ cover I_i.

Proposition 11.7 (one direction)**.** *In this setting let ω be an exterior configuration of R, $\nu = \nu_R^\omega$, $P_t = E_{V(R_t)}$, and suppose that $\rho^{\omega'}(Q_t^-, Q_t^+) \le c < \frac{1}{2}$ for every $t < M$ and every exterior configuration ω' of Q_t. Then every $F \in \mathcal{A}_R^\omega$ satisfies*

$$(1-2c)\operatorname{Var}_\nu(F) \le \sum_{t=0}^{M} \nu\big((F - P_t F)^2\big).$$

Proof. For $M = 0$, $P_0 F = \nu(F)$ and the claim is trivial; let $M \ge 1$. Put $\mathcal{K} = \mathcal{A}_R^\omega$. By Lemma 11.1(iii) each P_t preserves $\mathcal{K}$ and restricts to an orthogonal projection of $\mathcal{K}$; let $Q_t = I - P_t$ on $\mathcal{K}$.

Non-adjacent moves commute. Let $t' \ge t+2$. The base points of the links of $V(R_t)$ have i-th coordinate at most $\alpha + dt + 2d - 2$ and those of $V(R_{t'})$ at least $\alpha + dt + 2d$, so no plaquette meets both sets. Given σ off $V(R_t) \cup V(R_{t'})$, the conditional density of $(\sigma_{V(R_t)}, \sigma_{V(R_{t'})})$ is a product of factors each of which involves the links of only one of the two sets. So they are conditionally independent, and $P_t P_{t'} F$ and $P_{t'} P_t F$ are both the integral of F against the product of the two conditional laws. Hence $P_t P_{t'} = P_{t'} P_t$, and the Q_t commute.

Adjacent moves. Fix $t < M$ and write $a = V(Q_t^-)$, $b = V(Q_t^0)$, $c' = V(Q_t^+)$, $e = V(R) \setminus V(Q_t)$. By (11.1), $V(R) \setminus V(R_t) = c' \cup e$ and $V(R) \setminus V(R_{t+1}) = a \cup e$, so the ranges of P_t and of P_{t+1} in $\mathcal{K}$ are the elements of $\mathcal{K}$ measurable with respect to $(\sigma_{c'}, \sigma_e)$ and to (σ_a, σ_e). Let $\mathcal{N}$ be the elements of $\mathcal{K}$ measurable with respect to σ_e, a closed subspace of both ranges. We check the hypothesis of Lemma 11.4. Let x and y be in the ranges of P_t and P_{t+1} and orthogonal to $\mathcal{N}$. The conditional expectation E_e given σ_e is $E_{V(Q_t)}$, which preserves $\mathcal{K}$, so $E_e x \in \mathcal{N}$ and $\|E_e x\|^2 = \langle x, E_e x\rangle = 0$; likewise $E_e y = 0$. By Lemma 11.1(iii) for the box R take versions $x = \tilde{x}(\sigma_{c'}, \sigma_e)$ and $y = \tilde{y}(\sigma_a, \sigma_e)$ invariant at every point under $\mathcal{G}_R$. Every vertex interior to Q_t is interior to R, so $\mathcal{G}_{Q_t} \subset \mathcal{G}_R$, and $\mathcal{G}_{Q_t}$ fixes σ_e. Hence for each $\eta \in G^e$ the functions $X_\eta = \tilde{x}(\cdot, \eta)$ of $\sigma_{V(Q_t^+)}$ and $Y_\eta = \tilde{y}(\cdot, \eta)$ of $\sigma_{V(Q_t^-)}$ are invariant under $\mathcal{G}_{Q_t}$ at every point. The conditional law of $\sigma_{V(Q_t)}$ given $\sigma_e = \eta$ is $\nu_\eta = \nu_{Q_t}^{(\eta,\omega)}$, so for almost every η, $X_\eta, Y_\eta \in L^2(\nu_\eta)$, $\nu_\eta(X_\eta) = (E_e x)(\eta) = 0$, $\nu_\eta(Y_\eta) = 0$, and by the hypothesis with $\omega' = (\eta, \omega)$,

$$|\nu_\eta(X_\eta Y_\eta)| \le c\, \nu_\eta(X_\eta^2)^{1/2} \nu_\eta(Y_\eta^2)^{1/2}.$$

Integrating over η and using the Cauchy–Schwarz inequality, $|\langle x, y\rangle| \le c\|x\|\|y\|$, and Lemma 11.4 gives $Q_t Q_{t+1} + Q_{t+1} Q_t \ge -c(Q_t + Q_{t+1})$ on $\mathcal{K}$.

Conclusion. Lemma 11.5 applies. If $Hg = 0$ for $g \in \mathcal{K}$, then $\sum_t \|Q_t g\|^2 = \langle g, Hg\rangle = 0$, so $g = P_t g$ for all t and g is constant by Lemma 11.6. For $F \in \mathcal{K}$ the function $g = F - \nu(F)$ is orthogonal to the constants, hence to $\ker H$, and $\langle g, Hg\rangle = \sum_t \nu((F - P_t F)^2)$. □

11.4. **The patch gap and the cube condition.** For $d \ge 2$, $m \ge 4$ and $r \in \mathbb{Z}^4$ let $\Pi = r + [0, md-1)^4$, the box of base points of a patch, and for $j \in [0, m-2]^4$ let $B_j = V(r + dj + [0, 2d-1)^4)$, as in Appendix G. For an exterior configuration ω of Π write $\nu = \nu_\Pi^\omega$ and $P_j = E_{B_j}$.

Theorem 11.8 (patch gap from slab decorrelation)**.** *Let $d \ge 2$, $m \ge 4$, $0 \le c < \frac{1}{2}$, and assume* ($\mathrm{M}_{d,m}(c)$)*. Then for every r, every ω and every $F \in \mathcal{A}_\Pi^\omega$,*

$$(1-2c)^4 \operatorname{Var}_\nu(F) \le \sum_{j \in [0, m-2]^4} \nu\big((F - P_j F)^2\big).$$

Proof. For $0 \le k \le 4$ and $\tau \in [0, m-2]^k$ let $R_\tau^{(k)} = r + \prod_{i \le k}(d\tau_i + [0, 2d-1)) \times [0, md-1)^{4-k}$ and $P_\tau^{(k)} = E_{V(R_\tau^{(k)})}$, under ν. We prove by induction on k that

$$(1-2c)^k \operatorname{Var}_\nu(F) \le \sum_{\tau \in [0,m-2]^k} \nu\big((F - P_\tau^{(k)} F)^2\big) \qquad (F \in \mathcal{A}_\Pi^\omega).$$

For $k = 0$, $R^{(0)} = \Pi$ and $P^{(0)}F = \nu(F)$. Let $k < 4$, fix τ, write $R = R_\tau^{(k)}$, and split $\sigma = (\sigma', \omega')$ with $\sigma' = \sigma_{V(R)}$ and $\omega' = \sigma_{V(\Pi)\setminus V(R)}$. Take F invariant at every point under $\mathcal{G}_\Pi$ (Lemma 11.1(iii)). Since $\mathcal{G}_R \subset \mathcal{G}_\Pi$ fixes ω', each $F(\cdot, \omega')$ is invariant under $\mathcal{G}_R$. The conditional law of σ' given ω' is $\nu' = \nu_R^{(\omega', \omega)}$, so

$$\nu\big((F - P_\tau^{(k)} F)^2\big) = \int \operatorname{Var}_{\nu'}\big(F(\cdot, \omega')\big)\, d\nu(\omega').$$

In the coordinate $k+1$ the box R has width $md - 1 = ((m-2)+2)d - 1$. The boxes R_u and Q_u of Proposition 11.7 built from R with $i = k+1$ and $M = m-2$ are $R_u = R_{(\tau,u)}^{(k+1)}$ and boxes whose widths are $2d-1$ in the coordinates $\le k$, $3d-1$ in the coordinate $k+1$ and $md-1$ in the coordinates $> k+1$, split as in $(\mathrm{M}_{d,m}(c))$; all these widths lie in $[2d-1, md-1]$. By Proposition 11.7, for almost every ω',

$$(1-2c) \operatorname{Var}_{\nu'}(F(\cdot, \omega')) \le \sum_{u=0}^{m-2} \nu'\big((F(\cdot, \omega') - P_u' F(\cdot, \omega'))^2\big),$$

with P_u' the conditional expectation under ν' given σ' off $V(R_{(\tau,u)}^{(k+1)})$. The conditional law of the links of that set given the other links of Π is the same under ν and under ν', both having density proportional to $\prod_p \varphi(U_p)$ over the plaquettes meeting it; so $P_u' F(\cdot, \omega') = (P_{(\tau,u)}^{(k+1)} F)(\cdot, \omega')$, and integrating over ω' gives $\nu((F - P_{(\tau,u)}^{(k+1)} F)^2)$. Multiplying the induction hypothesis by $1 - 2c$ and inserting these bounds proves the case $k+1$. For $k = 4$, $V(R_\tau^{(4)}) = B_\tau$. □

Theorem D (mass generation implies the cube condition)**.** *Let $d \ge 2$, $0 \le c < \frac{1}{2}$ and $m \ge 4$ with $m - 1 \ge 148/(1-2c)^4$, and assume $(\mathrm{M}_{d,m}(c))$. Then for every even L with $d \mid L$ and $L \ge 2md$ the measure μ_L with weight φ satisfies*

$$\mathrm{BP}\Big(2d - 1, \frac{(1-2c)^4}{32}\Big).$$

Consequently, let (β_k, a_k) be a tuning with $a_k \to 0$ and Wilson weights, and $d_k \ge 2$ integers with $(2d_k - 1)a_k \le \ell$. If $(\mathrm{M}_{d_k,m}(c))$ holds for every k with $c < \frac{1}{2}$ and m as above independent of k, then statement (D) holds with $n_k = 2d_k - 1$ and $\varpi = (1-2c)^4/32$, and every continuum limit as in Theorem B along the tori with $d_k \mid L$ satisfies

$$\sigma(H) \subset \{0\} \cup \Big[\frac{(1-2c)^4}{1250\, e^2 \ell}, \infty\Big),$$

with 0 a simple eigenvalue; a nontrivial limit has a mass gap at least this large.

Proof. Theorem 11.8 says that the gauge-invariant patch gap of Definition G.3 satisfies $\gamma_{d,m} \ge (1-2c)^4$, and Theorem G.4 with $\gamma = (1-2c)^4$ gives $\mathrm{BP}(2d-1, (1-2c)^4/32)$. Along the tuning this holds on an infinite set of even tori with $L \ge 2(2d_k - 1) + 4$ for each k, with $(2d_k - 1)a_k \le \ell$, and Theorem B gives the gap $16\varpi/(625 e^2 \ell)$ with $\varpi = (1-2c)^4/32$. □

Corollary 11.9 (cylinders and anisotropic tori)**.** *Under the hypotheses of Theorem D, $\mathrm{BP}(2d-1, (1-2c)^4/32)$ holds for the measure with weight φ on every torus $\prod_{i=1}^4 (\mathbb{Z}/L_i\mathbb{Z})$ with $d \mid L_i$ and $L_i \ge 2md$ for every i; in particular on every cylinder $\Lambda_{L,M} = (\mathbb{Z}/L\mathbb{Z})^3 \times (\mathbb{Z}/M\mathbb{Z})$ with $d \mid L$, $d \mid M$ and $L, M \ge 2md$.*

Proof. The side of the torus enters the proof of Theorem D only through Appendix G, and there only coordinate by coordinate. With $K_i = L_i/d \geq 2m$ the coarse torus is $\prod_i (\mathbb{Z}/K_i\mathbb{Z})$ and the blocks B_j are as before. Lemma G.1(ii) needs $K_i \geq 4$ in the coordinate i in which $|v_i| \geq 2$. In Lemma G.2, two indices j, $j+v$ lie in a common patch only if $|v_i| \leq m-2$ for every i, which uses $K_i \geq 2m$ in each coordinate, and then for exactly $c_v = \prod_i (p - |v_i|)$ patches; the rest of that proof is algebra in p and γ. The patch inequality of Theorem G.4 needs the patch and its exterior to be disjoint, that is $L_i \geq md + 2$ in each coordinate; the averaging over the shifts $a \in [0,d)^4$ and the bound $L_i \geq 8d \geq 2n+4$ are unchanged, and Theorem 6.3 is stated for such tori. Parity of the sides is used nowhere. □

The patch gap obtained, $(1-2c)^4$, does not decay with m, whereas the finite-size criterion needs only $148/(m-1)$; this slack is what allows a large but fixed number m of blocks per patch. The threshold $c < \frac{1}{2}$ cannot be raised for this argument: for the massless Gaussian chain every adjacent pair of moves has maximal correlation exactly $\frac{1}{2}$, the non-adjacent moves commute, and the gap of the slab dynamics tends to zero (Proposition H.4).

11.5. **Decay of correlations under every boundary condition.**

Lemma 11.10. *Let Q be a box, ω an exterior configuration, i a coordinate, and $Y_0, \dots, Y_K$ the sets $V(Q \cap \{x_i \in J_k\})$ for consecutive nonempty integer intervals $J_0 < J_1 < \cdots < J_K$ whose union is the i-th side of Q. Under ν_Q^ω, for each k, the families $(\sigma_{Y_l})_{l<k}$ and $(\sigma_{Y_l})_{l>k}$ are conditionally independent given σ_{Y_k}.*

Proof. The links in $V(Q)$ of a plaquette are based at points whose i-th coordinates take at most two consecutive values, so they lie in one Y_l or in two consecutive ones. Grouping the factors of the density by the least l met, the density of ν_Q^ω with respect to λ is $\prod_l \psi_l(\sigma_{Y_l}, \sigma_{Y_{l+1}})$, with ψ_K a function of σ_{Y_K}, which gives the conditional independence. □

Proposition 11.11. *Assume $(\mathrm{M}_w^*(c))$ with $w \geq 1$ and $0 \leq c \leq 1$. Let Q be a box, ω an exterior configuration, $\nu = \nu_Q^\omega$, i a coordinate, $s \in \mathbb{Z}$, $r \geq 1$, and let $f, g \in \mathcal{A}_Q^\omega$ be measurable with respect to $\sigma_{V(Q \cap \{x_i \leq s\})}$ and $\sigma_{V(Q \cap \{x_i \geq s+r\})}$. Then*

$$|\operatorname{Cov}_\nu(f,g)| \leq c^{\lfloor r/(w+1) \rfloor} \|f - \nu(f)\| \, \|g - \nu(g)\|.$$

Proof. If one of the two parts of Q is empty, f or g is constant and the covariance vanishes. Otherwise the i-th side of Q contains $[s, s+r]$. Let $J = \lfloor r/(w+1) \rfloor$; for $J = 0$ the claim is the Cauchy–Schwarz inequality. Let $J \geq 1$ and split the i-th side into consecutive intervals: T_0, the values $\leq s$; for $1 \leq k \leq J$, $M_k = [s + (k-1)(w+1) + 1,\ s + (k-1)(w+1) + w]$; for $1 \leq k \leq J-1$, $T_k = \{s + k(w+1)\}$; and T_J, the values $\geq s + J(w+1)$, which contain the values $\geq s+r$. Use the same letters for the corresponding sets of links. By Lemma 11.10 the sequence $T_0, M_1, T_1, \dots, M_J, T_J$ is Markov, and by Lemma 11.1 the conditional expectations given σ_{T_k} preserve $\mathcal{A}_Q^\omega$.

For $1 \leq k \leq J$ let $u \in \mathcal{A}_Q^\omega$ be a centred function of $\sigma_{T_{k-1}}$ and $v = E[u \mid \sigma_{T_k}]$, which is centred and lies in $\mathcal{A}_Q^\omega$. The box Q splits in the coordinate i into the values $\leq s + (k-1)(w+1)$, the middle M_k of width w, and the values $\geq s + k(w+1)$; u is a function of the first part and v of the last, so by $(\mathrm{M}_w^*(c))$, $\|v\|^2 = \nu(uv) \leq c\|u\|\|v\|$, that is, $\|v\| \leq c\|u\|$. Let $f_0 = f - \nu(f)$ and $u_k = E[f_0 \mid \sigma_{T_k}]$. By the Markov property $u_k = E[u_{k-1} \mid \sigma_{T_k}]$, so $\|u_k\| \leq c\|u_{k-1}\|$ and $\|u_J\| \leq c^J \|f_0\|$. As g is σ_{T_J}-measurable, $\operatorname{Cov}_\nu(f,g) = \nu(u_J(g - \nu(g)))$, and the Cauchy–Schwarz inequality gives the claim. □

So $(\mathrm{M}_{d-1}^*(c))$ with $c < 1$, which contains $(\mathrm{M}_{d,m}(c))$ for every m, gives exponential decay of the covariances of gauge-invariant observables of a box, uniformly in the box and in the boundary condition, at rate $\log(1/c)/d$ per lattice spacing, that is, of order $\log(1/c)/\ell$ in physical units along the tuning. This is close to the decay hypothesis of Chatterjee [11, Def. 2.3], under which an area law for Wilson loops follows; there the observables are local functions of a cube with prescribed boundary, and the geometries differ, so neither condition

is claimed to imply the other. The converse direction is the weak point of any pairwise form: a bound $K_1 e^{-K_2 r}$ for pairs of local functions, summed over the pairs of two slabs of transverse size $(md)^3$, bounds ρ^ω only up to a factor of order $(md)^6$, which is not uniform in the lattice spacing.

11.6. **The hypothesis reduced to one layer, and its vacuum form.** Two reductions locate what $(\mathrm{M}_{d,m}(c))$ asks beyond a mass gap. The first says that the maximal correlation of the two outer parts of a box is the maximal correlation of their two inner layers, and that it is multiplicative over the layers in between; the second computes the same quantity, with the Dirichlet law of a box replaced by the vacuum of a cylinder, in terms of the spectral gap of the transfer matrix. Together they show that the content of the hypothesis is the uniformity in the boundary condition, and nothing else.

Let Q be a box, i a coordinate in which Q has width s, and for $0 \le j < s$ let Y_j be the set of links of $V(Q)$ whose base points have i-th coordinate equal to the j-th value of the i-th side of Q: the *layers* of Q in the coordinate i. For an exterior configuration ω, $\nu = \nu_Q^\omega$, and two layers Y, Y' put

$$\rho^\omega(Y, Y') = \sup |\nu(XX')|,$$

the supremum over $X, X' \in \mathcal{A}_Q^\omega$ measurable with respect to σ_Y and $\sigma_{Y'}$ respectively, with $\nu(X) = \nu(X') = 0$ and $\nu(X^2) = \nu(X'^2) = 1$. This is the maximal correlation of the gauge-invariant observables of the two layers; it is defined also for adjacent layers, which share plaquettes.

Lemma 11.12 (layers)**.** *Let Q, i, ω and the layers $Y_0, \dots, Y_{s-1}$ be as above.*

(i) *For a splitting $Q = Q_- \cup Q_0 \cup Q_+$ in the coordinate i as in Definition* 11.2*, with Q_- and Q_0 of widths s_- and s_0,*

$$\rho^\omega(Q_-, Q_+) = \rho^\omega(Y_{s_- - 1}, Y_{s_- + s_0}).$$

(ii) *For $0 \le j < j + r < s$,*

$$\rho^\omega(Y_j, Y_{j+r}) \le \prod_{l=j}^{j+r-1} \rho^\omega(Y_l, Y_{l+1}).$$

In particular, if $\rho^\omega(Y_l, Y_{l+1}) \le \theta$ for every box Q all of whose widths lie in $[2d-1, md-1]$, every coordinate, every l and every ω, then $(\mathrm{M}_{d,m}(\theta^d))$ holds.

Proof. Write $k = s_- - 1$ and $k' = s_- + s_0$, so that $V(Q_-) = Y_0 \cup \dots \cup Y_k$ and $V(Q_+) = Y_{k'} \cup \dots \cup Y_{s-1}$. By Lemma 11.10, applied to the layers themselves, the families $(\sigma_{Y_l})_{l<k}$ and $(\sigma_{Y_l})_{l>k}$ are conditionally independent given σ_{Y_k}. Hence for an integrable X measurable with respect to $\sigma(\sigma_{Y_l} : l \le k)$,

$$E[X \mid \sigma_{Y_l} : l \ge k] = E[X \mid \sigma_{Y_k}], \tag{11.2}$$

and symmetrically with the two sides exchanged, at every layer k.

(i) The inequality $\ge$ holds because a pair admissible for the right side is admissible for the left. For $\le$ let X, X' be admissible for $\rho^\omega(Q_-, Q_+)$ and put $\tilde X = E[X \mid \sigma_{Y_k}]$, $\tilde X' = E[X' \mid \sigma_{Y_{k'}}]$. Both are centred, of norm at most 1, and lie in $\mathcal{A}_Q^\omega$ by Lemma 11.1(iii), since each is a conditional expectation given the links of a subset of $V(Q)$. As X' is measurable with respect to $\sigma(\sigma_{Y_l} : l \ge k)$, (11.2) gives $\nu(XX') = \nu(\tilde X X')$, and as $\tilde X$ is measurable with respect to $\sigma(\sigma_{Y_l} : l \le k')$, the symmetric identity at the layer k' gives $\nu(\tilde X X') = \nu(\tilde X \tilde X')$. So $|\nu(XX')| \le \rho^\omega(Y_k, Y_{k'}) \|\tilde X\| \, \|\tilde X'\| \le \rho^\omega(Y_k, Y_{k'})$.

(ii) It suffices to treat $r \ge 2$ and to show $\rho^\omega(Y_j, Y_{j+r}) \le \rho^\omega(Y_j, Y_l)\, \rho^\omega(Y_l, Y_{j+r})$ for $j < l < j + r$; the product follows by induction on r. Let X, X' be admissible for $\rho^\omega(Y_j, Y_{j+r})$ and $Z = E[X' \mid \sigma_{Y_l}]$, which is centred and lies in $\mathcal{A}_Q^\omega$. By (11.2) at the layer l, $\nu(XX') = \nu(XZ)$ and $\nu(X'Z) = \nu(Z^2) = \|Z\|^2$. The first identity gives $|\nu(XX')| \le \rho^\omega(Y_j, Y_l)\|Z\|$, the second $\|Z\|^2 \le \rho^\omega(Y_l, Y_{j+r})\|Z\|$, that is, $\|Z\| \le \rho^\omega(Y_l, Y_{j+r})$, and the claim follows.

For the last assertion, in the splitting of Definition 11.3 the two layers of (i) are Y_{d-1} and Y_{2d-1}, at distance d, and (ii) gives $\rho^{\omega}(Q_-, Q_+) \le \theta^d$. □

So the hypothesis is a bound, uniform in the boundary condition, on the second singular value of the d-step transfer operator of the box between two of its layers, on gauge-invariant functions; and it follows from a bound θ on the one-step operator between adjacent layers whenever $\theta^d \le c$. In the vacuum of a cylinder the corresponding quantity is known exactly.

Proposition 11.13 (the vacuum form of the hypothesis)**.** *Let $L \ge 1$, let $\mathbb{T}$, λ_0, Ω_0 and μ_L^0 be as in Lemma 7.3, and let $\lambda_0\rho_1$ be the largest modulus of an eigenvalue $\lambda \ne \lambda_0$ of $\mathbb{T}$, so that $0 \le \rho_1 < 1$. For $s \ge 1$ let $\rho_L^{\mathrm{vac}}(s)$ be the supremum of $\mu_L^0(F\,\tau_s H)$ over the real bounded gauge-invariant functions F, H of the slice E_0 with $\mu_L^0(F) = \mu_L^0(H) = 0$ and $\mu_L^0(F^2) = \mu_L^0(H^2) = 1$. Then*

$$\rho_L^{\mathrm{vac}}(s) = \rho_1^s \qquad (s \ge 1).$$

Proof. By Lemma 7.3(v) with $w = w' = 0$, and $\vartheta\bar{F} = F$ for a real function of the slice E_0, $\mu_L^0(F\,\tau_s H) = \langle F\Omega_0, \mathbb{R}^s H\Omega_0\rangle$, $\mu_L^0(F) = \langle \Omega_0, F\Omega_0\rangle$ and $\mu_L^0(F^2) = \|F\Omega_0\|^2$, with $\mathbb{R} = \mathbb{T}/\lambda_0$ and $\|\Omega_0\| = 1$. Let $\mathcal{K}^G$ be the closed subspace of gauge-invariant elements of $\mathcal{K}$ and $\mathcal{K}_0^G = \mathcal{K}^G \ominus \Omega_0$; $\Omega_0 \in \mathcal{K}^G$ by Lemma 7.3(i). The map $F \mapsto u = F\Omega_0$ sends the admissible F to unit vectors of $\mathcal{K}_0^G$, since the product of two gauge-invariant functions is gauge invariant. Its image is dense in the unit sphere of $\mathcal{K}_0^G$: given a unit $u \in \mathcal{K}_0^G$, the function u/Ω_0 is gauge invariant, measurable, and in $L^2(\Omega_0^2\lambda)$; its truncations $F_n = (u/\Omega_0)\mathbf{1}\{|u/\Omega_0| \le n\}$ are bounded and gauge invariant, $F_n\Omega_0 \to u$ in $\mathcal{K}$ by dominated convergence, and $F_n' = F_n - \mu_L^0(F_n)$ is admissible after normalisation, with $F_n'\Omega_0 \to u - \langle\Omega_0, u\rangle\Omega_0 = u$. Hence

$$\rho_L^{\mathrm{vac}}(s) = \sup\{\langle u, \mathbb{R}^s v\rangle : u, v \in \mathcal{K}_0^G,\ \|u\| = \|v\| = 1\} = \|\mathbb{R}^s|_{\mathcal{K}_0^G}\|,$$

the supremum of a continuous function over a dense subset being its supremum over the set. By the proof of Lemma 7.3(i), $\mathbb{T}$ maps $\mathcal{K}$ into $\mathcal{K}^G$, and it is self-adjoint and maps Ω_0 to $\lambda_0\Omega_0$; so $\mathbb{R}$ maps $\mathcal{K}_0^G$ into itself, and its restriction is compact and self-adjoint. The nonzero eigenvalues of that restriction are the numbers λ/λ_0 for the eigenvalues $\lambda \ne \lambda_0$ of $\mathbb{T}$: every eigenvector with nonzero eigenvalue is gauge invariant by Lemma 7.3(i), and those with $\lambda \ne \lambda_0$ are orthogonal to Ω_0. The norm of the s-th power of a compact self-adjoint operator is the s-th power of its largest eigenvalue modulus, which is ρ_1. □

Under the vacuum law of the cylinder the maximal correlation of the gauge-invariant observables of two slices at distance s is thus $e^{-m_{\mathrm{lat}}s}$ with $m_{\mathrm{lat}} = -\log\rho_1$ the lattice gap, with no constant in front; a physical gap m gives $\rho_L^{\mathrm{vac}}(d) \le e^{-mda} < \frac{1}{2}$ as soon as $da > (\log 2)/m$, and for adjacent slices it gives exactly ρ_1, so that the product of Lemma 11.12(ii) is an equality in the vacuum.[21] Theorem D with Corollary 11.9 and Proposition 7.4 goes from $(\mathrm{M}_{d,m}(c))$ to a gap; the proposition goes from a gap back to the vacuum form of $(\mathrm{M}_{d,m}(c))$ without loss. The two do not close into a proof because the hypothesis is stated under every exterior configuration, and the vacuum is one law among them: what is missing is precisely that the second singular value of the one-layer transfer operator of a box of physical size ℓ stays below $c^{1/d}$ for every boundary condition, uniformly along the tuning. Appendix H shows in two dimensions that a twisted boundary condition can drive the correlation of a thin strip to 1, so the uniformity is not a formality; in four dimensions it is the statement that no boundary condition at the distance $\ell/2$ orders the gauge-invariant observables of the box, and it is here that the non-abelian input must enter.

[21] The two statements concern different laws. Proposition 11.13 is about the state μ_L^0 of the infinite cylinder, in which the slices outside $[0, s]$ are integrated against the vacuum; $(\mathrm{M}_{d,m}(c))$ is about the law ν_Q^{ω} of a box of bounded physical size whose exterior is held fixed at an arbitrary ω. Along the tuning of Section 12 the box has physical side between ℓ and $m\ell/2$, and d layers correspond to the physical distance $\ell/2$, so the vacuum form of the bound $c < \frac{1}{2}$ reads $m\ell > 2\log 2$.

The uniformity cannot fail at any fixed box: in the class of weights considered here no boundary condition locks a layer, so the question is asymptotic, in the size of the box at fixed coupling or along the tuning in boxes of fixed physical size.

Proposition 11.14 (no boundary condition locks a layer)**.** *Let G be compact, φ a positive continuous class function, Q a box and $Y, Y' \subset V(Q)$ disjoint. Then the supremum over all exterior configurations ω and all centred unit $X \in L^2(\nu_Q^\omega)$ measurable with respect to σ_Y and X' measurable with respect to $\sigma_{Y'}$ of $|\nu_Q^\omega(XX')|$ is less than* 1*. In particular* $\sup_\omega \rho^\omega(Y_k, Y_{k+1}) < 1$ *and* $\sup_\omega \rho^\omega(Q_-, Q_+) < 1$ *for every box and splitting, and* $(\mathrm{M}_{d,m}(c))$ *holds for some $c < 1$ for every d, m and φ.*

Proof. The density f_ω of ν_Q^ω with respect to the Haar measure λ of $G^{V(Q)}$ depends on finitely many exterior links, those of the plaquettes meeting $V(Q)$, so ω ranges over a compact set Ω, $\omega \mapsto f_\omega$ is continuous from Ω to $C(G^{V(Q)})$, and $m_0 \le f_\omega \le M_0$ with $m_0 = (\min\varphi/\max\varphi)^P > 0$, P the number of those plaquettes, and $M_0 = 1/m_0$. Let $k_\omega(y, y')$ be the joint density of $(\sigma_Y, \sigma_{Y'})$ with respect to $\lambda_Y \otimes \lambda_{Y'}$ and $k_\omega^{Y'}$ the marginal density of $\sigma_{Y'}$, integrals of f_ω over the other links, hence continuous in (ω, y, y') and between m_0 and M_0. The conditional expectation given $\sigma_{Y'}$ of a function g of σ_Y is $(B_\omega g)(y') = \int g(y) k_\omega(y, y')\, d\lambda_Y(y)/k_\omega^{Y'}(y')$, an operator from $L^2(\lambda_Y)$ to $L^2(\lambda_{Y'})$ with a continuous kernel depending continuously on ω in the uniform norm; so each B_ω is Hilbert–Schmidt, hence compact, and $\omega \mapsto B_\omega$ is continuous in the Hilbert–Schmidt norm.

Suppose the claim fails: there are $\omega_n \in \Omega$ and admissible X_n, X_n' with $\nu^{\omega_n}(X_n X_n') \to 1$. Pass to a subsequence with $\omega_n \to \omega$. In $L^2(\nu^{\omega_n})$, $\|X_n - X_n'\|^2 = 2 - 2\nu^{\omega_n}(X_n X_n') \to 0$, and since the densities lie in $[m_0, M_0]$ the X_n, X_n' are bounded in $L^2(\lambda)$ with $\|X_n - X_n'\|_{L^2(\lambda)} \to 0$. Pass to a further subsequence with $X_n \rightharpoonup X$ and $X_n' \rightharpoonup X'$ weakly in $L^2(\lambda)$; X is a function of σ_Y, X' one of $\sigma_{Y'}$, and $X = X'$ almost everywhere. Under the product measure λ a function of σ_Y that agrees almost everywhere with a function of the disjoint variables $\sigma_{Y'}$ is almost everywhere a constant κ, and $0 = \nu^{\omega_n}(X_n) = \int X_n f_{\omega_n}\, d\lambda \to \int X f_\omega\, d\lambda = \kappa$ by the weak convergence of X_n and the uniform convergence of f_{ω_n}. So $X = 0$. Now $\nu^{\omega_n}(X_n X_n') = \nu^{\omega_n}((B_{\omega_n} X_n) X_n') \le \|B_{\omega_n} X_n\|_{L^2(\nu^{\omega_n})} \le M_0^{1/2} \|B_{\omega_n} X_n\|_{L^2(\lambda)}$, and $B_{\omega_n} X_n = B_\omega X_n + (B_{\omega_n} - B_\omega) X_n \to B_\omega X = 0$ strongly, because B_ω is compact and $\|B_{\omega_n} - B_\omega\| \to 0$. This contradicts $\nu^{\omega_n}(X_n X_n') \to 1$. □

A model in which a boundary condition locks a layer must therefore have a weight with zeros, outside the class of Lemma 7.3; and the two-dimensional strip of Appendix H drives the correlation to 1 only in the limit along the tuning.

The supremum in Proposition 11.14 is attained, because the slab correlation depends continuously on the boundary condition.

Lemma 11.15 (continuity in the boundary condition)**.** *Fix a box Q and a splitting $Q_\pm$. The map $\omega \mapsto \rho^\omega(Q_-, Q_+)$ is uniformly continuous on the compact space Ω of the exterior configurations that enter ν_Q^ω. Consequently* $\sup_\omega \rho^\omega(Q_-, Q_+)$ *is attained, it equals the supremum over any dense set of ω, and $\{\omega : \rho^\omega(Q_-, Q_+) > c\}$ is open for every c.*

Proof. Write $\sigma = (x, z, y)$ with $x = \sigma_{V(Q_-)}$ and $y = \sigma_{V(Q_+)}$, let f^ω be the density of ν_Q^ω with respect to Haar measure, $j^\omega(x, y) = \int f^\omega\, dz$ the joint density of (x, y) with marginals $j_\pm^\omega$, and $k^\omega = j^\omega/(j_-^\omega \otimes j_+^\omega)^{1/2}$. As in the proof of Proposition 11.14 these are jointly continuous in (ω, x, y) and bounded above and below by positive constants. Let $I_- \subset L^2(dx)$ and $I_+ \subset L^2(dy)$ be the subspaces of functions invariant under the action of $\mathcal{G}_Q$ on the links of Q_- and of Q_+; they do not depend on ω, and $j_\pm^\omega$ are invariant by Lemma 11.1(i). The substitution $X = u/(j_-^\omega)^{1/2}$, $Y = v/(j_+^\omega)^{1/2}$ identifies the $\sigma_{V(Q_-)}$-measurable elements of $\mathcal{A}_Q^\omega$ with $u \in I_-$, and gives $\nu_Q^\omega(XY) = \langle v, K^\omega u\rangle$, $\nu_Q^\omega(X^2) = \|u\|^2$, $\nu_Q^\omega(X) = \langle u, (j_-^\omega)^{1/2}\rangle$, where K^ω is the integral operator with kernel k^ω. It commutes with the unitary action of $\mathcal{G}_Q$, so maps I_- into I_+; $\|K^\omega\| \le 1$ by the Cauchy–Schwarz inequality;

and $K^\omega(j_-^\omega)^{1/2} = (j_+^\omega)^{1/2}$, $K^{\omega *}(j_+^\omega)^{1/2} = (j_-^\omega)^{1/2}$. Hence $A^\omega = K^\omega|_{I_-}$ splits as $1 \oplus A'$ on $\mathbb{C}(j_-^\omega)^{1/2} \oplus (j_-^\omega)^{1/2\,\perp}$, and $\rho^\omega(Q_-, Q_+) = \|A'\| = s_2(A^\omega)$ is the second singular value of A^ω. By $s_k(A) = \inf\{\|A - R\| : \operatorname{rank} R < k\}$,

$$|s_2(A^\omega) - s_2(A^{\omega'})| \le \|K^\omega - K^{\omega'}\| \le \|k^\omega - k^{\omega'}\|_{L^2} \le \sup |k^\omega - k^{\omega'}|,$$

which tends to 0 uniformly by the uniform continuity of k on the compact set $\Omega \times G^{V(Q_-)} \times G^{V(Q_+)}$. □

Two consequences fix the status of the slab hypothesis among the hypotheses on the torus measure. First, when Q sits in a torus Λ_L with L at least the width of Q plus 2, the conditional law of $\sigma_{V(Q)}$ given the exterior links is ν_Q^ω, and the law of ω under μ has a density bounded below with respect to Haar measure, so every nonempty open set of boundary conditions has positive mass; by the lemma, $\sup_\omega \rho^\omega$ is the essential supremum over ω of the conditional maximal correlation of the two slabs under μ. The slab hypothesis is therefore a statement about the torus measure alone, a bound on a conditional correlation that holds almost surely, and not about exceptional boundary data of measure zero. Second, the vacuum form of Proposition 11.13 is an unconditional correlation, and conditioning can create correlation: for independent fair signs W and ω, the variables $X = W$ and $Y = \omega W$ are uncorrelated, while given ω their correlation is 1. So no inequality of general validity bounds the conditional correlation by the vacuum decay ρ_1^d, and the models below show that none holds in the class of reflection-positive systems with a gap.

A boundary condition can, however, push the slab correlation above $\frac{1}{2}$ at a fixed physical scale while the transfer matrix has a gap at that scale and the vacuum value of Proposition 11.13 is small. The following exact model has the layer structure of Lemma 11.10, a symmetric positive-definite transfer matrix with a simple vacuum and a physical gap Δ at every N, and it violates $(\mathrm{M}_{d,m}(c))$ for every $c \le \frac{1}{2}$ under one boundary condition. It is a spin system and not a gauge theory, so it decides only the abstract implication: no statement about the transfer matrix or about the vacuum alone can supply the slab hypothesis.

Lemma 11.16 (pinned two-state chain)**.** *Let $J > 0$, $h \in \mathbb{R}$ and $d \ge 1$, and let $s_0, \dots, s_{3d-2} \in \{\pm 1\}$ carry the weight $\prod_{k=-1}^{3d-2} w(s_k, s_{k+1})$ with $w(x, y) = e^{Jxy + h(x+y)/2}$ and the pinned boundary spins $s_{-1} = s_{3d-1} = -1$. Let $A = (s_0, \dots, s_{d-1})$ and $B = (s_{2d-1}, \dots, s_{3d-2})$. The matrix $T = (w(x, y))$ has eigenvalues $\lambda_0 > \lambda_1 > 0$ with orthonormal eigenvectors $\Omega = (\cos\alpha, \sin\alpha)$ and $\varphi = (-\sin\alpha, \cos\alpha)$ in the basis $(+, -)$, $\alpha \in (0, \pi/2)$. Put $Q = (\lambda_1/\lambda_0)^d$ and $t = \tan^2\alpha$. Then*

$$\rho(A; B) = C(t, Q) := \frac{Q(Q + t)}{(1 + Q)(Q^2 + t)}.$$

The function $t \mapsto C(t, Q)$ decreases from $1/(1+Q)$ to $Q/(1+Q)$ on $(0, \infty)$, and $C(t, Q) > \frac{1}{2}$ exactly when $t < Q^2$.

Proof. The weight is a nearest-neighbour product, so given (s_{d-1}, s_{2d-1}) the blocks A and B are conditionally independent of each other and of the spins between them, exactly as in the proof of Lemma 11.12(i); hence $\rho(A; B) = \rho(s_{d-1}; s_{2d-1})$, and for two-valued variables the centred square-integrable functions form a line, so this is the absolute correlation of s_{d-1} and s_{2d-1}. Write $S = \left(\begin{smallmatrix} \cos 2\alpha & -\sin 2\alpha \\ -\sin 2\alpha & -\cos 2\alpha \end{smallmatrix}\right)$ for the spin operator $\operatorname{diag}(1, -1)$ in the basis (Ω, φ), $D(x) = \operatorname{diag}(1, x)$ in that basis, and $a = (\sin\alpha, \cos\alpha)$ for the coordinates of the pinned vector e_-. Each of the three legs, from s_{-1} to s_{d-1}, from there to s_{2d-1} and from there to s_{3d-1}, carries d bonds, so after division by λ_0^{3d} the partition function, the one-spin and the two-spin sums are

$$Z = a^{\mathsf{T}} D(Q^3) a, \qquad M = a^{\mathsf{T}} D(Q) S D(Q^2) a = a^{\mathsf{T}} D(Q^2) S D(Q) a,$$
$$W = a^{\mathsf{T}} D(Q) S D(Q) S D(Q) a,$$

and the correlation is $(WZ-M^2)/(Z^2-M^2)$. With $\sigma=\sin\alpha$, $\gamma=\cos\alpha$, expanding the products gives $Z=\sigma^2+Q^3\gamma^2$, $M=-Q^3\gamma^4+(Q^3-2Q^2-2Q+1)\gamma^2\sigma^2-\sigma^4$,

$$WZ-M^2=4Q\sigma^2\gamma^2(1-Q)^2(Q\gamma^2+\sigma^2)^2,$$
$$Z^2-M^2=4\sigma^2\gamma^2(1-Q)^2(1+Q)(Q\gamma^2+\sigma^2)(Q^2\gamma^2+\sigma^2),$$

the first an identity of polynomials and the second one modulo $\sigma^2+\gamma^2=1$; dividing by γ^4 gives $C(t,Q)$. Since $\partial_t\frac{Q+t}{Q^2+t}=\frac{Q^2-Q}{(Q^2+t)^2}<0$ for $0<Q<1$, C decreases in t, with the limits $1/(1+Q)$ at $t=0$ and $Q/(1+Q)$ at $t=\infty$; and $2Q(Q+t)>(1+Q)(Q^2+t)$ is $(1-Q)(Q^2-t)>0$. □

Proposition 11.17 (a transfer gap does not imply the slab hypothesis)**.** *Fix $K_0\geq 1$, $\Delta>0$ and $r>0$, put $\kappa=\Delta/(2\sqrt{1+r^2})$ and $\eta=r\kappa$, and for $N\in K_0\mathbb{N}$ let $e^{-2J_N}=\kappa/N$, $h_N=\eta/N$ and $d=N/K_0$. On a layer of any finite set of sites let every site carry an independent chain as in Lemma* 11.16*, of width $3d-1$ in the coordinate of the splitting, with the exterior spins $\omega\equiv-1$. Then:*

(i) *the transfer matrix $T_N=e^{J_N}\left(\begin{smallmatrix}e^{h_N}&\epsilon_N\\ \epsilon_N&e^{-h_N}\end{smallmatrix}\right)$, $\epsilon_N=\kappa/N$, is symmetric and positive definite with a simple largest eigenvalue, its ratio is $\theta_N=(\cosh h_N-R_N)/(\cosh h_N+R_N)$ with $R_N=(\sinh^2 h_N+\epsilon_N^2)^{1/2}$, the vacuum slab correlation at the separation d is θ_N^d, and $\theta_N^d\to e^{-\Delta/K_0}$: the physical gap is Δ for every N;*

(ii) *$\rho^\omega(Q_-,Q_+)=C(t_N,\theta_N^d)$ exactly, with $t_N=\big((R_N-\sinh h_N)/\epsilon_N\big)^2\to t_\infty=\big(\sqrt{1+r^2}-r\big)^2$;*

(iii) *if $Q_0=e^{-\Delta/K_0}<\frac12$ and $\sqrt{1+r^2}+r>1/Q_0$, then $\rho^\omega(Q_-,Q_+)>\frac12$ for all large N, so that $(\mathrm{M}_{d,m}(c))$ fails for every $c\leq\frac12$ and every m, while the vacuum value Q_0 satisfies the hypotheses $c<\frac12$ and $m-1\geq 148/(1-2c)^4$ of Theorem* D *for m large.*

For $Q_0=\frac{1}{20}$ and $r=20$ the limit of ρ^ω is $0.7716\ldots$, against the vacuum value 0.05, which would satisfy Theorem D with $m=227$; as $r\to\infty$ the limit tends to $1/(1+Q_0)$.

Proof. (i) The eigenvalues of the bracketed matrix are $\cosh h_N\pm R_N$, both positive since its determinant is $1-\epsilon_N^2>0$. The vacuum slab correlation is $\rho_1^d=\theta_N^d$ by Proposition 11.13, whose proof applies verbatim to a two-state chain. Since $NR_N\to(\eta^2+\kappa^2)^{1/2}=\Delta/2$ and $\cosh h_N=1+O(N^{-2})$, $\log\theta_N=\log(1-2R_N/(\cosh h_N+R_N))=-\Delta/N+O(N^{-2})$, and $d\log\theta_N\to-\Delta/K_0$. (ii) The chains at different sites are independent, so by Lemma 7.15 ρ^ω is the common single-chain value, which is $C(t_N,\theta_N^d)$ by Lemma 11.16; there $\tan\alpha=(\cosh h_N+R_N-e^{h_N})/\epsilon_N=(R_N-\sinh h_N)/\epsilon_N$, and $N(R_N-\sinh h_N)\to(\eta^2+\kappa^2)^{1/2}-\eta=\kappa(\sqrt{1+r^2}-r)$. (iii) By Lemma 11.16, $C(t_\infty,Q_0)>\frac12$ exactly when $t_\infty<Q_0^2$, that is $\sqrt{1+r^2}+r>1/Q_0$, since $(\sqrt{1+r^2}-r)(\sqrt{1+r^2}+r)=1$; by continuity $\rho^\omega>\frac12$ for large N. For $Q_0=\frac{1}{20}$, $r=20$: $t_\infty=(\sqrt{401}-20)^2<6.25\cdot10^{-4}<Q_0^2$, $C(t_\infty,Q_0)=0.7716\ldots$, and $148/0.9^4=225.6\ldots$. □

The model fails the slab hypothesis through one boundary condition whose state is nearly orthogonal to the vacuum, $t\ll Q^2$; and since $C>Q$ whenever $t<1-Q-Q^2$, even the adjacent-layer bound $\rho^\omega(Y_l,Y_{l+1})\leq\theta_N$ of Lemma 11.12 fails for some l there. In a gauge theory the role of the boundary -1 against the field is played by the twisted boundary conditions of Proposition H.17; the slab hypothesis is a statement about the boundary-conditioned measures, uniform in the boundary condition, and Theorem D cannot be fed by the spectrum of the transfer matrix alone.

The model admits a sharper statement, in which the field h_N is free and the cube condition is tested on the same system. It shows that the gap and the uniqueness of the vacuum do not imply the slab hypothesis at any scale without (D_ℓ), and that inside this class (D_ℓ) does imply it, with explicit constants, by a tunnelling test function.

Proposition 11.18 (the field rings with an arbitrary field)**.** *Fix $K_0 \geq 1$ and $\Delta > 0$, let $r_N > 0$ be arbitrary, and put $\kappa_N = \Delta/(2\sqrt{1+r_N^2})$, $\eta_N = r_N\kappa_N$, $\epsilon_N = e^{-2J_N} = \kappa_N/N$, $h_N = \eta_N/N$, $d = N/K_0$, $Q_0 = e^{-\Delta/K_0}$ and $S_\Delta = \sinh\frac{\Delta}{2} + (\sinh^2\frac{\Delta}{2} + \frac{\Delta^2}{4})^{1/2}$; the chains are those of Proposition* 11.17*, with θ_N, R_N, t_N and $C(t,Q)$ as there.*

(i) *Uniformly in r_N: $d\log\theta_N = -\Delta/K_0 + O(N^{-1})$, and $t_N^{1/2} = \kappa_N/(NR_N + N\sinh h_N)$ satisfies $\kappa_N/S_\Delta \leq t_N^{1/2} \leq 2\kappa_N/\Delta \leq 1$.*

(ii) *Over the four exterior configurations $\omega = (s_{-1}, s_{3d-1}) \in \{\pm1\}^2$ of a chain of width $3d-1$,*

$$\rho^{(-,-)} = C(t_N, \theta_N^d) = \max_\omega \rho^\omega(Q_-, Q_+), \qquad \rho^{(+,+)} = C(1/t_N, \theta_N^d),$$
$$\rho^{(+,-)} = \rho^{(-,+)} = \big(\rho^{(+,+)}\rho^{(-,-)}\big)^{1/2}.$$

(iii) (Tunnelling.) *Let the chains run along the time axis of the cylinder Λ_{L,M_c} with $M_c = 2n+4$ and $n \leq \ell N$, $n \geq 2$. If $\mathrm{BP}(n,\varpi)$ holds for the product measure of the chains, then*

$$\varpi \leq 8\, e^{\Delta\ell}\big(\cosh((\ell+1)\kappa_N) - 1\big).$$

(iv) *If $r_N \to \infty$, then the physical gap is Δ and the vacuum is simple for every N, while at every scale K_0, $\max_\omega \rho^\omega(Q_-,Q_+) \to 1/(1+Q_0) > \frac12$, so $(\mathrm{M}_{d,m}(c))$ fails for every $c \leq \frac12$ and every m for large N, and (D_ℓ) fails for every ℓ and $\varpi > 0$.*

(v) *Conversely, if (D_ℓ) holds with the constant ϖ on the cylinders $M = 2n_N + 4$, define $\kappa_* > 0$ by $8e^{\Delta\ell}(\cosh((\ell+1)\kappa_*) - 1) = \varpi$ and $\tau = \kappa_*^2/S_\Delta^2$. Then for every K_0 with $Q_0^2 < \tau$ and every m, $(\mathrm{M}_{d,m}(c))$ holds for all large N with $c = \frac12(\frac12 + C(\tau, Q_0)) < \frac12$; and $c \to 0$ as $K_0 \to 0$, so the hypotheses of Theorem D are met at a scale depending only on ϖ, ℓ and Δ.*

Proof. (i) Since $\eta_N \leq \Delta/2$ and $N\sinh(x/N) = x + O(x^3N^{-2})$, $NR_N = (N^2\sinh^2 h_N + \kappa_N^2)^{1/2} = \Delta/2 + O(N^{-2})$ uniformly in r_N, and the first claim follows as in Proposition 11.17(i). From $R_N^2 - \sinh^2 h_N = \epsilon_N^2$, $t_N^{1/2} = (R_N - \sinh h_N)/\epsilon_N = \epsilon_N/(R_N + \sinh h_N)$, which is the stated expression. The denominator is at least $NR_N \geq (\eta_N^2 + \kappa_N^2)^{1/2} = \Delta/2$, because $N\sinh(x/N) \geq x$, and at most $(\sinh^2\eta_N + \kappa_N^2)^{1/2} + \sinh\eta_N \leq S_\Delta$, because $N\sinh(x/N) \leq \sinh x$ and $\eta_N, \kappa_N \leq \Delta/2$; and $2\kappa_N/\Delta \leq 1$.

(ii) Write T for the bracketed matrix, $u = T^d e_{s_{-1}}$, $v = T^d e_{s_{3d-1}}$. The joint law of (s_{d-1}, s_{2d-1}) is $P = D_u T^d D_v/\mathrm{tr}$ with $D_w = \mathrm{diag}(w_+, w_-)$, and for a 2×2 table the absolute correlation of two signs is $|\det P|$ divided by the square root of the product of the four marginals, which gives

$$\rho^\omega = \det(T^d)\,\big(g(u)g(v)\big)^{1/2}, \qquad g(w) = \frac{w_+w_-}{(T^dw)_+(T^dw)_-}.$$

So $\rho^{(+,-)} = (\rho^{(+,+)}\rho^{(-,-)})^{1/2}$; $\rho^{(-,-)} = C(t_N,\theta_N^d)$ is Lemma 11.16; the same lemma with the labels $\pm$ exchanged, that is $h \to -h$, has the vacuum vector $(\sin\alpha, \cos\alpha)$, so $\alpha \to \frac\pi2 - \alpha$ and $\rho^{(+,+)} = C(1/t_N, \theta_N^d)$; and C decreases in t with $t_N \leq 1$, so the maximum is $\rho^{(-,-)}$. For a splitting in another coordinate $\rho^\omega = 0$ by Lemma 7.15.

(iii) Let E be the event that one fixed chain is all $-$, and $f = \mathbf{1}_E$. A cube of side n meets that chain in a window w of n consecutive sites or not at all, and given the links outside the cube the chains are independent, so $\mathcal{P}_x f$ is the resampling of the window; each window belongs to exactly n^3 cubes, so $n^{-4}\sum_x \mu((f-\mathcal{P}_x f)^2) = n^{-1}\sum_w \mu((f-\mathcal{P}_w f)^2)$. Now $f - \mathcal{P}_w f \neq 0$ only when the chain is all $-$ outside w, and then, with q the probability that the window is all $-$ given its two pinned neighbours $-$, it equals $1-q$ with probability q and $-q$ otherwise: $\mu((f-\mathcal{P}_w f)^2) = \pi(E)(1-q)$. Since $\mathrm{Var}\, f = \pi(E)(1-\pi(E))$ and

$\pi(E) < \frac{1}{2}$, the all-+ configuration having the larger weight by the factor $e^{2h_N M_c}$,

$$\varpi \le \frac{M_c}{n}\,\frac{1-q}{1-\pi(E)} \le \frac{2M_c}{n}(1-q) \le 8(1-q).$$

A window configuration with $k \ge 1$ domains of + has $2k$ walls among its $n+1$ bonds, each costing the factor ϵ_N against a − bond, and a field factor at most $e^{2h_N n} \le e^{\Delta\ell}$; so $1-q \le \sum_{k\ge1} \binom{n+1}{2k} \epsilon_N^{2k} e^{\Delta\ell} \le e^{\Delta\ell}(\cosh((n+1)\epsilon_N) - 1)$, and $(n+1)\epsilon_N \le (\ell+1)\kappa_N$.

(iv) By (i), $t_N \le 4\kappa_N^2/\Delta^2 \to 0$ and $\theta_N^d \to Q_0$, so $\max_\omega \rho^\omega = C(t_N, \theta_N^d) \to C(0, Q_0) = 1/(1+Q_0)$ by the continuity of C; the gap and the simple vacuum are Proposition 11.17(i); and (iii) with $\kappa_N \to 0$ excludes every fixed ϖ.

(v) By (iii), $\kappa_N \ge \kappa_*$, hence $t_N \ge \tau$ by (i), and as C decreases in t, $\max_\omega \rho^\omega = C(t_N, \theta_N^d) \le C(\tau, \theta_N^d) \to C(\tau, Q_0)$, which is below $\frac{1}{2}$ exactly when $\tau > Q_0^2$ (Lemma 11.16) and tends to 0 with Q_0. □

The model is not a gauge theory and does not satisfy (W), its continuum limit being ultralocal across the sites; it decides two abstract questions. The gap at every N, a simple vacuum, reflection positivity and Markov layers do not imply the slab hypothesis at any scale, by (iv), so no proof of (M) can proceed from the spectrum of the transfer matrix without the cube condition; and in this class the cube condition excludes the bad boundary state by the tunnelling function $\mathbf{1}_E$, by (v). The bound (iii) is of the right order: for $K_0 = \Delta = 1$ the constant of $\mathrm{BP}(2d-1, \varpi)$ on periodic rings, computed by exact diagonalisation for $N \le 5$, lies within a factor three of the middle expression of (iii) for r between 1 and 100, and decays like $\kappa_N^2 \sim r^{-2}$ as (iii) predicts.

11.7. **The slab hypothesis as a block gap.** The slab hypothesis is a statement about two projections, and the cube condition it produces is a Poincaré inequality for many; the two are closer than this suggests. First, the slab hypothesis is itself a Poincaré inequality, exactly, for the chain with two overlapping blocks. For a splitting $Q = Q_- \cup Q_0 \cup Q_+$ in the coordinate i put $A_- = V(Q_- \cup Q_0)$ and $A_+ = V(Q_0 \cup Q_+)$, so that E_{A_-} projects onto the functions of $\sigma_{V(Q_+)}$ and E_{A_+} onto those of $\sigma_{V(Q_-)}$, and for $g \in \mathcal{A}_Q^\omega$ let

$$\mathcal{D}_Q^\omega(g) = \nu\big((g - E_{A_-}g)^2\big) + \nu\big((g - E_{A_+}g)^2\big),$$

the Dirichlet form of the chain that resamples, with probability $\frac{1}{2}$ each, everything but Q_+ or everything but Q_-.

Proposition 11.19 (the two-slab form)**.** *For every box Q, splitting, exterior configuration ω and $R \in [0,1]$,*

$$\rho^\omega(Q_-, Q_+) \le R \quad\Longleftrightarrow\quad \mathcal{D}_Q^\omega(g) \ge (1-R)\,\mathrm{Var}_\nu(g) \ \text{ for every } g \in \mathcal{A}_Q^\omega.$$

Hence $\rho^\omega(Q_-, Q_+) = \max\{0, 1 - \inf \mathcal{D}_Q^\omega(g)\}$, the infimum over $g \in \mathcal{A}_Q^\omega$ with $\mathrm{Var}_\nu(g) = 1$, and $(\mathrm{M}_{d,m}(c))$ is the inequality $\mathcal{D}_Q^\omega \ge (1-c)\,\mathrm{Var}_\nu$ on $\mathcal{A}_Q^\omega$ for all the boxes, splittings and ω of Definition 11.3(i).

Proof. On $\mathcal{K} = \mathcal{A}_Q^\omega$ let $P_1 = E_{A_-}$ and $P_2 = E_{A_+}$, orthogonal projections of $\mathcal{K}$ by Lemma 11.1(iii), and $\mathcal{N} = \mathbb{R}\mathbf{1}$. An element of $\operatorname{ran} P_1 \cap \operatorname{ran} P_2$ is almost surely a function of $\sigma_{V(Q_+)}$ and one of $\sigma_{V(Q_-)}$, hence constant by Lemma 11.6 with $R_0 = A_-$, $R_1 = A_+$; so $\operatorname{ran} P_1 \cap \operatorname{ran} P_2 = \mathcal{N}$. The pairs of Definition 11.2 are exactly the unit vectors $y \in \operatorname{ran} P_2 \ominus \mathcal{N}$ and $x \in \operatorname{ran} P_1 \ominus \mathcal{N}$, so $\rho^\omega(Q_-, Q_+) = \sup |\langle x, y\rangle|$ over them, and for $g \perp \mathbf{1}$, $\mathcal{D}_Q^\omega(g) = \langle g, Xg\rangle$ with $X = (I - P_1) + (I - P_2)$.

If $\rho^\omega(Q_-, Q_+) \le R$, Lemma 11.4 gives $X^2 \ge (1-R)X$, so the spectrum of $X \ge 0$ lies in $\{0\} \cup [1-R, \infty)$, and $Xg = 0$ forces $\|(I-P_1)g\|^2 + \|(I-P_2)g\|^2 = 0$, that is $g \in \mathcal{N}$. Hence $\mathcal{D}_Q^\omega(g) \ge (1-R)\|g\|^2$ for $g \perp \mathbf{1}$, and for general g apply this to $g - \nu(g)$, which changes neither side. Conversely, let x, y be admissible unit vectors with $r = \langle x, y\rangle \ge 0$, and $g = x + y$, so that $g \perp \mathbf{1}$ and $\|g\|^2 = 2 + 2r$. Then $P_1 g = x + P_1 y$ with $\|P_1 y\| \ge |\langle P_1 y, x\rangle| = r$, so $\|P_1 g\|^2 = 1 + 2r + \|P_1 y\|^2 \ge (1+r)^2$, and likewise

$\|P_2 g\|^2 \geq (1+r)^2$; as $\|P_1 g\|^2 + \|P_2 g\|^2 = 2\|g\|^2 - \mathcal{D}_Q^\omega(g) \leq (1+R)(2+2r)$, this gives $2(1+r)^2 \leq (1+R)(2+2r)$, that is $r \leq R$. The formula follows by taking $R = \rho^\omega(Q_-, Q_+)$ in the first direction and $R = \max\{0, 1 - \inf \mathcal{D}_Q^\omega\}$ in the second. □

So the hypothesis is a Poincaré inequality, uniform in ω, for the heat-bath chain with the two overlapping blocks $A_\pm$, and Lemma 11.25 says that a Lipschitz contraction of that chain implies its L^2 gap. The question is then whether the Poincaré inequality for the many small blocks of Theorem 11.8 implies the one for the two big blocks. The naive comparison goes the wrong way: $B \subset A$ gives $\nu((g - E_A g)^2) \geq \nu((g - E_B g)^2)$ for each small block, not for their sum. The answer is nevertheless yes, up to constants and a change of scale, and the mechanism is the finite speed of Section 6, run under a fixed boundary condition and in L^2.

For $d \geq 2$ and $m \geq 4$ call a box $Q = r + \prod_{i=1}^4 [0, (k_i+2)d - 1)$ with integers $0 \leq k_i \leq m-2$ a *d-box*; its *blocks* are $B_j = V(r + dj + [0, 2d-1)^4)$ for $j \in J_Q = \prod_i [0, k_i]$, $n_B = |J_Q| = \prod_i (k_i + 1) \leq (m-1)^4$ in number, and they cover $V(Q)$; put $P_j = E_{B_j}$ under ν_Q^ω. The patch box Π is the d-box with all $k_i = m - 2$.

$(\mathrm{B}_{d,m}(\gamma))$ for every d-box Q, every exterior configuration ω and every $F \in \mathcal{A}_Q^\omega$,

$$\gamma \operatorname{Var}_\nu(F) \leq \sum_{j \in J_Q} \nu((F - P_j F)^2).$$

Let $(\mathrm{M}_{d,m}^\flat(c))$ be $(\mathrm{M}_{d,m}(c))$ restricted to the boxes all of whose widths are $\equiv -1 \pmod d$, the d-boxes of width $3d-1$ in the coordinate i.

Lemma 11.20. (i) $(\mathrm{M}_{d,m}^\flat(c))$ *with* $c < \frac{1}{2}$ *implies* $(\mathrm{B}_{d,m}((1-2c)^4))$.

(ii) *Theorem 11.8, Theorem D, Corollary 11.9 and Proposition 7.13 hold with* $(\mathrm{M}_{d,m}(c))$ *replaced by* $(\mathrm{M}_{d,m}^\flat(c))$.

Proof. (i) The proof of Theorem 11.8, with k_i in place of $m-2$ in the coordinate i and Proposition 11.7 applied with $M = k_{k+1}$ (the case $M = 0$ being trivial), evaluates $\rho^{\omega'}$ only on boxes of widths $2d-1$, $3d-1$ and $(k_i+2)d-1$, all in $[2d-1, md-1]$ and all $\equiv -1 \pmod d$. (ii) Theorem D uses $(\mathrm{M}_{d,m}(c))$ only through Theorem 11.8, and the others only through Theorem D and the patch gap. □

Proposition 11.21 (block gap to slab bound at the same scale)**.** *Assume* $(\mathrm{B}_{d,m}(\gamma))$. *Let Q be a d-box with $k_i = 1$ in the coordinate i, split into Q_-, Q_0, Q_+ of widths d, $d-1$, d in that coordinate, and let $n_\perp = \prod_{i' \neq i}(k_{i'} + 1) \leq (m-1)^3$. Then for every ω*

$$\rho^\omega(Q_-, Q_+) \leq 1 - \frac{\gamma}{n_\perp} \leq 1 - \frac{\gamma}{(m-1)^3}.$$

In particular $(\mathrm{B}_{d,m}(\gamma))$ *implies* $(\mathrm{M}_{d,m}^\flat(1 - \gamma/(m-1)^3))$.

Proof. Write $j = (j_i, j')$ with $j_i \in \{0, 1\}$. The block $B_{(0,j')}$ has i-range $[0, 2d-1)$ relative to Q, so $B_{(0,j')} \subset A_-$, and $B_{(1,j')}$ has i-range $[d, 3d-1)$, so $B_{(1,j')} \subset A_+$. If $B \subset A$ then every function of $\sigma_{V(Q)\setminus A}$ is a function of $\sigma_{V(Q)\setminus B}$, so $E_A \leq E_B$ and $\nu((g - E_A g)^2) \geq \nu((g - E_B g)^2)$. Hence

$$\mathcal{D}_Q^\omega(g) \geq \max_{j'} \nu((g - P_{(0,j')} g)^2) + \max_{j'} \nu((g - P_{(1,j')} g)^2) \geq \frac{1}{n_\perp} \sum_{j \in J_Q} \nu((g - P_j g)^2),$$

which is at least $(\gamma / n_\perp) \operatorname{Var}_\nu(g)$, and Proposition 11.19 with $R = 1 - \gamma/n_\perp$ gives the claim. □

This is the weakest form of the comparison, at the same scale and with a constant below 1 only. The dynamics gives more. Fix a d-box Q, ω, $\nu = \nu_Q^\omega$, and let $\mathcal{K} = n_B^{-1} \sum_{j \in J_Q} P_j$, which is self-adjoint on $L^2(\nu)$ with $0 \leq \mathcal{K} \leq I$ and $\mathcal{K}\mathbf{1} = \mathbf{1}$ and preserves $\mathcal{A}_Q^\omega$, and $S_t = e^{-t(I - \mathcal{K})}$ for $t \geq 0$. For a block B let ∂B be the set of links of $V(Q)$ outside B that share a plaquette with a link of B, and write $j \rightsquigarrow j'$ if $B_j \cap \partial B_{j'} \neq \emptyset$. The links of $\partial B_{j'}$ have base points in

$r+dj'+[-1,2d-1]^4$ and those of B_j in $r+dj+[0,2d-2]^4$, so $j \rightsquigarrow j'$ implies $|j-j'|_\infty \le 1$, and at most $3^4=81$ indices j satisfy $j \rightsquigarrow j'$ for a given j'. For a set S of links of $V(Q)$ let $J_1(S)=\{j : B_j \cap S \neq \emptyset\}$ and $J_{l+1}(S)=J_l(S)\cup\{j : j \rightsquigarrow j' \text{ for some } j' \in J_l(S)\}$, $J_0(S)=\emptyset$; for $Y \subset J_Q$ let $\mathcal{K}_Y = n_B^{-1}(\sum_{j\in Y} P_j + (n_B-|Y|)I)$ and $S_t^Y = e^{-t(I-\mathcal{K}_Y)}$.

Lemma 11.22 (finite speed in L^2)**.** *Let $f \in L^2(\nu)$ be a function of the links of $S \subset V(Q)$, $m_0 \ge 0$ and $Y = J_{m_0}(S)$. Then $S_t^Y f$ is a function of the links of $S \cup \bigcup_{j\in Y} \partial B_j$, and for every $t \ge 0$*

$$\|S_t f - S_t^Y f\|^2_{L^2(\nu)} \le 4\,\|f\|^2_{L^2(\nu)}\,\frac{|J_1(S)|}{81}\sum_{m>m_0}\frac{(81\tau)^m}{m!}, \qquad \tau = \frac{t}{n_B}.$$

Proof. Realise S_t and S_t^Y by the continuous-time chains in which the points of a rate-one Poisson process on $[0,t]$ carry independent uniform labels in J_Q; at a point with label j the first chain resamples B_j from its conditional law given the other links, and the second does so only if $j \in Y$. Since $e^{-t(I-\mathcal{K})} = \sum_l e^{-t} t^l \mathcal{K}^l/l!$ and $\mathcal{K}^l$ is the average over l independent uniform labels of the product of the P_j, one has $S_t f(U) = E[f(\xi_t) \mid \xi_0 = U]$ and likewise for S_t^Y; the first claim follows by induction over the points, as in Lemma 6.2. Couple the two chains from the same U with the same labelled points, using one common sample when the label is in Y and the two configurations agree on ∂B_j, and independent samples otherwise. As in the proof of Lemma 6.2, a link where the two configurations differ at time t lies in B_{x_m} for labels $x_1,\dots,x_m$ carried by points $s_1<\dots<s_m\le t$ with $x_1 \notin Y$ and $x_l \rightsquigarrow x_{l+1}$; if such a link lies in S then $x_m \in J_1(S)$, $x_{m-l} \in J_{l+1}(S)$, hence $m > m_0$. Let $\mathcal{B}$ be the event that the labelled points contain such a chain with $x_m \in J_1(S)$ and $m > m_0$; it depends on the labelled points only. For a fixed sequence $x_1,\dots,x_m$ the expected number of increasing m-tuples of points with these labels is $t^m/(n_B^m m!)$, since the labels are independent marks, so $P(\mathcal{B}) \le \sum_{m>m_0} |J_1(S)|\, 81^{m-1} t^m/(n_B^m m!)$. Now $S_t f(U) - S_t^Y f(U) = E[f(\xi_t) - f(\xi_t') \mid U]$, so by Jensen's inequality, with $\xi_0 = \xi_0' \sim \nu$,

$$\|S_t f - S_t^Y f\|^2 \le E\big[(f(\xi_t)-f(\xi_t'))^2 \mathbf{1}_{\mathcal{B}}\big] \le 2E\big[(f(\xi_t)^2 + f(\xi_t')^2)\mathbf{1}_{\mathcal{B}}\big],$$

because off $\mathcal{B}$ the two configurations agree on S. Conditionally on the labelled points, $\xi_t \sim \nu$ and $\xi_t' \sim \nu$, every resampling preserving ν; so the last expectation is $4\nu(f^2)P(\mathcal{B})$. □

Theorem 11.23 (layer decay from the block gap)**.** *Assume* ($\mathrm{B}_{d,m}(\gamma)$). *Let Q be a d-box, ω an exterior configuration, i a coordinate, and Y_a, Y_b layers of Q in the coordinate i with $b-a \ge 2d$; put $k = \lfloor (b-a)/d \rfloor - 1 \ge 1$ and let $n_1 \le 2(m-1)^3$ be the number of blocks meeting Y_a. Then for every $T>0$*

$$\rho^\omega(Y_a,Y_b) \le e^{-\gamma T/n_B} + 2T\Big(\frac{n_1}{81}\Big)^{1/2}\Big(\sum_{m'\ge k}\frac{(81T/n_B)^{m'}}{m'!}\Big)^{1/2},$$

and with $T = k n_B/(81e^2)$

$$\rho^\omega(Y_a,Y_b) \le e^{-\gamma k/(81e^2)} + \frac{2k(m-1)^4}{81e^2}\Big(\frac{2(m-1)^3}{81}\Big)^{1/2} e^{-k(1-e^{-2})/2}.$$

By Lemma 11.12(i) *the same bounds hold for $\rho^\omega(Q_-,Q_+)$ for every splitting of Q in the coordinate i whose middle has width $b-a-1 \ge 2d-1$.*

Proof. Let $X, X' \in \mathcal{A}_Q^\omega$ be centred unit functions of σ_{Y_a} and σ_{Y_b}. For $h \in \mathcal{A}_Q^\omega \ominus \mathbf{1}$, $\langle h, (I-\mathcal{K})h\rangle = n_B^{-1}\sum_j \nu((h-P_j h)^2) \ge (\gamma/n_B)\|h\|^2$ by ($\mathrm{B}_{d,m}(\gamma)$), and $\mathcal{K}$ is self-adjoint and preserves $\mathcal{A}_Q^\omega \ominus \mathbf{1}$, so $\|S_t h\| \le e^{-\gamma t/n_B}\|h\|$ there. Since $X - S_T X = \int_0^T (I-\mathcal{K}) S_t X\, dt$,

$$\langle X, X'\rangle = \langle S_T X, X'\rangle + \int_0^T \langle S_t X, (I-\mathcal{K})X'\rangle\, dt, \qquad |\langle S_T X, X'\rangle| \le e^{-\gamma T/n_B}.$$

Now $(I-\mathcal{K})X' = n_B^{-1}\sum_{j\in J_1(Y_b)}(X' - P_j X')$, as $P_j X' = X'$ when $B_j \cap Y_b = \emptyset$, and $\langle h, X' - P_j X'\rangle = \langle h - P_j h, X' - P_j X'\rangle = 0$ whenever h does not depend on the links

of B_j. Put $m_0 = k-1$ and $Y = J_{m_0}(Y_a)$. In coordinates relative to Q, B_j meets Y_a if and only if $dj_i \le a \le dj_i + 2d - 2$, so $J_1(Y_a) \subset \{j_i \le \lfloor a/d \rfloor\}$, with at most two values of j_i, which gives $n_1 \le 2\prod_{i' \ne i}(k_{i'}+1)$; each step of $\rightsquigarrow$ changes j_i by at most one, so $J_k(Y_a) \subset \{j_i \le \lfloor a/d \rfloor + k - 1\}$, while $J_1(Y_b) \subset \{j_i \ge (b-2d+2)/d\}$. As $\lfloor a/d \rfloor + \lfloor (b-a)/d \rfloor - 2 \le b/d - 2 < (b-2d+2)/d$, the sets $J_k(Y_a)$ and $J_1(Y_b)$ are disjoint. Hence for $j \in J_1(Y_b)$, $B_j \cap Y_a = \emptyset$, and $B_j \cap \partial B_y = \emptyset$ for all $y \in Y$, since $j \rightsquigarrow y$ would put j in $J_k(Y_a)$; by Lemma 11.22, $S_t^Y X$ does not depend on the links of B_j, and $\langle S_t^Y X, (I-\mathcal{K})X' \rangle = 0$. Therefore, using $\|(I-\mathcal{K})X'\| \le 1$ and the lemma with $|J_1(Y_a)| = n_1$, monotonically in $t \le T$,

$$\Big| \int_0^T \langle S_t X, (I-\mathcal{K})X' \rangle \, dt \Big| \le \int_0^T \| S_t X - S_t^Y X \| \, dt \le 2T \Big(\frac{n_1}{81} \Big)^{1/2} \Big(\sum_{m' \ge k} \frac{(81T/n_B)^{m'}}{m'!} \Big)^{1/2}.$$

The supremum over X, X' gives the first bound. For the second put $z = 81T/n_B = k/e^2$: $\sum_{m' \ge k} z^{m'}/m'! \le (z^k/k!)e^z \le (ez/k)^k e^z = e^{-k} e^{k/e^2}$, and $\gamma T/n_B = \gamma k/(81e^2)$; finally $n_B \le (m-1)^4$ and $n_1 \le 2(m-1)^3$. □

Corollary 11.24 (the two hypotheses are equivalent up to constants)**.** *Let $d \ge 2$, $m \ge 4$, $\gamma > 0$, let $\kappa \ge 2$ be an integer with $m \ge 4\kappa$, and put $d' = \kappa d$, $m' = \lfloor m/\kappa \rfloor \ge 4$ and*

$$c'(\kappa) = e^{-\gamma(\kappa-1)/(81e^2)} + \frac{2(\kappa-1)(m-1)^4}{81e^2} \Big(\frac{2(m-1)^3}{81} \Big)^{1/2} e^{-(\kappa-1)(1-e^{-2})/2}.$$

(i) $(\mathrm{B}_{d,m}(\gamma))$ *implies* $(\mathrm{M}^{\flat}_{d',m'}(c'(\kappa)))$.

(ii) *For $\eta \in (0,1)$, $c'(\kappa) \le \eta$ as soon as $\kappa - 1 \ge \max\{600\log(2/\eta)/\gamma,\ 13\log(m-1) + 5\log(2/\eta) + 10\}$.*

(iii) *Hence, for $c < \frac{1}{2}$ and $\gamma = (1-2c)^4$, $(\mathrm{M}^{\flat}_{d,m}(c))$ implies $(\mathrm{B}_{d,m}(\gamma))$, which implies $(\mathrm{M}^{\flat}_{\kappa d, \lfloor m/\kappa \rfloor}(\eta))$ for every $\eta \in (0,1)$ and every κ as in* (ii) *with $m \ge 4\kappa$; and conversely $(\mathrm{B}_{d,m}(\gamma))$ implies $(\mathrm{M}^{\flat}_{\kappa d, m'}(\eta))$, which implies $(\mathrm{B}_{\kappa d, m'}((1-2\eta)^4))$ for $\eta < \frac{1}{2}$.*

Proof. (i) A box of $(\mathrm{M}^{\flat}_{d',m'})$ has widths $(k'+2)\kappa d - 1$ with $0 \le k' \le m'-2$, that is $(k+2)d-1$ with $k = (k'+2)\kappa - 2 \in [0, m-2]$, and width $3\kappa d - 1 \le md - 1$ in the coordinate i since $3\kappa \le m$; so it is a d-box, to which $(\mathrm{B}_{d,m}(\gamma))$ applies. Its splitting into widths d', $d'-1$, d' has $\rho^{\omega}(Q_-, Q_+) = \rho^{\omega}(Y_{d'-1}, Y_{2d'-1})$ by Lemma 11.12(i), with $b - a = \kappa d \ge 2d$ and $k = \kappa - 1$, and Theorem 11.23 gives the bound $c'(\kappa)$. (ii) Since $81e^2 < 600$, the first term is at most $\eta/2$ when $\kappa - 1 \ge 600\log(2/\eta)/\gamma$. For the second term write $k = \kappa - 1$ and note that $k\, e^{-k(1-e^{-2})/2}$ decreases for $k \ge 3$; at $k = 13\log(m-1) + 5\log(2/\eta) + 10$ one has $e^{k(1-e^{-2})/2} \ge (m-1)^{5.62}(2/\eta)^{2.16} e^{4.32}$, which exceeds the required $4\sqrt{2}\, k(m-1)^{5.5}/(729e^2\eta)$ for every $m \ge 4$ and $\eta \le 1$, since that amounts to $k \le \frac{729e^{6.32}}{2\sqrt{2}}(m-1)^{0.12}(2/\eta)^{1.16} = 1.43 \times 10^5 (m-1)^{0.12}(2/\eta)^{1.16}$, which the logarithmic k satisfies by a factor above 7000 at $m = 4$, $\eta = 1$, and the right side grows faster than k in both variables. (iii) is the composition of Lemma 11.20(i), (i) and (ii). □

So on the boxes with widths $\equiv -1 \pmod d$, the only ones on which the chain from (M) to the cube condition evaluates the hypothesis, the slab hypothesis at scale d with $c < \frac{1}{2}$ and the block gap $\gamma > 0$ at scale d, uniform in the boundary condition, are equivalent up to constants and a change of scale $d \to \kappa d$ with κ depending on γ and m only, not on d or on the lattice spacing: the block form is the input of Theorem 11.8 and of the finite-size criterion, the slab form that of Proposition 7.13, and the block gap even gives exponential decay of the layer correlation in the number of block steps, uniformly in ω, with a prefactor polynomial in m. They are one hypothesis in two coordinates, and neither is closer to the vacuum data of Proposition 11.13: what both ask, beyond the gap, is the uniformity in ω. A same-scale bound below $\frac{1}{2}$ does not follow from γ alone. On a d-box with $k_i = 1$ and $k_{i'} = 0$ for $i' \ne i$, so $n_\perp = 1$, the two blocks are $B_{(0)} = Q_- \cup Q_0$

and $B_{(1)} = Q_0 \cup Q_+$, so $P_{(0)} = E_{A_-}$, $P_{(1)} = E_{A_+}$, $\sum_j \nu((g - P_j g)^2) = \mathcal{D}^\omega_Q(g)$, and by Proposition 11.19 the best constant in $(\mathrm{B}_{d,m}(\gamma))$ on that box is exactly $1 - \rho^\omega(Q_-, Q_+)$: at the same scale the block gap and the slab bound are one number, $\gamma \le 1 - \rho^\omega(Q_-, Q_+)$, and a constant $c' < \frac{1}{2}$ needs $\gamma > \frac{1}{2}$ there, which $\gamma > 0$ does not give. A one-dimensional example in the manner of Proposition 7.17, with a pinning field, shows the two numbers apart with exact values: for spins $s_0, \dots, s_4 \in \{\pm 1\}$ with the weight $\prod_{j=-1}^{4} w(s_j, s_{j+1})$, the ends pinned at $s_{-1} = s_5 = -1$, $w(+,+) = 8$, $w(+,-) = \frac{1}{4}$, $w(-,-) = 2$ (the Ising weight $e^{Jss' + h(s+s')/2}$ with $J = \log 4$, $h = \log 2$), and $d = 2$, $Q_- = \{0,1\}$, $Q_0 = \{2\}$, $Q_+ = \{3,4\}$, the law is Markov, so the maximal correlation of Q_- and Q_+ is that of s_1 and s_3, which for two-valued variables is the modulus of their correlation coefficient, $(1143318969/2580030436)^{1/2} = 0.6657 > \frac{1}{2}$, while the two-block Poincaré constant on this chain is $1 - 0.6657 > 0$: the chain, pinned against its field at both ends, flips as a whole, and the two outer blocks are nearly functions of that one bit.

11.8. **What is known about the hypothesis.** *Strong coupling, uniformly in the box.* The hypothesis is not vacuous, and at strong coupling the layer bound of Lemma 11.12 holds with a constant that depends neither on the widths of the box nor on the boundary condition. The mechanism is the one of Proposition 8.7: a contraction of the heat-bath moves in the Wasserstein metric, turned into a bound in L^2 by the spectral theorem. Keep the metric δ of diameter D on G and the notion of partners of Section 8, and for $A \subset V(Q)$ and a configuration U of the other links of $V(Q)$, ω being fixed, let $\gamma_A(\cdot \mid U)$ be the conditional law of σ_A under ν^ω_Q, with density proportional to $\prod_{p \cap A \neq \emptyset} \varphi(\cdot_p)$ on G^A, and $d_A(\psi, \psi') = \sum_{e \in A} \delta(\psi_e, \psi'_e)$. A function h on $G^{V(Q)}$ is Lipschitz with constant $\mathrm{Lip}(h)$ if $|h(U) - h(U')| \le \mathrm{Lip}(h)\delta(U_i, U'_i)$ whenever U, U' differ only at the link i.

Lemma 11.25 (two projections)**.** *Let $\nu = \nu^\omega_Q$, let $Y, Y' \subset V(Q)$ be disjoint, $A = V(Q) \setminus Y'$, $A' = V(Q) \setminus Y$, and let E_A, $E_{A'}$ be the conditional expectations given $\sigma_{Y'}$ and given σ_Y, the orthogonal projections of $L^2(\nu)$ onto the functions of $\sigma_{Y'}$ and of σ_Y. Suppose that for some $R \ge 0$, for every link $i \in Y'$ and every pair U, U' differing only at i,*

$$W_1^{d_A}\big(\gamma_A(\cdot \mid U), \gamma_A(\cdot \mid U')\big) \le R\, \delta(U_i, U'_i),$$

and the same with (Y, A') in place of (Y', A). Then $|\nu(XX')| \le R$ for all centred unit vectors $X \in L^2(\nu)$ measurable with respect to σ_Y and X' measurable with respect to $\sigma_{Y'}$; in particular $\rho^\omega(Y, Y') \le R$ when Y, Y' are layers, and $\rho^\omega(Q_-, Q_+) \le R$ when $Y = V(Q_-)$, $Y' = V(Q_+)$.

Proof. Put $M = \frac{1}{2}(E_A + E_{A'})$, self-adjoint with $0 \le M \le I$ and $M\mathbf{1} = \mathbf{1}$. Let h be Lipschitz and U, U' differ only at i. The function $E_A h(U) = \int h(\psi, U_{Y'})\, \gamma_A(d\psi \mid U)$ depends on U only through $U_{Y'}$, so it does not move if $i \in A$; if $i \in Y'$, an optimal coupling π of $\gamma_A(\cdot \mid U)$ and $\gamma_A(\cdot \mid U')$, which exists because the cost is continuous on a compact space, gives

$$\begin{aligned} |E_A h(U) - E_A h(U')| &\le \int |h(\psi, U_{Y'}) - h(\psi', U'_{Y'})|\, d\pi \\ &\le \mathrm{Lip}(h) \int \big(d_A(\psi, \psi') + \delta(U_i, U'_i)\big) d\pi \le \mathrm{Lip}(h)(1 + R)\, \delta(U_i, U'_i). \end{aligned}$$

The same holds for $E_{A'}$ with Y in place of Y', and since $Y \cap Y' = \emptyset$ at most one of the two terms of M moves: $\mathrm{Lip}(Mh) \le \frac{1+R}{2} \mathrm{Lip}(h)$, hence $\mathrm{Lip}(M^t h) \le (\frac{1+R}{2})^t \mathrm{Lip}(h)$. If $\nu(h) = 0$ then $\nu(M^t h) = 0$ and $M^t h$ has oscillation at most $\mathrm{Lip}(M^t h) D |V(Q)|$, so $\|M^t h\|^2 \le (\frac{1+R}{2})^{2t} \mathrm{Lip}(h)^2 D^2 |V(Q)|^2$. With μ_h the spectral measure of M at h, $\|M^t h\|^2 = \int \lambda^{2t} d\mu_h \ge \lambda_1^{2t} \mu_h([\lambda_1, 1])$ for every $\lambda_1 > \frac{1+R}{2}$, so $\mu_h([\lambda_1, 1]) = 0$ and $\langle h, Mh \rangle \le \frac{1+R}{2} \|h\|^2$. Lipschitz functions are dense in $L^2(\nu)$, both sides are continuous in L^2, and M preserves the mean-zero functions, so the inequality holds for every mean-zero $h \in L^2(\nu)$.

Now let X, X' be as in the statement with $r = \nu(XX') \ge 0$, replacing X by $-X$ if necessary, and $g = X + X'$, $\|g\|^2 = 2 + 2r$. As X' is a function of $\sigma_{Y'}$, $E_A X' = X'$,

so $E_A g = E_A X + X'$ and $\|E_A g\|^2 = \|E_A X\|^2 + 2r + 1$, with $\|E_A X\| \geq |\langle E_A X, X'\rangle| = |\langle X, X'\rangle| = r$; hence $\|E_A g\|^2 \geq (1+r)^2$, and symmetrically $\|E_{A'} g\|^2 \geq (1+r)^2$. Since $E_A, E_{A'}$ are projections,

$$(1+r)^2 \leq \tfrac{1}{2}\big(\|E_A g\|^2 + \|E_{A'} g\|^2\big) = \langle g, Mg\rangle \leq \tfrac{1+R}{2}\,(2+2r),$$

that is, $r \leq R$. The last two assertions follow because the admissible pairs of Definition 11.2 and of Section 11.6 are among the pairs considered. □

Lemma 11.26 (response of a block to one link)**.** *For links e, k let C_{ek} be the supremum of $W_1(\gamma_e(\cdot \mid U), \gamma_e(\cdot \mid U'))/\delta(U_k, U'_k)$ over configurations differing only at k, where $\gamma_e(\cdot \mid U)$ is the conditional law of σ_e given all other links, with density proportional to $\prod_{p \ni e} \varphi(U_p)$; it vanishes unless k is a partner of e. Suppose $C_{ek} \leq c$ for all e, k with $18c < 1$. Let $A \subset V(Q)$ and let $i \in V(Q) \setminus A$ be a link with b partners in A. Then for U, U' differing only at i,*

$$W_1^{d_A}\big(\gamma_A(\cdot \mid U), \gamma_A(\cdot \mid U')\big) \leq \frac{c\,b}{1-18c}\,\delta(U_i, U'_i).$$

Proof. This is the proof of Lemma 8.4 without the gauge fixing. Let π minimise $\int d_A \, d\pi$ over couplings of the two laws and $\varrho(e) = \int \delta(\psi_e, \psi'_e)\, d\pi$ for $e \in A$. For a fixed e, drawing $(\psi_{A\setminus e}, \psi'_{A\setminus e})$ from π and then (ψ_e, ψ'_e) from a measurably chosen optimal coupling of $\gamma_e(\cdot \mid \psi, U)$ and $\gamma_e(\cdot \mid \psi', U')$ [58, Cor. 5.22] gives another coupling of the two laws, which differs from π only in the e-th coordinates; by minimality of π and the triangle inequality for W_1, changing the partners of e one at a time,

$$\varrho(e) \leq c \sum_{k \in A \text{ a partner of } e} \varrho(k) + c\,[i \text{ a partner of } e]\,\delta(U_i, U'_i).$$

Every link has eighteen partners, so summing over $e \in A$ gives $(1-18c)\sum_e \varrho(e) \leq c\,b\,\delta(U_i, U'_i)$, and $W_1^{d_A} \leq \int d_A \, d\pi = \sum_e \varrho(e)$. □

Proposition 11.27 (the layer bound at strong coupling)**.** *Let $C_{ek} \leq c$ for all links with $18c < 1$, and $\theta = 6c/(1-18c)$. For every box $Q \subset \mathbb{Z}^4$, every coordinate, every k, $r \geq 1$ and every exterior configuration ω,*

$$\rho^\omega(Y_k, Y_{k+1}) \leq \theta, \qquad \rho^\omega(Y_k, Y_{k+r}) \leq \theta^r,$$

so $(\mathrm{M}_{d,m}(\theta^d))$ holds for all $d \geq 2$, $m \geq 4$ and $(\mathrm{M}^*_w(\theta^{w+1}))$ for all $w \geq 1$. By Lemma 8.6, for $G = \mathrm{SU}(2)$ and the Wilson weight $c = |\beta_W|/4$ and $\theta = 3|\beta_W|/(2-9|\beta_W|)$, which is less than 1 for $|\beta_W| < \frac{1}{6}$; for G with semisimple Lie algebra and $30N|\beta| < \kappa_G$, $\theta = 6N|\beta|/(\kappa_G - 24N|\beta|)$.

Proof. Let $Y = Y_k$, $Y' = Y_{k+1}$ be adjacent layers in the coordinate i and $A = V(Q) \setminus Y_{k+1}$. A link (y, i) of Y_{k+1} lies in two plaquettes of each plane $\{i, \mu\}$, with base points y and $y - e_\mu$, whose links are (y, i), $(y + e_i, \mu)$, $(y + e_\mu, i)$, (y, μ) and $(y - e_\mu, i)$, $(y - e_\mu + e_i, \mu)$, (y, i), $(y - e_\mu, \mu)$; exactly two per plane are based in the layer $k+2$ and the others in Y_{k+1}, so (y, i) has at most six partners in A. A link (y, μ) with $\mu \neq i$ has all its partners in Y_{k+1} except in the plane $\{\mu, i\}$, where the plaquette at y contributes $(y + e_i, \mu)$ and the plaquette at $y - e_i$ contributes $(y - e_i, \mu)$, $(y - e_i + e_\mu, i)$, $(y - e_i, i)$: four partners in A. So $b \leq 6$ in Lemma 11.26, and symmetrically for $A' = V(Q) \setminus Y_k$ and the links of Y_k. Lemma 11.26 gives the hypothesis of Lemma 11.25 with $R = \theta$, and Lemma 11.25 the bound for adjacent layers; Lemma 11.12 gives the power and the two hypotheses. The single-link constant C_{ek} is the constant C^T_{ek} of Section 8 without gauge fixing, and the proof of Lemma 8.6 bounds it as stated, since it concerns the law of one link given its partners whatever the other links are. □

Theorem D then applies with $c = \theta^d < \frac{1}{2}$ as soon as $d > \log 2/\log(1/\theta)$, with m unrestricted. For $G = \mathrm{SU}(2)$ along any tuning with $\sup_N |\beta_{W,N}| \leq b < \frac{1}{6}$, the bound holds with $\theta_b = 3b/(2-9b)$ uniformly in N and in ω, twisted boundary conditions included: at $b = \frac{1}{10}$, $\theta_b = 0.2727\ldots$, $d = 2$ gives $c = \theta_b^2 = 0.07438\ldots$, $(1-2c)^4 = 0.5250\ldots$, $m = 283$ and $\mathrm{BP}(3, 0.0164)$, while at $b = 0.15$, $\theta_b = 0.6923\ldots$ and the same c needs $d \geq 8$. Nothing

physical is gained: at a fixed coupling the lattice gap is of order one in lattice units, and the range $|\beta_W| < \frac{1}{6}$ is below that of Theorem C. What the proposition shows is the only mechanism known to us that produces a bound uniform in the boundary condition, a Dobrushin contraction in lattice units, and the exact form in which the hypothesis would follow at weak coupling: by Lemma 11.25 applied to $Y = V(Q_-)$, $Y' = V(Q_+)$, with the links grouped into cells of lattice side of the order of d and δ replaced by any metric on the configurations of a cell inducing the product topology, for which the proof is unchanged, $(\mathrm{M}_{d,m}(\mathfrak{r}))$ follows from the response bound

(11.3)
$$W_1\Big(\nu_Q^\omega(\sigma_{V(Q_-\cup Q_0)} \in \cdot \mid \sigma_{V(Q_+)} = U),\, \nu_Q^\omega(\sigma_{V(Q_-\cup Q_0)} \in \cdot \mid \sigma_{V(Q_+)} = U')\Big) \le \mathfrak{r}\,\delta_B(U_B, U'_B)$$

for configurations U, U' of $V(Q_+)$ differing only on one cell B, and symmetrically, for every fat box and every ω: a Dobrushin condition for the degrees of freedom at the scale $\ell/2$ in every box of physical size at most $m\ell/2$, uniformly along the tuning. In lattice units the left side of (11.3) summed over the links of a box of side W is, for a nearly free field, of the order of W^2, so the condition can hold only after the blocking; whether it holds at the scale $\ell/2$ is the hypothesis in another form, and no result we know of bears on it. The expansions of Bałaban [3, 4, 5] and of Magnen, Rivasseau and Sénéor [43] control effective actions and the free energy in finite periodic volume while the running coupling is small, not Dirichlet boxes of fixed physical size at a scale where it is of order one, and the decay of correlations of Adhikari and Cao [1] is at a fixed lattice coupling for finite groups.

Two dimensions. In two dimensions the seam between the outer parts of a box is one-dimensional, and after an axial gauge the middle slab integrates out exactly to a convolution power of the weight. This proves the two-dimensional form of the hypothesis along the weak-coupling tuning, with $c \to 0$ as the physical size of the box grows, uniformly in the boundary condition (Theorem H.8 in Appendix H). In four dimensions the seam carries a three-dimensional layer of links, and $(\mathrm{M}_{d,m}(c))$ is a bound, uniform in the boundary condition, on the second singular value of the $(d-1)$-step transfer operator of the box on gauge-invariant functions. Boundary conditions with values in the centre, 't Hooft's twisted ones [57], leave a global conjugation symmetry of the box, and in the two-dimensional strip such a symmetry is broken at weak coupling (Proposition H.17); in four dimensions the hypothesis asks that the twisted sectors be disordered at the scale $\ell/2$, which is the form that confinement takes here. We know no boundary condition that refutes $(\mathrm{M}_{d,m}(c))$ along the four-dimensional tuning, and no proof of it.

The slab correlation as an operator norm. The number ρ^ω of Definition 11.2 is not a two-point function: it is the norm of a conditional expectation between the gauge-invariant observables of the two outer parts, equivalently the second singular value of the density of their joint law.

Proposition 11.28 (the slab correlation as an operator norm)**.** *Let Q, the splitting and ω be as in Definition 11.2, $\nu = \nu_Q^\omega$, let $\mathcal{H}_\pm \subset \mathcal{A}_Q^\omega$ be the closed subspaces of the centred elements measurable with respect to $\sigma_{V(Q_\pm)}$, and let $E_\pm$ be the conditional expectation given $\sigma_{V(Q_\pm)}$, the projection $E_{A_\mp}$ of Section 11.7.*

(i) $\rho^\omega(Q_-, Q_+) = \|E_+|_{\mathcal{H}_-}\| = \|E_-|_{\mathcal{H}_+}\| = \|E_-E_+|_{\mathcal{H}_-}\|^{1/2}$.

(ii) *Let k^ω be the density of the joint law of $(\sigma_{V(Q_-)}, \sigma_{V(Q_+)})$ under ν with respect to the product $\nu_- \otimes \nu_+$ of its two marginals. Then k^ω is continuous and bounded, $E_+X(\sigma_+) = \int X(\sigma_-)k^\omega(\sigma_-,\sigma_+)\,d\nu_-(\sigma_-)$ for $X \in \mathcal{H}_-$, and*

$$\rho^\omega(Q_-, Q_+)^2 \le \iint (k^\omega)^2\,d\nu_-\,d\nu_+ - 1.$$

Proof. (i) Let $X \in \mathcal{H}_-$ and $Y \in \mathcal{H}_+$ be unit vectors. Then $\nu(XY) = \nu((E_+X)Y) \le \|E_+X\|$ by the Cauchy–Schwarz inequality, with equality at $Y = E_+X/\|E_+X\|$ when $E_+X \neq 0$; this Y is admissible, being centred, $\sigma_{V(Q_+)}$-measurable and in $\mathcal{A}_Q^\omega$ by Lemma 11.1(iii). So

the supremum of Definition 11.2 is $\sup_X \|E_+X\| = \|E_+|_{\mathcal{H}_-}\|$, and the symmetry of $\nu(XY)$ gives the second equality. For the third, E_+ maps $\mathcal{H}_-$ into $\mathcal{H}_+$ and E_- maps $\mathcal{H}_+$ into $\mathcal{H}_-$, and for $X, X' \in \mathcal{H}_-$, $\langle X, E_-E_+X'\rangle = \langle E_+X, E_+X'\rangle$; so $E_-E_+|_{\mathcal{H}_-} = (E_+|_{\mathcal{H}_-})^*(E_+|_{\mathcal{H}_-})$ is a nonnegative self-adjoint operator on $\mathcal{H}_-$ whose norm is $\|E_+|_{\mathcal{H}_-}\|^2$.

(ii) The measure ν has a continuous density with respect to Haar measure on $G^{V(Q)}$, bounded above and below, since φ is positive and continuous on the compact group G. Integrating out the links of $V(Q_0)$ gives a density $j(\sigma_-, \sigma_+)$ of the joint law with respect to $\lambda_- \otimes \lambda_+$, continuous and bounded above and below, and the marginals have the densities $j_\pm = \int j\, d\lambda_\mp$ with the same properties; so $k^\omega = j/(j_-j_+)$ is continuous and bounded, and $E_+X(\sigma_+) = \int X(\sigma_-) j(\sigma_-, \sigma_+)\, d\lambda_-(\sigma_-)/j_+(\sigma_+) = \int X k^\omega\, d\nu_-$. Let $\mathcal{K}$ be the integral operator with the kernel k^ω from $L^2(\nu_-)$ to $L^2(\nu_+)$; it is Hilbert–Schmidt, with $\|\mathcal{K}\|_{HS}^2 = \iint (k^\omega)^2 d\nu_- d\nu_+$, and $\mathcal{K}\mathbf{1} = \mathbf{1}$ because $\int k^\omega d\nu_- = 1$. Choosing an orthonormal basis of $L^2(\nu_-)$ that contains $\mathbf{1}$, the restriction of $\mathcal{K}$ to the orthogonal complement of the constants has squared Hilbert–Schmidt norm $\|\mathcal{K}\|_{HS}^2 - 1$; as $\mathcal{H}_-$ lies in that complement and $E_+|_{\mathcal{H}_-}$ is the restriction of $\mathcal{K}$, the operator norm, which the Hilbert–Schmidt norm dominates, gives the bound. □

So ρ^ω is the second singular value of the density k^ω of the two slabs, and by Lemma 11.12 the slabs may be replaced by their two inner layers. It is determined by second moments for a Gaussian law only, and the two-point functions of the links bound it in no general way: for X uniform on $\{-1, 0, 1\}$ and $Y = X^2 - \frac{2}{3}$ the covariance vanishes while the maximal correlation is 1. The Hilbert–Schmidt bound of (ii) is what a convergent expansion of $k^\omega - 1$ would give, and it is not uniform in the transverse size of the box: if ν were a product of n independent pairs, each with $\iint k_j^2 - 1 = h_j$, the right side would be $\prod_j (1 + h_j) - 1 \geq n \min_j h_j$ while ρ^ω would be the largest of the n maximal correlations. A bound on ρ^ω uniform in the transverse widths must therefore come from the operator norm itself, as in Proposition 11.27, and not from the size of the density.

An area law does not suffice. The next proposition shows that the Wilson loops of the defining representation carry no information about the observables that (D) constrains.

Proposition 11.29 (central extensions)**.** *Let $\tilde{G} \subset \mathrm{U}(\tilde{N})$ and $G \subset \mathrm{U}(N)$ be compact, $\pi : \tilde{G} \to G$ a continuous surjective homomorphism with finite central kernel Z, φ a positive continuous class function on G and $\tilde{\varphi} = \varphi \circ \pi$. On Λ_L let $\tilde{\mu}$, μ be the measures with weights $\tilde{\varphi}$, φ, and $\Pi : \tilde{G}^E \to G^E$ the map π applied to every link.*

(i) *$\tilde{\varphi}$ is a positive continuous class function and $\Pi_* \tilde{\mu} = \mu$.*

(ii) *For every x and $F \in L^2(\mu)$, $\tilde{\mathcal{P}}_x(F \circ \Pi) = (\mathcal{P}_x F) \circ \Pi$ and $\mathrm{Var}_{\tilde{\mu}}(F \circ \Pi) = \mathrm{Var}_\mu(F)$; if F is gauge invariant, so is $F \circ \Pi$. Hence $\mathrm{BP}(n, \varpi)$ for $\tilde{\mu}$ implies $\mathrm{BP}(n, \varpi)$ for μ.*

(iii) *If some $z \in Z$ acts on $\mathbb{C}^{\tilde{N}}$ as a scalar $\zeta \neq 1$, then $\tilde{\mu}(\mathrm{Tr}\, \tilde{U}_C) = 0$ for every closed lattice path C that traverses some link exactly once.*

Proof. (i) $\pi(hgh^{-1}) = \pi(h)\pi(g)\pi(h)^{-1}$, so $\tilde{\varphi}$ is a class function. The image under π of Haar measure on $\tilde{G}$ is a left-invariant probability measure on G, hence Haar measure, so $\Pi_* \tilde{\lambda} = \lambda$. As π is a homomorphism, $\pi(\tilde{U}_p) = (\Pi \tilde{U})_p$, so $\tilde{\mu}$ has density $\Psi \circ \Pi$ with respect to $\tilde{\lambda}$, with $\Psi(U) = \tilde{Z}^{-1} \prod_p \varphi(U_p)$, and $\Pi_*((\Psi \circ \Pi)\tilde{\lambda}) = \Psi\lambda$; both being probability measures, $\Psi\lambda = \mu$.

(ii) $\tilde{\mathcal{P}}_x(F \circ \Pi)(\tilde{U})$ is the integral of $F(\Pi\tilde{\sigma}, \Pi\tilde{U}_{\mathrm{out}})$ against the density proportional to $\prod_{p \cap V_x \neq \emptyset} \varphi((\Pi\tilde{\sigma}, \Pi\tilde{U}_{\mathrm{out}})_p)$ on $\tilde{G}^{V_x}$; by the change of variables $\sigma = \Pi\tilde{\sigma}$ and (i) on V_x it equals $(\mathcal{P}_x F)(\Pi\tilde{U})$. The variance identity follows from (i), and $\Pi(\tilde{g}\tilde{U}) = \pi(\tilde{g})\Pi(\tilde{U})$ for a gauge transformation $\tilde{g}$ gives the invariance. Then for gauge-invariant F, $\varpi\, \mathrm{Var}_\mu(F) = \varpi\, \mathrm{Var}_{\tilde{\mu}}(F \circ \Pi) \leq n^{-4} \sum_x \tilde{\mu}((F \circ \Pi - \tilde{\mathcal{P}}_x(F \circ \Pi))^2) = n^{-4} \sum_x \mu((F - \mathcal{P}_x F)^2)$.

(iii) Let e be a link traversed once by C and T the map $\tilde{U}_e \mapsto z\tilde{U}_e$, the other links being fixed. It preserves $\tilde{\lambda}$, and as z is central with $\pi(z) = 1$, each plaquette variable through e becomes $z^{\pm 1}\tilde{U}_p$ with $\tilde{\varphi}(z^{\pm 1}\tilde{U}_p) = \tilde{\varphi}(\tilde{U}_p)$; so $\tilde{\mu}$ is T-invariant. Moving the central factor out of the product, $\mathrm{Tr}\, \tilde{U}_C \circ T = \zeta^{\pm 1}\, \mathrm{Tr}\, \tilde{U}_C$, so $\tilde{\mu}(\mathrm{Tr}\, \tilde{U}_C) = \zeta^{\pm 1}\tilde{\mu}(\mathrm{Tr}\, \tilde{U}_C) = 0$. □

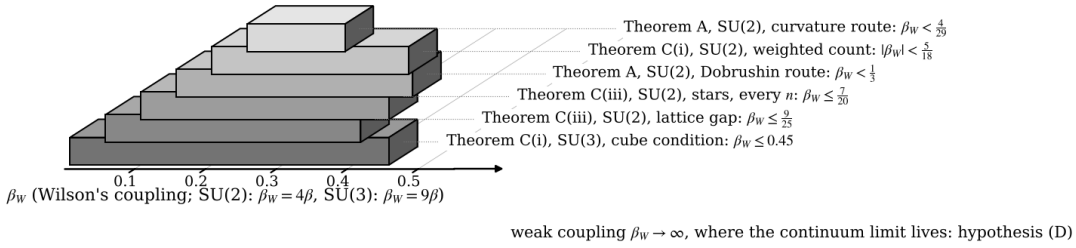


FIGURE 11. The ranges of Wilson's coupling β_W ($\beta_W = 4\beta$ for SU(2), 9β for SU(3)) covered by the theorems of this paper, from the curvature route of Theorem A to the cube condition of Theorem C(i) for SU(3); the continuum limit lives at $\beta_W \to \infty$, where (D) is a hypothesis.

For instance, with $\tilde{G} = G = \mathrm{U}(1)$, $\pi(z) = z^2$ and the Wilson weight φ of U(1), every Wilson loop of charge one under $\tilde{\varphi}(z) = e^{\beta \operatorname{Re} z^2}$ that traverses some link once vanishes for every β, so Wilson's area law holds with every string tension, while by part (ii) the cube condition for $\tilde{\varphi}$ is at least as hard as for the U(1) theory, in which it fails at weak coupling in the Coulomb phase (Section 12). The vanishing is kinematic, a consequence of a local symmetry and not of confinement; what the example shows is that an argument deriving (D) from an area law of the defining representation cannot be valid for all compact groups and weights. For SU(N) with the Wilson action no such example is known, and nothing is claimed about whether an area law and (D) hold together there.

12. Weak coupling: what a proof of the mass gap still needs

The continuum limit of four-dimensional Yang–Mills theory is taken along the tuning dictated by asymptotic freedom: for SU(N) the bare coupling $g^2 = 2N/\beta_W$ tends to zero like $1/(2b_0 \log(1/(a\Lambda)))$ with $b_0 = 11N/(48\pi^2)$, the first coefficient of the β-function [27, 49] and a fixed physical scale Λ, so $\beta_W \to \infty$ logarithmically in $1/a$. Theorems A and 8.8 hold for $\beta_W < N^2/29$ and $\beta_W < N^2/42$ respectively; at fixed β the lattice gap in physical units, $m_G(\beta)/a$, diverges as $a \to 0$, and no continuum limit exists at fixed β. The results of this paper therefore say nothing about the continuum theory without statement (D). Three facts show what (D) asks for.

(a) A cube condition is a mass. By Theorem 10.2 the best constant for a free field is at most $s^2/(s^2 + 16d/9)$ with $s = m\ell$. At weak coupling the leading approximation of the Wilson measure, in any gauge, is a massless Gaussian field, for which the constant vanishes as $a \to 0$ at fixed ℓ. Hence (D) does not follow from perturbation theory to any finite order: it holds only if the interaction generates a mass of order $1/\ell$, and it is a quantitative form of that mass generation.

(b) The cube must grow. Proposition H.24 shows that a spectral gap for blocks of a fixed lattice size, with a constant C_k such that $C_k a_k \to 0$, forces every continuum limit to be trivial; so a cube condition with n bounded along the tuning is incompatible with (E), and (D) can hold only with $n_k = \ell/a_k$ growing like a power of $1/a_k$ while β_k grows like $\log(1/a_k)$. Every argument at fixed cube size is excluded.

(c) The group must be non-abelian. For $G = \mathrm{U}(1)$ with the Villain action at large β the theory is in a massless phase [22]: the gauge-invariant two-point function of the field strength decays only like a power of the distance. By Theorem 6.3 a cube condition at any fixed cube size would make that correlation decay exponentially, so the cube condition

fails at every cube size in the Coulomb phase. Every proof of (D) must therefore use the non-abelian structure of G, exactly as a proof of the mass gap must.

The two ends. Bałaban's renormalisation group [3, 4, 5] controls the effective actions of the lattice theory from the scale a up to scales at which the running coupling is still small, uniformly in the lattice spacing, in finite volume. The theorems of Secs. 5–8 control the theory when the coupling is large: they need $N\beta$ small, that is, g large. The running coupling grows towards the infrared, so a proof of (D) would follow the renormalisation group from the weak-coupling regime, where Bałaban's estimates hold, through the regime of coupling of order one, to a scale ℓ at which the effective theory satisfies a strong-coupling criterion of the kind proved here, and would then transport the resulting cube estimate back to the fine lattice. Neither the middle step nor the final transport is available: the effective actions are not of Wilson form, Bałaban's analysis stops before the coupling is of order one, and no known criterion accepts the effective theory at that scale. This gap between the two ends is the Yang–Mills mass gap problem in the form isolated here.

Statement (E) is the construction problem. Bałaban's ultraviolet stability bounds give compactness of suitably smeared observables in finite volume, but the infinite-volume limit, Euclidean invariance and nontriviality of the limit are open [31]. Given (E), Theorem B reduces the mass gap to (D) with the explicit constant $\Delta \geq 16\varpi/(625e^2\ell)$.

13. Conclusion

Lattice Yang–Mills theory in four dimensions has a mass gap with the explicit lower bound $m_G(\beta)$ for every compact simple gauge group in the range $N\beta < \frac{2}{29}\kappa_G$, and for SU(2) up to $\beta_W = \frac{1}{3}$ with the bound $-\log(3\beta_W)$ and up to $\beta_W = \frac{9}{25}$ by resampling stars. A single spectral gap ϖ of the dynamics that resamples cubes of a fixed physical size ℓ, uniform along the continuum tuning, gives every nontrivial continuum limit the mass gap $16\varpi/(625e^2\ell)$. That spectral gap holds at strong coupling, for SU(2) up to $\beta_W = \frac{7}{20}$ uniformly in the cube size and for SU(3) up to $\beta_W = 0.45$, and for massive free fields; it fails for massless fields, at fixed cube size at weak coupling, and in the abelian Coulomb phase.

For a continuum theory built in the other order, by scaling limits on tori of fixed physical side, the same cube condition on cylinders, together with the independence of the volume limit from the order of its two limits, gives the gap $16\varpi/(625e^2\ell)$, a unique vacuum, and a single phase (Theorem E); the proof passes through the transfer matrix of the cylinder and the Euclidean vectors of the Wightman theory, and uses nothing beyond the three hypotheses. The independence of the order of the limits follows from a single renormalisation-free statement, that the partition function of the symmetric torus relative to the vacuum of the cylinder tends to one (Corollary 7.11), and the cube condition on cylinders follows from the slab decorrelation of Theorem D, which holds on cylinders (Proposition 7.13). With a construction of the torus theories taken as the existence input, the statement of the Clay problem for those theories reduces to these two conditions; neither is proved here.

All of these conditions follow from one. If, at a single time extent of fixed physical size, the thermal part of the free energy of the tori whose sides are between two and forty-eight times that extent is at most a small explicit multiple of their volume, then the ascent of Theorem 7.56 gives the gap on every larger torus and the vacuum dominance (Z), and Theorem F gives the Wightman theory the mass gap $0.172/t_0$ and a unique vacuum. The statement of the Clay problem for the theories of (W) is thereby reduced to (W) itself and to this one bound, which is an inequality between two partition functions. The bound contains the lattice gap at weak coupling and is open; and the constructions that would avoid it, by taking the transporters without renormalisation, by lifting the two-dimensional theory, or by keeping the lattice, give trivial theories or violate the axioms (Appendix I).

The cube gap follows from a decay of correlations, below the sharp threshold $\frac{1}{2}$, between the gauge-invariant observables on the two sides of a slab of physical width $\ell/2$, uniformly in the boundary condition. This hypothesis is a form of mass generation at one scale: it is

a bound on the maximal correlation of two layers of a box, a quantity which in the vacuum of a cylinder equals exactly $e^{-m_{\mathrm{lat}}s}$ at distance s (Proposition 11.13), so that its whole content is the uniformity in the boundary condition. It holds in two dimensions along the weak-coupling tuning and fails there for strips one link wide; in four dimensions it is not proved. Curvature arguments, with or without gauge fixing, stop at explicit couplings of order one; spectral gaps for blocks of fixed lattice size force a trivial continuum limit; and a lattice gap tending to zero at the right rate produces a gap in a continuum limit that is not identified with Yang–Mills theory. Statements (D) and (E) remain open, as do (U), (Z) and (D_ℓ) for the theories built on tori, and with them the Clay problem; (Z) holds at strong coupling and in two dimensions and follows from an entropy bound on the cylinder, and no correlation, gap, slab or cube hypothesis reaches it (Section 7.7, Proposition H.11), and it reduces exactly to an entropy bound at one physical scale together with a Casimir-energy condition on the halving of the spatial torus (Proposition 7.30), both of which are doubling defects of the free energies of finite tori, out of reach of the gap and of the free energy per site, and the second of which follows from a finite-size bound of massive type (Propositions 7.34, 7.35 and 7.36); given the gap on the tori of the halving chain, the Casimir condition reduces to the physical Casimir energy lying below the gap, and (C) to the exclusion of a linearly growing branch of thermal corrections (Theorem 7.39, Proposition 7.42), a branch that positivity and the gap exclude once the one-scale bound is taken on elongated boxes, so that (Z) then follows with no hypothesis at a second scale (Theorem 7.54, Corollary 7.55), and the free field shows that (C) is a statement about tori of about eighteen correlation lengths, violated by the free gluons at every scale (Corollary 10.7, Proposition 10.8); while the slab hypothesis is a block Poincaré inequality uniform in the boundary condition and the norm of a conditional expectation between the two sides (Section 11.7, Proposition 11.28), and it is not a consequence of a gap of the transfer matrix (Propositions 11.17 and 11.18), while the patch inequality that implies it is exactly the local Knabe inequality on the torus (Proposition G.5); what this paper contributes is an explicit strong-coupling theory for every gauge group, a reduction of the continuum gap to one estimate on one scale and of that estimate to a decay of correlations across one slab, and a map of what the present methods can reach.

Appendix A. The line field

Table 2 lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the values $\pi_x(\mu) = \sigma(x + e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions.

Table 2. The line field: x modulo two (digits $x_0x_1x_2x_3$), $\sigma(x)$, and $\sigma(x+e_\mu)$ for $\mu = 0, 1, 2, 3$.

x	σ	0	1	2	3	x	σ	0	1	2	3
0000	0	0	2	3	1	0001	1	2	1	3	0
1000	0	0	3	1	2	1001	2	1	3	2	0
0100	2	3	0	2	1	0101	1	3	1	0	2
1100	3	2	0	1	3	1101	3	1	2	0	3
0010	3	1	2	0	3	0011	3	2	0	1	3
1010	1	3	1	0	2	1011	2	3	0	2	1
0110	2	1	3	2	0	0111	0	0	3	1	2
1110	1	2	1	3	0	1111	0	0	2	3	1

Appendix B. The blocks of Lemma 5.2

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part $f_p = -\cos\phi\,(y_1+y_2-y_3-y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathcal{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1+y_2-y_3-y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4),$$

with $y_k = 0$, $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k = 1$ and $\cos\phi = 1$ the matrix in (y_2, y_3, y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

with leading principal minors $\frac{31}{10}$, $\frac{673}{200}$, $\frac{8421}{2000}$. In general the leading principal minors are $\frac{31}{10}, \frac{673}{200}, \frac{8421}{2000}$ (fixed link e_1 or e_4), $\frac{31}{10}, \frac{861}{100}, \frac{8421}{2000}$ (e_2) and $\frac{9}{4}, \frac{673}{200}, \frac{8421}{2000}$ (e_3) at $\cos\phi = 1$, and $\frac{11}{10}, \frac{53}{200}, \frac{21}{2000}$ (e_1, e_4), $\frac{11}{10}, \frac{21}{100}, \frac{21}{2000}$ (e_2) and $\frac{1}{4}, \frac{53}{200}, \frac{21}{2000}$ (e_3) at $\cos\phi = -1$; all are positive.

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m,\vartheta)$ of $\mathcal{F}$, $m = \cos\chi$, is affine in m with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m = 1$ suffices. For the fixed link e_2, in the variables (z_1, z_3, z_4) and with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0,1]$ its leading principal minors are

$$\tfrac{20u+11}{5}, \qquad \tfrac{(20u+1)(20u+21)}{25}, \qquad \tfrac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

Appendix C. The single-link law of SU(2)

This appendix proves the sharp quarter, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ for the one-link law of SU(2) in the chordal metric, which the proof of Theorem A for SU(2), Lemma 8.6(i) and the whole of Sec. 9 use, together with the exact value $\frac{1}{4}$ at $b = 0$, which shows that the single-link argument of Sec. 4 stops at $\beta_W = \frac{1}{3}$. It is given in full, so that the paper is self-contained. The computations it calls on are in Appendix D, and the facts about Bessel functions in Appendix E.

C.1. **The von Mises–Fisher law.** Identify SU(2) with the unit quaternions, that is with the unit sphere $S^3 \subset \mathbb{R}^4$, so that $\operatorname{Re}\operatorname{Tr} U = 2x_0(U)$ and $\operatorname{Re}\operatorname{Tr}(UV) = 2\langle u, \bar{v}\rangle$ for the Euclidean inner product of $\mathbb{R}^4$ and quaternion conjugation $v \mapsto \bar{v}$. Fix a link e and the field off e. Every plaquette through e can be rotated cyclically so that its trace reads $\operatorname{Re}\operatorname{Tr}(Q_e A_p)$ with $A_p \in \mathrm{SU}(2)$, the staple of e in p. The action therefore depends on Q_e only through

$$N\beta \sum_{p \ni e} \operatorname{Re}\operatorname{Tr}(Q_e A_p) = \langle x, b\rangle, \qquad b = 4\beta \sum_{p\ni e} \bar{A}_p \in \mathbb{R}^4,$$

with $x = Q_e \in S^3$. Since there are six staples, each of norm one,

$$\kappa := |b| \le y := 24|\beta|. \tag{C.1}$$

The conditional law of Q_e is the von Mises–Fisher law[22]

$$\nu_b(dx) = Z(\kappa)^{-1} e^{\langle x,b\rangle}\,\sigma(dx) \tag{C.2}$$

on the unit sphere, with σ the normalised surface measure. We use coordinates in which $b = \kappa e_1$. The density of $t = x_1$ is proportional to $(1-t^2)^{1/2}e^{\kappa t}$ on $[-1,1]$, so $Z(\kappa) = 2I_1(\kappa)/\kappa$ with the modified Bessel function I_1, normalised so that $Z(0) = 1$.

Lemma C.1. *For every $\kappa \geq 0$ the law ν_b on the unit sphere S^3 satisfies the Poincaré inequality $\mathrm{Var}_{\nu_b}(g) \leq \frac{1}{3}\int |\nabla g|^2 d\nu_b$, and the constant $\frac{1}{3}$ is attained at $\kappa = 0$.*

Proof. In the polar angle $\theta \in (0,\pi)$ from e_1 the generator of the reversible diffusion of ν_b is $L = \partial_\theta^2 + b(\theta)\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$ with $b(\theta) = 2\cot\theta - \kappa\sin\theta$. On spherical harmonics of degree ℓ on S^2 it acts as a Sturm–Liouville operator L_ℓ with potential $\ell(\ell+1)\sin^{-2}\theta$. A ground state comparison of Barta's type [6] with the trial function $\psi = \sin\theta\, e^{a\kappa\cos\theta}$ gives $-L_1\psi \geq 3\psi$ for $a = -\frac{1}{5}$, and the same trial function with $a = -\frac{2}{5}$ gives the bound 3 for the derivatives of zonal eigenfunctions; Appendix D carries out both computations. Hence every nonzero eigenvalue of $-L$ is at least 3. At $\kappa = 0$ the first eigenvalue of the Laplacian of the unit S^3 is 3. □

C.2. **Moments.** Let $A(\kappa) = \mathbb{E}_{\nu_b} x_1 = Z'(\kappa)/Z(\kappa) = I_2(\kappa)/I_1(\kappa)$.

Lemma C.2. *For every $\kappa > 0$, in coordinates with $b = \kappa e_1$,*

(i) $\mathbb{E}x_2^2 = \mathbb{E}x_3^2 = \mathbb{E}x_4^2 = A/\kappa$, $\mathbb{E}(1-x_1^2) = 3A/\kappa$, *the covariance matrix of x is* $\mathrm{diag}(A', A/\kappa, A/\kappa, A/\kappa)$, *and* $A' = 1 - A^2 - 3A/\kappa$;

(ii) $\dfrac{\kappa}{2+\sqrt{\kappa^2+4}} \leq A(\kappa) \leq \dfrac{\kappa}{1+\sqrt{\kappa^2+9}}$;

(iii) $A' \leq A/\kappa \leq \big(1+\sqrt{\kappa^2+9}\big)^{-1} \leq \frac{1}{4}$;

(iv) $A(\kappa) \geq \kappa/\big(\frac{3}{2} + \sqrt{\kappa^2 + 25/4}\big)$.

Proof. (i) Integrating by parts against $\frac{d}{dt}(1-t^2)^{3/2} = -3t(1-t^2)^{1/2}$ gives $\kappa\,\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, and the three coordinates x_2, x_3, x_4 are exchangeable under ν_b, with $\sum_i x_i^2 = 1$. The off-diagonal covariances vanish by the reflections $x_i \mapsto -x_i$, $i \geq 2$, and $A' = \mathrm{Var}\,x_1$ because $A = (\log Z)'$. (ii) By (i), A solves the Riccati equation $A' = 1 - A^2 - 3A/\kappa$ with $A(\kappa) = \kappa/4 + O(\kappa^3)$. Appendix D shows that the left side of (ii) is a subsolution and the right side a supersolution of this equation, with the same behaviour at 0, and concludes by comparison. (iii) The middle inequality is (ii). For the first, put $s = \sqrt{\kappa^2+4}$. By (ii), $4A/\kappa \geq 4/(2+s)$ and $1 - A^2 \leq 1 - \kappa^2/(2+s)^2$, and

$$\begin{aligned} 1 - \frac{\kappa^2}{(2+s)^2} \leq \frac{4}{2+s} &\iff (2+s)^2 - \kappa^2 \leq 4(2+s) \\ &\iff s^2 - \kappa^2 \leq 4, \end{aligned}$$

which holds with equality. So $A' = 1 - A^2 - 3A/\kappa \leq A/\kappa$. (iv) is the comparison of (ii) with a third barrier, which improves the lower bound of (ii); it is carried out in the same appendix. □

[22] On the sphere $S^{p-1} \subset \mathbb{R}^p$ the von Mises–Fisher law with mean direction μ and concentration κ has density proportional to $e^{\kappa\langle x,\mu\rangle}$ with respect to the surface measure, and its normalising constant is $(2\pi)^{p/2} I_{p/2-1}(\kappa)/\kappa^{p/2-1}$; it is the maximum entropy law with prescribed mean resultant, and it plays on the sphere the role of the Gaussian in $\mathbb{R}^p$. For $p = 4$ the Bessel index is 1. The mean resultant length $\mathbb{E}\langle x,\mu\rangle = I_{p/2}(\kappa)/I_{p/2-1}(\kappa)$ is the function A of Lemma C.2. See Mardia and Jupp [42], Ch. 9, for these facts and for the tangent-normal decomposition used in Appendix D.

C.3. **The Lipschitz rate of the family.** Let δ be the chordal metric $\delta(x,x')=|x-x'|$ on S^3 and W_1 the Kantorovich distance for δ. For $c\in\mathbb{R}^4$ and a unit vector $h\in\mathbb{R}^4$ put

$$L(c,h)=\sup_{g}\operatorname{Cov}_{\nu_c}\bigl(g,\langle h,x\rangle\bigr), \tag{C.3}$$

the supremum over 1-Lipschitz functions g for δ. Since $\frac{d}{dt}\nu_{c+th}(g)=\operatorname{Cov}_{\nu_{c+th}}(g,\langle h,x\rangle)$, the Kantorovich–Rubinstein duality gives, for all b,b',

$$W_1(\nu_b,\nu_{b'})\le|b-b'|\sup_{c\in[b,b']}L\Bigl(c,\frac{b'-b}{|b'-b|}\Bigr). \tag{C.4}$$

By rotation invariance $L(c,h)$ depends only on $\kappa=|c|$ and on the angle φ between h and $c^{\perp}$, and we write $L(\kappa,\varphi)$.

Lemma C.3. *For all c and h, $L(c,h)\le 1/(2\sqrt{3})$. Moreover:*

(i) *at $c=0$, $L(0,h)=\frac14$ for every h;*
(ii) *if $h\perp c$, then $L(c,h)=\mathbb{E}_{\nu_c}x_2^2=A(\kappa)/\kappa\le\frac14$;*
(iii) *if $\kappa\ge\sqrt{88}/3\approx 3.127$, then $L(c,h)\le\frac14$ for every h.*

Proof. A 1-Lipschitz function for δ is 1-Lipschitz for the geodesic distance, which is larger, so $|\nabla g|\le 1$ almost everywhere, and Lemma C.1 gives $\operatorname{Var} g\le\frac13$. By Lemma C.2(i), (iii) the largest eigenvalue of the covariance of x is $A/\kappa\le\frac14$, so $\operatorname{Var}\langle h,x\rangle\le A/\kappa$. The Cauchy–Schwarz inequality gives

$$L(c,h)\le\bigl(\tfrac13 A/\kappa\bigr)^{1/2}\le\Bigl(3\bigl(1+\sqrt{\kappa^2+9}\bigr)\Bigr)^{-1/2}\le\frac{1}{2\sqrt{3}}. \tag{C.5}$$

(i) and (ii) are reflection arguments. Let R be the reflection in the hyperplane $h^{\perp}$, and suppose ν_c is R-invariant, which is the case when $c=0$ or $h\perp c$. Then $\mathbb{E}\langle h,x\rangle=0$ and

$$\begin{aligned}\operatorname{Cov}(g,\langle h,x\rangle)&=\mathbb{E}\bigl[(g(x)-g(Rx))\langle h,x\rangle;\ \langle h,x\rangle>0\bigr]\\&\le\mathbb{E}\bigl[2\langle h,x\rangle^2;\ \langle h,x\rangle>0\bigr]=\mathbb{E}\langle h,x\rangle^2,\end{aligned}$$

since $\delta(x,Rx)=2|\langle h,x\rangle|$. Equality holds for $g=\langle h,x\rangle$, which is 1-Lipschitz. At $c=0$, $\mathbb{E}\langle h,x\rangle^2=\frac14$; for $h\perp c$, $\mathbb{E}\langle h,x\rangle^2=A/\kappa$. (iii) By (C.5), $L\le\frac14$ as soon as $1+\sqrt{\kappa^2+9}\ge\frac{16}{3}$, that is $\kappa^2\ge\frac{88}{9}$. □

The reflection settles the directions $h\perp c$ and the point $c=0$; for an oblique h the reflection no longer preserves ν_c, and the Cauchy–Schwarz bound (C.5) is too weak for $\kappa<\sqrt{88}/3$. The next theorem closes this case.

Theorem C.4 (The sharp quarter)**.** *For every $c\in\mathbb{R}^4$ and every unit vector $h\in\mathbb{R}^4$,*

$$L(c,h)\le\tfrac14,$$

with equality at $c=0$. Equivalently, $W_1(\nu_b,\nu_{b'})\le\frac14|b-b'|$ for all $b,b'\in\mathbb{R}^4$, and the constant $\frac14$ is the best possible.

The equivalence is (C.4) in one direction and the derivative formula above it in the other; the sharpness is Lemma C.3(i). The metric matters.[23] The proof has four steps. A Lipschitz function on S^3 extends to the unit ball $B\subset\mathbb{R}^4$, and Green's formula in B bounds the covariance by a negative Sobolev norm of the density (Lemma C.5). This norm splits into a zonal and a transverse part, both explicit series in modified Bessel functions (Lemma C.6). The transverse part is at most $\frac14$ because a certain power series has nonnegative coefficients (Proposition C.8), and the zonal part is at most $\frac14$ by a Riccati comparison (Proposition C.9).

[23] In the geodesic metric of S^3 the statement is false already at $c=0$: the function $\arcsin x_2$ is geodesically 1-Lipschitz and its covariance with x_2 under the uniform law is $\frac{8}{9\pi}\approx 0.283>\frac14$. A proof therefore has to use that the chordal distance between two points is shorter than their geodesic distance, that is, it has to see the sphere from the ball it bounds. The proof below does exactly this: it extends a chordally Lipschitz function to the ball and integrates by parts there.

C.4. **Harmonic extension.** Let σ be the normalised surface measure of S^3 and H_k the space of spherical harmonics of degree k, the restrictions to S^3 of harmonic homogeneous polynomials of degree k on $\mathbb{R}^4$. The spaces H_k are orthogonal in $L^2(\sigma)$ and span it; we write u_k for the component of u in H_k, and for u with $\int u\,d\sigma = 0$ we put

$$\mathcal{Q}(u) = \sum_{k\geq 1} \frac{1}{k}\,\|u_k\|^2_{L^2(\sigma)}. \tag{C.6}$$

This is $\langle u, \Lambda^{-1}u\rangle$ for the Dirichlet-to-Neumann operator Λ of B, which acts on H_k as multiplication by k.

Lemma C.5. *Let u be a real-analytic function on S^3 with $\int u\,d\sigma = 0$ and let g be 1-Lipschitz for the chordal metric. Then*

$$\int_{S^3} g\,u\,d\sigma \leq \tfrac{1}{2}\,\mathcal{Q}(u)^{1/2}.$$

Proof. The chordal metric is the Euclidean metric of $\mathbb{R}^4$ restricted to S^3, so McShane's extension $G(y) = \inf_{x\in S^3}\big(g(x) + |x-y|\big)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals g on S^3 [40]; by Rademacher's theorem $|\nabla G| \leq 1$ almost everywhere. Let $\psi = \sum_{k\geq 1} k^{-1} r^k u_k(\omega)$ in polar coordinates $y = r\omega$. Each term is a harmonic polynomial, the series converges with all derivatives on the closed ball because u is real-analytic,[24] and $\partial_r \psi = u$ on S^3. Green's formula for the Lipschitz function G and the smooth harmonic function ψ reads $\int_{S^3} G\,\partial_r\psi\,dS = \int_B \nabla G\cdot\nabla\psi\,dy$. With $|S^3| = 2\pi^2$ and $|B| = \pi^2/2$ the Cauchy–Schwarz inequality gives

$$\int g\,u\,d\sigma = \frac{1}{2\pi^2}\int_B \nabla G\cdot\nabla\psi\,dy \leq \frac{|B|^{1/2}}{2\pi^2}\Big(\int_B |\nabla\psi|^2 dy\Big)^{1/2}.$$

Green's formula once more gives $\int_B |\nabla\psi|^2 = \int_{S^3}\psi\,\partial_r\psi\,dS = 2\pi^2\sum_k k^{-1}\|u_k\|^2 = 2\pi^2\mathcal{Q}(u)$, and the right side becomes $\frac{1}{2}\mathcal{Q}(u)^{1/2}$. □

The lemma is a form of Thomson's principle. By Kantorovich–Rubinstein duality, $\sup_g \int gu\,d\sigma$ is the least cost of a flow in $\mathbb{R}^4$ that carries the positive part of $u\,d\sigma$ to its negative part, the cost being the total length travelled by the mass.[25] The harmonic field $\nabla\psi$ is one such flow, through the interior of the ball, and the lemma computes its cost. At $c = 0$ the flow is optimal: for $u = \langle h, x\rangle$ one has $\psi = \langle h, y\rangle$, the field $\nabla\psi = h$ is constant, the Cauchy–Schwarz inequality is an equality, and $\mathcal{Q}(u) = \|u\|^2 = \frac{1}{4}$.

C.5. **Spectral decomposition.** Fix $c = \kappa e_1$ with $\kappa > 0$ and write $h = \cos\varphi\,e_1 + \sin\varphi\,e_2$, which is no loss of generality by rotation invariance. Let $\rho = d\nu_c/d\sigma = e^{\kappa x_1}/Z(\kappa)$, $Z(\kappa) = \int e^{\kappa x_1}d\sigma = 2I_1(\kappa)/\kappa$, and

$$u = \big(\langle h, x\rangle - \mathbb{E}_{\nu_c}\langle h, x\rangle\big)\rho,$$

so that $\mathrm{Cov}_{\nu_c}(g, \langle h, x\rangle) = \int gu\,d\sigma$ and $\int u\,d\sigma = 0$. Let U_n be the Chebyshev polynomials of the second kind and put

$$d_n(\kappa) = \frac{(n+1)\,I_{n+1}(\kappa)}{I_1(\kappa)}, \qquad n \geq 0. \tag{C.7}$$

[24] For real-analytic u the norms $\|u_k\|$ decay geometrically in k, and the supremum of a function in H_k is at most $(\dim H_k)^{1/2} = k+1$ times its $L^2(\sigma)$ norm, so the series for ψ and for each of its derivatives converge uniformly on $\bar B$.

[25] This is the Beckmann formulation of the transport problem with cost $|x-y|$: the optimal cost equals the least $\int |V|$ over vector measures V on $\mathbb{R}^4$ with divergence $u\,d\sigma$. Restricting the flow to the closed ball does not change the infimum, since projecting onto $\bar B$ shortens paths. The gradient $\nabla\psi$ on B is one admissible flow. It minimises the energy $\int |V|^2$ rather than the cost $\int |V|$, and the Cauchy–Schwarz inequality converts one into the other at the price of the factor $|B|^{1/2}$.

Lemma C.6. *With this notation*

$$\mathcal{Q}(u) = \cos^2\varphi\, \mathcal{Q}_z(\kappa) + \sin^2\varphi\, \mathcal{Q}_x(\kappa),$$

where

$$\mathcal{Q}_z = \sum_{n\geq 1} \frac{d_n'(\kappa)^2}{n}, \qquad \mathcal{Q}_x = \frac{1}{3\kappa^2}\sum_{n\geq 1}(n+2)\, d_n(\kappa)^2.$$

Proof. The functions $U_n(x_1)$, $n \geq 0$, are the zonal harmonics: $U_n(x_1) \in H_n$, and they are orthonormal in $L^2(\sigma)$ (Appendix E). The expansion $e^{\kappa t} = \frac{2}{\kappa}\sum_n (n+1) I_{n+1}(\kappa) U_n(t)$ of the same appendix gives $\rho = \sum_n d_n(\kappa) U_n(x_1)$, with $d_0 = 1$.

Write $u = \cos\varphi\, u_z + \sin\varphi\, u_x$ with $u_z = (x_1 - A)\rho$ and $u_x = x_2\rho$; here $\mathbb{E}_{\nu_c} x_1 = A$ and $\mathbb{E}_{\nu_c} x_2 = 0$. The reflection $x_2 \mapsto -x_2$ maps each H_k to itself, fixes u_z and reverses u_x, so the components of u_z and u_x in each H_k are orthogonal and the cross term in $\mathcal{Q}(u)$ vanishes.

Since $A = Z'/Z$, we have $u_z = \partial_\kappa \rho = \sum_n d_n'(\kappa) U_n(x_1)$, and the component of u_z in H_n is $d_n' U_n(x_1)$, of norm $|d_n'|$. This gives $\mathcal{Q}_z$.

For u_x, let $y(t) = \cos t\, e_1 + \sin t\, e_2$. The law $\nu_{\kappa y(t)}$ has density $\rho_t = e^{\kappa\langle y(t), x\rangle}/Z(\kappa) = \sum_n d_n(\kappa) U_n(\langle y(t), x\rangle)$, and $\partial_t \rho_t|_{t=0} = \kappa x_2 \rho$. The function $U_n(\langle y(t), x\rangle)$ lies in H_n for every t, hence so does its derivative $x_2 U_n'(x_1)$ at $t = 0$, and the component of u_x in H_n is $\kappa^{-1} d_n x_2 U_n'(x_1)$. Its norm follows from the eigenvalue $n(n+2)$ of the Laplacian of S^3 on H_n. For $f = U_n(x_1)$,

$$\begin{aligned} n(n+2) = n(n+2)\|f\|^2 &= \int |\nabla_{S^3} f|^2 d\sigma \\ &= \int U_n'(x_1)^2 (1 - x_1^2)\, d\sigma, \end{aligned}$$

since the tangential gradient of x_1 has squared length $1 - x_1^2$. Given x_1, the point (x_2, x_3, x_4) is uniform on the sphere of radius $(1 - x_1^2)^{1/2}$, so $\mathbb{E}[x_2^2 \mid x_1] = \frac{1}{3}(1 - x_1^2)$ and $\|x_2 U_n'(x_1)\|^2 = \frac{1}{3} n(n+2)$. Summing $n^{-1}\kappa^{-2} d_n^2 \cdot \frac{1}{3} n(n+2)$ over $n \geq 1$ gives $\mathcal{Q}_x$. □

By Lemmas C.5 and C.6,

$$L(\kappa, \varphi) \leq \tfrac{1}{2} \max\big(\mathcal{Q}_z(\kappa), \mathcal{Q}_x(\kappa)\big)^{1/2}, \tag{C.8}$$

and Theorem C.4 follows once both parts are at most $\frac{1}{4}$. Both are $\frac{1}{4}$ at $\kappa = 0$, by continuity from the values $d_1 \sim \kappa/2$, $d_1' \to \frac{1}{2}$ and $d_n = O(\kappa^n)$.

C.6. **Three Bessel sums.** The series of Lemma C.6 reduce to closed forms. Write I_j for $I_j(\kappa)$.

Lemma C.7. *For every $\kappa > 0$,*

$$\sum_{j\geq 1} j^2 I_j^2 = \frac{\kappa}{4} I_1(2\kappa), \qquad \sum_{j\geq 1} j^3 I_j^2 = \frac{\kappa^2}{4}(I_0^2 + I_1^2),$$
$$I_1(2\kappa) = 2\sum_{j\geq 1} I_{j-1} I_j.$$

The first and third identities follow from Parseval's formula for $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$, and the second from the recurrences of I_j by telescoping; Appendix E gives the proofs.

C.7. **The transverse part.**

Proposition C.8. *$\mathcal{Q}_x(\kappa) \leq \frac{1}{4}$ for every $\kappa \geq 0$.*

Proof. By (C.7), $\mathcal{Q}_x = (3\kappa^2 I_1^2)^{-1} \sum_{j\geq 2} j^2(j+1) I_j^2$, and the sum equals $\sum_{j\geq 1}(j^3 + j^2) I_j^2 - 2I_1^2$. By Lemma C.7, $\mathcal{Q}_x \leq \frac{1}{4}$ is equivalent to

$$\mathcal{D}(\kappa) := (8 + 2\kappa^2) I_1^2 - \kappa^2 I_0^2 - \kappa I_1(2\kappa) \geq 0. \tag{C.9}$$

Each term is an entire function of κ that is even after the factor κ^2 is removed, so $\mathcal{D}(\kappa)=\sum_{n\ge 0}\mathcal{D}_n\kappa^{2n}$. Lemma E.2 of Appendix E computes the coefficients: $\mathcal{D}_0=\mathcal{D}_1=0$, and for $n=m+1\ge 2$

$$\mathcal{D}_n=\frac{N_m}{4^m\,m!\,(m+1)!\,(m+1)(m+2)},$$

with

$$N_m=\binom{2m}{m}\bigl(4(2m+1)+(m-1)(m+1)(m+2)\bigr)\\ -4^m(m+1)(m+2).$$

The numerator N_m vanishes for $m=0,1,2$ and is positive for $m\ge 3$ (Lemma E.2). So every coefficient of $\mathcal{D}$ is nonnegative and $\mathcal{D}\ge 0$. □

The first nonzero coefficient is $\mathcal{D}_4=\frac{1}{2304}$, so $\frac14-\mathcal{Q}_x=\mathcal{D}/(12\kappa^2I_1^2)$ is of order κ^4 at 0: the transverse part leaves the value $\frac14$ very slowly, which is why no estimate that loses a constant can prove the proposition.

C.8. **The zonal part.** Let $A=I_2/I_1$ as before, so that $d_1=2A$ and, by Lemma C.2(i), $A'=1-A^2-3A/\kappa$.

Proposition C.9. *For $0<\kappa\le\sqrt{88}/3$,*

$$\mathcal{Q}_z(\kappa)\le\frac14+4A'^2+\frac{A^2}{4}-\frac{A}{2\kappa}-\frac{2A^2}{\kappa^2}<\frac14.$$

Proof. The higher modes. With $r_\nu=I_{\nu+1}/I_\nu$ the recurrence $I_\nu'=I_{\nu+1}+\frac{\nu}{\kappa}I_\nu$ gives

$$\frac{d_n'}{d_n}=\frac{I_{n+1}'}{I_{n+1}}-\frac{I_1'}{I_1}=\frac{n}{\kappa}+r_{n+1}-r_1.$$

By Lemma E.3 of Appendix E, $r_{n+1}\le r_1$, and $r_1=A\le\kappa/4$ by Lemma C.2(iii). Hence $-n/\kappa\le n/\kappa-\kappa/4\le d_n'/d_n\le n/\kappa$ for $n\ge 2$ and $\kappa^2\le 16$, so $d_n'^2\le n^2d_n^2/\kappa^2$. With $d_1'=2A'$,

$$\mathcal{Q}_z\le 4A'^2+\frac{1}{\kappa^2}\sum_{n\ge2}n\,d_n^2=4A'^2+\frac{S_3-S_2-4I_2^2}{\kappa^2I_1^2},$$

where $S_2=\sum_{j\ge1}j^2I_j^2$ and $S_3=\sum_{j\ge1}j^3I_j^2$; the second form uses $n\,d_n^2I_1^2=(j-1)j^2I_j^2$ with $j=n+1$.

The closed form. The recurrences $I_0-I_2=\frac{2}{\kappa}I_1$ and $I_1-I_3=\frac{4}{\kappa}I_2$ give $I_0/I_1=2/\kappa+A$ and $I_2I_3/I_1^2=A-4A^2/\kappa$. By Lemma C.7, $S_3/(\kappa^2I_1^2)=\frac14(2/\kappa+A)^2+\frac14$, and, keeping three terms of the third identity,

$$\frac{S_2}{\kappa^2I_1^2}\ge\frac{I_0I_1+I_1I_2+I_2I_3}{2\kappa I_1^2}=\frac{1}{2\kappa}\Bigl(\frac{2}{\kappa}+3A-\frac{4A^2}{\kappa}\Bigr).$$

Collecting terms, $\mathcal{Q}_z\le\frac14+4A'^2+T(A)$ with $T(a)=\frac{a^2}{4}-\frac{a}{2\kappa}-\frac{2a^2}{\kappa^2}$, which is the first inequality.

The barrier. Let $s=\sqrt{\kappa^2+25/4}$ and $A_\flat=\kappa/(\frac32+s)$. By Lemma C.2(iv), $A_\flat\le A\le\kappa/4$. The function $a\mapsto 1-a^2-3a/\kappa$ decreases on $a\ge0$, and $A'\ge0$ because $A'=\operatorname{Var}x_1$, so $0\le A'\le\alpha:=1-A_\flat^2-3A_\flat/\kappa$. For $\kappa^2\le12$ the quadratic T decreases on $[0,\kappa/4]$, since

$$T(a)-T(a')=(a'-a)\,\frac{2\kappa-(a+a')(\kappa^2-8)}{4\kappa^2}$$

and $(a+a')(\kappa^2-8) \le \frac{\kappa}{2}(\kappa^2-8) \le 2\kappa$ there. So $\mathcal{Q}_z \le \frac{1}{4} + 4\alpha^2 + T(A_\flat)$. With $\kappa^2 = s^2 - \frac{25}{4}$ and $A_\flat/\kappa = 2/(2s+3)$ both terms are rational in s: $\alpha = 16/(2s+3)^2$, and

$$4\alpha^2 + T(A_\flat) = \frac{(2s-5)\,p(s)}{4(2s+3)^4}, \qquad p(s) = 8s^3 + 28s^2 - 98s - 695. \tag{C.10}$$

For $0 < \kappa \le \sqrt{88}/3$ we have $\frac{5}{2} < s \le \sqrt{577}/6 < 4.004$. On $s \ge \frac{5}{2}$ the cubic p increases, since $p'(s) = 24s^2 + 56s - 98 > 0$, and $p(4.004) < -120$. So the right side of (C.10) is negative. □

C.9. **Proof of Theorem C.4.** At $c = 0$ the theorem is Lemma C.3(i), and for $|c| \ge \sqrt{88}/3$ it is Lemma C.3(iii). For $0 < |c| < \sqrt{88}/3$, Propositions C.8 and C.9 give $\mathcal{Q}_x, \mathcal{Q}_z \le \frac{1}{4}$, and (C.8) gives $L \le \frac{1}{4}$. □

Appendix D. Computations for the single-link law

D.1. **Barta's inequality.** Let $\rho(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$ on $(0,\pi)$, let V be a continuous potential, and let

$$T = -\rho^{-1}\partial_\theta\,\rho\,\partial_\theta + V.$$

If $\psi > 0$ is smooth on $(0,\pi)$ with $\rho\psi^2 = O(\sin^4\theta)$ at the end points and $T\psi \ge \lambda_0\psi$, then

$$\int (|h'|^2 + Vh^2)\rho\, d\theta \ge \lambda_0 \int h^2 \rho\, d\theta$$

for every smooth h vanishing to first order at 0 and π. Indeed, writing $h = \psi u$ and integrating by parts, the boundary terms $\rho\psi\psi' u^2$ vanish and

$$\int (|h'|^2 + Vh^2)\rho = \int \rho\psi^2 |u'|^2 + \int \frac{T\psi}{\psi}\, h^2\rho \;\ge\; \lambda_0 \int h^2\rho.$$

This is the inequality of Barta [6] in the form of a ground state transformation.

D.2. **The computations of Lemma C.1.** The generator of the reversible diffusion of ν_b on S^3 is $L = \Delta_{S^3} + \kappa \nabla x_1 \cdot \nabla$. In the polar angle θ from e_1 the round metric is $d\theta^2 + \sin^2\theta\, g_{S^2}$, so $\Delta_{S^3} = \partial_\theta^2 + 2\cot\theta\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$, and $\nabla x_1 \cdot \nabla = -\sin\theta\,\partial_\theta$. On $h(\theta)Y_\ell$ with a spherical harmonic Y_ℓ of degree ℓ on S^2 the operator acts as $L_\ell = \partial_\theta^2 + b\,\partial_\theta - \ell(\ell+1)\sin^{-2}\theta$, $b = 2\cot\theta - \kappa\sin\theta = \rho'/\rho$, so $-L_\ell = T$ with $V = \ell(\ell+1)\sin^{-2}\theta$.

Degree $\ell \ge 1$. Take $\psi = \sin\theta\, e^{a\kappa\cos\theta}$. Then $\psi'/\psi = \cot\theta - a\kappa\sin\theta$ and $\psi''/\psi = (\psi'/\psi)^2 + (\psi'/\psi)'$, and a short computation gives

$$\frac{-L_1\psi}{\psi} = 3 + (1+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta.$$

For $a = -\frac{1}{5}$ this is $3 + \frac{4}{25}\kappa^2\sin^2\theta \ge 3$. Since $V = \ell(\ell+1)\sin^{-2}\theta \ge 2\sin^{-2}\theta$ for $\ell \ge 1$, Barta's inequality gives $-L_\ell \ge 3$ on every sector with $\ell \ge 1$.

Degree $\ell = 0$. Let $g(\theta)$ be a zonal eigenfunction, $-L_0 g = \lambda g$, g not constant, so that $g'(0) = g'(\pi) = 0$. Differentiating $g'' + bg' = -\lambda g$ gives, for $h = g'$,

$$-(h'' + bh') - b'h = \lambda h, \qquad -b' = \frac{2}{\sin^2\theta} + \kappa\cos\theta,$$

so h is an eigenfunction of T with $V = 2\sin^{-2}\theta + \kappa\cos\theta$, with eigenvalue λ, and $h \not\equiv 0$ vanishes at both ends. With the same ψ,

$$\frac{T\psi}{\psi} = 3 + (2+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta,$$

which for $a = -\frac{2}{5}$ is $3 + \frac{6}{25}\kappa^2\sin^2\theta \ge 3$. Barta's inequality applied to h gives $\lambda \ge 3$. Together with the case $\ell \ge 1$ every nonzero eigenvalue of $-L$ is at least 3, which is Lemma C.1; the spectrum is discrete because S^3 is compact and the drift is smooth.

D.3. **Moments.** The density of $t = x_1$ under ν_b is proportional to $(1-t^2)^{1/2}e^{\kappa t}$. For every smooth ϕ on $[-1,1]$,

$$\frac{d}{dt}\Big[(1-t^2)^{3/2}e^{\kappa t}\Big] = (1-t^2)^{1/2}e^{\kappa t}\big(\kappa(1-t^2)-3t\big),$$

and integration by parts gives $\mathbb{E}[\phi'(t)(1-t^2)] = \mathbb{E}\big[\phi(t)\big(3t-\kappa(1-t^2)\big)\big]$. With $\phi = 1$ this is $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, the identity of Lemma C.2(i). With $\phi = t^k$ it gives the recursion

$$m_{k+2} = m_k - \frac{k+3}{\kappa}\,m_{k+1} + \frac{k}{\kappa}\,m_{k-1}, \qquad m_k = \mathbb{E}x_1^k,$$

from $m_0 = 1$ and $m_1 = A$; all polynomial moments of ν_b are rational in κ and $A(\kappa)$.[26]

D.4. **The Riccati barriers.** By Lemma C.2(i), A solves $A' = \mathcal{F}(\kappa, A) := 1 - A^2 - 3A/\kappa$ on $(0,\infty)$, and $A(\kappa) = \frac{\kappa}{4} - \frac{\kappa^3}{96} + O(\kappa^5)$ from the series of I_1 and I_2. Let $s = \sqrt{\kappa^2+4}$, $u = \sqrt{\kappa^2+9}$, $\phi_- = \kappa/(2+s)$ and $\phi_+ = \kappa/(1+u)$. Direct differentiation gives

$$\phi_-' - \mathcal{F}(\kappa,\phi_-) = -\frac{\kappa^2}{s(s+2)^2} \le 0,$$

$$\phi_+' - \mathcal{F}(\kappa,\phi_+) = \frac{(u-3)^2}{u(1+u)^2} \ge 0,$$

and $\phi_\pm = \frac{\kappa}{4} + O(\kappa^3)$.[27] The comparison is singular at 0 because of the term $3A/\kappa$, and we carry it out with an integrating factor. Let $D = A - \phi_-$. Then

$$\begin{aligned} D' &= \mathcal{F}(\kappa,A) - \mathcal{F}(\kappa,\phi_-) - (\phi_-' - \mathcal{F}(\kappa,\phi_-)) \\ &= -\Big(A + \phi_- + \frac{3}{\kappa}\Big)D + R, \end{aligned}$$

with $R = \kappa^2/(s(s+2)^2) \ge 0$. Put $\mu(\kappa) = \kappa^3 \exp\int_0^\kappa (A+\phi_-)$. Then $(\mu D)' = \mu R \ge 0$, and $\mu D = O(\kappa^6) \to 0$ as $\kappa \to 0$, so $\mu D \ge 0$ and $A \ge \phi_-$. The same argument with $D = \phi_+ - A$ and $R = (u-3)^2/(u(1+u)^2)$ gives $A \le \phi_+$. This is Lemma C.2(ii).

For (iv) let $s = \sqrt{\kappa^2 + 25/4}$ and $\phi_\flat = \kappa/(\frac{3}{2}+s)$. Since $s^2 - \frac{9}{4} = \kappa^2 + 4$, the same differentiation gives

$$\phi_\flat' - \mathcal{F}(\kappa,\phi_\flat) = -\frac{5\,(s-\frac{5}{2})}{2\,s\,(\frac{3}{2}+s)^2} \le 0,$$

so $\phi_\flat$ is a subsolution, and $\phi_\flat = \frac{\kappa}{4} - \frac{\kappa^3}{80} + O(\kappa^5)$.[28] Hence $A - \phi_\flat = \frac{\kappa^3}{480} + O(\kappa^5)$, and the integrating-factor argument above, with $D = A - \phi_\flat$, $R = \frac{5(s-5/2)}{2s(3/2+s)^2} \ge 0$ and $\mu = \kappa^3 \exp\int_0^\kappa (A+\phi_\flat)$, gives $\mu D = O(\kappa^6)$ at 0, $(\mu D)' = \mu R \ge 0$, and so $A \ge \phi_\flat$. Since $\frac{3}{2} + \sqrt{\kappa^2+25/4} < 2 + \sqrt{\kappa^2+4}$ for $\kappa > 0$, this improves the lower bound of (ii).

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[26]For instance $m_2 = 1 - 3A/\kappa$, $m_3 = A - 3/\kappa + 12A/\kappa^2$ and $m_4 = m_2 - 5m_3/\kappa + 2A/\kappa$. The joint moments with x_2, x_3, x_4 follow from exchangeability: $\mathbb{E}[\phi(x_1)x_2^2] = \frac{1}{3}\mathbb{E}[\phi(x_1)(1-x_1^2)]$.

[27]For the first identity write $\phi_- = (s-2)/\kappa$, which follows from $s^2 - 4 = \kappa^2$; then $\phi_-' = (\kappa^2/s - s + 2)/\kappa^2$ and $1 - \phi_-^2 - 3\phi_-/\kappa = (\kappa^2 - (s-2)^2 - 3(s-2))/\kappa^2$, and the difference simplifies with $s^2 = \kappa^2 + 4$. The second is the same computation with $\phi_+ = \kappa(u-1)/(u^2-1) = \kappa(u-1)/(\kappa^2+8)$ in place of ϕ_- and $u^2 = \kappa^2 + 9$ in place of $s^2 = \kappa^2 + 4$. After clearing the denominator $\kappa^2(1+u)^2$ the numerator becomes $\kappa^2(u-3)^2/u$, which is the stated residual.

[28]With $\phi_\flat = \kappa(s - \frac{3}{2})/(\kappa^2+4)$ one has $\phi_\flat' = (\frac{3}{2}s + \frac{25}{4})/(s(\frac{3}{2}+s)^2)$, using $ds/d\kappa = \kappa/s$; and $\mathcal{F}(\kappa,\phi_\flat) = 1 - \phi_\flat^2 - 3/(\frac{3}{2}+s)$. Clearing the denominator $s(\frac{3}{2}+s)^2$ and replacing κ^2 by $s^2 - \frac{25}{4}$ leaves $-\frac{5}{2}s + \frac{25}{4}$ in the numerator, which is the stated residual. The expansion follows from $s = \frac{5}{2} + \frac{\kappa^2}{5} + O(\kappa^4)$.

Appendix E. Bessel functions

This appendix proves the auxiliary statements used in Appendix C for Theorem C.4. Throughout, $I_j = I_j(\kappa)$ with $\kappa > 0$, and $I_{-j} = I_j$ for integers j. We use the series

$$I_\nu(\kappa) = \sum_{i\geq 0} \frac{(\kappa/2)^{2i+\nu}}{i!\,(i+\nu)!},$$

the integral $I_j(\kappa) = \pi^{-1}\int_0^\pi e^{\kappa\cos\theta}\cos j\theta\,d\theta$, the Fourier series $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$ and the recurrences

$$I_{j-1} - I_{j+1} = \frac{2j}{\kappa}\,I_j, \quad I_j' = I_{j+1} + \frac{j}{\kappa}\,I_j, \tag{E.1}$$

all of which are in Watson's treatise [60], §§3.7, 3.71, 2.22. All series below converge absolutely, because $0 < I_j \leq (\kappa/2)^j e^\kappa/j!$.

E.1. **Zonal harmonics.** Under σ the coordinate x_1 has the density $\frac{2}{\pi}(1-t^2)^{1/2}$ on $[-1,1]$, the area of the two-sphere $\{x_1 = t\}$ being proportional to $1-t^2$ and its normal speed to $(1-t^2)^{-1/2}$. The Chebyshev polynomials $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$ satisfy

$$\int U_n U_m\,d\sigma = \frac{2}{\pi}\int_0^\pi \sin((n+1)\theta)\sin((m+1)\theta)\,d\theta = \delta_{nm},$$

and $U_n = C_n^{(1)}$ is the Gegenbauer polynomial of index $\frac{4-2}{2} = 1$, so $U_n(\langle y,x\rangle)$ is the zonal harmonic of degree n on S^3 with pole y: $|x|^n U_n(\langle y,x\rangle/|x|)$ is a harmonic polynomial on $\mathbb{R}^4$ [55], Ch. IV, §2.

Lemma E.1. *$e^{\kappa t} = \frac{2}{\kappa}\sum_{n\geq 0}(n+1)I_{n+1}(\kappa)\,U_n(t)$ in $L^2(\frac{2}{\pi}(1-t^2)^{1/2}dt)$, and uniformly on $[-1,1]$.*

Proof. With $t = \cos\theta$ and $2\sin\theta\sin((n+1)\theta) = \cos n\theta - \cos(n+2)\theta$, the n-th coefficient is

$$\begin{aligned}\frac{2}{\pi}\int_0^\pi e^{\kappa\cos\theta}\sin((n+1)\theta)\sin\theta\,d\theta &= I_n - I_{n+2}\\ &= \frac{2(n+1)}{\kappa}I_{n+1},\end{aligned}$$

by (E.1). Since $|U_n| \leq n+1$ and I_{n+1} decays factorially, the series converges uniformly. □

E.2. **Proof of Lemma C.7.** Let $F(\theta) = e^{\kappa\cos\theta} = \sum_j I_j e^{ij\theta}$. Then

$$F'(\theta) = -\kappa\sin\theta\,F(\theta) = \sum_j ijI_j e^{ij\theta},$$

and Parseval's formula gives

$$\begin{aligned}\sum_{j\in\mathbb{Z}} j^2 I_j^2 &= \frac{\kappa^2}{2\pi}\int_0^{2\pi}\sin^2\theta\,e^{2\kappa\cos\theta}d\theta\\ &= \frac{\kappa^2}{2}\big(I_0(2\kappa) - I_2(2\kappa)\big) = \frac{\kappa}{2}I_1(2\kappa),\end{aligned}$$

the last step by (E.1) at the argument 2κ. Halving gives the first identity. The coefficient of $e^{i\theta}$ in $F^2 = e^{2\kappa\cos\theta}$ is $I_1(2\kappa) = \sum_{j\in\mathbb{Z}} I_j I_{1-j}$, and the terms with $j \leq 0$ repeat those with $j \geq 1$; this is the third identity.

For the second, $jI_j = \frac{\kappa}{2}(I_{j-1} - I_{j+1})$ by (E.1), so

$$\sum_{j\geq 1} j^3 I_j^2 = \frac{\kappa^2}{4}\sum_{j\geq 1} j(I_{j-1}^2 + I_{j+1}^2 - 2I_{j-1}I_{j+1}).$$

Shifting indices gives

$$\sum_{j\geq 1} j(I_{j-1}^2 + I_{j+1}^2) = I_0^2 + \sum_{m\geq 1} 2mI_m^2, \qquad \sum_{m\geq 1} 2mI_m^2 = \kappa\sum_{m\geq 1}(I_{m-1}I_m - I_mI_{m+1}) = \kappa I_0 I_1,$$

the last by telescoping. For the mixed term write $j=\frac12((j-1)+(j+1))$ and apply (E.1) to $(j-1)I_{j-1}$ and to $(j+1)I_{j+1}$, using $I_{-1}=I_1$ at $j=1$:

$$jI_{j-1}I_{j+1}=\frac{\kappa}{4}\big[(I_{j-1}I_j-I_jI_{j+1})\\ +(I_{j-2}I_{j+1}-I_{j-1}I_{j+2})\big].$$

Both brackets telescope, to I_0I_1 and to $I_{-1}I_2=I_1I_2$, so $\sum_{j\ge1}jI_{j-1}I_{j+1}=\frac{\kappa}{4}(I_0I_1+I_1I_2)$. Altogether

$$\sum_{j\ge1}j^3I_j^2=\frac{\kappa^2}{4}\Big(I_0^2+\frac{\kappa}{2}I_1(I_0-I_2)\Big)=\frac{\kappa^2}{4}(I_0^2+I_1^2).\qquad\square$$

E.3. **The coefficients of $\mathcal{D}$.**

Lemma E.2. *Write $\mathcal{D}(\kappa)=\sum_{n\ge0}\mathcal{D}_n\kappa^{2n}$ for the function of* (C.9)*. Then $\mathcal{D}_0=\mathcal{D}_1=0$, and for $m\ge1$*

$$\mathcal{D}_{m+1}=\frac{N_m}{4^m\,m!\,(m+1)!\,(m+1)(m+2)},$$

with $N_m=\binom{2m}{m}\big(4(2m+1)+(m-1)(m+1)(m+2)\big)-4^m(m+1)(m+2)$. Moreover $N_m=0$ for $m\le2$ and $N_m>0$ for $m\ge3$.

Proof. The coefficients. Multiplying the series of I_0 and I_1 and collecting powers,

$$I_0^2=\sum_{m\ge0}\alpha_m\kappa^{2m},\qquad I_1^2=\sum_{m\ge0}\beta_m\kappa^{2m+2},$$

$$\kappa I_1(2\kappa)=\sum_{m\ge0}\frac{\kappa^{2m+2}}{m!\,(m+1)!},$$

where

$$\alpha_m=\frac{1}{4^m}\sum_{i=0}^{m}\frac{1}{i!^2(m-i)!^2},$$

$$\beta_m=\frac{1}{4^{m+1}}\sum_{i=0}^{m}\frac{1}{i!\,(i+1)!\,(m-i)!\,(m-i+1)!}.$$

By Vandermonde's identity, $\sum_i\binom{m}{i}^2=\binom{2m}{m}$ and $\sum_i\binom{m}{i}\binom{m+2}{i+1}=\binom{2m+2}{m+1}$, so

$$\alpha_m=\frac{\binom{2m}{m}}{4^m\,m!^2},\qquad\beta_m=\frac{\binom{2m+2}{m+1}}{4^{m+1}m!\,(m+2)!}.$$

The coefficient of κ^{2m+2} in $\mathcal{D}$ is $\mathcal{D}_{m+1}=8\beta_m+2\beta_{m-1}-\alpha_m-1/(m!(m+1)!)$, with $\beta_{-1}=0$; for $m=0$ it is $\frac84-1-1=0$, and $\mathcal{D}_0=0$ because every term of $\mathcal{D}$ is $O(\kappa^2)$. For $m\ge1$ put $\Delta=4^m\,m!\,(m+1)!\,(m+1)(m+2)$ and $b=\binom{2m}{m}$. With $\binom{2m+2}{m+1}=\frac{2(2m+1)}{m+1}b$,

$$8\beta_m=\frac{4(2m+1)\,b}{\Delta},\qquad\alpha_m=\frac{(m+1)^2(m+2)\,b}{\Delta},$$

$$2\beta_{m-1}=\frac{2m(m+1)(m+2)\,b}{\Delta},$$

$$\frac{1}{m!(m+1)!}=\frac{4^m(m+1)(m+2)}{\Delta},$$

and $2m-(m+1)=m-1$ gives $\mathcal{D}_{m+1}=N_m/\Delta$.

The sign. Directly, $N_0=N_1=N_2=0$ ($2-2$, $24-24$, $192-192$), $N_3=80$, $N_4=1140$ and $N_5=10416$. For $m\ge6$ we first show $16^m\le4m\binom{2m}{m}^2$ for $m\ge1$ by induction: it is an

equality at $m=1$, and $(m+1)\binom{2m+2}{m+1} = 2(2m+1)\binom{2m}{m}$ gives

$$4(m+1)\binom{2m+2}{m+1}^2 = \frac{16(2m+1)^2}{m+1}\binom{2m}{m}^2 \geq \frac{16(2m+1)^2}{4m(m+1)}16^m \geq 16^{m+1}.$$

Since $(m-1)^2 \geq 4m$ for $m \geq 6$, it follows that $\binom{2m}{m}^2(m-1)^2 \geq 16^m$, that is $\binom{2m}{m}(m-1) \geq 4^m$, and

$$N_m = \Big(\binom{2m}{m}(m-1) - 4^m\Big)(m+1)(m+2) + 4(2m+1)\binom{2m}{m} > 0.$$

□

E.4. **Monotonicity of Bessel ratios.**

Lemma E.3. *For $n \geq 1$ and $\kappa > 0$, $I_{n+2}(\kappa)/I_{n+1}(\kappa) \leq I_2(\kappa)/I_1(\kappa)$.*

Proof. We use a lemma of Biernacki and Krzyż [9]: if $f = \sum a_j x^j$ and $g = \sum b_j x^j$ have positive coefficients and a_j/b_j is nonincreasing in j, then f/g is nonincreasing on $x > 0$. Indeed

$$f'g - fg' = \sum_{i,j}(i-j)a_i b_j x^{i+j-1} = \frac{1}{2}\sum_{i,j}(i-j)b_i b_j\Big(\frac{a_i}{b_i} - \frac{a_j}{b_j}\Big)x^{i+j-1} \leq 0.$$

Take $x = \kappa^2$, $f = I_{n+1}(\kappa)/\kappa^{n+1}$ and $g = I_1(\kappa)/\kappa$. Then $a_j/b_j = 2^{-n}(j+1)!/(j+n+1)!$ decreases in j, so $I_{n+1}/(\kappa^n I_1)$ is nonincreasing in κ. Its logarithmic derivative is, by the second recurrence of (E.1),

$$\frac{n+1}{\kappa} + \frac{I_{n+2}}{I_{n+1}} - \frac{n}{\kappa} - \frac{1}{\kappa} - \frac{I_2}{I_1} = \frac{I_{n+2}}{I_{n+1}} - \frac{I_2}{I_1} \leq 0.$$

□

Appendix F. The single-link law of SU(3)

This appendix[29] proves the one-link estimate used in Theorem 8.11(ii). Throughout $G = \mathrm{SU}(3)$, the distance is $d(U,V) = \|U - V\|_{\mathrm{HS}}$, which is at most the geodesic distance of the metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$ on $\mathfrak{su}(3)$, K is the closed convex hull of SU(3) in $\mathbb{C}^{3\times 3}$, $A(M) = \frac{1}{2}(M - M^*)$ is the anti-Hermitian part of a matrix and P removes the trace, $PX = X - \frac{1}{3}(\operatorname{Tr} X)I$. With this metric $\mathrm{Ric} = \frac{3}{2}$, the Casimir operator acts on the irreducible representation of highest weight (p,q) by

$$C(p,q) = \tfrac{2}{3}(p^2 + q^2 + pq + 3p + 3q), \tag{F.1}$$

so that $\lambda_F = C(1,0) = \frac{8}{3}$ on the fundamental representation and $C(1,1) = 6$ on the adjoint [21, Sec. 15.3].

[29] Throughout this appendix the Lie algebra $\mathfrak{su}(3)$ carries the metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$ of the defining representation, the irreducible representations are labelled by their highest weight (p,q) with respect to the fundamental weights, of dimension $\frac{1}{2}(p+1)(q+1)(p+q+2)$, and the Casimir operator is the Laplace–Beltrami operator of the bi-invariant metric acting on matrix coefficients; the defining representation is $(1,0)$ and the adjoint is $(1,1)$.

F.1. **Reduction to one link.** For $b \in \mathbb{C}^{3\times 3}$ let $\nu_b(dU) \propto e^{\operatorname{Re}\operatorname{Tr}(Ub)}dU$ on SU(3). In the setting of Theorem 8.11(ii) the conditional law of a free link e is ν_b with $b = N\beta\sum_{p\ni e} S_p$, where S_p is the staple of e in p, a product of three matrices each of which is an element of SU(3) or the identity, or the adjoint of such a product; so $b \in rK$ with $r = 6N\beta = 2\beta_W$. For a matrix h put $\varphi_h(U) = \operatorname{Re}\operatorname{Tr}(Uh)$ and

$$L_h(b) = \sup\{\operatorname{Cov}_{\nu_b}(g,\varphi_h) : g \text{ 1-Lipschitz for } d\}, \qquad \Lambda(r) = \sup_{b\in rK}\ \sup_{\|h\|_{\mathrm{HS}}=1} L_h(b).$$

Changing a partner k of e by $d(Q_k, Q_k')$ changes one staple by a matrix of Hilbert–Schmidt norm $d(Q_k, Q_k')$, since multiplication by unitaries preserves that norm, hence b by h with $\|h\|_{\mathrm{HS}} \le N\beta\, d(Q_k, Q_k')$. Along the segment $b_s = b + sh$, $0 \le s \le 1$, which stays in rK by convexity, $\frac{d}{ds}\nu_{b_s}(g) = \operatorname{Cov}_{\nu_{b_s}}(g,\varphi_h)$, so by Kantorovich–Rubinstein duality

$$C_{ek}^T \le N\beta\,\Lambda(r) = \frac{r}{6}\,\Lambda(r). \tag{F.2}$$

Since K is convex and contains 0, the mean of I, ωI and $\omega^2 I$ with $\omega = e^{2\pi i/3}$, $r'K \subset rK$ for $r' \le r$ and Λ is nondecreasing. The rest of the appendix bounds $\Lambda(r)$.

F.2. **Gradients and one-forms.**

Lemma F.1. *In the left trivialisation, $\nabla\varphi_c(U) = -PA(cU)$; if $V = \varphi_b$ then $\operatorname{Hess} V(X,X) = \operatorname{Re}\operatorname{Tr}(UX^2 b)$; and $\Delta\varphi_c = -\lambda_F\varphi_c$.*

Proof. $\frac{d}{dt}\operatorname{Re}\operatorname{Tr}(Ue^{tX}c)|_{t=0} = \operatorname{Re}\operatorname{Tr}(cUX) = -\langle PA(cU), X\rangle$ for $X \in \mathfrak{su}(3)$, because $\operatorname{Re}\operatorname{Tr}(MX) = \operatorname{Re}\operatorname{Tr}(A(M)X) = -\langle A(M), X\rangle$ and X is traceless. On a group with a bi-invariant metric $\nabla_X X = 0$ for a left-invariant field X, so the Hessian is the second derivative along $t \mapsto Ue^{tX}$. The function φ_c is a real part of a matrix coefficient of the fundamental representation, on which the Casimir acts by λ_F. □

Lemma F.2. *The bottom of the spectrum of the rough Laplacian $\nabla^*\nabla$ on L^2 one-forms of* SU(3) *is $\mu_0 = \frac{7}{6}$.*

Proof. Trivialise one-forms on the left, so that a one-form is a function $f : \mathrm{SU}(3) \to \mathfrak{su}(3)$. For an orthonormal basis X_a acting as left-invariant vector fields, $\nabla_{X_a}Y = \frac{1}{2}[X_a, Y]$ and $\nabla_{X_a}X_a = 0$, so $\nabla^*\nabla = -\sum_a(\rho(X_a) + \frac{1}{2}\operatorname{ad} X_a)^2$, with ρ the right regular representation. On the isotypic part $V_\pi \otimes V_\pi^* \otimes \mathfrak{su}(3)$ of $L^2 \otimes \mathfrak{su}(3)$, given by the Peter–Weyl theorem, put $A_a = \rho_\pi(X_a)$, $B_a = \operatorname{ad} X_a$ and $T_a = A_a + B_a$, the action on the tensor product. Since A_a and B_a commute, $A_a + \frac{1}{2}B_a = \frac{1}{2}(A_a + T_a)$ and $A_aT_a + T_aA_a = A_a^2 + T_a^2 - B_a^2$, so on a component $\sigma \subset \pi \otimes \mathrm{ad}$

$$\nabla^*\nabla = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{1}{4}C_{\mathrm{ad}} = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{3}{2},$$

with C as in (F.1). The components of $(p,q)\otimes(1,1)$ are $(p+1,q+1)$, $(p+2,q-1)$, $(p-1,q+2)$, $(p+1,q-2)$, $(p-2,q+1)$, $(p-1,q-1)$ and (p,q), the last with multiplicity two, one or zero as both, one or none of p,q are positive, and only those with nonnegative labels occur [21, Sec. 13.2]. For $\pi = (0,0)$ the only component is $(1,1)$, with value $\frac{3}{2}$. For $\pi = (1,0)$ the components are $(2,1)$, $(0,2)$, $(1,0)$, with values $\frac{31}{6}$, $\frac{19}{6}$, $\frac{7}{6}$, and $(0,1)$ is conjugate. For $p+q \ge 2$ the smallest values occur at $(p-1,q-1)$, (p,q), $(p-2,q+1)$ or $(p+1,q-2)$, and each is at least $\frac{1}{3}(2(p^2+q^2+pq)+3(p+q)-3) - \frac{3}{2} \ge \frac{3}{2}$. □

F.3. **The gap of the one-link measure.**

Lemma F.3. *For $b \in rK$ the spectral gap of $L = \Delta + \langle\nabla V, \nabla\cdot\rangle$, $V = \varphi_b$, on $L^2(\nu_b)$ is at least*

$$\gamma(r) = \tfrac{39}{64}\mu_0 + \tfrac{3}{2} - \tfrac{r}{2} = \tfrac{283}{128} - \tfrac{r}{2}.$$

Proof. Let $-Lf = \lambda f$ with $\lambda > 0$, $\omega = df$ and $\eta = e^{V/2}\omega$; norms are in $L^2(dU)$. Bochner's formula integrated against ν_b gives $\lambda\nu_b|\omega|^2 = \nu_b|\nabla\omega|^2 + \nu_b((\mathrm{Ric} - \mathrm{Hess}\, V)(\omega,\omega))$, and since $e^{V/2}\nabla\omega = \nabla\eta - \frac{1}{2}dV\otimes\eta$,

$$\lambda\|\eta\|^2 = \|\nabla\eta - \tfrac{1}{2}dV\otimes\eta\|^2 + \int (\mathrm{Ric} - \mathrm{Hess}\, V)(\eta,\eta). \tag{F.3}$$

Put $\mathcal{A} = \nabla\eta$ and $\mathcal{B} = \frac{1}{2}dV\otimes\eta$. Then

$$\langle \mathcal{A}, \mathcal{B}\rangle = \tfrac{1}{4}\int \langle \nabla V, \nabla|\eta|^2\rangle = \tfrac{\lambda_F}{4}\int V|\eta|^2,$$

because $-\Delta V = \lambda_F V$ by Lemma F.1. Writing $\|\mathcal{A}-\mathcal{B}\|^2 = \|\frac{5}{8}\mathcal{A}-\mathcal{B}\|^2 + \frac{39}{64}\|\mathcal{A}\|^2 - \frac{3}{4}\langle\mathcal{A},\mathcal{B}\rangle$ and dropping the first term,

$$\|\mathcal{A}-\mathcal{B}\|^2 \geq \tfrac{39}{64}\mu_0\|\eta\|^2 - \tfrac{1}{2}\int V|\eta|^2$$

by Lemma F.2. With $\mathrm{Ric} = \frac{3}{2}$, (F.3) gives $\lambda\|\eta\|^2 \geq (\frac{39}{64}\mu_0 + \frac{3}{2})\|\eta\|^2 - \int(\mathrm{Hess}\, V(\eta,\eta) + \frac{1}{2}V|\eta|^2)$, and the claim follows from the pointwise bound

$$\mathrm{Re}\,\mathrm{Tr}(MX^2) + \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}(M)\,|X|^2 \leq \tfrac{r}{2}|X|^2, \qquad M = bU \in rK, \tag{F.4}$$

since $\mathrm{Hess}\, V(X,X) = \mathrm{Re}\,\mathrm{Tr}(MX^2)$ and $V = \mathrm{Re}\,\mathrm{Tr}\, M$. The maximum over $|X| = 1$ of the left side is convex in M, so it suffices to take $M = rW$ with $W \in \mathrm{SU}(3)$ and, after a conjugation, $W = \mathrm{diag}(e^{i\theta_k})$ with $\sum_k \theta_k = 0$; put $c_k = \cos\theta_k$. Write $X = iH$ with H Hermitian, traceless, $\mathrm{Tr}\, H^2 = 1$, with diagonal d_k and entries H_{jk}. Then

$$\mathrm{Re}\,\mathrm{Tr}(WX^2) = -\sum_k c_k d_k^2 - \sum_{j<k}(c_j + c_k)|H_{jk}|^2,$$

with $\sum_k d_k^2 + 2\sum_{j<k}|H_{jk}|^2 = 1$, so the maximum is the larger of $q_{\mathrm{off}} = \max_{j<k} -\frac{1}{2}(c_j + c_k)$ and $q_{\mathrm{diag}} = \max\{-\sum c_k d_k^2 : \sum d_k = 0,\ \sum d_k^2 = 1\}$. If l is the third index, $q_{\mathrm{off}} + \frac{1}{2}\sum_k c_k = \frac{1}{2}c_l \leq \frac{1}{2}$. The value q_{diag} is minus the smaller eigenvalue of $\mathrm{diag}(c)$ compressed to the plane $\sum d_k = 0$; the eigenvalues of the compression solve $\sum_k (c_k - \mu)^{-1} = 0$, that is $3\mu^2 - 2e_1\mu + e_2 = 0$ with e_1, e_2 the elementary symmetric functions of the c_k, so $q_{\mathrm{diag}} = \frac{1}{3}(\sqrt{e_1^2 - 3e_2} - e_1)$. The inequality $q_{\mathrm{diag}} + \frac{1}{2}e_1 \leq \frac{1}{2}$ is $\sqrt{e_1^2 - 3e_2} \leq \frac{1}{2}(3 - e_1)$, that is $e_1^2 - 4e_2 + 2e_1 - 3 \leq 0$. When $\sum\theta_k = 0$, $\sum c_k^2 = 1 + 2c_1c_2c_3$, and with this identity the left side equals $-2(1-c_1)(1-c_2)(1-c_3) \leq 0$. □

The weight $\frac{5}{8}$ is the one that gives the coefficient $\frac{1}{2}$ of V, and (F.4) is an equality at $W = I$.

F.4. **Haar integrals.** For $h \in \mathbb{C}^{3\times 3}$ let $E_2(h) = \frac{1}{2}((\mathrm{Tr}\, h)^2 - \mathrm{Tr}\, h^2)$, the sum of the principal 2×2 minors of h. All integrals are over $\mathrm{SU}(3)$ with Haar measure; put $t = \mathrm{Tr}\, U$,

$$Z_*(r) = \int e^{r\,\mathrm{Re}\,\mathrm{Tr}\, U}dU, \qquad m_k = \int (\mathrm{Re}\,\mathrm{Tr}\, U)^k dU.$$

Lemma F.4. (i) *The integral of a monomial of degree j in the entries of U and k in those of $\bar U$ vanishes unless $j \equiv k$ modulo three; $\int u_{ij}\bar u_{kl}dU = \frac{1}{3}\delta_{ik}\delta_{jl}$ and $\int u_{ab}u_{cd}u_{ef}dU = \frac{1}{6}\varepsilon_{ace}\varepsilon_{bdf}$.*

(ii) *For every h the integrals of $\mathrm{Tr}(hUhU)$ and $(\mathrm{Tr}\, hU)^2$ vanish, $\int|\mathrm{Tr}\, hU|^2dU = \frac{1}{3}\|h\|_{\mathrm{HS}}^2$,*

$$\int \mathrm{Tr}(hUhU)\,\mathrm{Tr}\, U\, dU = -\tfrac{1}{3}E_2(h), \qquad \int (\mathrm{Tr}\, hU)^2\,\mathrm{Tr}\, U\, dU = \tfrac{1}{3}E_2(h),$$

and $|E_2(h)| \leq \|h\|_{\mathrm{HS}}^2$.

(iii) $\int t^a \bar{t}^b dU$ *vanishes unless* $a \equiv b$ *modulo three, does not change when* a *and* b *are exchanged, and equals* 1 *at* $(a,b) = (0,0)$, $(1,1)$, $(3,0)$, 2 *at* $(2,2)$, 3 *at* $(4,1)$, 5 *at* $(6,0)$, 6 *at* $(3,3)$, 11 *at* $(5,2)$, 21 *at* $(7,1)$ *and* 23 *at* $(4,4)$. *Hence* $m_0 = 1$, $m_1 = 0$,

$$(m_2, \dots, m_8) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{3}{4}, \tfrac{15}{16}, \tfrac{65}{32}, \tfrac{231}{64}, \tfrac{973}{128}),$$

and $|m_k| \le 3^{k-8} m_8$ *for* $k \ge 8$.

Proof. (iii) By Schur–Weyl duality [21, Thm. 6.3], $t^a = \sum_{\lambda \vdash a} f^\lambda s_\lambda(U)$ with f^λ the number of standard Young tableaux of shape λ, and on SU(3) the Schur polynomial s_λ vanishes when λ has more than three rows and is otherwise the irreducible character of highest weight $(\lambda_1 - \lambda_3, \lambda_2 - \lambda_3)$. By Schur orthogonality [10, Ch. 2], $\int t^a \bar{t}^b dU$ is the sum of $f^\lambda f^\mu$ over the pairs $\lambda \vdash a$, $\mu \vdash b$ with at most three rows and $\lambda - \mu$ a multiple of $(1,1,1)$; such pairs exist only when $a \equiv b$ modulo three, and the sum is symmetric in a, b. For $a \ge b$ and $a + b \le 8$ the pairs are $((1),(1))$; $((1,1,1),\varnothing)$; $\lambda = \mu \in \{(2),(1,1)\}$; $((2,1,1),(1))$; $((2,2,2),\varnothing)$; $\lambda = \mu \in \{(3),(2,1),(1,1,1)\}$; $((3,1,1),(2))$ and $((2,2,1),(1,1))$; $((3,2,2),(1))$; and $\lambda = \mu \in \{(4),(3,1),(2,2),(2,1,1)\}$. The hook length formula gives $f^{(2,1)} = 2$, $f^{(3,1)} = f^{(2,1,1)} = 3$, $f^{(2,2)} = 2$, $f^{(2,2,2)} = 5$, $f^{(3,1,1)} = 6$, $f^{(2,2,1)} = 5$ and $f^{(3,2,2)} = 21$, hence the values, the last one being $1 + 9 + 4 + 9$. Since $(\operatorname{Re}\operatorname{Tr} U)^k = 2^{-k} \sum_a \binom{k}{a} t^a \bar{t}^{k-a}$,

$$m_2 = \tfrac{2}{4}, \quad m_3 = \tfrac{2}{8}, \quad m_4 = \tfrac{6\cdot 2}{16}, \quad m_5 = \tfrac{2\cdot 5\cdot 3}{32},$$
$$m_6 = \tfrac{2\cdot 5 + 20\cdot 6}{64}, \quad m_7 = \tfrac{2\cdot 21\cdot 11}{128}, \quad m_8 = \tfrac{2\cdot 8\cdot 21 + 70\cdot 23}{256}.$$

The eigenvalues of U have modulus one, so $|\operatorname{Re}\operatorname{Tr} U| \le 3$ and $|\operatorname{Re}\operatorname{Tr} U|^k \le 3^{k-8} (\operatorname{Re}\operatorname{Tr} U)^8$ for $k \ge 8$.

(i) The map $U \mapsto \omega U$, $\omega = e^{2\pi i/3}$, preserves Haar measure and multiplies the monomial by ω^{j-k}. For a unitary representation π of SU(3), $\int \pi(U) dU$ is the orthogonal projection onto the invariant vectors, whose dimension is the mean of the character of π. On $\mathbb{C}^3 \otimes \overline{\mathbb{C}^3}$ the invariant vectors are the multiples of the identity, which gives the second formula. On $(\mathbb{C}^3)^{\otimes 3}$ the character is t^3, whose mean is 1 by (iii), and ε is invariant because $\det U = 1$; the projection onto the line of ε, whose squared norm is 6, gives the third formula.

(ii) The first two integrals have degree two in U and none in $\bar{U}$, and $\int |\operatorname{Tr} hU|^2 dU = \frac{1}{3} \sum |h_{ji}|^2$ by (i). By (i), $\int h_{pa} u_{aq} h_{qc} u_{cp} u_{ee} dU = \frac{1}{6} h_{pa} h_{qc} \varepsilon_{ace} \varepsilon_{qpe}$, summed over repeated indices, and $\sum_e \varepsilon_{ace} \varepsilon_{qpe} = \delta_{aq}\delta_{cp} - \delta_{ap}\delta_{cq}$, which gives $\frac{1}{6}(\operatorname{Tr} h^2 - (\operatorname{Tr} h)^2)$; the second integral is the same with ε_{pqe} in place of ε_{qpe}, which changes the sign. Finally $E_2(h)$ is the trace of the second exterior power of h, whose singular values are the products $\sigma_i \sigma_j$, $i < j$, of those of h; so $|E_2(h)| \le \sum_{i<j} \sigma_i \sigma_j = \frac{1}{2}((\sum_i \sigma_i)^2 - \sum_i \sigma_i^2) \le \sum_i \sigma_i^2$. □

F.5. **The modulus.**

Lemma F.5. *Let* $r \ge 0$ *with* $\gamma(r) > 0$, *put*

$$\epsilon(r) = 1 - \frac{5\, e^{-r/5}}{9\, Z_*(r)},$$

and let $b \in rK$ *and* $\|h\|_{\mathrm{HS}} = 1$. *Then*

$$\nu_b |\nabla \varphi_h|^2 \le \epsilon(r), \quad \operatorname{Var}_{\nu_b}(\varphi_h) \le \frac{\epsilon(r)}{\gamma(r)}, \quad L_h(b) \le \frac{\sqrt{\epsilon(r)}}{\gamma(r)},$$

so that $\Lambda(r) \le \sqrt{\epsilon(r)}/\gamma(r)$ *and, by* (F.2), $C^T_{ek} \le r\sqrt{\epsilon(r)}/(6\gamma(r))$. *The function* ϵ *is nondecreasing on* $[0, \infty)$.

Proof. By Lemma F.1, $\nabla \varphi_h(U) = -PA(M)$ with $M = hU$, and $\|M\|_{\mathrm{HS}} = 1$. The anti-Hermitian part has $|A(M)|^2 = \frac{1}{2}(\|M\|^2 - \operatorname{Re}\operatorname{Tr} M^2)$, and P removes from it the multiple

$\frac{1}{3}\operatorname{Tr} A(M) = \frac{i}{3}\operatorname{Im}\operatorname{Tr} M$ of the identity, whose squared norm is $\frac{1}{3}(\operatorname{Im}\operatorname{Tr} M)^2$. Hence

$$|\nabla\varphi_h|^2 = 1 - \tfrac{1}{2}F_h,$$
$$F_h(U) = 1 + \operatorname{Re}\operatorname{Tr}(hUhU) + \tfrac{2}{3}(\operatorname{Im}\operatorname{Tr} hU)^2,$$

and $F_h \geq 0$ because $|\operatorname{Re}\operatorname{Tr} M^2| \leq \|M\|^2$. For complex numbers X, Y, $\operatorname{Re} X \operatorname{Re} Y = \frac{1}{2}\operatorname{Re}(XY) + \frac{1}{2}\operatorname{Re}(X\bar{Y})$ and $(\operatorname{Im} X)^2 = \frac{1}{2}|X|^2 - \frac{1}{2}\operatorname{Re} X^2$. With these identities, the vanishing of the integrals of degree two in U and one in $\bar{U}$, or the converse, and Lemma F.4(ii),

$$\int F_h\, dU = 1 + \tfrac{2}{3}\cdot\tfrac{1}{6} = \tfrac{10}{9}, \quad \int F_h \operatorname{Re}\operatorname{Tr} U\, dU = -\tfrac{2}{9}\operatorname{Re} E_2(h),$$

the second as the sum of $-\frac{1}{6}\operatorname{Re} E_2(h)$ and $\frac{2}{3}\cdot(-\frac{1}{12})\operatorname{Re} E_2(h)$. For $W \in \mathrm{SU}(3)$ the substitution $U \mapsto W^{-1}U$ and Lemma F.4(ii) give

$$\int F_h \operatorname{Re}\operatorname{Tr}(WU)\, dU = \int F_{hW^{-1}} \operatorname{Re}\operatorname{Tr} U\, dU \geq -\tfrac{2}{9}.$$

Write $b = r\sum_i \lambda_i W_i$ as a convex combination of elements of $\mathrm{SU}(3)$ and $V = \operatorname{Re}\operatorname{Tr}(Ub)$. Under the probability measure $\pi = \frac{9}{10}F_h\, dU$ the mean of V is at least $-\frac{9}{10}\cdot\frac{2}{9}r = -\frac{r}{5}$, so Jensen's inequality gives $\int F_h e^V dU = \frac{10}{9}\pi(e^V) \geq \frac{10}{9}e^{-r/5}$. The function $c \mapsto \log\int e^{\operatorname{Re}\operatorname{Tr}(Uc)}dU$ is convex by Hölder's inequality and equals $\log Z_*(r)$ at every $c = rW$, $W \in \mathrm{SU}(3)$, by the invariance of Haar measure; hence $\int e^V dU \leq Z_*(r)$ and $\nu_b(F_h) \geq \frac{10}{9}e^{-r/5}/Z_*(r)$, which is the first bound.

The spectral gap of ν_b is at least $\gamma(r)$ by Lemma F.3. The Poincaré inequality gives $\operatorname{Var}_{\nu_b}(\varphi_h) \leq \nu_b|\nabla\varphi_h|^2/\gamma(r)$, the second bound, and, for g 1-Lipschitz for d, hence for the geodesic distance, $\operatorname{Var}_{\nu_b}(g) \leq 1/\gamma(r)$; so $\operatorname{Cov}_{\nu_b}(g, \varphi_h) \leq \operatorname{Var}_{\nu_b}(g)^{1/2}\operatorname{Var}_{\nu_b}(\varphi_h)^{1/2} \leq \sqrt{\epsilon(r)}/\gamma(r)$. Finally $\log Z_*$ is convex in r with derivative $m_1 = 0$ at $r = 0$, so Z_* and ϵ are nondecreasing on $[0, \infty)$. □

The losses in this bound are Jensen's inequality, the Poincaré inequality for φ_h, and the gap $\gamma(r)$ itself: at $b = 0$ the variance of φ_h is $\frac{1}{6}$ by Lemma F.4(ii) and the gap is $\lambda_F = \frac{8}{3}$, while the lemma uses $\epsilon(0) = \frac{4}{9}$, which is exact, and $\gamma(0) = \frac{283}{128}$, so at $b = 0$ only the gap is lost.

Appendix G. A finite-size criterion for the cube condition

This appendix proves the finite-size criterion used in Sec. 11: a gap for the dynamics that resamples the blocks of one patch of m^4 blocks, uniform in the configuration outside the patch, gives the cube condition on every large torus. Blocks that share no plaquette give commuting projections, and this replaces the disjointness used in the finite-size criteria of Knabe [34] and of Gosset and Mozgunov [25] for quantum spin systems.[30]

Setting. The notation is that of Sec. 6 and of the setting of Sec. 11. Fix integers $d \geq 2$ and $m \geq 4$ with $d \mid L$, put $K = L/d$, $n = 2d - 1$, and fix a shift $a \in [0, d)^4$. For j in the coarse torus $T_K = (\mathbb{Z}/K\mathbb{Z})^4$ let $r_j = a + dj$, let $B_j = V_{r_j}$ be the set of links with base point in $r_j + [0, n)^4$, and write $\mathcal{P}_j = \mathcal{P}_{r_j}$. Displacements $v = j' - j$ in T_K are represented in $(-K/2, K/2]^4$. Two sets A, A' of links are *non-interacting* if no plaquette contains a link of each; for a set A of links, P_A is the conditional expectation under μ given the links outside A.

Lemma G.1. (i) *For non-interacting sets A, A' of links, $P_A P_{A'} = P_{A'} P_A = P_{A\cup A'}$, and $(I - P_A)(I - P_{A'})$ is an orthogonal projection.*
(ii) *If $K \geq 4$ and $|j' - j|_\infty \geq 2$, then B_j and $B_{j'}$ are non-interacting.*
(iii) *The blocks B_j, $j \in T_K$, cover E.*

[30] In those criteria the local terms of a Hamiltonian act on overlapping sets of sites, and the gap of the whole system is bounded below by the gap of a finite patch minus a correction that decays with the size of the patch. Here the local terms are the complements $I - \mathcal{P}_j$ of conditional expectations, and two of them commute as soon as the blocks are at distance at least two.

(iv) *If $f \in L^2(\mu)$ and $\mathcal{P}_j f = f$ for every $j \in T_K$, then f is almost surely constant.*

Proof. (i) Every link lies on a plaquette, so $A \cap A' = \emptyset$. Given the links outside $A \cup A'$, the law of $(U_A, U_{A'})$ has density proportional to $\prod_{p \cap (A \cup A') \neq \emptyset} \varphi(U_p)$, and each factor involves the links of exactly one of A, A'. The density is therefore a product of a function of U_A and a function of $U_{A'}$, with coefficients depending on the links outside $A \cup A'$. Hence U_A and $U_{A'}$ are conditionally independent, the conditional law γ_A of U_A given the links outside A does not depend on $U_{A'}$, and likewise for A'. So $P_A P_{A'} f$ and $P_{A'} P_A f$ are both the integral of f against $\gamma_A \otimes \gamma_{A'}$, which is $P_{A \cup A'} f$, and $(I - P_A)(I - P_{A'}) = I - P_A - P_{A'} + P_{A \cup A'}$ is self-adjoint and idempotent.

(ii) Let $|v_i| \geq 2$ for $v = j' - j$. In the coordinate i the base points of the links of B_j and of $B_{j'}$ lie in $dj_i + a_i + [0, 2d-2]$ and in $dj_i + a_i + dv_i + [0, 2d-2]$ modulo L. If $v_i \geq 2$, their distance in $\mathbb{Z}/L\mathbb{Z}$ is at least $\min\{(v_i - 2)d + 2, (K - v_i - 2)d + 2\} \geq 2$, since $v_i \leq K/2$ and $K \geq 4$; the case $v_i \leq -2$ is symmetric. A plaquette meeting both blocks would contain two links whose base points differ by at least two in the coordinate i, while the base points of the links of a plaquette differ by at most one in each coordinate.

(iii) The sets $r_j + [0, d)^4$ partition Λ_L, and $[0, d) \subset [0, n)$.

(iv) The density of μ with respect to the Haar measure h of G^E is bounded above and below by positive constants, so μ and h have the same null sets and $L^2(\mu) = L^2(h)$. If $\mathcal{P}_j f = f$, then f is almost surely a function of $U_{E \setminus B_j}$, and it is fixed by the conditional expectation Q_j of h given $U_{E \setminus B_j}$, which integrates out U_{B_j}. By Fubini's theorem the Q_j commute and their product integrates out the links of $\bigcup_j B_j = E$, so $f = \int f \, dh$ almost surely. □

In four dimensions the displacements v with $|v|_\infty = 1$ are 80 in number, and those with $|v|_\infty \geq 2$ give commuting projections. For $s \in T_K$ let $S_s = s + [0, m-1]^4$ be the *patch* at s and $S_s^\circ = s + [0, m-2]^4$ the set of its $(m-1)^4$ block indices, and put

$$H = \sum_{j \in T_K} (I - \mathcal{P}_j), \qquad H_s = \sum_{j \in S_s^\circ} (I - \mathcal{P}_j).$$

These are bounded nonnegative operators on $L^2(\mu)$ and preserve the subspace $\mathcal{A}$ of gauge-invariant functions.

Lemma G.2. *Let $m \geq 4$, $p = m - 1$, $K \geq 2m$, let $\mathcal{K}$ be $L^2(\mu)$ or $\mathcal{A}$, and suppose that $H_s^2 \geq \gamma H_s$ on $\mathcal{K}$ for every $s \in T_K$, with $\gamma > 0$. Then $H^2 \geq \varepsilon H$ on $\mathcal{K}$ with*

$$\varepsilon = 1 - \frac{(1-\gamma)p^4 + 72p^3 - 168p^2 + 96p - 16}{p^3(p-2)} \geq \frac{\gamma p - 74}{p - 2}.$$

In particular $\varepsilon \geq \gamma/2$ if $m - 1 \geq 148/\gamma$.

Proof. All operators below preserve $\mathcal{K}$, and a restriction of an orthogonal projection to an invariant closed subspace is an orthogonal projection; so it suffices to argue on $L^2(\mu)$ with inequalities read on $\mathcal{K}$. Write $\mathbf{P}_j = I - \mathcal{P}_j$ and $A_v = \sum_j \mathbf{P}_j \mathbf{P}_{j+v}$. Since $K \geq 2m$, two indices j and $j + v$ lie in a common S_s° only if $|v_i| \leq m - 2$ for all i, and then for exactly $c_v = \prod_{i=1}^4 (p - |v_i|)$ points s. Let O be the set of the 80 displacements with $|v|_\infty = 1$, D the set of those with $|v|_\infty \geq 2$, and D' the set of $v \in D$ with all $|v_i| \leq m - 2$. For $v \in D$, $\mathbf{P}_j \mathbf{P}_{j+v}$ is an orthogonal projection by Lemma G.1(i),(ii), so $A_v \geq 0$. Summing the hypothesis over s,

$$\sum_s H_s^2 = c_0 H + \sum_{v \in O} c_v A_v + \sum_{v \in D'} c_v A_v \geq \gamma \sum_s H_s = \gamma c_0 H.$$

Put $c_* = p^3(p-2)$; then $c_v \leq c_*$ for $v \in D'$, since one factor of c_v is at most $p - 2$ and the others are at most p. For orthogonal projections $\mathbf{P}, \mathbf{P}'$, $(\mathbf{P} \pm \mathbf{P}')^2 \geq 0$ gives $-(\mathbf{P} + \mathbf{P}') \leq \mathbf{P}\mathbf{P}' + \mathbf{P}'\mathbf{P} \leq \mathbf{P} + \mathbf{P}'$, so $-2H \leq A_v + A_{-v} \leq 2H$. As O is symmetric and

$c_{-v} = c_v$,

$$\sum_{v\in O}(c_* - c_v)A_v \geq -\Sigma H, \qquad \Sigma = \sum_{v\in O}|c_* - c_v|,$$

and therefore

$$c_* \sum_{v\in O} A_v = \sum_{v\in O} c_v A_v + \sum_{v\in O}(c_* - c_v)A_v \geq -\big((1-\gamma)c_0 + \Sigma\big)H - \sum_{v\in D'} c_v A_v.$$

Since $H^2 = \sum_v A_v = H + \sum_{v\in O} A_v + \sum_{v\in D} A_v$,

$$H^2 \geq \Big(1 - \frac{(1-\gamma)c_0 + \Sigma}{c_*}\Big)H + \sum_{v\in D'}\Big(1 - \frac{c_v}{c_*}\Big)A_v + \sum_{v\in D\setminus D'} A_v,$$

and the last two sums are nonnegative. It remains to compute Σ. A displacement $v \in O$ with k nonzero coordinates has $c_v = p^{4-k}(p-1)^k$, and there are 8, 24, 32 and 16 of them for $k = 1, 2, 3, 4$. The differences $c_v - c_*$ are

$$p^3, \qquad p^2, \qquad -p(p^2 - 3p + 1), \qquad -2p(p-1)(p-2) + 1,$$

with the signs shown since $p \geq 3$. Hence $\Sigma = 8p^3 + 24p^2 + 32p(p^2 - 3p + 1) + 16(2p(p-1)(p-2) - 1) = 72p^3 - 168p^2 + 96p - 16$, and with $c_0 = p^4$ this is the formula for ε. As $-168p^2 + 96p - 16 < 0$, $\varepsilon \geq 1 - ((1-\gamma)p + 72)/(p-2) = (\gamma p - 74)/(p-2)$, and if $\gamma p \geq 148$ this is at least $(\gamma p - 74)/p \geq \gamma/2$. □

For a patch S_s let $\Pi_s = \bigcup_{j\in S_s^\circ} B_j$, the set of links with base point in $r_s + [0, md-1)^4$. For $L \geq md + 2$ the patch and the plaquettes meeting it embed in $\mathbb{Z}^4$, and the conditional law of U_{Π_s} under μ given the links outside Π_s is the measure ν_Q^ω of Sec. 11 for the box $Q = r_s + [0, md-1)^4$, with ω the configuration outside. For $j \in S_s^\circ$ write $\mathcal{P}_j^\omega$ for the conditional expectation under ν_Q^ω given the links of $\Pi_s \setminus B_j$.

Definition G.3. The *gauge-invariant patch gap* $\gamma_{d,m}$ is the largest $\gamma \geq 0$ such that

$$\gamma \operatorname{Var}_{\nu_Q^\omega}(F) \leq \sum_{j\in S_s^\circ} \nu_Q^\omega\big((F - \mathcal{P}_j^\omega F)^2\big)$$

for every patch box $Q = r + [0, md-1)^4$ of $\mathbb{Z}^4$, every exterior configuration ω and every $F \in \mathcal{A}_Q^\omega$.

By translation invariance $\gamma_{d,m}$ depends only on G, φ, d and m. It is the gap, on gauge-invariant functions, of the dynamics that resamples the $(m-1)^4$ blocks of one patch, uniform in the configuration outside the patch.

Theorem G.4 (finite-size criterion)**.** *Let $d \geq 2$, $m \geq 4$, $d \mid L$, L even, $L \geq 2md$, and suppose that $\gamma_{d,m} \geq \gamma > 0$ and that the number ε of Lemma G.2 is positive. Then*

$$\frac{\varepsilon}{16}\operatorname{Var}_\mu(f) \leq \frac{1}{n^4}\sum_{x\in\Lambda_L}\mu\big((f - \mathcal{P}_x f)^2\big) \qquad \textit{for every real } f \in \mathcal{A},$$

that is, $\mathrm{BP}(2d-1, \varepsilon/16)$*. If $m - 1 \geq 148/\gamma$, then $\varepsilon/16 \geq \gamma/32$.*

Proof. Fix a. Since $L \geq 2md$, $K \geq 2m \geq 8$.

The patch inequality on the torus. Let $F \in \mathcal{A}$; replacing F by $\int F(gU)\,dg$, the integral over all gauge transformations of Λ_L with Haar measure, we may take F invariant at every point. Write $U = (\sigma, \omega)$ with σ the links of Π_s and ω the others. Under μ the conditional law of σ given ω is ν_Q^ω, and for $j \in S_s^\circ$ the conditional law of U_{B_j} given the other links is the same under μ and under ν_Q^ω, so $(\mathcal{P}_j F)(\cdot, \omega) = \mathcal{P}_j^\omega(F(\cdot, \omega))$ for almost every ω and H_s acts on each fibre as $H_s^\omega = \sum_{j\in S_s^\circ}(I - \mathcal{P}_j^\omega)$. A gauge transformation of Λ_L that is trivial at the vertices not interior to Q fixes ω, so each fibre $F(\cdot, \omega)$ lies in $\mathcal{A}_Q^\omega$. On $\mathcal{A}_Q^\omega$ the operator H_s^ω is self-adjoint and nonnegative, preserves the constants and their orthogonal complement (Lemma 11.1(iii)), and by Definition G.3 is at least γ on that complement. So its spectrum

there lies in $\{0\} \cup [\gamma, \infty)$ and $(H_s^\omega)^2 \geq \gamma H_s^\omega$. Integrating over ω, $\|H_s F\|^2 \geq \gamma \langle F, H_s F\rangle$ for $F \in \mathcal{A}$, that is, $H_s^2 \geq \gamma H_s$ on $\mathcal{A}$.

Conclusion. By Lemma G.2 with $\mathcal{K} = \mathcal{A}$, $H^2 \geq \varepsilon H$ on $\mathcal{A}$, so the spectrum of H on $\mathcal{A}$ lies in $\{0\} \cup [\varepsilon, \infty)$. If $F \in \mathcal{A}$ and $HF = 0$, then $\sum_j \|F - \mathcal{P}_j F\|^2 = \langle F, HF\rangle = 0$, so F is constant by Lemma G.1(iv). Hence for real $f \in \mathcal{A}$ the function $g = f - \mu(f) \in \mathcal{A}$ is orthogonal to the kernel of H on $\mathcal{A}$, and

$$\varepsilon \operatorname{Var}_\mu(f) \leq \langle g, Hg\rangle = \sum_{j \in T_K} \mu((f - \mathcal{P}_{r_j} f)^2).$$

As a runs over $[0, d)^4$ and j over T_K, $r_j = a + dj$ runs over Λ_L once. Averaging over a gives $\varepsilon d^4 \operatorname{Var}_\mu(f) \leq \sum_x \mu((f - \mathcal{P}_x f)^2)$, and $d^4 \geq n^4/16$. Since $L \geq 8d \geq 2n + 4$ this is $\mathrm{BP}(2d - 1, \varepsilon/16)$, and the last claim is Lemma G.2. □

The passage from the patch to the torus costs a factor: at $\varphi \equiv 1$ the links are independent, $\gamma_{d,m} \geq 1$ by the Efron–Stein inequality [20], and the theorem gives $\mathrm{BP}(2d - 1, \varepsilon/16)$ with $\varepsilon < 1$, against $\mathrm{BP}(n, 1)$ from Proposition 8.1.

The hypothesis of Theorem G.4 is a statement about every boundary condition of the patch. It is in fact equivalent to the local inequality $H_s^2 \geq \gamma H_s$ on the torus, so that it is not a stronger hypothesis than the one the proof uses, and it need only be imposed on the functions that are even under the centre. Let $\mathcal{G}_{\mathrm{ext}}$ be the group of gauge transformations of Λ_L that are trivial at the vertices interior to Q; it fixes the links of Π_s with both end points interior, acts on the other links of Π_s and on ω, and commutes with $\mathcal{G}_Q$, and every gauge transformation of Λ_L is a product of one element of each. For $x \in G$ let $c_x \in \mathcal{G}_{\mathrm{ext}}$ be equal to x at every non-interior vertex; it acts on ω by conjugation, so $c_z \omega = \omega$ for z in the centre Z of G, while c_z multiplies the links of Π_s with one interior end point by $z^{\pm 1}$. Write $\mathrm{Stab}(\omega) = \{g \in \mathcal{G}_{\mathrm{ext}} : g\omega = \omega\} \ni c_z$, $H^\omega = \sum_{j \in S_s^\circ} (I - \mathcal{P}_j^\omega)$ on $L^2(\nu_Q^\omega)$ and

$$\Phi_\omega(F) = \|H^\omega F\|^2 - \gamma \langle F, H^\omega F\rangle \qquad (F \in L^2(\nu_Q^\omega)).$$

A function $F \in \mathcal{A}_Q^\omega$ is *Z-even* if $F \circ c_z = F$ for all $z \in Z$.

Proposition G.5 (the patch inequality fibre by fibre)**.** *Let $d \geq 2$, $m \geq 4$, $d \mid L$, L even, $L \geq 2md$, and $\gamma > 0$. The following are equivalent.*

(a) *$H_s^2 \geq \gamma H_s$ on $\mathcal{A}$ for every $s \in T_K$.*
(b) *$\Phi_\omega(F) \geq 0$ for every ω and every Z-even $F \in \mathcal{A}_Q^\omega$.*
(c) *$\Phi_\omega(F) \geq 0$ for every ω and every $\mathrm{Stab}(\omega)$-invariant $F \in \mathcal{A}_Q^\omega$.*
(d) (b) *holds for a dense set of ω, or for μ-almost every ω.*

In particular Theorem G.4 holds with Definition G.3 restricted to Z-even functions, and no condition on a set of boundary conditions that is not dense can replace (b).

Lemma G.6. (i) *For a continuous function F of σ, $\omega \mapsto \Phi_\omega(F)$ is continuous and depends only on the exterior links of the plaquettes meeting $V(Q)$.*
(ii) *For $g \in \mathcal{G}_{\mathrm{ext}}$, $\Phi_\omega(F \circ g) = \Phi_{g\omega}(F)$.*
(iii) *$|\Phi_\omega(F + R) - \Phi_\omega(F)| \leq C_m \|R\|_\infty (2\|F\|_\infty + \|R\|_\infty)$ with $C_m = (m-1)^8 + \gamma(m-1)^4$.*
(iv) *The exterior graph, whose vertices are the non-interior vertices and whose edges are the links outside Π_s, is connected. For a spanning tree $\mathcal{T}$ with root $\mathfrak{r}$ there is a unique continuous $k(\omega) \in \mathcal{G}_{\mathrm{ext}}$ with $k_{\mathfrak{r}} = 1$ and $(k(\omega)\omega)_e = 1$ for $e \in \mathcal{T}$, and $k(h\omega) = c_{h_{\mathfrak{r}}} k(\omega) h^{-1}$ for $h \in \mathcal{G}_{\mathrm{ext}}$.*
(v) *In a compact Lie group there are finitely many elements $g_1, \dots, g_k$ whose centralisers intersect in the centre; for $\mathrm{SU}(2)$ two non-commuting elements suffice.*

Proof. (i) $\mathcal{P}_j^\omega F(\sigma) = \int F(\sigma_{-B_j}, \tau) p_j^\omega(\tau \mid \sigma_{-B_j})\, d\tau$ with a density p_j^ω proportional to $\prod_{p \cap B_j \neq \emptyset} \varphi(U_p)$, jointly continuous and bounded below, so $\mathcal{P}_j^\omega F$ is jointly continuous in (σ, ω); expanding $\Phi_\omega(F)$ into the terms $\nu_Q^\omega((F - \mathcal{P}_j F)(F - \mathcal{P}_{j'} F))$ and $\nu_Q^\omega((F - \mathcal{P}_j F)^2)$,

each is the integral of a jointly continuous function against the jointly continuous density of ν_Q^ω, and only the plaquettes meeting $V(Q)$ enter. (ii) $\sigma \mapsto g\sigma$ preserves Haar measure, conjugates every plaquette variable, hence carries ν_Q^ω to $\nu_Q^{g\omega}$ since φ is a class function, and acts link by link, so it intertwines $\mathcal{P}_j^\omega$ with $\mathcal{P}_j^{g\omega}$. (iii) $\|H^\omega\| \le (m-1)^4$, $\|H(F+R)\|^2 - \|HF\|^2 = \langle HR, H(2F+R)\rangle$, likewise for the second term, and $\|\cdot\|_{L^2(\nu)} \le \|\cdot\|_\infty$. (iv) The vertices outside the open box of side $md-1$ form a connected subgraph of the torus since $L \ge 2md$, and each vertex of the lower faces of Q is joined to its neighbour outside by an exterior link. Let k_x be the inverse of the holonomy of ω along the tree path from $\mathfrak{r}$ to x; uniqueness is clear, and $c_{h_\mathfrak{r}} k(\omega) h^{-1}$ trivialises $h\omega$ on $\mathcal{T}$ and is 1 at $\mathfrak{r}$. (v) For closed subgroups $H' \subsetneq H$ either $\dim H' < \dim H$ or $H'_0 = H_0$ and H' has fewer components. Choose a finite set S for which $(\dim C(S), |\pi_0 C(S)|)$ is lexicographically least; then $C(S \cup \{g\}) = C(S)$ for every g, so $C(S) \subset \bigcap_g C(g) = Z \subset C(S)$. In SU(2) two distinct maximal tori meet in $\{\pm 1\}$. □

Proof of Proposition G.5. (b)⇒(a) and (c)⇒(a). Continuous elements are dense in $\mathcal{A}$, by averaging continuous approximants over the compact gauge group of Λ_L. For continuous $F \in \mathcal{A}$ and every ω the fibre $F_\omega = F(\cdot, \omega)$ lies in $\mathcal{A}_Q^\omega$, since $\mathcal{G}_Q$ fixes ω, and is $\mathrm{Stab}(\omega)$-invariant, since $F(g\sigma, \omega) = F(g\sigma, g\omega) = F(\sigma, \omega)$; in particular it is Z-even. As in the proof of Theorem G.4, $(H_s F)_\omega = H^\omega F_\omega$, so $\|H_s F\|^2 - \gamma\langle F, H_s F\rangle = \int \Phi_\omega(F_\omega)\, d\mu(\omega) \ge 0$, and (a) follows by density.

(b)⇒(c) and (b)⇒(d) are trivial, since $\mathrm{Stab}(\omega)$-invariant functions are Z-even.

(d)⇒(b). Suppose $\Phi_{\omega_0}(F_0) < 0$ for a Z-even $F_0 \in \mathcal{A}_Q^{\omega_0}$. Averaging continuous approximants in $L^2(\nu_Q^{\omega_0})$ over the compact group generated by $\mathcal{G}_Q$ and c_Z, which preserves $\nu_Q^{\omega_0}$ by Lemma 11.1(i) and Lemma G.6(ii), is an orthogonal projection, so F_0 may be taken continuous, $\mathcal{G}_Q$-invariant pointwise and Z-even; these properties do not depend on ω. By Lemma G.6(i), $\Phi_\omega(F_0) < 0$ on a neighbourhood of ω_0, so the set where (b) holds is closed, and (b) on a dense set gives (b) everywhere; and since the law of ω under μ has a density bounded below, almost every ω is a dense set.

(a)⇒(b). Suppose $\Phi_{\omega_0}(F_0) < 0$ with F_0 continuous, $\mathcal{G}_Q$-invariant and Z-even, as just arranged. *Generic stabiliser.* Take $g_1, \dots, g_k$ from Lemma G.6(v), exterior links $e_1, \dots, e_k$ on no plaquette meeting $V(Q)$, and the tree $\mathcal{T}$ avoiding them; this is possible since the exterior graph at distance at least 2 from Q contains cycles through any k prescribed links, and removing links that lie on cycles keeps the graph connected. Let ω_1 agree with ω_0 except that $(k(\omega_0)\omega_1)_{e_i} = g_i$; then $k(\omega_1) = k(\omega_0)$, since k depends on the tree links only, and $\Phi_{\omega_1}(F_0) = \Phi_{\omega_0}(F_0)$ by Lemma G.6(i). The slice point $\hat\omega_1 = k(\omega_1)\omega_1$ has $\{x : c_x\hat\omega_1 = \hat\omega_1\} = \bigcap_i C(g_i) = Z$. *Transport.* Put $F_1 = F_0 \circ k(\omega_1)^{-1}$; it is $\mathcal{G}_Q$-invariant, since $\mathcal{G}_{\mathrm{ext}}$ commutes with $\mathcal{G}_Q$, and Z-even, since c_z commutes with $k(\omega_1)$, and by Lemma G.6(ii), $\Phi_{\hat\omega_1}(F_1) = \Phi_{\omega_1}(F_0) = -2\delta < 0$. *Neighbourhoods.* By Lemma G.6(i) there is an open $U_1 \ni \hat\omega_1$ with $\Phi_{\omega'}(F_1) \le -\delta$ on U_1. By uniform continuity there is an open $W_0 \ni 1$ in G such that $\varepsilon = \sup\{|F_1(c_w\sigma) - F_1(\sigma)| : w \in W_0,\ \sigma\}$ satisfies $C_m\varepsilon(2\|F_1\|_\infty + \varepsilon) \le \delta/2$. There is an open $U \subset U_1$ containing $\hat\omega_1$ such that $c_x U \cap U \ne \emptyset$ implies $x \in W_0 Z$: otherwise there are $x_n \notin W_0 Z$ and $\omega_n \to \hat\omega_1$ with $c_{x_n}\omega_n \to \hat\omega_1$, a subsequence $x_n \to x$ has $c_x\hat\omega_1 = \hat\omega_1$, so $x \in Z \subset W_0 Z$, which is open, against $x_n \notin W_0 Z$. Let $\chi \ge 0$ be continuous, supported in U, with $\chi(\hat\omega_1) > 0$. *The test function.* Define

$$F(\sigma, \omega) = \int_G \chi(c_x k(\omega)\omega)\, F_1(c_x k(\omega)\sigma)\, dx.$$

It is bounded, measurable and $\mathcal{G}_Q$-invariant, and for $h \in \mathcal{G}_{\mathrm{ext}}$, Lemma G.6(iv) gives $k(h\omega)h = c_{h_\mathfrak{r}} k(\omega)$, so $F(h\sigma, h\omega)$ is the same integral with x replaced by $xh_\mathfrak{r}$, which is $F(\sigma, \omega)$ by the invariance of Haar measure; hence $F \in \mathcal{A}$. *Its fibres.* Put $\hat\omega = k(\omega)\omega$ and $a(\omega) = \int \chi(c_x\hat\omega)\, dx$, continuous with $a(\omega_1) > 0$. If $a(\omega) = 0$, then $F_\omega = 0$. Otherwise choose x_0 with $c_{x_0}\hat\omega \in U$; whenever $\chi(c_x\hat\omega) \ne 0$, both $c_{x_0}\hat\omega$ and $c_{x x_0^{-1}}(c_{x_0}\hat\omega)$ lie in U, so $x = wzx_0$ with $w \in W_0$, $z \in Z$, and by the centrality of z and the Z-evenness of F_1, $F_1(c_x\tau) = F_1(c_w c_{x_0}\tau)$ differs from

$F_1(c_{x_0}\tau)$ by at most ε. Hence $F_\omega = a(\omega)[F_1 \circ (c_{x_0}k(\omega)) + R_\omega]$ with $\|R_\omega\|_\infty \le \varepsilon$, and by Lemma G.6(ii),(iii) and the homogeneity of Φ_ω,

$$\Phi_\omega(F_\omega) \le a(\omega)^2\big(\Phi_{c_{x_0}\hat\omega}(F_1) + \delta/2\big) \le -a(\omega)^2\delta/2.$$

The set $\{a > 0\}$ is open and nonempty, hence of positive μ-mass, so $\|H_sF\|^2 - \gamma\langle F, H_sF\rangle = \int \Phi_\omega(F_\omega)\, d\mu(\omega) < 0$, against (a).

The last claims: the proof of Theorem G.4 uses only (b)⇒(a); and (a) forces (b) at every ω. □

At a central boundary condition, one on which c_x acts trivially for all x, $\mathrm{Stab}(\omega) \supset \{c_x\}$ and (c) asks the inequality for colour-singlet fibres only. The proposition identifies the hypothesis of Theorem G.4 with the local Knabe inequality on the torus, in the sense that the one holds for one $L \ge 2md$ if and only if the other holds for every boundary condition, hence for every such L; the passage from the global inequality $\mathrm{BP}(n_N, \varpi)$ of (D_ℓ) to that local inequality, the step that would make the slab hypothesis a consequence of the cube condition, is the one implication that is not proved here, and Proposition 11.18(v) shows it only in the class of the field rings.

Appendix H. The threshold one half, two-dimensional models and limits of methods

This appendix proves the statements about the hypothesis of Section 11 and about other approaches to (D) that are used in the text. Section H.1 shows that the threshold $c < \frac{1}{2}$ of Theorem D is the exact limit of the slab mechanism, the boundary value being attained by a massless Gaussian chain. Sections H.2 and H.3 treat two dimensions: the slab hypothesis for fat boxes holds along the weak-coupling tuning, uniformly in the boundary condition, while the same condition for strips of width one fails. Sections H.4 to H.6 prove that four other routes cannot give (D) without an input of the same strength: curvature of the Wilson measure, entropy factorisation, tensorisation over blocks of fixed lattice size, and the transfer matrix taken alone.

H.1. **The Gaussian chain.** For a probability measure and σ-algebras $\mathcal{F}_1, \mathcal{F}_2$ let $\rho(\mathcal{F}_1, \mathcal{F}_2) = \sup|E[XY]|$, the supremum over centred $X \in L^2(\mathcal{F}_1)$ and $Y \in L^2(\mathcal{F}_2)$ with $E[X^2] = E[Y^2] = 1$; this is the maximal correlation of Definition 11.2, without the constraint of gauge invariance.

Lemma H.1. *Let (U, V, M, W, X) be random variables with values in standard Borel spaces whose joint law has the form $\psi_1(u, v)\psi_2(v, m, w)\psi_3(w, x)\, d\lambda$ for a product measure λ and nonnegative functions ψ_i. Then $\rho(\sigma(U, V), \sigma(W, X)) \le \rho(\sigma(V), \sigma(W))$.*

Proof. By the factorisation, the conditional law of U given (V, M, W, X) has density proportional to $\psi_1(\cdot, V)$ and that of X given (U, V, M, W) has density proportional to $\psi_3(W, \cdot)$, and given (V, M, W) the variables U and X are conditionally independent. Let $f = f(U, V)$ and $g = g(W, X)$ be centred and square integrable. Then $E[f \mid V, M, W, X] = \tilde f(V)$ with $\tilde f = E[f \mid V]$, and $E[g \mid U, V, M, W] = \tilde g(W)$ with $\tilde g = E[g \mid W]$. Hence $E[fg] = E[E[fg \mid V, M, W]] = E[\tilde f(V)\tilde g(W)]$, where $\tilde f$ and $\tilde g$ are centred with $\|\tilde f\| \le \|f\|$ and $\|\tilde g\| \le \|g\|$. □

Lemma H.2 (Gebelein)**.** *If (V, W) is a nondegenerate Gaussian vector in $\mathbb{R}^2$ with correlation coefficient r, then $\rho(\sigma(V), \sigma(W)) = |r|$.*

Proof. The choice $X = V$, $Y = \pm W$, standardised, gives $\rho \ge |r|$. For the converse standardise V and W. Let H_n be the Hermite polynomials with generating function $\sum_n H_n(x)s^n/n! = e^{sx - s^2/2}$; then $h_n = H_n/\sqrt{n!}$ is a complete orthonormal system of $L^2(\mathbb{R}, \gamma)$, γ the standard Gaussian law [44, Sec. 1.1.1]. From $E[e^{sV - s^2/2}e^{uW - u^2/2}] = e^{rsu}$ and a comparison of coefficients one gets $E[h_j(V)h_k(W)] = \delta_{jk}r^j$; the double series converges

absolutely, since $H_n(x) = E(x+iZ)^n$ with $Z \sim \gamma$ gives $|H_n(x)| \le E(|x|+|Z|)^n$ and $Ee^{t(|V|+|W|+2|Z|)} < \infty$ for every t. For centred $f = \sum_{j\ge1} a_j h_j$ and $g = \sum_{j\ge1} b_j h_j$ in $L^2(\gamma)$ the form $(f,g) \mapsto E[f(V)g(W)]$ is bounded by $\|f\|\|g\|$, so passing to the limit in the partial sums, $E[f(V)g(W)] = \sum_{j\ge1} a_j b_j r^j$, whose modulus is at most $|r| \sum_j |a_j b_j| \le |r|\|f\|\|g\|$. □

The inequality $\rho \le |r|$ is Gebelein's theorem [23].

The chain. Let $\mu \ge 0$ and define $\kappa \ge 0$ by $2\cosh\kappa = 2+\mu^2$. For integers $p < q$ with $q-p \ge 2$ and real boundary values z_p, z_q let ν be the Gaussian law of $(z_{p+1}, \dots, z_{q-1})$ with density proportional to

$$\exp\Big(-\tfrac12 \sum_{j=p}^{q-1} (z_{j+1}-z_j)^2 - \tfrac12 \mu^2 \sum_{j=p+1}^{q-1} z_j^2\Big),$$

a nearest-neighbour Gaussian chain, massless for $\mu = 0$. Its precision matrix on the $q-p-1$ interior sites is $\mathsf{L} = \mathrm{tridiag}(-1, 2+\mu^2, -1)$, independent of z_p and z_q, which only shift the mean, and its conditional laws on subintervals are chains of the same form.

Lemma H.3. *Put $T = q-p$ and, for $0 \le i \le T$, $u(i) = \sinh(\kappa i)$ and $v(i) = \sinh(\kappa(T-i))$ if $\mu > 0$, $u(i) = i$ and $v(i) = T-i$ if $\mu = 0$. For $0 < i \le j < T$, $\mathrm{Cov}_\nu(z_{p+i}, z_{p+j}) = A\,u(i)v(j)$ with $A > 0$ independent of i and j; for $\mu = 0$, $A = 1/T$. Hence the correlation of z_{p+i} and z_{p+j} is $(u(i)v(j)/(v(i)u(j)))^{1/2}$.*

Proof. Both u and v satisfy $(2+\mu^2)f(l) = f(l-1) + f(l+1)$ (for $\mu > 0$ because $\sinh(x-\kappa) + \sinh(x+\kappa) = 2\cosh\kappa\sinh x$), $u(0) = v(T) = 0$, and both are positive on $(0,T)$. The covariance matrix is $G = \mathsf{L}^{-1}$, which is symmetric. For fixed j the column $G(\cdot, j)$, extended by 0 at 0 and T, satisfies the recursion at every $l \ne j$, so $G(i,j) = \alpha_j u(i)$ for $i \le j$ and $G(i,j) = \beta_j v(i)$ for $i \ge j$: a solution of the second-order recursion on $[0,j]$ that vanishes at 0 is a multiple of u, and similarly for v. For $i < j$, $\alpha_j u(i) = G(i,j) = G(j,i) = \beta_i v(j)$, so $\alpha_j/v(j) = \beta_i/u(i)$ for all $i < j$, a constant A; and $\alpha_j u(j) = \beta_j v(j)$ gives $G(j,j) = Au(j)v(j)$, so $A > 0$. For $\mu = 0$ the j-th entry of $\mathsf{L}G(\cdot, j)$ equals one, that is $A[2j(T-j) - (j-1)(T-j) - j(T-j-1)] = AT = 1$. □

Slab geometry. Let $s \ge 1$, $s \le w \le 2s-1$ and $M \ge 1$. Take the sites $I = [0, sM+w)$ with the two exterior sites -1 and $sM+w$, the blocks $R_t = [st, st+w)$ for $0 \le t \le M$, and, for $t < M$, $Q_t = R_t \cup R_{t+1}$ split into $Q_t^- = R_t \setminus R_{t+1} = [st, st+s)$, $Q_t^0 = R_t \cap R_{t+1} = [st+s, st+w)$ and $Q_t^+ = R_{t+1} \setminus R_t = [st+w, st+s+w)$. The condition $w \le 2s-1$ says that R_t and R_{t+2} contain no adjacent sites, so that, exactly as in Proposition 11.7, the resampling moves of non-adjacent blocks commute; Section 11 uses $s = d$ and $w = 2d-1$. Let ν be the chain on I with any exterior values, P_t the conditional expectation given the sites outside R_t, and ρ_t the supremum, over all values outside Q_t, of $\rho(\sigma(z_{Q_t^-}), \sigma(z_{Q_t^+}))$ under the conditional law of z_{Q_t}.

Proposition H.4 (the Gaussian chain)**.**

(i) *For every $t < M$, $\rho_t = \sinh(\kappa s)/\sinh(\kappa(w+1))$ if $\mu > 0$, and $\rho_t = s/(w+1)$ if $\mu = 0$. For $w = 2s-1$ this is $1/(2\cosh(\kappa s))$, and $\frac12$ for $\mu = 0$. For $\mu = 0$, $\rho_t \ge \frac12$ in every admissible geometry, with equality only for $w = 2s-1$.*

(ii) *For $\mu = 0$ the best constant γ_M in the inequality $\gamma_M \mathrm{Var}_\nu(F) \le \sum_{t=0}^{M} \nu((F - P_t F)^2)$ for all $F \in L^2(\nu)$ satisfies $\gamma_M \le 16/(3M)$.*

(iii) *For a subinterval split into consecutive nonempty intervals of A', r and C' sites, with the sites outside it conditioned, the maximal correlation of the two outer intervals is*

$$\Big(\frac{\sinh(\kappa A')\sinh(\kappa C')}{\sinh(\kappa(A'+r+1))\sinh(\kappa(C'+r+1))}\Big)^{1/2} \le e^{-\kappa(r+1)} \quad \text{if } \mu > 0,$$

with supremum $e^{-\kappa(r+1)}$ over A' and C', and $(A'C'/((A'+r+1)(C'+r+1)))^{1/2}$ if $\mu = 0$, with supremum 1.

Proof. (i) Given the values outside Q_t, the law of z_{Q_t} is the chain with $p = st-1$, $q = st+s+w$, $T = s + w + 1$. Its density factorises as in Lemma H.1 with $U = z_{[st,st+s-1)}$, $V = z_{st+s-1}$, $M = z_{Q_t^0}$, $W = z_{st+w}$ and $X = z_{(st+w,st+s+w)}$, the interactions being between nearest neighbours; so $\rho_t \le \rho(\sigma(V), \sigma(W))$, with equality since V and W are functions of the two outer parts. By Lemma H.2 this is the modulus of the correlation of V and W, which by Lemma H.3 with $i = s$, $j = w+1$ and $T = s+w+1$ equals $(u(s)v(w+1)/(v(s)u(w+1)))^{1/2}$. Here $v(w+1) = u(s)$ and $v(s) = u(w+1)$, which gives the formulas; they do not depend on the conditioning values. For $w = 2s-1$, $\sinh(\kappa s)/\sinh(2\kappa s) = 1/(2\cosh \kappa s)$. For $\mu = 0$, $w + 1 \le 2s$ gives $s/(w+1) \ge \frac{1}{2}$, with equality only when $w = 2s - 1$.

(ii) Let $T = sM+w+1 \ge sM \ge 2$ and take $F = z_j$ with $j = \lfloor T/2 \rfloor - 1 \in I$. By Lemma H.3 with $p = -1$, $\mathrm{Var}_\nu(F) = i(T-i)/T$ with $i = \lfloor T/2 \rfloor$, so $\mathrm{Var}_\nu(F) \ge (T^2-1)/(4T) \ge 3T/16$. If $j \notin R_t$, then $P_t F = F$. If $j \in R_t$, then $\nu((F - P_t F)^2)$ is the mean conditional variance of z_j given z_{st-1} and z_{st+w}, the variance of a chain of $w+1$ steps at an interior site, which is at most $(w+1)/4 \le s/2$ by Lemma H.3. The t with $j \in R_t$ satisfy $(j-w)/s < t \le j/s$, an interval of length $w/s < 2$, so there are at most two. Hence $\sum_t \nu((F - P_t F)^2) \le s$, and $\gamma_M \le 16s/(3T) \le 16/(3M)$.

(iii) Put the left conditioned site at 0. The outer parts are $[1, A']$ and $[A'+r+1, A'+r+C']$, and the right conditioned site is $T = A' + r + C' + 1$. As in (i), by Lemmas H.1 and H.2, the maximal correlation is the modulus of the correlation of the inner ends $z_{A'}$ and $z_{A'+r+1}$, which by Lemma H.3 is $(u(A')v(A'+r+1)/(v(A')u(A'+r+1)))^{1/2}$. With $u_0(x) = \sinh \kappa x$, respectively $u_0(x) = x$, one has $u(i) = u_0(i)$ and $v(i) = u_0(T-i)$, so $v(A'+r+1) = u_0(C')$ and $v(A') = u_0(C'+r+1)$, which gives the formulas. For $\mu > 0$ the inequality $\sinh x/\sinh(x+y) \le e^{-y}$ for $x, y \ge 0$, which is equivalent to $e^{-x-y} \le e^{-x+y}$, bounds both factors by $e^{-\kappa(r+1)}$, and both tend to $e^{-\kappa(r+1)}$ as $A', C' \to \infty$. For $\mu = 0$ the expression tends to 1. □

Three consequences explain the role of the chain in Section 11. First, for $w = 2s-1$ the massless chain satisfies the hypothesis of Proposition 11.7 with $c = \frac{1}{2}$ for every exterior condition, its non-adjacent moves commute, and yet by (ii) no inequality $\gamma \,\mathrm{Var}_\nu(F) \le \sum_t \nu((F - P_t F)^2)$ holds with $\gamma > 0$ independent of M; in any geometry with commuting non-adjacent moves the same holds for every $c \ge s/(w+1) \ge \frac{1}{2}$. So the threshold $\frac{1}{2}$ of Lemma 11.5 cannot be raised by an argument that uses only these two properties, and no slab geometry of this kind improves it. Second, for $w = 2s-1$ the slab constant of the chain is $1/(2\cosh \kappa s)$, which is below $\frac{1}{2}$ if and only if $\mu > 0$: along a tuning $s = \ell/(2a)$ with lattice mass $\mu = \mu_{\mathrm{phys}} a$ one has $\kappa/\mu \to 1$, so the constant tends to $1/(2\cosh(\mu_{\mathrm{phys}}\ell/2)) < \frac{1}{2}$. In the chain the hypothesis is exactly a mass at the scale $\ell/2$. Third, the condition $(\mathrm{M}_r^*(c))$ with a middle of width r has in the chain the best constant $e^{-\kappa(r+1)}$ by (iii), and for $\mu = 0$ it fails for every $c < 1$ and every r; the iteration of Proposition 11.11 reproduces the true decay up to the integer part of the ratio of widths, so it loses nothing, but it only transports a mass present at the starting scale.

H.2. **Two dimensions: fat boxes.** In this section and the next, G is a compact group with normalised Haar measure λ, also on products of copies of G, and φ is a positive continuous class function on G with $\varphi(g^{-1}) = \varphi(g)$; for $G = \mathrm{SU}(2)$ the last condition holds for every class function, since every element of $\mathrm{SU}(2)$ is conjugate to its inverse. Multiplying φ by a constant changes no measure, so we normalise $\int \varphi \, d\lambda = 1$. For class functions put $(\alpha * \beta)(g) = \int \alpha(gy^{-1})\beta(y)\, dy$ and let α^{*k} be the k-fold convolution power.

Links of $\mathbb{Z}^2$ are (y, ν), $\nu \in \{1, 2\}$, from y to $y + e_\nu$, and the plaquette with base point z has holonomy $U_p = U_{(z,1)} U_{(z+e_1,2)} U_{(z+e_2,1)}^{-1} U_{(z,2)}^{-1}$. Boxes $Q = ([\alpha_1, \alpha_1 + s_1) \times [\alpha_2, \alpha_2 + s_2)) \cap \mathbb{Z}^2$, the sets $V(Q)$, the measures ν_Q^ω, the groups $\mathcal{G}_Q$, the spaces $\mathcal{A}_Q^\omega$, splittings and the maximal correlation $\rho^\omega(Q_-, Q_+)$ are defined as in Section 11 and Definition 11.2, with $\mathbb{Z}^2$ in place of $\mathbb{Z}^4$; a vertex v is interior to Q when its four incident links lie in $V(Q)$, that is, when v, $v - e_1$ and $v - e_2$ lie in Q. The *two-dimensional form* of $(\mathrm{M}_{d,m}(c))$ is the condition $\rho^\omega(Q_-, Q_+) \le c$

for every box of $\mathbb{Z}^2$ whose two widths lie in $[2d-1, md-1]$, every coordinate in which it has width $3d-1$, the splitting into widths d, $d-1$, d in that coordinate, and every exterior configuration ω.

Lemma H.5 (merging)**.** *Let α, β be continuous class functions on G.*

(i) *$\alpha * \beta$ is a continuous class function, positive if α and β are, and $\int \alpha(Ax)\beta(x^{-1}B)\,dx = (\alpha * \beta)(AB)$ for all $A, B \in G$.*

(ii) *Let $\alpha_0, \dots, \alpha_M$ be continuous class functions, $M \geq 1$, and $g_j, c_j, V_0, V_{M+1} \in G$. Then*

$$\int \prod_{j=0}^{M} \alpha_j(V_{j+1} g_j V_j^{-1} c_j)\, dV_1 \cdots dV_M = (\alpha_0 * \cdots * \alpha_M)(V_{M+1}\, g_M \cdots g_0\, V_0^{-1}\, c_0 \cdots c_M).$$

Proof. (i) The substitution $y = x^{-1}B$ preserves Haar measure and turns the integral into $\int \alpha(ABy^{-1})\beta(y)\,dy$. For $h \in G$ the substitution $y = hy'h^{-1}$ gives $(\alpha * \beta)(hgh^{-1}) = \int \alpha(hgy'^{-1}h^{-1})\beta(hy'h^{-1})\,dy' = (\alpha * \beta)(g)$. Continuity and positivity are clear.

(ii) For $M = 1$: since α_0 and α_1 are class functions, $\alpha_0(V_1 g_0 V_0^{-1} c_0) = \alpha_0(g_0 V_0^{-1} c_0 V_1)$ and $\alpha_1(V_2 g_1 V_1^{-1} c_1) = \alpha_1(V_1^{-1} c_1 V_2 g_1)$, so by (i) the integral over V_1 is $(\alpha_0 * \alpha_1)(g_0 V_0^{-1} c_0 c_1 V_2 g_1) = (\alpha_0 * \alpha_1)(V_2 g_1 g_0 V_0^{-1} c_0 c_1)$. For $M \geq 2$ integrate $V_1, \dots, V_{M-1}$ first; by induction the result is $\beta(V_M g_{M-1} \cdots g_0 V_0^{-1} c_0 \cdots c_{M-1})$ with $\beta = \alpha_0 * \cdots * \alpha_{M-1}$, and the case $M = 1$ with $g_{M-1} \cdots g_0$ and $c_0 \cdots c_{M-1}$ in place of g_0 and c_0 gives the claim. □

Fix a box, after translation $Q = [0, s_1) \times [0, s)$ with $s_1 = 3d-1$, $d \geq 2$, $s \geq 1$, split in the first coordinate into $Q_- = [0, d) \times [0, s)$, $Q_0 = [d, 2d-1) \times [0, s)$ and $Q_+ = [2d-1, 3d-1) \times [0, s)$. Write $h(x, y) = \sigma_{((x,y),1)}$ and $v(x, y) = \sigma_{((x,y),2)}$. The interior vertices are $I = [1, s_1 - 1] \times [1, s-1]$. Let

$$F' = \{((x, y), 2) : 1 \leq x \leq s_1 - 1,\ 0 \leq y \leq s - 2\},$$

the vertical links entering the interior vertices from below ($F' = \emptyset$ if $s = 1$), and $E' = V(Q) \setminus F'$: all $h(x, y)$, all $v(0, y)$, and $v(x, s-1)$ for $x \geq 1$.

Lemma H.6 (axial gauge)**.** *For $\sigma \in G^{V(Q)}$ define $H(\sigma) \in \mathcal{G}_Q$ recursively in the second coordinate by $H_u = \sigma_{(u-e_2,2)}^{-1} H_{u-e_2}$ for $u \in I$, with $H_w = 1$ for $w \notin I$, and put $\tau(\sigma) = (H(\sigma)^{-1}\sigma)|_{E'}$.*

(i) *$(H(\sigma)^{-1}\sigma)_e = 1$ for $e \in F'$, and $H(\sigma)$ depends only on $\sigma_{F'}$.*

(ii) *Under λ on $G^{V(Q)}$, $\tau(\sigma)$ has law λ on $G^{E'}$.*

(iii) *Under $\nu = \nu_Q^\omega$, $\tau(\sigma)$ has density proportional to $\mathcal{W}(\tau) = \prod_{p \cap V(Q) \neq \emptyset} \varphi((\tau, 1_{F'}, \omega)_p)$.*

(iv) *If $X \in \mathcal{A}_Q^\omega$ is measurable with respect to $\sigma_{V(Q_-)}$, there is a Borel function $\bar{X}$ with $X = \bar{X}(\tau(\sigma)_{E' \cap V(Q_-)})$ ν-almost surely; likewise for Q_+.*

Proof. (i) For $u \in I$, $(H^{-1}\sigma)_{(u-e_2,2)} = H_{u-e_2}^{-1} \sigma_{(u-e_2,2)} H_u = 1$ by the definition of H_u. The recursion terminates, since $u - e_2$ has a smaller second coordinate, and it uses only the links $(u - je_2, 2)$, $1 \leq j \leq u_2$, which lie in F'.

(ii) Under λ the families $\sigma_{F'}$ and $\sigma_{E'}$ are independent with laws λ. Given $\sigma_{F'}$, H is fixed, and for a link e of E' from x to y, $\tau_e = H_x^{-1} \sigma_e H_y$ is a two-sided translate of σ_e. Translations preserve λ, so the conditional law of τ given $\sigma_{F'}$ is λ, and so is its law.

(iii) Let $w(\sigma) = \prod_{p \cap V(Q) \neq \emptyset} \varphi((\sigma, \omega)_p)$. Every $h \in \mathcal{G}_Q$ fixes the links outside $V(Q)$, whose end points are not interior, so $(h\sigma, \omega) = h \cdot (\sigma, \omega)$ and each plaquette holonomy is conjugated; as φ is a class function, $w(h\sigma) = w(\sigma)$. Hence $w(\sigma) = w(H(\sigma)^{-1}\sigma) = \mathcal{W}(\tau(\sigma))$ by (i), and for bounded Borel Ψ, by (ii), $\int \Psi(\tau(\sigma)) w(\sigma)\, d\lambda(\sigma) = \int \Psi(\tau) \mathcal{W}(\tau)\, d\lambda(\tau)$.

(iv) The group $\mathcal{G}_Q \cong G^I$ acts on $G^{V(Q_-)}$ link by link. Let $X = f(\sigma_{V(Q_-)})$ almost surely with f Borel, and put $\bar{f}(\eta) = \int f(h\eta)\, dh$ where the integral converges absolutely and $\bar{f} = 0$ elsewhere; by invariance of Haar measure both the set of convergence and $\bar{f}$ are invariant under every $k \in \mathcal{G}_Q$. Since ν and λ have the same null sets, the density being bounded above and below, and $f(h\eta) = f(\eta)$ for ν-almost every configuration for each h, Fubini's

theorem gives $\bar f = f$ almost surely. Then $X = \bar f(\sigma_{V(Q_-)}) = \bar f((H(\sigma)^{-1}\sigma)_{V(Q_-)})$ almost surely, and by (i) the right side is a function of $\tau(\sigma)_{E'\cap V(Q_-)}$, the coordinates in F' being 1. □

Lemma H.7 (near-product densities)**.** *Let (S_1,μ_1) and (S_2,μ_2) be probability spaces and π a probability measure on $S_1\times S_2$ with density r with respect to $\mu_1\otimes\mu_2$, $|r-1|\le\delta<1$. If $X\in L^2(\pi)$ depends only on s_1, $Y\in L^2(\pi)$ only on s_2, and $\pi(X)=\pi(Y)=0$, then $|\pi(XY)|\le\frac{\delta+\delta^2}{1-\delta}\|X\|_{L^2(\pi)}\|Y\|_{L^2(\pi)}$. If $r=k/Z'$ with $1-\varepsilon\le k\le 1+\varepsilon$ and $Z'=\int k\,d(\mu_1\otimes\mu_2)$, then $\delta=2\varepsilon/(1-\varepsilon)$ is admissible, and for $\varepsilon\le\frac{1}{10}$ the bound is at most 4ε.*

Proof. The marginal density $m_1(s_1)=\int r(s_1,\cdot)\,d\mu_2$ lies in $[1-\delta,1+\delta]$, so $\int X^2d\mu_1\le(1-\delta)^{-1}\|X\|^2_{L^2(\pi)}$; and $\int X\,d\mu_1=\int X(1-m_1)\,d\mu_1$ since $\int Xm_1\,d\mu_1=\pi(X)=0$, so $|\int X\,d\mu_1|\le\delta\int|X|\,d\mu_1$. The same holds for Y. Now $\pi(XY)=\iint XY(r-1)\,d\mu_1d\mu_2+\int X\,d\mu_1\int Y\,d\mu_2$, and by the Cauchy–Schwarz inequality the two terms together are at most $(\delta+\delta^2)(\int X^2d\mu_1\int Y^2d\mu_2)^{1/2}$, which gives the first claim. If $r=k/Z'$, then $Z'\in[1-\varepsilon,1+\varepsilon]$ and $r\in[\frac{1-\varepsilon}{1+\varepsilon},\frac{1+\varepsilon}{1-\varepsilon}]$, so $|r-1|\le2\varepsilon/(1-\varepsilon)$. For $\varepsilon\le\frac{1}{10}$, $\delta\le\frac{20}{9}\varepsilon\le\frac{2}{9}$ and $\frac{\delta(1+\delta)}{1-\delta}\le\frac{11}{7}\delta\le\frac{220}{63}\varepsilon<4\varepsilon$. □

For $G=\mathrm{SU}(2)$ write χ_j for the character of spin $j\in\frac12\mathbb{Z}_{\ge0}$, $d_j=2j+1$ and $c_j=j(j+1)$, and let

$$p_t=\sum_j d_je^{-tc_j}\chi_j\qquad(t>0)$$

be the heat kernel. It is a uniformly convergent series of real class functions, $\int p_t=1$, $p_t*p_{t'}=p_{t+t'}$ because $\chi_i*\chi_j=\delta_{ij}\chi_j/d_j$, and $p_t>0$, as the heat kernel of a compact connected Riemannian manifold [26, Chs. 7–8]. It is the weight of the *heat-kernel action*, for which the lattice theory in two dimensions reproduces the continuum Yang–Mills measure on every lattice [24, 19, 37], and the two-dimensional tuning is $t=g^2a^2$ with a coupling g of the dimension of a mass. With $n=2j+1$, $c_j=(n^2-1)/4$ and $|\chi_j|\le d_j$,

$$\|p_\theta-1\|_\infty\le\sum_{n\ge2}n^2e^{-\theta(n^2-1)/4}\le\sum_{k\ge1}(k+1)^2q^k\le\frac{4q}{(1-q)^3},\qquad q=e^{-3\theta/4},\tag{H.1}$$

using $n^2-1\ge3(n-1)$ and $\sum_{k\ge1}(k+1)^2q^k=(4q-3q^2+q^3)/(1-q)^3$. For $\theta\ge6$, $q\le e^{-4.5}<\frac{1}{20}$ and the right side is at most $5q\le0.06$.

Theorem H.8 (two dimensions, fat boxes)**.**

(i) *Let $d\ge2$, $s\ge1$, let Q be a box of $\mathbb{Z}^2$ of widths $3d-1$ and s, in either order, split in the coordinate of width $3d-1$ into widths d, $d-1$, d, and put $N=1+(d-1)(s+1)$ and $\varepsilon_N=\|\varphi^{*N}-1\|_\infty$. If $\varepsilon_N\le\frac{1}{10}$, then $\rho^\omega(Q_-,Q_+)\le4\varepsilon_N$ for every exterior configuration ω.*

(ii) *Let $G=\mathrm{SU}(2)$, $\varphi=p_t$ with $t=g^2a^2$, $d\ge2$ and $gda\ge\sqrt6$. For every box of widths $3d-1$ and $s\ge2d-1$ and every ω, $\rho^\omega(Q_-,Q_+)\le20e^{-3Nt/4}$ with $N=1+(d-1)(s+1)$. In particular the two-dimensional form of $(\mathrm{M}_{d,m}(c))$ holds for every $m\ge4$ with*

$$c=20\,e^{-3g^2(da)^2/4}.$$

Proof. (i) *Reduction to the split in the first coordinate.* The exchange of the two coordinates maps links to links, boxes to boxes, interior vertices to interior vertices and $\mathcal{G}_Q$ to $\mathcal{G}_{Q'}$, and it maps the holonomy of the plaquette at z to the inverse of the holonomy of the plaquette at the image of z; since $\varphi(g^{-1})=\varphi(g)$, it maps ν^ω_Q to $\nu^{\omega'}_{Q'}$. So we may take Q as above.

Reduction to the axial model. By Lemma H.6(iii) and (iv), $\rho^\omega(Q_-,Q_+)$ is at most the supremum of $|\pi(\bar X\bar Y)|$ over centred unit vectors $\bar X\in L^2(\pi)$ depending on $\tau_-=\tau_{E'\cap V(Q_-)}$ and $\bar Y$ depending on $\tau_+=\tau_{E'\cap V(Q_+)}$, where π is the law on $G^{E'}$ with density proportional to $\mathcal{W}$.

Which plaquettes couple the three parts. Every plaquette meeting $V(Q)$ contains a link of E': the plaquette at $z \in Q$ contains $h(z)$, the plaquette at $(-1, y)$ contains $v(0, y)$, and the plaquette at $(x, -1)$ contains $h(x, 0)$. The plaquette at (x, y) contains $h(x, y)$, $h(x, y+1)$ and $v(x, y)$, which lie in the part containing the column x, and $v(x+1, y)$, which lies in the part containing the column $x+1$ and belongs to F', so is fixed to 1, unless $y = s-1$. Hence the only plaquettes that contain free links of two different parts are the two seam plaquettes at $(d-1, s-1)$, joining Q_- and Q_0, and at $(2d-2, s-1)$, joining Q_0 and Q_+, and no plaquette contains free links of both Q_- and Q_+. So $\mathcal{W} = \mathcal{W}_-(\tau_-)\mathcal{W}_+(\tau_+)\widehat{\mathcal{W}}$, where $\widehat{\mathcal{W}}$ is the product of the factors of the plaquettes that contain a free link of Q_0.

Integration of the middle. Write $H_x = h(x, s-1)$, $V_x = v(x, s-1)$ and $e_x = \sigma_{((x,s),1)}$, an exterior link. For a column $x \in [d, 2d-2]$ the factors of $\widehat{\mathcal{W}}$ are: the bottom plaquette at $(x, -1)$, which equals $\varphi(h(x, 0)^{-1}c_x)$ with c_x a product of exterior links, by the cyclic invariance of φ; the plaquettes at (x, y), $0 \le y \le s-2$, which equal $\varphi(h(x, y)h(x, y+1)^{-1})$, both vertical links being in F'; and the top plaquette $\varphi(H_x V_{x+1} e_x^{-1} V_x^{-1})$. Integrating $h(x, 0), \dots, h(x, s-2)$ successively with Lemma H.5(i), each step having the form $\int \varphi(h_{y+1}^{-1}h_y)\psi(h_y^{-1}c_x)\, dh_y$, gives $\varphi^{*s}(H_x^{-1}c_x)$, and then $\int \varphi(V_{x+1}e_x^{-1}V_x^{-1}H_x)\varphi^{*s}(H_x^{-1}c_x)\, dH_x = \varphi^{*(s+1)}(V_{x+1}e_x^{-1}V_x^{-1}c_x)$, again by Lemma H.5(i). The seam plaquette at $(d-1, s-1)$ is $\varphi(H_{d-1}V_d e_{d-1}^{-1}V_{d-1}^{-1}) = \varphi(V_d e_{d-1}^{-1}V_{d-1}^{-1}H_{d-1})$, with H_{d-1} and V_{d-1} free links of Q_-, and the seam plaquette at $(2d-2, s-1)$ is the top plaquette of the column $2d-2$, which contains V_{2d-1}, a free link of Q_+. Lemma H.5(ii) with $M = d-1$, $\alpha_0 = \varphi$, $\alpha_j = \varphi^{*(s+1)}$ for $j \ge 1$, $V_0 = V_{d-1}$, $c_0 = H_{d-1}$, $g_j = e_{d-1+j}^{-1}$ and $c_j = c_{d-1+j}$ gives, after a cyclic rearrangement,

$$\int \widehat{\mathcal{W}}\, d\lambda(\tau_0) = \varphi^{*N}(\Phi_-\Phi_+), \qquad \Phi_- = V_{d-1}^{-1}H_{d-1}C, \quad \Phi_+ = V_{2d-1}\Gamma,$$

with $C = c_d \cdots c_{2d-2}$ and $\Gamma = e_{2d-2}^{-1} \cdots e_{d-1}^{-1}$ exterior, and $N = 1 + (d-1)(s+1)$.

Conclusion. Under π the law of the pair (τ_-, τ_+) has a density proportional to

$$\mathcal{W}_-(\tau_-)\, \mathcal{W}_+(\tau_+)\, \varphi^{*N}(\Phi_-\Phi_+).$$

Let μ_1 and μ_2 be the probability measures proportional to $\mathcal{W}_-\lambda$ and to $\mathcal{W}_+\lambda$. Then the law has density k/Z' with respect to $\mu_1 \otimes \mu_2$, where $k = \varphi^{*N}(\Phi_-\Phi_+)$ takes values in $[1-\varepsilon_N, 1+\varepsilon_N]$, and Lemma H.7 gives the bound $4\varepsilon_N$.

(ii) For $s \ge 2d-1$, $N \ge 1 + 2d(d-1) \ge d^2$, so $Nt \ge g^2 d^2 a^2 \ge 6$. Since $\varphi^{*N} = p_{Nt}$, (H.1) gives $\varepsilon_N \le 5e^{-3Nt/4} \le 0.06$, and (i) gives $\rho^\omega \le 20e^{-3Nt/4} \le 20e^{-3g^2d^2a^2/4}$. □

The heat-kernel weight has string tension $\frac{3}{4}g^2$, and Nt is g^2 times the area of the middle slab up to boundary terms, so the bound reads $\rho \le 20e^{-\frac{3}{4}g^2 \cdot \text{area}}$: in two dimensions the slab decorrelation of Section 11 is the confinement of the middle slab, and it holds uniformly in the boundary condition with $c \to 0$ as the physical size da of the box grows. The proof rests on one fact. After the axial gauge the plaquettes straddling each seam contain a single free link, so the middle slab integrates out exactly to a function $\varphi^{*N}(\Phi_-\Phi_+)$ of one group element, uniformly close to 1 once the middle slab is confined. In four dimensions the same gauge leaves on each seam the free links of a three-dimensional layer, and integrating out the middle produces the $(d-1)$-step transfer operator of the box, a kernel on of order $(md)^3$ copies of G; the hypothesis $(\mathrm{M}_{d,m}(c))$ is a bound on its second singular value on gauge-invariant functions, uniform in the boundary condition, and the argument above has no four-dimensional counterpart.

Vacuum dominance in two dimensions. Hypothesis (Z) of Definition 7.9 can also be decided in two dimensions, and exactly. For integers $L \ge 1$, $M \ge 2$ let $\Lambda_{L,M} = (\mathbb{Z}/L\mathbb{Z}) \times (\mathbb{Z}/M\mathbb{Z})$, the first coordinate being space, carry the Wilson measure of Section 2 for a compact group $G \subset \mathrm{U}(N_c)$ (we write N_c for the rank of the unitary group in this paragraph, since N is the mesh index), with the one-plaquette weight $f(g) = e^{N_c\beta \operatorname{Re}\operatorname{Tr} g}$. The time slice consists of the L spatial links, $\mathcal{K} = L^2(G^L)$, the gauge group G^L acts by

$(g\cdot Q)_x = g_x Q_x g_{x+1}^{-1}$, there are no spatial plaquettes, so $S_s \equiv 0$, and the kernel (7.2) is $\mathbb{T}(Q,Q') = \int \prod_x f(Q_x U_{x+1} Q'^*_x U^*_x) \prod_x dU_x$. The proof of Lemma 7.3 uses nothing about the dimension, so its statements, the quantity $\mathcal{Z}_L(M) = \operatorname{Tr}\mathbb{R}^M$ of Section 7.7, hypothesis (Z) with $\mathcal{Z}_{N,p} = \mathcal{Z}_{NL_p}(NL_p)$, and the energies ε_i, the counting function $\mathcal{N}_{N,p}$ and the bound (7.8) of Proposition 7.20 are read on $\Lambda_{L,M}$ without change. For $R \in \hat{G}$, the irreducible unitary representations up to equivalence, with dimension d_R and character χ_R, put

$$b_R(\beta) = \frac{1}{d_R}\int_G f\,\overline{\chi_R}\,d\lambda, \qquad \text{so that} \qquad f = \sum_{R\in\hat{G}} d_R\, b_R\, \chi_R;$$

$b_0 = \int_G f > 0$ is the coefficient of the trivial representation, and $b_R \geq 0$ for every R by the proof of Lemma 7.3(vii). We write $\kappa_0 = \log(1+\sqrt{2}) = \operatorname{asinh} 1$, and I_ν for the modified Bessel function[31] of order ν.

Lemma H.9 (ratios of Bessel functions)**.** *Let $\nu \geq 0$, $x > 0$, $r_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$, and put $u = \nu + \frac12$, $v = \nu + 1$, $w = \nu + \frac32$. Then*

$$\max\Big\{\frac{x}{v+\sqrt{v^2+x^2}},\ \frac{x}{u+\sqrt{w^2+x^2}}\Big\} < r_\nu(x) < \frac{x}{u+\sqrt{u^2+x^2}} < 1,$$

that is, $e^{-\operatorname{asinh}(v/x)} < r_\nu(x) < e^{-\operatorname{asinh}(u/x)}$, and $-\log r_\nu(x) \leq \frac{u}{x} + \frac{w^2}{2x^2}$. Consequently, for integers $n \geq 2$ and $\Phi_n(x) = -\log(I_n(x)/I_1(x)) = \sum_{\nu=1}^{n-1}(-\log r_\nu(x))$,

$$\begin{aligned} &\kappa_0\,\frac{n^2-1}{2(x+n)} \leq \Phi_n(x) \leq \frac{(n-1)(n+2)}{2x}, \qquad \Phi_n(x) \leq (n-1)\log\Big(1+\frac{2n}{x}\Big), \\ &\Big|\Phi_n(x) - \frac{n^2-1}{2x}\Big| \leq \frac{n^3}{2x^2} + \frac{n^4}{24x^3}. \end{aligned} \tag{H.2}$$

Proof. The two recurrences of the footnote give the Riccati equation[32]

$$r'_\nu = \frac{I'_{\nu+1}}{I_\nu} - r_\nu\frac{I'_\nu}{I_\nu} = 1 - \frac{2\nu+1}{x}\,r_\nu - r_\nu^2, \tag{H.3}$$

and the series gives $r_\nu(x) = \frac{x}{2v}\big(1 - \frac{x^2}{4v(\nu+2)} + O(x^4)\big)$ as $x \to 0$. We use the following comparison principle: if A is C^1 on $(0,\infty)$, $D = r_\nu - A$ is positive on some interval $(0,\delta)$ and $D'(x_c) > 0$ at every $x_c > 0$ with $D(x_c) = 0$, then $D > 0$ on $(0,\infty)$; for otherwise $x_c = \inf\{x : D(x) \leq 0\} \geq \delta$ has $D(x_c) = 0$ and $D > 0$ on $(0,x_c)$, while $D(x) = D'(x_c)(x-x_c) + o(x-x_c) < 0$ for $x < x_c$ close to x_c. The same holds with all signs reversed.

Upper bound. $B(x) = x/(u+\sqrt{u^2+x^2}) = 1/(u/x + \sqrt{1+u^2/x^2})$ is the positive root of $\rho^2 + \frac{2u}{x}\rho - 1 = 0$, strictly increasing, and less than 1. By (H.3), $r'_\nu = 0$ wherever $r_\nu = B$, so $(r_\nu - B)' = -B' < 0$ at every zero of $r_\nu - B$; near 0, $B(x) = \frac{x}{2u}(1+O(x^2))$ and $r_\nu(x) = \frac{x}{2v}(1+O(x^2))$ with $u < v$, so $r_\nu < B$ on some $(0,\delta)$, and the reversed principle gives $r_\nu < B$ everywhere.

First lower bound. $A_1(x) = x/(v+\sqrt{v^2+x^2})$ is the positive root of $\rho^2 + \frac{2v}{x}\rho - 1 = 0$, so at a zero x_c of $r_\nu - A_1$, (H.3) gives $r'_\nu(x_c) = \frac{2(v-u)}{x_c}A_1(x_c) = A_1(x_c)/x_c$, while $A_1(x)/x = 1/(v+\sqrt{v^2+x^2})$ is strictly decreasing, so $A_1' < A_1/x$ and $(r_\nu - A_1)'(x_c) > 0$. Near 0, $A_1(x) = \frac{x}{2v}\big(1 - \frac{x^2}{4v^2} + O(x^4)\big)$, so $r_\nu - A_1 = \frac{x^3}{8v^2}\big(\frac{1}{v} - \frac{1}{\nu+2}\big) + O(x^5) > 0$, and the principle applies.

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[31] For $\nu \geq 0$ and $x > 0$, $I_\nu(x) = \sum_{k\geq 0}(x/2)^{2k+\nu}/(k!\,\Gamma(\nu+k+1))$, a series of positive terms, so $I_\nu > 0$ on $(0,\infty)$; differentiating termwise gives $I'_\nu = I_{\nu+1} + \frac{\nu}{x}I_\nu$ and $I'_{\nu+1} = I_\nu - \frac{\nu+1}{x}I_{\nu+1}$, whence $I_{\nu-1} - I_{\nu+1} = \frac{2\nu}{x}I_\nu$; and for integers $n \geq 0$, $I_n(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos(n\theta)\,d\theta$ [60, §§3.7, 3.71]. Only these four facts are used below.

[32] The Riccati equation and the comparison with its constant-coefficient roots is the method of Amos [2] and Segura [51], where the bounds of the lemma appear; the proof here is self-contained.

Second lower bound. Put $S=\sqrt{w^2+x^2}$ and $A_2=x/(u+S)$. Then $1-\frac{2u}{x}A_2-A_2^2=\frac{S^2-u^2-x^2}{(u+S)^2}=\frac{w^2-u^2}{(u+S)^2}=\frac{2v}{(u+S)^2}$ and $A_2'=\frac{u+S-x^2/S}{(u+S)^2}=\frac{u+w^2/S}{(u+S)^2}$, so at a zero x_c of $r_\nu-A_2$, by (H.3) and $2v-u=w$, $(r_\nu-A_2)'(x_c)=\frac{w(1-w/S)}{(u+S)^2}>0$, as $S>w$. Near 0, $S=w+\frac{x^2}{2w}+O(x^4)$ and $u+w=2v$ give $A_2=\frac{x}{2v}(1-\frac{x^2}{4vw}+O(x^4))$, so $r_\nu-A_2=\frac{x^3}{8v^2}(\frac{1}{w}-\frac{1}{\nu+2})+O(x^5)>0$, and the principle applies.

The exponential forms. $B=\sqrt{1+(u/x)^2}-u/x=e^{-\operatorname{asinh}(u/x)}$ and likewise $A_1=e^{-\operatorname{asinh}(v/x)}$. For the last inequality of the first part, $S\le x+\frac{w^2}{2x}$ gives $-\log r_\nu<\log\frac{u+S}{x}\le\log(1+\frac{u}{x}+\frac{w^2}{2x^2})\le\frac{u}{x}+\frac{w^2}{2x^2}$.

The sums. Since asinh is concave with $\operatorname{asinh}0=0$ and $\operatorname{asinh}1=\kappa_0$, one has $\operatorname{asinh}s\ge\kappa_0\min\{s,1\}\ge\kappa_0\,s/(1+s)$ for $s\ge0$; with $s=(\nu+\frac12)/x$ and $\nu+\frac12<n$ this gives $\Phi_n\ge\kappa_0\sum_{\nu=1}^{n-1}\frac{\nu+\frac12}{x+n}=\kappa_0\frac{n^2-1}{2(x+n)}$. Next, $\operatorname{asinh}s\le s$ gives $\Phi_n\le\sum_{\nu=1}^{n-1}\frac{\nu+1}{x}=\frac{(n-1)(n+2)}{2x}$. Since $(\operatorname{asinh}s)'=(1+s^2)^{-1/2}\ge1-\frac{s^2}{2}$, $\operatorname{asinh}s\ge s-\frac{s^3}{6}$, and $\sum_{\nu=1}^{n-1}(\nu+\frac12)^3\le\int_1^n t^3\,dt\le\frac{n^4}{4}$ by the midpoint rule for a convex function, so $\Phi_n\ge\frac{n^2-1}{2x}-\frac{n^4}{24x^3}$; and the bound $-\log r_\nu\le\frac{u}{x}+\frac{w^2}{2x^2}$ with $\sum_{\nu=1}^{n-1}(\nu+\frac32)^2\le\int_2^{n+1}t^2\,dt\le n^3$ for $n\ge3$ (and $\frac{25}{4}\le8$ for $n=2$) gives $\Phi_n\le\frac{n^2-1}{2x}+\frac{n^3}{2x^2}$. Finally $\sqrt{v^2+x^2}\le v+x$ gives $-\log r_\nu\le\log(1+2v/x)\le\log(1+2n/x)$ for $\nu\le n-1$. □

Lemma H.10 (character coefficients of SU(2))**.** *For $G=\mathrm{SU}(2)$, $f(g)=e^{2\beta\operatorname{Re}\operatorname{Tr}g}$ with $\beta>0$, and $j\in\frac12\mathbb{Z}_{\ge0}$,*

$$b_j=\frac{I_{2j+1}(4\beta)}{2\beta},\qquad \frac{b_j}{b_0}=\frac{I_{2j+1}(4\beta)}{I_1(4\beta)}=e^{-\Phi_{2j+1}(4\beta)}.$$

Proof. Every element is conjugate to $\operatorname{diag}(e^{i\theta},e^{-i\theta})$, $\theta\in[0,\pi]$, with $\operatorname{Re}\operatorname{Tr}=2\cos\theta$ and $\chi_j=\sin((2j+1)\theta)/\sin\theta$, and the Weyl integration formula reads $\int_G F\,d\lambda=\frac{2}{\pi}\int_0^\pi F(\theta)\sin^2\theta\,d\theta$ for class functions. With $2\sin((2j+1)\theta)\sin\theta=\cos(2j\theta)-\cos((2j+2)\theta)$ and the integral representation of the footnote,

$$d_jb_j=\frac{2}{\pi}\int_0^\pi e^{4\beta\cos\theta}\sin((2j+1)\theta)\sin\theta\,d\theta=I_{2j}(4\beta)-I_{2j+2}(4\beta)=\frac{2j+1}{2\beta}\,I_{2j+1}(4\beta),$$

by the recurrence $I_{n-1}-I_{n+1}=\frac{2n}{x}I_n$; and $d_j=2j+1$. □

Proposition H.11 (vacuum dominance in two dimensions)**.**

(i) *For every compact $G\subset\mathrm{U}(N_c)$, $\beta>0$ and $L\ge1$, the functions $\psi_R(Q)=\chi_R(Q_1Q_2\cdots Q_L)$, $R\in\hat G$, are orthonormal gauge invariant eigenfunctions of $\mathbb{T}$ with $\mathbb{T}\psi_R=b_R^L\psi_R$, and $\mathbb{T}$ vanishes on their orthogonal complement. Hence $\lambda_0=b_0^L$, $\Omega_0=1$, $0\le b_R<b_0$ for $R\ne0$, and for $M\ge2$*

$$Z(\Lambda_{L,M})=\sum_{R\in\hat G}b_R^{LM},\qquad \mathcal{Z}_L(M)=\sum_{R\in\hat G}\Big(\frac{b_R}{b_0}\Big)^{LM};$$

the nonzero eigenvalues of $\mathbb{R}$ are $(b_R/b_0)^L$, one for each R, and with $a_N=1/N$ and $L=NL_p$ the excitation energies are $\varepsilon_R=-N^2L_p\log(b_R/b_0)$, $R\ne0$, each once.

(ii) *Let $G=\mathrm{SU}(2)$. For all $j\in\frac12\mathbb{N}$ and $\beta>0$,*

$$\exp\Big(-\frac{3j(j+1)}{4\beta}\Big)\le\frac{b_j}{b_0}\le\exp\Big(-\frac{2\kappa_0\,j(j+1)}{4\beta+2j+1}\Big),$$

$$\Big|\log\frac{b_j}{b_0}+\frac{j(j+1)}{2\beta}\Big|\le\frac{(2j+1)^3}{32\beta^2}+\frac{(2j+1)^4}{1536\,\beta^3}.$$

In particular $b_j/b_0\le e^{-\kappa_0 j(j+1)/(4\beta)}$ when $2j+1\le4\beta$ and $b_j/b_0\le e^{-\kappa_0 j/2}$ when $2j+1\ge4\beta$; and no bound $b_j/b_0\le e^{-Kj(j+1)/\beta}$ with $K>0$ holds for all j, since $-\log(b_j/b_0)\le2j\log(1+(2j+1)/(2\beta))$.

(iii) *Let* $G = \mathrm{SU}(2)$, $\beta_{N,p} = \kappa N^2$ *with a fixed* $\kappa > 0$, $L = M = NL_p$, *and* $\gamma = \kappa_0 \min\{3/(8\kappa), \frac{1}{2}\}$. *Then for all* $N \geq 1$ *and all* p*:*

(a) $\varepsilon_j \geq \gamma L_p j$ *for every* $j \geq \frac{1}{2}$, *so* $\Delta_{N,p} = \min_j \varepsilon_j \geq \gamma L_p/2$, *and* $\bigl|\varepsilon_j - L_p j(j+1)/(2\kappa)\bigr| \leq L_p((2j+1)^3/(32\kappa^2 N^2) + (2j+1)^4/(1536\kappa^3 N^4))$;

(b) $\mathcal{N}_{N,p}(\varepsilon) \leq \lfloor 2\varepsilon/(\gamma L_p)\rfloor$ *for all* $\varepsilon > 0$, *and* $\mathcal{N}_{N,p}(\varepsilon) = 0$ *for* $\varepsilon < \gamma L_p/2$;

(c) $e^{-9L_p^2/(16\kappa)} \leq \mathcal{Z}_{N,p} - 1 \leq \bigl(e^{\gamma L_p^2/2} - 1\bigr)^{-1}$; *hence* (Z) *holds, along every progression and every sequence of meshes;*

(d) *for every* $\delta \in (0,1)$, *the bound* (7.8) *holds for all* $\varepsilon > 0$ *and all* $N \geq 1$ *as soon as* $L_p^2 \geq 2/(\gamma(1-\delta))$; *it fails for* $\delta = 1$.

The tuning of (iii) is the Wilson form of the heat-kernel tuning $t = g^2 a^2$ of Theorem H.8: by (ii), $b_j/b_0 = e^{-j(j+1)/(2\beta) + O(\beta^{-2})}$, to be compared with the coefficient $e^{-tj(j+1)}$ of p_t, so $t = 1/(2\beta)$ and $\kappa = 1/(2g^2)$; for the heat-kernel weight itself (i) gives $b_j/b_0 = e^{-tj(j+1)}$ exactly, hence $\varepsilon_j = L_p j(j+1)/(2\kappa)$ and $\mathcal{Z}_{N,p} - 1 = \sum_{j \geq 1/2} e^{-L_p^2 j(j+1)/(2\kappa)}$, independent of N. All statements of (iii) hold for every $\kappa > 0$.

Proof. (i) *The operator is* $K\Pi$. Let K be the operator on $\mathcal{K}$ with the continuous kernel $K(Q, Q') = \prod_x f(Q_x Q_x'^*)$ and $\Pi = \int_{G^L} R_U \, dU$ the orthogonal projection on the gauge-invariant functions. With $(U \cdot Q')_x = U_x Q'_x U_{x+1}^*$ one has $Q_x U_{x+1} Q_x'^* U_x^* = Q_x (U \cdot Q')_x^*$, so $\mathbb{T}(Q, Q') = \int K(Q, U \cdot Q') \, dU$ and, by the invariance of Haar measure, $\mathbb{T} = K\Pi$: this is the identity $\mathbb{T} = DK\Pi D$ of the proof of Lemma 7.3(vii) with $D = 1$.

K on matrix coefficients. $K = K_1^{\otimes L}$ on $\mathcal{K} = L^2(G)^{\otimes L}$, where $(K_1\phi)(q) = \int f(qq'^{-1})\phi(q') \, dq'$. Let D^R_{ab} be the matrix coefficients of R in a unitary basis and M_R their span. The substitution $q' = y^{-1}q$ gives $K_1 D^R_{ab} = \sum_c F_{ac} D^R_{cb}$ with $F = \int f(y) D^R(y^{-1}) \, dy$; F commutes with every $D^R(g)$ because f is a class function, so $F = \lambda I$ by Schur's lemma, and $\lambda d_R = \operatorname{Tr} F = \int f \, \overline{\chi_R} = d_R b_R$. Thus $K_1 = b_R$ on M_R, and K is the scalar $\prod_x b_{R_x}$ on the block $B_{\mathbf{R}} = \bigotimes_x M_{R_x}$, $\mathbf{R} = (R_x)_{x \in \mathbb{Z}/L\mathbb{Z}}$. By the Peter–Weyl theorem the blocks are pairwise orthogonal and span a dense subspace.

The gauge invariants of a block. For $g \in G^L$ and $\phi(y) = \xi(D^R(y)v)$ with $\xi \in V_R^*$, $v \in V_R$, one has $\phi(g_x y g_{x+1}^{-1}) = (\xi \circ D^R(g_x))(D^R(y) D^R(g_{x+1})^{-1} v)$. So R_g preserves every M_{R_x}, hence every block, and as a G^L-module $B_{\mathbf{R}} \cong \bigotimes_x W_x$ with $W_x = V_{R_x}^* \otimes V_{R_{x-1}}$ carrying the action of the factor G_x alone. The invariants of a product group in a tensor product of modules of its factors are the tensor product of the invariants, the averaging projection being the tensor product of the averaging projections, so

$$\dim B_{\mathbf{R}}^{G^L} = \prod_x \dim \mathrm{Hom}_G(V_{R_x}, V_{R_{x-1}}) = \prod_x \delta_{R_x, R_{x-1}}$$

by Schur's lemma: $\Pi B_{\mathbf{R}} = 0$ unless all R_x equal one R, and then $\Pi B_{\mathbf{R}}$ is one-dimensional. The function $\psi_R(Q) = \sum_{a_1, \dots, a_L} D^R_{a_1 a_2}(Q_1) D^R_{a_2 a_3}(Q_2) \cdots D^R_{a_L a_1}(Q_L)$ lies in the block $(R, \dots, R)$ and is gauge invariant, the holonomy of $g \cdot Q$ being $g_1(Q_1 \cdots Q_L) g_1^{-1}$; so $\Pi B_{(R,\dots,R)} = \mathbb{C}\psi_R$. The ψ_R are orthonormal, because $(Q_1, \dots, Q_L) \mapsto Q_1 \cdots Q_L$ pushes the product Haar measure to Haar measure and the irreducible characters are orthonormal. Hence the range of Π is the closed span of the ψ_R.

Spectrum and traces. By the three steps, $\mathbb{T}\psi_R = K\psi_R = b_R^L \psi_R$ and $\mathbb{T}\phi = 0$ for $\phi \perp \{\psi_R\}$, so $\mathbb{T}$ is diagonal in an orthonormal basis formed by the ψ_R and a basis of $\ker \Pi$, with eigenvalues b_R^L and 0. The constant $\psi_0 = 1$ is a positive eigenfunction with eigenvalue b_0^L; by Lemma 7.3(i) the largest eigenvalue is simple with a positive eigenfunction Ω_0, and since $\langle \Omega_0, 1 \rangle > 0$ and eigenfunctions of a self-adjoint operator for distinct eigenvalues are orthogonal, $\lambda_0 = b_0^L$ and $\Omega_0 = 1$; every other eigenvalue is smaller, so $b_R < b_0$ for $R \neq 0$. For $M \geq 2$, $\mathbb{T}^M$ is trace class and its trace is the sum of its eigenvalues with multiplicity, which gives the two formulas by Lemma 7.3(ii). The eigenvalues of $\mathbb{R}$ are 1 and $(b_R/b_0)^L$, $R \neq 0$, one for each R (coinciding values, as for R and $\bar{R}$, are counted once per representation), and 0 on $\ker \Pi$, of energy $+\infty$, which $\mathcal{N}_{N,p}$ does not count; so $\varepsilon_R = -N \log(b_R/b_0)^{NL_p}$.

(ii) By Lemma H.10, $b_j/b_0 = e^{-\Phi_n(x)}$ with $n = 2j+1 \geq 2$, $x = 4\beta$ and $n^2 - 1 = 4j(j+1)$, and the claims are the four bounds of (H.2): the first gives $\Phi_n \geq \frac{2\kappa_0 j(j+1)}{4\beta+2j+1}$, whose denominator is at most 8β when $2j+1 \leq 4\beta$ and at most $2(2j+1)$ when $2j+1 \geq 4\beta$, where $j(j+1)/(2j+1) \geq j/2$; the second gives $\Phi_n \leq \frac{j(2j+3)}{4\beta} \leq \frac{3j(j+1)}{4\beta}$; the third is $\Phi_n \leq 2j\log(1+(2j+1)/(2\beta))$, which is $o(j^2/\beta)$ as $j/\beta \to \infty$; and the last is the two-sided bound.

(iii) Throughout $\beta = \kappa N^2$ and, by (i), $\varepsilon_j = N^2 L_p \Phi_{2j+1}(4\kappa N^2)$ for $j \in \frac{1}{2}\mathbb{N}$, each once.

(a) By (ii), $\varepsilon_j \geq N^2 L_p \frac{2\kappa_0 j(j+1)}{4\kappa N^2+2j+1}$, and since $4\kappa N^2 + 2j + 1 \leq 2\max\{4\kappa N^2, 2j+1\}$,

$$\varepsilon_j \geq L_p \kappa_0 \min\Big\{\frac{j(j+1)}{4\kappa},\ \frac{N^2 j(j+1)}{2j+1}\Big\} \geq L_p \kappa_0\, j \min\Big\{\frac{j+1}{4\kappa}, \frac{1}{2}\Big\} \geq \gamma L_p j,$$

using $N \geq 1$ and $j+1 \geq \frac{3}{2}$. The bound is least at $j = \frac{1}{2}$, whence $\Delta_{N,p} \geq \gamma L_p/2$. The error bound is the two-sided inequality of (ii) multiplied by $N^2 L_p$.

(b) By (a), $\varepsilon_j \leq \varepsilon$ forces $2j \leq 2\varepsilon/(\gamma L_p)$, and the number of such $j \in \frac{1}{2}\mathbb{N}$ is $\lfloor 2\varepsilon/(\gamma L_p)\rfloor$, which is 0 for $\varepsilon < \gamma L_p/2$.

(c) By (i), $\mathcal{Z}_{N,p} - 1 = \sum_{j\geq 1/2}(b_j/b_0)^{N^2 L_p^2} = \sum_{j\geq 1/2} e^{-L_p \varepsilon_j}$. By (a) this is at most $\sum_{m\geq 1} e^{-\gamma L_p^2 m/2} = (e^{\gamma L_p^2/2} - 1)^{-1}$, and the term $j = \frac{1}{2}$ alone is at least $\exp(-\frac{3}{4}\cdot\frac{3}{4}\cdot N^2 L_p^2/(\kappa N^2)) = e^{-9L_p^2/(16\kappa)}$ by (ii). The upper bound does not depend on N and tends to 0 as $L_p \to \infty$.

(d) Let $\delta \in (0,1)$ and $y_0 = \gamma L_p^2/2 \geq 1/(1-\delta)$. For $\varepsilon < \gamma L_p/2$ the left side of (7.8) vanishes by (b). For $\varepsilon \geq \gamma L_p/2$ put $y = L_p\varepsilon \geq y_0$; by (b), $\mathcal{N}_{N,p}(\varepsilon) \leq y/y_0$, and $h(y) = e^{(1-\delta)y} - y/y_0$ has $h(y_0) \geq 0$ and $h'(y) \geq (1-\delta) - 1/y_0 \geq 0$ for $y \geq y_0$, so $\mathcal{N}_{N,p}(\varepsilon) \leq e^{(1-\delta)L_p\varepsilon}$. For $\delta = 1$ the right side of (7.8) is 1 while $\mathcal{N}_{N,p}(\varepsilon) \geq 2$ for $\varepsilon \geq \varepsilon_1$. □

Through Proposition 7.20, (iii)(a) and (d) give $\mathcal{Z}_{N,p} - 1 \leq \delta^{-1} e^{-\delta\gamma L_p^2/2}$, the Gaussian rate of (c): the gap of the two-dimensional cylinder grows linearly with its circumference, and the thermal correction is the Boltzmann weight $e^{-L_p\cdot\sigma L_p}$ of one flux line of string tension $\sigma \approx \frac{3}{8\kappa} = \frac{3}{4}g^2$, the string tension of the heat-kernel weight, wound once around the time circle. What the two-dimensional case shows about (Z) is this. The gauge-invariant part of the slice space is the space of class functions of the holonomy, $\cong \ell^2(\hat{G})$: one state per irreducible representation, no state carrying a position or a momentum, the energy of the state R being that of an electric flux line in the representation R stretched along the spatial circle, $\varepsilon_R \to L_p\, g^2 C_2(R)/2$ with $C_2(j) = 2j(j+1)$, string tension times length. The multiplicity of every level is one, independently of the mesh N, and since every energy is proportional to L_p the count $\mathcal{N}_{N,p}(\varepsilon) \leq 2\varepsilon/(\gamma L_p)$ *decreases* with the physical size: the number of states per unit of physical volume below any fixed energy is zero for large L_p. This is why (7.8) holds for every $\delta < 1$ and (Z) holds at the rate $e^{-cL_p^2}$ rather than the rate $L_p^4 e^{-\Delta L_p}$ of a gapped theory with particles; both facts are the absence of local degrees of freedom. The mechanism is equally two-dimensional: a slice has no spatial plaquettes, so the factor D of Lemma 7.3(vii) is the identity and $\mathbb{T} = K\Pi$ with K a tensor product of one-link convolutions, which the characters diagonalise exactly. In four dimensions D is a multiplication operator that does not commute with K, the gauge-invariant slice functions are functions of all spatial Wilson loops of a three-dimensional lattice of $N^3 L_p^3$ sites, the excitations are local, and a relativistic gapped theory with finitely many species has $\mathcal{N}_{N,p}(\varepsilon)$ of order $\exp(c(L_p^3\varepsilon^3)^{1/4})$, growing with the volume. Hypothesis (Z), equivalently the entropy bound, asks that the multiplicities at fixed physical energy stay bounded as $N \to \infty$ at fixed L_p, with no flat band of lattice-volume size as in the rings of Proposition 7.17: it concerns the density of states per unit of physical volume. In two dimensions that density is zero, and the proposition gives no information on the four-dimensional case, where it is exactly what is unknown. For $G \neq \mathrm{SU}(2)$ part (i) still reduces (Z) to the single sequence $b_R(\kappa N^2)/b_0(\kappa N^2)$,

$R \in \hat{G}$; for $G = \mathrm{U}(1)$ the same argument, with $b_n = I_{|n|}(\beta)$ and Lemma H.9 from $\nu = 0$, gives (iii) with $\gamma' = \kappa_0 \min\{1/(4\kappa), \frac{1}{4}\}$ and a factor 2 for the two signs of n; the quantitative bound on b_R/b_0 for a general compact group is not carried out here.

H.3. **Two dimensions: thin strips.** The restriction of $(\mathrm{M}_{d,m}(c))$ to boxes all of whose widths are at least $2d-1$ is needed. We show that for strips of width one the same bound fails along the two-dimensional tuning for every $c < 1$.

Identify $\mathrm{SU}(2)$ with the unit quaternions $g = g_0 + g_1\mathbf{i} + g_2\mathbf{j} + g_3\mathbf{k}$; then $\chi := \chi_{1/2}$ satisfies $\chi(g) = \operatorname{Tr} g = 2g_0$, $\chi(g^{-1}) = \chi(g)$, and $\langle g, h\rangle_{\mathbb{R}^4} = \frac{1}{2}\chi(g^{-1}h)$. For $\epsilon \in \{1, -1\}$ and $a, b, c > 0$ with $T = a + b + c$ let $\mu^{\epsilon}_{a,b,c}$ be the probability measure on $\mathrm{SU}(2)^2$ with density

$$p_a(x)\, p_b(x^{-1}y)\, p_c(\epsilon y)/p_T(\epsilon)$$

with respect to $\lambda\otimes\lambda$: a Brownian bridge on $\mathrm{SU}(2)$ from 1 to ϵ observed at the times a and $a+b$. The normalisation follows from $\int p_a(x)p_b(x^{-1}y)\,dx = p_{a+b}(y)$ and $\int p_{a+b}(y)p_c(\epsilon y)\,dy = p_T(\epsilon)$, both instances of Lemma H.5(i).

Lemma H.12 (the strip is a bridge)**.** *Let $Q = [0, 3d-1) \times \{0\}$, $d \geq 2$, split in the first coordinate, let $\varphi = p_t$, and let ω_ϵ be the exterior configuration with all links equal to 1 except $\sigma_{((3d-1,0),2)} = \epsilon$. Then $\mathcal{G}_Q$ is trivial, so $\mathcal{A}_Q^{\omega_\epsilon} = L^2(\nu_Q^{\omega_\epsilon})$, and with $h_x = h(x,0)$ and $v_x = v(x,0)$ the pair $(x,y) = (v_{d-1}^{-1}h_{d-1}, v_{2d-1}^{-1})$, formed by a function of $\sigma_{V(Q_-)}$ and a function of $\sigma_{V(Q_+)}$, has law $\mu^{\epsilon}_{a,b,c}$ with $a = c = 2dt$ and $b = (2d-1)t$.*

Proof. No vertex v of Q has $v - e_2 \in Q$, so there are no interior vertices. The plaquettes meeting $V(Q)$ are those at $(-1, 0)$, with holonomy v_0; at $(x, -1)$, with holonomy h_x^{-1}; and at $(x, 0)$, $0 \leq x \leq 3d-2$, with holonomy $h_x v_{x+1} v_x^{-1}$, where $v_{3d-1} := \epsilon$. So the density of ν is proportional to $p_t(v_0)\prod_x p_t(h_x)\prod_x p_t(h_x v_{x+1} v_x^{-1})$. For $x \neq d-1$, $\int p_t(h)p_t(hg)\,dh = (p_t * p_t)(g) = p_{2t}(g)$ by Lemma H.5(i) and $p_t(h^{-1}) = p_t(h)$. Integrating then $v_0, \dots, v_{d-2}$, $v_{2d}, \dots, v_{3d-2}$ and $v_d, \dots, v_{2d-2}$ with Lemma H.5(i), in the form $\int \alpha(gy^{-1})\beta(yh)\,dy = (\alpha * \beta)(gh)$, leaves the density of $(v_{d-1}, h_{d-1}, v_{2d-1})$ proportional to

$$p_{(2d-1)t}(v_{d-1})\, p_t(h_{d-1})\, p_{(2d-1)t}(v_{2d-1} v_{d-1}^{-1} h_{d-1})\, p_{2dt}(\epsilon v_{2d-1}^{-1}).$$

Substituting $x = v_{d-1}^{-1}h_{d-1}$ for h_{d-1}, which preserves Haar measure for fixed v_{d-1}, and integrating v_{d-1} gives $(p_{(2d-1)t} * p_t)(x) = p_{2dt}(x)$. With $y = v_{2d-1}^{-1}$ the third factor is $p_{(2d-1)t}(y^{-1}x) = p_b(x^{-1}y)$, and the last is $p_{2dt}(\epsilon y)$. □

Lemma H.13 (moments of the bridge)**.** *Under $\mu^{\epsilon}_{a,b,c}$, $E[x_1] = E[y_1] = 0$ and*

$$E[x_1 y_1] = \tfrac{1}{3}(\tfrac{1}{2}D - \tfrac{1}{4}C), \quad E[x_1^2] = \tfrac{1}{3}(1 - \tfrac{1}{4}A), \quad E[y_1^2] = \tfrac{1}{3}(1 - \tfrac{1}{4}B),$$

where $D = E\chi(x^{-1}y)$, $C = E[\chi(x)\chi(y)]$, $A = E\chi(x)^2$ and $B = E\chi(y)^2$ are given, with $q_\theta(n) = e^{-\theta(n^2-1)/4}$ and $Z = p_T(\epsilon)$, by

$$\begin{aligned}
Z &= \tfrac{1}{2}\textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n^2 q_T(n), \qquad DZ = \sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+1) q_{a+c}(n) q_b(n+1),\\
CZ &= \textstyle\sum_{n\in\mathbb{Z}} \epsilon^n \big[(n+1)^2 q_a(n+1) q_c(n+1) + (n^2-1) q_a(n+1) q_c(n-1)\big] q_b(n),\\
AZ &= 2Z + \textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+2) q_a(n+2) q_{b+c}(n),\\
BZ &= 2Z + \textstyle\sum_{n\in\mathbb{Z}} \epsilon^{n-1} n(n+2) q_c(n+2) q_{a+b}(n).
\end{aligned}$$

Proof. Symmetry. Conjugation $(x, y) \mapsto (gxg^{-1}, gyg^{-1})$ preserves μ^{ϵ}, since ϵ is central and the p_θ are class functions. It fixes x_0 and y_0 and rotates (x_1, x_2, x_3) and (y_1, y_2, y_3) by the same element of $\mathrm{SO}(3)$. Conjugation by $\mathbf{j}$ maps $\mathbf{i}$ to $-\mathbf{i}$, so $E[x_1] = E[y_1] = 0$; conjugation by $\frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$ permutes $\mathbf{i}, \mathbf{j}, \mathbf{k}$ cyclically, so $E[x_1 y_1] = \frac{1}{3}E\sum_k x_k y_k$ and $E[x_1^2] = \frac{1}{3}E\sum_k x_k^2$. Finally $\sum_k x_k y_k = \langle x, y\rangle - x_0 y_0 = \frac{1}{2}\chi(x^{-1}y) - \frac{1}{4}\chi(x)\chi(y)$ and $\sum_k x_k^2 = 1 - \frac{1}{4}\chi(x)^2$.

Reduction to character sums. Write class functions as $\sum_{n\geq 1} F_n \chi_{(n-1)/2}$. Then p_θ has coefficients $nq_\theta(n)$, and $p_\theta(\epsilon\,\cdot)$ has coefficients $\epsilon^{n-1} n q_\theta(n)$, since $\chi_j(\epsilon g) = \epsilon^{2j}\chi_j(g)$; moreover $\int(\sum F_n\chi)(\sum G_n\chi) = \sum F_n G_n$, and $(\alpha * p_\theta * \gamma)(1) = \sum_n \alpha_n \gamma_n q_\theta(n)$ because $\chi_i * \chi_j = \delta_{ij}\chi_j/d_j$

and $\chi_j(1) = d_j$. By the Clebsch–Gordan rule $\chi\chi_j = \chi_{j-1/2} + \chi_{j+1/2}$ for $j \geq \frac{1}{2}$ and $\chi\chi_0 = \chi$, multiplication by χ acts on coefficients by $F_n \mapsto F_{n+1} + F_{n-1}$ with $F_0 := 0$. The marginal of x is $p_a(x)p_{b+c}(\epsilon x)/Z$ and that of y is $p_{a+b}(y)p_c(\epsilon y)/Z$; substituting $z = x^{-1}y$, the law of z is $p_b(z)p_{a+c}(\epsilon z)/Z$; and $C = ((\chi p_a) * p_b * \gamma)(1)/Z$ with $\gamma(w) = \chi(w)p_c(\epsilon w)$. All series converge absolutely. Hence, with $P_n^{(\theta)} = nq_\theta(n)$ and $\alpha_n^{(\theta)} = P_{n+1}^{(\theta)} + P_{n-1}^{(\theta)}$,

$$Z = \sum_{n\geq 1} \epsilon^{n-1} n^2 q_T(n), \quad DZ = \sum_{n\geq 1} \epsilon^{n-1} \alpha_n^{(b)}\, n q_{a+c}(n), \quad CZ = \sum_{n\geq 1} \epsilon^n \alpha_n^{(a)} \alpha_n^{(c)} q_b(n),$$
$$AZ = \sum_{n\geq 1} \epsilon^{n-1} (\alpha_{n+1}^{(a)} + \alpha_{n-1}^{(a)})\, n q_{b+c}(n), \quad BZ = \sum_{n\geq 1} \epsilon^{n-1} (\alpha_{n+1}^{(c)} + \alpha_{n-1}^{(c)})\, n q_{a+b}(n),$$

where for CZ we used that the coefficients of γ are $\sum_{\pm} \epsilon^{n\pm 1-1} P_{n\pm 1}^{(c)} = \epsilon^n \alpha_n^{(c)}$, and $\alpha_0 := 0$.

Symmetrisation. Extend $P_n^{(\theta)} = nq_\theta(n)$ to $n \in \mathbb{Z}$; it is odd, so $\alpha^{(\theta)}$, now defined for all n, is odd with $\alpha_0^{(\theta)} = 0$ and agrees with the above for $n \geq 1$, and so is $\alpha_{n+1} + \alpha_{n-1}$. Each summand above is then an even function of n vanishing at 0, as $\epsilon^{n\pm 1}$ is even in n, so $\sum_{n\geq 1} = \frac{1}{2}\sum_{n\in\mathbb{Z}}$. Expanding the products and using the substitution $n \mapsto -n$, under which $P_{n-1} \mapsto -P_{n+1}$ and $P_{n-2} \mapsto -P_{n+2}$, to identify the terms in pairs gives the stated formulas; for AZ the middle term $2P_n^{(a)} \cdot nq_{b+c}(n)$ gives $2Z$, since $q_a q_{b+c} = q_T$. □

Lemma H.14 (Poisson summation)**.** *Let $0 < T \leq 1$, $\lambda = T/4$, $\gamma = \sqrt{4\pi/T}$, $\mu \in \mathbb{R}$, $\widetilde{Q}(m) = q_2 m^2 + q_1 m + q_0$ real, and $f(x) = \widetilde{Q}(x+\mu)e^{-\lambda(x+\mu)^2}$.*

(i) $|\sum_{n\in\mathbb{Z}} f(n) - \gamma(q_0 + 2q_2/T)| \leq 30\gamma(|q_0| + 4|q_1|/T + 16|q_2|/T^2)e^{-4\pi^2/T}$.

(ii) *If $q_2 = 1$, $|q_1| \leq 2$ and $|q_0| \leq 1$, then $\sum_{n\in\mathbb{Z}} (-1)^{n-1} f(n) = \Lambda_0(\cos\pi\mu - \mathcal{E})$ with $\Lambda_0 = \frac{8\pi^2}{T^2}\gamma e^{-\pi^2/T}$ and $|\mathcal{E}| \leq 0.22\,T$.*

Proof. f is a Schwartz function. With $\widehat{f}(\xi) = \int f(x)e^{-2\pi i\xi x}dx$, the substitution $y = x + \mu$ and the formulas $\int e^{-\lambda y^2 - i\omega y}dy = \gamma e^{-\omega^2/(4\lambda)} =: G(\omega)$, $\int y e^{-\lambda y^2 - i\omega y}dy = iG'(\omega)$, $\int y^2 e^{-\lambda y^2 - i\omega y}dy = -G''(\omega)$, $G' = -\frac{\omega}{2\lambda}G$ and $G'' = (\frac{\omega^2}{4\lambda^2} - \frac{1}{2\lambda})G$ give

$$\widehat{f}(\xi) = e^{2\pi i\xi\mu}\gamma e^{-\pi^2\xi^2/\lambda}\Big[q_0 - iq_1\frac{\pi\xi}{\lambda} + q_2\Big(\frac{1}{2\lambda} - \frac{\pi^2\xi^2}{\lambda^2}\Big)\Big].$$

(i) By the Poisson summation formula [53, Ch. 5], $\sum_n f(n) = \sum_k \widehat{f}(k)$, and $\widehat{f}(0) = \gamma(q_0 + 2q_2/T)$. For $k \neq 0$, since $\lambda \leq \frac{1}{4}$, each of 1, $\pi|k|/\lambda$ and $\frac{1}{2\lambda} + \frac{\pi^2 k^2}{\lambda^2}$ is at most $M_k\lambda^{-e}$ with $e = 0, 1, 2$ respectively and $M_k = 1 + \pi|k| + \pi^2k^2$; so $|\widehat{f}(k)| \leq \gamma e^{-\pi^2k^2/\lambda}M_k(|q_0| + |q_1|/\lambda + |q_2|/\lambda^2)$, and $\sum_{k\neq 0} e^{-\pi^2k^2/\lambda}M_k \leq 2e^{-\pi^2/\lambda}(M_1 + \sum_{k\geq 2} M_k e^{-12\pi^2(k-1)}) \leq 30e^{-\pi^2/\lambda}$, using $k^2 - 1 \geq 3(k-1)$ and $M_1 < 14.1$. Put $\lambda = T/4$.

(ii) Poisson summation for $f(x)e^{i\pi x}$ gives $\sum_n (-1)^n f(n) = \sum_{\xi\in\frac{1}{2}+\mathbb{Z}} \widehat{f}(\xi)$. The terms $\xi = \pm\frac{1}{2}$ are complex conjugate and add up to $2\gamma e^{-\pi^2/(4\lambda)}[\cos(\pi\mu)(q_0 + \frac{1}{2\lambda} - \frac{\pi^2}{4\lambda^2}) + \sin(\pi\mu)\frac{q_1\pi}{2\lambda}]$. Hence $\sum(-1)^{n-1}f(n) = \Lambda_0(\cos\pi\mu - \mathcal{E})$ with $\Lambda_0 = \frac{\pi^2}{2\lambda^2}\gamma e^{-\pi^2/(4\lambda)}$, the stated value, and $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$, where $\mathcal{E}_1 = \frac{2\lambda^2}{\pi^2}[\cos(\pi\mu)(q_0 + \frac{1}{2\lambda}) + \sin(\pi\mu)\frac{q_1\pi}{2\lambda}]$ and $\mathcal{E}_2 = \Lambda_0^{-1}\sum_{|\xi|\geq 3/2}\widehat{f}(\xi)$. First, $|\mathcal{E}_1| \leq \frac{2}{\pi^2}(\lambda^2 + \frac{\lambda}{2} + \pi\lambda) \leq \frac{2}{\pi^2}(\frac{1}{4} + \frac{1}{2} + \pi)\lambda < 0.79\lambda$. Second, for $|\xi| \geq \frac{3}{2}$, $|\widehat{f}(\xi)|/\Lambda_0 \leq e^{-\pi^2(\xi^2-1/4)/\lambda}(\frac{2\lambda^2}{\pi^2} + \frac{4\lambda|\xi|}{\pi} + \frac{\lambda}{\pi^2} + 2\xi^2) \leq 3\xi^2 e^{-\pi^2(\xi^2-1/4)/\lambda}$, and with $\xi = \pm(k + \frac{1}{2})$, $\xi^2 - \frac{1}{4} = k^2 + k \geq 2k$, so $|\mathcal{E}_2| \leq 6\sum_{k\geq 1}(k + \frac{1}{2})^2 e^{-2\pi^2k/\lambda} \leq 14e^{-2\pi^2/\lambda} \leq 56e^{-8\pi^2}\lambda$, as $e^{-2\pi^2/\lambda}/\lambda$ increases on $(0, \frac{1}{4}]$. So $|\mathcal{E}| \leq 0.8\lambda \leq 0.22\,T$. □

Lemma H.15 (the bridge from 1 to 1)**.** *Let $0 < T \leq 1$ and suppose that $\kappa := ac/T$, $\tau_x := a(b+c)/T$ and $\tau_y := c(a+b)/T$ are all at least $T/10$. Under $\mu^1_{a,b,c}$,*

$$\mathrm{Corr}(x_1, y_1) \geq r_G(1 - T), \qquad r_G = \Big(\frac{ac}{(a+b)(b+c)}\Big)^{1/2}.$$

Here r_G is the correlation of a Euclidean Brownian bridge on $[0,T]$ at the times a and $a+b$.

Proof. Main terms. Complete the squares in the exponents of Lemma H.13 with $\epsilon=1$: $q_T(n)=e^{T/4}e^{-Tn^2/4}$; $q_{a+c}(n)q_b(n+1)=e^{E_D}e^{-T(n+\mu_D)^2/4}$ with $\mu_D=b/T$ and $E_D=\frac{b^2}{4T}+\frac{a+c}{4}$; $q_a(n+1)q_c(n+1)q_b(n)=e^{E_1}e^{-T(n+\mu_1)^2/4}$ with $\mu_1=\frac{a+c}{T}$ and $E_1=\frac{(a+c)^2}{4T}+\frac{b}{4}$; $q_a(n+1)q_c(n-1)q_b(n)=e^{E_2}e^{-T(n+\mu_2)^2/4}$ with $\mu_2=\frac{a-c}{T}$ and $E_2=\frac{(a-c)^2}{4T}+\frac{b}{4}$; and $q_a(n+2)q_{b+c}(n)=e^{E_A}e^{-T(n+\mu_A)^2/4}$ with $\mu_A=\frac{2a}{T}$ and $E_A=\frac{a^2}{T}-\frac{3a-b-c}{4}$. In the variable $m=n+\mu$ the polynomial factors become m^2 for Z, $m^2+(1-2\mu_D)m+\mu_D^2-\mu_D$ for D, $(m+1-\mu_1)^2$ with $1-\mu_1=b/T$ and $(m-\mu_2)^2-1$ for C, and $(m-\mu_A)(m-\mu_A+2)$ for A. The main terms $\gamma(q_0+2q_2/T)$ of Lemma H.14(i), divided by the main term $\gamma e^{T/4}/T$ of Z, give, after the simplifications $T-b=a+c$, $(a-c)^2-T(a+c)=-4ac-b(a+c)$ and $a^2/T-a=-\tau_x$,

$$D_0=e^{-u}(2-4u),\qquad C_0=e^{-u}\Big(2+\frac{b^2}{T}\Big)+e^{-u-\kappa}\Big(2+\frac{(a-c)^2}{T}-T\Big),$$
$$A_0=2+e^{-\tau_x}(2-4\tau_x),\qquad B_0=2+e^{-\tau_y}(2-4\tau_y),\qquad u=\frac{b(a+c)}{4T}.$$

Put $P=\frac{b(T+a+c)}{4T}$. Using $\frac{(a-c)^2-T^2}{4T}=-\frac{(2a+b)(2c+b)}{4T}$ and $\frac{(2a+b)(2c+b)}{4T}=P+\kappa$, one finds

$$X_0:=\tfrac12 D_0-\tfrac14 C_0=e^{-u}\big[(\tfrac12-P)(1-e^{-\kappa})+\kappa e^{-\kappa}\big],\qquad V_0^x:=1-\tfrac14 A_0=\tfrac12(1-e^{-\tau_x})+\tau_x e^{-\tau_x}.$$

Since $ac\le T^2/4$ and $b(a+c)\le T^2/4$, we have $\kappa\le T/4$, $u\le T/16$ and $P\le b/2\le T/2\le\frac12$. With $1-e^{-\kappa}\ge\kappa-\kappa^2/2$, $e^{-\kappa}\ge1-\kappa$ and $e^{-u}\ge1-u$,

$$X_0\ge(1-u)\kappa(\tfrac32-P-\tfrac54\kappa)\ge\tfrac32\kappa(1-\tfrac{T}{16})(1-\tfrac{13}{24}T)\ge\tfrac32\kappa(1-\tfrac{29}{48}T).$$

Also $V_0^x\le\frac32\tau_x$ and $V_0^y:=1-\frac14 B_0\le\frac32\tau_y$.

Errors. Each of the five sums is its main term times $1+\theta$ with $|\theta|\le\eta:=750e^{-4\pi^2/T}/T$. This follows from Lemma H.14(i), with $T\le1$, the coefficient bounds $q_2=1$, $|q_1|\le2$ and $|q_0|\le1$ read off above, and the lower bounds $2/T-1/4$, $2/T$, $2/T-1$ and $2/T-1\ge1/T$ for the brackets q_0+2q_2/T of the main terms: the relative errors are at most $240e^{-4\pi^2/T}/T$ for Z and $750e^{-4\pi^2/T}/T$ for the others. As $e^{-4\pi^2/T}/T^3$ increases on $(0,1]$, $\eta\le750e^{-4\pi^2}T^2<10^{-14}T^2$. Since $|D_0|\le2$, both parts of C_0 are at most 3 in absolute value, and $0\le A_0-2\le2$, we get $|D-D_0|\le6\eta$, $|C-C_0|\le18\eta$ and $|A-A_0|,|B-B_0|\le6\eta$, hence $|X-X_0|\le7.5\eta$ and $|V^x-V_0^x|,|V^y-V_0^y|\le1.5\eta$, where $X=\frac12D-\frac14C$, $V^x=1-\frac14A$ and $V^y=1-\frac14B$.

Conclusion. By Lemma H.13, $\mathrm{Corr}(x_1,y_1)=X/(V^xV^y)^{1/2}$. Using $\kappa,\tau_x,\tau_y\ge T/10$,

$$\mathrm{Corr}(x_1,y_1)\ge\frac{\frac32\kappa(1-\frac{29}{48}T-50\eta/T)}{\frac32(\tau_x\tau_y)^{1/2}(1+10\eta/T)}\ge r_G\Big(1-\frac{29}{48}T-60\frac{\eta}{T}\Big)\ge r_G(1-T),$$

since $\kappa/(\tau_x\tau_y)^{1/2}=r_G$ and $60\eta/T<10^{-12}T$. □

Lemma H.16 (the bridge from 1 to -1)**.** *Let $s_0>0$ and $0<T\le\min(1,s_0^2/4)$, and suppose $\sin(\pi a/T)\ge s_0$ and $\sin(\pi c/T)\ge s_0$. Under $\mu_{a,b,c}^{-1}$,*

$$\mathrm{Corr}(x_1,y_1)\ge1-\frac{5T}{2s_0^2}.$$

Proof. Write $r_a=a/T$, $r_b=b/T$, $r_c=c/T$, $s_a=\sin\pi r_a$ and $s_c=\sin\pi r_c$. The sums of Lemma H.13 with $\epsilon=-1$ have the exponents and polynomial factors of the proof of Lemma H.15, all with $q_2=1$, $|q_1|\le2$ and $|q_0|\le1$, and carry the sign $(-1)^{n-1}$, except CZ,

which carries $(-1)^n$. Lemma H.14(ii) gives

$$Z = \tfrac{1}{2}e^{T/4}\Lambda_0(1-\mathcal{E}_Z), \qquad D = 2e^{-u}\frac{\cos\pi r_b - \mathcal{E}_D}{1-\mathcal{E}_Z},$$
$$C = -2\,\frac{e^{-u}(\cos\pi(r_a+r_c)-\mathcal{E}_1) + e^{-u-\kappa}(\cos\pi(r_a-r_c)-\mathcal{E}_2)}{1-\mathcal{E}_Z},$$
$$A = 2 + 2e^{-\tau_x}\frac{\cos 2\pi r_a - \mathcal{E}_A}{1-\mathcal{E}_Z}, \qquad B = 2 + 2e^{-\tau_y}\frac{\cos 2\pi r_c - \mathcal{E}_B}{1-\mathcal{E}_Z},$$

with all $|\mathcal{E}_\bullet| \le 0.22T$ and $u, \kappa, \tau_x, \tau_y$ as before; the exponential prefactors are those of the main terms in the proof of Lemma H.15. Each quotient has the form $w(\cos\theta - \mathcal{E})/(1-\mathcal{E}_Z)$ with $w \in [1-T/4, 1]$, as $u+\kappa, \tau_x, \tau_y \le T/4$; since $|1/(1-\mathcal{E}_Z)| \le 1.29$ and $|1/(1-\mathcal{E}_Z)-1| \le 0.29T$, it differs from $\cos\theta$ by at most $0.25T\cdot 1.29 + 0.29T + 0.22T\cdot 1.29 < T$. Hence

$$\left|\tfrac{D}{2} - \cos\pi r_b\right| \le T, \quad \left|\tfrac{C}{4} + \cos\pi r_a \cos\pi r_c\right| \le T, \quad \left|\tfrac{A}{4} - \cos^2\pi r_a\right| \le \tfrac{T}{2}, \quad \left|\tfrac{B}{4} - \cos^2\pi r_c\right| \le \tfrac{T}{2},$$

using $\frac{1}{2}(\cos\pi(r_a+r_c) + \cos\pi(r_a-r_c)) = \cos\pi r_a\cos\pi r_c$. Since $r_b = 1 - r_a - r_c$, $\cos\pi r_b = -\cos\pi(r_a+r_c)$, so $\cos\pi r_b + \cos\pi r_a\cos\pi r_c = s_a s_c$. Thus $X = \frac{1}{2}D - \frac{1}{4}C \ge s_a s_c - 2T > 0$, and $0 < V^x = 1 - \frac{1}{4}A \le s_a^2 + \frac{T}{2}$ and $0 < V^y \le s_c^2 + \frac{T}{2}$, positivity coming from Lemma H.13 and $T \le s_0^2/4$. By Lemma H.13 and $\sqrt{(1+x)(1+y)} \le 1 + \frac{x+y}{2}$,

$$\operatorname{Corr}(x_1, y_1) \ge \frac{s_a s_c - 2T}{s_a s_c(1 + \frac{T}{4s_a^2} + \frac{T}{4s_c^2})} \ge \frac{1 - 2T/s_0^2}{1 + T/(2s_0^2)} \ge 1 - \frac{5T}{2s_0^2}. \qquad \square$$

Proposition H.17 (thin strips)**.** *Let $G = \mathrm{SU}(2)$, $\varphi = p_t$, $d \ge 2$, let $Q = [0, 3d-1) \times \{0\}$ be split in the first coordinate into widths d, $d-1$, d, put $T = (6d-1)t$, and let $\omega_{\pm 1}$ be as in Lemma H.12.*

(i) *If $T \le 1$, then $\rho^{\omega_1}(Q_-, Q_+) \ge \frac{2d}{4d-1}(1-T)$.*
(ii) *If $T \le \frac{3}{16}$, then $\rho^{\omega_{-1}}(Q_-, Q_+) \ge 1 - \frac{10}{3}T$.*
(iii) *Along the tuning $t_k = g^2 a_k^2$, $a_k \to 0$, with integers $d_k \ge 2$ and $d_k a_k \le \ell$, for every $m \ge 4$ and every $c < 1$ the bound $\rho^{\omega}(Q_-, Q_+) \le c$ fails for all large k for the strip of widths $3d_k - 1$ and 1, both at most $md_k - 1$; already the trivial boundary condition gives $\rho \ge \frac{1}{2}(1 - 6g^2\ell a_k)$.*

Proof. By Lemma H.12 the functions $X = x_1$ and $Y = y_1$ of $\sigma_{V(Q_-)}$ and $\sigma_{V(Q_+)}$ lie in $\mathcal{A}_Q^{\omega_\epsilon}$, and their joint law is that of $\mu^{\epsilon}_{a,b,c}$ with $a = c = 2dt$ and $b = (2d-1)t$; so $\rho^{\omega_\epsilon} \ge |\operatorname{Corr}(x_1, y_1)|$. (i) Here $\kappa/T = 4d^2/(6d-1)^2 \ge \frac{1}{9}$, $\tau_x/T = \tau_y/T = 2d(4d-1)/(6d-1)^2 \ge \frac{1}{5}$ for $d \ge 2$, and $r_G = 2d/(4d-1)$; apply Lemma H.15. (ii) $a/T = c/T = 2d/(6d-1) \in (\frac{1}{3}, \frac{1}{2})$, so $\sin(\pi a/T) \ge \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2} =: s_0$ with $s_0^2/4 = \frac{3}{16}$; apply Lemma H.16. (iii) $T_k = (6d_k - 1)t_k \le 6g^2(d_k a_k)a_k \le 6g^2\ell a_k \to 0$, so for large k, by (ii), $\rho^{\omega_{-1}} \ge 1 - \frac{10}{3}T_k > c$, and by (i), $\rho^{\omega_1} \ge \frac{2d_k}{4d_k-1}(1-T_k) \ge \frac{1}{2}(1 - 6g^2\ell a_k)$. $\square$

The mechanism is a bridge. The fixed boundary pins the holonomy of the loop surrounding the strip, and both outer parts read off the same collective variable. For the condition ω_{-1} the measure is invariant under the global conjugation $\sigma \mapsto g\sigma g^{-1}$ of all links of $V(Q)$, the exterior links being central, and part (ii) shows that this global SU(2) symmetry is broken at small T: the bridge from 1 to -1 concentrates near the two-sphere of minimising geodesics, whose point is read off by both outer parts, and the slab correlation tends to one. Theorem H.8 and Proposition H.17 together show that in two dimensions, along the weak-coupling tuning of a non-abelian theory, the slab hypothesis holds for fat boxes uniformly in the boundary condition with $c \to 0$, and fails for strips of width one for every $c < 1$. In four dimensions the boundary conditions with values in the centre are 't Hooft's twisted ones [57], and for fat boxes they are the natural test of $(\mathrm{M}_{d,m}(c))$; we have no estimate for them along the four-dimensional tuning.

H.4. **Curvature.** Theorem A is a curvature argument on the gauge-fixed measure, and its range is limited by the Hessian bound of Theorem 5.3. We show that no curvature argument applied to the Wilson measure itself, with or without the gauge, can be uniform along the weak-coupling tuning. We use the notation of Section 5: $G = \mathrm{SU}(N)$ with the metric $\langle X, Y\rangle = -\operatorname{Tr}(XY)$, tangent vectors $v = (X_e Q_e)$, the transported vectors $Y_1, \dots, Y_4$ of a plaquette, and $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$. For a probability measure with density proportional to e^{-V} on a compact Riemannian manifold, the *Bakry–Émery constant* is $\inf(\mathrm{Ric} + \operatorname{Hess} V)(v,v)/|v|^2$ over all points and nonzero tangent vectors; by Bochner's formula it is the largest K with $\Gamma_2(f) \geq K\Gamma(f)$ pointwise for all smooth f, and it controls the Poincaré and logarithmic Sobolev constants of the Langevin dynamics [8]. For a plaquette put $Z_p = Y_1 + Y_2 - Y_3 - Y_4$.

Lemma H.18 (central plaquettes)**.** *Let Q be a field and p a plaquette with $Q_p = wI$ central. Along the geodesics $Q_e(t) = e^{tX_e} Q_e$,*

$$\frac{d^2}{dt^2}\Big|_{t=0} \operatorname{Re}\operatorname{Tr} Q_p(t) = -\operatorname{Re}(w)\,|Z_p|^2.$$

If φ is a smooth class function on G, then $\frac{d^2}{dt^2}|_0 \varphi(Q_p(t)) = H_w(Z_p, Z_p)$ for an Ad*-invariant quadratic form H_w on $\mathfrak{g}$; for $G = \mathrm{SU}(2)$, $H_{\pm 1} = h_\pm |\cdot|^2$ with real constants $h_\pm$.*

Proof. As in Section 5, $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$ with $W = Q_p$. Since $W = wI$ is central, $Q_p(t) = w\,c(t)$ with $c(t) = e^{tY_1} e^{tY_2} e^{-tY_3} e^{-tY_4}$. With $a = (1, 1, -1, -1)$, $c''(0) = \sum_i a_i^2 Y_i^2 + 2\sum_{i<j} a_i a_j Y_i Y_j$, and as $\operatorname{Tr}(Y_i Y_j) = \operatorname{Tr}(Y_j Y_i)$, $\operatorname{Tr} c''(0) = \operatorname{Tr}(Z_p^2) = -|Z_p|^2$. Hence $\frac{d^2}{dt^2} \operatorname{Re}\operatorname{Tr} Q_p = \operatorname{Re}(w \operatorname{Tr} Z_p^2) = -\operatorname{Re}(w)|Z_p|^2$, the trace of a product of two skew-Hermitian matrices being real. For φ, the curve $\gamma(t) = Q_p(t)$ has $\gamma(0) = wI$ and $\gamma'(0) = Z_p wI$, and $(\varphi\circ\gamma)''(0) = \operatorname{Hess}\varphi(\gamma'(0), \gamma'(0)) + d\varphi(\nabla_t \gamma'(0))$. The gradient of φ at wI is YwI with Y fixed by every Ad_g, because conjugation by g is an isometry that fixes wI and preserves φ; the centre of $\mathfrak{su}(N)$ is trivial, so $Y = 0$ and the second term vanishes. The same isometries show that $H_w(Z, Z) = \operatorname{Hess}\varphi_{wI}(ZwI, ZwI)$ is Ad-invariant, and Ad is irreducible on $\mathfrak{su}(2)$, so $H_{\pm 1}$ is a multiple of the metric. □

The configuration

$$U^\pi_{x,\mu} = (-1)^{\sum_{\nu<\mu} x_\nu} I \tag{H.4}$$

has every plaquette equal to $-I$: for $\mu < \nu$ the link $(x + e_\mu, \nu)$ sees the coordinate x_μ raised by one and the link $(x + e_\nu, \mu)$ does not, so the four signs of the plaquette at x multiply to -1. It is defined on every box and on Λ_L for even L, and lies in $\mathrm{SU}(N)$ for even N. Put $s(x) = (-1)^{|x|}$ with $|x| = \sum_\mu x_\mu$.

Theorem H.19 (full space)**.** *Let N be even, $L \geq 4$ even, and μ_L the* $\mathrm{SU}(N)$ *Wilson measure on Λ_L with density proportional to $e^{N\beta S_0}$. Its Bakry–Émery constant satisfies*

$$K_{\mathrm{full}} \leq \frac{N}{2} - 16N|\beta| = \frac{N}{2} - \frac{16|\beta_W|}{N}.$$

For $\mathrm{SU}(2)$*, $K_{\mathrm{full}} \leq 1 - 8|\beta_W|$.*

Proof. By Lemma 5.4 and Table 1, $\mathrm{Ric} = \frac{N}{2}\langle\cdot,\cdot\rangle$. Let $c \in \mathbb{R}^4$ with $\sum_\mu c_\mu = 0$ and $c \neq 0$, let $E \in \mathfrak{su}(N)$ with $|E| = 1$, and put $X_{x,\mu} = c_\mu s(x) E$. For $\beta \geq 0$ take the field U^π, at which all transports are $\mathrm{Ad}_{\pm I} = \mathrm{id}$. For the plaquette at x in the plane $\mu < \nu$, $Y_1 = c_\mu s(x)E$, $Y_2 = -c_\nu s(x)E$, $Y_3 = -c_\mu s(x)E$ and $Y_4 = c_\nu s(x)E$, so $Z_p = 2s(x)(c_\mu - c_\nu)E$. By Lemma H.18 with $w = -1$,

$$\operatorname{Hess} S_0(v,v) = \sum_p |Z_p|^2 = 4L^4 \sum_{\mu<\nu} (c_\mu - c_\nu)^2 = 4L^4\Big(4\sum_\mu c_\mu^2 - \Big(\sum_\mu c_\mu\Big)^2\Big) = 16|v|^2.$$

With $V = -N\beta S_0$ this gives $(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v) = (\frac{N}{2} - 16N\beta)|v|^2$. For $\beta < 0$ take the field $Q \equiv I$, where Lemma H.18 with $w = 1$ gives $\mathrm{Hess}\, S_0(v,v) = -16|v|^2$ for the same v, hence $(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v) = (\frac{N}{2} - 16N|\beta|)|v|^2$. Finally $\beta_W = N^2\beta$, and $\beta_W = 4\beta$ for SU(2). □

Theorem H.19 says that without the gauge no Bakry–Émery constant exceeds $K_{\mathrm{full}} = \frac{N}{2} - 16N|\beta|$. The Ricci curvature grows with N, but so does the Hessian, and $K_{\mathrm{full}} > 0$ exactly for $|\beta_W| < N^2/32$; at fixed N along the tuning $\beta_W \to \infty$ the constant tends to $-\infty$ linearly. The Bakry–Émery constant is also the best rate of contraction of the Langevin dynamics in the quadratic Kantorovich distance [59], so no synchronous coupling of the dynamics does better. The gauge of Section 3 improves the constant 16 to $\frac{29}{2}$, and the next theorem shows that it cannot improve it below $\frac{38}{3}$.

Theorem H.20 (the gauge slice)**.** *Let $G = \mathrm{SU}(2)$, $m \geq 1$, $\Lambda_m = [0, 2m]^4 \cap \mathbb{Z}^4$ with free boundary, and let $K_F(\Lambda_m)$ be the Bakry–Émery constant of the gauge-fixed measure $\mu^F_{\Lambda_m}$ of Lemma 4.1. Then*

$$K_F(\Lambda_m) \leq 1 - 2|\beta|\rho_m, \qquad \rho_m = \frac{304\, m^4}{24\, m^4 + 8(2m+1)^3} \xrightarrow[m\to\infty]{} \frac{38}{3}.$$

Hence the constant $\frac{29}{2}$ of Theorem 5.3 cannot be replaced by any number below $\frac{38}{3}$, and asymptotically in m

$$1 - \tfrac{29}{4}|\beta_W| \leq K_F \leq 1 - \tfrac{19}{3}|\beta_W|,$$

so no Bakry–Émery argument for the gauge-fixed measure reaches $|\beta_W| = \frac{3}{19}$.

Proof. The field. For $\beta \geq 0$ start from U^π on Λ_m and apply the gauge transformation of the proof of Lemma 4.1 along the forest $F \cap E(\Lambda_m)$; the recursion $g_{x_{i+1}} = g_{x_i} U_{(x_i, x_{i+1})}$ produces $g_x \in \{\pm I\}$, so the result Q takes values in $\{\pm I\}$, equals I on F, and has all plaquettes conjugate to those of U^π, that is $-I$. For $\beta < 0$ take $Q \equiv I$. For v vanishing on F the Hessian of the gauge-fixed action is $\mathrm{Hess}\, S_{\Lambda_m}(v,v)$, which by Lemma H.18 equals $\pm \sum_{p \in P(\Lambda_m)} |Z_p|^2$, with $+$ for $\beta \geq 0$ and $-$ for $\beta < 0$; all transports are trivial.

The vector. Let $c = (1, -1, 0, 0)$, $|E| = 1$, and put $X_{x,\mu} = c_\mu s(x) E$ on the active links of $E(\Lambda_m)$ and $X = 0$ on F. Let p be a plaquette at x in the plane $\mu < \nu$ with all its links in $E(\Lambda_m)$; it contains exactly one link of F. Without that link the computation of the proof of Theorem H.19 gives $Z_p = 2s(x)(c_\mu - c_\nu)E$, the links e_1, e_3 of direction μ contributing $s(x)c_\mu E$ each and the links e_2, e_4 of direction ν contributing $-s(x)c_\nu E$ each. Removing the link of F gives $Z_p = s(x)(c_\mu - 2c_\nu)E$ if that link has direction μ, and $Z_p = s(x)(2c_\mu - c_\nu)E$ if it has direction ν.

Counting. Let $B = [0, 2m-1]^4$; every plaquette with base point in B has its links in $E(\Lambda_m)$. Fix $\mu < \nu$. The link of F of the plaquette at x has direction μ exactly when $\sigma(x) = \mu$ or $\sigma(x + e_\nu) = \mu$, and the two cases exclude each other. By Table 2, σ has period two and takes each value at four of the sixteen residues. B and $B + e_\nu$ are disjoint unions of m^4 translates of $\{0,1\}^4$, each containing every residue once, so each case occurs for $4m^4$ base points in B; of the $16m^4$ plaquettes of the plane with base point in B, exactly $8m^4$ have their link of F in direction μ and $8m^4$ in direction ν. Hence, with $|c|^2 = 2$ and $\sum_\mu c_\mu = 0$,

$$\sum_{p \in P(\Lambda_m)} |Z_p|^2 \geq 8m^4 \sum_{\mu<\nu} \left[(c_\mu - 2c_\nu)^2 + (2c_\mu - c_\nu)^2\right] = 8m^4\Big(19|c|^2 - 4\Big(\sum_\mu c_\mu\Big)^2\Big) = 304 m^4,$$

using $\sum_{\mu<\nu}(5c_\mu^2 + 5c_\nu^2 - 8c_\mu c_\nu) = 15|c|^2 - 4((\sum_\mu c_\mu)^2 - |c|^2)$. On the other side, for $\mu \in \{0,1\}$ the links (x,μ) with $x \in B$ lie in $E(\Lambda_m)$, and exactly $16m^4 - 4m^4 = 12m^4$ of them are active; they contribute $24m^4$ to $|v|^2$. The remaining links of $E(\Lambda_m)$ with $X \neq 0$ have direction 0 or 1 and base point in $[0,2m]^4 \setminus B$, at most $(2m+1)^4 - (2m)^4 \leq 4(2m+1)^3$ in each direction, and each contributes at most 1. So $|v|^2 \leq 24m^4 + 8(2m+1)^3$, and $|\mathrm{Hess}\, S_{\Lambda_m}(v,v)| \geq \rho_m |v|^2$ with the sign that gives $\mathrm{Hess}\, V(v,v) \leq -2|\beta|\rho_m |v|^2$, $V = -2\beta S_{\Lambda_m}$.

Conclusion. $(\mathrm{Ric} + \mathrm{Hess}\, V)(v,v) \le (1 - 2|\beta|\rho_m)|v|^2$. If Theorem 5.3 held with a constant $c' < \frac{38}{3}$, then $\rho_m \le c'$ for all m, contradicting $\rho_m \to \frac{38}{3}$. The lower bound for K_F is Lemma 5.5 with $\lambda = 0$, and $\beta_W = 4\beta$. □

Proposition H.21 (plaquette actions at a coarse scale)**.** *Let $L \ge 4$ be even and $\nu(dU) \propto \exp(\sum_p \phi(U_p)) \prod_e dU_e$ on $\mathrm{SU}(2)^{E(\Lambda_L)}$ with $\phi(U) = g(\frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\, U)$, $g \in C^2([-1,1])$. The Bakry–Émery constant K of ν satisfies*

$$K \le 1 - 8\max(g'(-1), -g'(1)).$$

Proof. $\frac{1}{2}\mathrm{Re}\,\mathrm{Tr}(\pm e^{tZ}) = \pm(1 - \frac{t^2}{4}|Z|^2) + O(t^3)$, since $\mathrm{Tr}\, Z^2 = -|Z|^2$; so in Lemma H.18 $h_+ = -\frac{1}{2}g'(1)$ and $h_- = \frac{1}{2}g'(-1)$. The potential is $-\sum_p \phi(U_p)$. At U^π, with the vector of the proof of Theorem H.19, $\mathrm{Hess} \sum_p \phi(U_p)(v,v) = h_- \sum_p |Z_p|^2 = 16h_-|v|^2$, and at $Q \equiv I$ it is $16h_+|v|^2$. Hence $K \le 1 - 16h_\pm$. □

For the Wilson weight $g(s) = 4\beta s$ this is Theorem H.19. For an effective plaquette action at a coarse scale the Bakry–Émery route needs $|g'(\pm 1)| < \frac{1}{8}$, an effective theory deep in strong coupling, and producing such an effective measure from the weak-coupling lattice theory, and transporting the resulting estimate back, are the two missing steps of Section 12.

H.5. **Entropy, link covariances and blocks of fixed size.** In the setting of Section 6, say that μ satisfies *block entropy factorisation* with constant C if $\mathrm{Ent}_\mu(F) \le Cn^{-4} \sum_x \mu(\mathrm{Ent}_{\gamma^0_{V_x}} F)$ for all bounded gauge-invariant $F \ge 0$, where $\mathrm{Ent}_{\gamma^0_{V_x}} F$ is the entropy of F under $\gamma^0_{V_x}(\cdot \mid U)$.

Proposition H.22. *Block entropy factorisation with constant C implies* $\mathrm{BP}(n, 1/C)$.

Proof. Let $\Phi(s) = s\log s$. For $|u| \le \frac{1}{2}$, $\Phi(1+u) = u + \frac{u^2}{2} + r(u)$ with $r(u) = \sum_{k\ge 3}(-u)^k/(k(k-1))$ and $|r(u)| \le |u|^3 \sum_{k\ge 3} 2^{3-k}/(k(k-1)) \le |u|^3$. Let g be real, bounded and gauge invariant, and $0 < \varepsilon \le 1/(2\|g\|_\infty)$. Then $\mathrm{Ent}_\mu(1+\varepsilon g) = \mu\Phi(1+\varepsilon g) - \Phi(1+\varepsilon\mu g) = \frac{\varepsilon^2}{2}\mathrm{Var}_\mu\, g + R$ with $|R| \le 2\varepsilon^3\|g\|^3_\infty$, and in the same way $\mathrm{Ent}_{\gamma^0_{V_x}}(1+\varepsilon g) = \frac{\varepsilon^2}{2}\mathrm{Var}_{\gamma^0_{V_x}} g + R_x$ with $|R_x| \le 2\varepsilon^3\|g\|^3_\infty$. Inserting $F = 1 + \varepsilon g$, dividing by $\varepsilon^2/2$ and letting $\varepsilon \to 0$ gives $\mathrm{Var}_\mu\, g \le Cn^{-4}\sum_x \mu(\mathrm{Var}_{\gamma^0_{V_x}} g) = Cn^{-4}\sum_x \mu((g - \mathcal{P}_x g)^2)$. Truncations of $f \in \mathcal{A}$ are bounded and gauge invariant and converge in $L^2(\mu)$, and both sides are continuous on $L^2(\mu)$. □

So entropy factorisation at the physical scale, uniform along the tuning, implies (D) with $\varpi = 1/C$, and every obstruction to the cube condition applies to it: it fails for U(1) at weak coupling and at fixed cube size (Section 12). The methods that derive block factorisation from spatial mixing uniformly in the boundary condition [13] need, at the physical scale, an input of the strength of (D) itself.

Proposition H.23 (link covariances)**.** *Let μ be a gauge-invariant probability measure on $\mathrm{SU}(2)^E$, E the links of a box or of Λ_L with $L \ge 3$, and identify U_e with a unit vector $\sigma_e \in \mathbb{R}^4$. Then $E\sigma_e = 0$, $E\sigma_e\sigma_e^T = \frac{1}{4}I_4$ and $E\sigma_e\sigma_{e'}^T = 0$ for $e \ne e'$.*

Proof. Distinct links have distinct sets of end points, so e has an end point x that is not an end point of e'. The gauge transformation g at x alone maps σ_e to $\ell_g\sigma_e$, where ℓ_g is left multiplication by g or right multiplication by g^{-1}, an orthogonal map, and fixes $\sigma_{e'}$. By invariance $E\sigma_e\sigma_{e'}^T = \int \ell_g\, dg\; E\sigma_e\sigma_{e'}^T = 0$, since ℓ_g is linear in g and $\int g\, dg = 0$. The same argument gives $E\sigma_e = 0$ and shows that the law of σ_e is invariant under multiplication, hence Haar measure, so $E\sigma_e\sigma_e^T = \frac{1}{4}I_4$. □

The influence and covariance matrices that spectral independence and localisation schemes ask to be bounded are therefore exactly $\frac{1}{4}I$ for the Wilson measure at every β, weak coupling included, and carry no information. Those methods need the bound for all tilted measures, and the tilts break gauge invariance. Their output is a tensorisation inequality over blocks

of bounded lattice size, uniform in the volume, and the next proposition shows that such an inequality cannot hold uniformly along the tuning when the continuum limit is nontrivial.

Proposition H.24 (fixed blocks force triviality)**.** *Let $a_k = a_0 2^{-k}$ and β_k be given, and suppose that for every k there is $C_k < \infty$ such that the Wilson measure $\mu_{\beta_k,L}$ satisfies*

$$\mathrm{Var}_\mu(f) \le C_k \sum_{e\in E} \mu(\mathrm{Var}_e f) \qquad (f \in \mathcal{A})$$

for all L in an infinite set $\mathcal{L}_k$ of even integers $L \ge 8$, where $\mathrm{Var}_e f$ is the conditional variance given the links other than e. Then $\mathrm{BP}(2, 1/(64C_k))$ holds on these tori, every torus limit along $\mathcal{L}_k$ has transfer-matrix gap at least $1/(5000e^2C_k)$, and every continuum limit of such torus limits in the sense of Definition 6.5 has $\sigma(H) \subset \{0\} \cup [E_*, \infty)$ with $E_* = \limsup_k (5000e^2C_k a_k)^{-1}$. In particular, if $C_k a_k \to 0$, every continuum limit is trivial.

Proof. For $e \in V_x$ the σ-algebra of the links off V_x is contained in that of the links other than e, so $\mu(\mathrm{Var}_e f) = \|f - E[f \mid e^c]\|^2 \le \|f - \mathcal{P}_x f\|^2$. With $n = 2$ each link lies in 16 cubes V_x and $|V_x| = 4 \cdot 2^4 = 64$, so

$$\sum_e \mu(\mathrm{Var}_e f) \le \frac{1}{16} \sum_x \sum_{e\in V_x} \mu((f - \mathcal{P}_x f)^2) = 4 \sum_x \mu((f - \mathcal{P}_x f)^2),$$

and $\mathrm{Var}_\mu f \le 64C_k \cdot 2^{-4} \sum_x \mu((f-\mathcal{P}_x f)^2)$, which is $\mathrm{BP}(2, 1/(64C_k))$ on tori with $L \ge 8 = 2 \cdot 2+4$. Corollary 6.4 gives the gap $m_k = \varpi/(2e^2 q_2)$ with $q_2 = \frac{625}{16}$, that is $m_k = 1/(5000e^2C_k)$. For a continuum limit, as in the proof of Theorem B, for $f \in \mathcal{T}$, $\varphi = \psi(f) - \langle \Omega, \psi(f)\rangle \Omega$ and dyadic $\tau > 0$, $\langle \varphi, e^{-\tau H}\varphi\rangle = \lim_k \langle \varphi_k, \mathbb{T}_k^{\tau/a_k}\varphi_k\rangle \le \liminf_k e^{-m_k \tau/a_k}\|\varphi_k\|^2 = e^{-E_*\tau}\|\varphi\|^2$, using $\|\varphi_k\|^2 \to \|\varphi\|^2$, the case $\tau = 0$ of the definition. These φ are dense in $\Omega^\perp$, and the spectral argument of the proof of Theorem B gives $\sigma(H|_{\Omega^\perp}) \subset [E_*, \infty)$. If $E_* = \infty$, then $\langle \varphi, e^{-\tau H}\varphi\rangle = 0$ for all dyadic $\tau > 0$, and strong continuity of $e^{-\tau H}$ at $\tau = 0$ gives $\varphi = 0$, so $\Omega^\perp = \{0\}$. □

The same holds for blocks of any fixed lattice size b in place of single links, with $\mathrm{BP}(b, \cdot)$. So the tensorisation constants of single links or of bounded blocks must grow at least like $1/a_k$ along the tuning whenever the continuum limit is nontrivial; a usable version must work with blocks of physical size, and is then a form of (D).

H.6. **The transfer matrix alone.** The next results show what the continuum gap needs and that a lattice gap tending to zero at the right rate is, by itself, a kinematic statement. In this section Definition 6.5 is used with $a_k = a_0 2^{-j_k}$ for a strictly increasing sequence of integers j_k, with τ in $\mathbb{D} = \{a_0 p 2^{-q} : p, q \in \mathbb{Z}_{\ge 0}\}$, and with an arbitrary complex vector space $\mathcal{T}$ and linear maps O_k into the gauge-invariant local observables at nonnegative times; the proof of Theorem B uses nothing more. The states ω_k are translation invariant and reflection positive, with transfer matrices $0 \le \mathbb{T}_k \le 1$ on $\mathcal{H}_k$, vacua Ω_k and maps Ψ_k from observables at nonnegative times onto dense subspaces, as in Section 2. Put $\gamma_k = \|\mathbb{T}_k|_{\Omega_k^\perp}\|$, $m_k = -\log \gamma_k \in [0, \infty]$, and $\mathrm{C}_k(f; s) = \omega_k(\vartheta(\overline{O_k f})\tau_{s/a_k} O_k f) - |\omega_k(O_k f)|^2$.

Proposition H.25 (the spectral input)**.** *Let $(\mathcal{H}, \Omega, H, \psi)$ be a continuum limit of (ω_k, O_k) and $\Delta > 0$. The following are equivalent.*

(i) *$\sigma(H|_{\Omega^\perp}) \subset [\Delta, \infty)$.*
(ii) *For every $f \in \mathcal{T}$ there is $C_f < \infty$ with $\lim_k \mathrm{C}_k(f; s) \le C_f e^{-\Delta s}$ for every $s \in \mathbb{D}$.*

Proof. For $f \in \mathcal{T}$ let $\phi_f = \psi f - \langle \Omega, \psi f\rangle \Omega$. Since $e^{-sH}\Omega = \Omega$ and e^{-sH} is self-adjoint, $\langle \phi_f, e^{-sH}\phi_f\rangle = \langle \psi f, e^{-sH}\psi f\rangle - |\langle \Omega, \psi f\rangle|^2$, so $\lim_k \mathrm{C}_k(f; s)$ exists and equals $\langle \phi_f, e^{-sH}\phi_f\rangle$. If (i) holds, the spectral theorem gives (ii) with $C_f = \|\phi_f\|^2$. Conversely, let μ_f be the spectral measure of H at ϕ_f. If $\mu_f([0, \Delta - \eta]) = c > 0$ for some $\eta > 0$, then $\langle \phi_f, e^{-sH}\phi_f\rangle \ge ce^{-(\Delta-\eta)s} > C_f e^{-\Delta s}$ for large $s \in \mathbb{D}$, which contradicts (ii); so the spectral projection P of H for $[0, \Delta)$ satisfies $P\phi_f = 0$. The ϕ_f are dense in $\Omega^\perp$, so $P = 0$ on $\Omega^\perp$. □

So the continuum gap needs exponential decay, at physical times and only asymptotically in k, of the autocorrelations of the observables that generate the limit; it does not need a lattice gap, since lattice states of physical energy below Δ are allowed if they decouple from every $O_k(f)$.

Lemma H.26. *Let $(\mathcal{H}, \Omega, H, \psi)$ be a continuum limit of (ω_k, O_k) along a subsequence Λ and $\bar{E} = \limsup_{k\in\Lambda} m_k/a_k \in [0,\infty]$. Then $\langle\phi, e^{-sH}\phi\rangle \le e^{-\bar{E}s}\|\phi\|^2$ for all $\phi \in \Omega^\perp$ and $s \ge 0$; so $\sigma(H|_{\Omega^\perp}) \subset [\bar{E},\infty)$, and $\Omega^\perp = \{0\}$ if $\bar{E} = \infty$.*

Proof. For $f \in \mathcal{T}$ and $k \in \Lambda$ put $O = O_k f$ and $\phi_k = \Psi_k(O) - \omega_k(O)\Omega_k \perp \Omega_k$. Since $\mathbb{T}_k \ge 0$ is self-adjoint and fixes Ω_k, it preserves $\Omega_k^\perp$, and for $s \in \mathbb{D}$ and $t = s/a_k \in \mathbb{Z}$, $\mathrm{C}_k(f;s) = \langle\phi_k, \mathbb{T}_k^t\phi_k\rangle \le \gamma_k^t\|\phi_k\|^2 = e^{-(m_k/a_k)s}\|\phi_k\|^2$. Along a further subsequence $m_k/a_k \to \bar{E}$, $\mathrm{C}_k(f;s) \to \langle\phi_f, e^{-sH}\phi_f\rangle$ and $\|\phi_k\|^2 = \mathrm{C}_k(f;0) \to \|\phi_f\|^2$, with $e^{-\infty s} = 0$ for $s > 0$. The ϕ_f are dense in $\Omega^\perp$, both sides are continuous in ϕ, and $s \mapsto \langle\phi, e^{-sH}\phi\rangle$ is continuous, so the inequality holds on $\Omega^\perp$ for all $s \ge 0$. The spectral statement follows as in Proposition H.25; if $\bar{E} = \infty$, then $\langle\phi, e^{-sH}\phi\rangle = 0$ for $s > 0$, so $\phi = 0$ by strong continuity. □

Theorem H.27 (gap tuning)**.** *Suppose $0 < m_k < \infty$ for all k and $m_k \to 0$, and let $E > 0$ and $a_0 > 0$. For each k let j_k be the integer with $E \le m_k 2^{j_k}/a_0 < 2E$, pass to a subsequence on which j_k is strictly increasing, and put $a_k = a_0 2^{-j_k}$.*

(a) *Every continuum limit of (ω_k, O_k) along any subsequence, for any $\mathcal{T}$ and O_k, has $\sigma(H|_{\Omega^\perp}) \subset [E,\infty)$.*

(b) *There are a subsequence, a space $\mathcal{T}$ with maps O_k, and a nontrivial continuum limit along it with $\inf\sigma(H|_{\Omega^\perp}) = \lim_k m_k/a_k \in [E, 2E]$.*

Proof. The integer j_k exists and is unique, because the condition reads

$$2^{j_k} \in [Ea_0/m_k,\ 2Ea_0/m_k),$$

and $j_k \to \infty$ since $m_k \to 0$. Part (a) is Lemma H.26, since $m_k/a_k \ge E$.

(b) Pass to a subsequence with $m_k/a_k \to \bar{E} \in [E, 2E]$ and fix k. As $\gamma_k > 0$, $\Omega_k^\perp \ne \{0\}$, and γ_k, the norm of the positive operator $\mathbb{T}_k|_{\Omega_k^\perp}$, lies in its spectrum; so the spectral projection P_k of $\mathbb{T}_k|_{\Omega_k^\perp}$ for $[\gamma_k e^{-a_k/k}, \gamma_k]$ is nonzero. Choose a unit vector v_k in its range. The vectors $\Psi_k(F - \omega_k(F))$ are the projections of $\Psi_k(F)$ onto $\Omega_k^\perp$, hence dense in it; choose F_k' with $\|u_k' - v_k\| \le \frac{1}{10}$ for $u_k' = \Psi_k(F_k' - \omega_k(F_k'))$, and put $F_k = (F_k' - \omega_k(F_k'))/\|u_k'\|$ and $u_k = \Psi_k(F_k)$. Then $\|u_k\| = 1$, $u_k \perp \Omega_k$ and $\|P_k u_k\| \ge (1 - \frac{1}{10})/\frac{11}{10} = \frac{9}{11}$. Let $c_k(t) = \langle u_k, \mathbb{T}_k^t u_k\rangle$. Since $\mathbb{T}_k^t \ge 0$ commutes with P_k and has norm γ_k^t on $\Omega_k^\perp$,

$$\tfrac{81}{121} e^{-(m_k + a_k/k)t} \le \langle P_k u_k, \mathbb{T}_k^t P_k u_k\rangle \le c_k(t) \le e^{-m_k t}. \tag{H.5}$$

Let σ_k be the spectral measure of $\mathbb{T}_k$ at u_k, a probability measure on $[0,1]$, and ρ_k its image under $x \mapsto -a_k^{-1}\log x$, a probability measure on the compact space $[0,\infty]$, with $0 \mapsto \infty$. Then $c_k(s/a_k) = \int e^{-sx}\, d\rho_k(x)$ for $s \in \mathbb{D}$, with $e^{-s\infty} = 0$ for $s > 0$. Pass to a subsequence with $\rho_k \to \rho$ weakly on $[0,\infty]$. For $s > 0$ the function $x \mapsto e^{-sx}$ is continuous on $[0,\infty]$, so with $\rho' = \rho|_{[0,\infty)}$,

$$B(s) := \lim_k c_k(s/a_k) = \int_{[0,\infty)} e^{-sx}\, d\rho'(x) \qquad (s \in \mathbb{D},\ s > 0).$$

Let $\mathcal{T}$ be the free complex vector space on the symbols $\mathbf{1}$ and j_s, $s \in \mathbb{D}$, $s > 0$; put $O_k(\mathbf{1}) = 1$ and $O_k(j_s) = \tau_{s/a_k} F_k$ for the k with $s/a_k \in \mathbb{Z}$, extended linearly. Let $\mathcal{H} = \mathbb{C}\Omega \oplus L^2([0,\infty), \rho')$ with $\Omega = (1,0)$, $H = 0 \oplus x$, $\psi(\mathbf{1}) = \Omega$ and $\psi(j_s) = (0, e^{-sx})$. Since $\Psi_k(O_k j_s) = \mathbb{T}_k^{s/a_k} u_k$, for $s, s' > 0$ and r in $\mathbb{D}$,

$$\omega_k\big(\vartheta(\overline{O_k j_s})\tau_{r/a_k} O_k j_{s'}\big) = c_k((s + s' + r)/a_k) \to B(s + s' + r) = \langle \psi j_s, e^{-rH}\psi j_{s'}\rangle;$$

moreover $\omega_k(\tau_{r/a_k} O_k j_s) = \langle \Omega_k, \mathbb{T}_k^{(r+s)/a_k} u_k \rangle = \langle \Omega_k, u_k \rangle = 0 = \langle \Omega, e^{-rH} \psi j_s \rangle$, the same holds with the roles exchanged, and $\omega_k(1) = 1 = \langle \Omega, e^{-rH}\Omega \rangle$. By sesquilinearity the identities of Definition 6.5 hold on $\mathcal{T}$. The span of $\{e^{-sx} : s \in \mathbb{D}, s > 0\}$ is an algebra of functions in $C_0([0,\infty))$ that separates points and vanishes nowhere, hence dense in $C_0([0,\infty))$ by the Stone–Weierstrass theorem and dense in $L^2(\rho')$; so ψ has dense range, and $H \geq 0$ is self-adjoint with $H\Omega = 0$.

By (H.5) with $t = s/a_k$, $\frac{81}{121} e^{-\bar{E}s} \leq B(s) \leq e^{-\bar{E}s}$ for $s \in \mathbb{D}$, $s > 0$. The lower bound gives $\rho' \neq 0$, so $\mathcal{H} \neq \mathbb{C}\Omega$. The upper bound gives $\rho'([0, \bar{E} - \eta]) = 0$ for every $\eta > 0$, and the lower bound gives $\rho'([\bar{E}, \bar{E} + \eta]) > 0$ for every $\eta > 0$. Since $\sigma(H|_{\Omega^\perp})$ is the support of ρ', its infimum is $\bar{E}$. □

Theorem H.27 uses no property of the gauge theory: it applies equally to the free scalar field with lattice masses tending to zero. A nontrivial continuum limit with a gap, in the sense of Definition 6.5, therefore exists as soon as the lattice gaps $m(\beta_k)$ are positive and tend to zero along a sequence $\beta_k \to \infty$, and neither property is known for four-dimensional SU(N). The limit so obtained is generated by one sequence of observables chosen through spectral projections; nothing identifies it with Yang–Mills theory, since it carries no Euclidean invariance, no locality and no Wilson loops. The physical content of (E) is the convergence of fixed physical observables, smeared fields and Wilson loops of fixed physical size, and by Lemma H.26 a gap coming from the transfer matrices requires m_k/a_k to stay bounded below along a subsequence: along the asymptotic-freedom tuning, the lattice gap must decay at exactly the asymptotic-freedom rate.

Appendix I. Degenerate constructions

A statement of the form "there is a quantum field theory on $\mathbb{R}^4$ with a mass gap" is easy to satisfy if nothing ties the theory to the Yang–Mills action: a generalised free field whose Källén–Lehmann measure is supported in $[\Delta^2, \infty)$ obeys the Wightman axioms and has $\sigma(H) \subset \{0\} \cup [\Delta, \infty)$. The problem as posed by Jaffe and Witten [31] excludes such answers by asking for the local gauge-invariant fields of the Yang–Mills theory and for its short-distance behaviour. This appendix records what happens to three natural shortcuts that stay inside the lattice theory: the limit of the parallel transporters themselves, without renormalisation; a theory built from two-dimensional Yang–Mills theory on parallel planes; and a lattice theory at fixed spacing. Each gives either a trivial theory or one that fails the axioms, and the first two statements are theorems. None of them is used elsewhere in the paper.

Throughout, $G \subset \mathrm{U}(N)$ is the gauge group, $\|\cdot\|$ is the operator norm, μ_L is the Wilson measure of the torus Λ_L with weight $\exp(N\beta \sum_p \operatorname{Re} \operatorname{Tr} Q_p)$, $\beta \geq 0$, as in Section 2, and each link lies in six plaquettes.

I.1. **One link.** Let π be a unitary representation of G on a finite-dimensional space V such that no nonzero vector is annihilated by $d\pi(\mathfrak{g})$; every nontrivial irreducible representation has this property, since a vector annihilated by $d\pi(\mathfrak{g})$ is fixed by the connected group G and spans an invariant line. Put

$$D_\pi = \sup_{\|X\| \leq 1} \|d\pi(X)\|, \qquad \kappa_\pi = \inf_{|v|=1} \sup_{\|X\| \leq 1} |d\pi(X)v|,$$

the suprema over $X \in \mathfrak{g}$. The function $v \mapsto \sup_{\|X\| \leq 1} |d\pi(X)v|$ is continuous and positive on the unit sphere, so $\kappa_\pi > 0$, and $\kappa_\pi \leq D_\pi$. Let

$$s_\pi = \min\Big\{1, \frac{\kappa_\pi}{2D_\pi^2}\Big\}, \qquad c_\pi = \frac{s_\pi^2 \kappa_\pi^2}{16(1+e^2)}.$$

Lemma I.1. *Let $\varphi\colon G \to \mathbb{R}$ satisfy $|\varphi(Qg) - \varphi(Q)| \le \Lambda\|g-1\|$ for all $Q, g \in G$, and let ν be the probability measure proportional to $e^{\varphi(Q)}dQ$. Then*

$$\Big\| \int \pi(Q)\, d\nu(Q) \Big\| \le 1 - \frac{c_\pi}{\max(1,\Lambda)^2}.$$

Proof. Write $M = \int \pi\, d\nu$, let u, v be unit vectors and $I = \int |\pi(Q)v - u|^2 d\nu(Q)$, so that $\mathrm{Re}\langle u, Mv\rangle = 1 - I/2$. For $g \in G$ and every Q,

$$|\pi(g)v - v| = |\pi(Qg)v - \pi(Q)v| \le |\pi(Qg)v - u| + |\pi(Q)v - u|,$$

hence $|\pi(g)v - v|^2 \le 2\int |\pi(Qg)v - u|^2 d\nu(Q) + 2I$. Haar measure is right invariant, so for $F \ge 0$

$$\int F(Qg) e^{\varphi(Q)} dQ = \int F(Q) e^{\varphi(Qg^{-1})} dQ \le e^{\Lambda\|g-1\|} \int F(Q) e^{\varphi(Q)} dQ,$$

where $\|g^{-1} - 1\| = \|g - 1\|$ was used. Therefore

$$|\pi(g)v - v|^2 \le 2\big(1 + e^{\Lambda\|g-1\|}\big) I. \tag{I.1}$$

Choose $X \in \mathfrak{g}$ with $\|X\| \le 1$ and $|d\pi(X)v| \ge \kappa_\pi$, put $s = s_\pi / \max(1,\Lambda) \le 1$ and $g = e^{sX}$. Then $\|g-1\| \le e^s - 1 \le 2s$, so $\Lambda\|g-1\| \le 2$. Since $sD_\pi \le \kappa_\pi/(2D_\pi) \le \frac12$ and $e^x - 1 - x \le x^2$ on $[0,1]$,

$$|\pi(g)v - v - s\, d\pi(X)v| \le e^{sD_\pi} - 1 - sD_\pi \le s^2 D_\pi^2,$$

and $s \le \kappa_\pi/(2D_\pi^2)$ gives $|\pi(g)v - v| \ge s\kappa_\pi - s^2 D_\pi^2 \ge s\kappa_\pi/2$. With (I.1), $I \ge s^2\kappa_\pi^2/(8(1+e^2))$, that is, $\mathrm{Re}\langle u, Mv\rangle \le 1 - c_\pi/\max(1,\Lambda)^2$. The supremum over u and v is $\|M\|$. □

I.2. **Straight segments.**

Proposition I.2. *Let $e_1, \dots, e_n$, $n \ge 3$, be consecutive collinear links of Λ_L, $L \ge n+1$, and $U = Q_{e_1} \cdots Q_{e_n}$. Let π be as in Lemma I.1, and let $\Phi(A; Q')$ be linear in $A \in \mathrm{End}(V)$ and bounded measurable in the links $Q' = (Q_e)_{e \ne e_2, \dots, e_{n-1}}$, with $|\Phi(A;Q')| \le C\|A\|$. Then*

$$|\mu_L(\Phi(\pi(U); Q'))| \le C\Big(1 - \frac{c_\pi}{\max(1, 6N^2\beta)^2}\Big)^{n-2}.$$

Proof. Two distinct links lie in a common plaquette only if they are perpendicular and share a site, or are parallel and differ by one step in a transverse direction; collinear links never do. So every plaquette contains at most one of the links $e_2, \dots, e_{n-1}$, and conditionally on Q' these links are independent, the link e having the law proportional to $\exp(N\beta\, \mathrm{Re}\,\mathrm{Tr}(QA_e))dQ$ with A_e a sum of six unitary matrices that depend on Q'. For $g \in G$,

$$|\,\mathrm{Tr}(Q(g-1)A_e)| \le \|g-1\|\ \mathrm{Tr}\,|A_e Q| \le 6N\|g-1\|,$$

so the exponent satisfies the hypothesis of Lemma I.1 with $\Lambda = 6N^2\beta$. The function $\Phi(\pi(Q_{e_1})\pi(Q_{e_2})\cdots\pi(Q_{e_n}); Q')$ is linear in each $\pi(Q_e)$, $e \in \{e_2, \dots, e_{n-1}\}$, separately, so its conditional expectation given Q' is $\Phi(\pi(Q_{e_1})M_{e_2}\cdots M_{e_{n-1}}\pi(Q_{e_n}); Q')$ with $M_e = \int \pi\, d\nu_e$, and its modulus is at most $C\prod_e \|M_e\|$. The lemma bounds each factor. □

I.3. **The limit of the transporters.** Let $a_k = 2^{-k}a_0$ and let $\beta_k \ge 0$ be couplings with

$$a_k \max(1, \beta_k)^2 \longrightarrow 0. \tag{I.2}$$

This holds whenever β_k grows at most like a power of $\log(1/a_k)$, in particular for the tuning $\beta_k \sim \log(1/a_k)$ suggested by asymptotic freedom in four dimensions. Fix j and a finite set B of bonds of the lattice $a_j\mathbb{Z}^4$. For $k \ge j+2$ a bond $b \in B$ is a straight segment of 2^{k-j} links of $a_k\mathbb{Z}^4$; let $U_b^{(k)}$ be the ordered product of the link variables along it, and let $\mu^{(k)}$ be the Wilson measure at β_k on a torus of any side large enough to contain B.

Theorem I.3. *Under* (I.2)*, for every continuous* $F\colon G^B \to \mathbb{C}$*,*

$$\mu^{(k)}\big(F((U_b^{(k)})_{b\in B})\big) \longrightarrow \int F \prod_{b\in B} dQ_b \qquad (k\to\infty).$$

The parallel transporters along the bonds of any fixed lattice become independent and Haar distributed.

Proof. By the Peter–Weyl theorem the finite linear combinations of the products of matrix coefficients $\prod_b \langle u_b, \pi_b(Q_b)v_b\rangle$, with π_b irreducible and u_b, v_b unit vectors, are uniformly dense in $C(G^B)$, and both sides are continuous for the uniform norm; it is enough to treat one product. If every π_b is trivial the two sides are the same constant. Otherwise let π_{b_0} be nontrivial; the Haar integral vanishes. Distinct bonds of $a_j\mathbb{Z}^4$ share no link of $a_k\mathbb{Z}^4$, so Proposition I.2 applies to the segment b_0 with $n = 2^{k-j}$,

$$\Phi(A;Q') = \langle u_{b_0}, Av_{b_0}\rangle \prod_{b\neq b_0} \langle u_b, \pi_b(U_b^{(k)})v_b\rangle, \qquad C = 1,$$

and bounds the left side by

$$\Big(1 - \frac{c}{\max(1, 6N^2\beta_k)^2}\Big)^{2^{k-j}-2} \le \exp\Big(-\frac{c\,(a_j/a_k - 2)}{\max(1, 6N^2\beta_k)^2}\Big), \qquad c = c_{\pi_{b_0}},$$

which tends to 0 by (I.2). □

Gauge invariance of F was not used. For the trace of the holonomy of a lattice loop the theorem says that every Wilson loop of fixed physical size, in every nontrivial irreducible representation, tends to its value at $\beta = 0$, which is zero for a simple loop. This is the perimeter divergence in rigorous form, and it is the reason why the fields of a continuum limit cannot be the transporters or the loops themselves: in the limit they carry no correlation at all. Condition (I.2) is where the dimension enters; in two dimensions the coupling grows like a^{-2} and Wilson loops have nontrivial limits.

I.4. **Theories constant along a plane.** Two-dimensional Yang–Mills theory is exactly solvable and has a gap, and one may try to read it as a theory on $\mathbb{R}^4$ that does not depend on two of the coordinates.

Proposition I.4. *Let* $(\mathcal{H}, U, \Omega, \phi)$ *be a Wightman theory on* $\mathbb{R}^{1+3}$ *with a unique vacuum that is cyclic for the fields, and suppose* $\phi(x + c) = \phi(x)$ *for every* c *in a spacelike two-plane* E*. Then* $\mathcal{H} = \mathbb{C}\Omega$*.*

Proof. For $c \in E$ the unitary $U(c)$ commutes with the fields and fixes Ω, so $U(c) = 1$ by cyclicity. The joint spectrum Σ of the energy and momentum is then contained in the two-plane $E^\perp$, which in suitable coordinates is $\{p_2 = p_3 = 0\}$. It is also a closed Lorentz-invariant subset of the closed forward cone. If $p \in \Sigma$ and $p \neq 0$, then either $p_1 \neq 0$, and a rotation in the plane of the first and second coordinates moves p out of $\{p_2 = 0\}$, or $p_1 = 0$ and $p_0 > 0$, and a boost in the second direction does. Hence $\Sigma = \{0\}$, every vector is translation invariant, and the uniqueness of the vacuum gives $\mathcal{H} = \mathbb{C}\Omega$. □

Independent copies of the two-dimensional theory on the parallel planes $E^\perp + z$ fare no better: the average $\epsilon^2 \sum_z g(\epsilon z)X_z$ of independent identically distributed variables against a test function has variance $\epsilon^4 \sum_z g(\epsilon z)^2 \operatorname{Var} X \to 0$, so every field smeared across the planes is a constant, and no rotation mixes the planes.

I.5. **Fixed lattice spacing.** The lattice theory of Theorem A has a gap at a fixed spacing a, and its fields may be regarded as distributions on $\mathbb{R}^4$ supported on $a\mathbb{Z}^4$. The Schwinger functions are then invariant under the lattice translations and the hyperoctahedral group only. Averaging over a uniform shift restores translation invariance, but not rotation invariance, and it destroys reflection positivity. The mechanism is visible in one dimension. Let $(\xi_n)_{n\in\mathbb{Z}}$ be the stationary Gaussian sequence with covariance $\lambda^{|n-m|}$, $0 < \lambda < 1$, which is

reflection positive, and let $\phi_u = \sum_n \xi_n \delta_{n+u}$ with u uniform on $[0,1)$. The averaged two-point function is $\sum_{k\in\mathbb{Z}} \lambda^{|k|}\delta(t-s-k)$, and for a test function f supported in $(0,1)$ the reflection form is

$$\sum_k \lambda^{|k|} \int \overline{f(t)} f(-t-k)\,dt = \lambda \int_0^1 \overline{f(t)} f(1-t)\,dt,$$

only $k=-1$ contributing. For f odd about $\frac{1}{2}$ this equals $-\lambda \int |f|^2 < 0$.

The three shortcuts fail for one reason. The mass gap is a statement about the renormalised local fields at a distance fixed in physical units, and a construction that avoids renormalisation, or avoids four dimensions, or avoids the continuum, does not produce those fields. What is left open in this paper is the statement of Theorem F: the Wightman limit (W) and the one-scale bound (B*) of Section 7.11.

Appendix J. Summary tables

The four tables of this appendix collect what is spread over the text: the hypotheses with their status, the constants, and the contents of the two supporting checks.

Table 3. The hypotheses of the paper and their status. "Strong coupling" refers to the ranges of Theorems A and C.

Name	Statement	Status
(W)	the torus construction has a limit that is a mixture of Wightman theories with unique vacua	input from [45, 62], unrefereed; not proved here
(D)	cube condition BP(n_k, ϖ) with $n_k a_k \le \ell$ along the tuning	proved at strong coupling and for massive free fields; open at weak coupling
(D_ℓ)	the same on cylinders	follows from (M) (Proposition 7.13); open at weak coupling
(M)	slab decorrelation below $\frac{1}{2}$, uniform in the boundary condition	proved at strong coupling and in two dimensions for fat boxes; open in four dimensions at weak coupling
(U), (U_w)	the two orders of the volume limit agree	follow from (Z) (Corollary 7.11)
(Z)	$\mathcal{Z}_{N,p} \to 1$	proved at strong coupling and in two dimensions; follows from (B*) (Theorem F)
(G_{24}), (B_{24})	gap and pressure bound on elongated tori	imply (Z) (Theorem 7.54); both follow from (B*) (Theorem 7.56)
(B*), (B**)	$\log \mathcal{Z}_s(T_0) \le \gamma V(s)$ at one time extent; $Z(s,T_0)^2 \le e^{\gamma V(s)} Z(s,2T_0)$	equivalent up to a factor two in γ (Proposition 7.57); open at weak coupling
(E)	a nontrivial continuum limit exists	open

TABLE 4. The principal constants.

Constant	Meaning	Where
$m_G(\beta)$	lattice gap $\operatorname{arccosh}\big(1+\frac{1}{12}(\frac{\kappa_G}{N\lvert\beta\rvert}-\frac{29}{2})\big)$, $N\lvert\beta\rvert<\frac{2}{29}\kappa_G$	Theorem A
$-\log(3\beta_W)$	lattice gap of SU(2), $\beta_W<\frac{1}{3}$	Theorem A
$16\varpi/(625e^2\ell)$	continuum gap from the cube condition	Theorems B, E
$\frac{1}{2}$	sharp threshold of the slab decorrelation	Theorem D
$y_0\le\frac{1}{20}$, $y_c=0.0589\ldots$	normalised pressure at the base scale and the fixed point of $y\mapsto 16(e^y-1)^2$	Theorem 7.56
$u_0=16.83\,y_0\le 0.8415$	contraction per base time	Theorem 7.56
$0.172/t_0$	lower bound of the gap $\log(1/u_0)/t_0$	Theorem F
$1.9\cdot 10^{-11}$	prefactor of $\mathcal{Z}_n(n)-1$	Theorem 7.56(c)
$\varepsilon(c)=12(\log c+5.42)/c$	mean plaquette action of SU(2) at $c=2\beta$	Theorem 7.59
$c_\pi/\max(1,6N^2\beta)^2$	loss per link of a straight segment	Proposition I.2

TABLE 5. The Lean 4 project: files, numbers of theorems, contents. The total is 265 theorems in twenty files.

File	Theorems	Contents
`AveragedCondition`	38	counting of cubes and translates, SU(2) trace identity, constants of $\mathrm{BP}(n,\varpi)$
`CumulantCounterexample`	3	bounded local terms give no volume-uniform cumulant bound
`GapFree`	11	chord bound, the recursion $16(e^y-1)^2$, constants of Theorem F, Haar ball, base scale
`HolleyStroock`	11	perturbation lemma on a finite state space
`LimitRatio`	1	the critical scaling of the lattice gap
`OneScale`	88	one-scale reduction, doubling, rings, Casimir defect, pressure bound, ascent
`RunningCoupling`	5	one-loop running coupling and the intermediate window
`SpectralNoGo`	9	no gap uniform in lattice units along a continuum limit
`WilsonAction`	9	bounds on the Wilson action and its oscillation
`StrongCoupling/BlockGauge`	5	block-rooted axial gauge on blocks of side two
`StrongCoupling/CouplingMinimality`	4	discrepancy recurrence from minimality
`StrongCoupling/CouplingMinimizer`	5	a continuous cost attains its minimum on a compact space
`StrongCoupling/CubeOpen`	11	arithmetic of perturbations of the cube condition
`StrongCoupling/DobrushinComparison`	4	arithmetic core of the Wasserstein comparison
`StrongCoupling/GaugeFix`	7	one fixed link in every plaquette, twelve partners
`StrongCoupling/IslandLayer`	14	layer bound and block-map growth
`StrongCoupling/QxCoefficients`	9	coefficient positivity for the sharp quarter
`StrongCoupling/Thresholds`	7	Dobrushin threshold of SU(2)
`StrongCoupling/WassersteinBridge`	15	resolvent comparison and coupling interface
`StrongCoupling/ZonalBound`	9	Riccati comparison for the zonal bound
`YangMills (root)`	0	imports the nineteen files above

TABLE 6. The Julia script: groups of checks and their numbers. All 234 checks pass.

Group	Checks
Theorem A: the lattice gap $m_G(\beta)$ for SU(N), $\kappa_G = N/2$	3
Theorem B: the constant 16/625	3
Theorem C(i): SU(2) weighted count and SU(3) range	2
Appendix H: the single-link law of SU(3)	23
Theorem D and its cylinder form	7
Thermal discrepancy lemma: the sharp two-state example	4
Theorem E: the cylinder gap and $m_n N \geq \Delta$	1
Closed polymers, the strong-coupling range and the one-scale reduction	35
Doubling defects, symmetrised rings and the finite-size bound	22
The gap against the Casimir defect	22
The pinned two-state chain and the field rings	11
The Casimir defect of the free field	17
The pressure bound at one scale	33
Extensivity and the ascent in time	12
The field rings with an arbitrary field	12
Strong coupling, uniform in the boundary condition	5
The ascent without a gap hypothesis (Theorem F)	14
The gap closes at a fixed lattice size (SU(2))	5
Appendix: the one-link constant	3

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The constructive framework is that of Osterwalder, Schrader and Seiler, the curvature criterion that of Bakry and Émery, and the uniqueness criterion that of Dobrushin. The maximal correlation of Gaussian variables is Gebelein's, the finite-size criterion follows Knabe and Gosset–Mozgunov, the two-dimensional analysis rests on the heat kernel of the continuum measure of Driver, Gross, King, Sengupta and Lévy, and the twisted boundary conditions are 't Hooft's; the transfer matrix of a cylinder is built as in Lüscher's construction, and the positivity argument for its leading eigenvector is Jentzsch's. The Lean 4 formalisation of the strong-coupling analysis described in the introduction is the work of I. J. Sessford, and Lean files for the lattice part of the project were written by J. M. Lockwood.

AUTHOR CONTRIBUTIONS

Deep Bhattacharjee, who leads the project, posed the problem in the form treated here, carried out the research and wrote the paper. The proofs are his: the lattice gap and its constants, the cube condition and Theorems B to E, the transfer matrix of the cylinder, the reduction to one scale, the ascent and Theorem F, and the appendices. He made all the computations: the Lean 4 project of Table 5, the Julia script of Table 6 with its exact rational evaluations, the numerical checks of the constants, and the figures; and he brought the work to its present state. James Montgomery Lockwood wrote Lean developments for the lattice part and proposed the block-projection metrics examined in earlier versions. Tristan Romanov contributed to the strong-coupling analysis, in particular the layer bound and the block responses, ran numerical experiments on the cube gap of SU(2), and supplied computing tools. Ikailo John Sessford formalised the strong-coupling analysis on periodic lattices in Lean 4, in the separate development described in the introduction. Ushashi Bhattacharya read the manuscript and verified its arguments and references. Sanjeevan Singha Roy read the manuscript. All authors approved the final version.

OPEN GLUON AND ANAXIA

Two tools built by D. B. stand behind the way this project was organised. Open Gluon is a workspace he assembled and maintains, which keeps pinned copies of the mathematical

libraries, the source record of the project and the checks in one reproducible place. Anaxia, the engine of the eBLLM project, is his construction engine: it stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact rational arithmetic. He wrote both and developed them from the constructions of his earlier papers. Three of their constructions were used here. The first is exact certification: the Julia script recomputes the constants of this paper, in exact rational arithmetic wherever the value is rational. The second is the transfer of a boundary response between a block law and its restriction to the complement of a rare event, which was developed inside this project in its earlier versions, where the cube condition of Section 6 originated. The third is tightness with observables that do not escape to infinite energy, the route whose failure test, that a unit normalisation does not prevent the mass from escaping, is illustrated by Theorem I.3. The engine proposes routes and tests them; every statement of this paper is proved in the text.

Declarations

Data availability. No data were generated. The paper, the Lean 4 project and the Julia script are archived at doi:10.5281/zenodo.24147967. The Lean project builds over Mathlib without unproved statements and without added axioms; its audit script lists 168 named results, of which 153 depend on the three standard axioms of Lean (propositional extensionality, choice and quotient soundness), 10 on propositional extensionality alone, and 5 on none. It covers the finite and algebraic parts of the arguments, file by file as in Table 5; the analytic core, that is, the curvature bounds, Theorems B to E, the transfer matrix of a cylinder and Appendix F, is not formalised. The Julia script recomputes the explicit constants of the paper in the groups of Table 6, and its 234 checks pass. Nothing in the paper rests on either: every statement has its analytical proof in the text.

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